VU MEDICAL ZONE

ADMIN:

AMAAN KHAN

BIOCHEMISTRY-I

BIO 202

LECTURE 1 TO 112

MIDTERM MERGE PPT

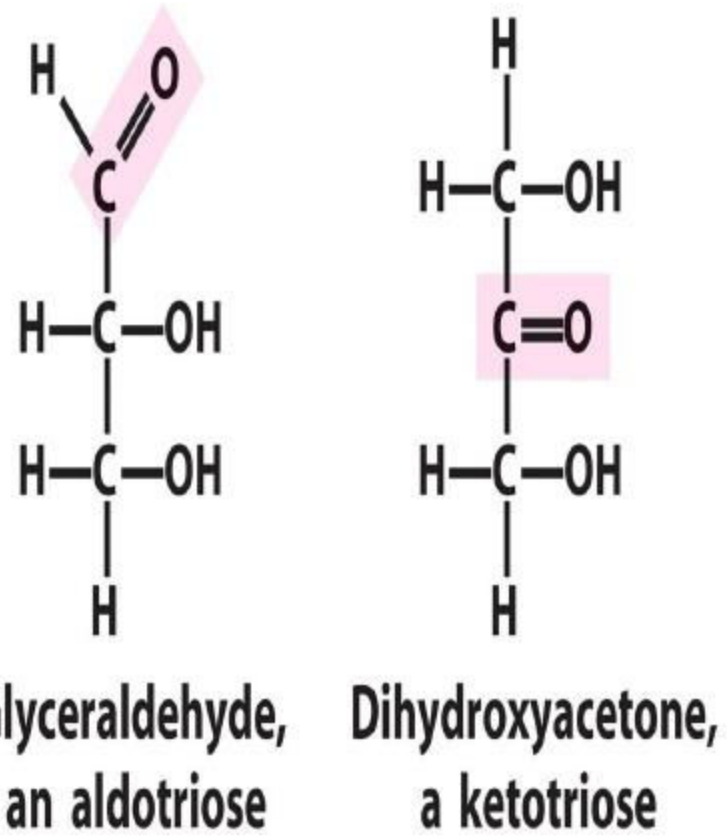
Carbohydrates

LECTURE NO 1

Carbohydrates

- Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis

- Many, but not all, carbohydrates have the empirical formula $(CH_2O)_n$; some also contain nitrogen, phosphorus or sulfur



Carbohydrates

Classification

- Carbohydrates are classified into **four** major groups:

– Monosaccharides

– Oligosaccharides

- Disaccharides

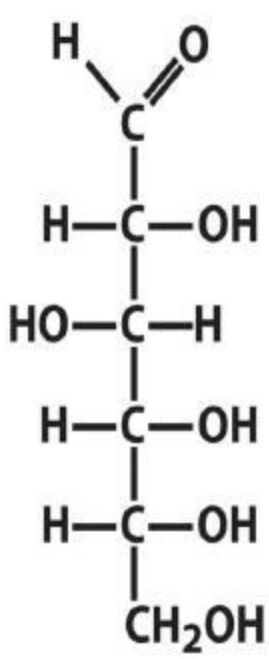
– Polysaccharides

Carbohydrates

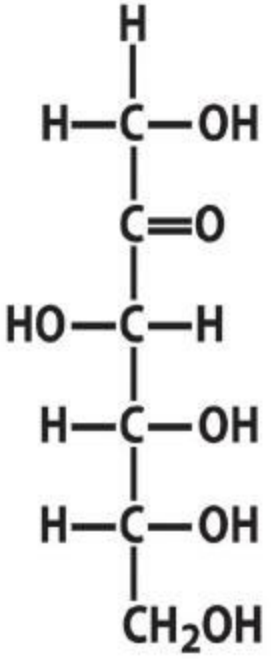
- **Monosaccharides** (simple sugars) are those which cannot be hydrolyzed further into simpler forms
- The backbones of common monosaccharides are *unbranched* carbon chains in which all the carbon atoms are linked by single bonds

Carbohydrates

- In **the open-chain form,** one of the carbon atoms is double-bonded to an oxygen atom to form a ***carbonyl group***
- Each of the other carbon atoms has a hydroxyl group



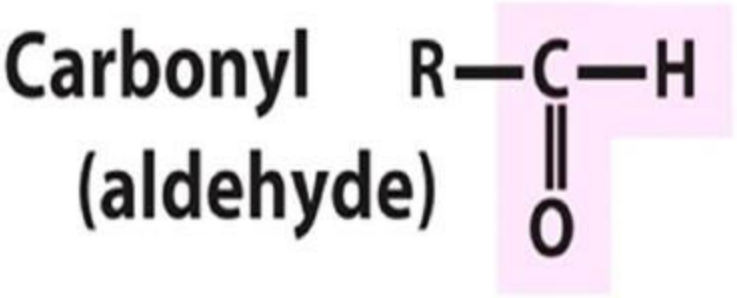
D-Glucose,
an aldohexose



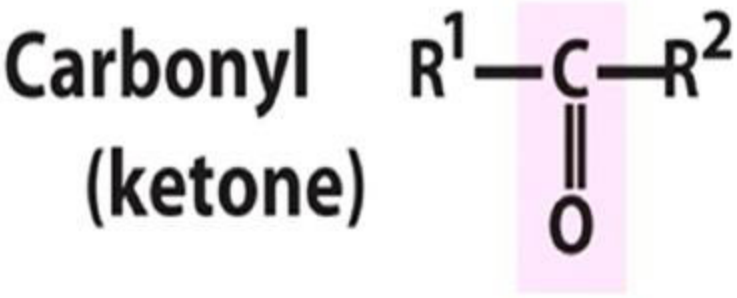
D-Fructose,
a ketohexose

Carbohydrates

- If the carbonyl group is at an end of the carbon chain (that is, in an aldehyde group) the monosaccharide is an ***aldose***

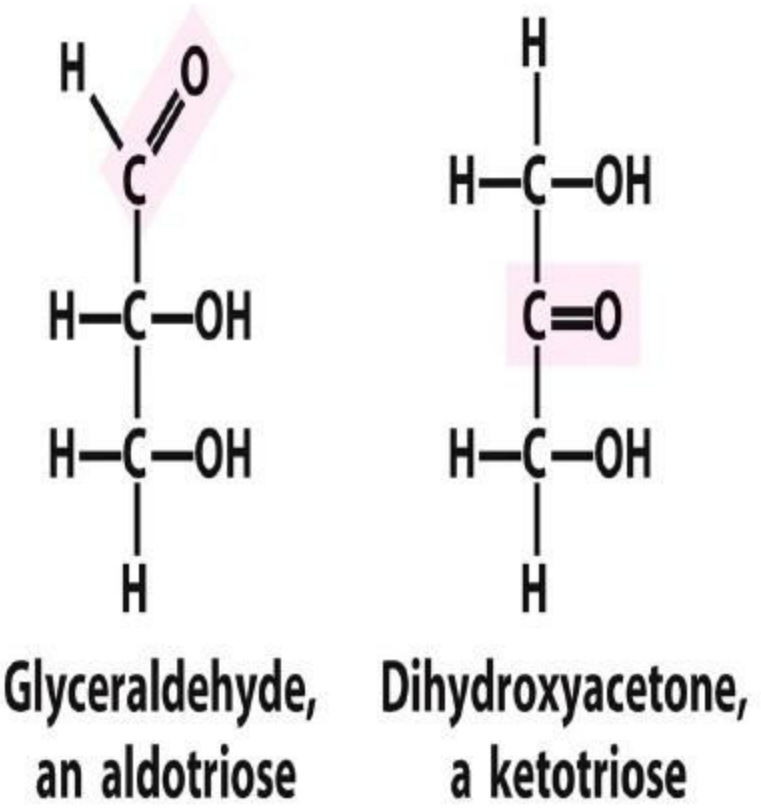


- If the carbonyl group is at any other position (in a ketone group) the monosaccharide is a ***ketose***



Carbohydrates

- **The simplest** monosaccharides are the two three-carbon trioses:
  - Glyceraldehyde,an aldotriose
  - Dihydroxyacetone, a ketotriose



Carbohydrates

- The carbons of a sugar are numbered beginning at the end of the chain nearest the carbonyl group

- All common monosaccharides and disaccharides have names ending with the suffix “-ose.”

Carbohydrates

- Ketoses are designated by inserting "ul" into the name of a corresponding aldose
- For example, ribulose is the ketopentose corresponding to the aldopentose ribose
- There are a few exceptions however, for example, fructose and dihydroxyacetone

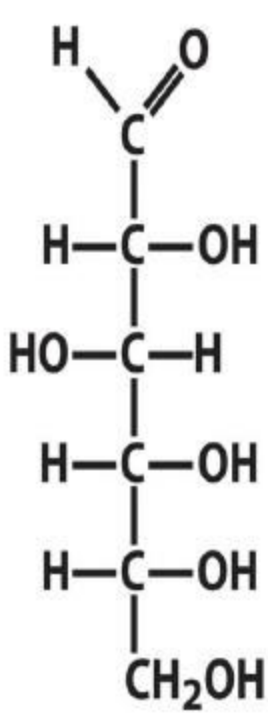
Carbohydrates

- Monosaccharides can be subdivided further depending upon:
  - Number of carbon atoms
  - Whether aldehyde or ketone groups are present as *aldoses* or *ketoses*

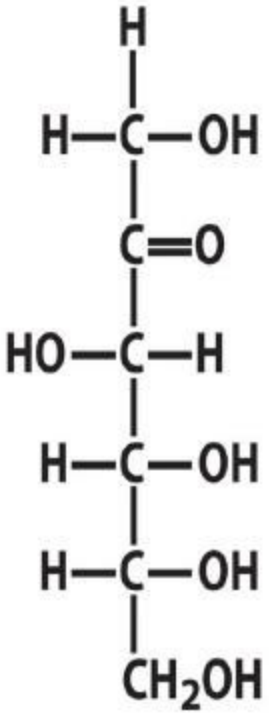
General formula	Aldosugars	Ketosugars
Trioses $C_3H_6O_3$	Glyceraldehyde	Dihydroxyacetone
Tetroses $C_4H_8O_4$	Erythrose	Erythrulose
Pentoses $C_5H_{10}O_5$	Ribose	Ribulose
Hexoses $C_6H_{12}O_6$	Glucose	Fructose

Carbohydrates

- The **hexoses** which include the aldohexose D-glucose and the ketohexose D-fructose, are the most common monosaccharides in nature



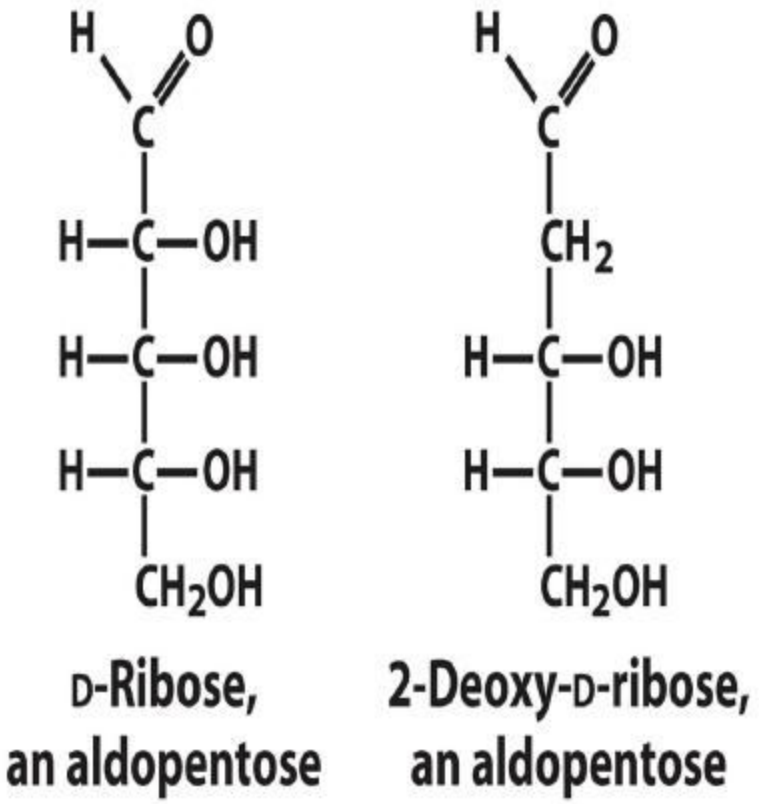
D-Glucose, an aldohexose



D-Fructose, a ketohexose

Carbohydrates

- The **aldopentoses** D-ribose and 2-deoxy-D-ribose are components of nucleotides and nucleic acids

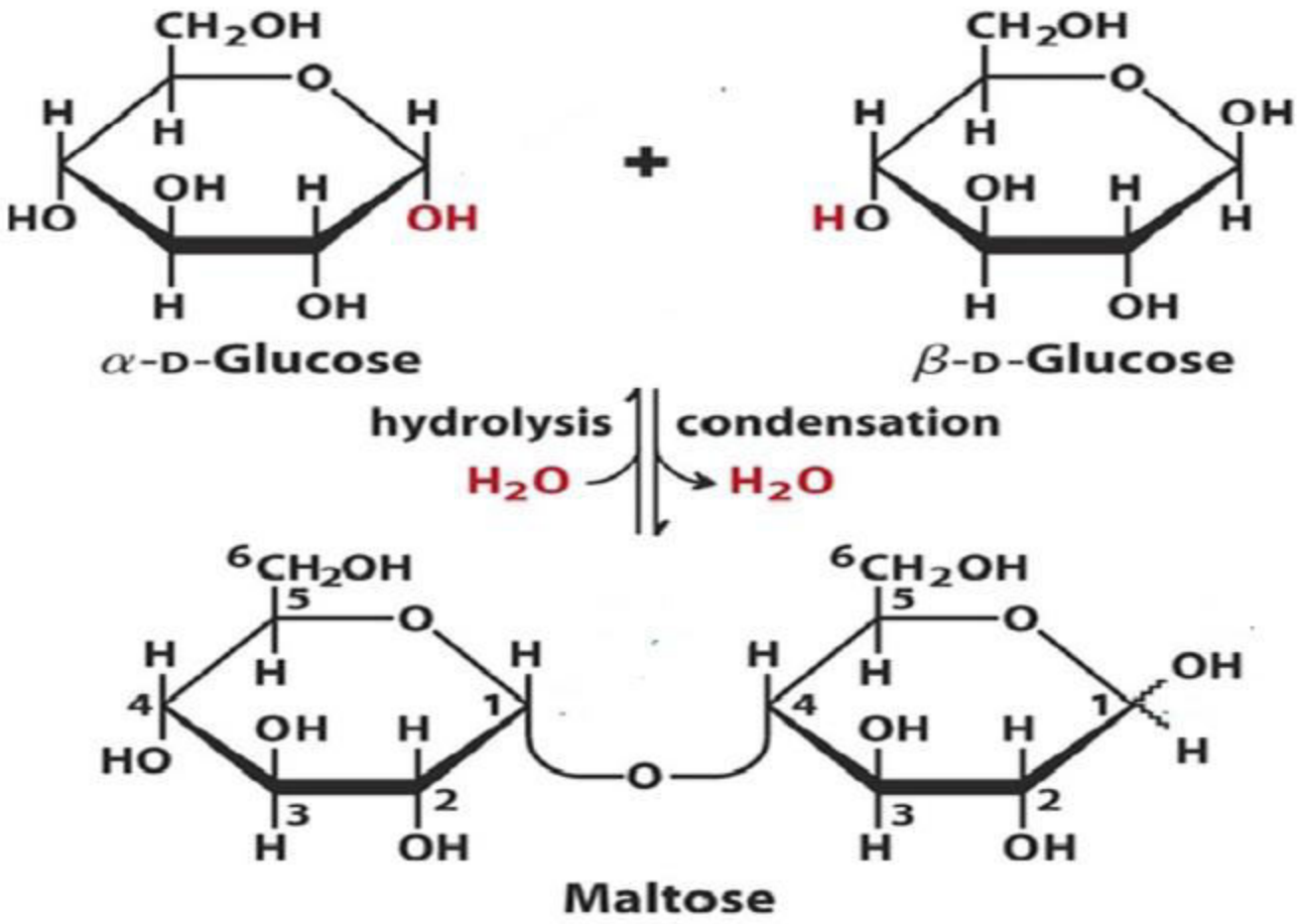


Carbohydrates

LECTURE NO 2

- **Disaccharides** are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis
  - Maltose - 2 molecules of glucose on hydrolysis

Carbohydrates

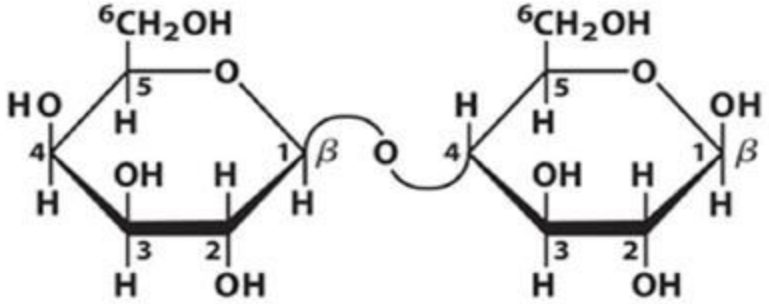


Carbohydrates

- Lactose – 1 glucose & 1 galactose on hydrolysis

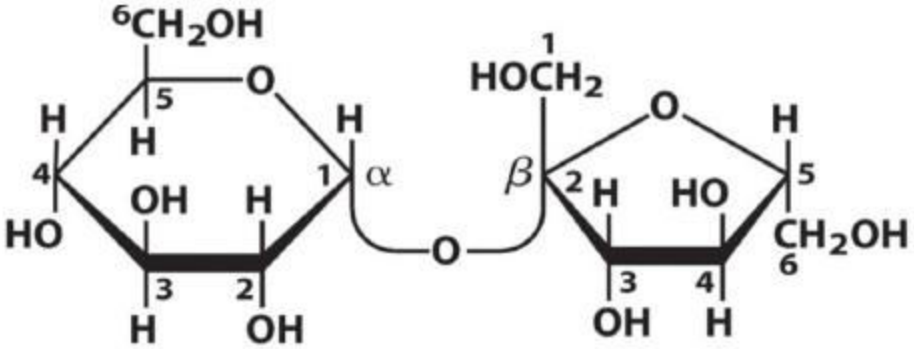
- Sucrose – 1 glucose & 1 fructose on hydrolysis

Carbohydrates



Lactose

Galactose Glucose



Sucrose

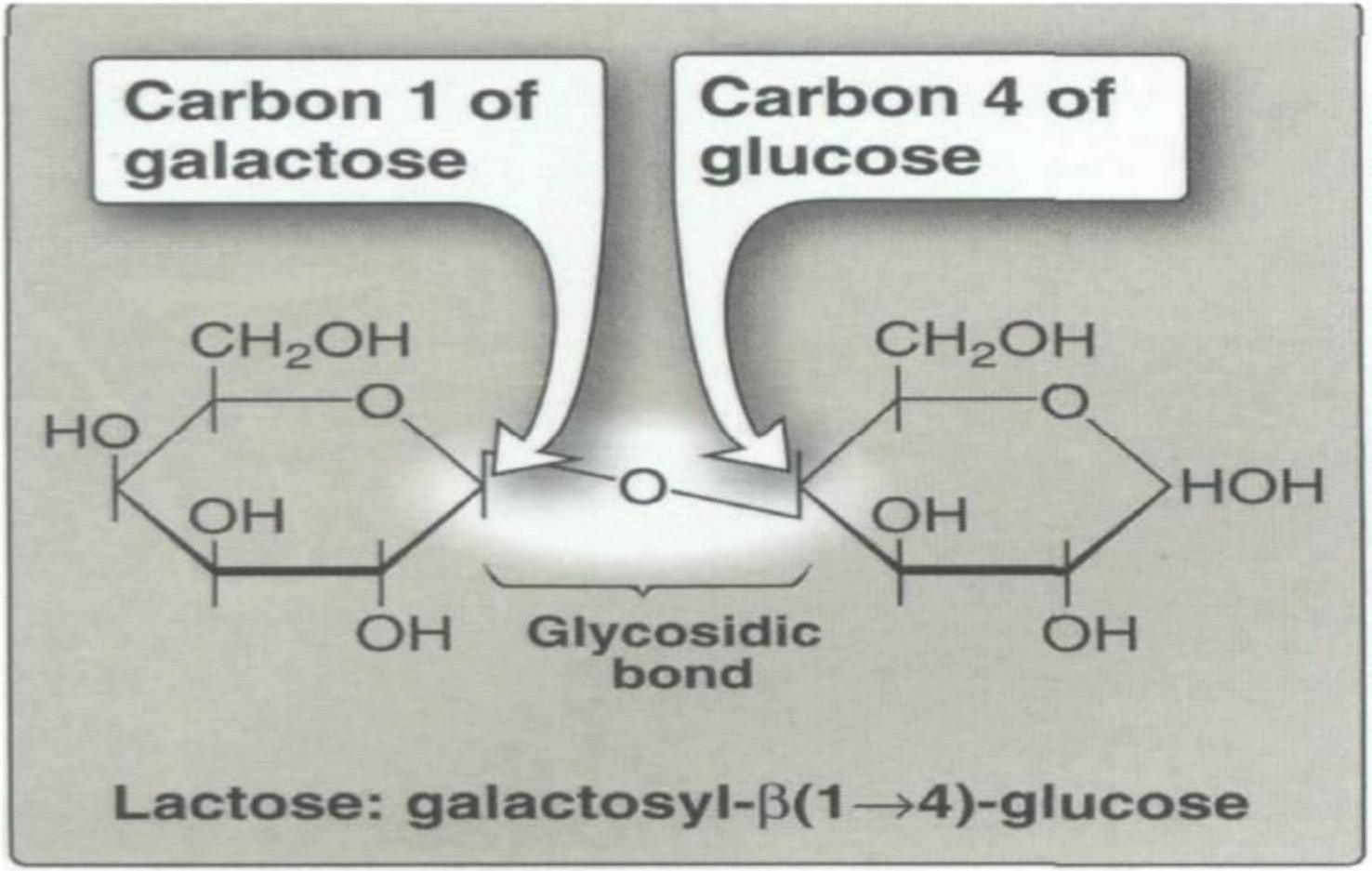
α-D-glucopyranosyl β-D-fructofuranoside

Glc(α1↔2β)Fru

Carbohydrates

- **Oligosaccharides** are short chains of monosaccharide units or residues (3-10) joined by characteristic linkages called glycosidic linkage

Carbohydrates

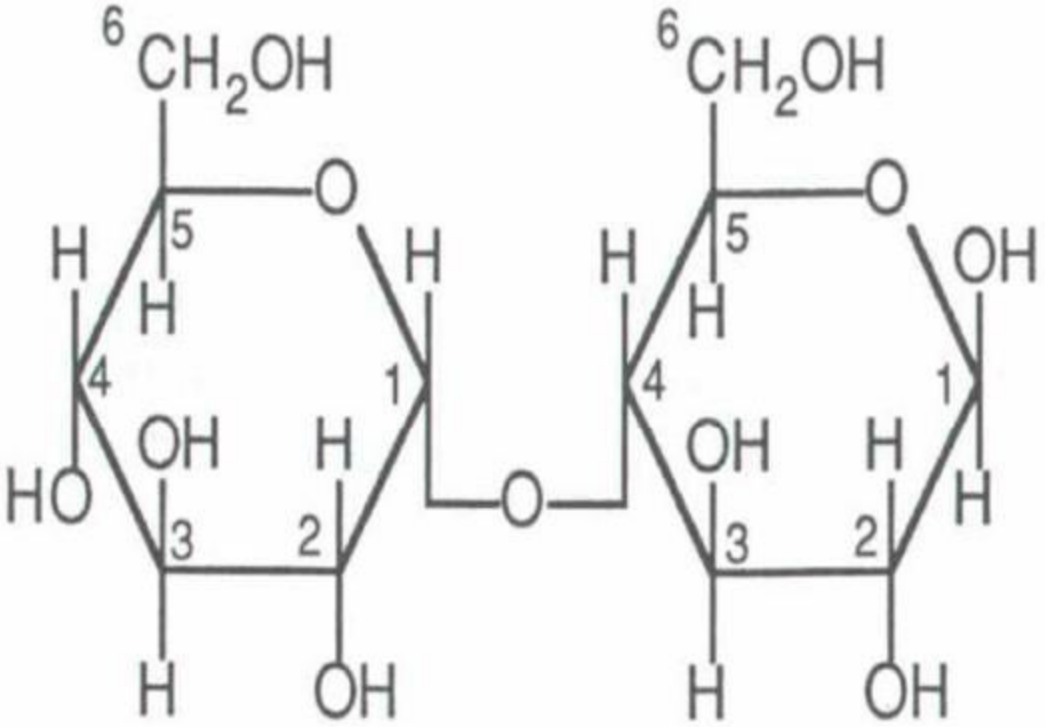


Carbohydrates

Glycosidic bond

of maltose is formed between the OH of carbon 1 and carbon 4 of 2nd glucose monomers. Thereforer, it forms an alpha (1– 4) glycosidic bond.

Carbohydrates



Maltose

Carbohydrates

LECTURE NO 3

- **Polysaccharides** (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis
- Some have hundreds or thousands of units
- Some polysaccharides, such as ***cellulose***, are linear chains; others, such as ***glycogen***, are branched

Carbohydrates

- Both glycogen and cellulose consist of recurring units of D-glucose, but they differ in the type of glycosidic linkage and consequently have strikingly different properties and biological roles

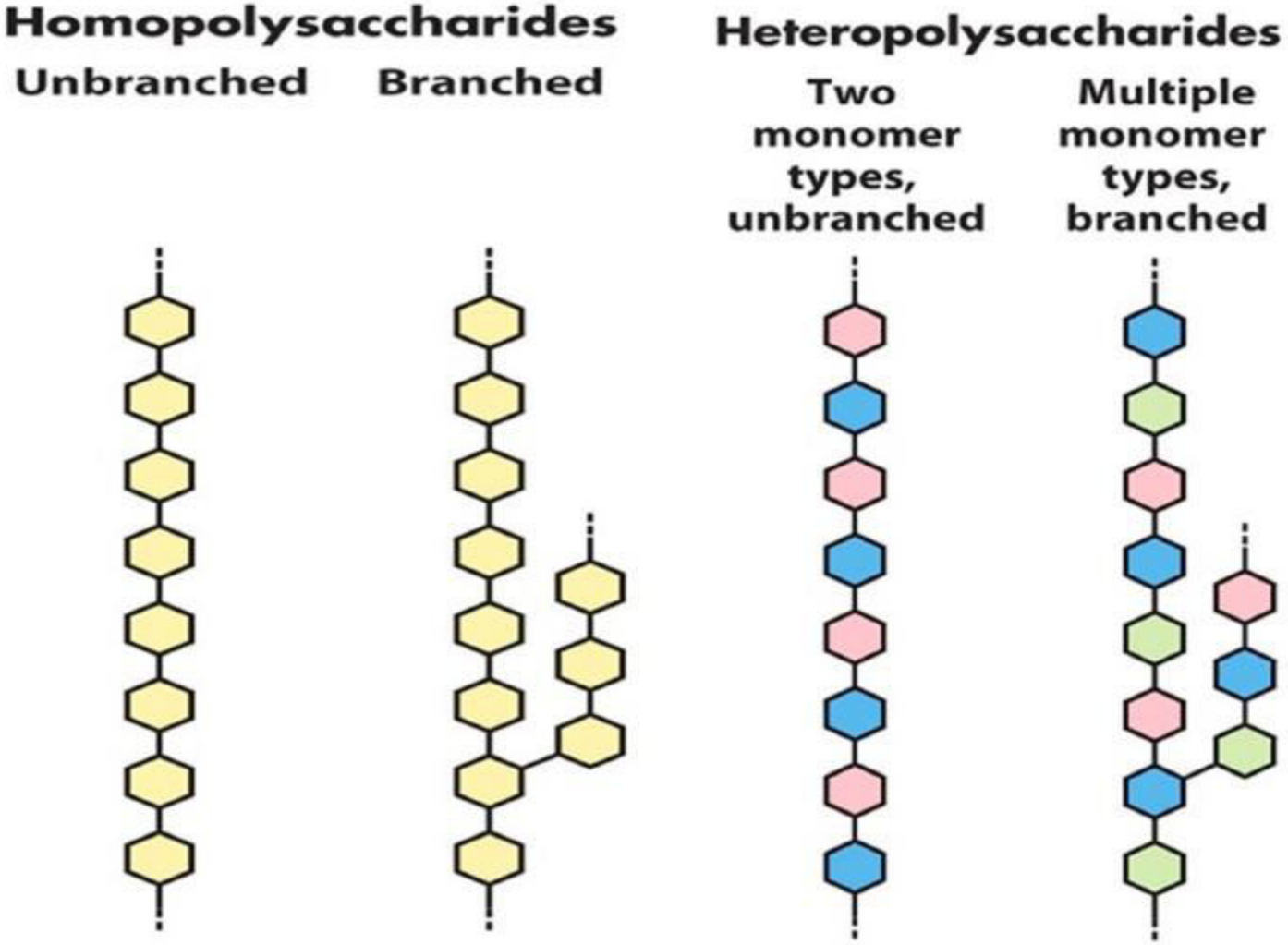
Carbohydrates

- Polysaccharides are further divided into two groups:

1. Homopolysaccharides (homoglycans):
Polymers of same monosaccharide units e.g. starch, glycogen, inulin, cellulose, dextrins, dextrans

2. Heteropolysaccharides (heteroglycans):
Polymer of different monosaccharide units or their derivatives e.g. Mucopolysaccharides (glycosaminoglycans)

Carbohydrates



Carbohydrates

Biomedical Importance

- Chief source of energy
- Constituents of compound lipids and conjugated proteins
- Lactose , the principal sugar of milk
- Degradation products utilized for synthesis of other substances such as fatty acids, amino acids, cholesterol, etc

Carbohydrates

- Constituents of mucopolysaccharides which form the ground substance of mesenchymal tissues
- Certain carbohydrate derivatives are used as drugs like **cardiac glycosides.**

Carbohydrates

- Inherited deficiency of certain enzymes in metabolic pathways of different carbohydrates can cause diseases e.g. **galactosemia, glycogen storage diseases, lactose intolerance,** etc
- Derangement of glucose metabolism is seen in **Diabetes mellitus**

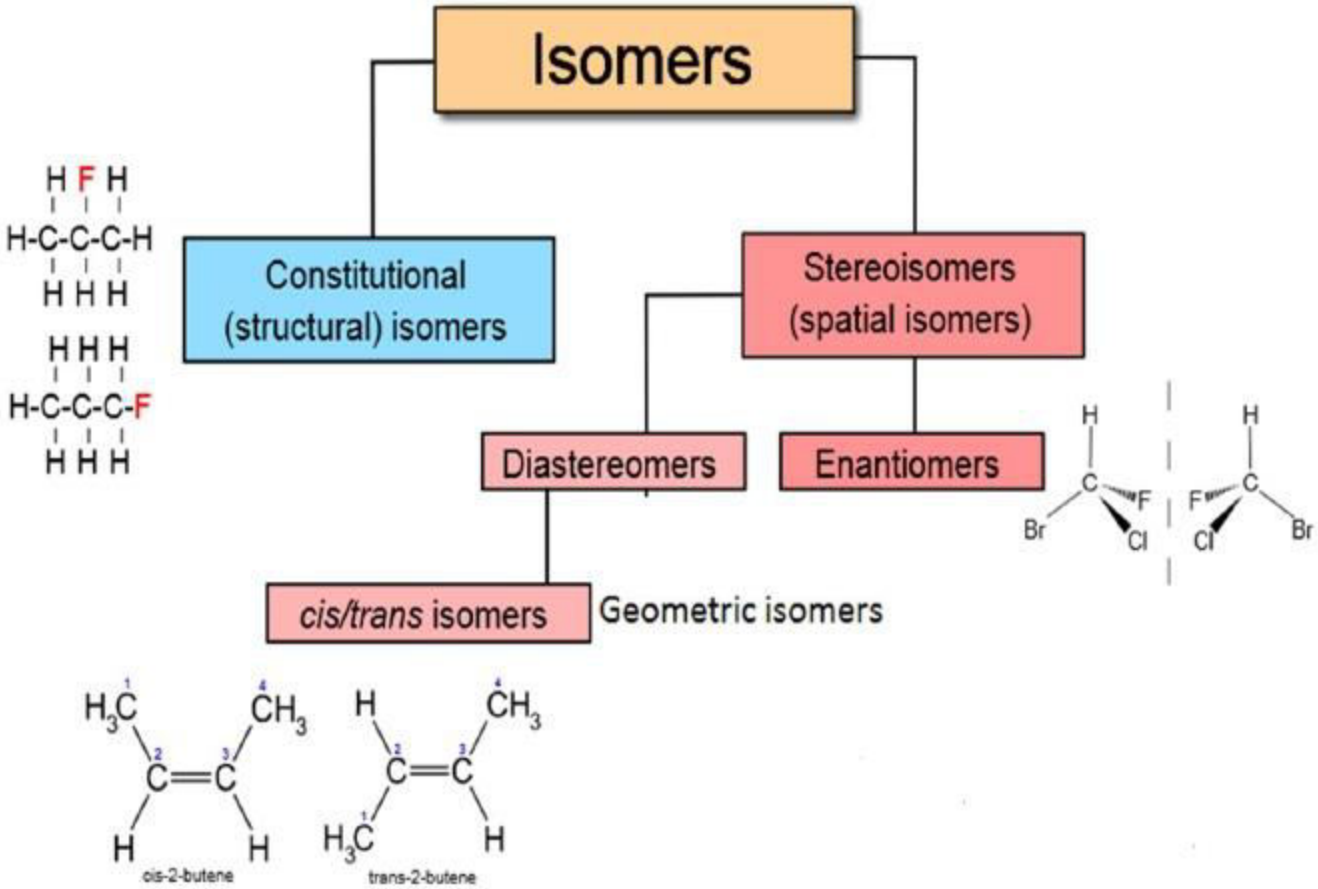
Carbohydrates

LECTURE NO 4

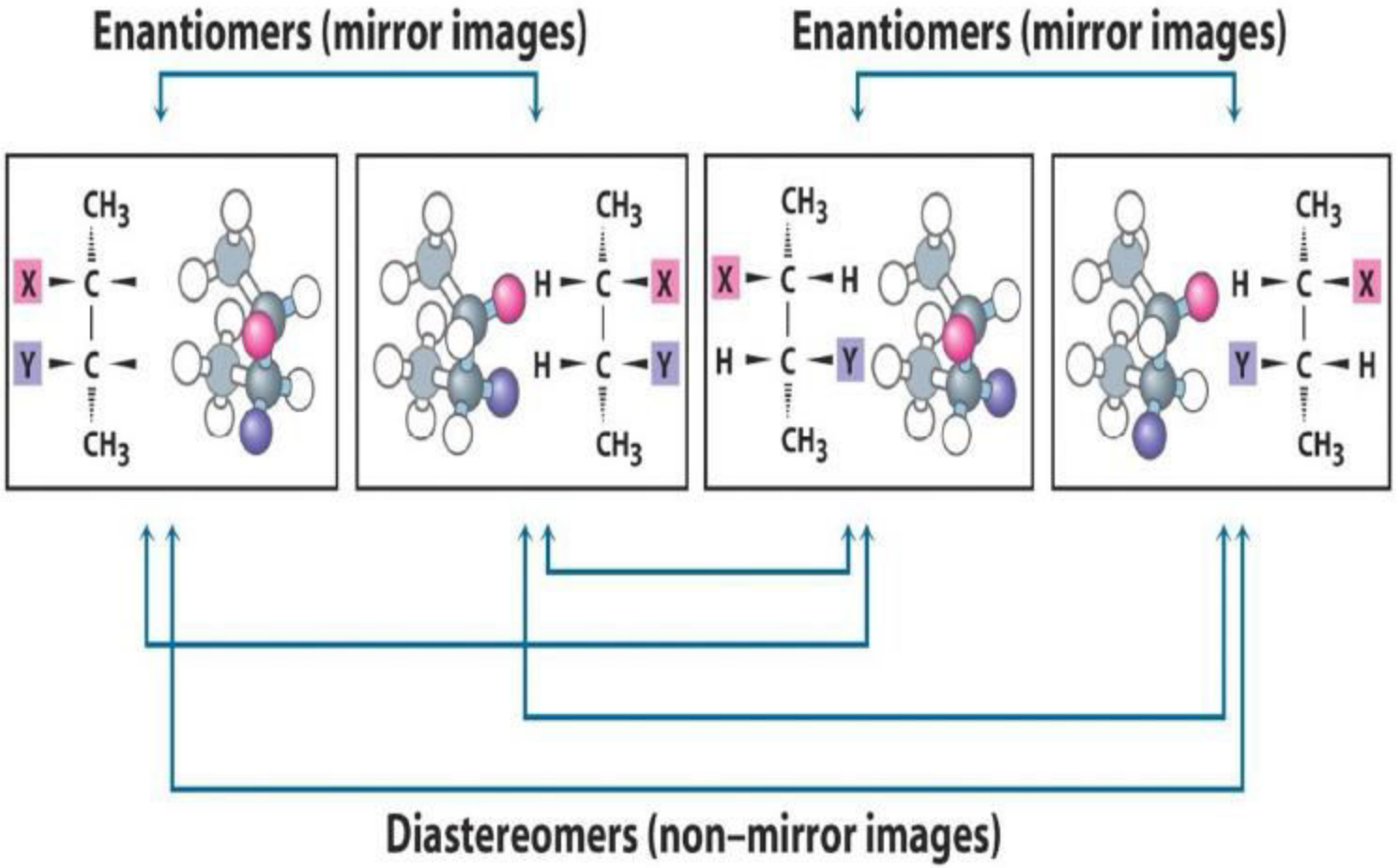
Isomerism

- The existence of two or more compounds having the same molecular formula but a different arrangement of atoms within the molecule

Carbohydrates



Carbohydrates



Carbohydrates

Isomerism

- Compounds which are identical in composition and differ only in configuration (the fixed spatial arrangement of atoms) are called ***stereoisomers***

Carbohydrates

- Stereoisomers that are mirror images of each other are called ***enantiomers***
- Pairs of stereoisomers that are not mirror images of each other are called ***diastereomers***

Carbohydrates

Isomerism

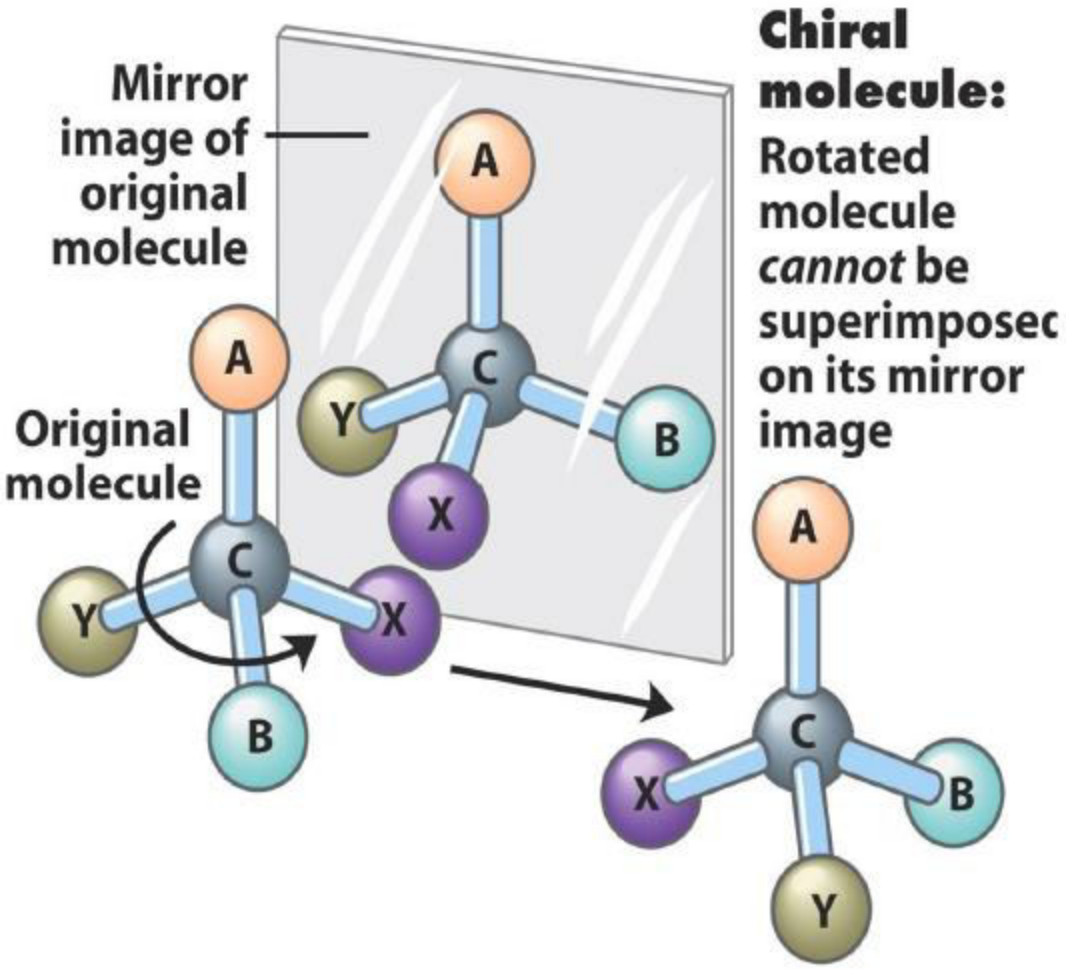
- The identifying characteristic of stereoisomers is that they cannot be interconverted without temporarily breaking one or more covalent bonds

Carbohydrates

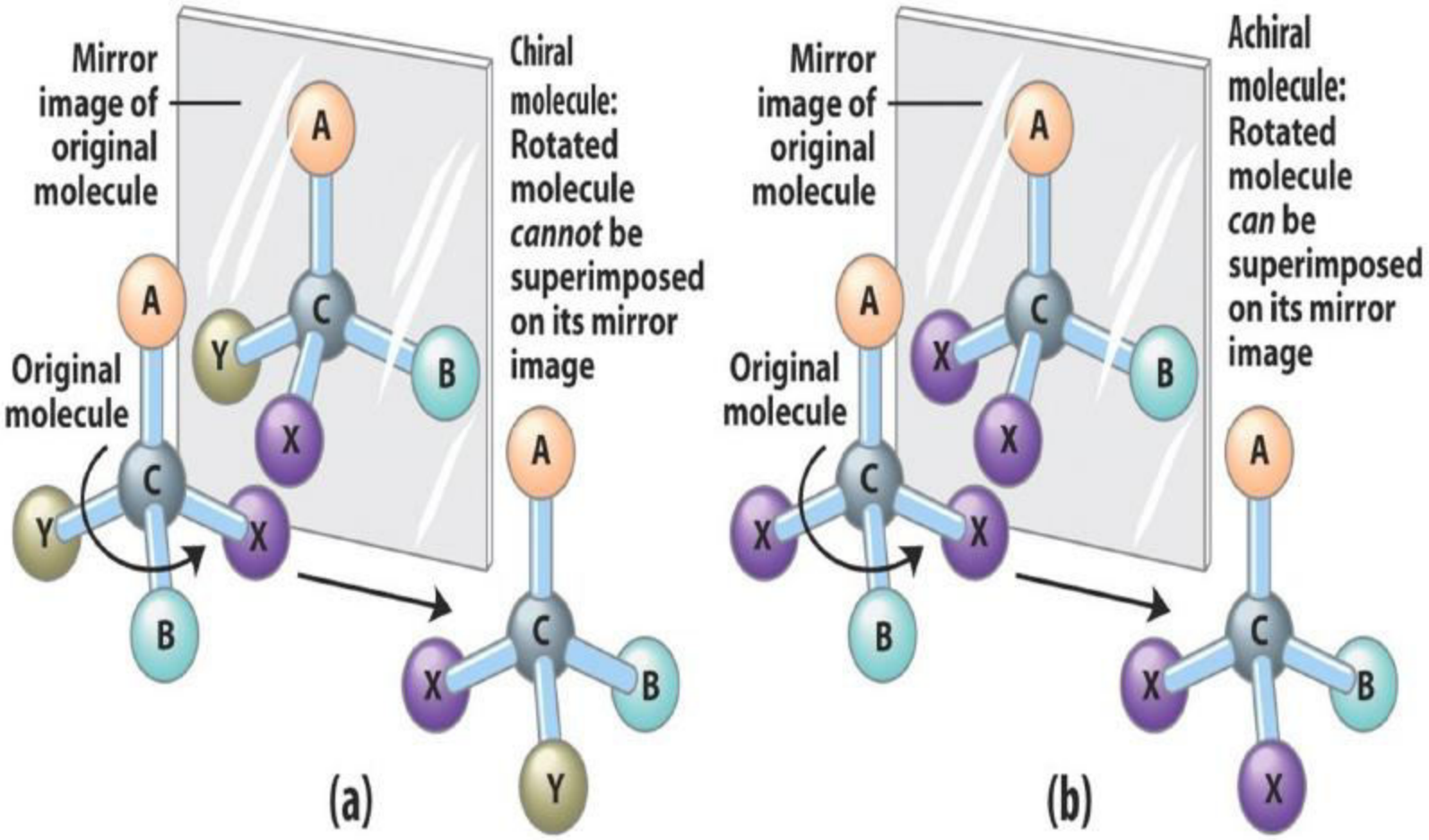
LECTURE NO 5

- A carbon atom to which four different atoms or groups of atoms are attached is said to be **asymmetric** or ***chiral***

Carbohydrates



Carbohydrates

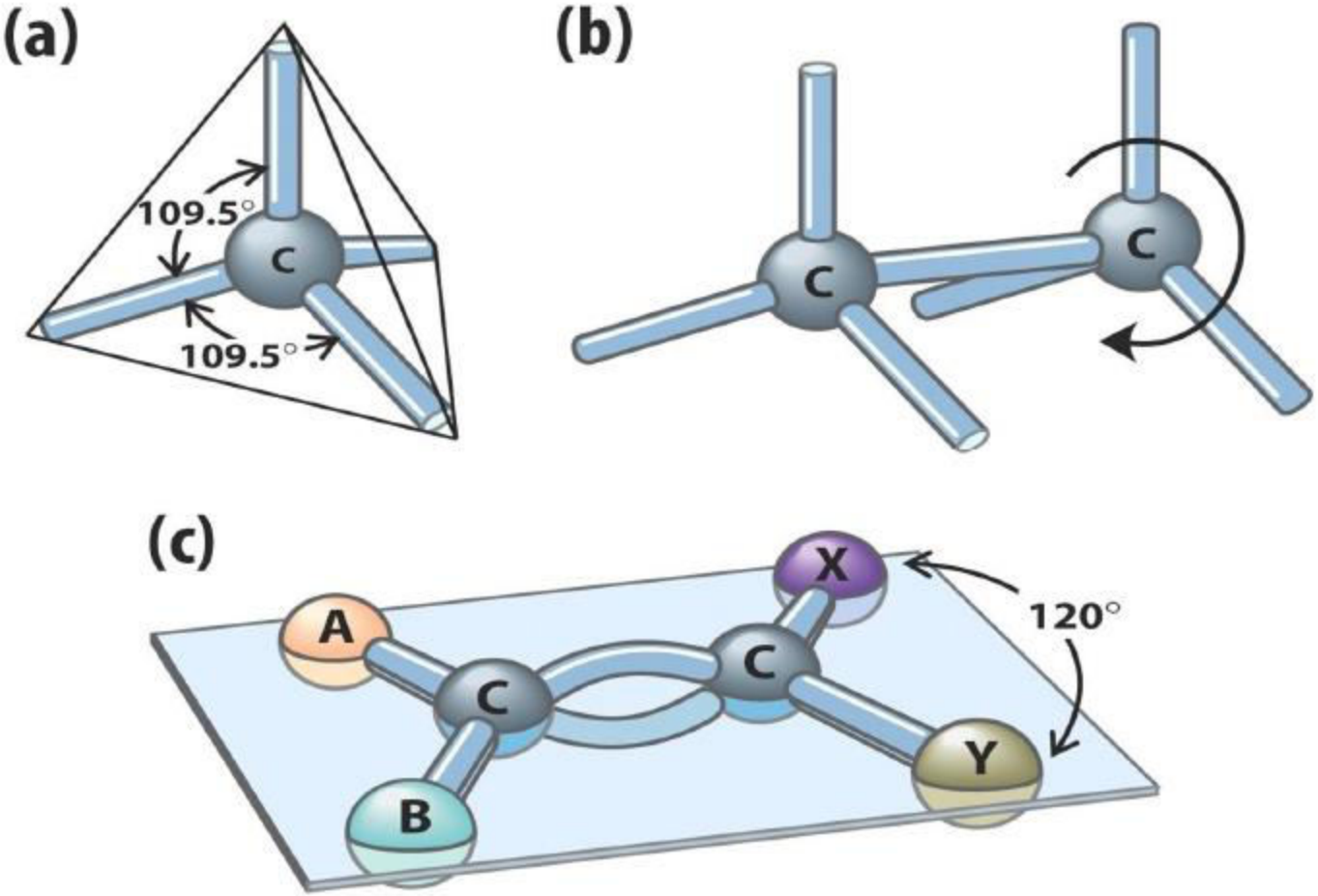


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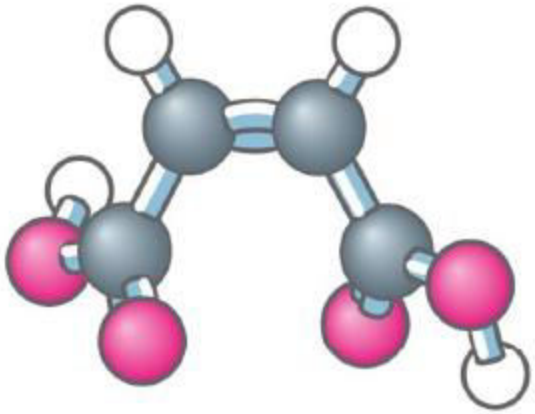
Carbohydrates

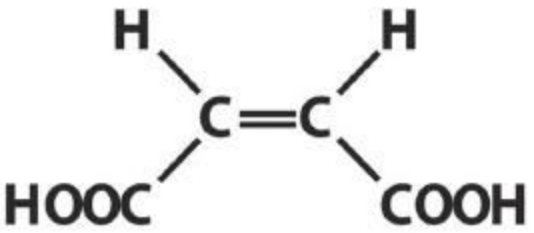
- Configuration is conferred by the presence of either
  - (a) ***chiral centers,*** around which substituent groups are arranged in a specific orientation
  - (b) ***double bonds,*** around which there is no freedom of rotation

Carbohydrates

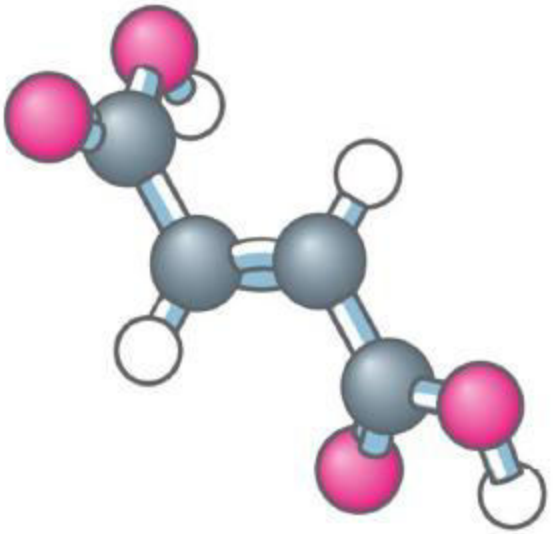


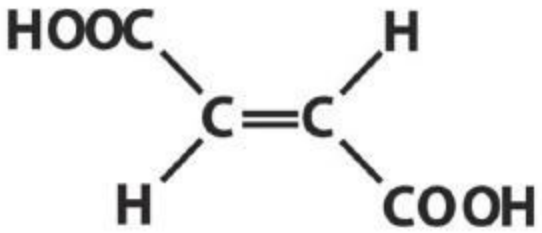
Carbohydrates





Maleic acid (cis)





Fumaric acid (trans)

Carbohydrates

- **Geometric isomers;** differ in the arrangement of their substituent groups with respect to the nonrotating double bond (each is a well-defined compound that can be separated from the other, and each has *its own unique chemical properties*)

Carbohydrates

- The identifying characteristic of configurational isomers is that they cannot be interconverted without temporarily breaking one or more covalent bonds.

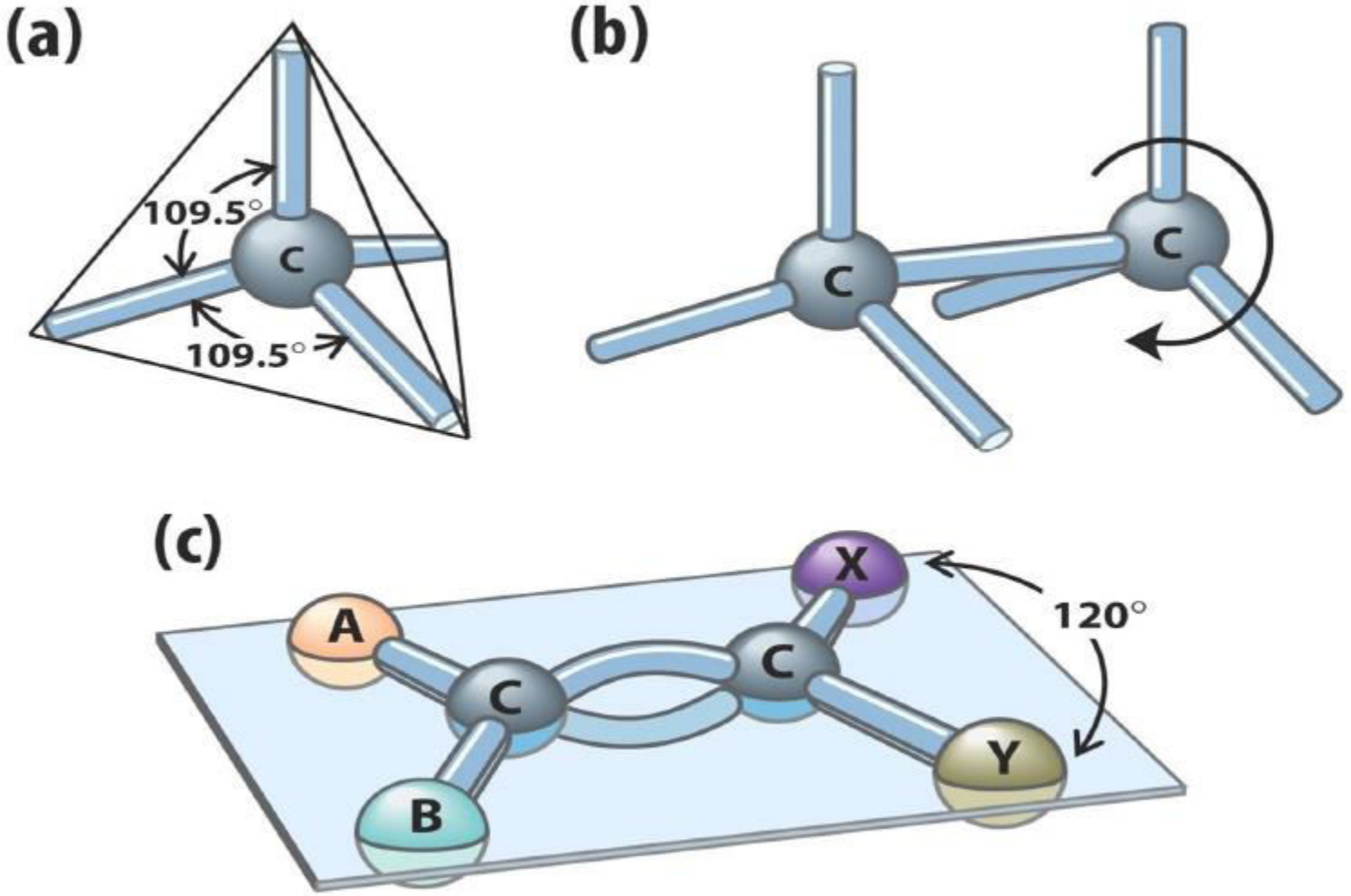
Carbohydrates

- The configurations of maleic acid and its isomer, fumaric acid.
- These compounds are **geometric,** or **cis-trans, isomers;**
- They differ in the arrangement of their substituent groups with respect to the nonrotating double bond

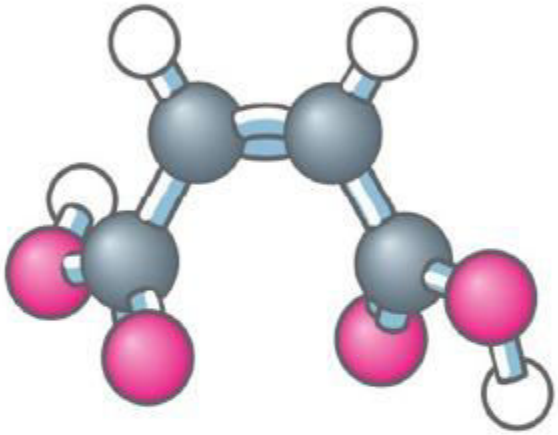
Carbohydrates

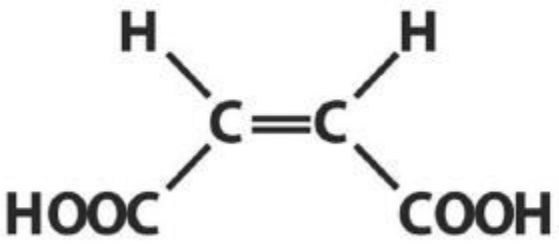
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Carbohydrates

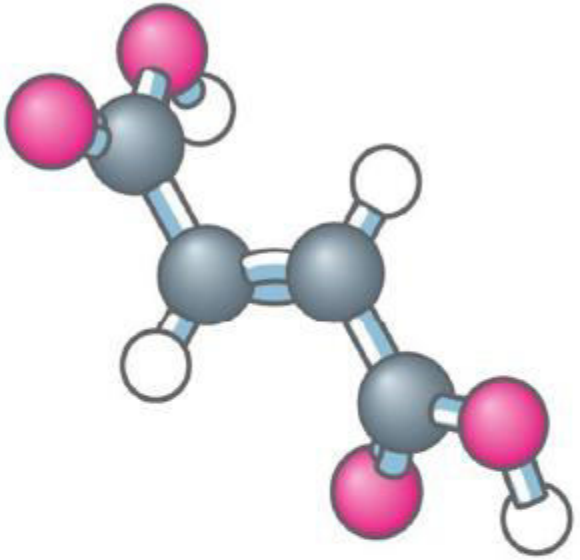


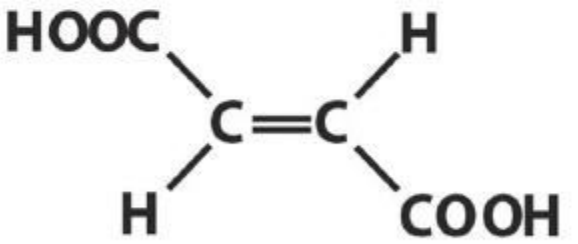
Carbohydrates





Maleic acid (cis)





Fumaric acid (trans)

Carbohydrates

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Carbohydrates

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Carbohydrates

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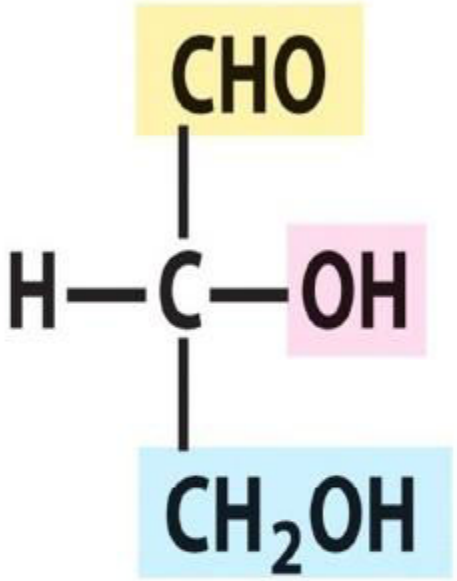
Carbohydrates

LECTURE NO 6

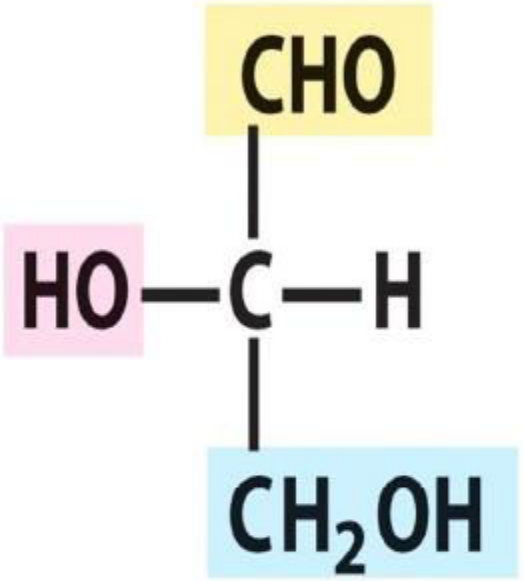
D and L isomerism:

- The designation of a sugar isomer as the D form or as the L form is determined by its spatial relationship to the parent compound of the carbohydrates, the glyceraldehyde

Carbohydrates



D-Glyceraldehyde



L-Glyceraldehyde

Fischer projection formulas

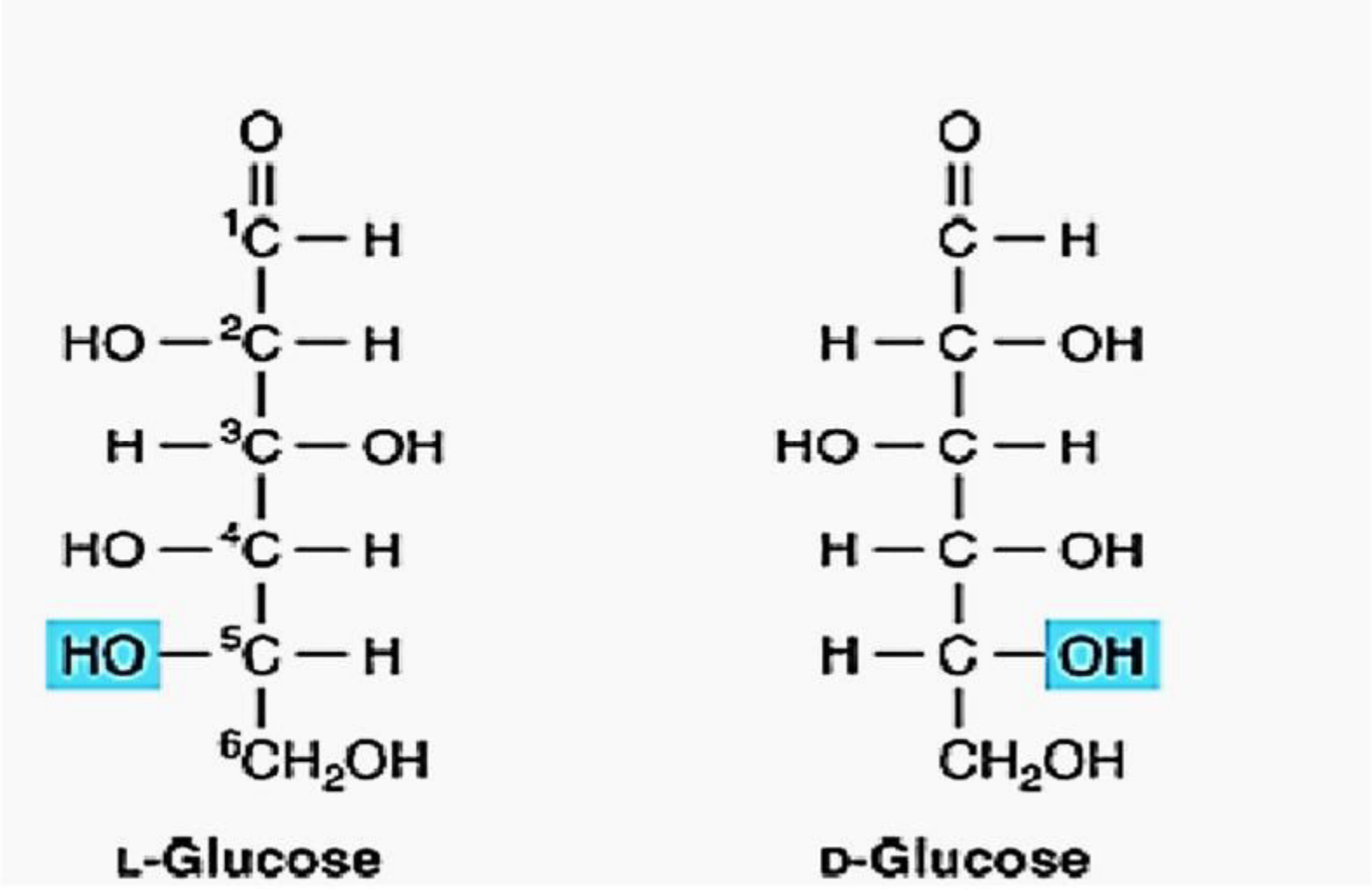
Carbohydrates

- The orientation of the —H and —OH groups around the ***carbon atom adjacent to the terminal primary alcohol or carbon most distant from the carbonyl carbon – the penultimate carbon***

Carbohydrates

- ***(carbon 5 in glucose)*** determines whether the sugar belongs to the D or L series.
- When the —OH group on this carbon is on the right the sugar is the D isomer;
- when it is on the left, it is the L isomer

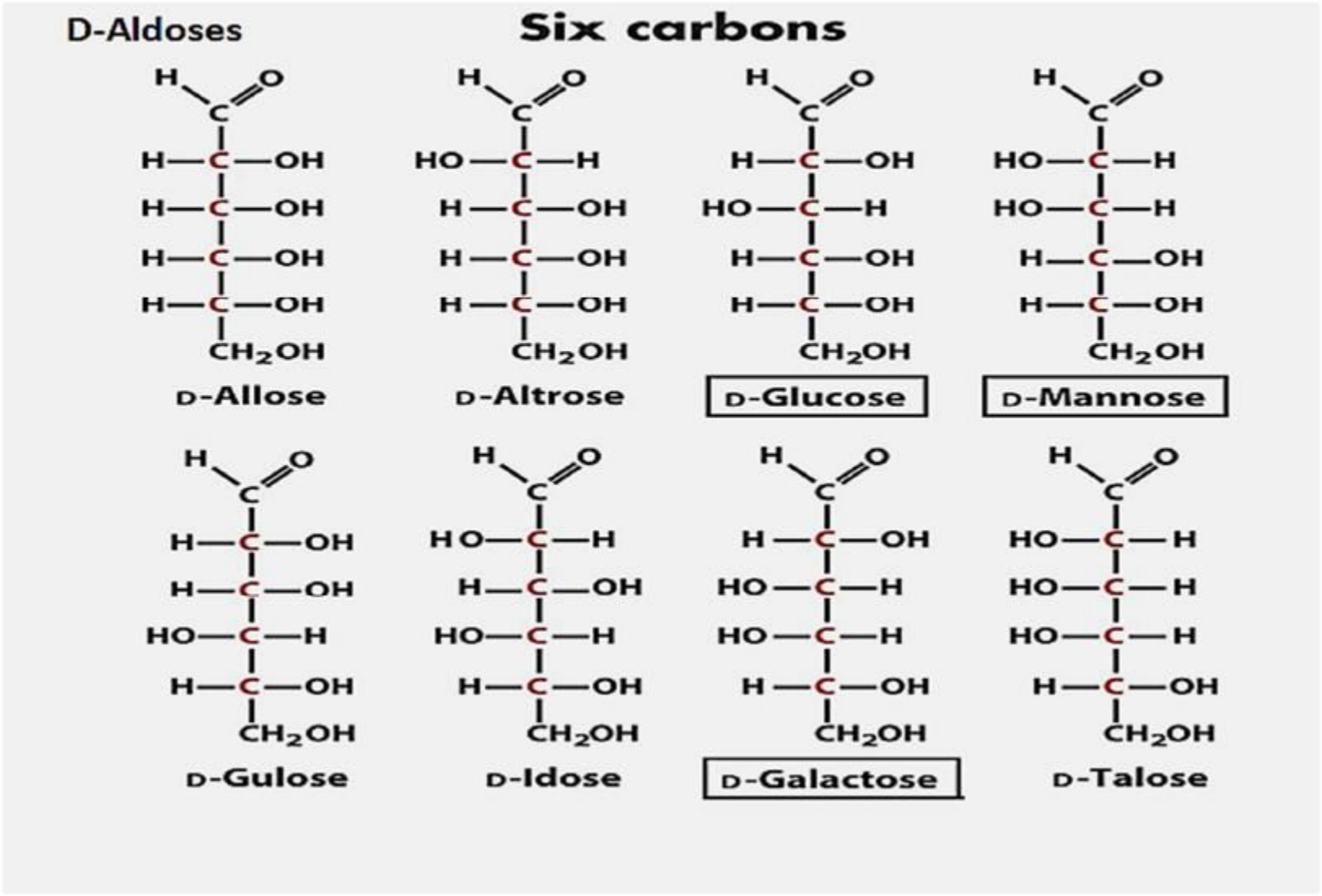
Carbohydrates



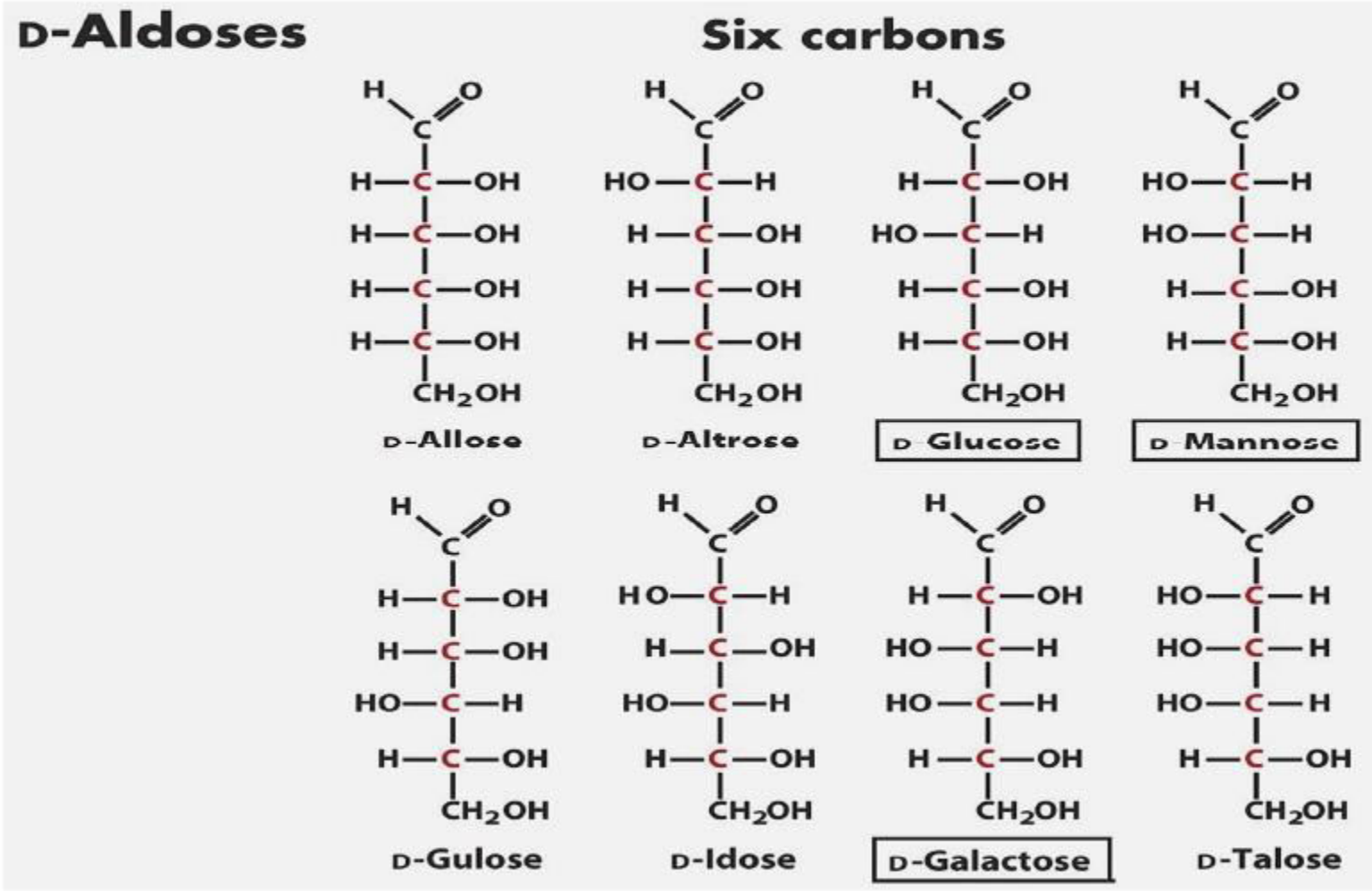
Carbohydrates

- Of the 16 possible aldohexoses eight are D forms and eight are L
- Most of the hexoses of living organisms are D isomers and the enzymes responsible for their metabolism are specific for this configuration

Carbohydrates



Carbohydrates



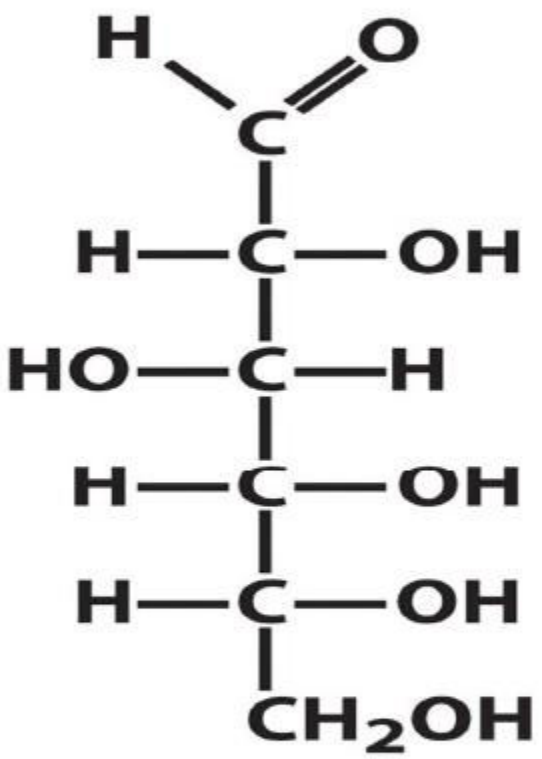
Carbohydrates

- Four different substituents bonded to a chiral carbon atom may be arranged in two different ways in space, that is, have two stereoisomers (with similar or identical chemical properties ***but differing in certain physical and biological properties)***

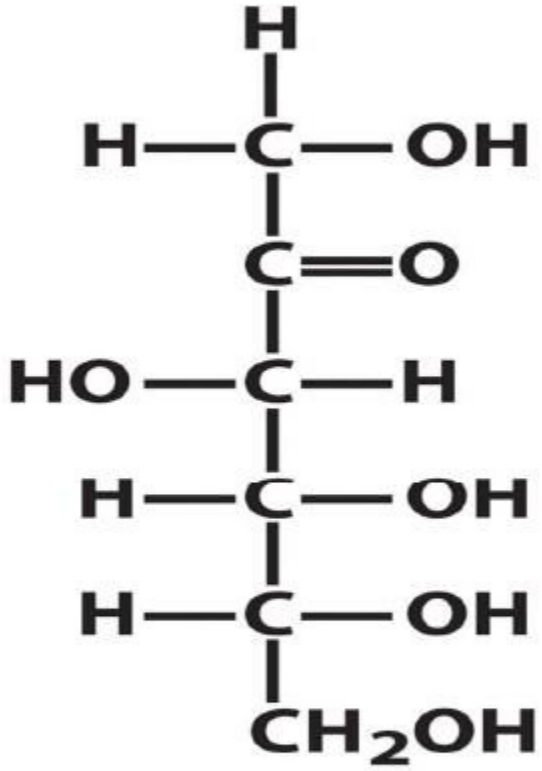
Carbohydrates

- A molecule with only one chiral carbon can have two stereoisomers; when two or more (n) chiral carbons are present, there can be 2^n stereoisomers

Carbohydrates



D-Glucose,
an aldohexose



D-Fructose,
a ketohexose

Carbohydrates

LECTURE NO 7

- **Enantiomers** have nearly identical chemical properties but differ in a characteristic physical property:
  - their interaction with plane-polarized light

Carbohydrates

- Thus the presence of asymmetric carbon atoms also confers **optical activity** on the compound

Carbohydrates

- When a beam of plane-polarized light is passed through a solution of an **optical isomer,** it rotates either to the right, dextrorotatory (+), or to the left, levorotatory (–)

Carbohydrates

- The direction of rotation of plane polarized light is independent of the stereochemistry of the sugar, so it may be designated D(–), D(+), L(–), or L(+)

Carbohydrates

- For example, the naturally occurring form of fructose is the D(–) isomer
- In solution, glucose is dextrorotatory, and glucose solutions are sometimes known as **dextrose**

Carbohydrates

- When equal amounts of dextrorotatory and levorotatory isomers are present, the resulting mixture has no optical activity

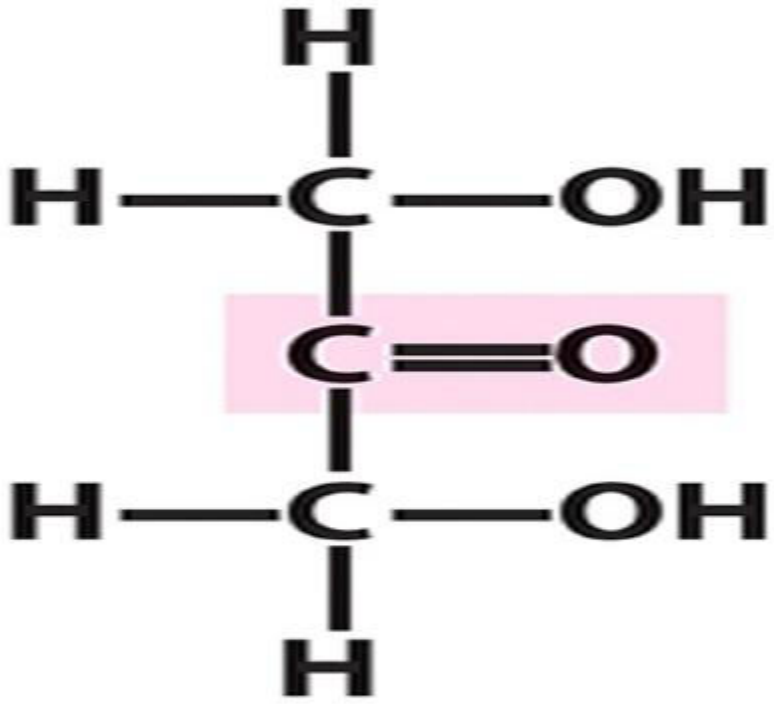
Carbohydrates

- Such a mixture is said to be ***racemic*** and separation of optically active isomers from a racemic mixture is called ***resolution***

Carbohydrates

- All the monosaccharides except dihydroxyacetoe contain one or more asymmetric (chiral) carbon atoms and thus occur in optically active isomeric forms

Carbohydrates



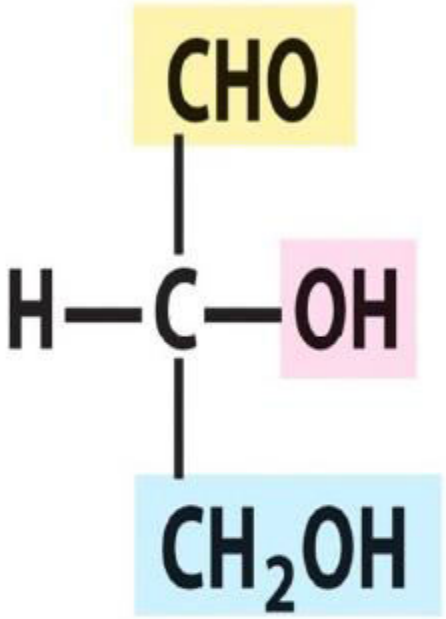
Dihydroxyacetone, a ketotriose

Carbohydrates

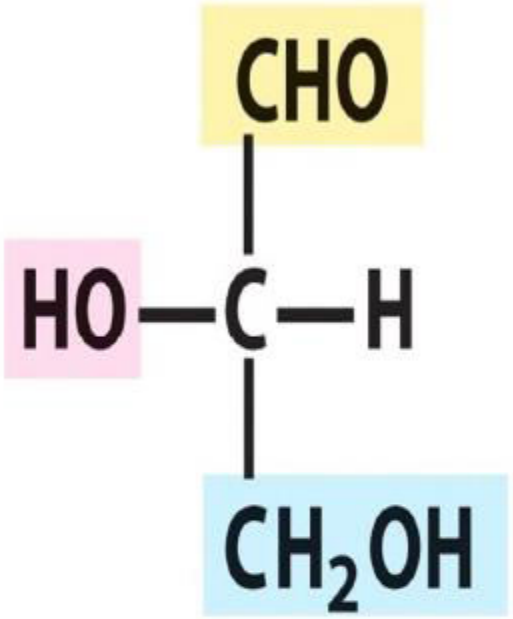
LECTURE NO 8

- The simplest aldose, glyceraldehyde contains one chiral center (the middle carbon atom) and therefore has two different optical isomers

Carbohydrates



D-Glyceraldehyde



L-Glyceraldehyde

Fischer projection formulas

Carbohydrates

LECTURE NO 9

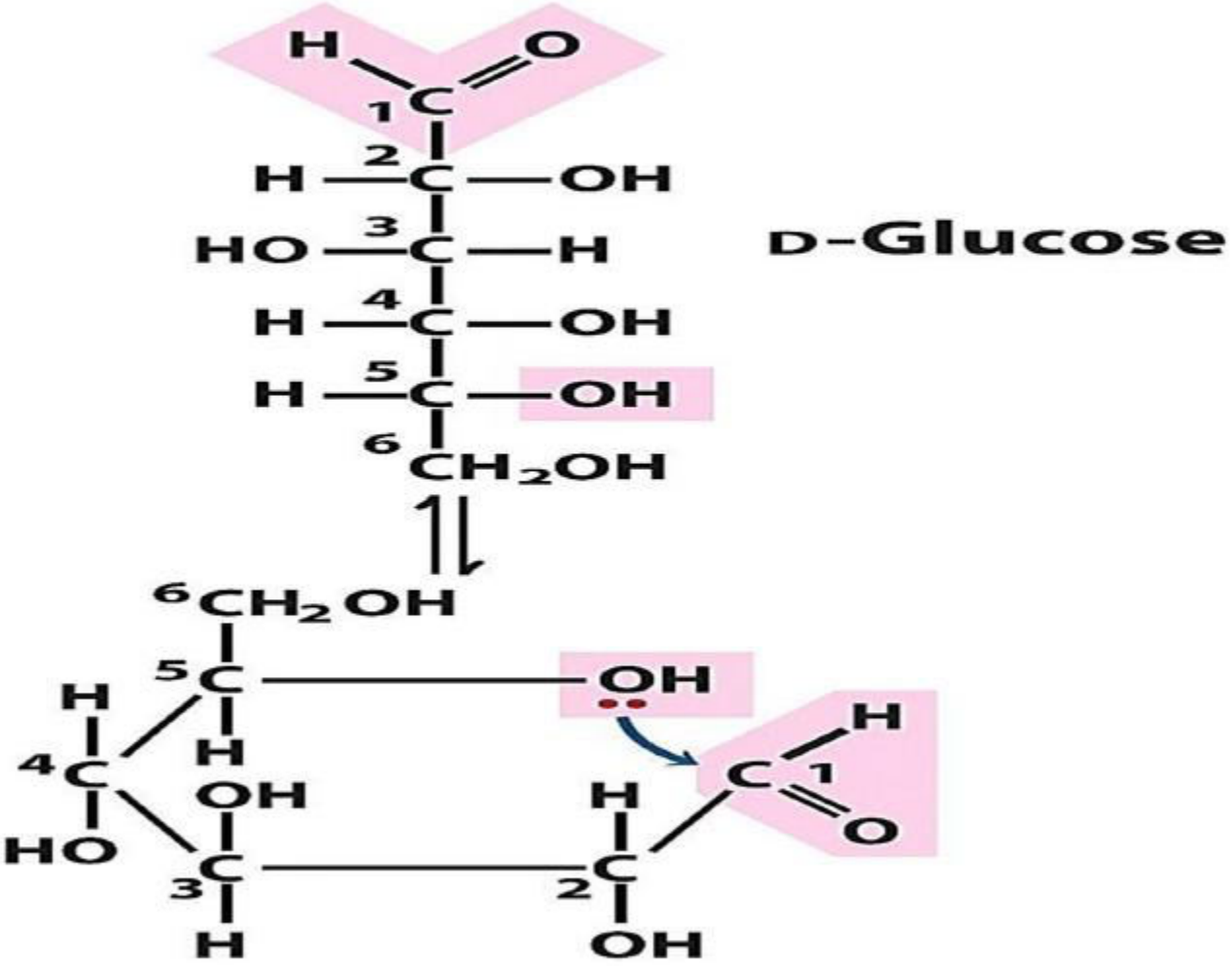
- In aqueous solution, aldotetroses and all monosaccharides with five or more carbon atoms in the backbone occur predominantly as ***cyclic (ring) structures***.

Carbohydrates

- In the ring structure the carbonyl group has formed a **covalent bond** with the oxygen of a hydroxyl group along the chain.

- Therefore forming an ether linkage

Carbohydrates



Carbohydrates

- As the two reacting groups, the aldehyde and the alcoholic group, belong to the same molecule, a cyclic structure forms

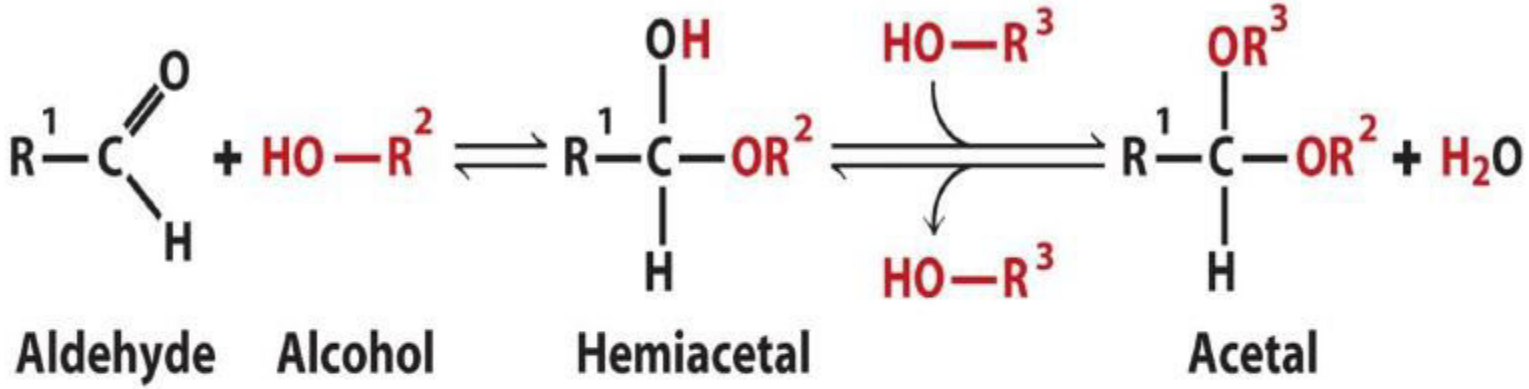
Carbohydrates

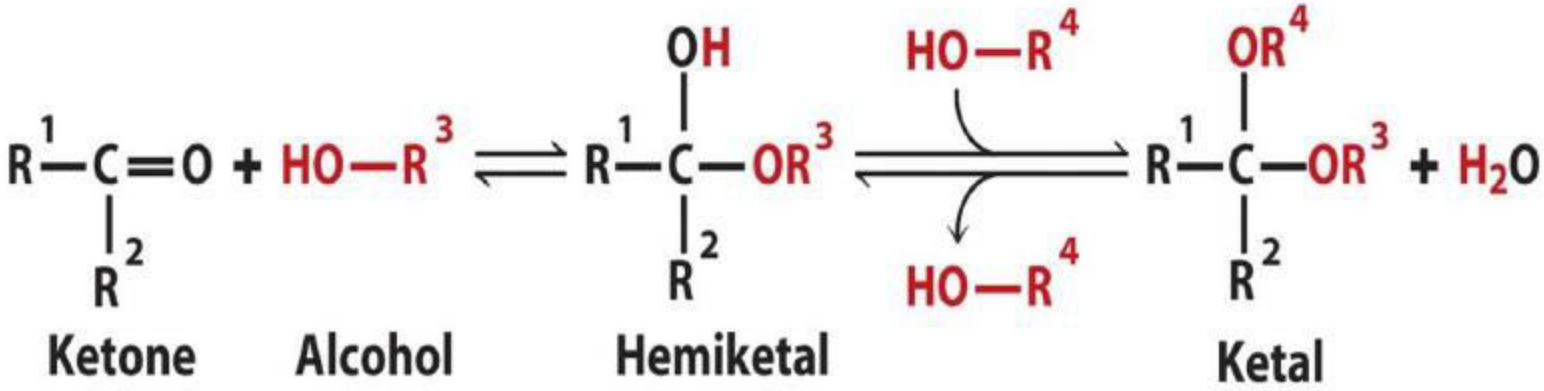
- The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called ***hemiacetals or hemiketals.***

Carbohydrates

- Which contain an additional ***asymmetric carbon atom*** and thus can exist in two stereoisomeric forms

Carbohydrates





Carbohydrates

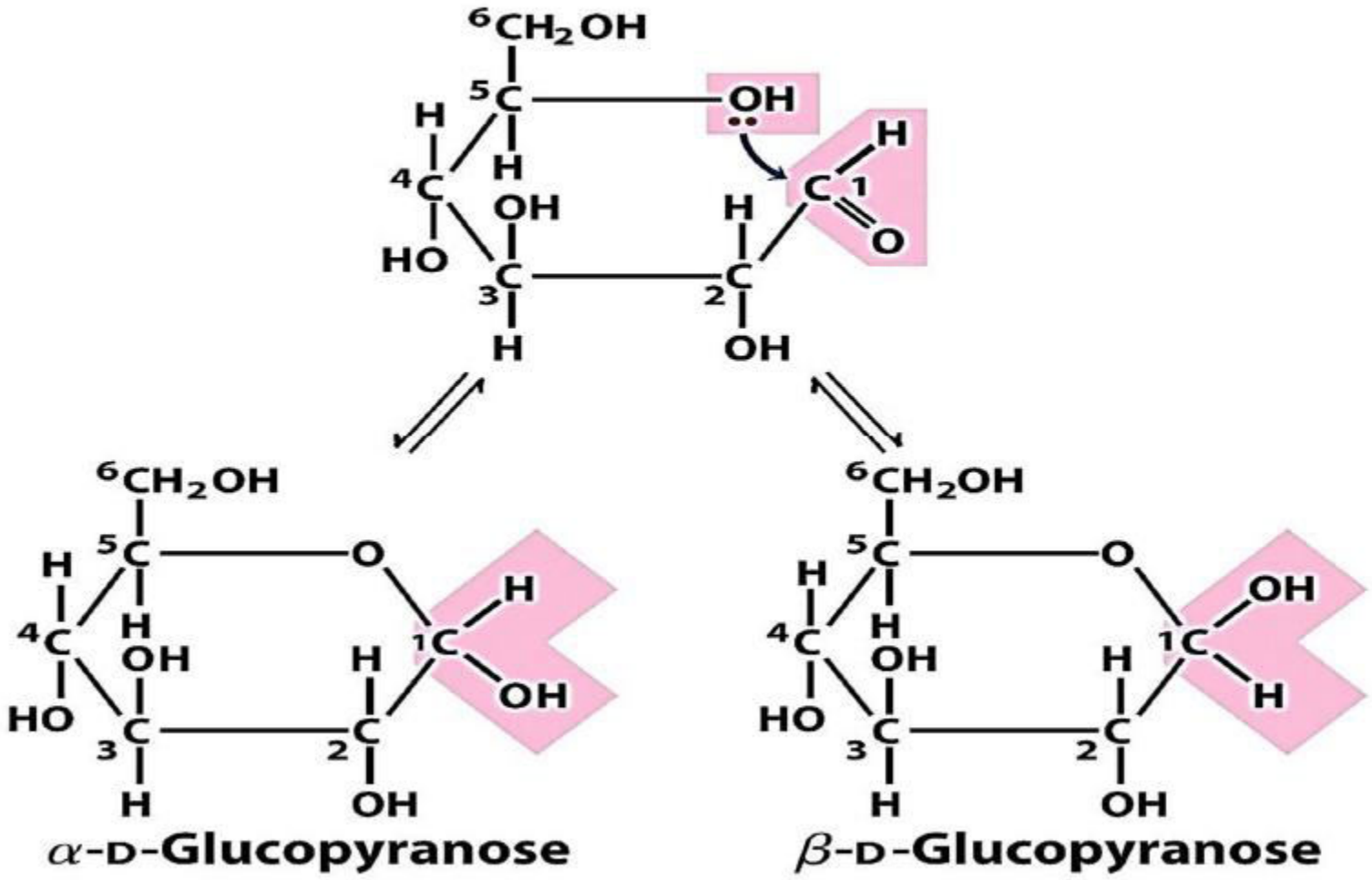
- For example,D-glucose exists in solution as an **intramolecular hemiacetal** in which the free hydroxyl group at C-5 has reacted with the aldehydic C-l,

Carbohydrates

- Therefore rendering the latter carbon (aldehydic C-l)

  asymmetric
- And producing two stereoisomers, designated ***α and β***

Carbohydrates

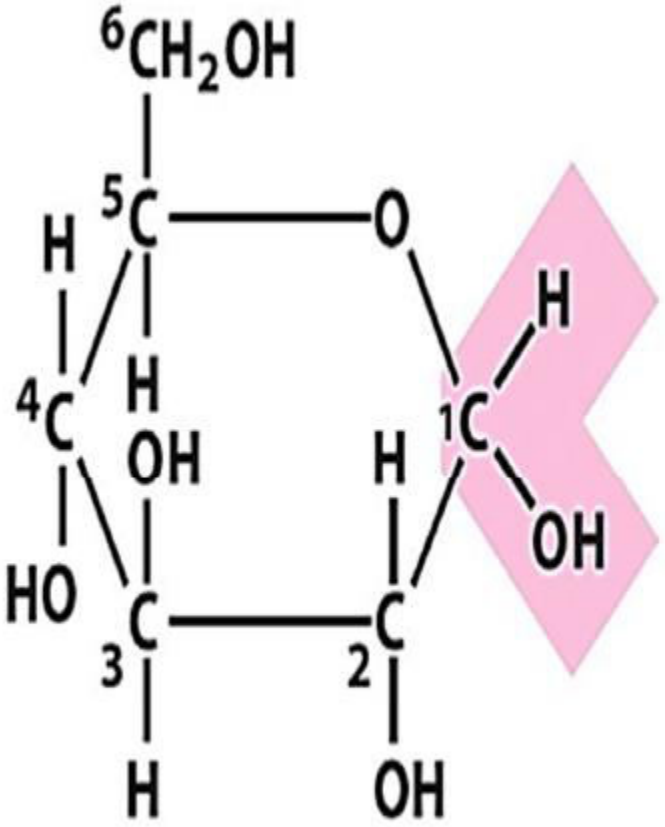


Carbohydrates

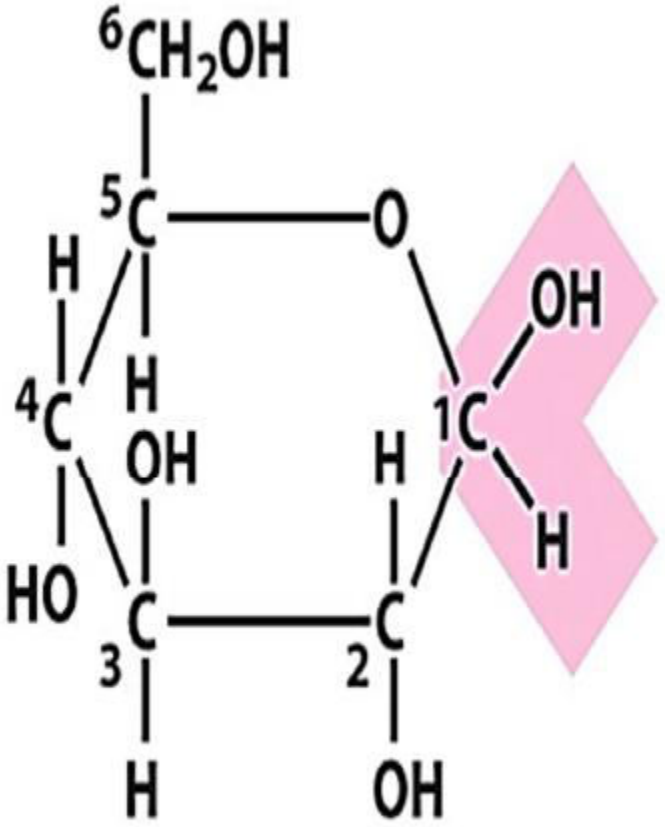
- The designation **α** indicates that the hydroxyl group at the C-1 is **trans** to the CH_2O.
- Whereas **β** indicates that the hydroxyl group at the C-1 is **cis** to the CH_2O

Carbohydrates

- The ordinary crystalline glucose happens to be in the α form



α-D-Glucopyranose



β-D-Glucopyranose

Carbohydrates

- Carbon 1, after cyclization has four different groups attached to it and thus it also becomes ‘asymmetric’

Carbohydrates

LECTURE NO 10

- *Is*omeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**

Carbohydrates

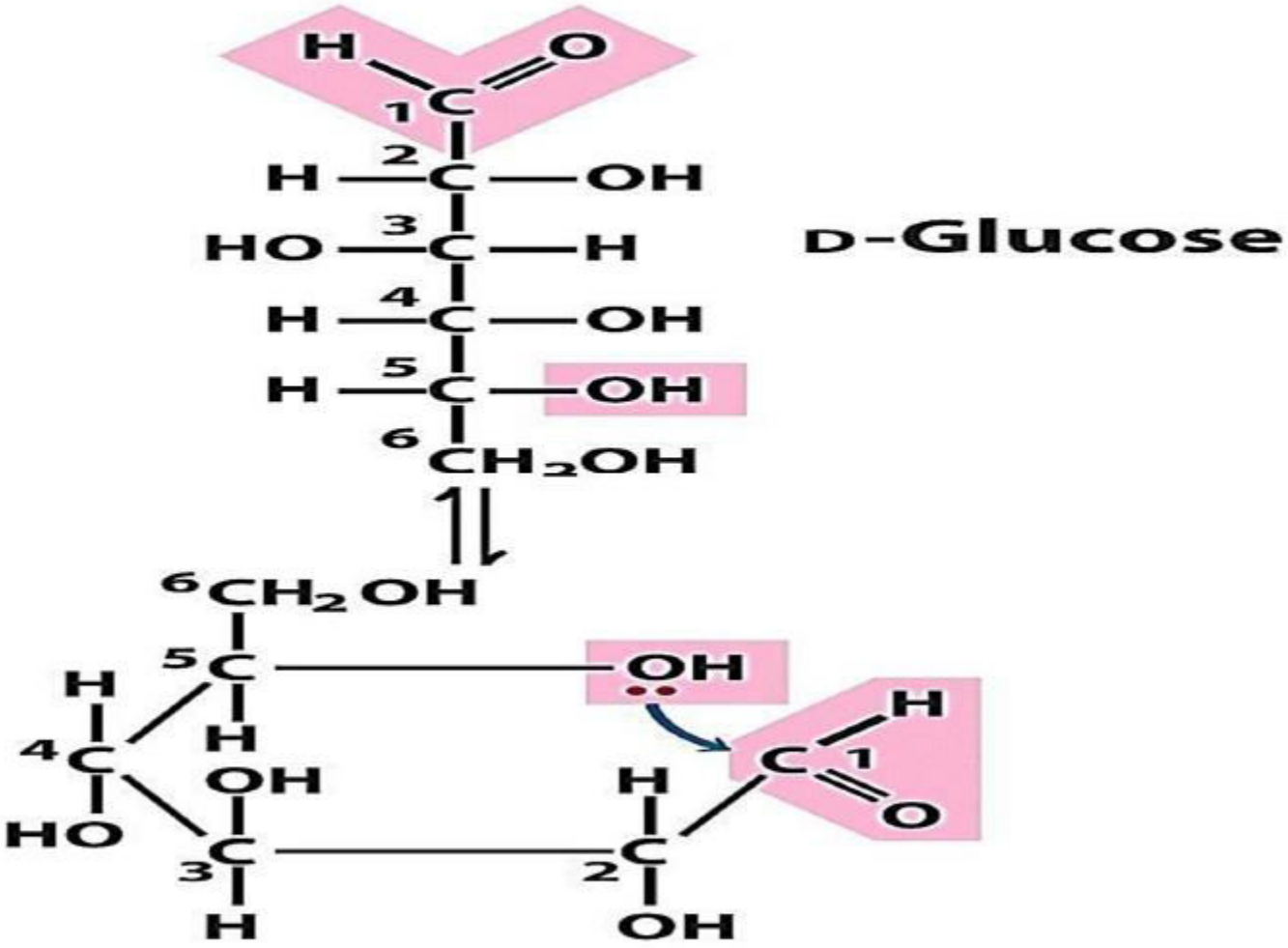
- The carbonyl carbon atom is called the ***anomeric carbon***

Carbohydrates

Ring Structure of Carbohydrates

- For example,D-glucose exists in solution as an **intramolecular hemiacetal** in which the free hydroxyl group at C-5 has reacted with the aldehydic C-l,

Carbohydrates



Carbohydrates

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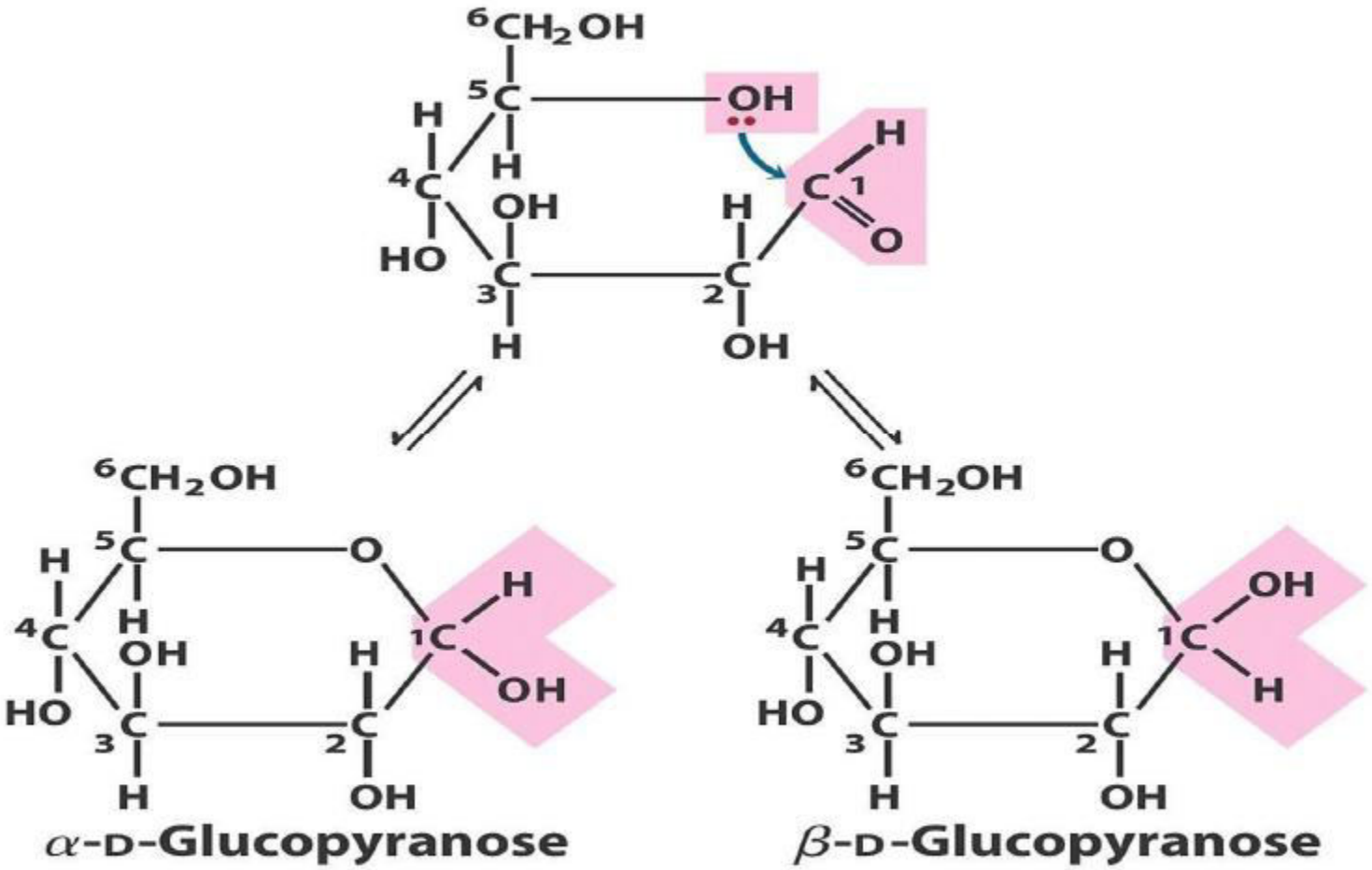
Carbohydrates

- Carbon 1, after cyclization has four different groups attached to it and thus it also becomes ‘asymmetric’
- The newly created assymetric or chiral carbon is called the ***anomeric carbon***

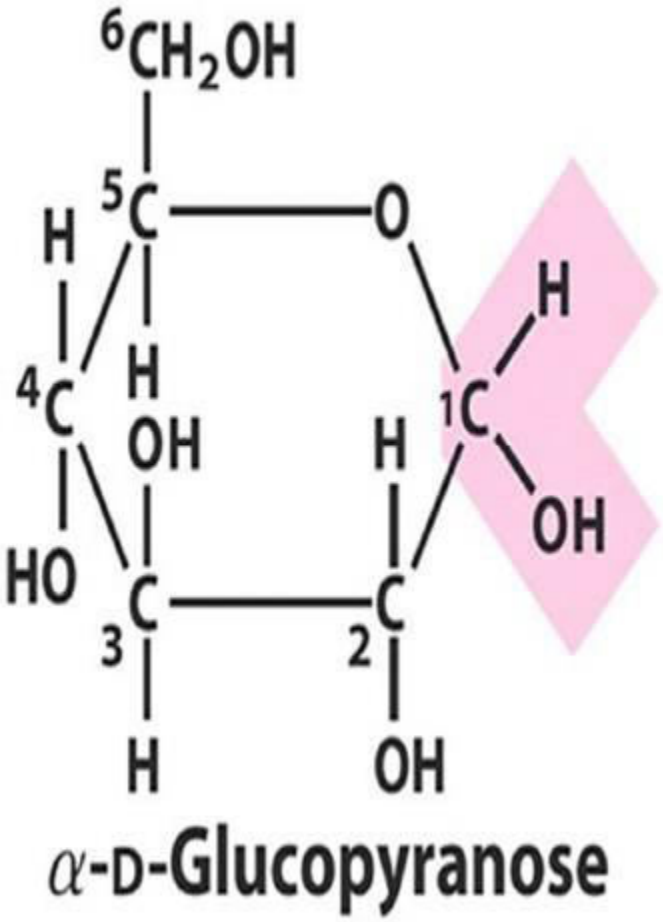
Carbohydrates

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Carbohydrates



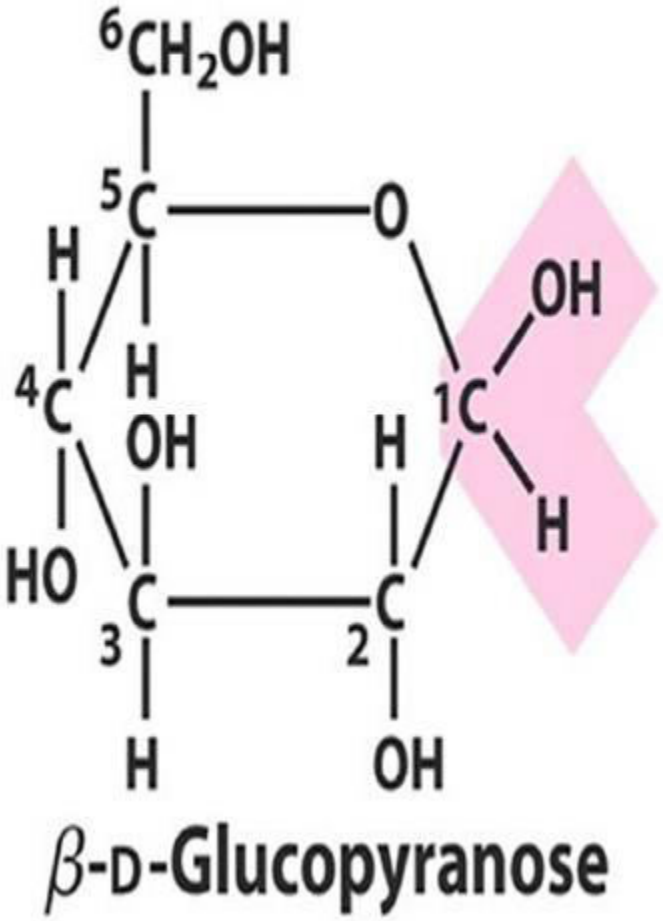
Carbohydrates



α-D-Glucopyranose

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Carbohydrates



β-D-Glucopyranose

- Whereas ββindicates that the hydroxyl group at the C-1 is is **cis** to tothe CH_2O O

Carbohydrates

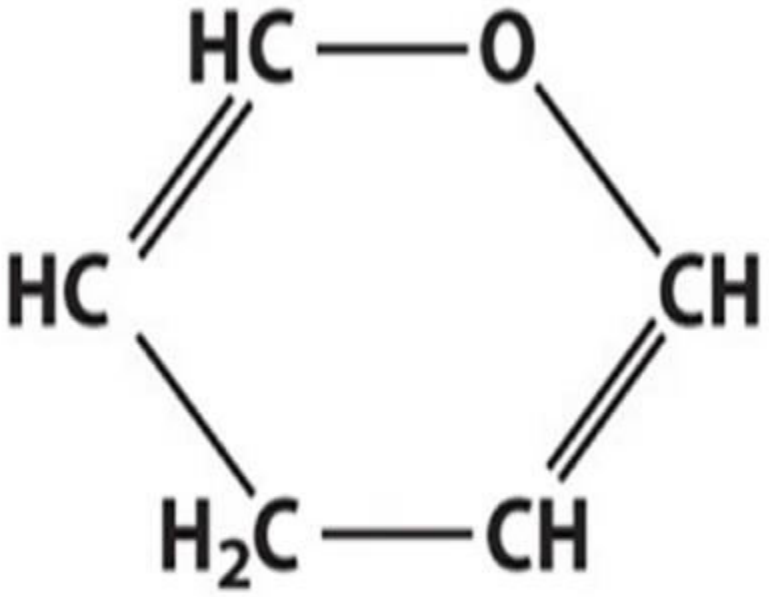
- The ordinary crystalline glucose happens to be in the α form

Carbohydrates

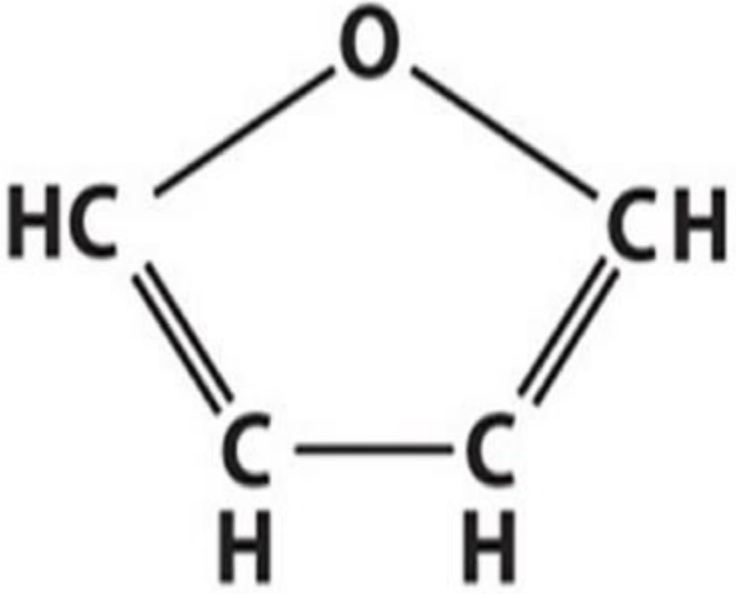
LECTURE NO 11

- The ring structures of monosaccharides are similar to the ring structures of either pyran (six-membered ring) and are called ***pyranoses*** or furan (a five-membered ring), which are known as ***furanoses***

Carbohydrates



Pyran

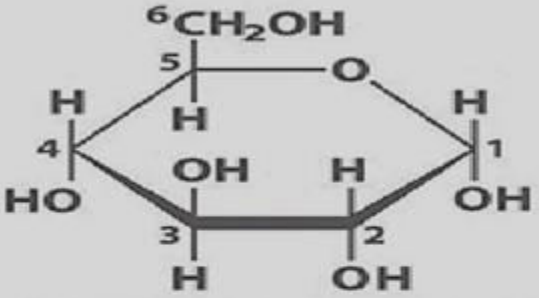


Furan

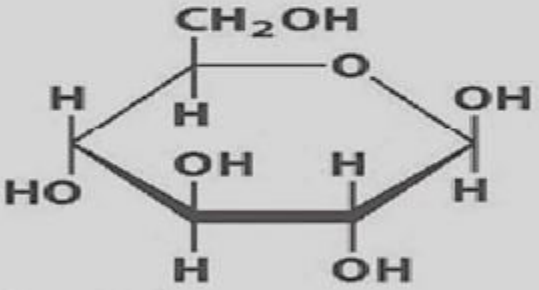
Carbohydrates

- The six-membered aldopyranose ring is much more stable than the aldofuranose ring
- Therefore predominates in aldohexose and aldopentose solutions
- Similarly Crystalline glucose is **α -D - glucopyranose**

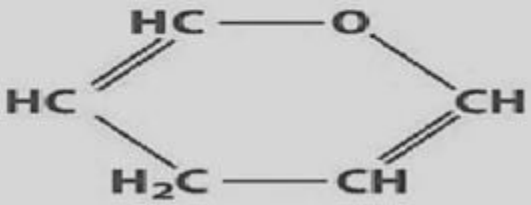
Carbohydrates



α-D-Glucopyranose



β-D-Glucopyranose



Pyran

Carbohydrates

- The α and *β* anomers of D-glucose interconvert in aqueous solution by a process called **mutarotation.**
- Thus, a solution of *α*-D-glucose and a solution of *β*-D-glucose eventually form identical equilibrium mixtures having identical optical properties.

Carbohydrates

- This mixture consists of about;
- one-third α-D-glucose,
- two-thirds β-D-glucose, and very small amounts of the linear and five-membered ring (glucofuranose) forms.

Carbohydrates

- α and β forms are NOT mirror images and are referred to as ***diastereomers***
- Enzymes are able to distinguish between these two structures and use one or the other preferentially

Carbohydrates

- For example, glycogen is synthesized from α-D-glucopyranose, whereas cellulose is synthesized from β-D-glucopyranose

Carbohydrates

- Ketohexoses also occur in α and β anomeric forms
- In these compounds the hydroxyl group at C-5 (or C-6) reacts with the keto group at C-2, forming a furanose (or pyranose) ring containing a hemiketal linkage

Carbohydrates

- D-Fructose readily forms the furanose ring ; the more common anomer of this sugar in combined forms or in derivatives is β-D-fructofuranose

Carbohydrates

LECTURE NO 12

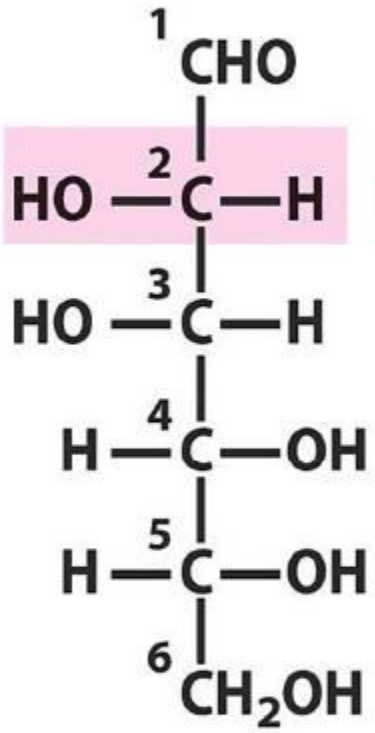
Epimers

- Two sugars which differ from one another only in configuration around a single carbon atom are termed ***epimers***
- *Glucose and Galactose are an example of an epimeric pair which differ only with respect to C_4*

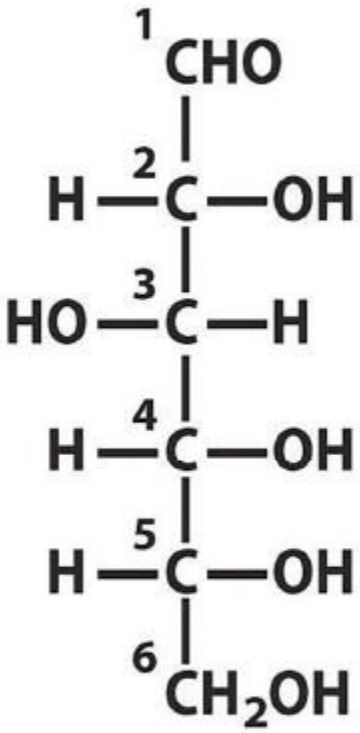
Carbohydrates

- *Similarly Glucose and mannose are epimers with respect to C_2*
- However, Galactose and mannose are NOT epimers as they differ in the position of –OH groups at two carbons (2 and 4), and are therefore defined only as isomers

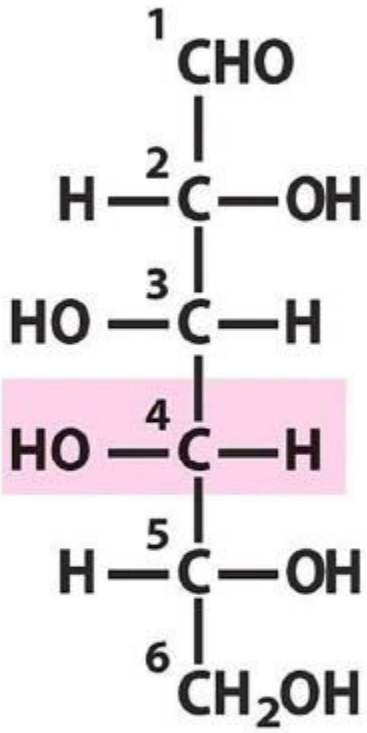
Carbohydrates



D-Mannose
(epimer at C-2)



D-Glucose



D-Galactose
(epimer at C-4)

Carbohydrates

***Trioses*:**

- Both **D-Glyceraldehyde & Dihydroxyacetone** occur as intermediates in glycolysis
- They are also the precursors of glycerol

Carbohydrates

Tetroses:

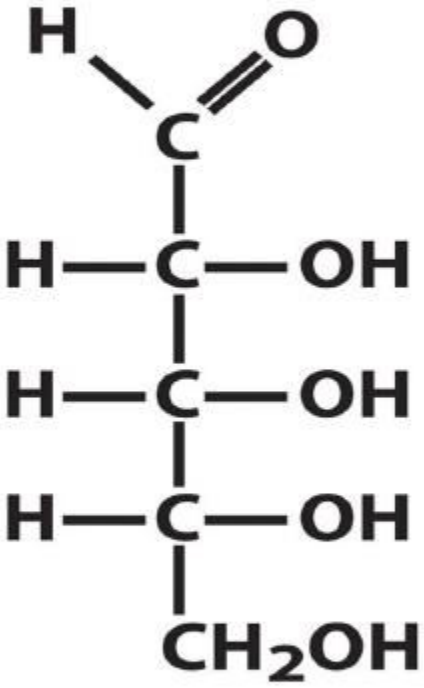
- **Erythrose** occurs as an intermediate in the Hexose Mono Phosohate shunt which is an alternative pathway for glucose oxidation

Carbohydrates

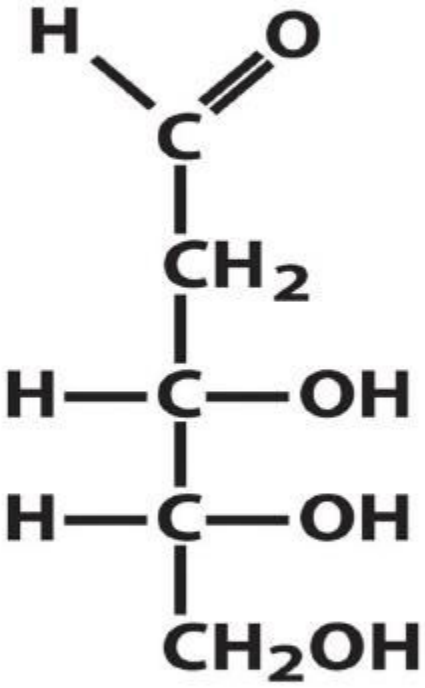
Pentoses:

- D-Ribose is a constituent of nucleic acid *RNA* and also certain co-enzymes like *FAD, NAD & co-enzyme A*
- D-2-Deoxyribose is a constituent of *DNA*

Carbohydrates



D-Ribose,
an aldopentose



2-Deoxy-D-ribose,
an aldopentose

Carbohydrates

Hexoses

- D-Glucose (Dextrose, Grape sugar):
- Most dietary carbohydrate is absorbed into the bloodstream as glucose formed by hydrolysis of dietary starch and disaccharides, and other sugars are converted to glucose in the liver

Carbohydrates

- Sources include fruit juices, hydrolysis of starch, cane or beet sugar, maltose and lactose
- Glucose is the major metabolic fuel of mammals

Carbohydrates

- It is the chief physiological sugar present in normal blood at a fairly constant level
- All tissues utilize glucose for energy
- Erythrocyte utilize glucose solely for energy purposes and Brain cells rely heavily on glucose

Carbohydrates

- It is the precursor for synthesis of all the other carbohydrates in the body, including
  - **glycogen** for storage;
  - **ribose** and **deoxyribose** in nucleic acids;
  - **galactose** in lactose of milk, in glycolipids,

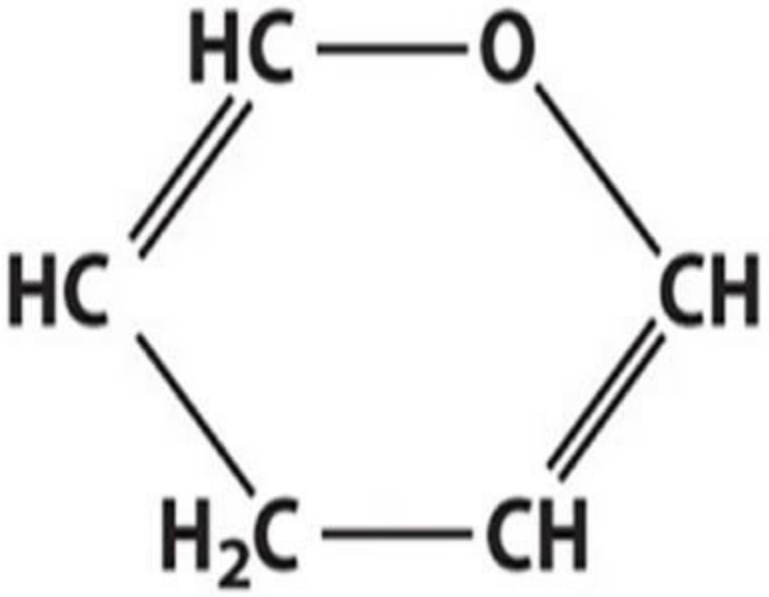
Carbohydrates

- And in combination with protein in glycoproteins and proteoglycans
- Stored as Glycogen in liver and muscles only
- Excreted in the urine (glucosuria) in poorly controlled diabetes mellitus as a result of hyperglycemia

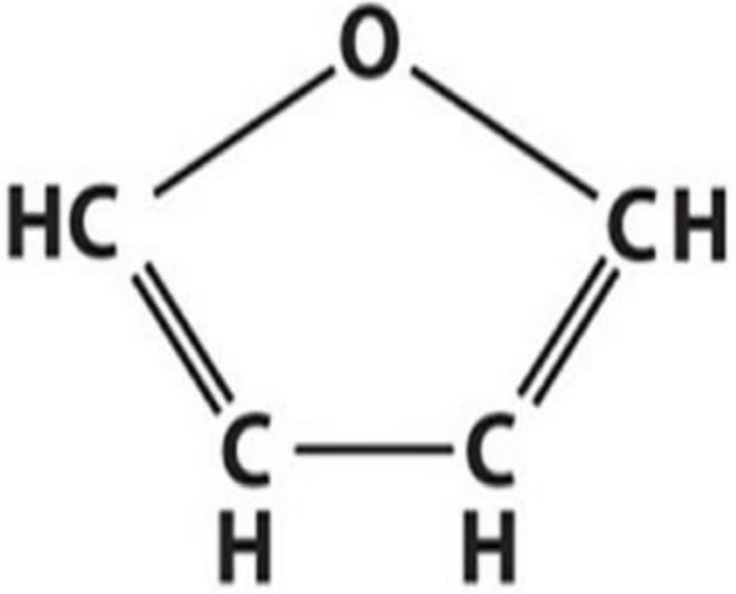
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Carbohydrates



Pyran

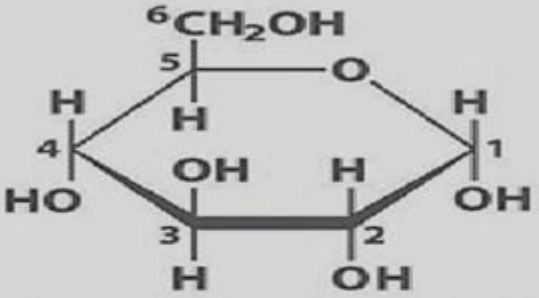


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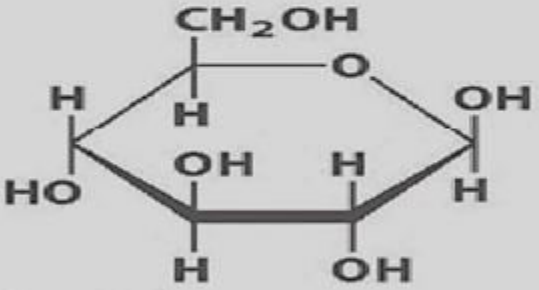
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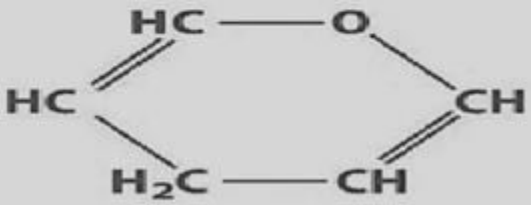
Carbohydrates



α-D-Glucopyranose



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Carbohydrates

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Carbohydrates

- This mixture consists of about;
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Carbohydrates

- α and β forms are NOT mirror images and are referred to as ***diastereomers***
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Carbohydrates

- For example, glycogen is synthesized from α-D-glucopyranose, whereas cellulose is synthesized from β-D-glucopyranose

Carbohydrates

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- In these compounds the hydroxyl group at C-5 (or C-6) reacts with the keto group at C-2, forming a furanose (or pyranose) ring containing a hemiketal linkage

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Carbohydrates

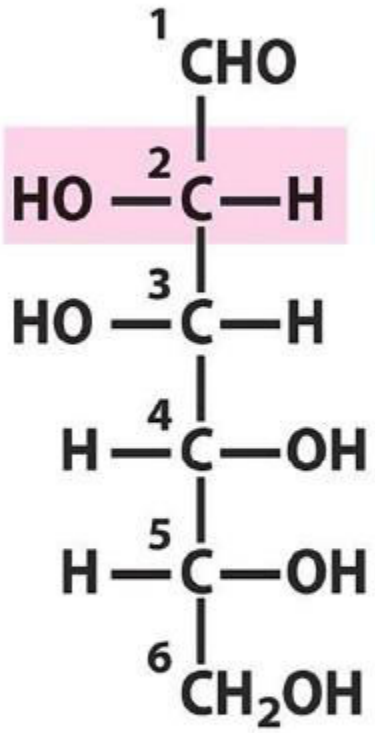
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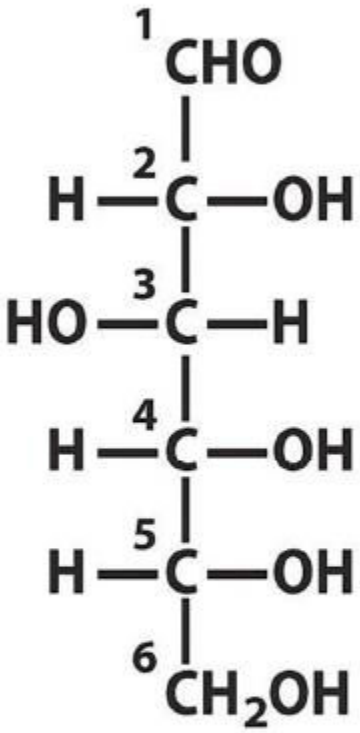
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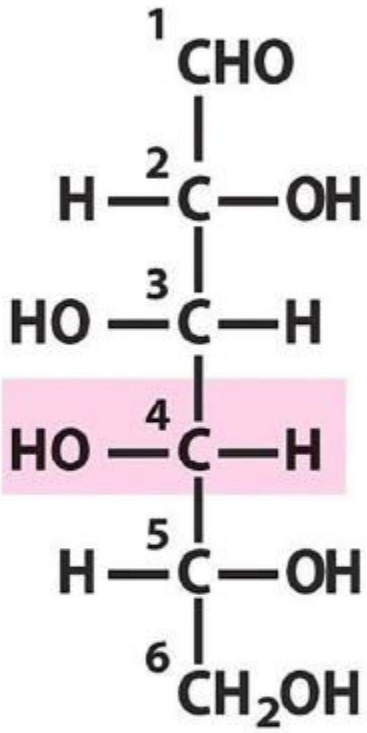
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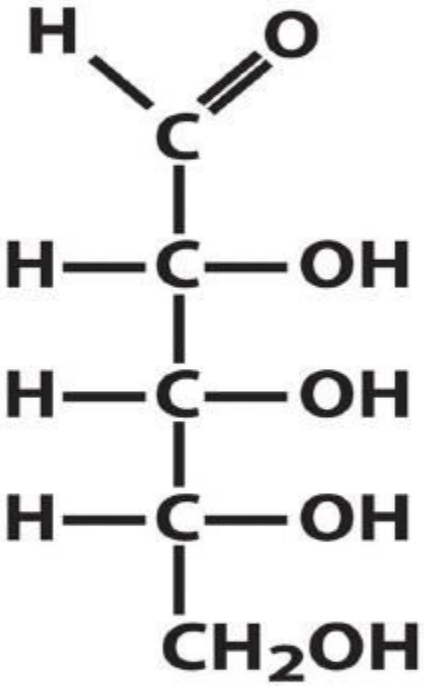
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Carbohydrates

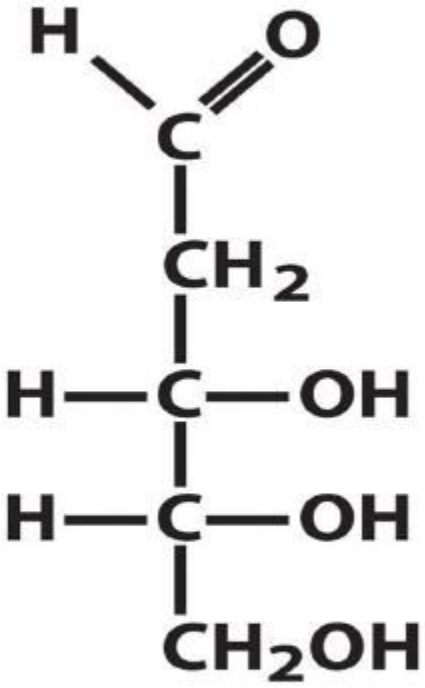
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Carbohydrates



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2-Deoxy-D-ribose,
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Carbohydrates

Hexoses

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Carbohydrates

- Sources include fruit juices, hydrolysis of starch, cane or beet sugar, maltose and lactose
- Glucose is the major metabolic fuel of mammals

Carbohydrates

- It is the chief physiological sugar present in normal blood at a fairly constant level
- All tissues utilize glucose for energy
- Erythrocyte utilize glucose solely for energy purposes and Brain cells rely heavily on glucose

Carbohydrates

- It is the precursor for synthesis of all the other carbohydrates in the body, including
  - **glycogen** for storage;
  - **ribose** and **deoxyribose** in nucleic acids;
  - **galactose** in lactose of milk, in glycolipids,

Carbohydrates

- And in combination with protein in glycoproteins and proteoglycans
- Stored as Glycogen in liver and muscles only
- Excreted in the urine (glucosuria) in poorly controlled diabetes mellitus as a result of hyperglycemia

Carbohydrates

LECTURE NO 13

D-Galactose:

- Occurs as a constituent of milk sugar lactose
- It is an epimer of glucose and differs in orientation of H and OH around C-4
- Readily metabolized to glucose;

Carbohydrates

- Synthesized in the mammary gland for synthesis of lactose in milk
- A constituent of glycolipids and glycoproteins
- **Hereditary galactosemia** as a result of failure to metabolize galactose leads to cataracts

Carbohydrates

D-Fructose:

- Commonly known as *fruit sugar* as it occurs free in fruits, honey etc
- Also occurs as a constituent of sucrose
- It is levorotatory and hence is also called Levulose

Carbohydrates

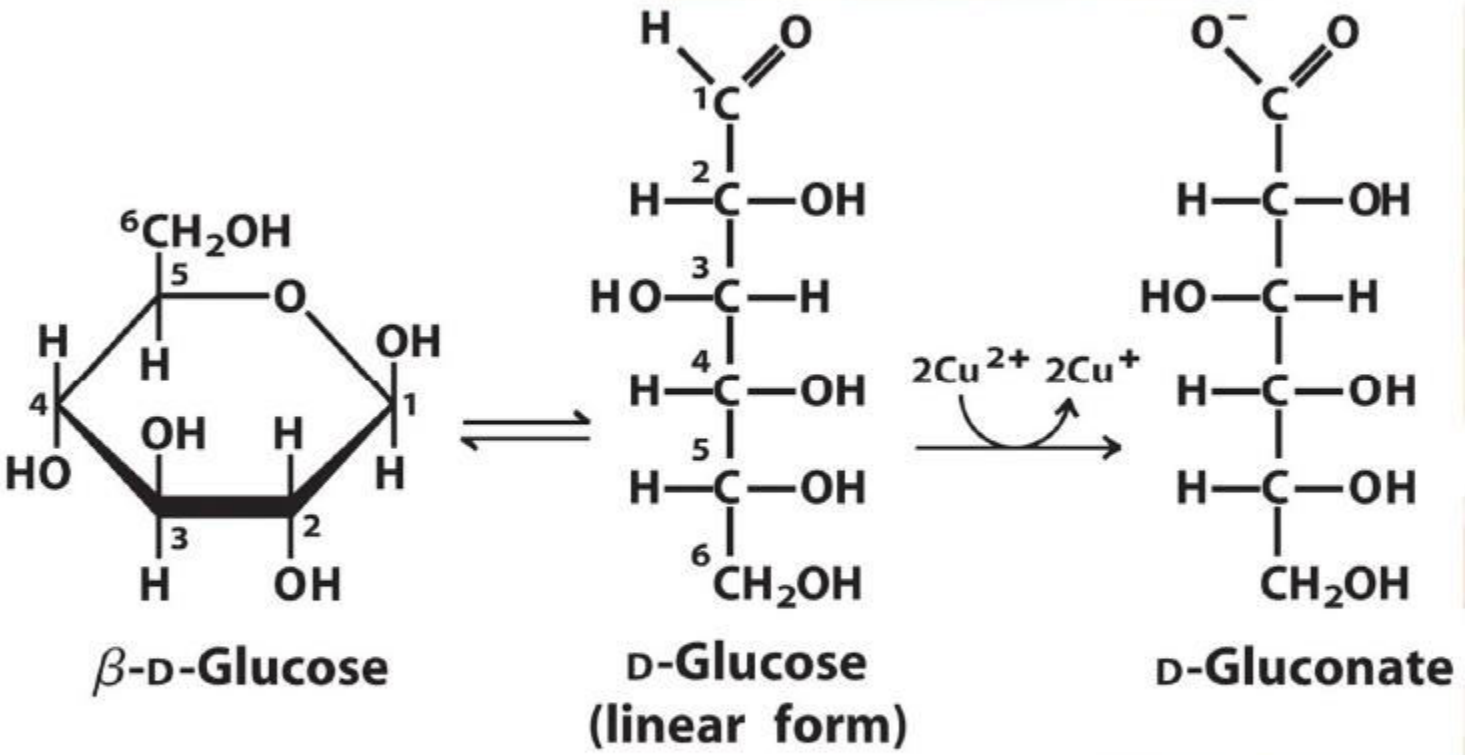
- *Seminal fluid is rich in fructose and sperm utilize fructose for energy*
- Readily metabolized either via glucose or directly
- **Hereditary fructose intolerance** leads to fructose accumulation

Carbohydrates

LECTURE NO 14

- **Properties of Monosaccharides**
- **1. Oxidation** of the carbonyl (aldehyde) carbon of glucose to the carboxyl level produces ***gluconic acid***
- Other aldoses yield other ***aldonic acids***

Carbohydrates



Carbohydrates

- Oxidation of the carbon at the other end of the carbon chain---C-6 of glucose, galactose, or mannose-forms the corresponding ***uronic acid,*** glucuronic, galacturonic, or mannuronic acid

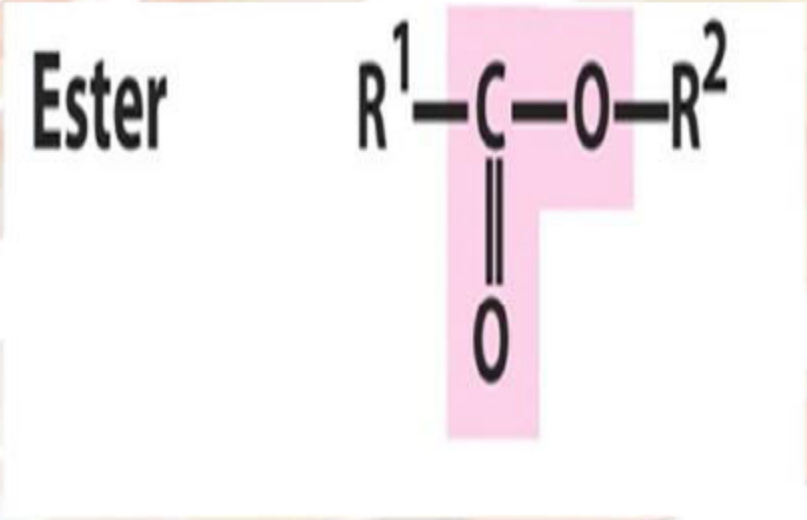
Carbohydrates

- The carboxylic acid groups of the acidic sugar derivatives are ionized at pH 7, and the compounds are therefore correctly named as the carboxylates-glucuronate, galacturonate, and so forth

Carbohydrates

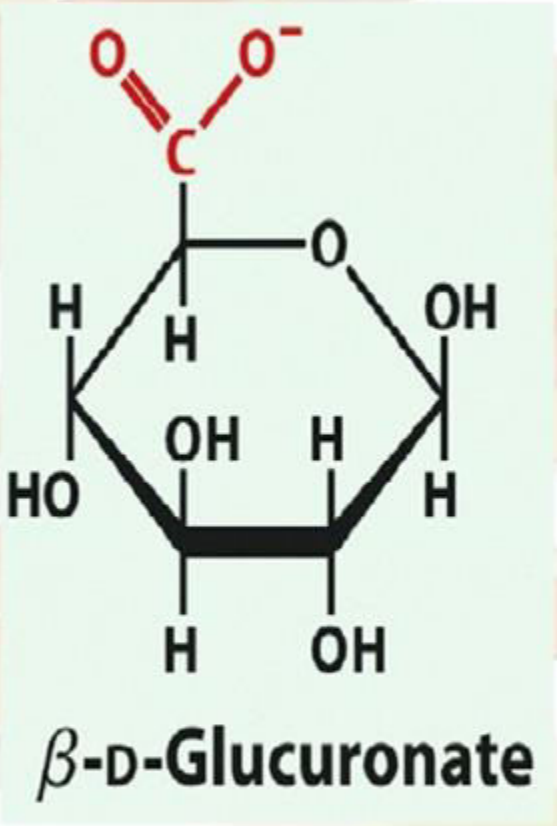
- Both aldonic and uronic acids form stable intramolecular esters called ***lactones***
- D-Glucono-δ-lactone results from formation of an ester linkage between the C-l carboxylate group and the C-5 (also known as the δ carbon) hydroxyl group of D-gluconate

Carbohydrates



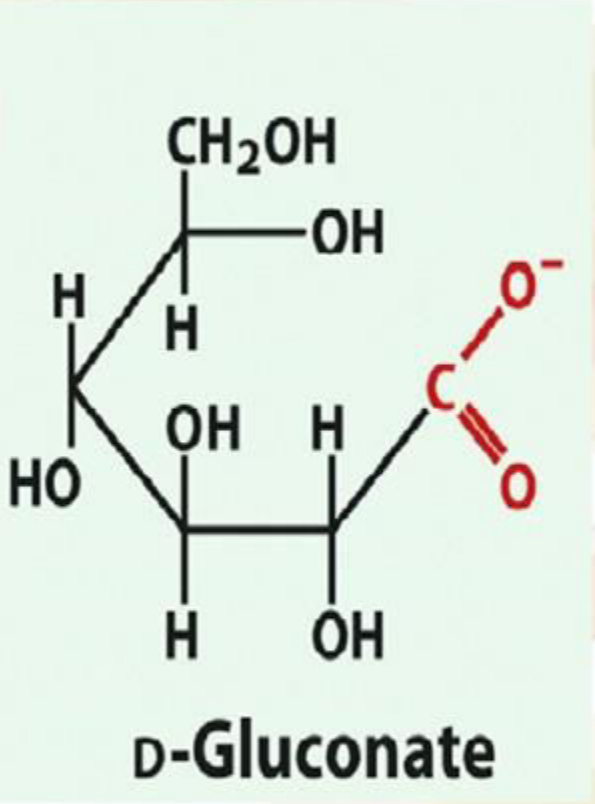
Carbohydrates

Glucuronic acid(Oxidation at C-1)



Carbohydrates

Gluconic acid(Oxidation at C-1)



Carbohydrates



Carbohydrates

- **Properties of Monosaccharides**
- Glucono delta-lactone (GDL) is commonly found in honey, fruit juices, personal lubricants, and wine . GDL is neutral.

LECTURE NO 15

Carbohydrates

- But hydrolyses in water to gluconic acid which is acidic, adding a tangy taste to foods
- It is metabolized to glucose; one gram of GDL yields roughly the same amount of metabolic energy as one gram of sugar.

Carbohydrates

- In the body, D-Glucuronic acid is formed from glucose in liver
- It occurs as a constituent of certain polysaccharides
- It is of importance in that it conjugates toxic substances, drugs, hormones and even bilirubin.

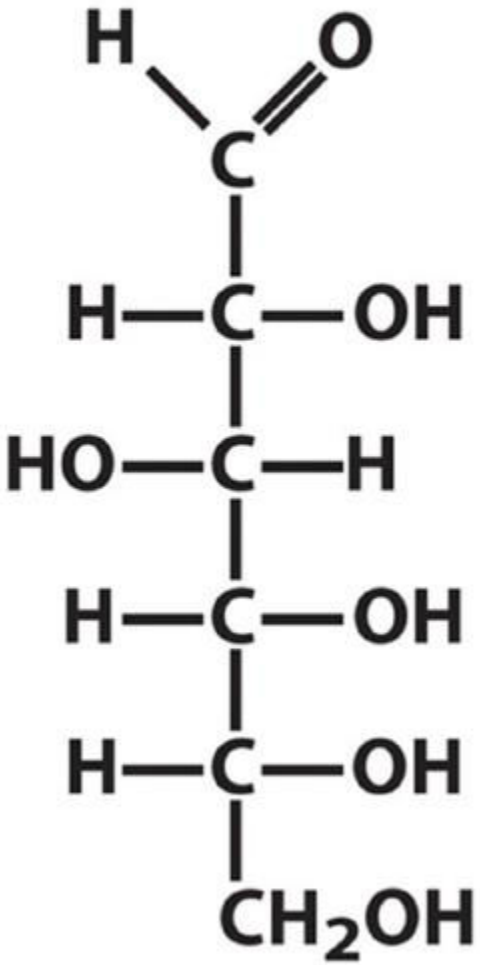
Carbohydrates

- And converts them to a soluble non toxic substance, a glucuronide, which is excreted in urine
- Oxidation of the carbon at the both ends of the carbon chain.

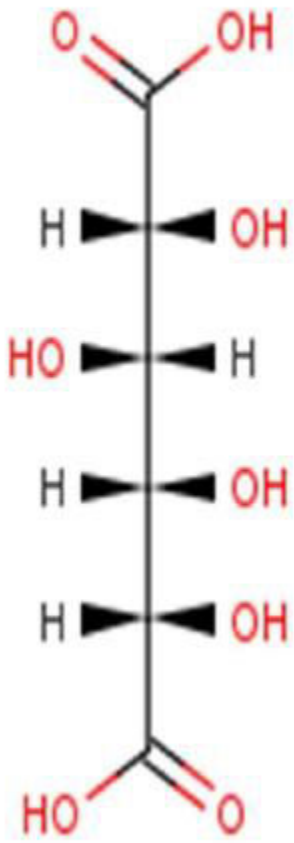
Carbohydrates

- **The terminal methyl & the carbonyl carbon** of glucose, galactose, or mannose- forms the corresponding **aldaric acid**
- Glucose forms **Glucaric acid**

Carbohydrates



D-Glucose,



Carbohydrates

- Monosacharides can be oxidized by relatively mild oxidizing agents such as cupric (Cu^{2+}) ion
- The carbonyl carbon is oxidized to a carboxyl group
- Glucose and other sugars capable of reducing cupric ion are called ***reducing sugars***

Carbohydrates

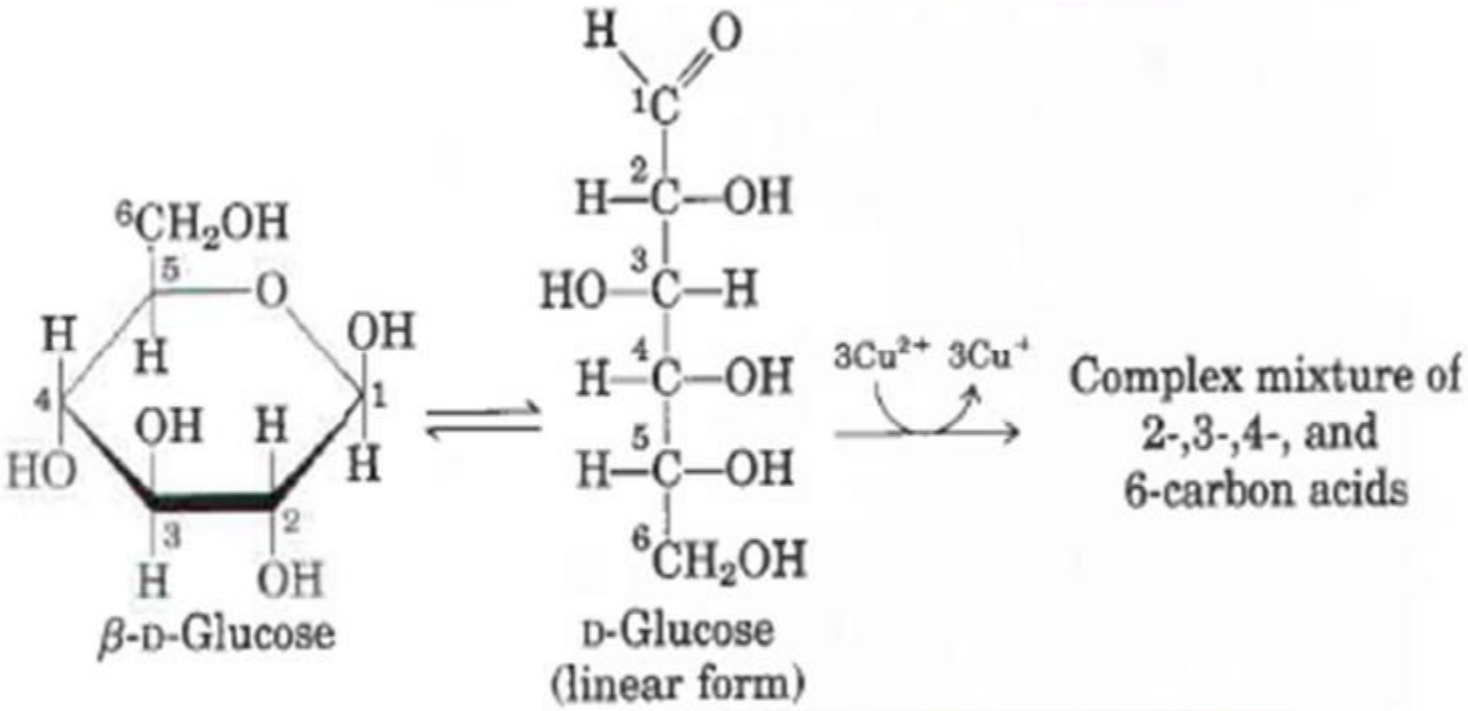
Sugars as reducing agents

- Oxidation of the anomeric carbon (and probably the neighboring carbon) of glucose and other sugars under alkaline conditions is the basis for
- Fehling's reaction
- Benedicts reaction

Carbohydrates

- The cuprous ion (Cu^{+}) produced forms a red cuprous oxide precipitate

Carbohydrates



Carbohydrates

- By measuring the amount of oxidizing agent reduced by a solution of a sugar, it is also possible to estimate the concentration of that sugar

Carbohydrates

- For many years this test was used to detect and measure elevated glucose levels in blood and urine in the diagnosis of diabetes mellitus

Carbohydrates

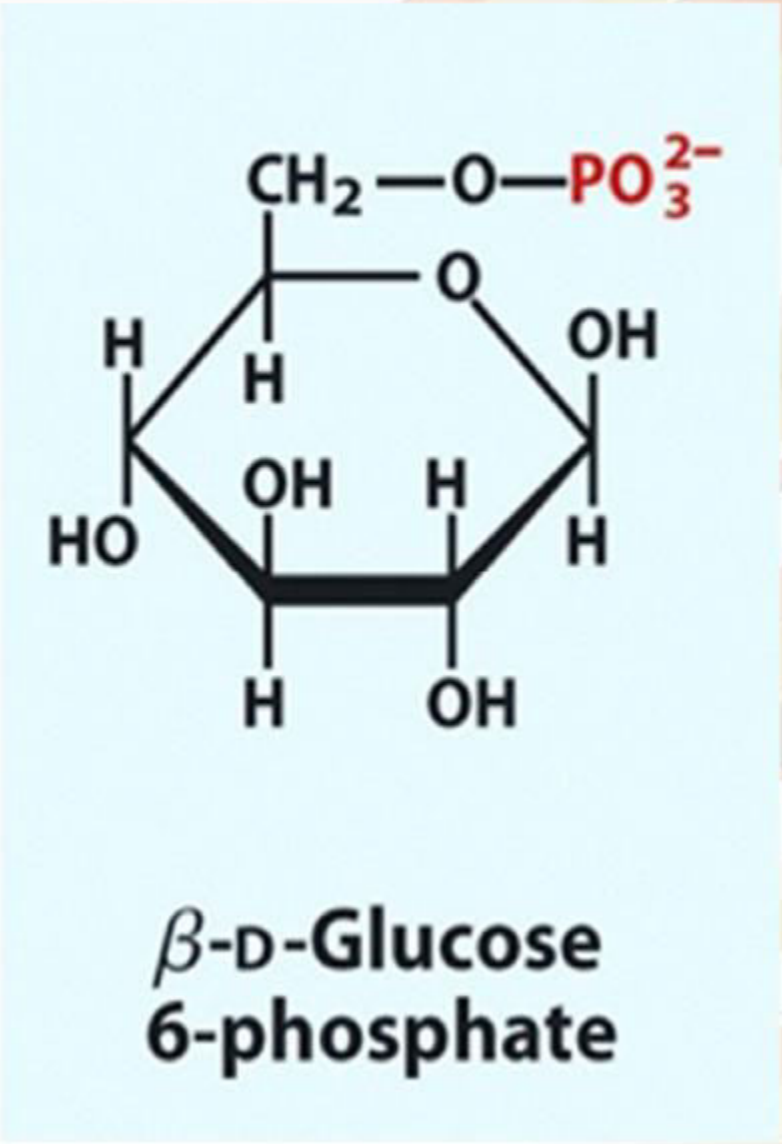
- **Properties of Monosaccharides**
- 2. Condensation of phosphoric acid with one of the hydroxyl groups of a sugar forms a ***phosphate ester***, as in glucose 6-phosphate

LECTURE NO 16

Carbohydrates

- Sugar phosphates are relatively stable at neutral pH and bear a negative charge

Carbohydrates



Carbohydrates

- In fact, one of the strategies of glycolysis is to form three carbon intermediates, that can transfer their phosphate groups to ADP, to achieve a net synthesis of ATP

Carbohydrates

- In the synthesis and metabolism of carbohydrates, the intermediates are very often not the sugars themselves but their phosphorylated derivatives

Carbohydrates

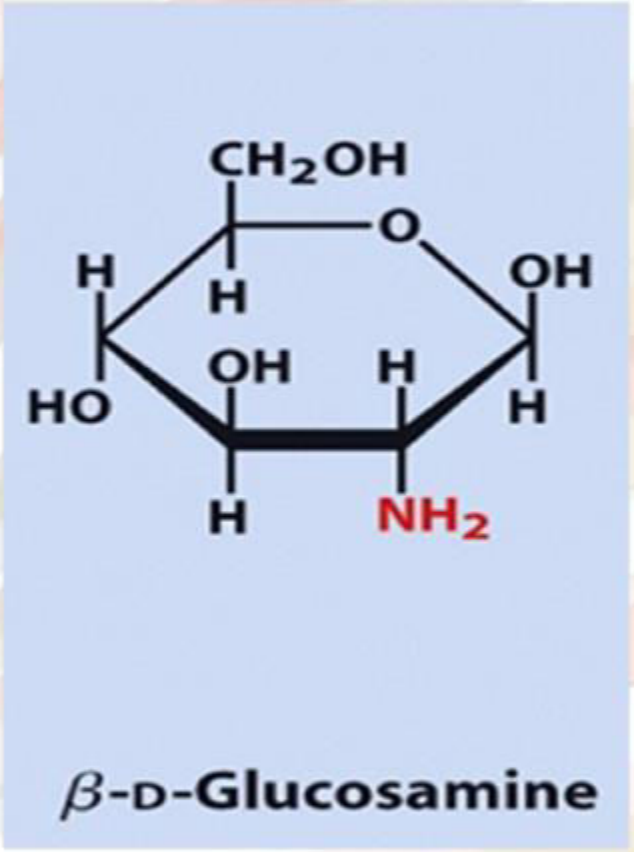
- Another function of phosphorylation is the creation of reactive intermediates for the formation of O- and N-glycosidic linkages

Carbohydrates

- **Properties of Monosaccharides**
- 3.Synthesis of Amino Sugar
- If the hydroxyl at C-2 of the parent monosaccharide is replaced with an amino group, the corresponding ***amino sugar*** is formed

LECTURE NO 17

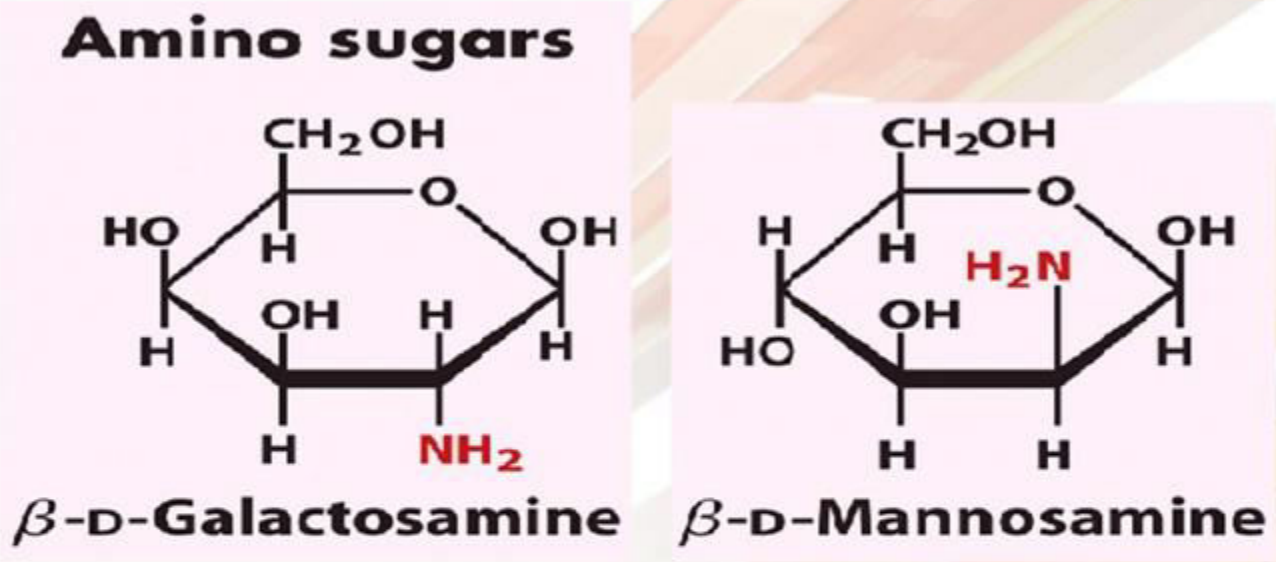
Carbohydrates



Carbohydrates

- The amino sugars include:
  - ***D -glucosamine***, a constituent of hyaluronic acid
  - ***D -galactosamine*** (also known as *chondrosamine*), a constituent of chondroitin
  - ***D -mannosamine***

Carbohydrates



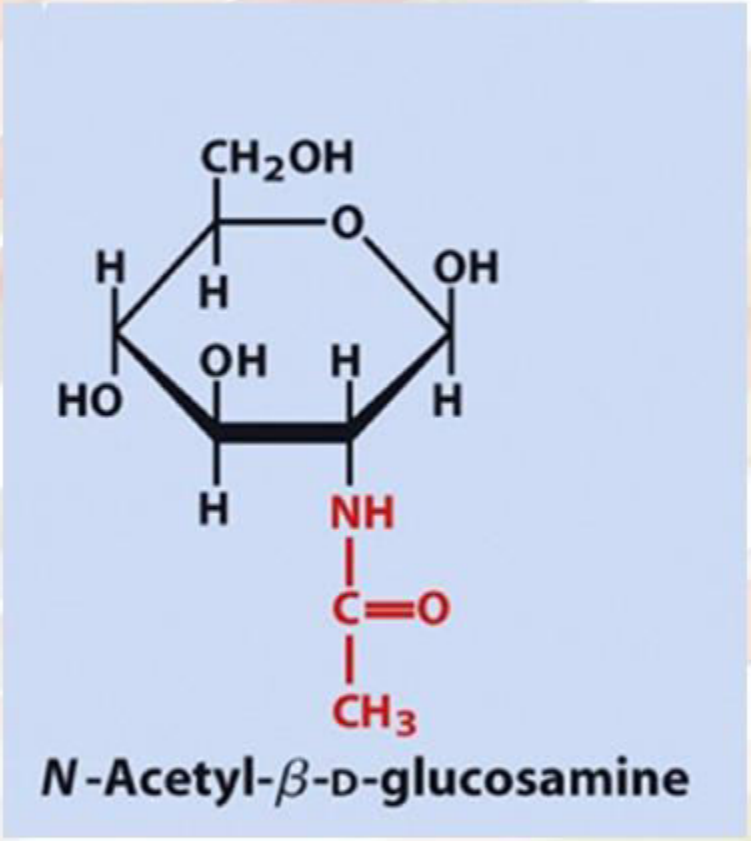
Carbohydrates

- The amino group is nearly always condensed with ***acetic acid***, as in N-acetylglucosamine
- Both glucosamine and galactosamine occur as *N*-acetyl derivatives in more complex carbohydrates.

Carbohydrates

- For example glycoproteins and is part of many structural polymers, including those of the bacterial cell wall

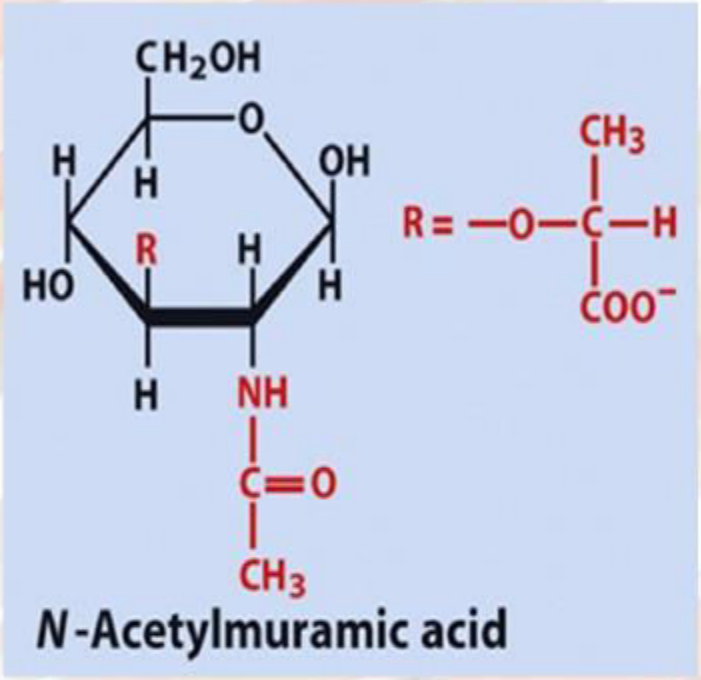
Carbohydrates



Carbohydrates

- Bacterial cell walls also contain a derivative of glucosamine, ***N-acetylmuramic acid***, in which lactic acid (a three carbon carboxylic acid) is ether-linked to the oxygen at C-3 of N-acetylglucosamine

Carbohydrates



Carbohydrates

- The Amino Sugars (Hexosamines) are Components of Glycoproteins, Gangliosides, & Glycosaminoglycans
- Several **antibiotics** (e.g, erythromycin) contain amino sugars, which are important for their antibiotic activity

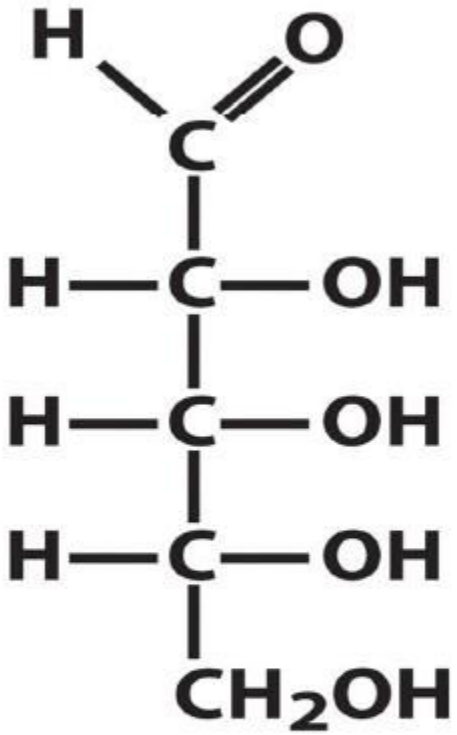
Properties of Monosaccharides

- 4. When oxygen of a –OH group is removed in a monosaccharide, leaving behind only hydrogen, ***deoxy sugars*** are formed

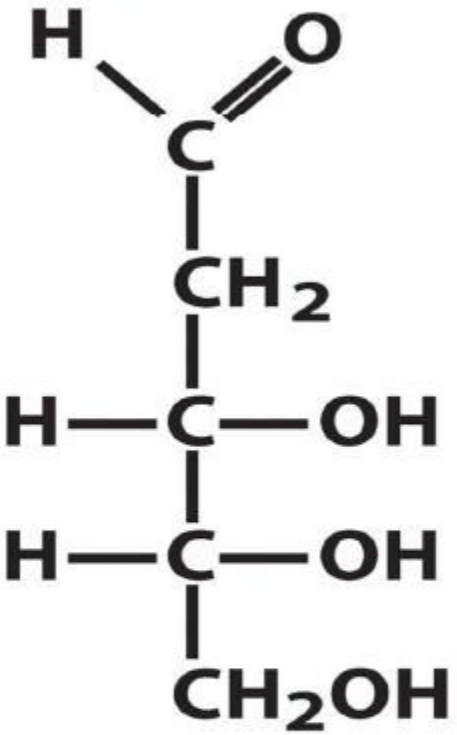
Properties of Monosaccharides

- 2-Deoxy-D-Ribose is derived from D-Ribose and is an important constituent of DNA

Properties of Monosaccharides



D-Ribose,
an aldopentose



2-Deoxy-D-ribose,
an aldopentose

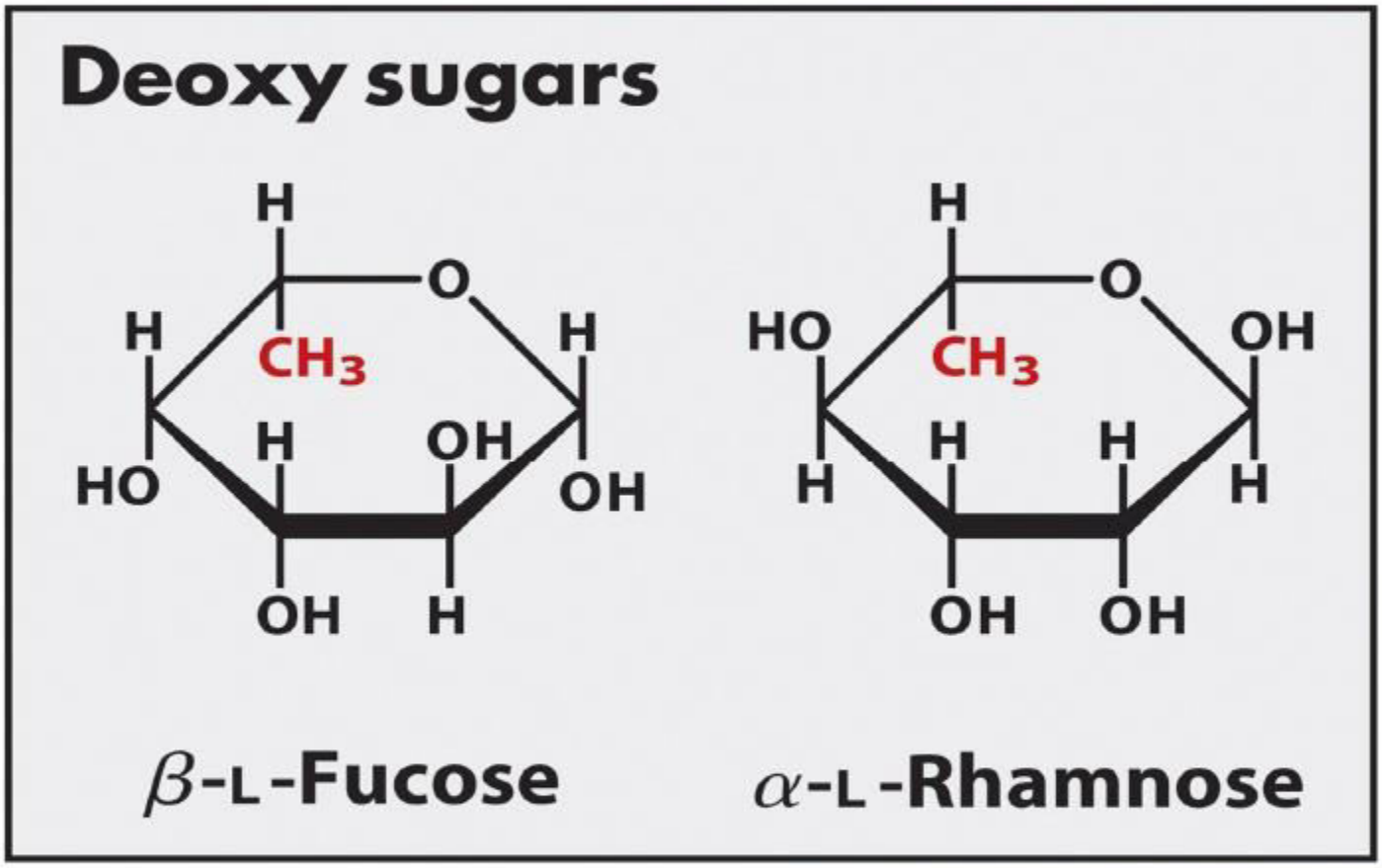
Properties of Monosaccharides

- The substitution of a hydrogen for the hydroxyl group at C-6 of ;
- L-galactose &
- L-mannose produces;
- L-fucose and
- L-rhamnose, respectively.

Properties of Monosaccharides

- ***L-Fucose*** is found in the complex oligosaccharide components of
- glycoproteins and glycolipids;
- **L-Rhamnose** is found in plant polysaccharide

Properties of Monosaccharides



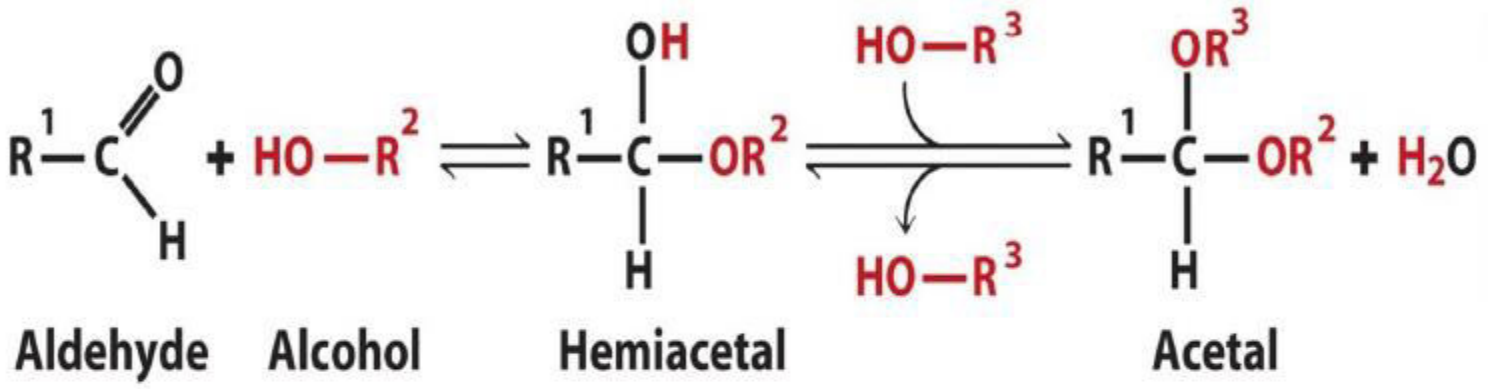
Properties of Monosaccharides

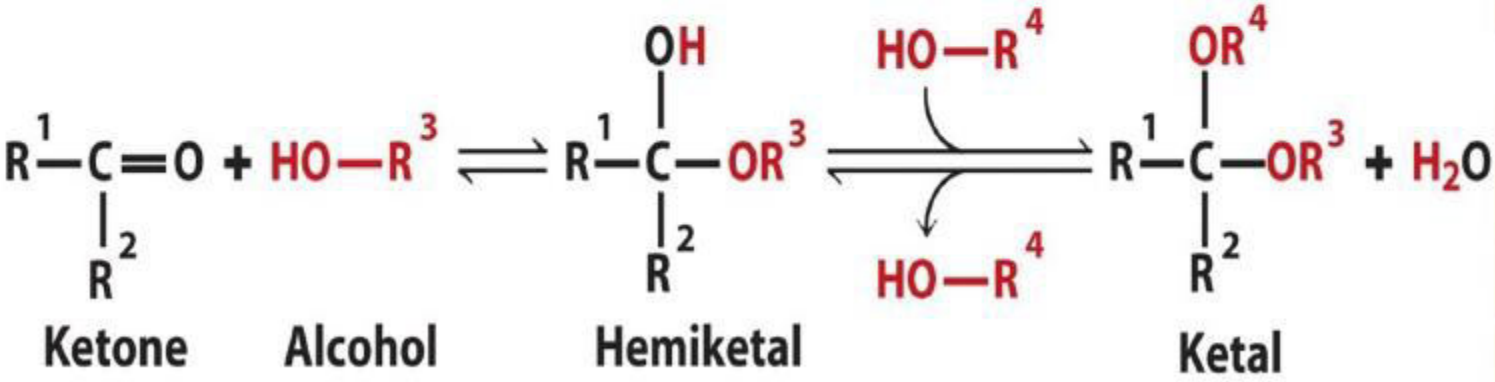
- **5. Glycosides** are formed by condensation between the hydroxyl group of the anomeric carbon of a monosaccharide, and a second compound that may or may not (in the case of an **aglycone**) be another monosaccharide

Properties of Monosaccharides

- If the second group is a ***hydroxyl,*** the ***O-glycosidic*** bond is an **acetal link** because it results from a reaction between a hemiacetal group (formed from an aldehyde and an —OH group) and another —OH group

Properties of Monosaccharides

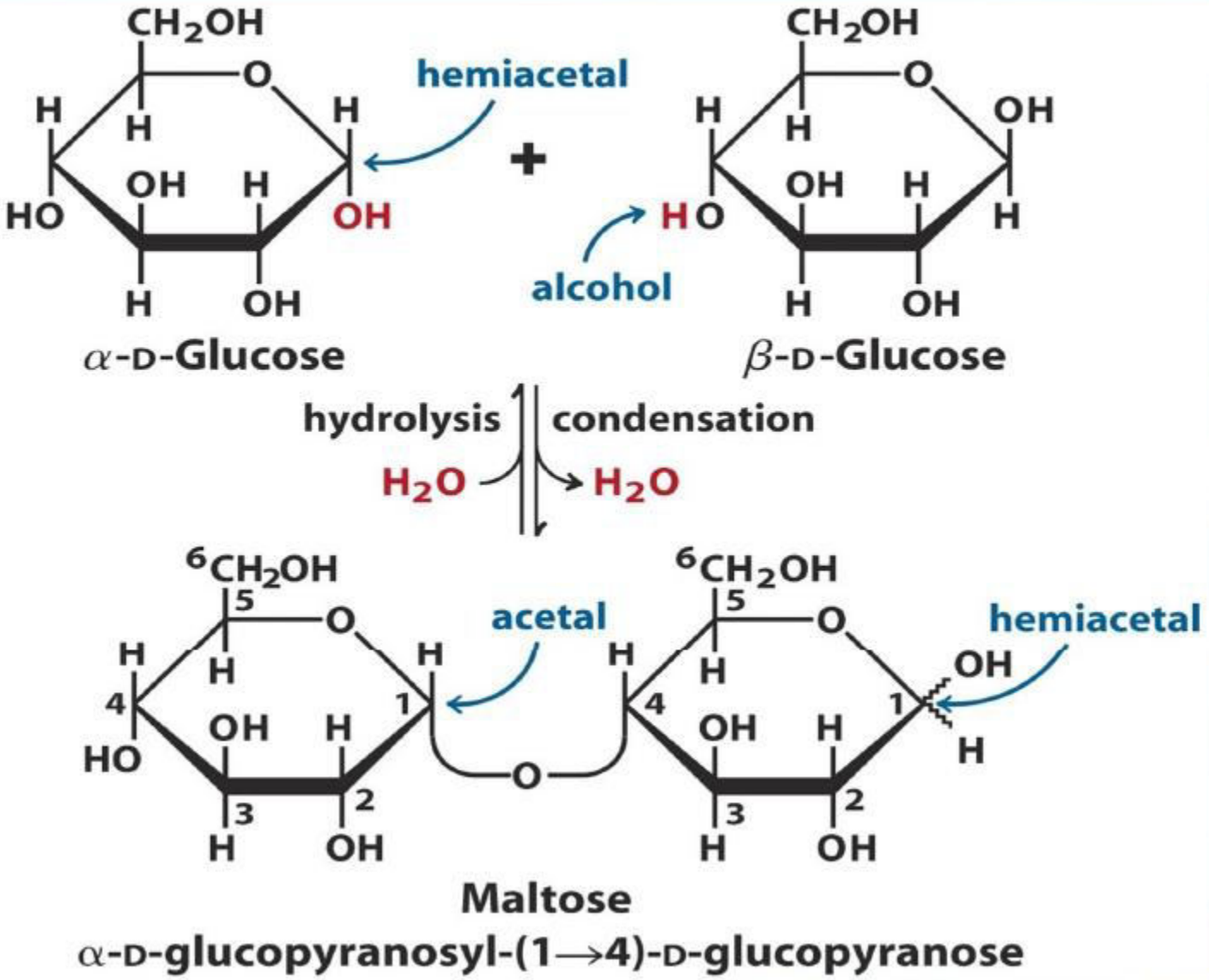




Properties of Monosaccharides

- If the hemiacetal portion is glucose, the resulting compound is a **glucoside**
- If galactose, a **galactoside,** and so on

Properties of Monosaccharides



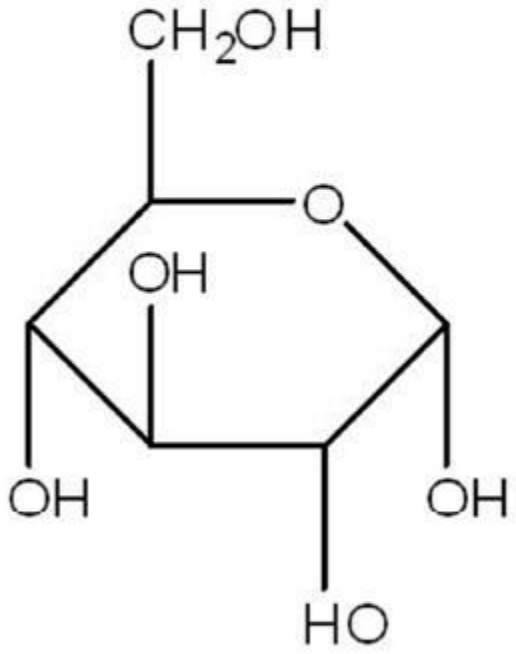
Properties of Monosaccharides

6. The monosaccharides may be reduced to form their ***corresponding alcohols*** by reducing agents;

Properties of Monosaccharides

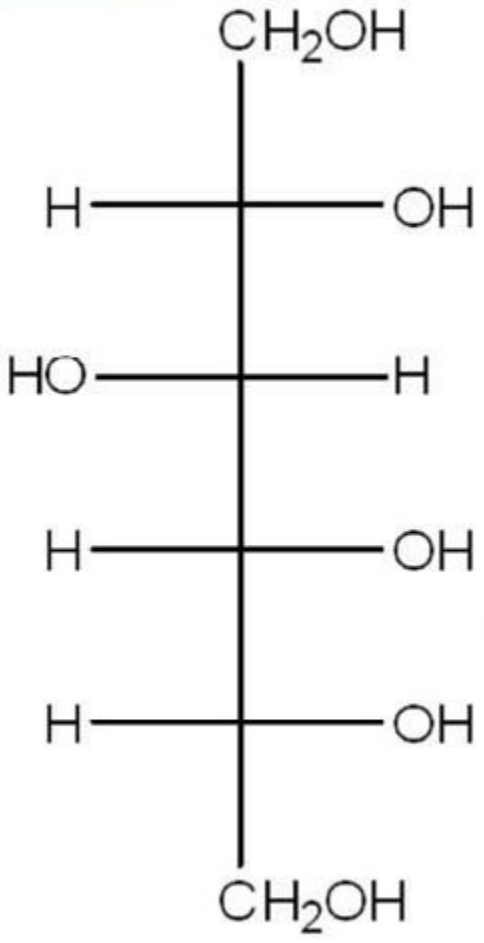
- Examples are
  - D-Glucose yields D-Sorbitol
  - D-Galactose yields D-Dulcitol
  - D-Mannose yields D-Mannitol
  - Ribose to ribitol

Properties of Monosaccharides



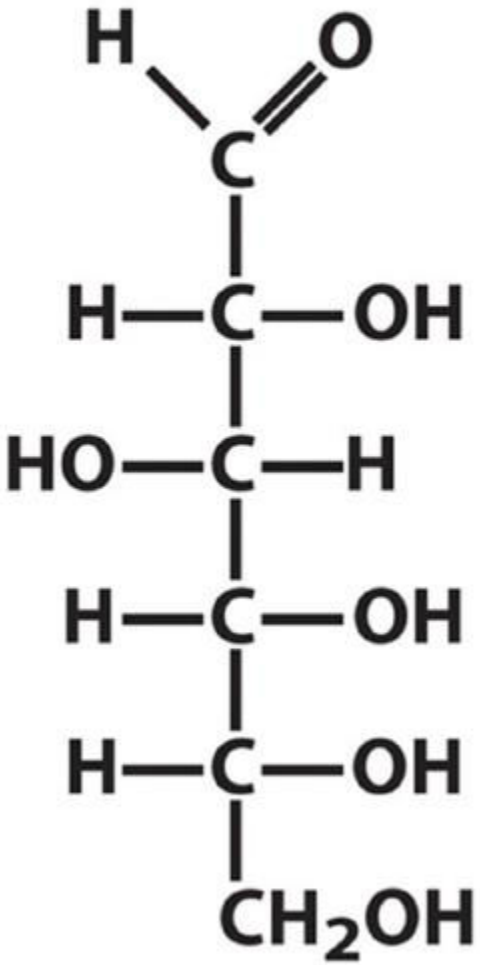
D-Glucose



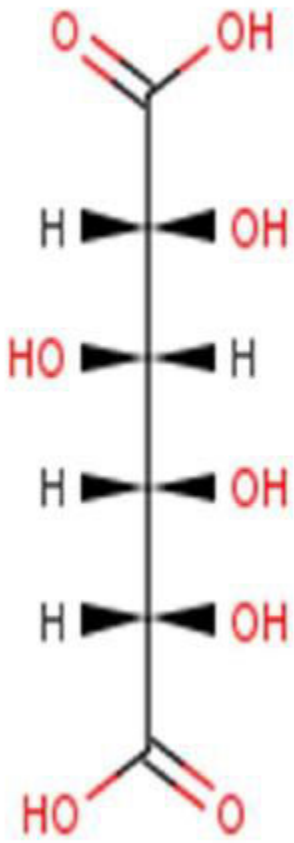


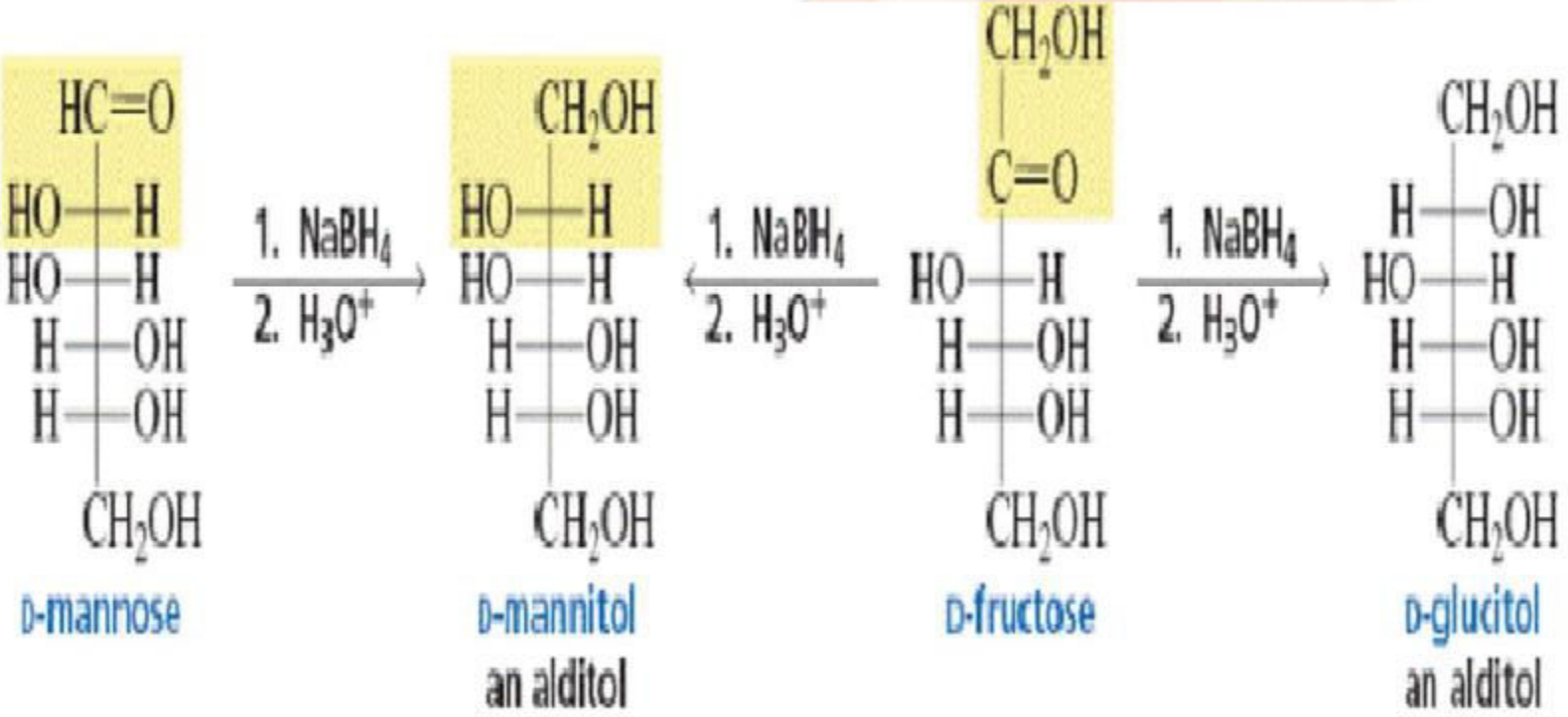
D-Sorbitol

Carbohydrates



D-Glucose,




HC=O
HO—H
HO—H
H—OH
H—OH
CH2OH
D-mannose
1. NaBH4
2. H3O+
CH2OH
HO—H
HO—H
H—OH
H—OH
CH2OH
D-mannitol
an alditol
1. NaBH4
2. H3O+
CH2OH
C=O
HO—H
H—OH
H—OH
CH2OH
D-fructose
1. NaBH4
2. H3O+
CH2OH
H—OH
HO—H
H—OH
H—OH
CH2OH
D-glucitol
an alditol

Properties of Monosaccharides

Sorbitol:

- 35-60% sweetness of glucose
- Used as an artificial sweetner
- Accumulates in tissues such as the eye lens in diabetes mellitus

Properties of Monosaccharides

Mannitol:

- Administered intravenously as an osmotic diuretic in acute renal failure
- Also used to relieve raised intracranial pressure by forced diuresis

Properties of Monosaccharides

7. Tautomerization

- The process of shifting a hydrogen atom from one carbon atom to another to produce **enediols** is known as tautomerization

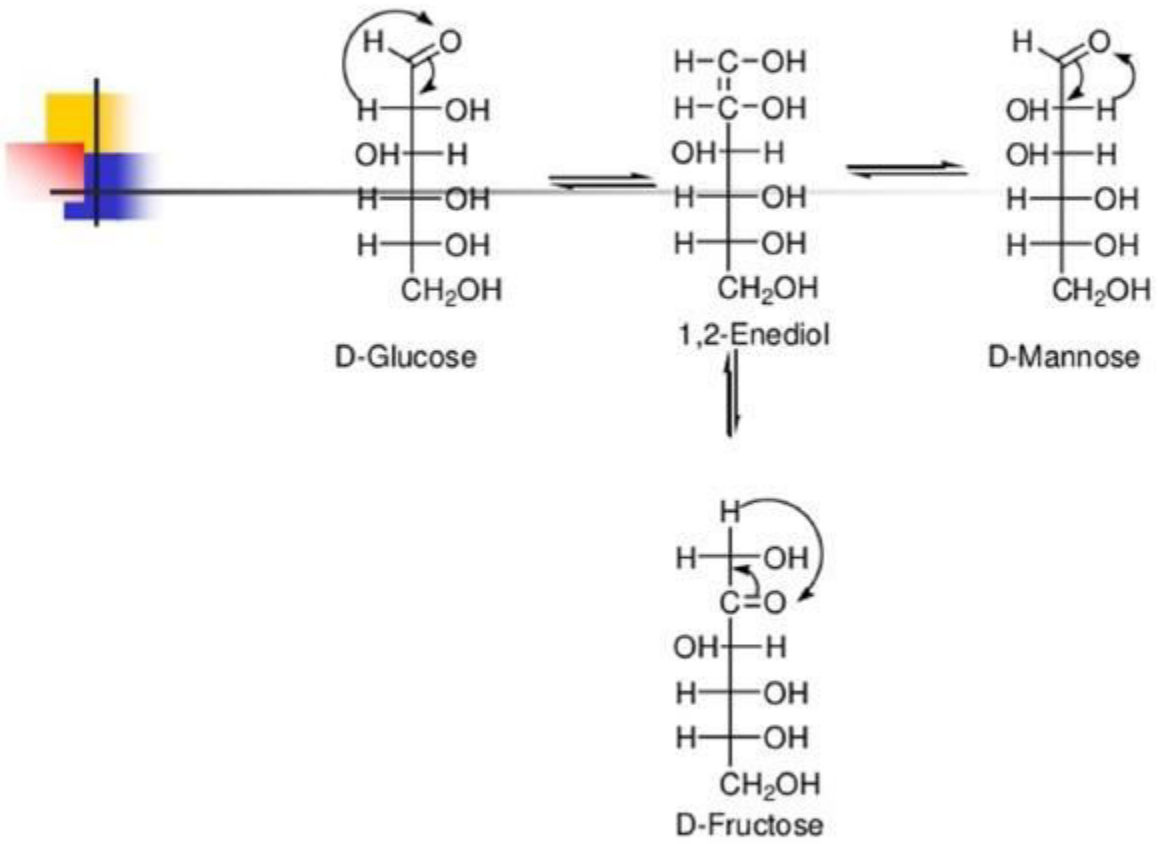
Properties of Monosaccharides

Sugars possessing anomeric carbon atom undergo

tautomerization in alkaline solutions

- When glucose is kept in alkaline solution for several hours, it undergoes isomerization to form D-fructose and D-mannose

Properties of Monosaccharides



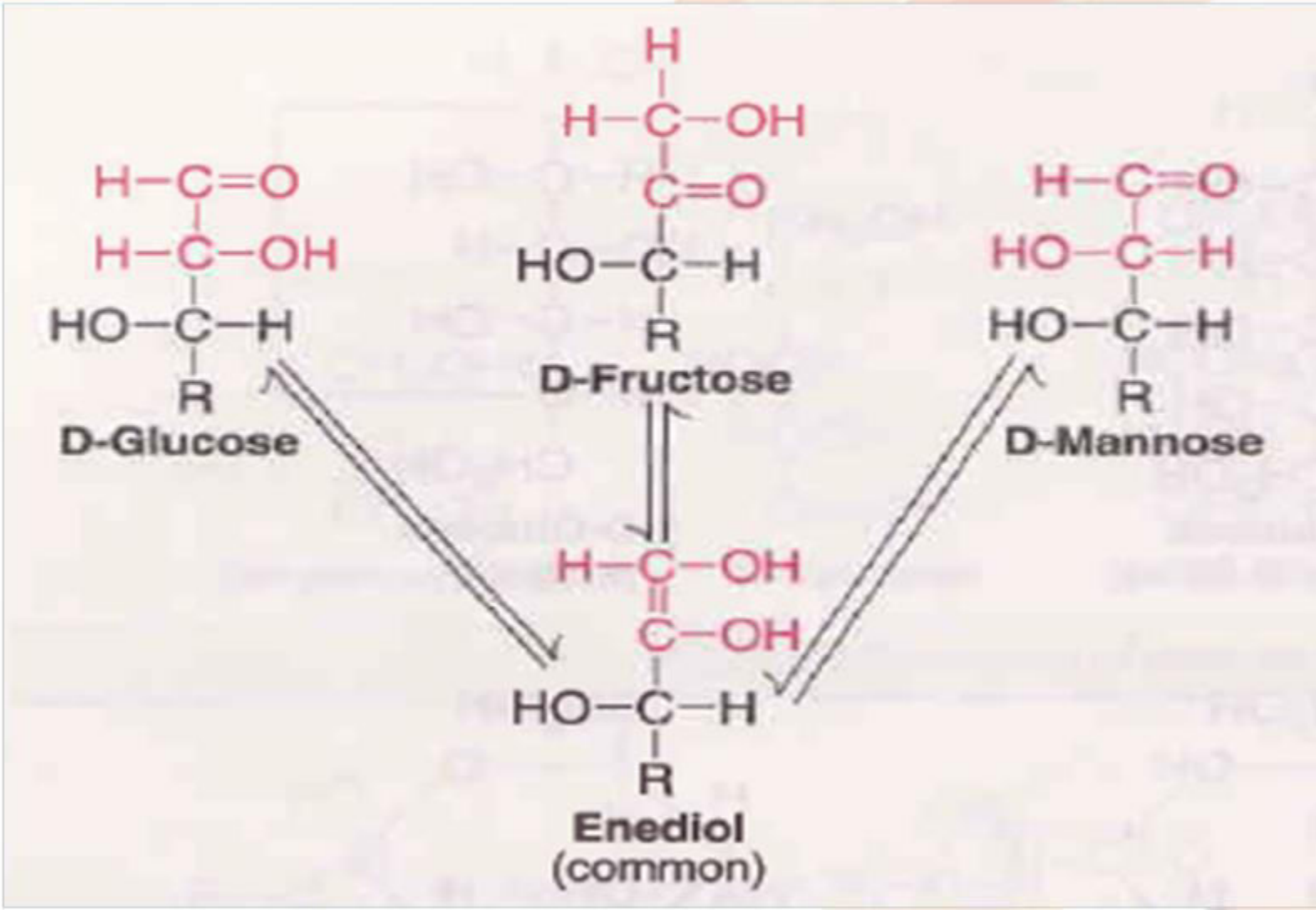
Properties of Monosaccharides

This reaction results in the formation of a

common intermediate-namely enediol--for all the three sugars

- The enediols are highly reactive, hence sugars in alkaline solution are powerful reducing agents

Properties of Monosaccharides



Properties of Monosaccharides

8. ***Osazone formation***:

- It is a useful means of preparing *crystalline derivatives* of sugars called ***osazones***
- They are obtained by adding a mixture of ***phenyl hydrazine hydrochloride*** and ***sodium acetate*** to a sugar solution and heating in water bath for 45 mins

Properties of Monosaccharides

- Crystals are formed after the solution is cooled slowly
- Reaction involves only the carbonyl carbon and the adjacent carbon

Properties of Monosaccharides

- First phenyl hydrazone is formed and then reacts with an additional molecule of phenylhydrazine to form the osazones.
- As is evident from the reaction, the first two carbons (C1 and C2) are involved in osazone formation

Properties of Monosaccharides

- The sugars that differ in their configuration on these two carbons give the same type of osazones, since the difference is masked by binding with phenylhydrazine

Properties of Monosaccharides

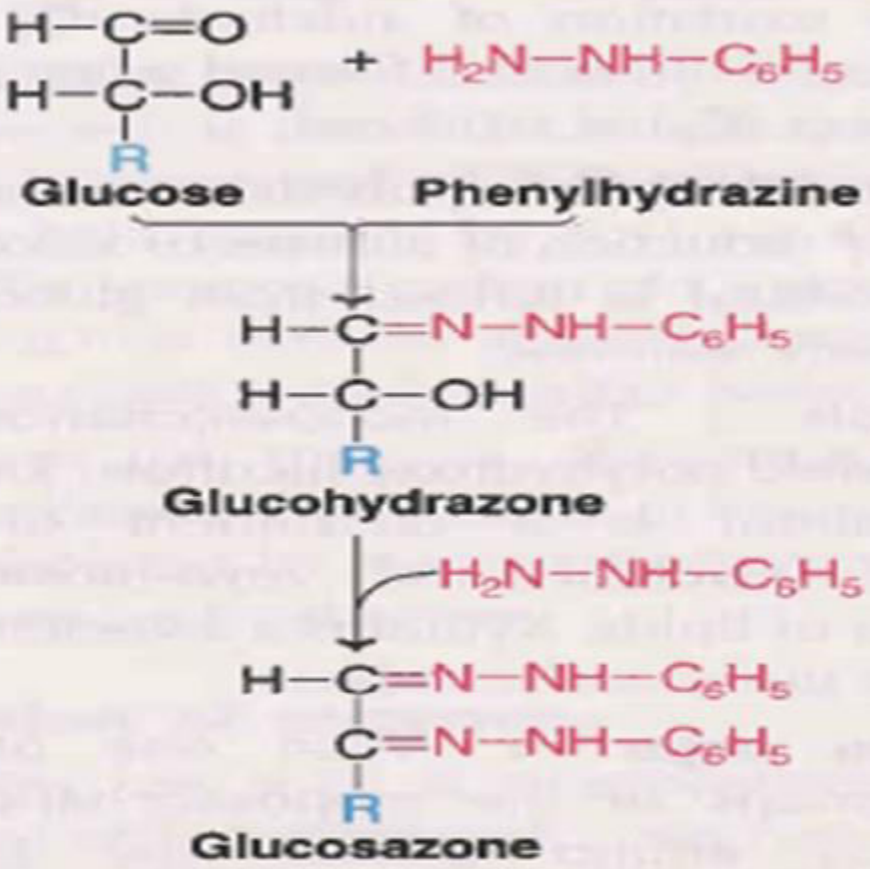
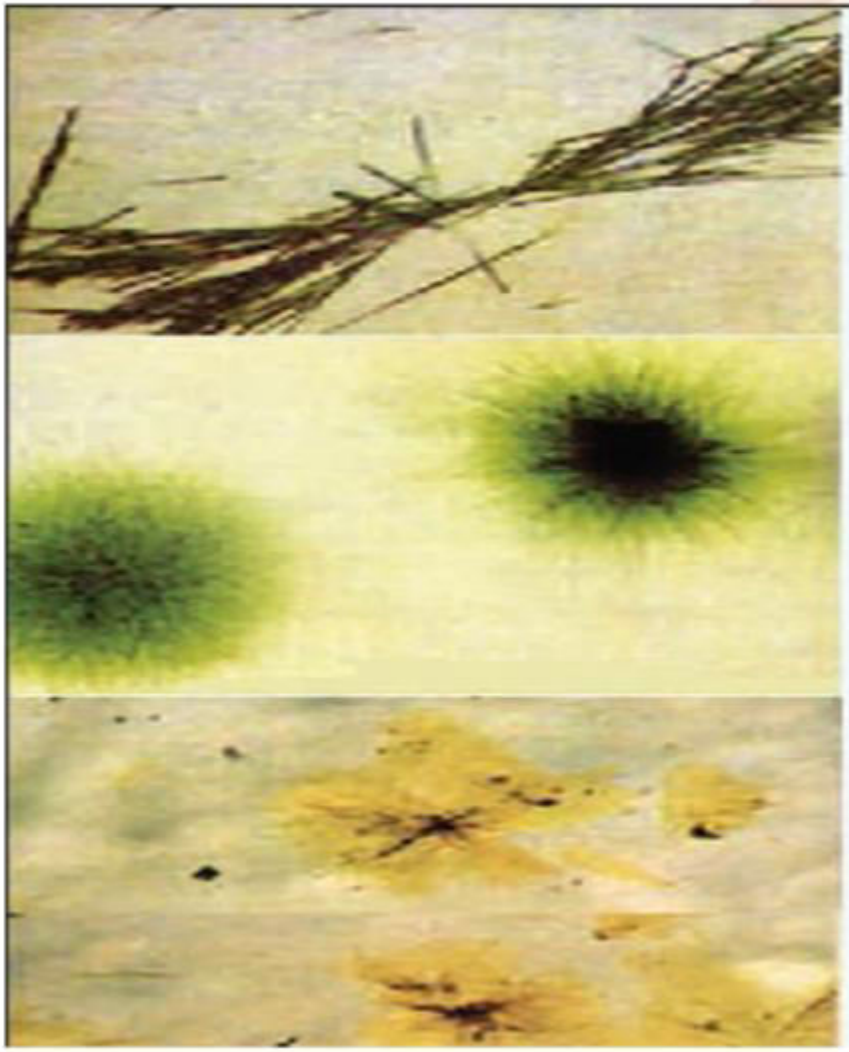


Fig. 2.10 : A summary of osazone formation (R represents C_3 to C_6 of glucose).

Properties of Monosaccharides

- Thus glucose, fructose and mannose give the same type (needle-shaped) osazones
- Reducing disaccharides also give osazones
- Maltose sunflower-shaped
- And lactose powder puff shaped

Properties of Monosaccharides



Needle shaped crystals arranged like a broom
Glucososazone

Hedgehog or "pincushion with pins" or flower of "touch-me-not-plant"
Lactososazone

Sunflower shaped or petal shaped crystals of
Maltosazone

Properties of Monosaccharides

- Osazones have characteristic
  - Melting points
  - Crystal structures
  - Precipitation time
  - and thus are used in identifying the sugars

Properties of Monosaccharides

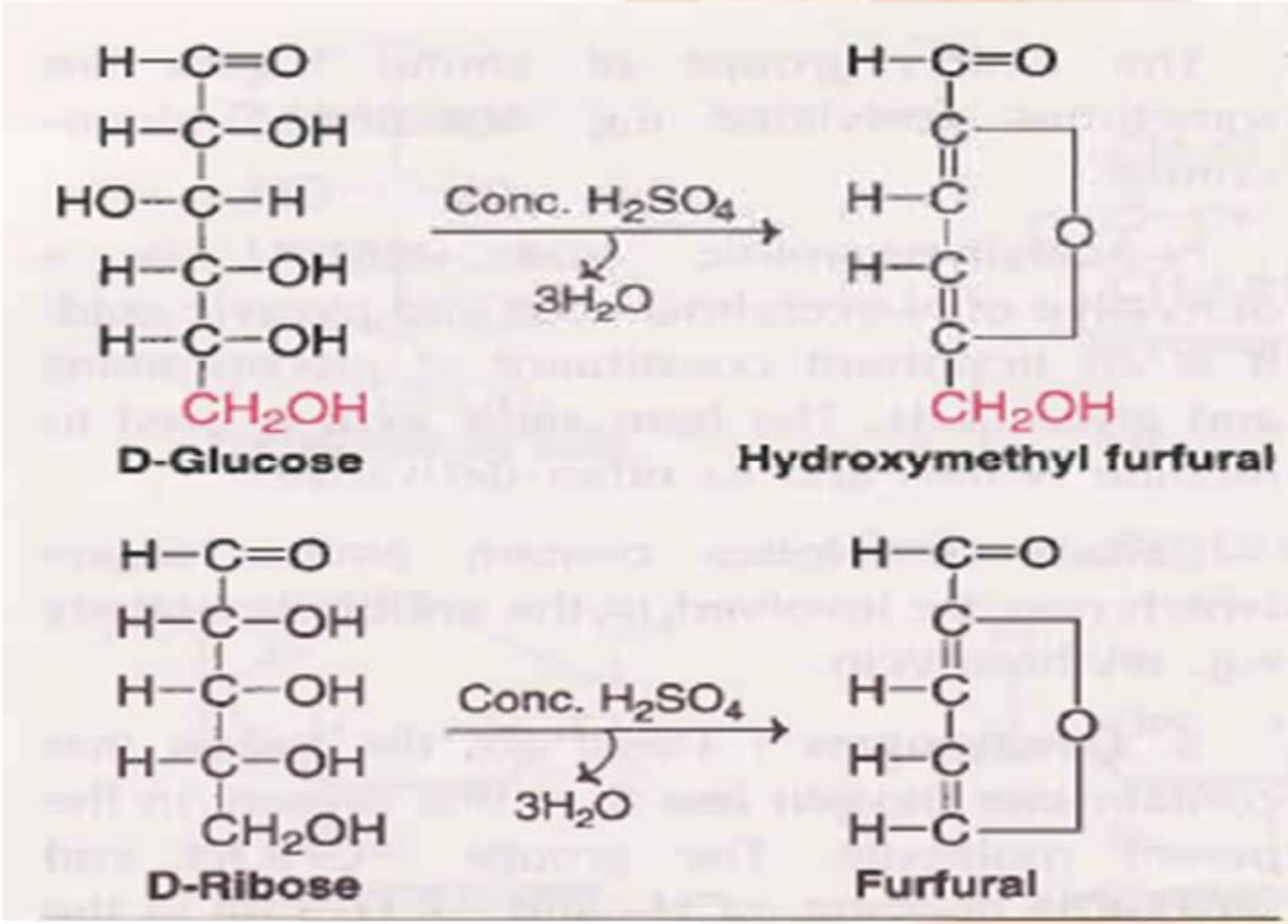
9. Action of acids on carbohydrates

- Polysaccharides are hydrolyzed into their constituent monosaccharides by boiling with **dilute** acids
- With **concentrated** acids the monosaccharides are decomposed

Properties of Monosaccharides

- Pentoses yield a cyclic aldehyde *furfural* with 12% HCL
- Hexoses are decomposed by hot strong acids to give *hydroxy methyl furfural*

Properties of Monosaccharides



Properties of Monosaccharides

- The furfural products thus formed can condense with certain organic phenols (α-naphthol) to form compounds having characteristic colours
- This forms basis of certain tests used for detection of sugars

Properties of Monosaccharides

- Molisch,s and
- Selivanoff,s are examples of such tests

Properties of Monosaccharides

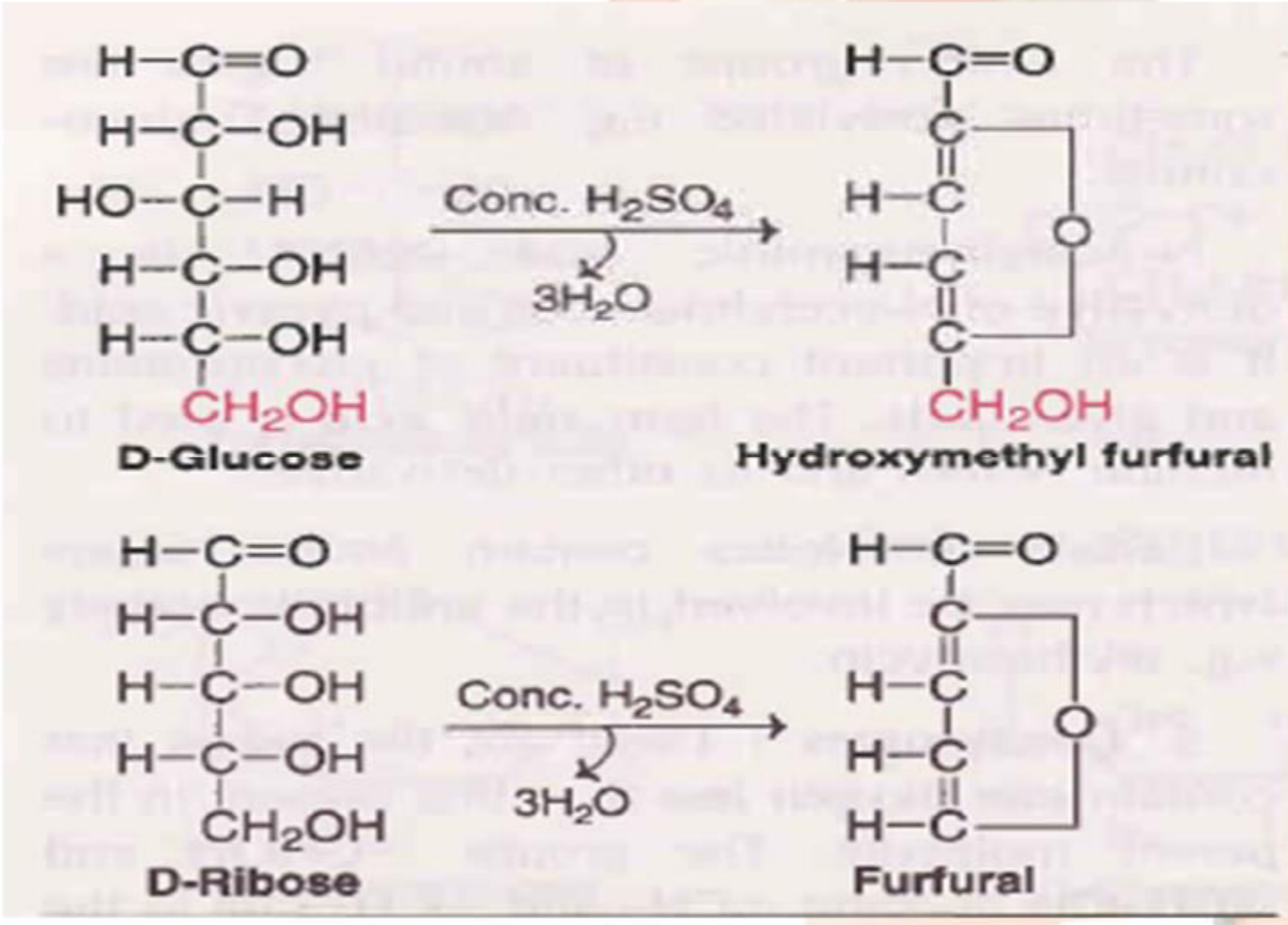
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- Selivanoff,s are examples of such tests

Disacharrides

- **Disaccharides** (such as maltose, lactose, and sucrose) consist of two monosaccharides joined covalently by an **O-glycosidic bond.**
- Which is formed when a hydroxyl group of one sugar reacts with the anomeric carbon of the other

LECTURE NO 22

Disacharrides

- When the anomeric carbon is involved in **a glycosidic bond**, that sugar residue is not capable of reducing cupric ions and therefore becomes a non-reducing sugar

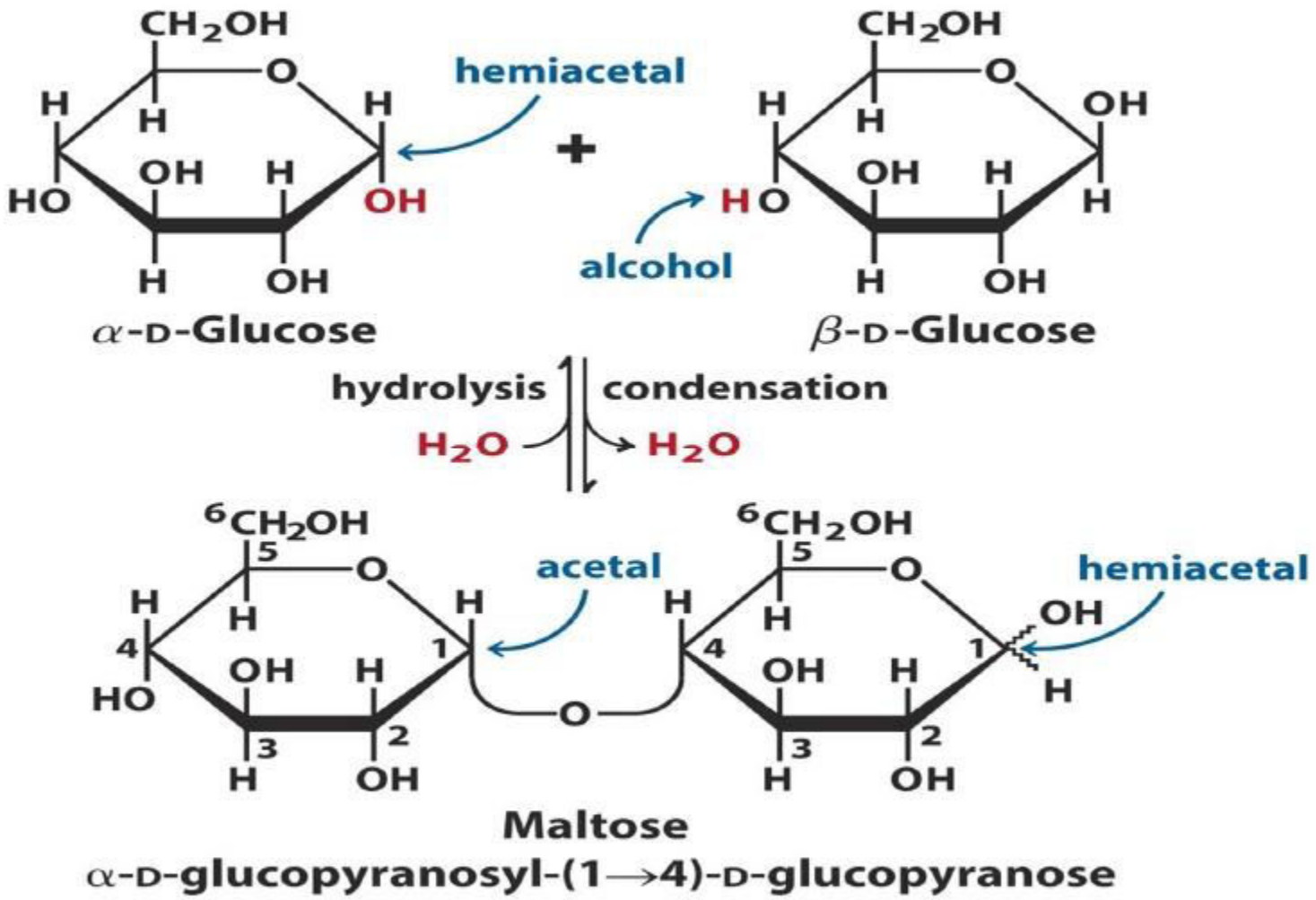
Disacharrides

- In describing disaccharides or polysaccharides, the end of a chain with a free anomeric carbon (one not involved in a glycosidic bond) is commonly called the reducing end

Disacharrides

- **Maltose** has two glucose units joined by an **α(1→4)** glycosidic linkage
- Since it has one aldehyde ‘free’, it has ***reducing properties*** and also exhibits mutarotation

Disacharrides



Disacharrides

- Maltose comes from the hydrolysis of starch (by **amylase**)
- It is in turn hydrolysed to glucose by **maltase**, which is located on the intestinal brush border

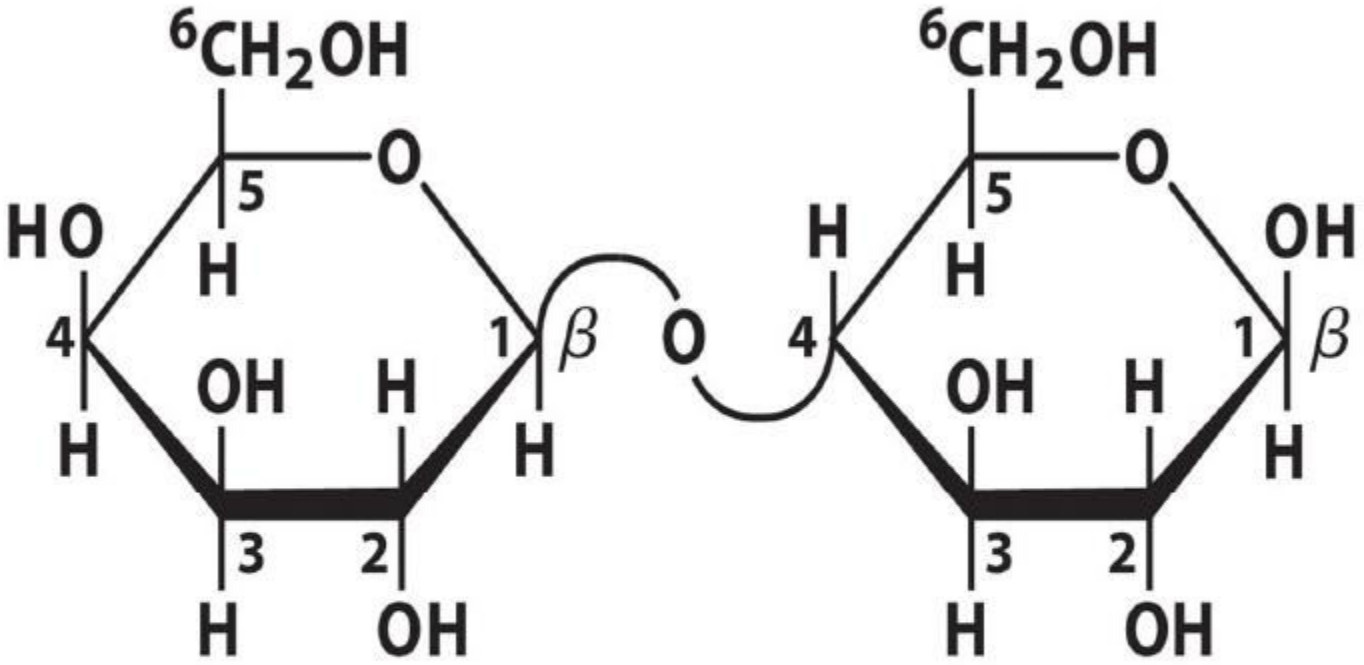
Disacharrides

- From nutritional point of view, maltose is easily digestible
- Various food preparations, such as baby and invalid foods, are produced by hydrolysis of grains and contain large amounts of maltose

Disacharrides

- **Lactose,** the disaccharide of milk consists of galactose joined to glucose by **β (1→4)** glycosidic linkage
- It is dextrorotatory
- As one of the aldehyde group is free, it has reducing properties and exhibits mutarotation

Disacharrides



Lactose (β form)

β-D-galactopyranosyl-(1$\rightarrow$4)-β-D-glucopyranose

Gal(β1$\rightarrow$4)Glc

Disacharrides

- In lactating mammary gland, the lactose is synthesized from glucose by the duct epithelium
- Many organisms that are found in milk, e.g., *E.Coli* convert lactose of milk to lactic acid (by ***β*** **galactosidase**) thus causing souring of milk

Disacharrides

- **Lactase**, the specific enzyme which hydrolyses lactose is present in the intestinal brush border

Disacharrides

- **Lactose intolerance** refers to a condition of pain, nausea, and flatulence after the ingestion of foods containing lactose, most notably dairy products
- Although it is often caused by
  - low levels of lactase
  - it also can because by intestinal injury

Disacharrides

- In a **lactase deficient** person, lactose accumulates in the lumen of the small intestine after ingestion of milk

Disacharrides

- The large osmotic effect of the unabsorbed lactose leads to an influx of fluid into the small intestine

Disacharrides

Lactose Intolerance



- Lactose transits through the small intestine undigested due to insufficient lactase activity on the intestinal surface
- Undigested lactose transits to the large intestine where unadapted colonic bacteria attempts to ferment (metabolize) the lactose
- Lactose fermentation results in large quantities of gas, causing flatulence, distention and acute diarrhea

Lactose
Glucose
Galactose
Bacteria in colon
Methane
Hydrogen

Disacharrides

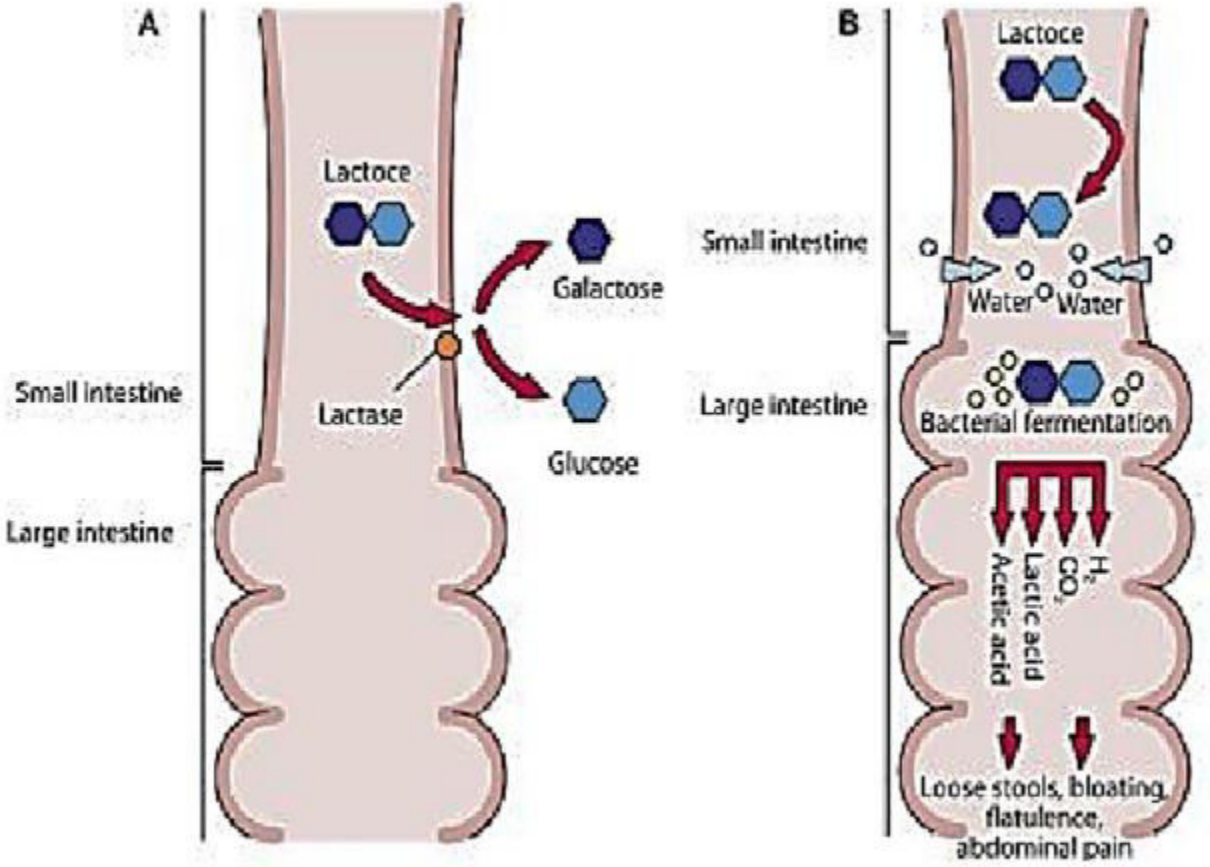


Figure 3. Picture A: Lactase degrades lactose. No symptoms of lactose intolerance. Picture B: Unabsorbed lactose in the large intestine causes lactose intolerance symptoms.

Disacharrides

- Hence, the clinical symptoms of **lactose intolerance** are:
  - abdominal distention,
  - nausea,
  - cramping,
  - pain
  - a watery diarrhea

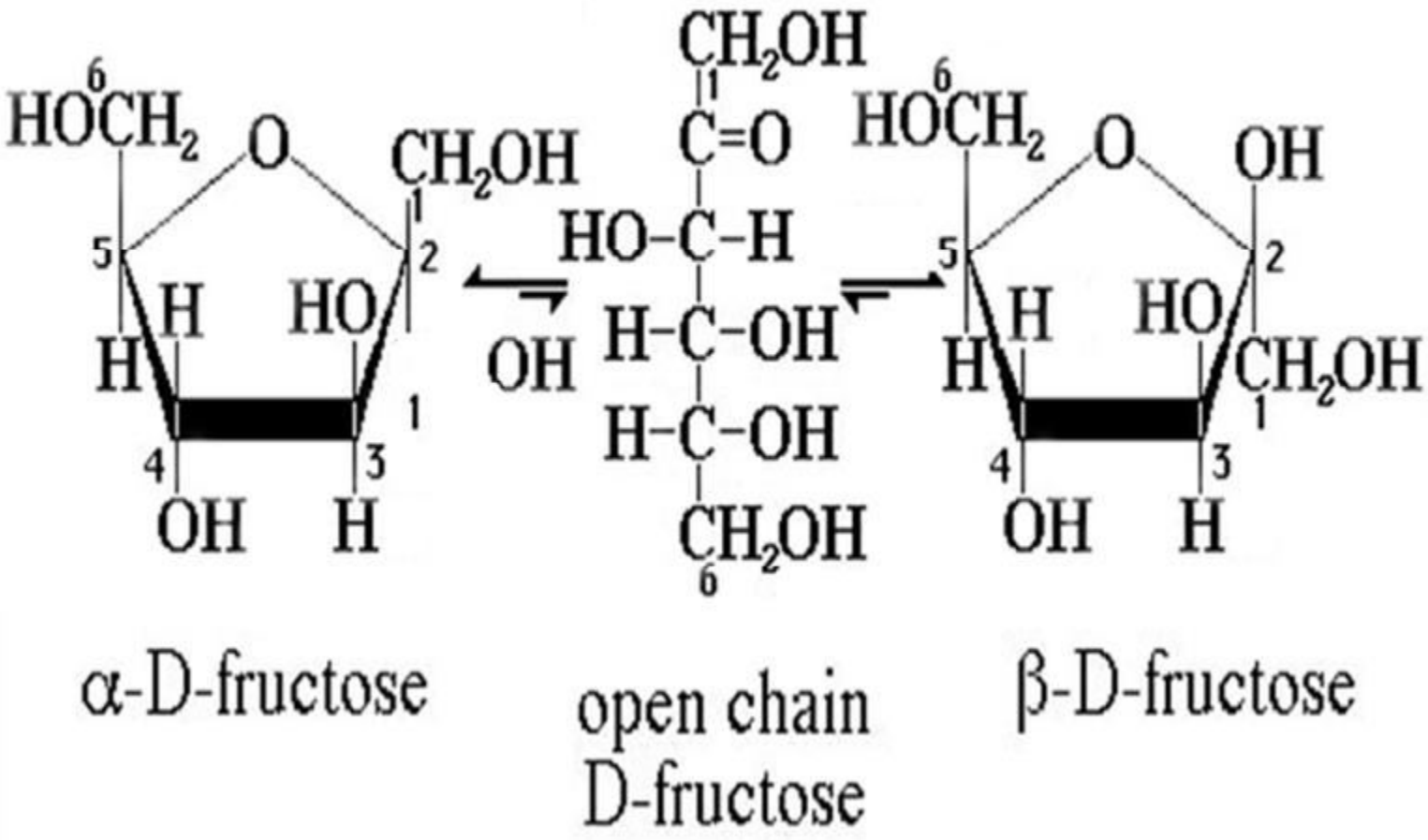
End

Disacharrides

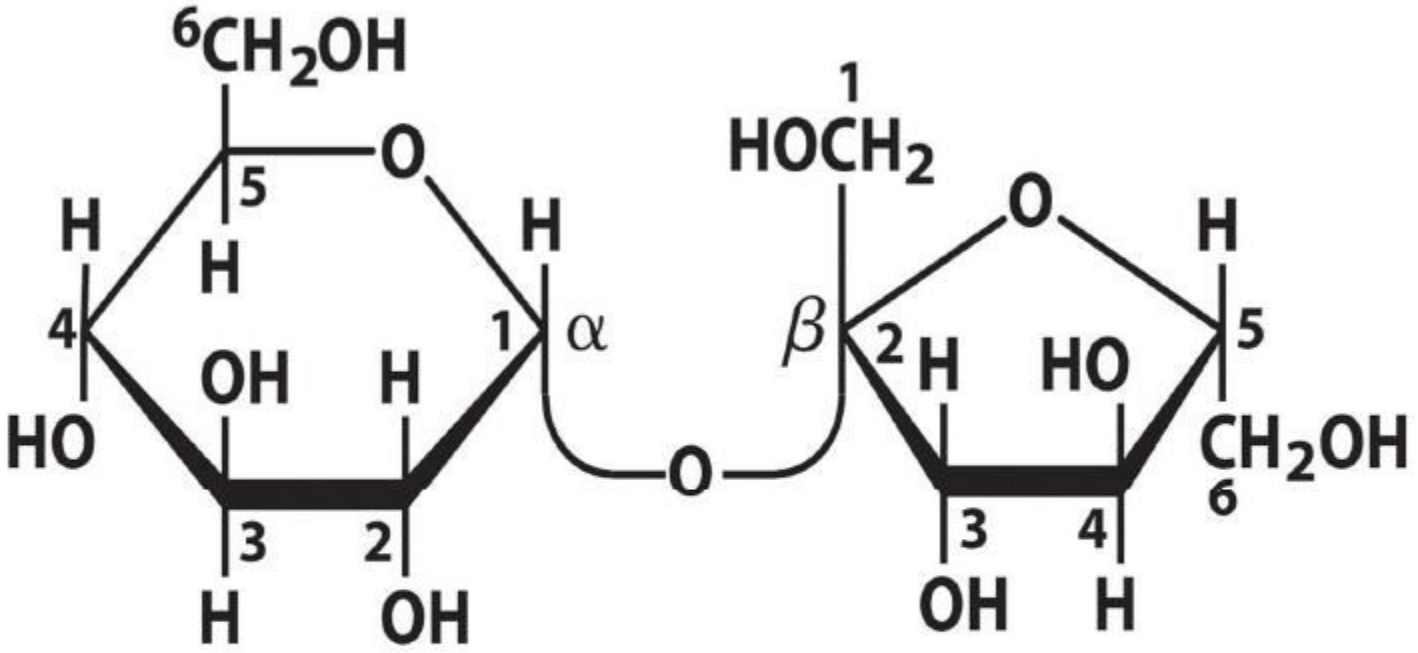
- **Sucrose** (Ordinary table sugar) is obtained commercially from cane or beet
- The anomeric carbon atoms of a glucose unit and a fructose unit are joined in this disaccharide and the configuration of this glycosidic linkage is α for glucose and β for fructose

LECTURE NO 24

Disacharrides



Disacharrides



Sucrose

α-D-glucopyranosyl β-D-fructofuranoside

Glc(α1↔2β)Fru

Disacharrides

- Consequently, sucrose lacks a free reducing group (an aldehyde or ketone end), in contrast with most other sugars
- Sucrose is therefore a non-reducing sugar and does not exhibit mutarotation

Disacharrides

- Sucrose is dextrorotatory but its hydrolytic products are levorotatory as fructose has a levorotation greater than the dextrorotation of glucose

Disacharrides

- As the hydrolytic products invert the rotation, the resulting mixture is called as **invert sugar** and the process is called as **inversion**

Disacharrides

- The hydrolysis of sucrose to glucose and fructose is catalyzed by **sucrase** (also called **invertase**), which is also present in the intestinal brush border like lactase and maltase

Disacharrides

- Honey is largely invert sugar and the presence of fructose accounts for the greater sweetness of honey

END

Oligosaccharides

- **Oligosaccharides** are short polymers of three to ten monosaccharides joined by glycosidic bonds.
- Most are not digested by human enzymes

LECTURE NO 25

Oligosaccharides

- At one end of the chain, the reducing end, is a monosaccharide unit with its anomeric carbon not involved in a glycosidic bond

Oligosaccharides

Biomedical Importance

- Integral membrane proteins contain covalently attached carbohydrate units, oligosaccharides, on their extracellular face
- Many secreted proteins such as antibodies and coagulation factors also contain oligosaccharide units

Oligosaccharides

- The oligosaccharides participate in molecular targeting and cell to cell recognition

- They also mark the passage of time and determine when the proteins should be taken out of circulation

Oligosaccharides

- The human ABO blood groups illustrate the effects of glycosyl-transferases

- Carbohydrates are attached to glycoproteins and glycolipids on the surfaces of red blood cells

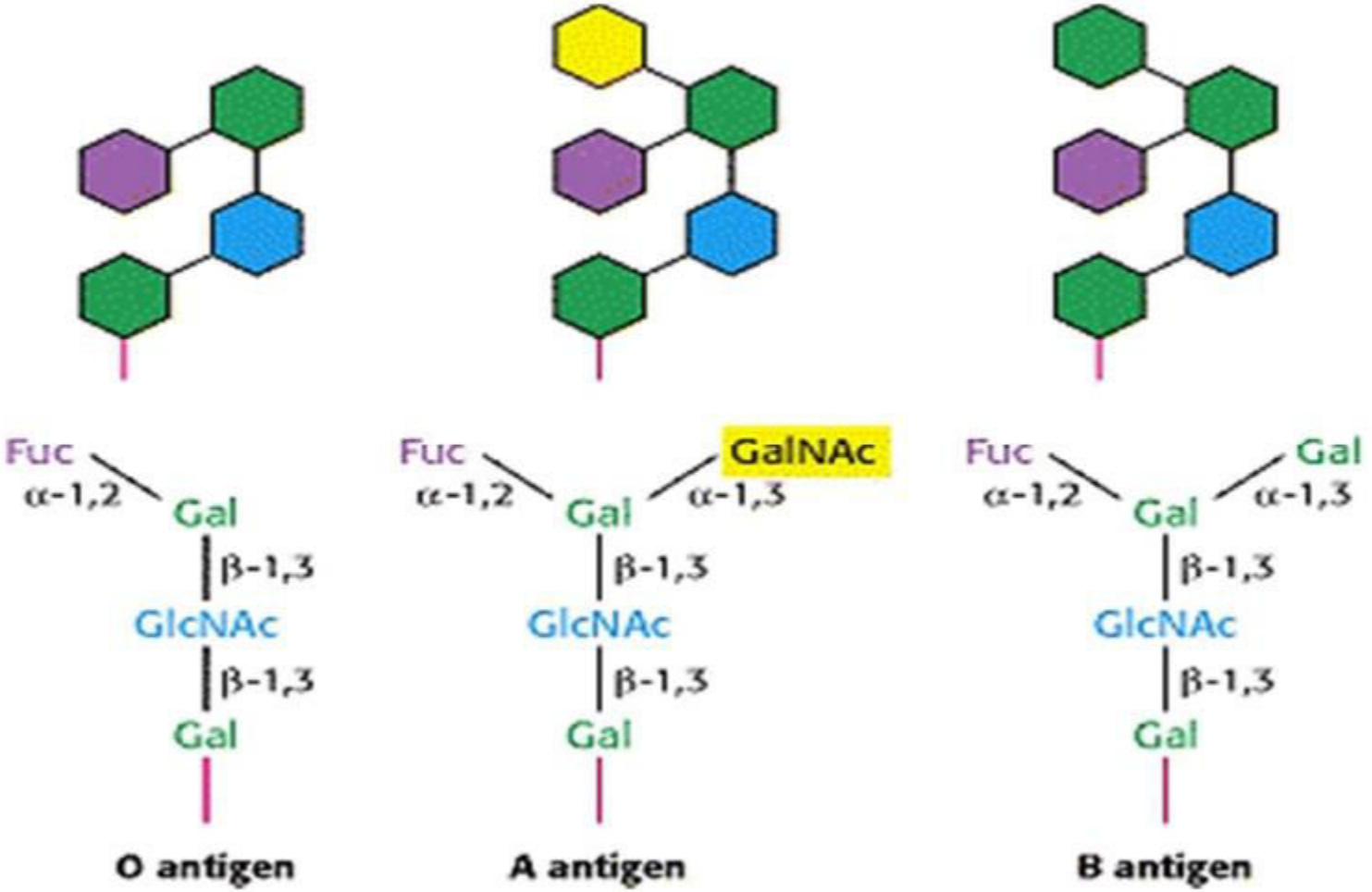
Oligosaccharides

- For one type of blood group, one of the three different structures, termed A, B, and O, may be present
- These structures have in common an oligosaccharide foundation called the O (or sometimes H) antigen

Oligosaccharides

- The A and B antigens differ from the O antigen by the addition of one extra monosaccharide, either *N*-acetylgalactosamine (for A) or galactose (for B) through an α-1,3 linkage to a galactose moiety of the O antigen.

Oligosaccharides



Polysaccharides

Polysaccharides

- Most carbohydrates found in nature occur as polysaccharides, polymers of medium to high molecular weight

LECTURE NO 26

Polysaccharides

Polysaccharides also called **glycans**, differ from each other

- in the identity of their recurring monosaccharide units
- in the length of their chains
- in the types of bonds Iinking the units
- in the degree of branching

Polysaccharides

- **Homopolysaccharides** contain only a single monomeric species
- **Heteropolysaccharides** contain two or more different kinds

Polysaccharides

- Some **homopolysaccharides** serve as storage forms of monosaccharides that are used as fuels
- Other homopolysaccharides (cellulose and chitin) serve as structural elements in plants

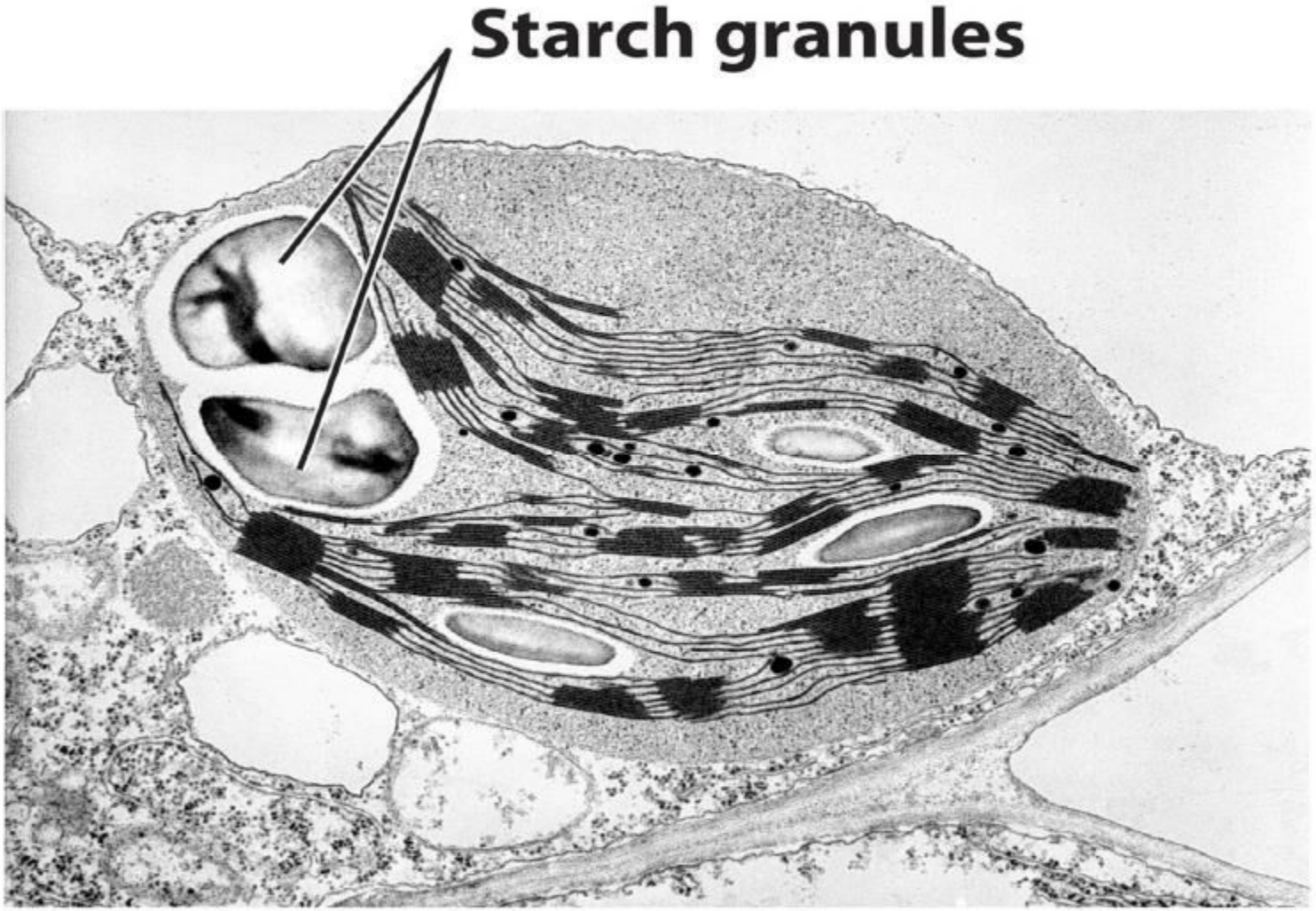
Polysaccharides

- The most important storage polysaccharides are **starch in plant cells and glycogen in animal cells**
- Both polysaccharides occur intracellularly as large clusters or granules

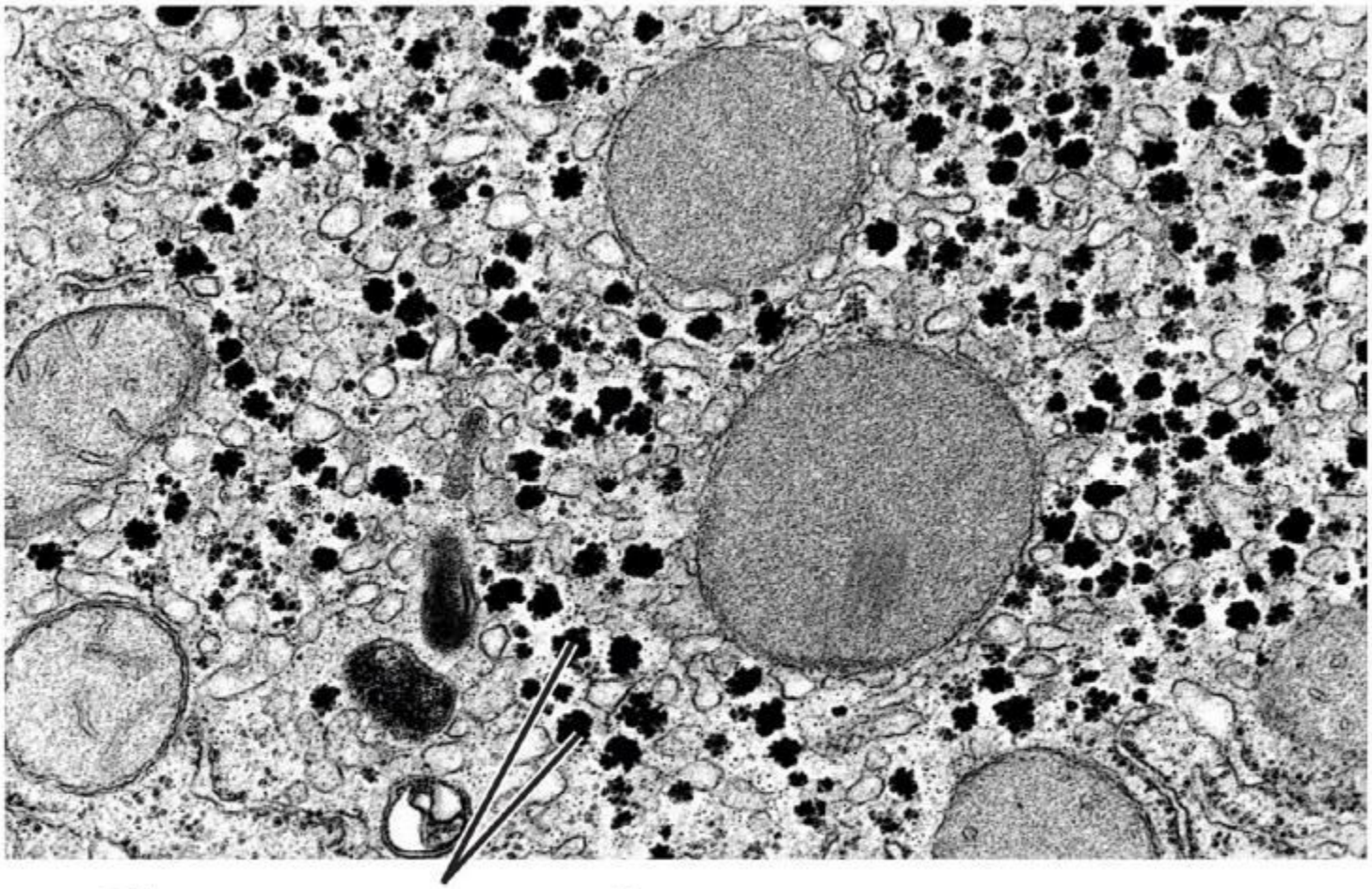
Polysaccharides

- Most plant cells have the ability to form starch and starch storage is especially abundant in tubers (underground stems), such as potatoes and in seeds

Polysaccharides



Polysaccharides



Polysaccharides

- **Heteropolysaccharides** provide extracellular support for organisms of all kingdoms
- For example, the rigid layer of the bacterial cell envelope (the peptidoglycan) is composed in part of a heteropolysaccharide built from two alternating monosaccharide units

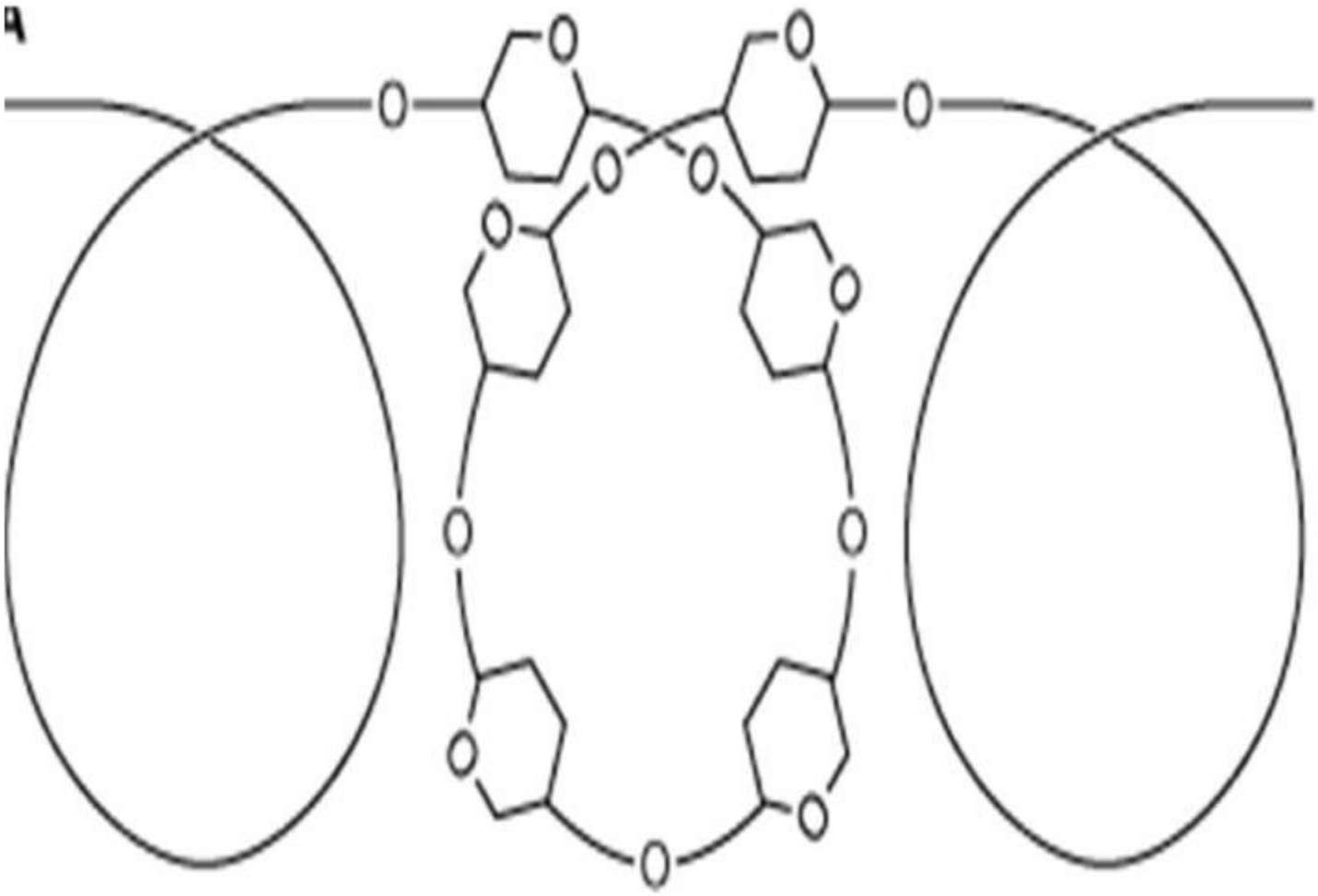
Polysaccharides

- **Starch** contains two types of glucose polymers:
  - Amylose
  - amylopectin
- It is the most important dietary carbohydrate in cereals, potatoes, legumes, and other vegetables

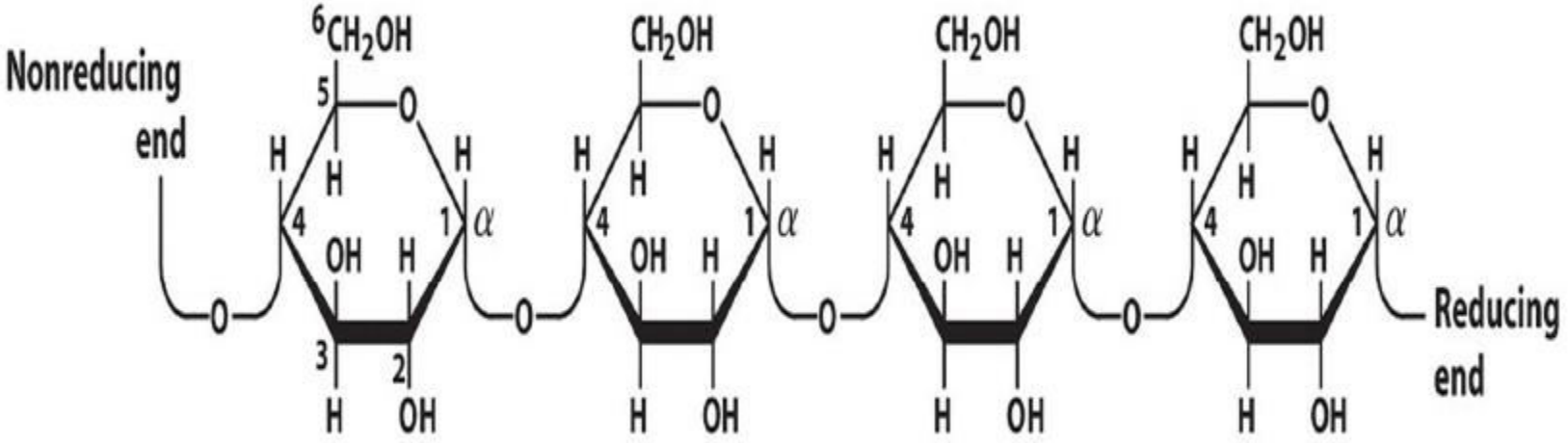
Polysaccharides

- **Amylose** (13–20%), consists of long, unbranched helical structure of D-glucose residues connected by (α1→4) linkages (as in maltose)

Polysaccharides



Polysaccharides



Proteins

- Proteins mediate virtually every process that takes place in a cell, exhibiting an almost endless diversity of functions.

LECTURE NO 27

Proteins

- To explore the molecular mechanism of a biological process, a biochemist almost inevitably studies one or more protein

Proteins

- Proteins are the most abundant biological macromolecules occurring ;
- in all cells and
- all parts of cells

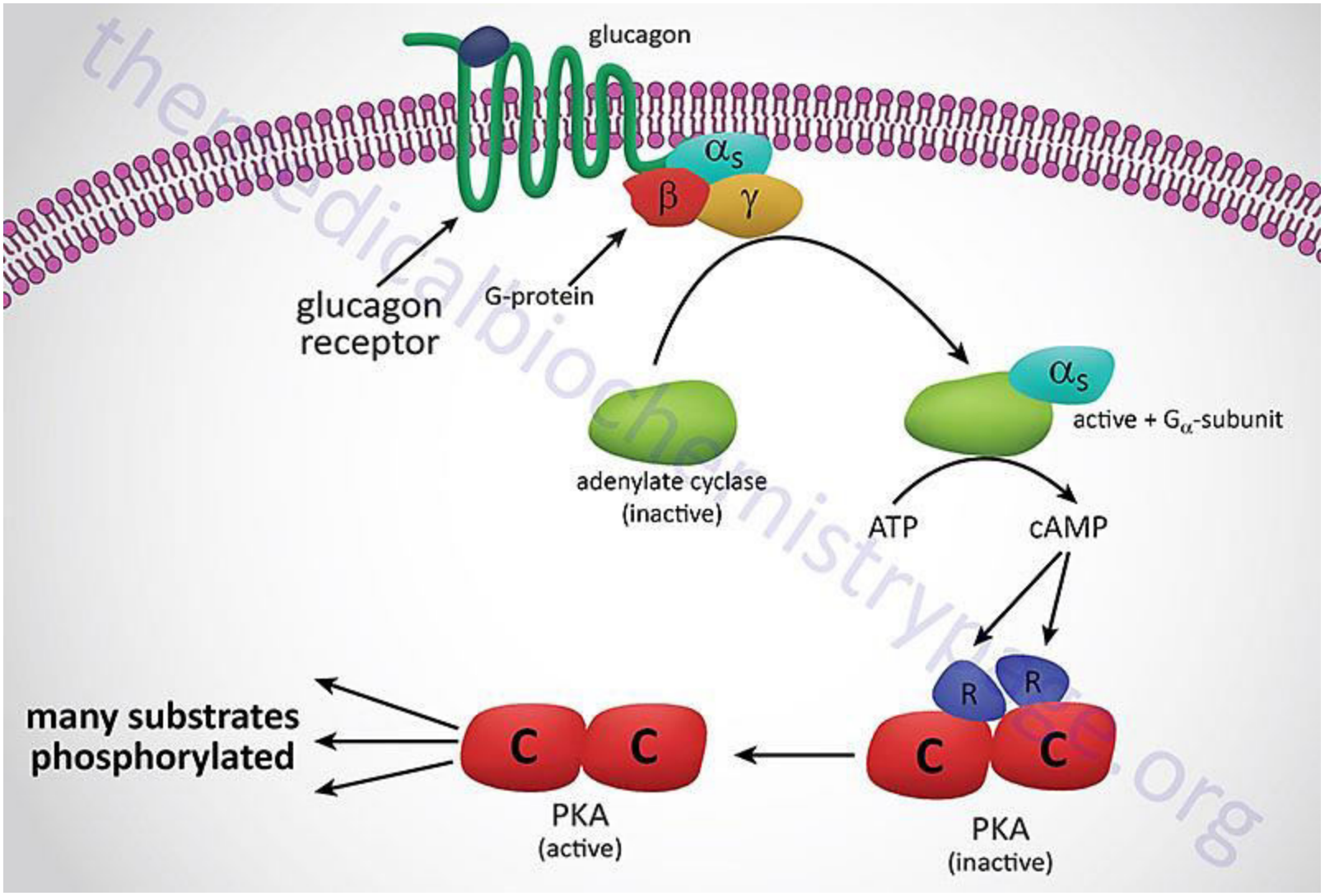
Proteins

- Proteins also occur in great variety; thousands of different kinds may be found in a single cell

Proteins

- Moreover, proteins exhibit enormous diversity of biological function and are the most important final products of the information pathways
- Example signal transduction

Proteins


glucagon
α_s
β
γ
glucagon receptor
G-protein
adenylate cyclase (inactive)
α_s
active + G_α-subunit
ATP
cAMP
R
R
C
C
C
C
many substrates phosphorylated
PKA (active)
PKA (inactive)

Biomedical importance

- Biochemical catalysts known as **enzymes** are proteins
- **Immunoglobulins** that serve as the first line of defense against bacteria and viruses and other foreign agents are proteins

Biomedical importance

- Several **hormones** are proteins
- **Structural proteins** provide **mechanical support** e.g collagen
- **Contractile proteins** help in the **movement** of muscle fiber and microvilli

Biomedical importance

- Some proteins present in the cell membrane, cytoplasm and nucleus act as **receptors**
- Certain other proteins in the cell membrane act as **channels and transporters**

Biomedical importance

- **The transport proteins** carry out the function of transporting specific substances across the membrane or in the body fluids
- **Storage proteins** bind with specific substances and store them

Biomedical importance

- Under certain conditions, proteins can be catabolized to **provide energy**
- Proteins help in the maintenance of water and electrolyte balance in the body by **exerting osmotic pressure**

Biomedical importance

- Hence proteins are the most versatile macromolecules in living systems and serve crucial functions in essentially all biological processes

Biomedical importance

- They function as catalysts, they transport and store other molecules such as oxygen,
- They provide mechanical support and immune protection

Biomedical importance

- They generate movement, and they control growth and differentiation

END

Properties of proteins

- Several key properties enable proteins to participate in such a wide range of functions

1. Proteins are linear polymers built of monomer units called amino acids

LECTURE NO 28

Properties of proteins

- The function of a protein is directly dependent on its three dimensional structure

Properties of proteins

- Remarkably, proteins spontaneously fold up into three-dimensional structures that are determined by the sequence of amino acids in the protein polymer

Properties of proteins

2. Proteins contain a wide range of functional groups

These functional groups include

- alcohols,
- thiols,
- thioethers,
- carboxylic acids,
- carboxamides,
- and a variety of basic groups

Properties of proteins

- When combined in various sequences, this array of functional groups accounts for the broad spectrum of protein function

Properties of proteins

3. Proteins can interact with one another and with other biological macromolecules to form complex assemblies

Properties of proteins

- The proteins within these assemblies can act synergistically to generate capabilities not afforded by the individual component proteins

Properties of proteins

- These assemblies include macromolecular machines that carry out;
- the accurate replication of DNA,
- the transmission of signals within cells, and many other essential processes

Properties of proteins

4. Some proteins are quite rigid, whereas others display limited flexibility

- Rigid units can function as structural elements in the cytoskeleton (the internal scaffolding within cells) or in connective tissue

Properties of proteins

- Parts of proteins with limited flexibility may act as
- hinges,
- springs, and
- levers

Properties of proteins

- That are crucial to protein function, to the assembly of proteins with one another and with other molecules into complex units
- And to the transmission of information within and between cells

END

Composition of Proteins

- All proteins, whether from the most ancient lines of bacteria or from the most complex forms of life, are constructed from the same set of;
- 20 amino acids,

LECTURE NO 29

Composition of Proteins

- Which are covalently linked in characteristic linear sequence.
- Each of these amino acids has a side chain with distinctive chemical properties

Composition of Proteins

- This group of 20 precursor molecules may be regarded as the alphabet in which the language of protein structure is written

Composition of Proteins

- Proteins are found in a wide range of sizes,
- from relatively small peptides with just a few amino acid residues to huge polymers with molecular weights in the millions

Composition of Proteins

- Cells can produce proteins with strikingly different properties and activities by joining the same 20 amino acids in many different combinations and sequence

Composition of Proteins

- From these building blocks different organisms can make such widely diverse products as
  - enzymes,
  - hormones,
  - antibodies,
  - transporters,
  - muscle fibers,

Composition of Proteins

- the lens protein of the eye,
- feathers,
- spider webs,
- rhinoceros horn,
- milk proteins,
- antibiotics,
- mushroom poisons etc

Composition of Proteins

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase



Composition of Proteins

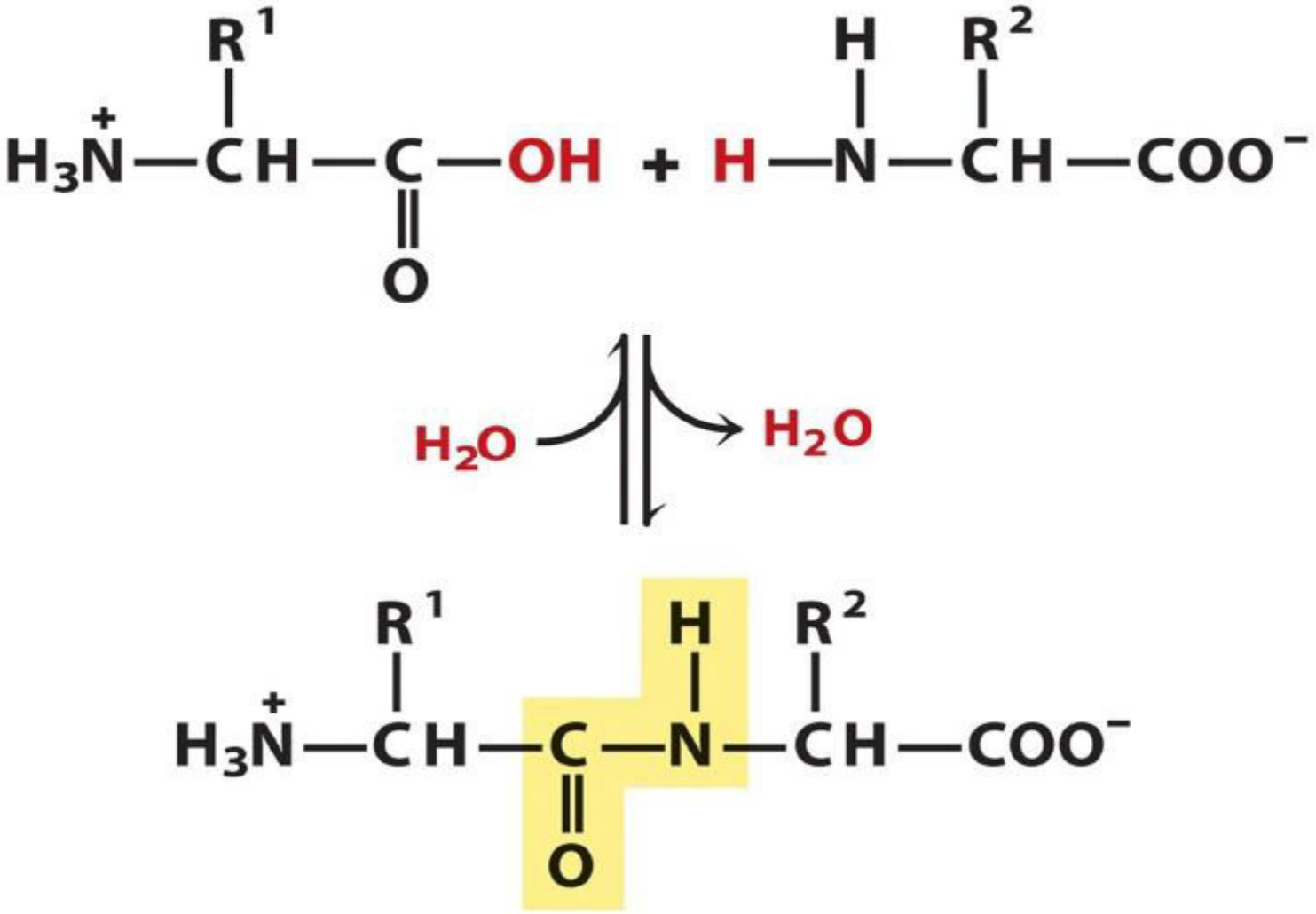




Composition of Proteins

- Proteins are polymers of amino acids, with each amino acid residue joined to its neighbor by a specific type of covalent bond termed the **peptide bond**

Composition of Proteins



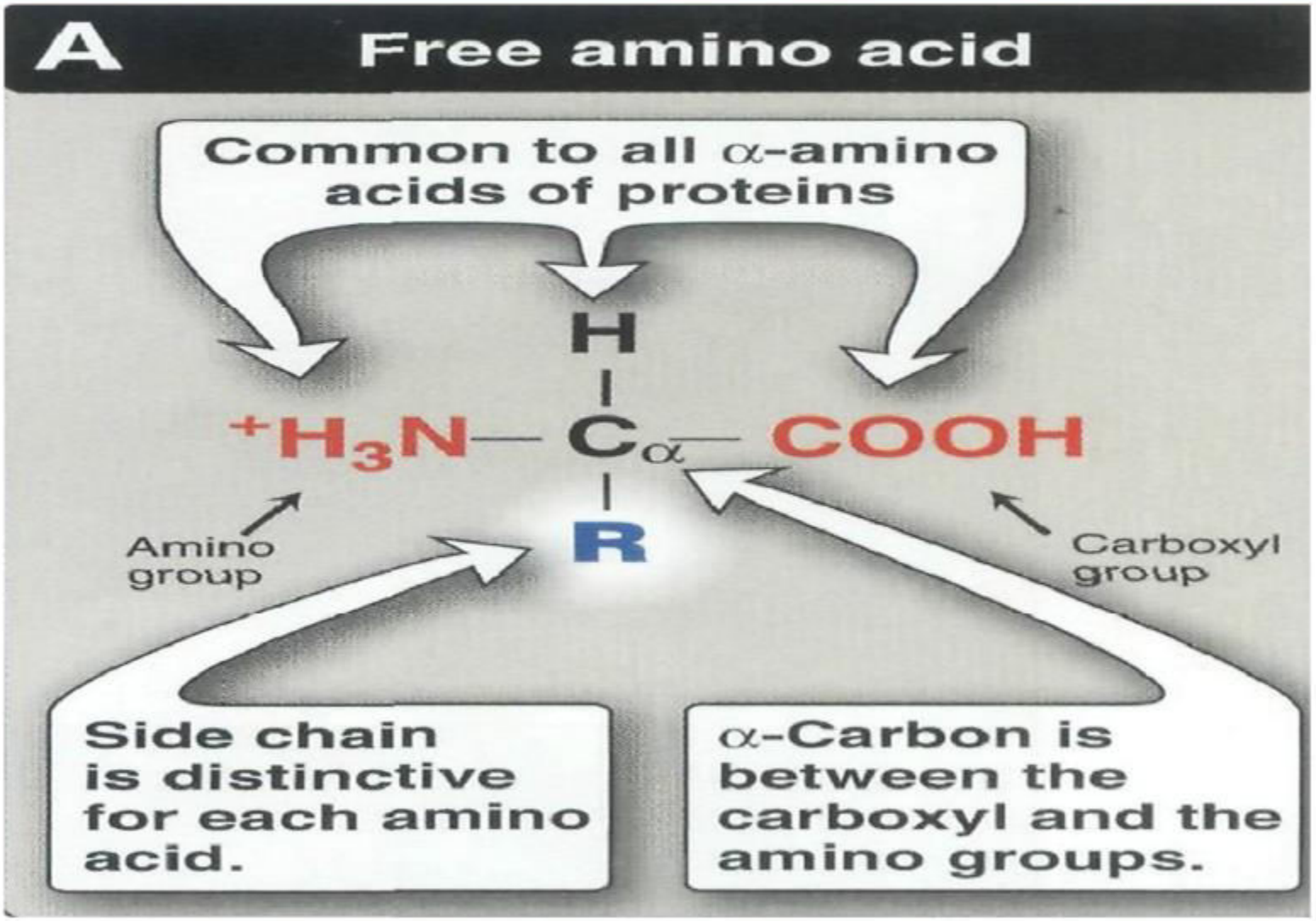
Composition of Proteins

- The term "residue" reflects the loss of the elements of water when one amino acid is joined to another
- Twenty different amino acids are commonly found in proteins

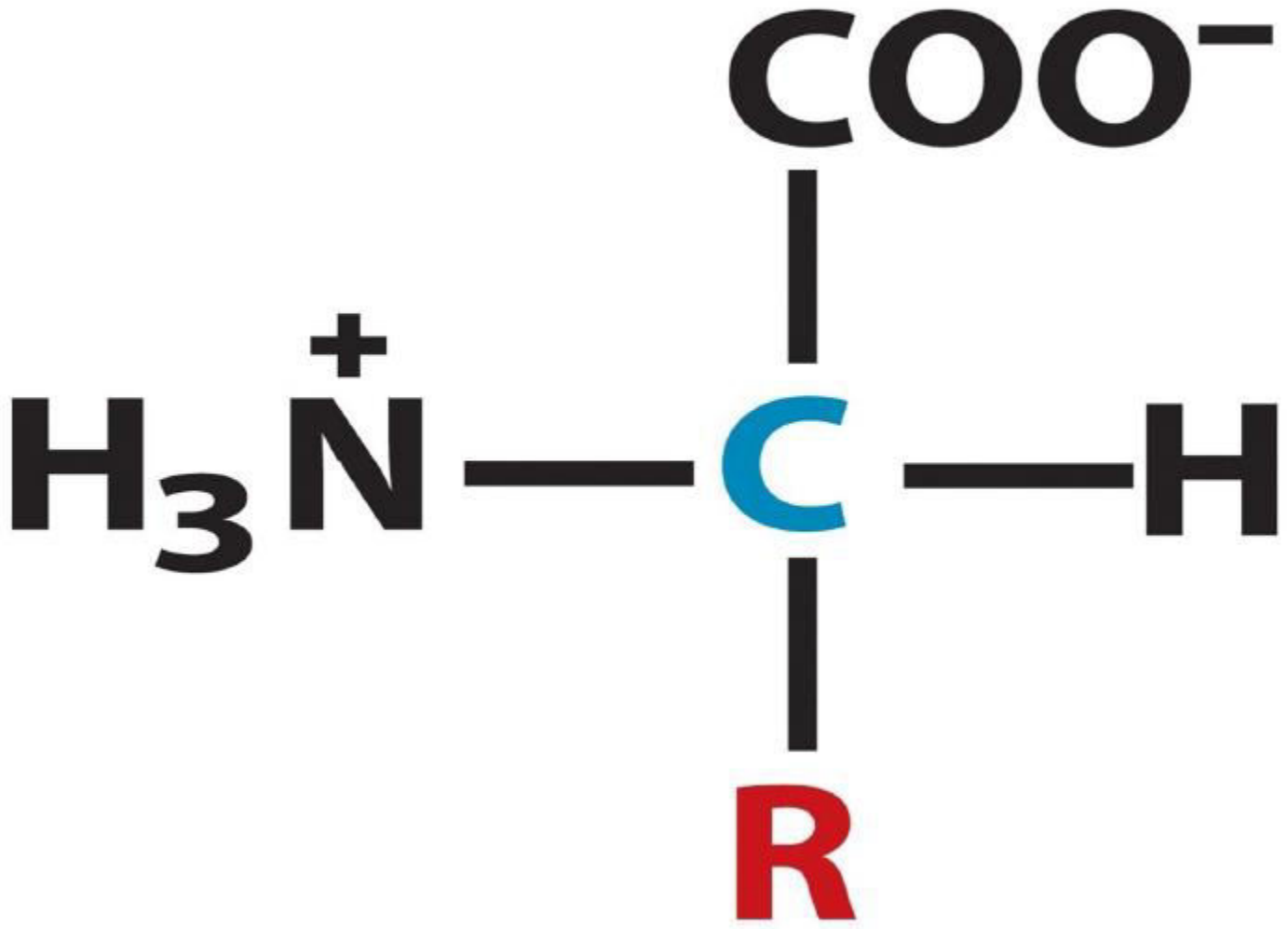
Composition of Proteins

- All of the 20 common amino acids are **α amino acids**
- They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon)

Composition of Proteins



Composition of Proteins



Composition of Proteins

- They differ from each other in their side chains, or R groups, which vary in
  - structure,
  - size,
  - electric charge,
  - and which influence the solubility of the amino acids in water

Composition of Proteins

Standard amino acids

- The 20 amino acids that constitute the monomer units of proteins are **the standard amino acids** as the genetic code specifies only these 20 L -α-amino acids

Composition of Proteins

- In addition to these 20 amino acids there are many less common ones i.e., **the non standard amino acids**

END

Composition of Proteins

- In addition to the 20 common amino acids, proteins may contain residues created by modification of common residues already incorporated in to a polypeptide

LECTURE NO 30

Composition of Proteins

- Some proteins contain additional amino acids that arise by modification of an amino acid already present in a peptide i.e., after the protein has been synthesized

Composition of Proteins

- Examples include
  - conversion of peptidyl proline and lysine to
  - 4-hydroxyproline and
  - 5-hydroxylysine
  - the conversion of peptidyl glutamate to carboxyglutamate

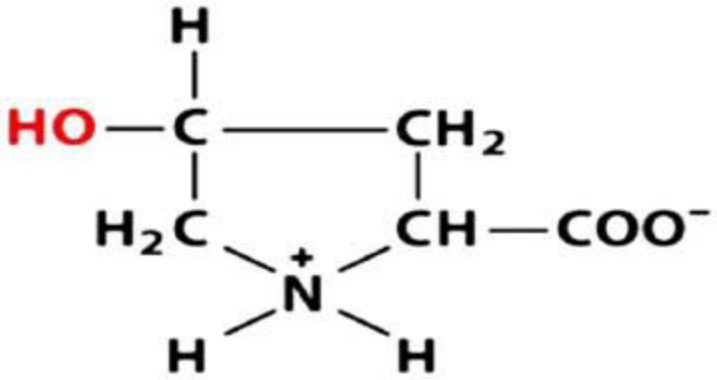
Composition of Proteins

- the methylation,
- formylation,
- acetylation,
- prenylation, and
- phosphorylation of certain aminoacyl residues

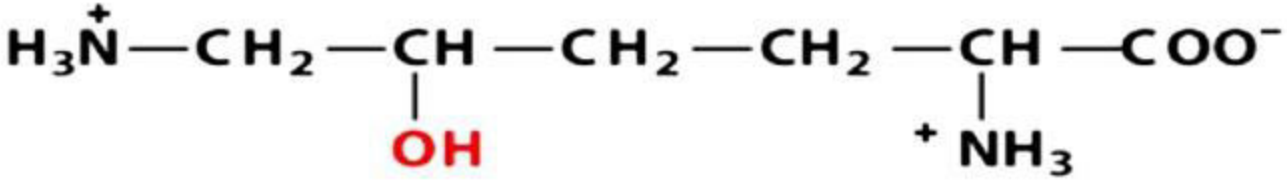
Composition of Proteins

- Among these uncommon amino acids
  - **4-hydroxyproline**, a derivative of proline
  - **5-hydroxylysine**, derived from lysine
- The former is found in plant cell wall proteins, and both are found in collagen

Composition of Proteins



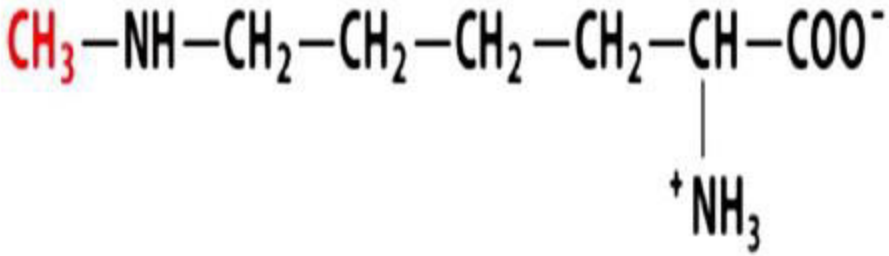
4-Hydroxyproline



5-Hydroxylysine

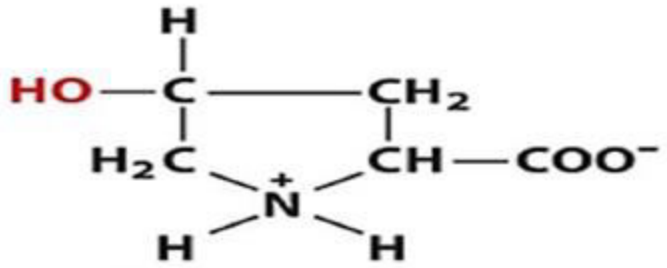
Composition of Proteins

- **6-N-Methyllysine** is a constituent of myosin, a contractile protein

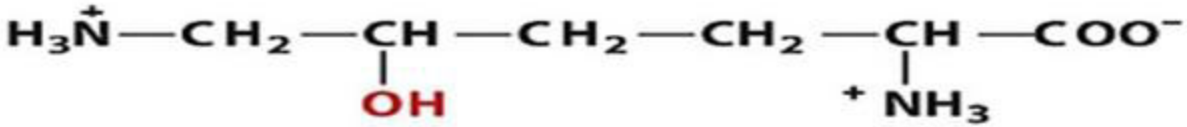


6-*N*-Methyllysine

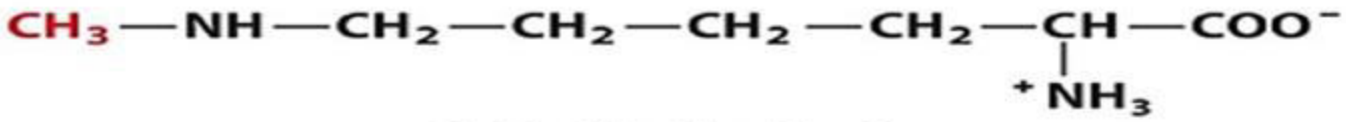
Composition of Proteins



4-Hydroxyproline



5-Hydroxylysine

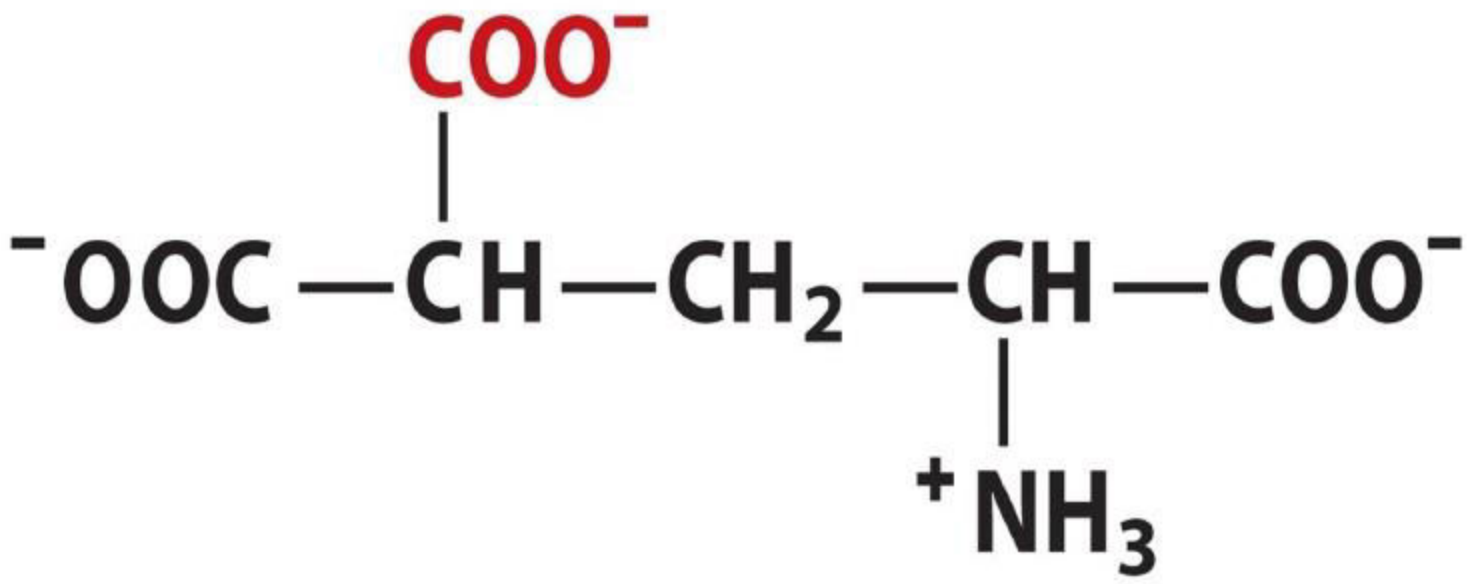


6-*N*-Methyllysine

Composition of Proteins

- Another important uncommon amino acid is;
- **γ -carboxyglutamate,** found in the blood clotting protein prothrombin and in certain other proteins that bind Ca^{2+} as part of their biological function

Composition of Proteins



γ-Carboxyglutamate

Composition of Proteins

- Some 300 additional amino acids have been found in cells but not as constituents of proteins
- Examples include
  - Ornithine and citrulline (intermediates in urea cycle)

Composition of Proteins

- Taurine (found in bile acids)
- δ-Aminolevulinic acid (intermediate in haem synthesis)

Composition of Proteins

D- AMINO ACIDS

- D-amino acids are also non standard amino acids that occur **naturally** and include
  - free D- serine, and D-aspartate in brain tissue

Composition of Proteins

  - D-alanine & D-glutamate in the cell walls of gram positive bacteria
- D-amino acids are also found in some **antibiotics**

Composition of Proteins

Selenocysteine, the 21st L-α-Amino Acid?

- Selenocysteine is an L-α-amino acid found in a handful of proteins, including certain **peroxidases and reductases** where it participates in the catalysis of electron transfer reactions

Composition of Proteins

- Its synthesis is not a post translational modification, but a modification to a **serine** that occurs while the serine is bound to a unique RNA
- The hydroxyl group of serine is replaced by a selenium atom

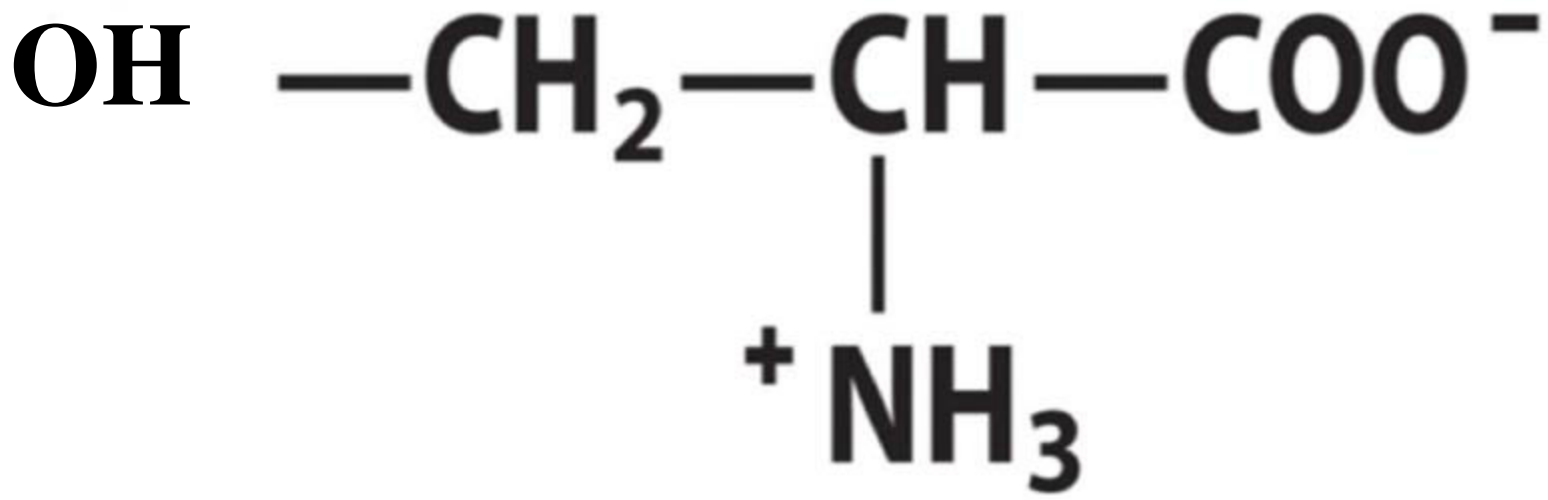
Composition of Proteins

- The selenocysteine is then inserted into a protein as it is being synthesized
- Since selenocysteine is inserted into polypeptides during translation, it is commonly referred to as the "21st amino acid"

Composition of Proteins

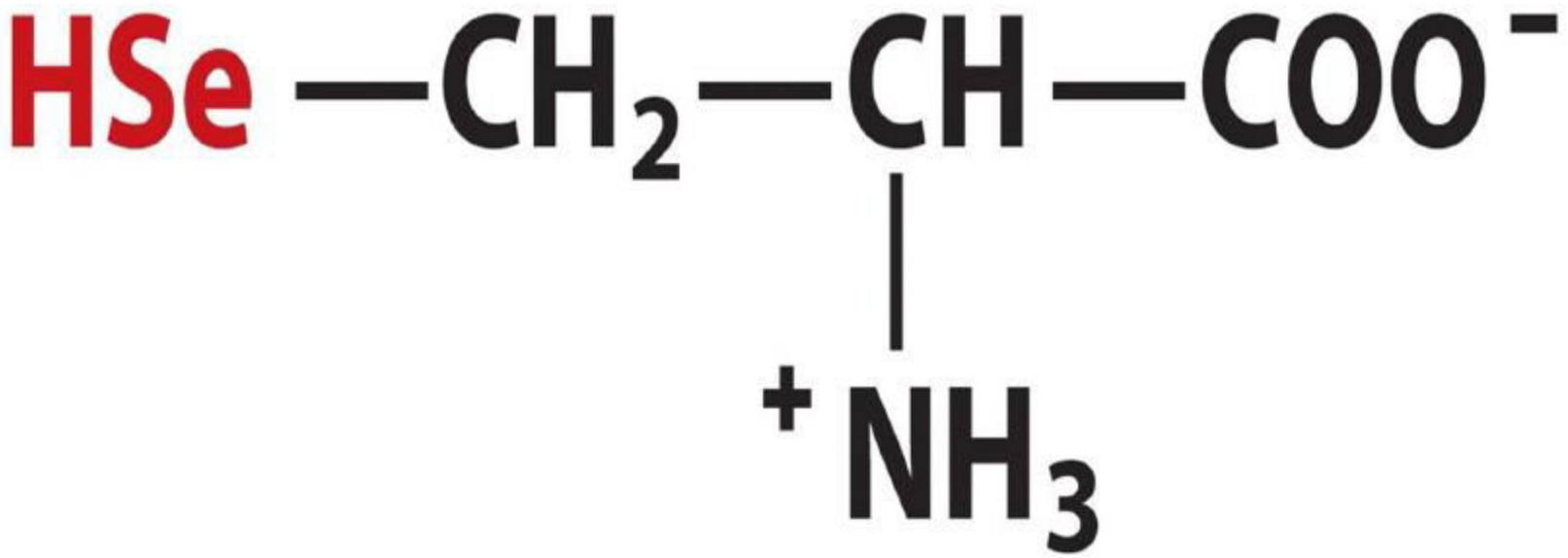
- However, unlike the other 20 genetically encoded amino acids, selenocysteine is not specified by a simple three-letter codon

Composition of Proteins



Serine

Composition of Proteins



Selenocysteine

Composition of Proteins

- **22nd Amino Acid**
- Pyrrolysine (abbreviated as Pyl or O) is a naturally occurring, genetically coded amino acid used by some methanogenic archaea.

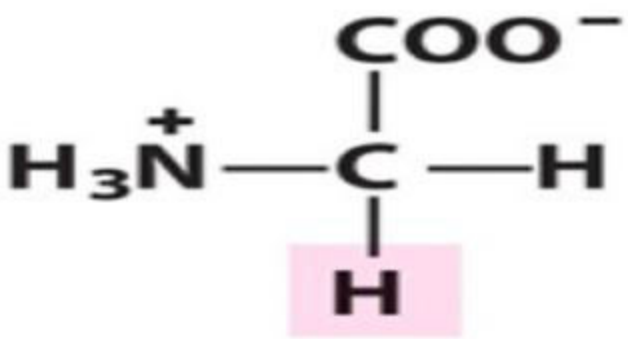
END

Composition of Proteins

Chiral centers and isomerism

- For all the common amino acids except glycine (the R group is another hydrogen atom), the α carbon is bonded to four different groups:

Composition of Proteins

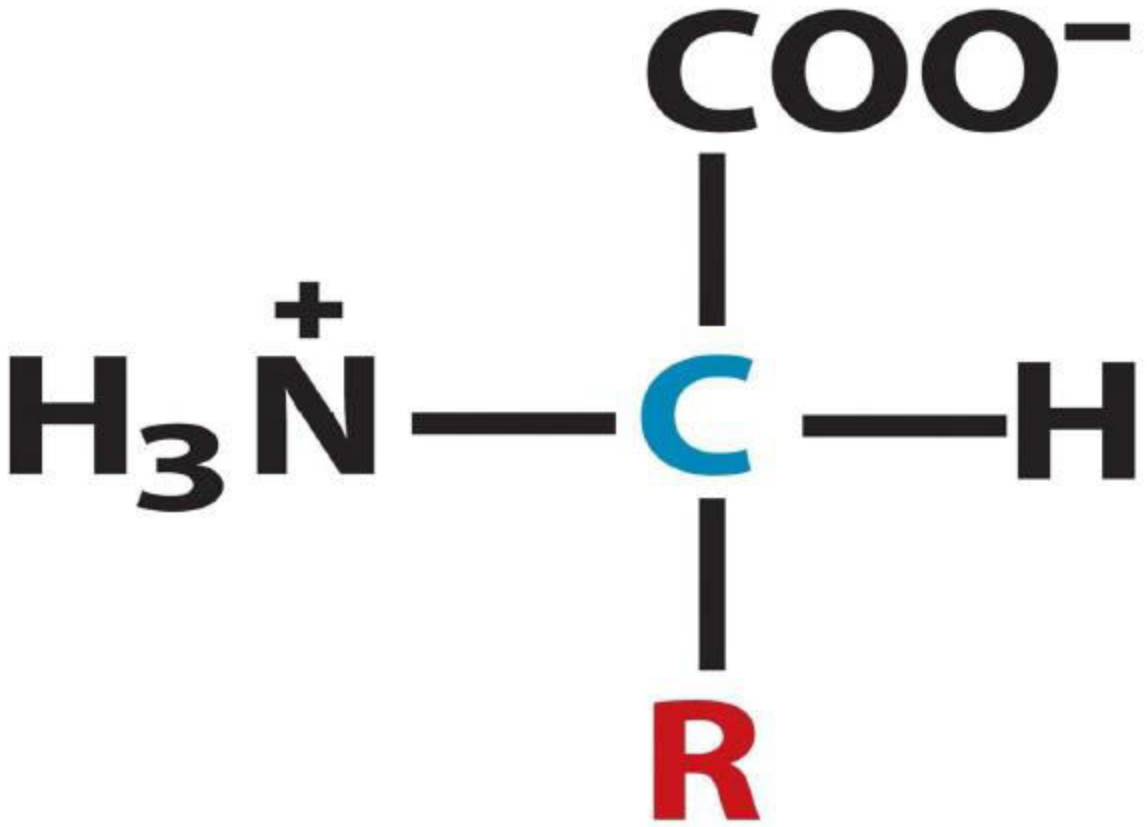


Glycine

Composition of Proteins

- a carboxyl group,
- an amino group,
- an R group,
- a hydrogen atom

Composition of Proteins



Composition of Proteins

- The α-carbon atom is thus a chiral center and thus amino acids have two possible stereoisomers
- All molecules with a chiral center are also optically active-that is, they rotate plane polarized light

Composition of Proteins

- Special nomenclature has been developed to specify the absolute configuration of the four substituents of asymmetric carbon atom

Composition of Proteins

- The absolute configurations of simple sugars and amino acids are specified by the D, L system, based on the absolute configuration of the three-carbon sugar glyceraldehyde

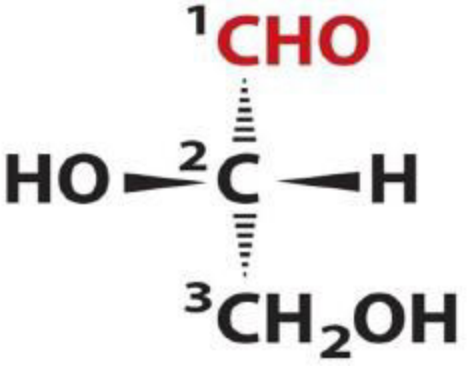
Composition of Proteins

- Thus the amino group of L-alanine occupies the same position about the chiral carbon as does the hydroxyl group of L-glyceraldehyde

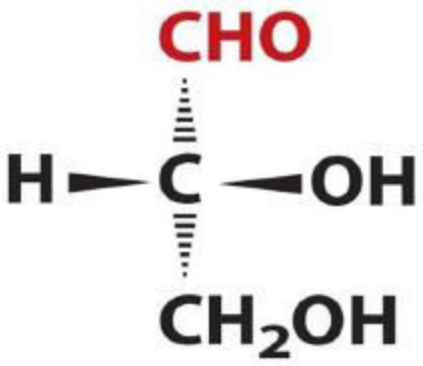
Composition of Proteins

- Since they are nonsuperposable mirror images of each other, the two forms represent a class of stereoisomers called **enantiomers**

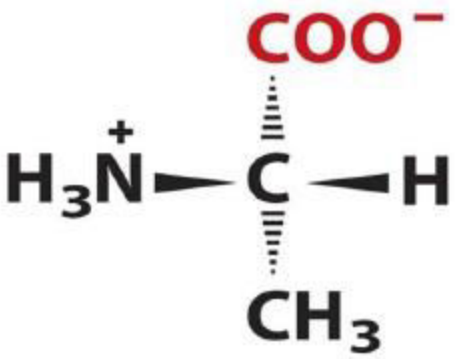
Composition of Proteins



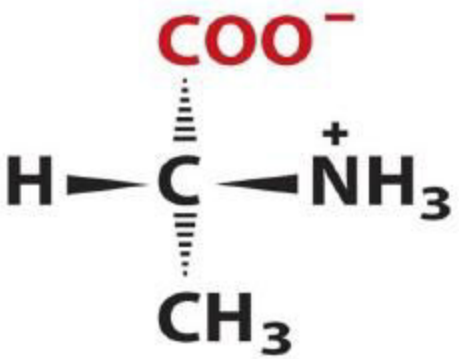
L-Glyceraldehyde



D-Glyceraldehyde



L-Alanine



D-Alanine

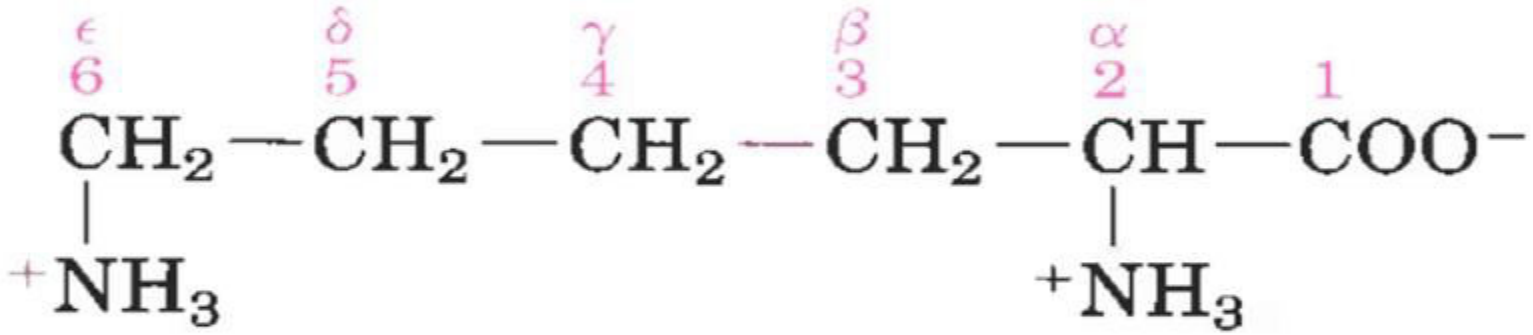
END

Conventions used to identify the carbons

Two conventions are used to identify the carbons in an amino acid

The additional carbons in an R group are commonly designated **β, γ, δ** and so forth, proceeding out from the **α carbon**

Conventions used to identify the carbons



Lysine

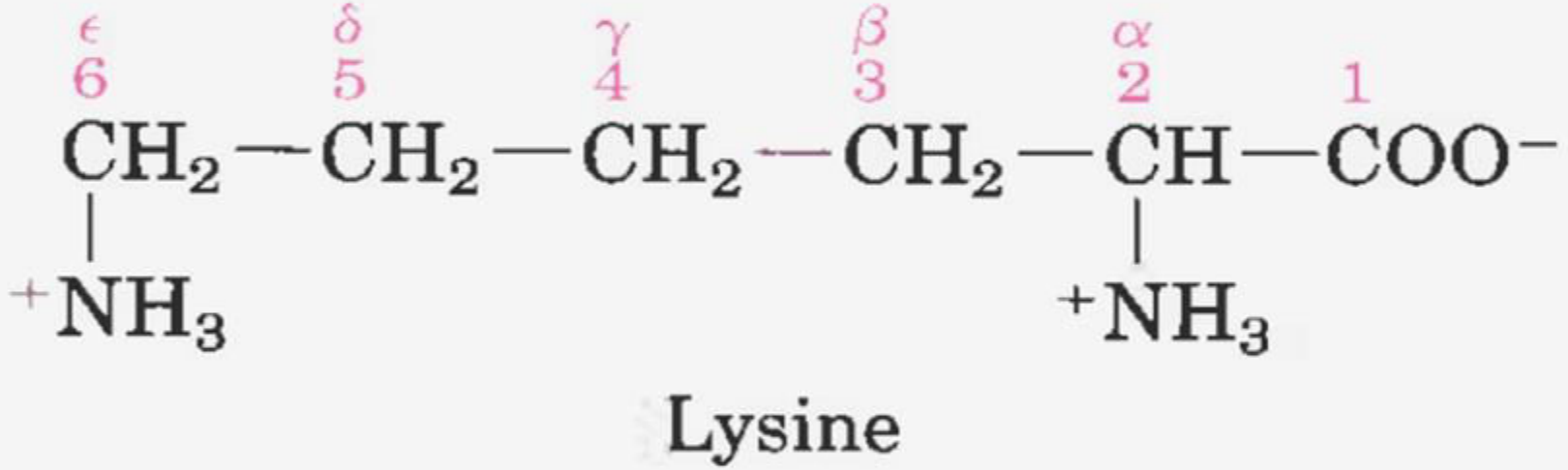
Conventions used to identify the carbons

- For most other organic molecules, carbon atoms are simply numbered from one end, giving highest priority (C-1) to the carbon with the substituent containing the atom of **highest atomic number**

Conventions used to identify the carbons

- Within this latter convention, the carboxyl carbon of an amino acid would be **C-1** and the **α** carbon would be **C-2**

Conventions used to identify the carbons

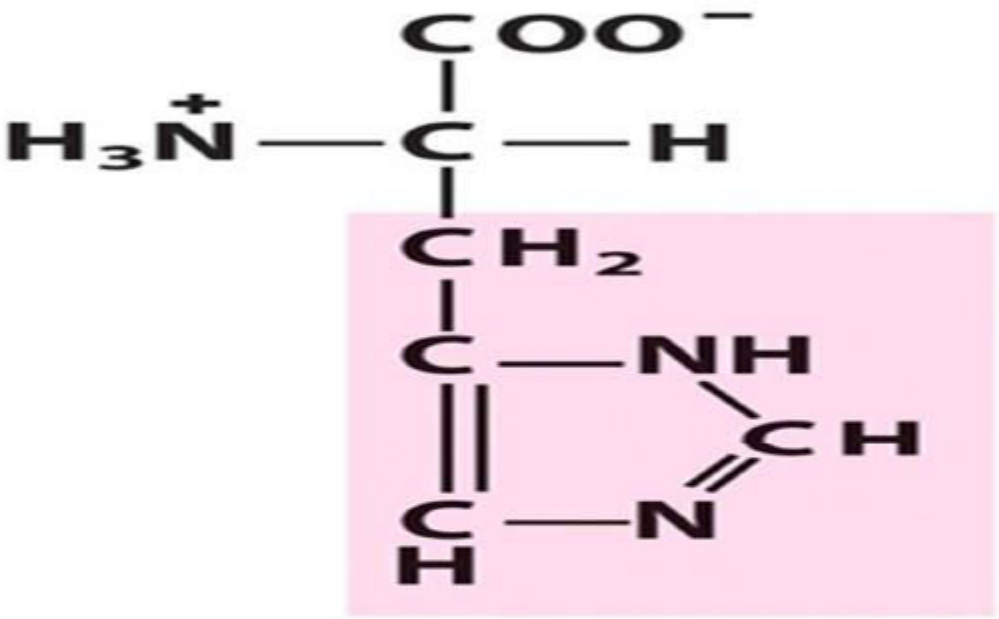


Lysine

Conventions used to identify the carbons

- In some cases, such as amino acids with heterocyclic R groups (such as histidine), the Greek lettering system is ambiguous and the numbering convention is therefore used

Conventions used to identify the carbons



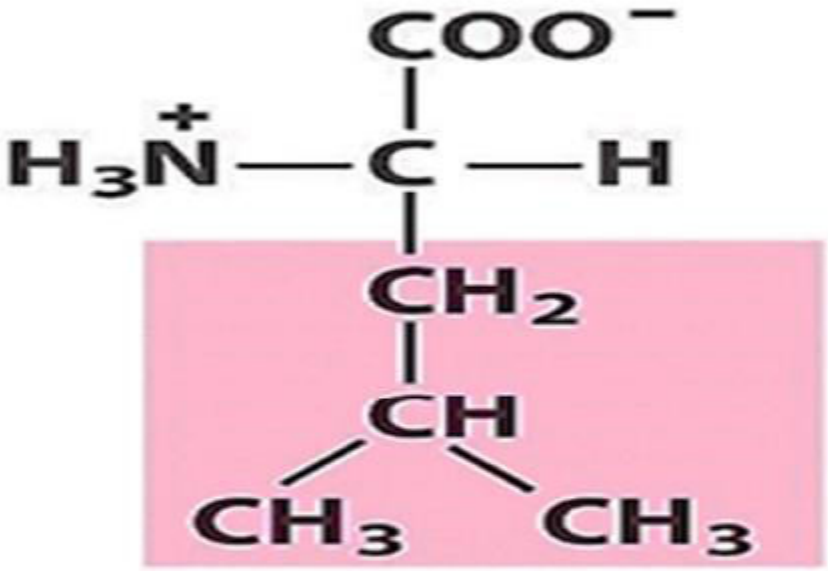
Histidine

Conventions used to identify the carbons

- For branched amino acid side chains, equivalent carbons are given numbers after the Greek letters

- Leucine thus has δ1 and δ2 carbons

Conventions used to identify the carbons



Leucine

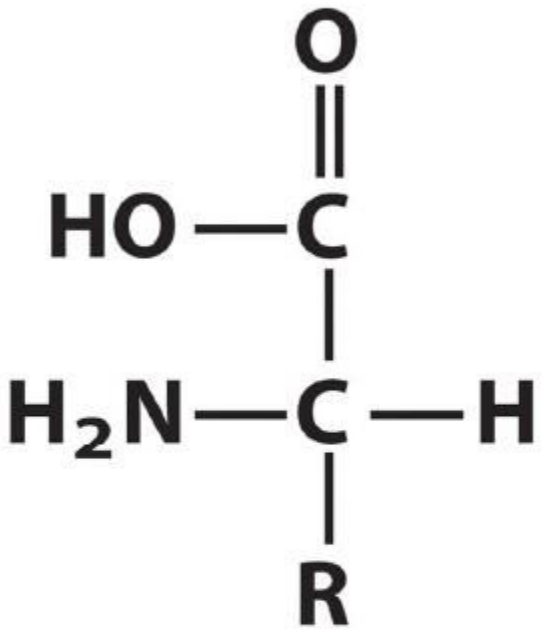
Acidic and basic properties of amino acids

- The form of an amino acid that has both a positive and a negative charge is called a **zwitterion**

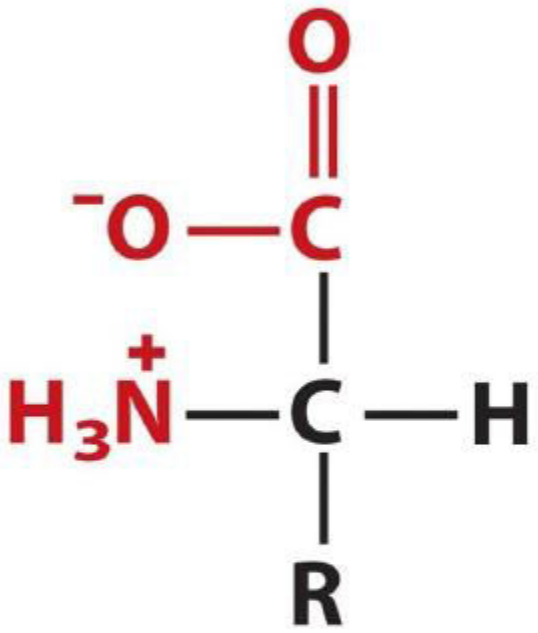
Acidic and basic properties of amino acids

- Amino acids in solution at neutral pH exist predominantly as dipolar ions **(zwitterions)**

Acidic and basic properties of amino acids



Nonionic form



Zwitterionic form

Acidic and basic properties of amino acids

- In this form, the amino group is protonated ($–NH_3^+$)
- and the carboxyl group is deprotonated ($–COO^-$)
- A zwitterion can act either as an acid (proton donor) or as a base (proton acceptor)

Acidic and basic properties of amino acids

- Substances having this dual nature are **amphoteric** and are often called **ampholytes**

Acidic and basic properties of amino acids

Acidic and basic properties of amino acids

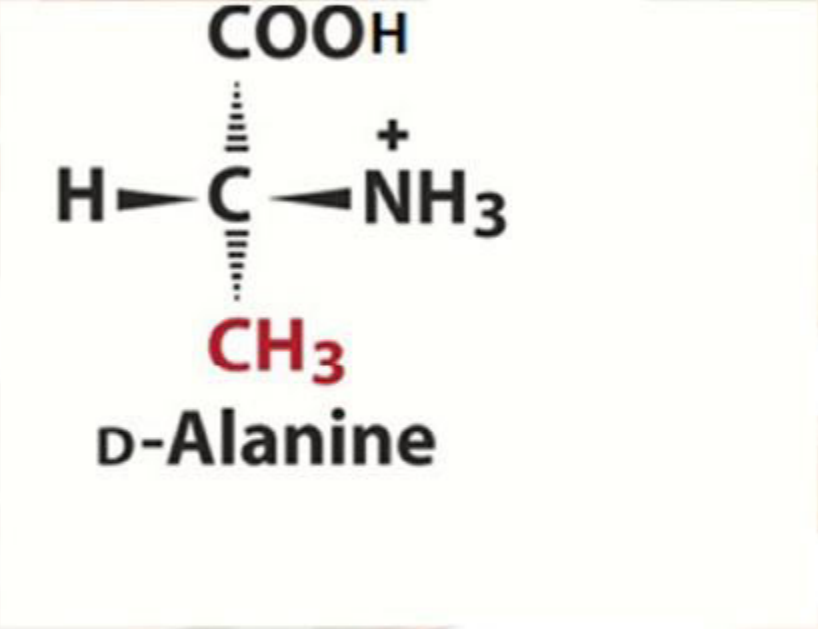
Acidic and basic properties of amino acids

- The ionization state of an amino acid varies with pH
- At physiological pH, the carboxyl group is dissociated , forming the negatively charged carboxylate ion ($-COO^-$) and the amino group is protonated (NH_3^+)

Acidic and basic properties of amino acids

- A simple monoamino monocarboxylic -amino acid, such as alanine, is a diprotic acid when fully protonated—it has two groups, the-COOH group and the - NH_{3+} group, that can yield protons

Acidic and basic properties of amino acids



END

Acidic and basic properties of amino acids

- Each acid has a characteristic tendency to lose its proton in an aqueous solution
- The stronger the acid, the greater its tendency to lose its proton

Acidic and basic properties of amino acids

- The tendency of any acid (HA) to lose a proton and form its conjugate base (A^-) is defined by the equilibrium constant (K_{eq}) for the reversible reaction
- $HA \rightleftharpoons A^- + H^+$

Acidic and basic properties of amino acids

- Equilibrium constants for ionization reactions are usually called ionization constants or acid dissociation constants, often designated pK_a

Acidic and basic properties of amino acids

- The stronger the tendency to dissociate a proton, **the stronger is the acid and the lower its pK_a**
- The p*K*a expresses, on a logarithmic scale, the relative strength of a weak acid or base:

Acidic and basic properties of amino acids

- $$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

- p*K*a = log — =–

- The stronger the acid, the lower its p*K*a;

Acidic and basic properties of amino acids

- The p*K*a can be determined experimentally;
- it is the pH at the midpoint of the titration curve for the acid or base

Acidic and basic properties of amino acids

- At the midpoint of the titration, the concentrations of the proton donor and proton acceptor are equal, and the pH is numerically equal to the p*K*a.

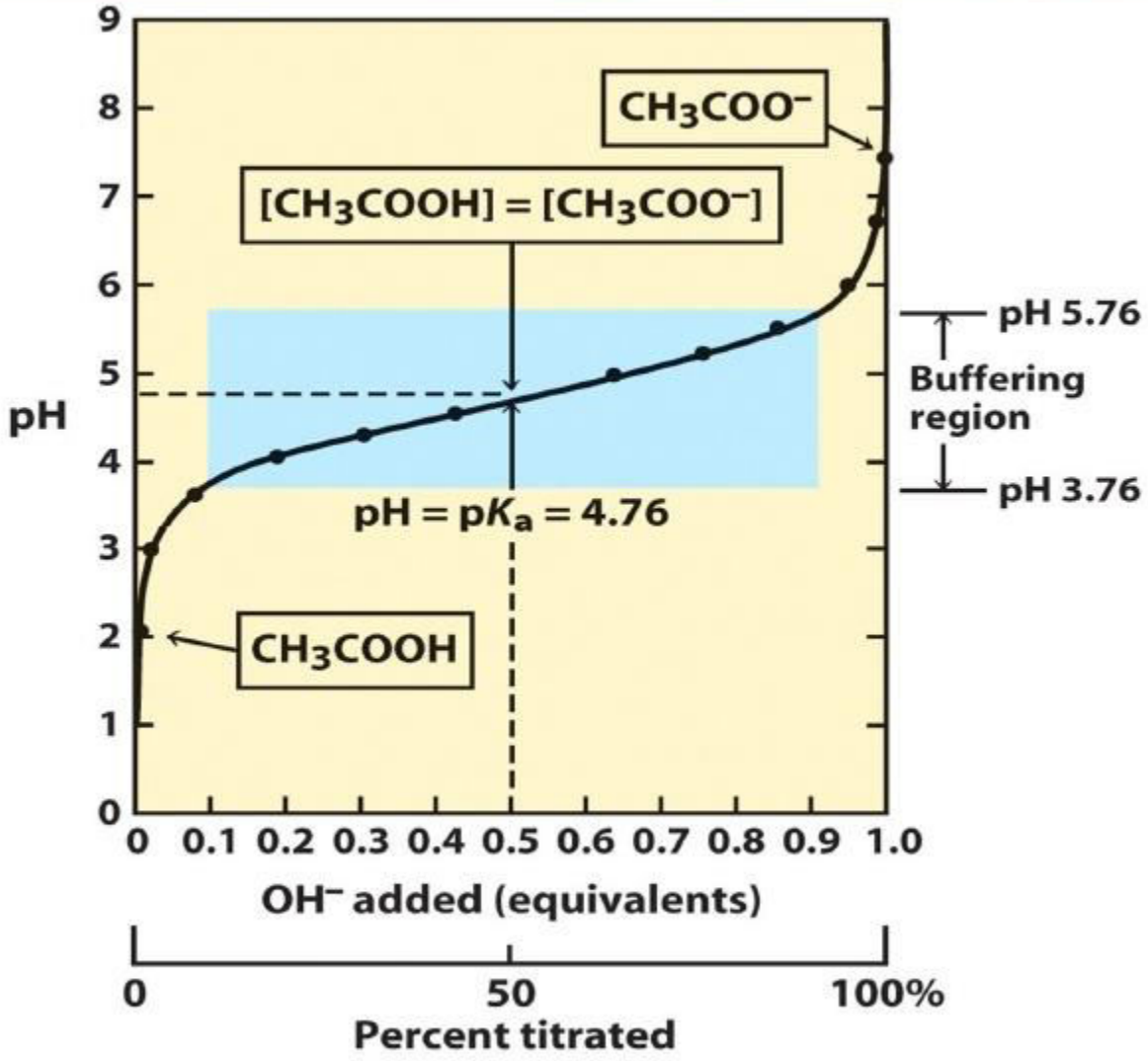
Acidic and basic properties of amino acids

- $$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

- At that point, [HA] equals [A], and

- pH = p*K*a + log 1
- = p*K*a + 0 = p*K*a

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- pKa is the measure of the tendency of a group to give up (donate) a proton with the tendency decreasing ten-fold as the pKa increases by one unit

Acidic and basic properties of amino acids

- Each titratable group has a pK_a that is numerically equal to the pH at which exactly one half of the protons have been removed from that group

Acidic and basic properties of amino acids

- The pK_a for the most acidic group (COOH) is pK_1, whereas the pK_a for the next most acidic group (NH_{3+}) is pK_2
- **In acid solution** (e.g., pH 1), the amino group is protonated ($–NH_{3+}$) and the carboxyl group is not dissociated (–COOH)

Acidic and basic properties of amino acids

- As the pH is raised, the carboxylic acid is the first group to give up a proton, as its p*Ka is near 2*

- The dipolar form persists until the pH approaches 9, when the protonated amino group loses a proton

END

Acidic and basic properties of amino acids

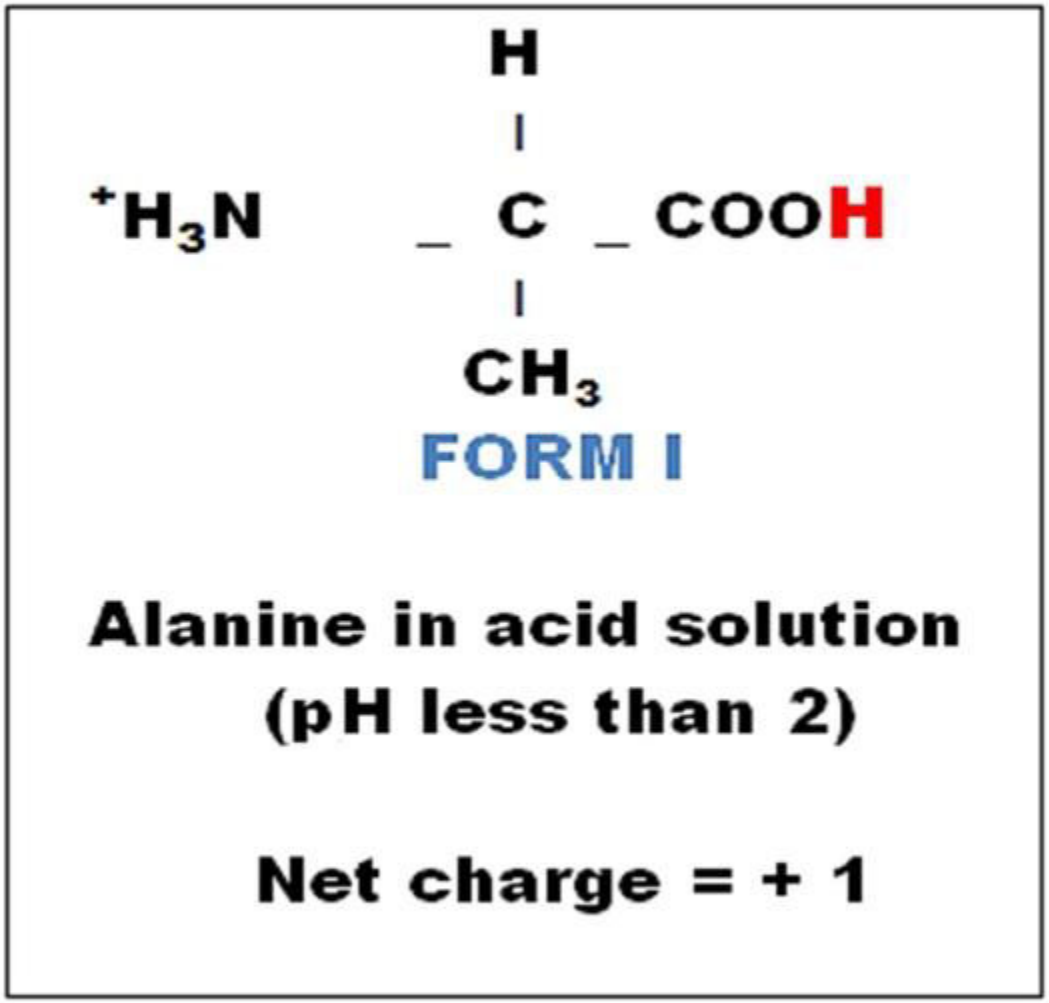
Titration of an amino acid Dissociation of the carboxyl group:

- The titration curve of an amino acid can be analyzed
- Consider alanine, for example, which contains both an α-carboxyl and an α-amino group

Acidic and basic properties of amino acids

- At a low (acidic) pH, both of these groups are protonated

Acidic and basic properties of amino acids



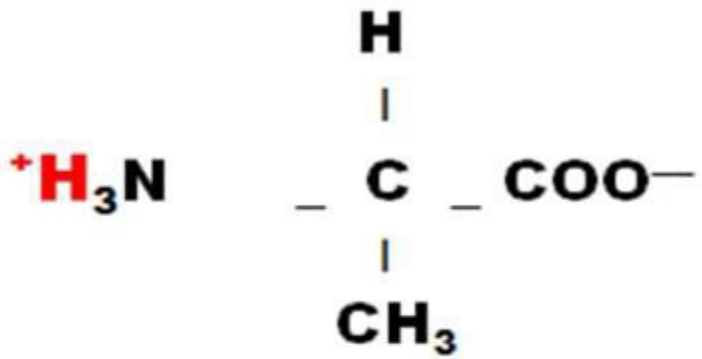
Acidic and basic properties of amino acids

- As the pH of the solution is raised, the -COOH group of form I can dissociate by donating a proton to the medium
- The release of a proton results in the formation of the carboxylate group –COO-

Acidic and basic properties of amino acids

- This structure is shown as form II, which is the **dipolar form of the molecule**
- This form, also called a **zwitterion,** is the **isoelectric form** of alanine
- That is, it has an overall charge of zero

Acidic and basic properties of amino acids



FORM II

Alanine in neutral solution (pH approximately 6)

Net charge = 0 (isoelectric form)

Acidic and basic properties of amino acids

- The dissociation constant of the carboxyl group of an amino acid is called K_1 rather than K_2 because the molecule contains a second titratable group

Acidic and basic properties of amino acids

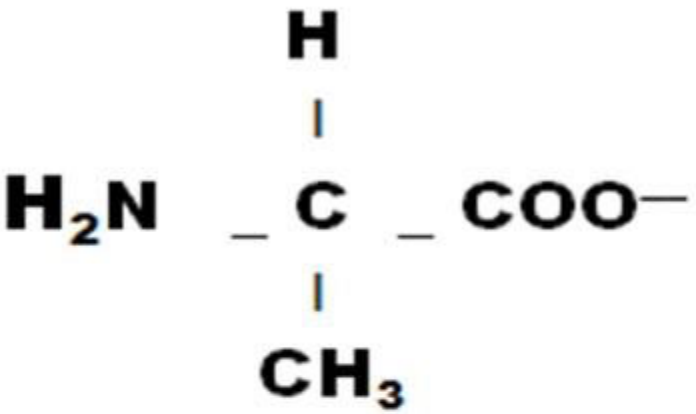
Dissociation of the amino group:

- The second titratable group of alanine is the amino group
- This is a much weaker acid than the -COOH group and, therefore, has a much smaller dissociation constant, K_2

Acidic and basic properties of amino acids

- Release of a proton from the protonated amino group of form II results in the fully deprotonated form of alanine, form III

Acidic and basic properties of amino acids

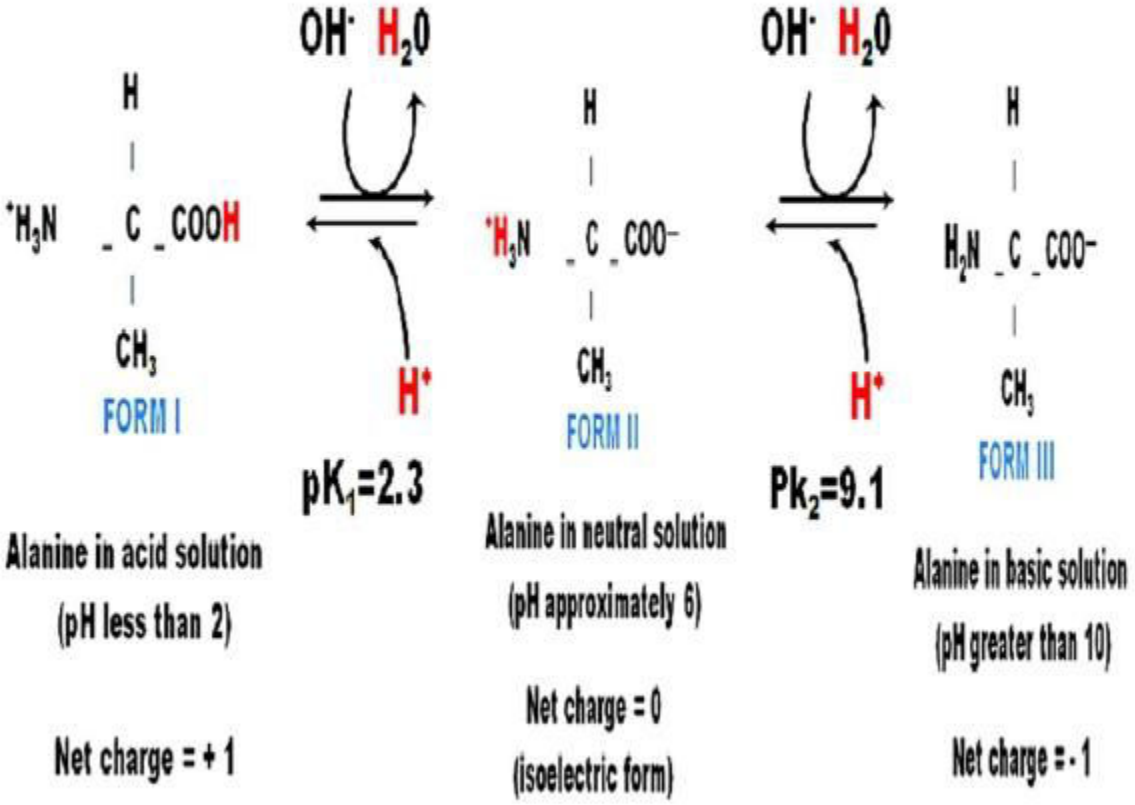


FORM III

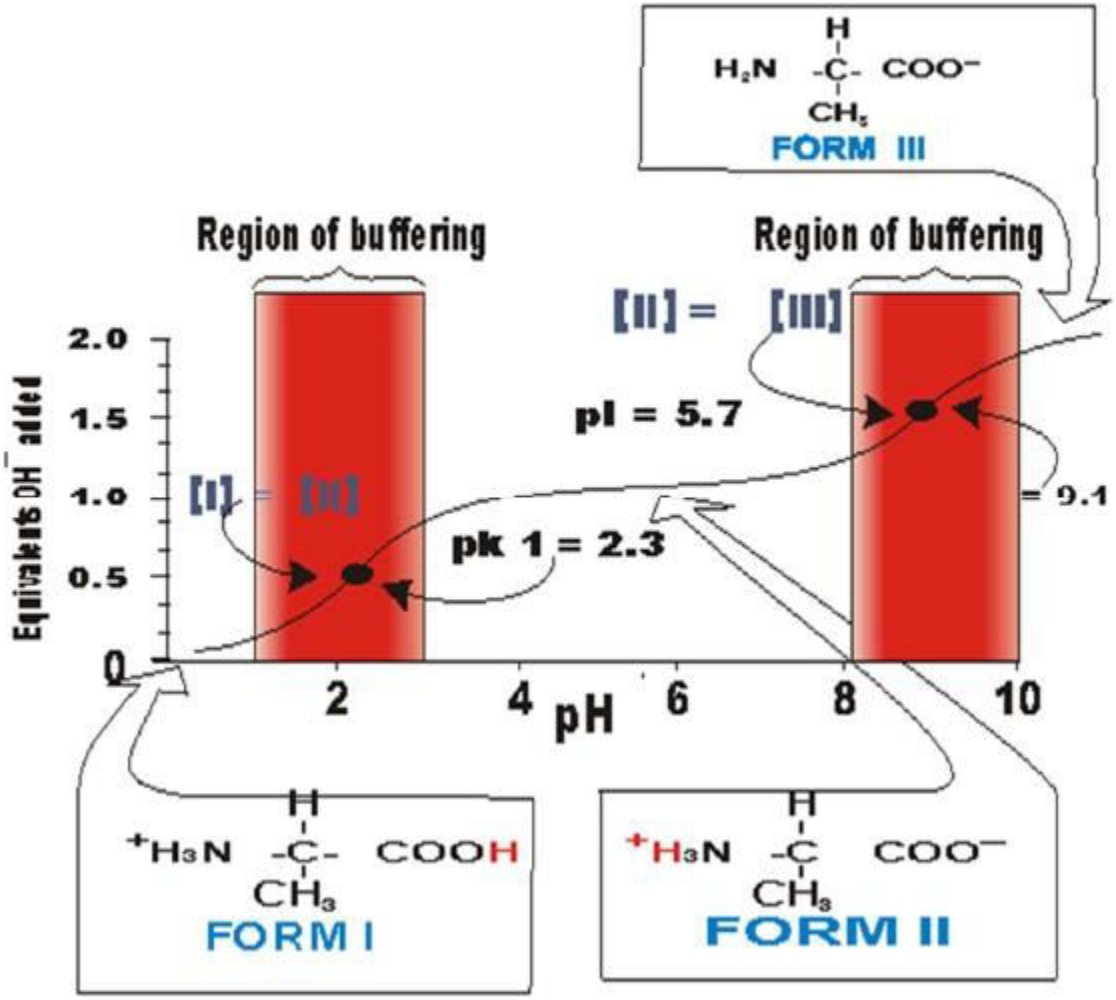
Alanine in basic solution
(pH greater than 10)

Net charge = - 1

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- A simple amino acid e.g. **glycine** is a diprotic acid when fully protonated
- This means that it has two groups, the COOH and the NH_{3+} group that can yield protons

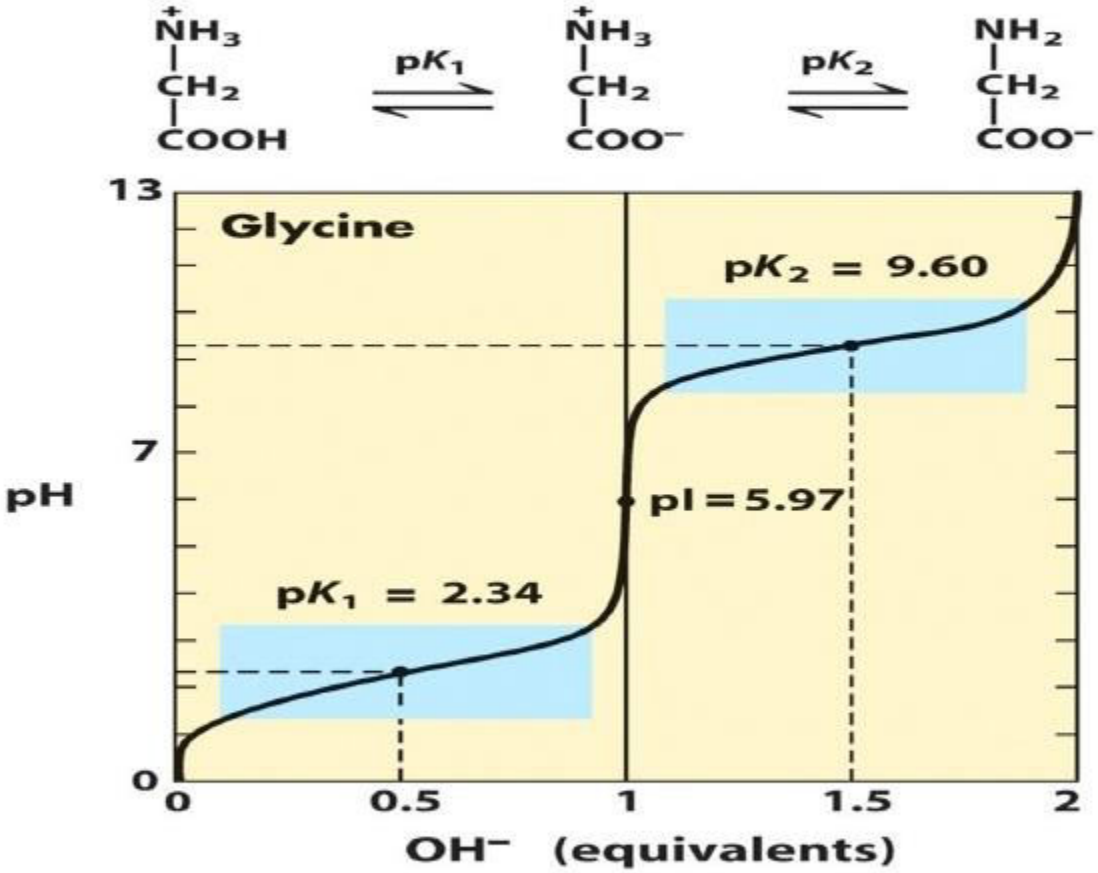
Acidic and basic properties of amino acids

- Figure shows the titration curve of the diprotic form of glycine
- The two ionizable groups of glycine, the carboxyl group and the amino group, are titrated with a strong base

Acidic and basic properties of amino acids

- The plot has two distinct stages, corresponding to deprotonation of two different groups on glycine

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- At very low pH, the predominant ionic species of glycine is **the fully protonated form,**

$^{+}H_3N\text{-}CH_2\text{-}COOH$

Acidic and basic properties of amino acids

- **At the midpoint** of any titration, a point is reached where the **pH is equal to the pK_a** of the protonated group being titrated

Acidic and basic properties of amino acids

- At the midpoint in the first stage of the titration, in which the --COOH group of glycine loses its proton, equimolar concentrations of the protonated ($_{+}H_3N$-CH_2-COOH) and the deprotonated ($_{+}H_3N$-CH_2-COO^-) species are present

Acidic and basic properties of amino acids

- As the titration proceeds, another important point is reached at pH 5.97
- This is a point, at which removal of the first proton is essentially complete and removal of the second has just begun

Acidic and basic properties of amino acids

- At this pH glycine is present largely as the dipolar ion (zwitterion)

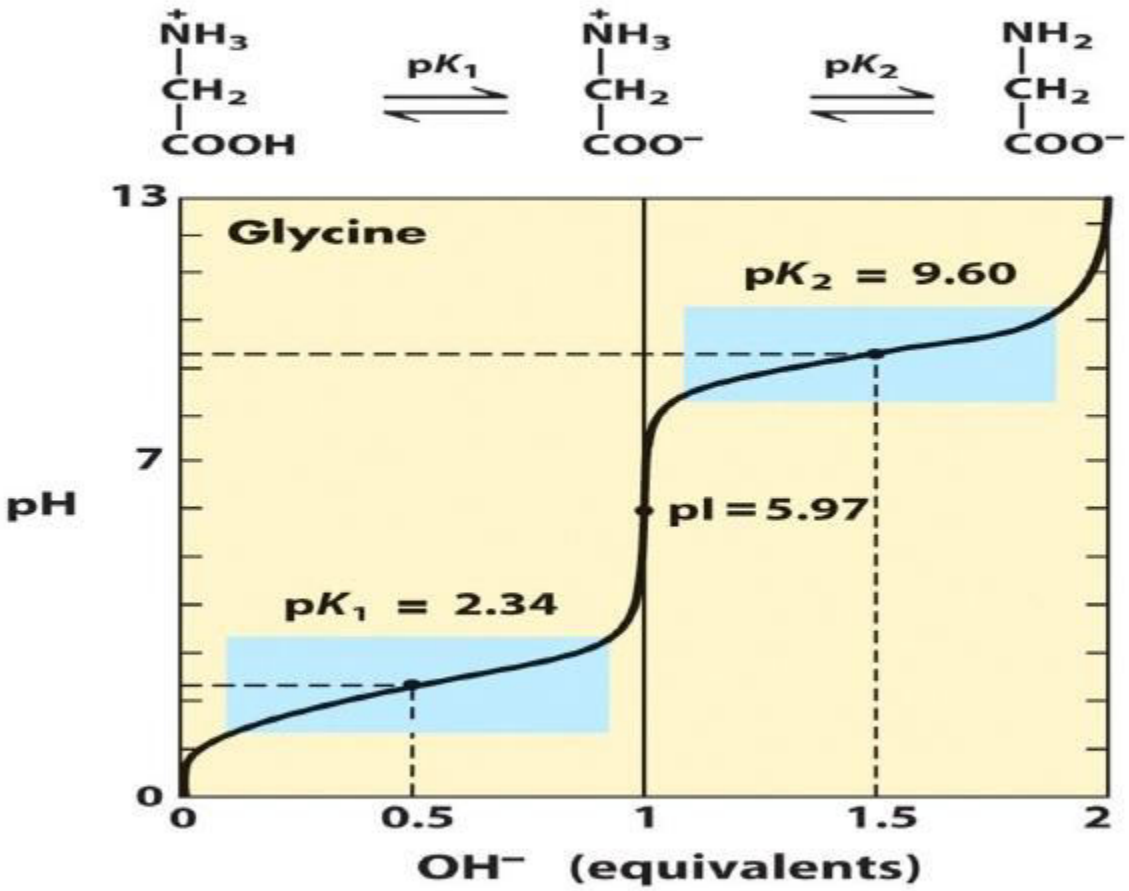
  $^{+}H_3N\text{-}CH_2\text{-}COO^{-}$

- The second stage of the titration corresponds to the removal of a proton from the $-NH_3^{+}$ group of glycine

Acidic and basic properties of amino acids

- The pH at the midpoint of this stage is 9.60, equal to the pK_a for the $–NH_3$ group
- The titration is essentially complete at a pH of about 12, at which point the predominant form of glycine is
- $H_2N\text{-}CH_2\text{-}COO^-$

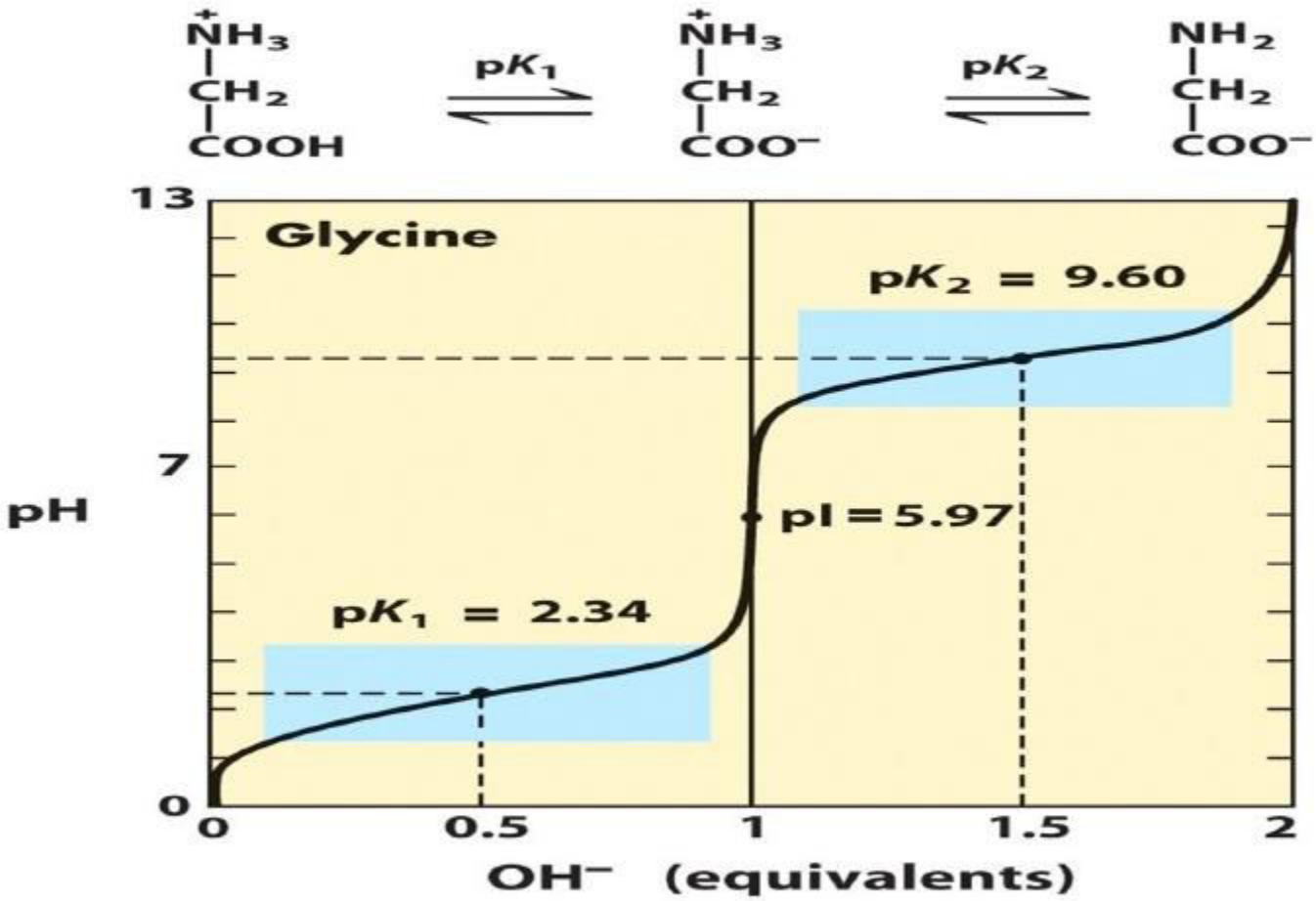
Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- From the titration curve of glycine we can derive several important pieces of information

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- First, it gives a quantitative measure of the pK_a, of each of the two ionizing groups:
  - 2.34 for the -COOH group
  - 9.60 for the $-NH_3^+$ group

Acidic and basic properties of amino acids

- The second piece of information is that this amino acid has **two regions of buffering power**

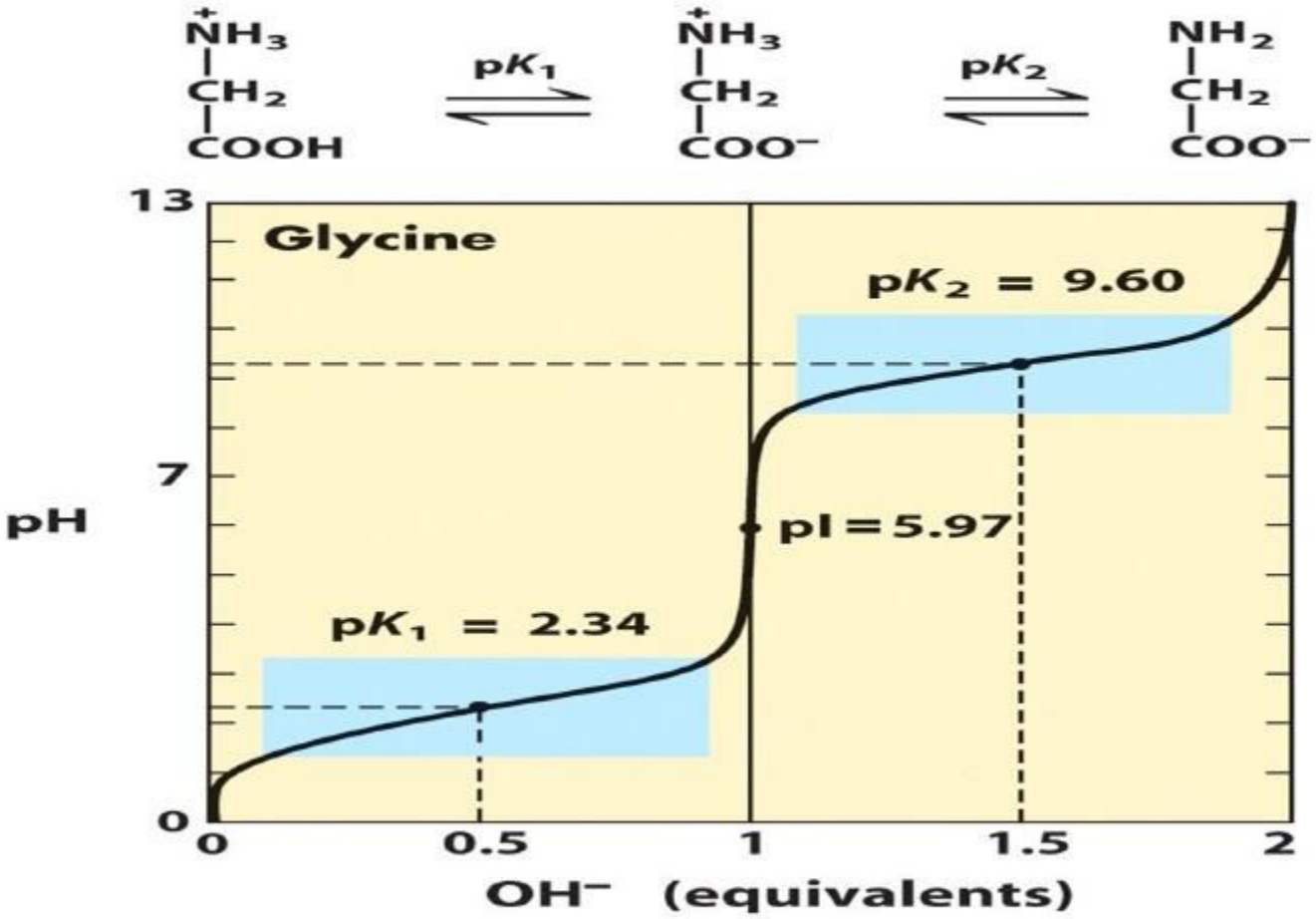
Acidic and basic properties of amino acids

- One of these is the relatively flat portion of the curve, extending for approximately 1 pH unit on either side of the first pKa, of 2.6, indicating that glycine is a good buffer near this pH

Acidic and basic properties of amino acids

- The other buffering zone is centered around pH 9.60
- Note that glycine is not a good buffer at the pH of intracellular fluid or blood, about 7.4

Acidic and basic properties of amino acids



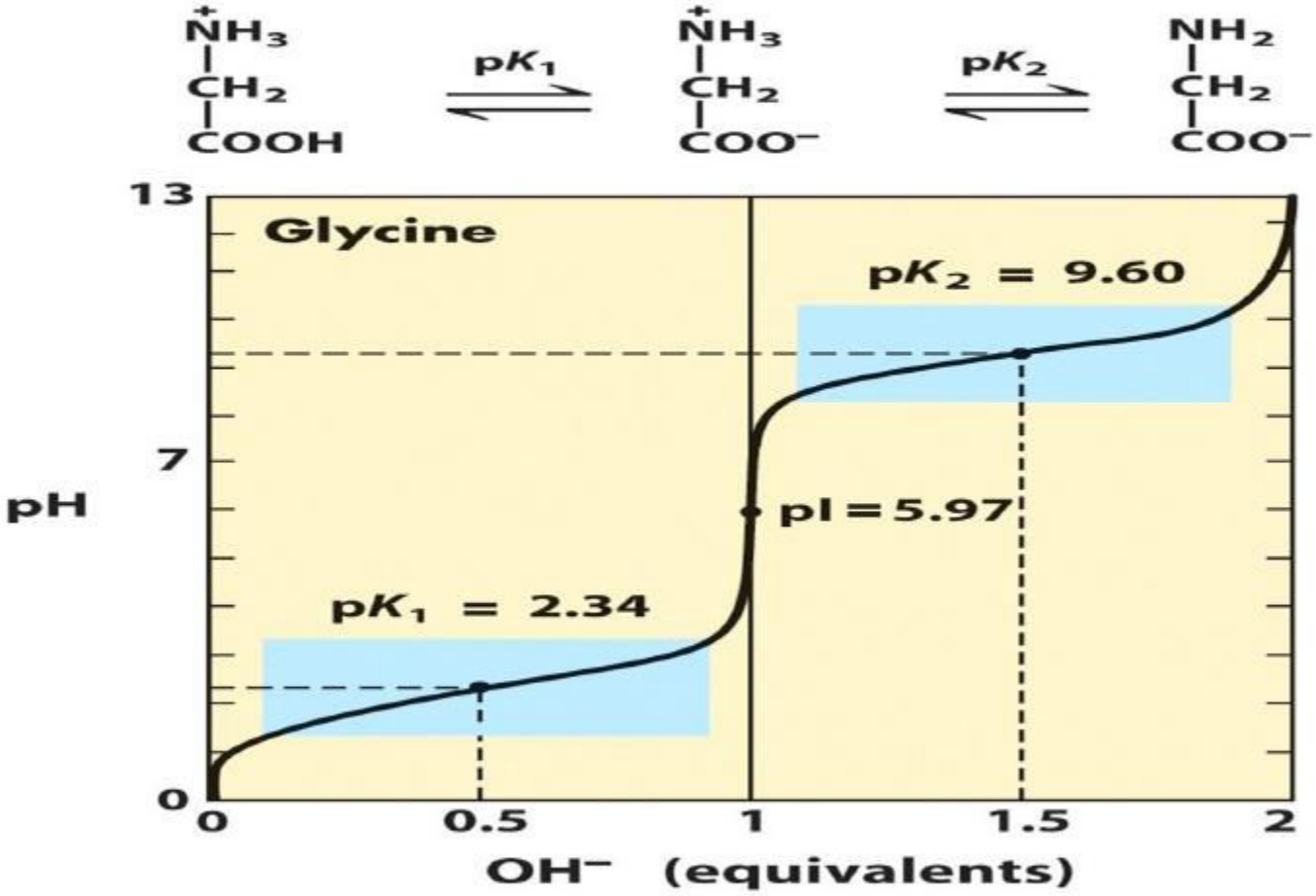
Acidic and basic properties of amino acids

- Another important piece of information derived from the titration curve of an amino acid is the relationship between its net charge and the pH of the solution

Acidic and basic properties of amino acids

- At pH 5.97, glycine is present predominantly as its dipolar form, fully ionized but with no net electric charge

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- The characteristic pH at which the net electric charge is zero is called the **isoeleetric point or isoelectric pH, designated pI**

Acidic and basic properties of amino acids

- For glycine, which has no ionizable group in its side chain;
- the pI is simply the arithmetic mean of the two pKa values
- pI = ½ (pK1+pK2)
- = ½ (2.34+9.60) = 5.97

Acidic and basic properties of amino acids

- The farther the pH of a glycine solution is from its pI, the greater the net electric charge of the population of glycine molecules

Acidic and basic properties of amino acids

- At any pH below its pI, glycine has a **net positive charge** and will move toward the negative electrode (cathode)

Acidic and basic properties of amino acids

- At any pH above its pI, glycine will have a net negative charge and thus will move towards the positive electrode (anode)

END

Acidic and basic properties of amino acids

- All amino acids with a
- single α-amino group,
- a single α-carboxyl group, and
- an R group that does not ionize
- have titration curves resembling that of glycine

Acidic and basic properties of amino acids

- These amino acids have very similar, although not identical, pKa values:
- pKa of the –COOH group in the range of 1.8 to 2.4

Acidic and basic properties of amino acids

- pKa of the -NH3+ group in the range of 8.8 to 1 1
- The differences in these pKa values reflect the effects of the R groups
- Alanine has only two dissociable hydrogens (one from α carboxyl & one from α amino group)

Acidic and basic properties of amino acids

- Its pI can be calculated as follows:
- pI = pK1 + pK2 ÷ 2
- pK1 = 2.3 (for carboxyl group)
- pK2 = 9.1 (for amino group)
- pI = 2.3 + 9.1 ÷ 2 = 5.7

Acidic and basic properties of amino acids

- Amino acids with an ionizable R group have more complex titration curves, with three stages corresponding to the three possible ionization steps and thus they have **three pKa values**

Acidic and basic properties of amino acids

- The additional stage for the titration of the ionizable R group merges to some extent with the other two
- In addition, each of the acidic and basic amino acids contains an ionizable group in its side chain

Acidic and basic properties of amino acids

- Thus, both free amino acids and some amino acids combined in peptide linkages can act as **buffers**
- A buffer is a solution that resists change in pH following the addition of an acid or base

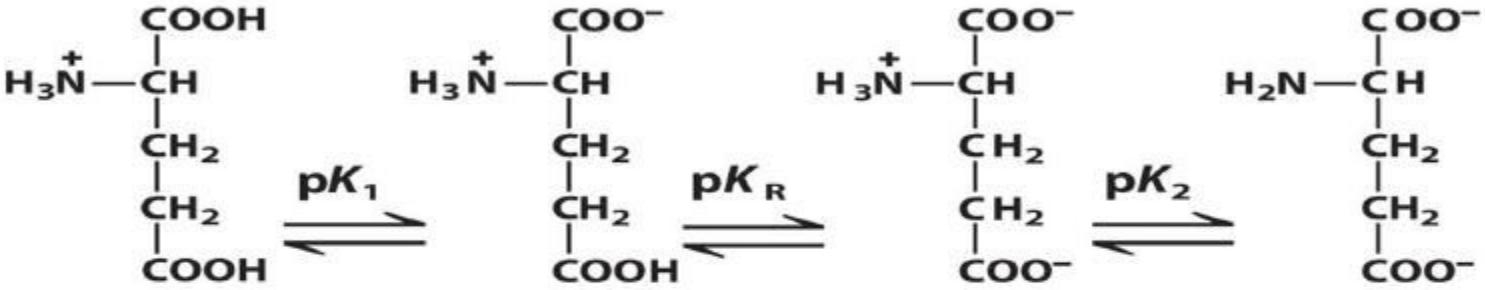
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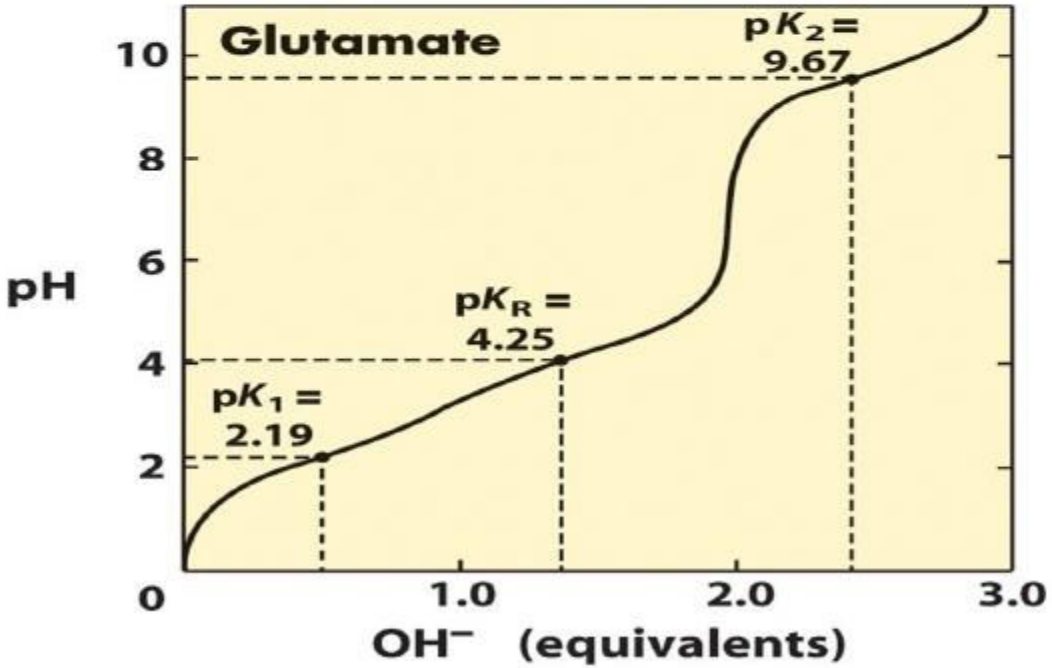
- The isoelectric points reflect the nature of the ionizing R groups present
- For example, glutamate has a pI of 3.22, considerably lower than that of glycine

Acidic and basic properties of amino acids

- This is due to the presence of two carboxyl groups, with an average of their pKa values of 3.22

Acidic and basic properties of amino acids





Acidic and basic properties of amino acids

- Histidine has got 3 dissociable hydrogens
  - one from carboxyl, pK1=1.8,
  - one from imidazole group, pK2=6.0
  - one from amino group, pK3=9.2

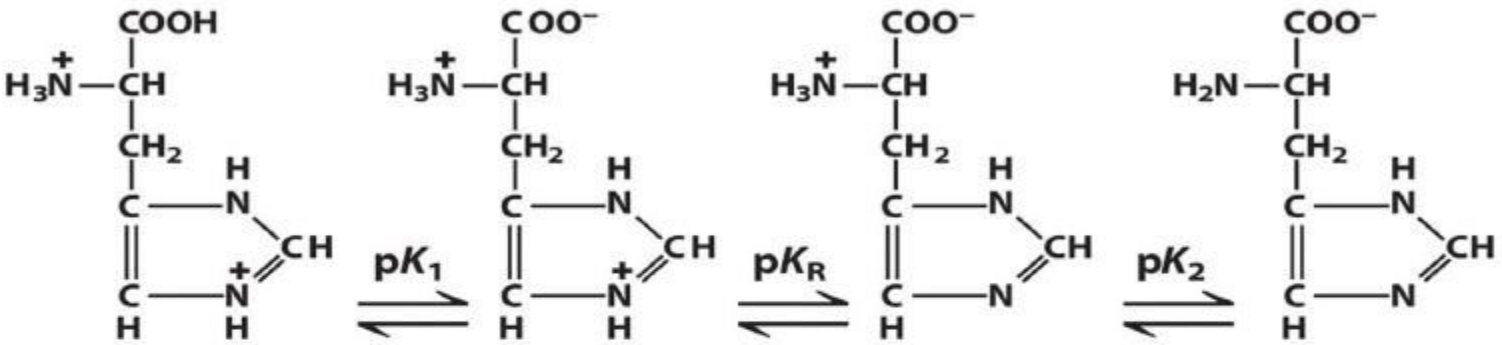
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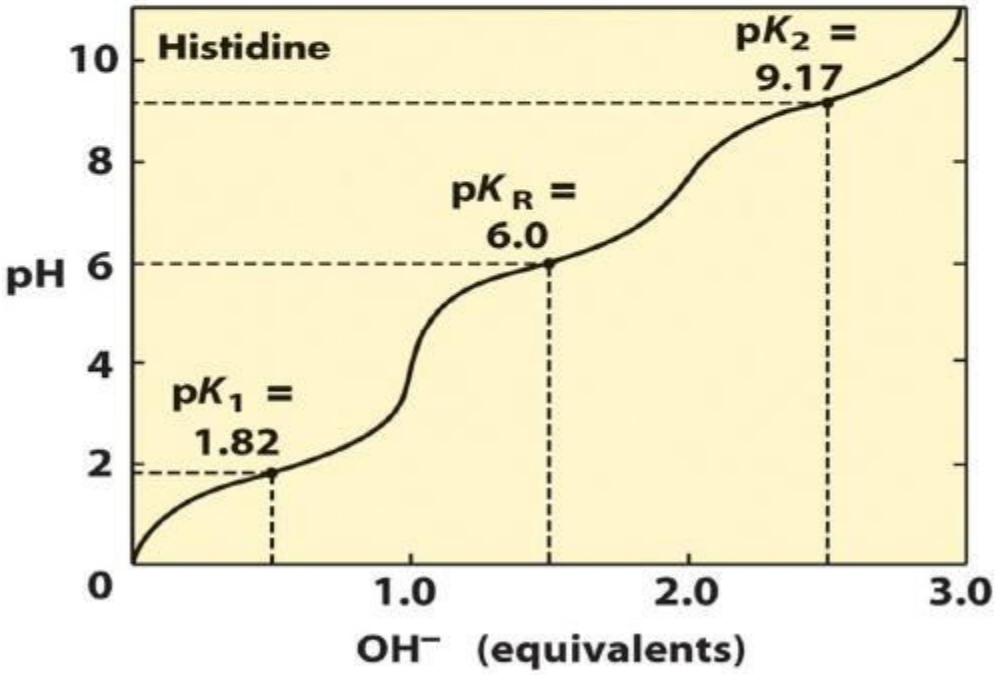
- The pI of Histidine, with two groups that are positively charged when protonated, is 7.59 (the average of the pKa values of the amino and imidazole groups)

Acidic and basic properties of amino acids

- Only histidine has an R group (pKa = 6.0) providing significant buffering power near the neutral pH usually found in the intracellular and extracellular fluids of most animals

Acidic and basic properties of amino acids





Acidic and basic properties of amino acids

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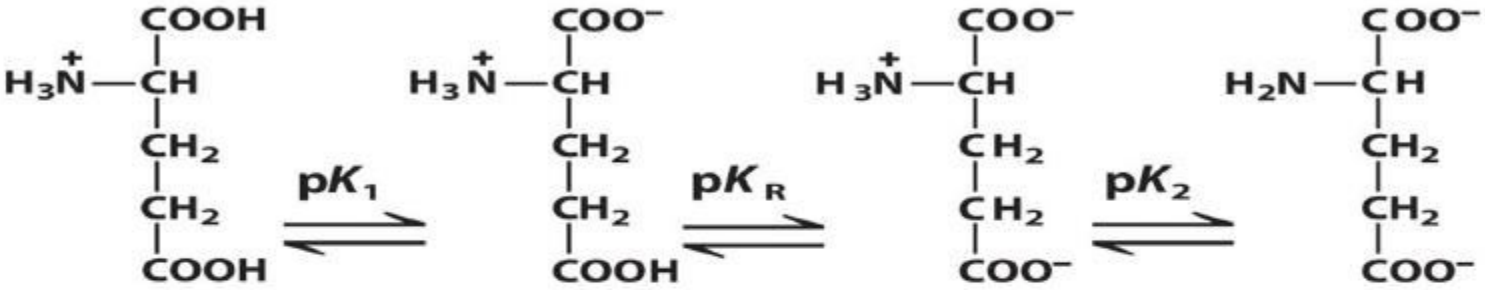
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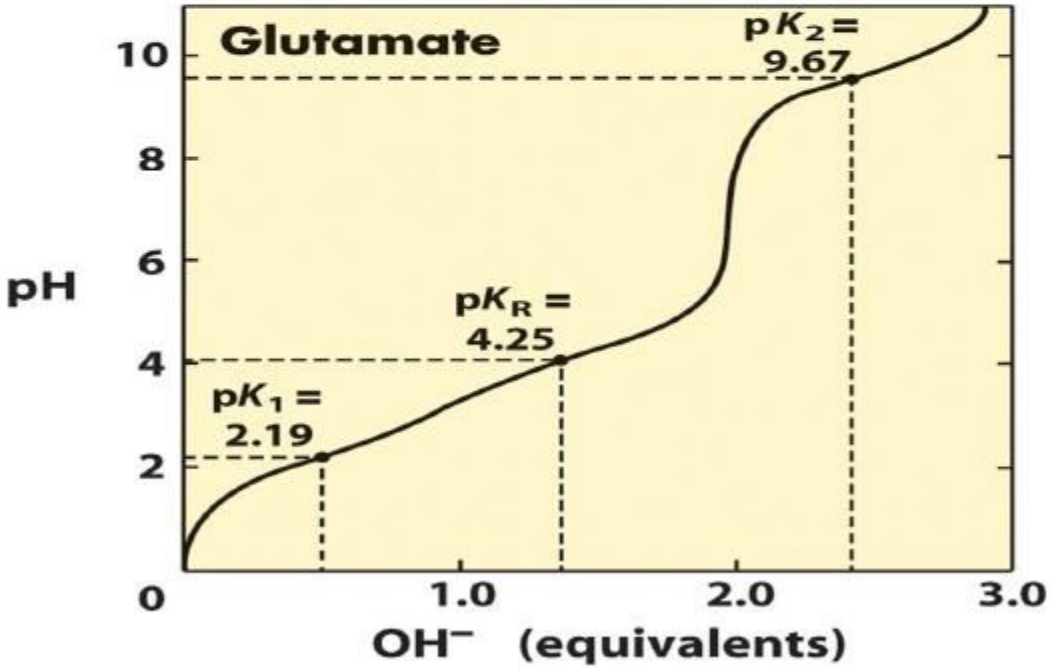
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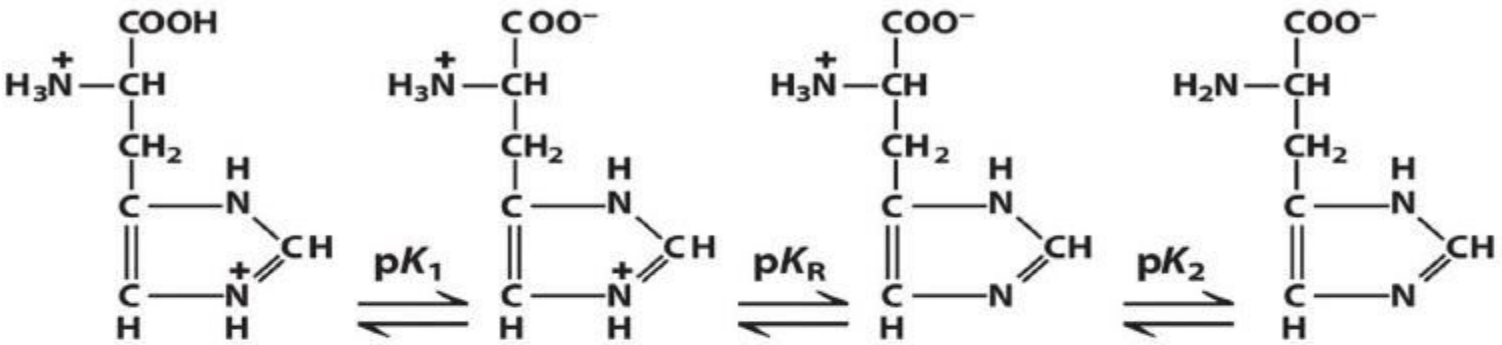
Acidic and basic properties of amino acids

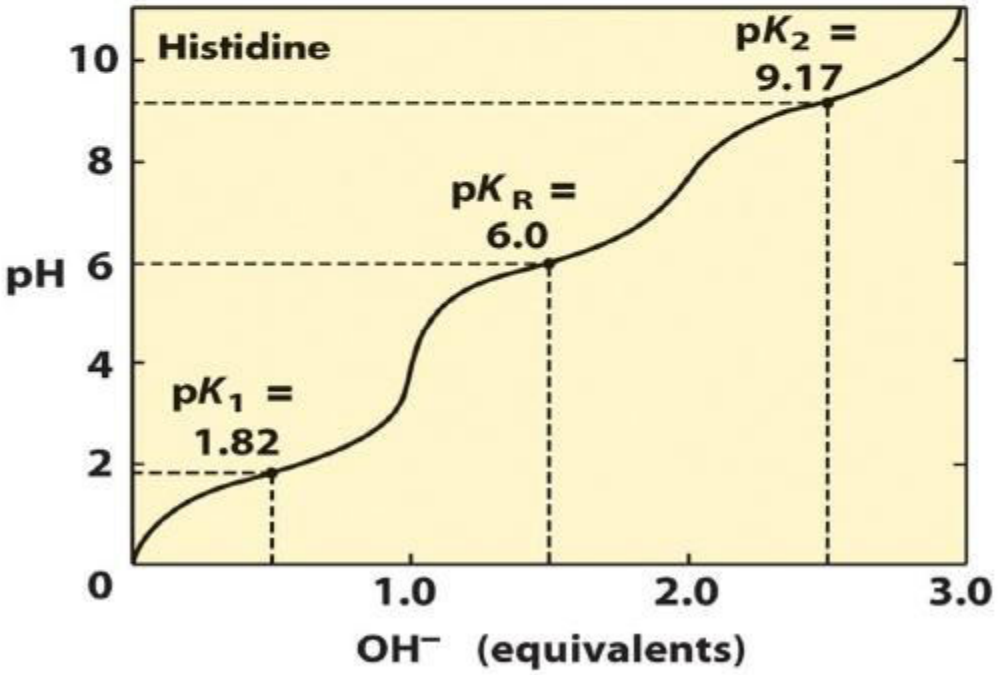
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Acidic and basic properties of amino acids





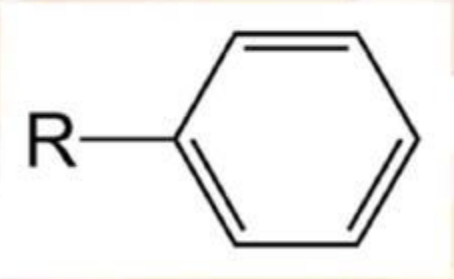
Classification of amino acids

B. Aromatic Amino Acids

- The aromatic amino acids have been grouped together because they all contain ring structures with similar properties, **but their polarity differs**

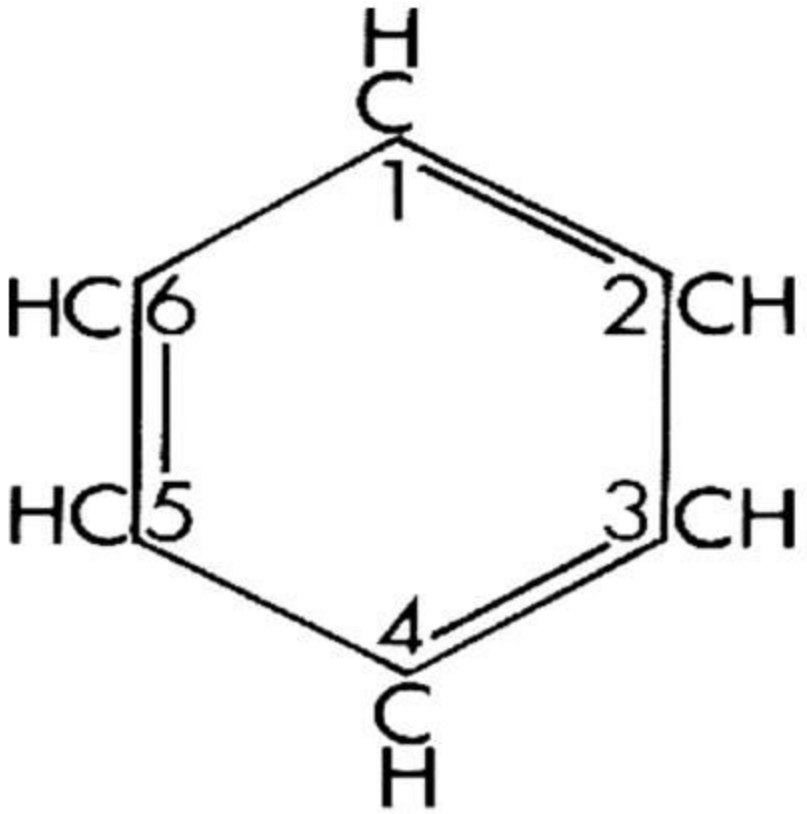
Classification of amino acids

- The aromatic ring is a six-membered carbon–hydrogen ring with three conjugated double bonds (the benzene ring or phenyl group)



Classification of amino acids

- Benzene is an organic chemical compound with the chemical formula C6H6. Its molecule is composed of 6 carbon atoms joined in a ring, with 1 hydrogen atom attached to each carbon atom



Classification of amino acids

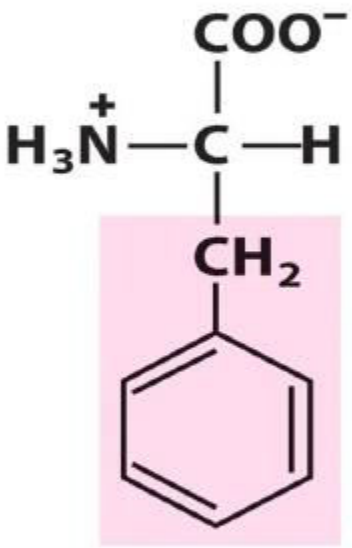
- The substituents on this ring determine whether the amino acid side chain engages in polar or hydrophobic interactions

Classification of amino acids

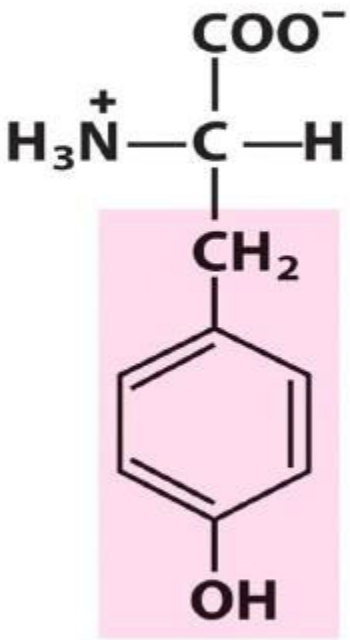
- This group includes
  - Phenylalanine
  - Tyrosine
  - Tryptophan
  - with their aromatic side chains, are relatively nonpolar (hydrophobic).
  - All can participate in hydrophobic interactions.

Classification of amino acids

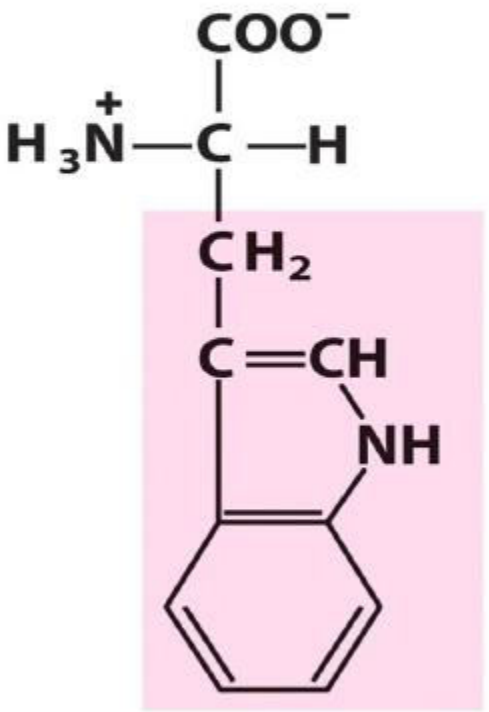
Aromatic R groups



Phenylalanine



Tyrosine



Tryptophan

Classification of amino acids

- In the amino acid **phenylalanine**, the ring contains no substituents, and the electrons are shared equally between the carbons in the ring, resulting in *a very* **nonpolar hydrophobic** structure in which the rings can stack on each other

Classification of amino acids

- In **tyrosine**, a hydroxyl group on the phenyl ring engages in hydrogen bonds, and the side chain is therefore more polar and **more hydrophilic**

Classification of amino acids

- The more complex ring structure in **tryptophan** is an indole ring with a nitrogen that can engage in hydrogen bonds

Classification of amino acids

- Tryptophan is therefore also more polar than phenylalanine
- Tyrosine and tryptophan are significantly more polar than phenylalanine, because of the tyrosine hydroxyl group and the nitrogen of the **tryptophan** indole ring

END

Classification of amino acids

C. Aliphatic, Polar, Uncharged R Groups

- The R groups of these amino acids are more soluble in water, or more hydrophilic, than those of the nonpolar amino acids.

Classification of amino acids

- Because they contain functional groups that form hydrogen bonds with water.

Classification of amino acids

- This class of amino acids includes
- **serine, threonine,**
- **cysteine,**
- **asparagine,** and
- **glutamine.**

Classification of amino acids

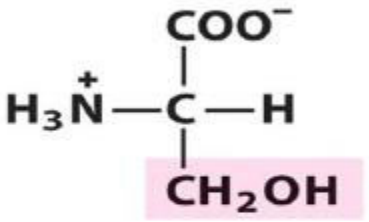
- These Amino acids have zero net charge at neutral pH
  - **Serine & threonine** of this class contain a polar hydroxyl group

Classification of amino acids

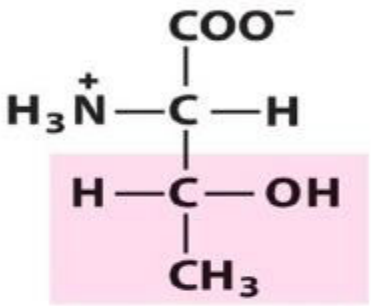
- Similarly side chains of **asparagine & glutamine** contain a carbonyl group & amide group, both of which can participate in hydrogen bonding

Classification of amino acids

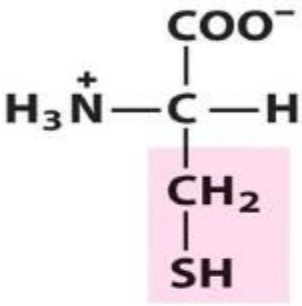
Polar, uncharged R groups



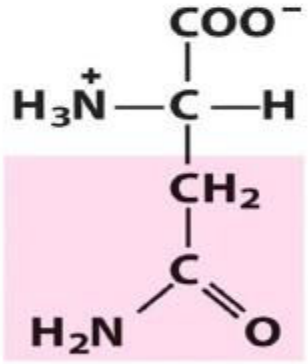
Serine



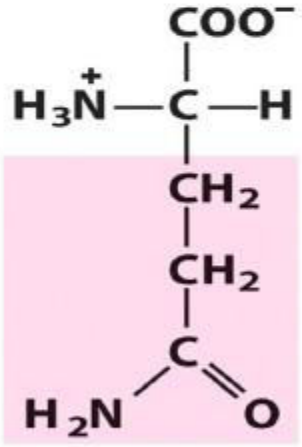
Threonine



Cysteine



Asparagine



Glutamine

Classification of amino acids

- These side chains are sites of attachment for other compounds
- The polar hydroxyl groups of **serine & threonine** serve as a site of attachment for phosphate groups

Classification of amino acids

- The amide group of **asparagine** & the hydroxyl group of **serine & threonine** serve as a site of attachment for oligosaccharide chains in glycoproteins

Classification of amino acids

- Amino acids with side chains that contain an amide group (asparagine and glutamine) or a hydroxyl group (serine and threonine) can be classified as **aliphatic, polar, uncharged amino acids**

Classification of amino acids

- Asparagine and glutamine are amides of the amino acids aspartate and glutamate
- As a consequence of their hydrophilicity, these amino acids are frequently found on the surface of water-soluble globular proteins

Classification of amino acids

- Cysteine, which is sometimes included in this class of amino acids, has been separated into the class of sulfur-containing amino acids

END

Classification of amino acids

D. Positively Charged (Basic) R Groups

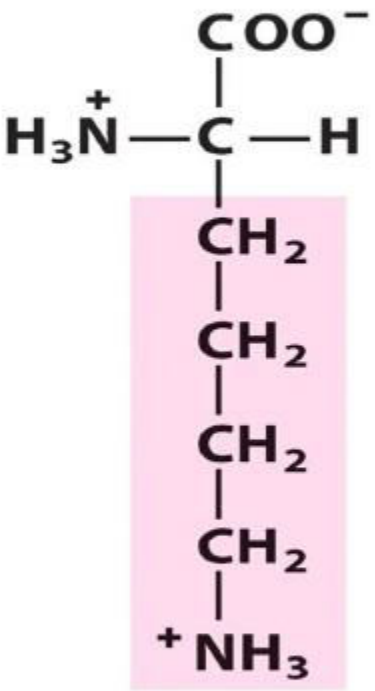
- The side chains of these amino acids are proton acceptors
- The amino acids in which the R groups have significant **positive charge** at pH 7.0 are:

Classification of amino acids

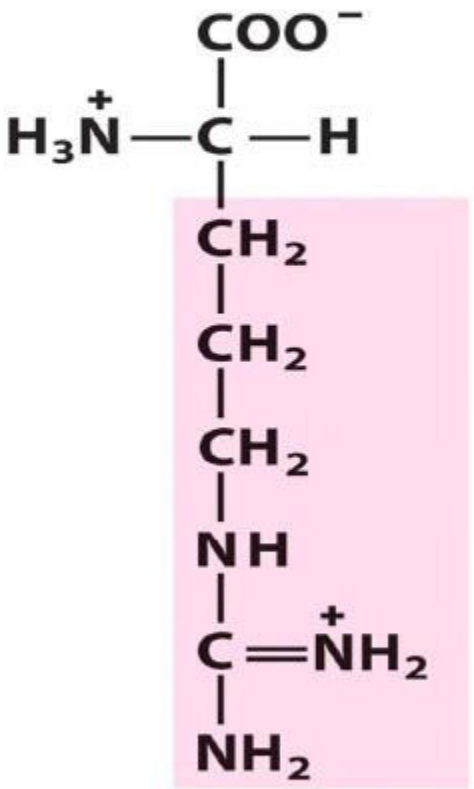
- **Lysine**, which has a second primary amino group at the ε position on its aliphatic chain
- **Arginine,** which has a positively charged guanidinium group
- **Histidine,** which has an aromatic imidazole group

Classification of amino acids

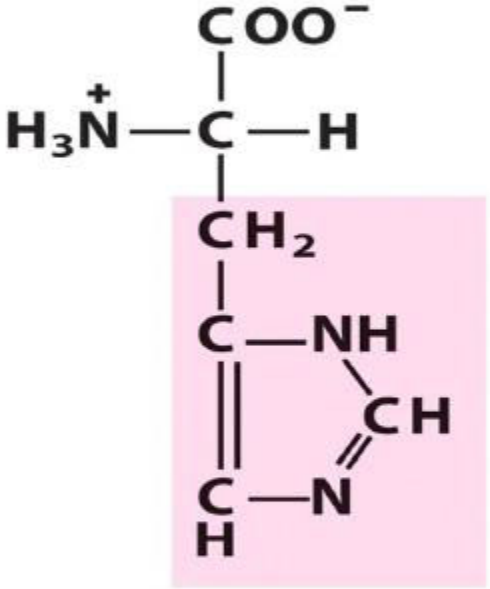
Positively charged R groups



Lysine



Arginine



Histidine

Classification of amino acids

- The side chains of the two basic amino acids, arginine and lysine, have pK_a values above 10, so that the positively charged form always predominates at physiologic pH

Classification of amino acids

- The side chain of histidine (pK_a ~ 6.0) dissociates near physiologic pH, so only a portion of the histidine side chains carry a positive charge

Classification of amino acids

- **Histidine** is therefore weakly basic & the *free* amino acid is mainly uncharged at physiologic pH

Classification of amino acids

- The amino acid side chains might have very different pK_as than those of the free amino acids if they are involved in hydrogen or ionic bonds with other amino acid side chains

Classification of amino acids

- Therefore, when histidine is incorporated into a protein, its side chain can be either positively charged or neutral depending on the ionic environment provided by the polypeptide chains of the protein

END

Classification of amino acids

E. Negatively Charged (Acidic) R Groups

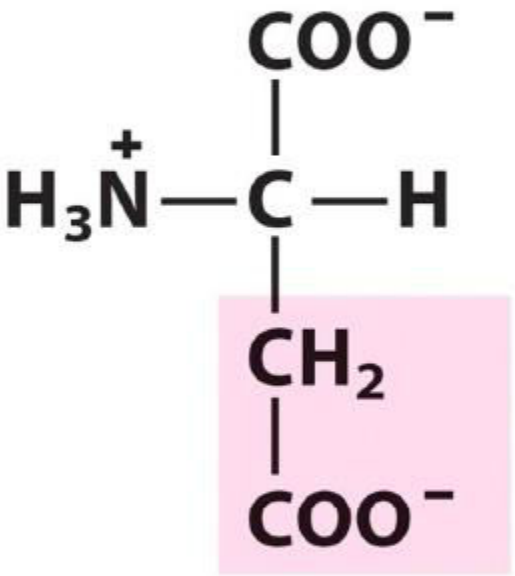
- These amino acids are proton donors
- At neutral pH, the side chains of these amino acids are fully ionized, containing a negatively charged carboxylate group
- (--COO^-)

Classification of amino acids

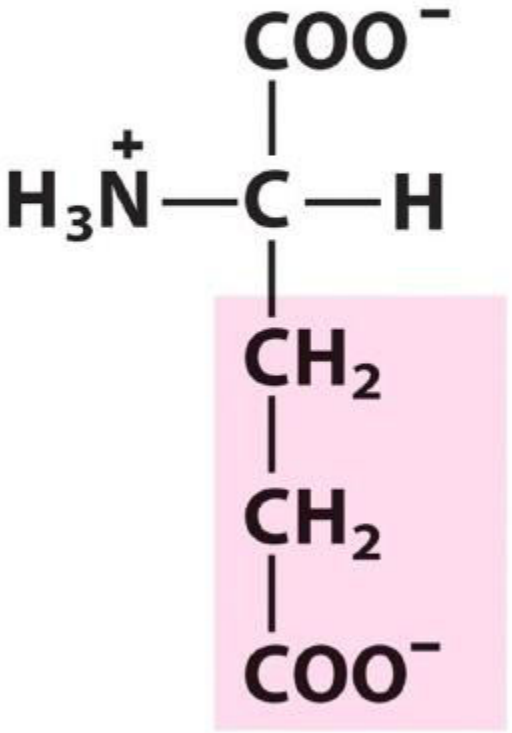
- Therefore these amino acids are negatively charged at phyisologic pH
- The two amino acids having R groups with a net negative charge at pH 7.0 are **aspartate and glutamate**, each of which has a second carboxyl group

Classification of amino acids

Negatively charged R groups



Aspartate



Glutamate

Classification of amino acids

- Humans have no dietary requirement for protein, per se, but, the protein in food does provide essential amino acids.

Classification of amino acids

- Nine of the twenty amino acids needed for the synthesis of body proteins are essential-that is, they cannot be synthesized in humans at an adequate rate

Classification of amino acids

- **Essential amino acids** can not be made by the body. As a result, they must come from food. The nine **essential amino acids** are: histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

Classification of amino acids

- Of these nine, some amino acids are essential at all times, whereas some are required only during periods of rapid tissue growth characteristic of childhood or recovery from illness

Classification of amino acids

	Glucogenic	Glucogenic And Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essentia	Histidine Methionine Valine	Threonine Isoleucine Phenyl-alanine Tryptophan	Leucine Lysine

Classification of amino acids

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Classification of amino acids

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Classification of amino acids

- Amino acids can be classified as **glucogenic or ketogenic** based on what intermediates are produced during their catabolism

Classification of amino acids

A. Glucogenic amino acids

- Amino acids whose catabolism yields **pyruvate are termed glucogenic or glycogenic**

Classification of amino acids

- These intermediates are substrates for **gluconeogenesis** and, therefore, can give rise to the net formation of glucose or glycogen in the liver and glycogen in the muscle

Classification of amino acids

B. Ketogenic amino acids

- Amino acids whose catabolism yields either **acetoacetate or its precursor, (acetyl CoA or acetoacetyl CoA)** are termed **ketogenic**

Classification of amino acids

- Acetoacetate is one of the ketone bodies which also include 3-hydroxybutyrate and acetone
- **Leucine and lysine** are the only exclusively ketogenic amino acids found in proteins

Classification of amino acids

- Their carbon skeletons are not substrates for gluconeogenesis and, therefore, cannot give rise to the net formation of glucose or glycogen in the liver, or glycogen in the muscles

Classification of amino acids

	Glucogenic	Glucogenic And Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essential	Histidine Methionine Valine	Threonine Isoleucine Phenyl-alanine Tryptophan	Leucine Lysine

Classification of amino acids

TABLE 14–4 Glucogenic Amino Acids, Grouped by Site of Entry

Pyruvate
Alanine
Cysteine
Glycine
Serine
Threonine
Tryptophan*

α-Ketoglutarate
Arginine
Glutamate
Glutamine
Histidine
Proline

Succinyl-CoA
Isoleucine*
Methionine
Threonine
Valine

Fumarate
Phenylalanine*
Tyrosine*

Oxaloacetate
Asparagine
Aspartate

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis. *These amino acids are also ketogenic (see Fig. 18–21).

Classification of amino acids

TABLE 14–4 Glucogenic Amino Acids, Grouped by Site of Entry

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Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis. *These amino acids are also ketogenic (see Fig. 18–21).

Classification of Proteins

- Different classifications of proteins are based on their:

1. Shape and Size
2. Biological actions/ Functions
3. Solubility and physical properties
4. Quality

Functional Classification of Proteins

Catalytic Proteins:

These specialized proteins are called **enzymes** which catalyze the biochemical reactions

Functional Classification of Proteins

2. Protective Proteins

a) Immunoglobins (Igs)

- These freely circulating proteins protect the body from invading microbes such as bacteria or viruses by inactivating or killing them through various mechanisms.

Functional Classification of Proteins

b) Fibrinogen

- This forms fibrin clot and stops bleeding from wounds

Functional Classification of Proteins

3. Regulatory Proteins:

- Hormones control genetic expression, cellular signalling *and* biochemical reactions catalyzed by enzymes. Enzymes are either activated or inactivated through modification of their structure.

Functional Classification of Proteins

- Examples of protein hormones are
  - growth hormone
  - Insulin
  - Glucagon
  - Somatostatin

Functional Classification of Proteins

4. Structural Proteins:

- These proteins form various body structures e.g.
  - Collagen
  - Elastin
  - Keratin

Functional Classification of Proteins

5. Transport Proteins:

- These proteins transport various substances from one part of the body to the other e.g.
  - Hemoglobin transport O_2 from lungs to tissues and CO_2 from tissues to lungs
  - Transferrin transports iron

Functional Classification of Proteins

6. Contractile Proteins:

- These proteins are involved in muscle contraction and relaxation
  - Myosin of thick filaments
  - Actin of thin filaments of skeletal muscles

END

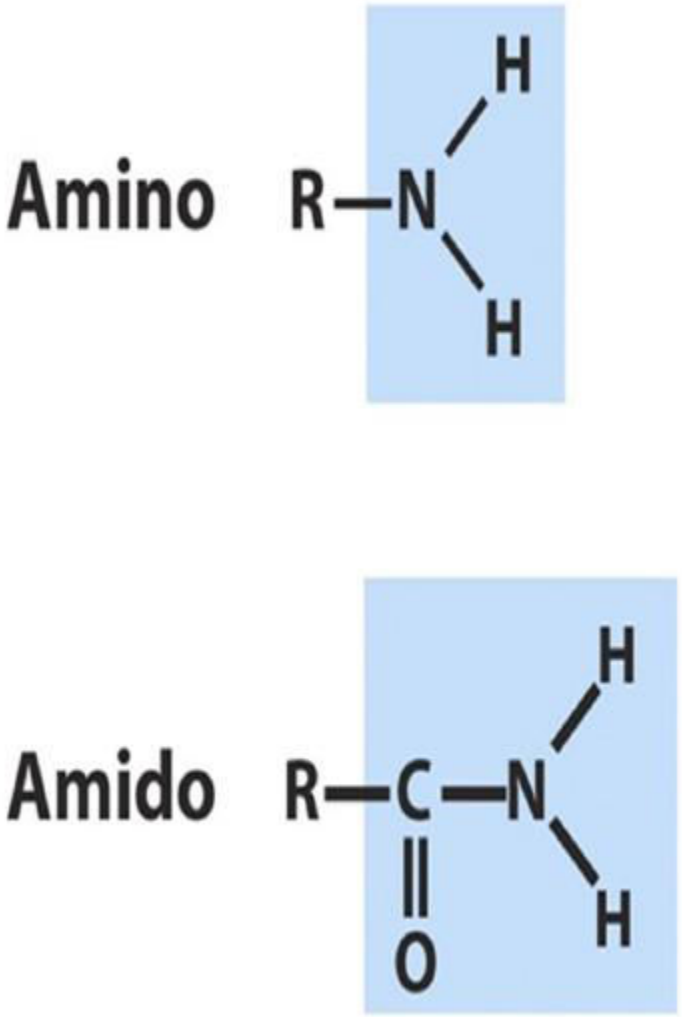
PEPTIDE BOND

- We now turn to polymers of amino acids, the peptides and proteins
- Biologically occurring polypeptides range in size from small to very large, consisting of two or three to thousands of linked amino acid residues

PEPTIDE BOND

- Two amino acid molecules can be covalently joined through **an amide linkage,** termed **a peptide bond**

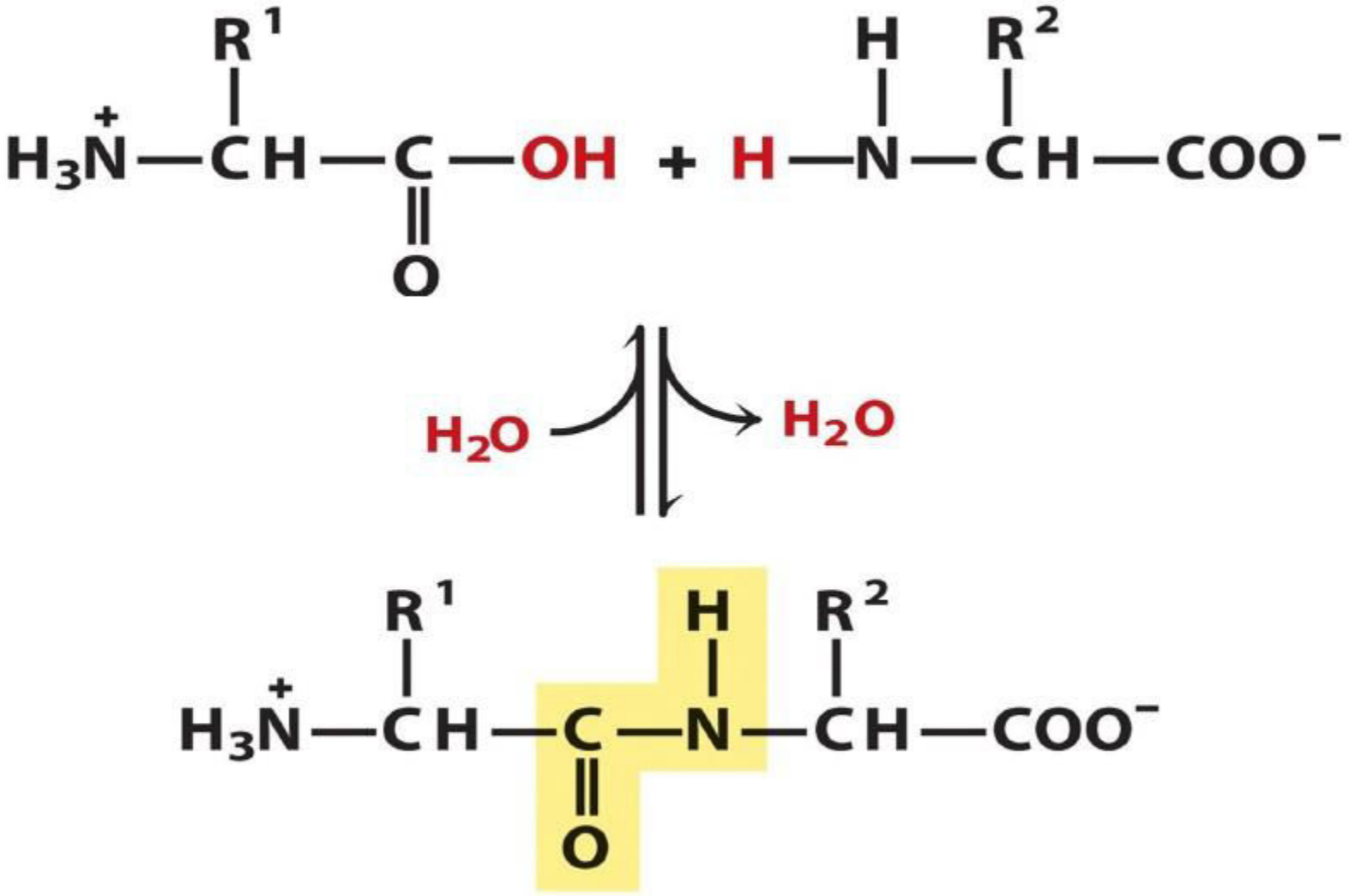
PEPTIDE BOND



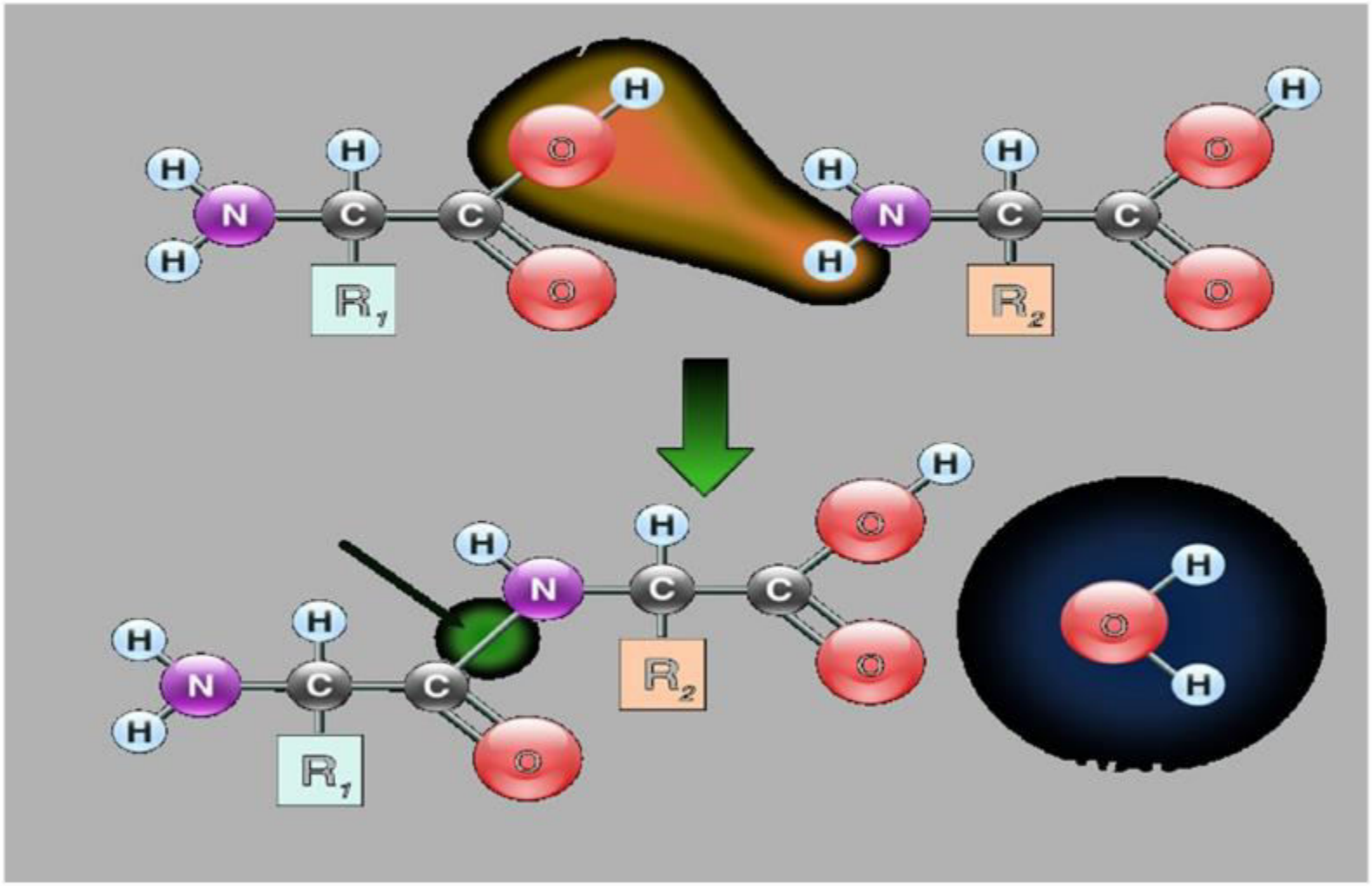
PEPTIDE BOND

- This linkage is formed by removal of the elements of water from the α-carboxyl group of one amino acid and the α-amino group of another

PEPTIDE BOND



PEPTIDE BOND



PEPTIDE BOND

- Two configurations are possible for a planar peptide bond
- In the trans configuration, the two -carbon atoms are on opposite sides of the peptide bond

PEPTIDE BOND

- In the cis con-, these groups are on the same side of the peptide bond.
- *Almost all peptide bonds in proteins are trans*

PEPTIDE BOND

- This preference for trans over cis can be explained by the fact that steric clashes between groups attached to the -carbon atoms hinder formation of the cis form but do not occur in the trans configuration

PEPTIDE BOND

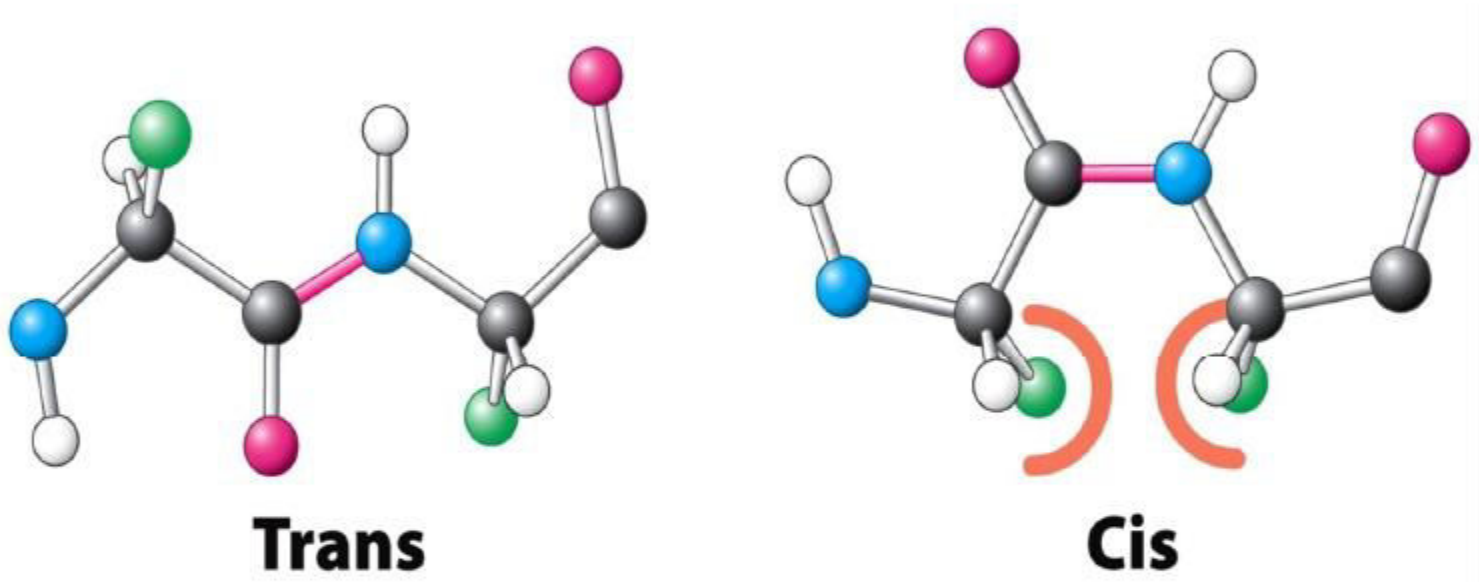


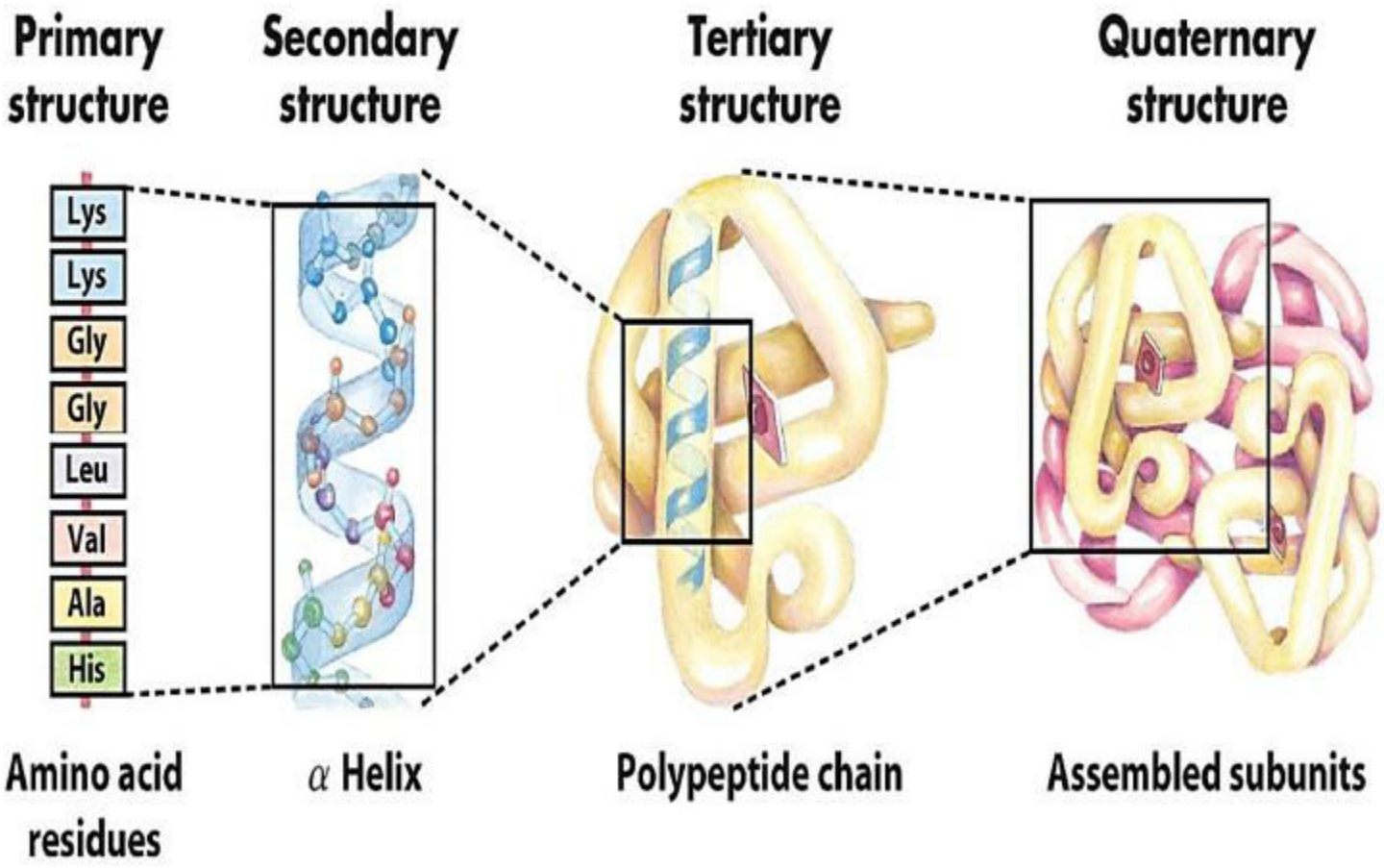
Figure 2.20
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Structural Hierarchy of Proteins

The Structure of Proteins

- Four levels of protein structure are commonly defined

Structural Hierarchy of Proteins



Structural Hierarchy of Proteins

The Primary Structure

- A description of all covalent bonds (mainly **peptide bonds** and **disulfide bonds**) linking amino acid residues in a polypeptide chain is its primary structure

Structural Hierarchy of Proteins

- The most important element of primary structure is the **sequence** of amino acid residues

Structural Hierarchy of Proteins

Secondary structure

- Secondary structure refers to **particularly stable arrangements** of amino acid residues giving rise to recurring structural patterns

Structural Hierarchy of Proteins

Tertiary structure

- Tertiary structure describes all aspects of the **three-dimensional** folding of a polypeptide

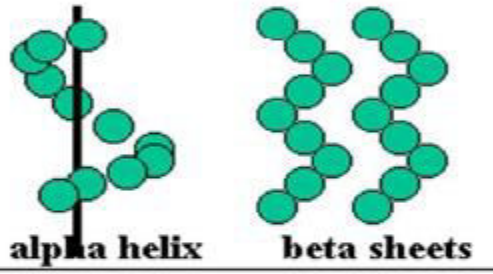
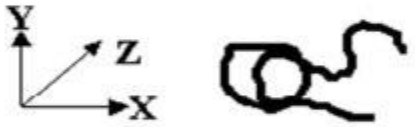
Structural Hierarchy of Proteins

Quaternary structure

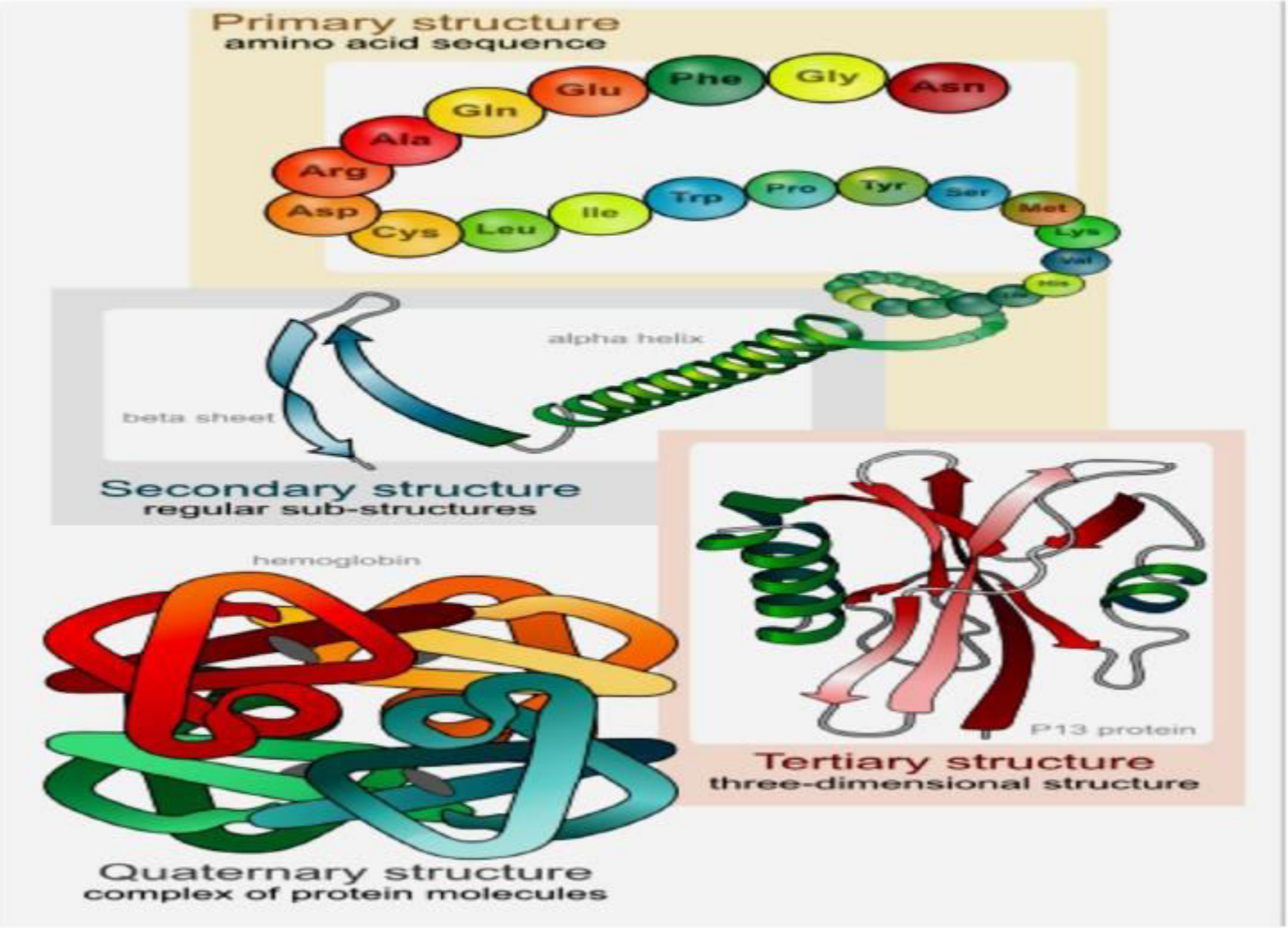
- When a protein has **two or more polypeptide** subunits, their arrangement in space is referred to as quaternary structure

Structural Hierarchy of Proteins

Protein Structure(Summary)

•Primary	The amino acid sequence	Glu-Arg-Phe-Gly
•Secondary	Characteristic structures that occur in many proteins (E.g. alpha helix , beta sheets)	
•Tertiary	Three dimensional structure of proteins	
•Quaternary	Three dimensional structure of proteins composed of multiple subunits	

Structural Hierarchy of Proteins



The Primary Structure

- The primary structure of a protein refers to the **linear sequence** of amino acids in the polypeptide chain.

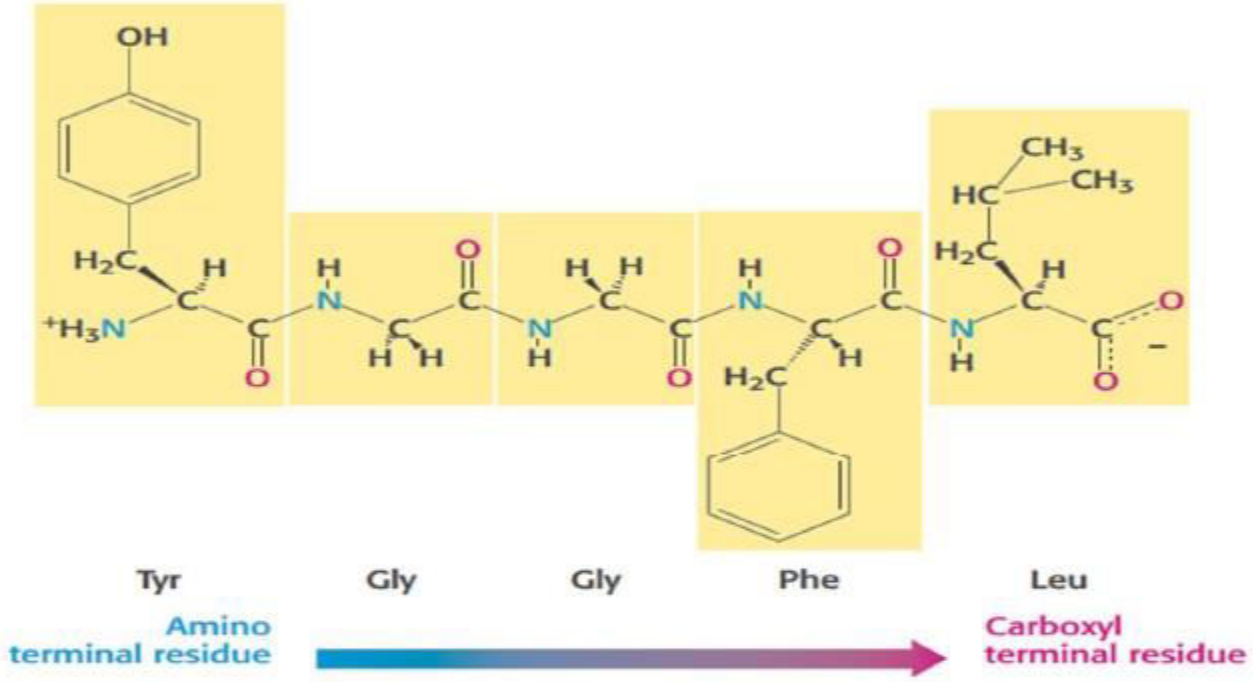
The Primary Structure

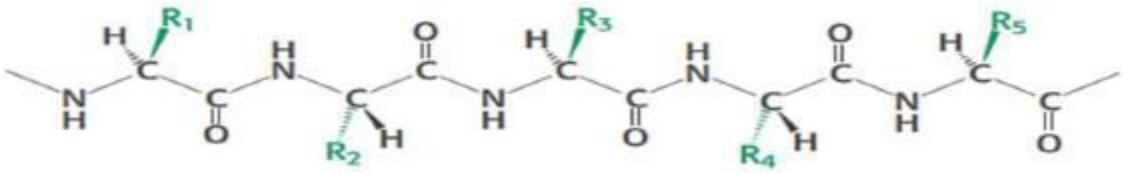
- The primary structure is held together by covalent bonds such as peptide bonds, which are made during the process of protein biosynthesis or translation.

The Primary Structure

- The two ends of the polypeptide chain are referred to as the **carboxyl terminus** (C-terminus) and the **amino terminus** (N-terminus) based on the nature of the free group on each extremity.

The Primary Structure

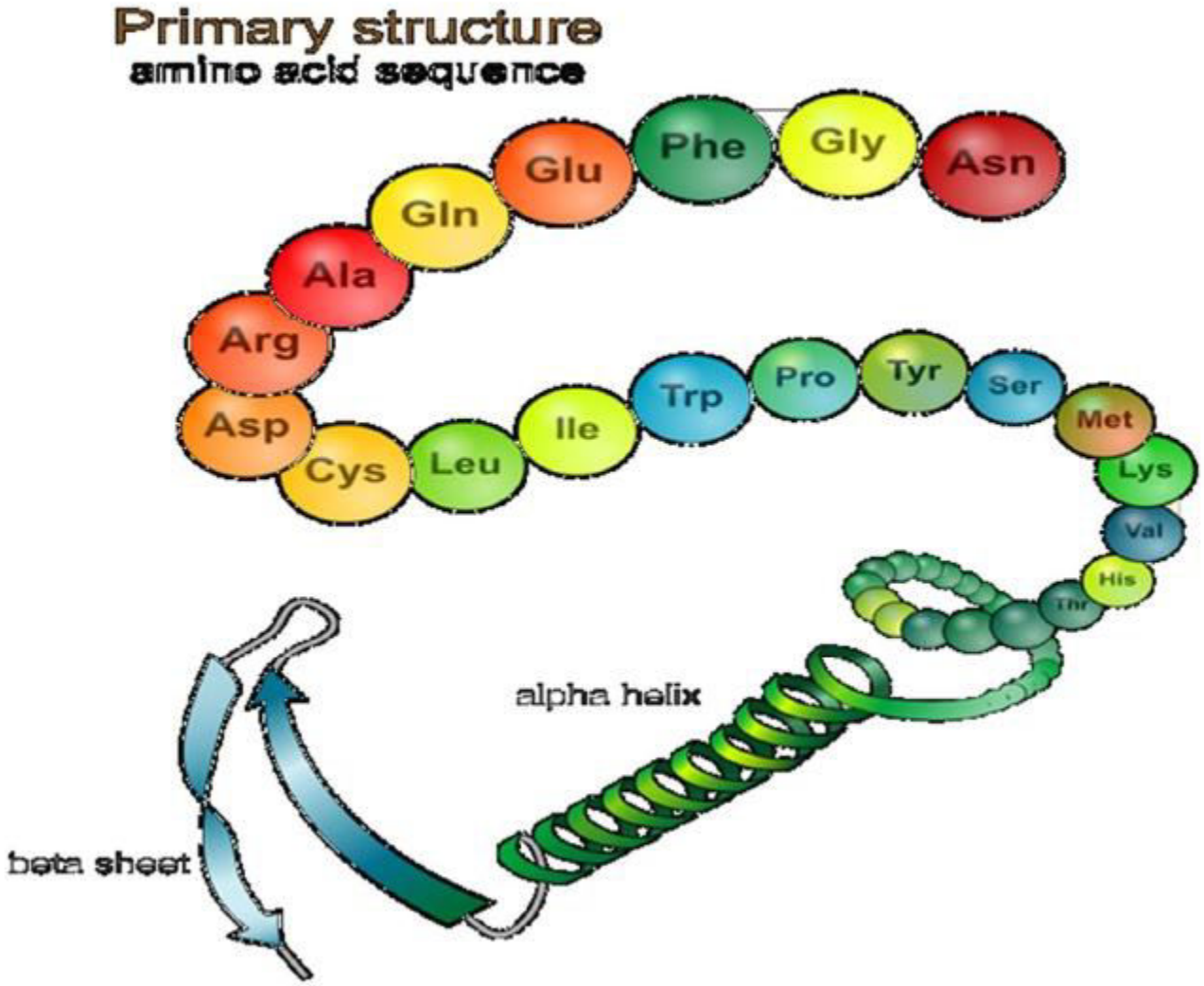




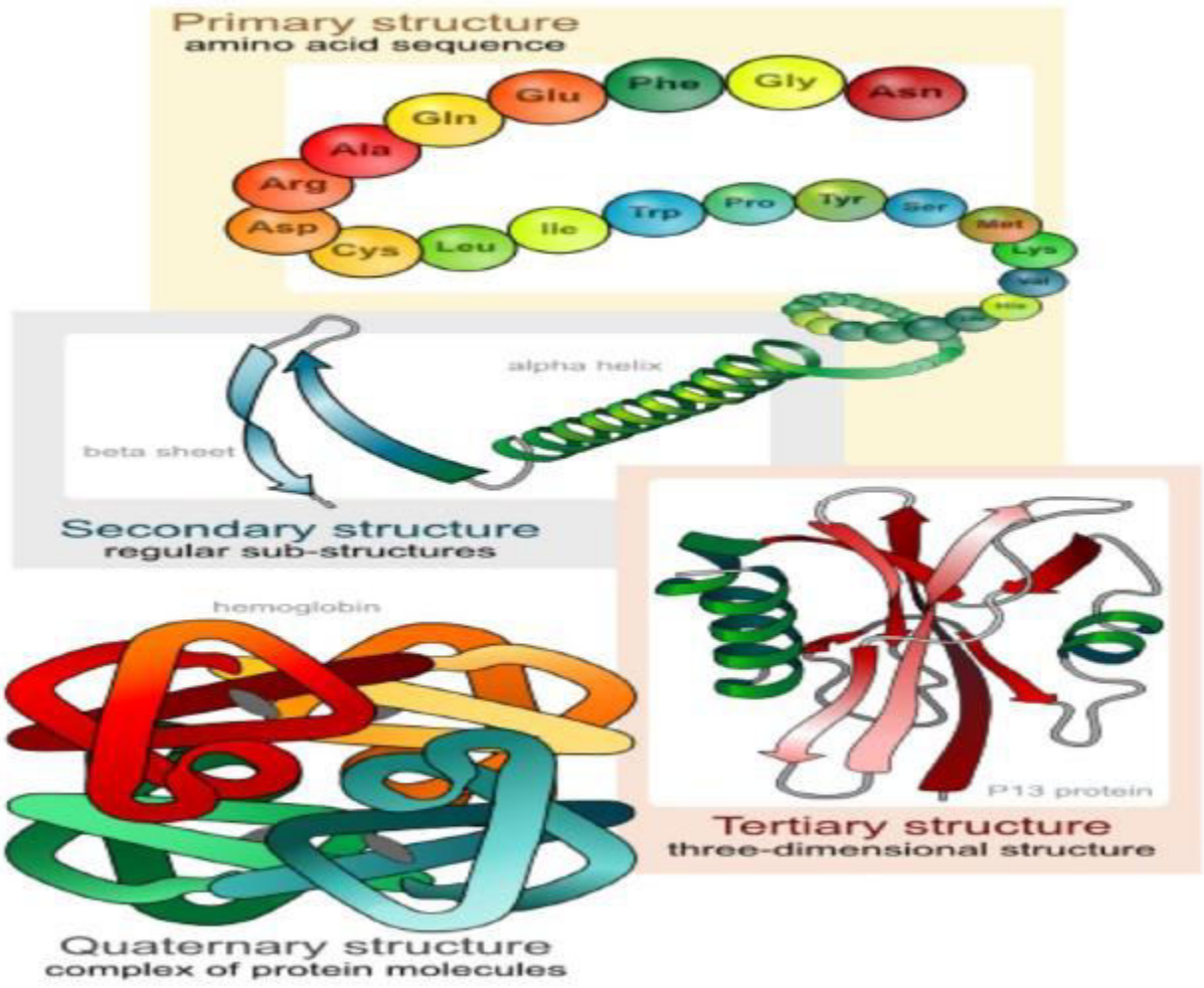
The Primary Structure

- The primary structure of a protein determines **how it folds up** into its unique three-dimensional structure, and this in turn determines the function of the protein

The Primary Structure



The Primary Structure



The Primary Structure

- The function of a protein therefore depends on its amino acid sequence

The Primary Structure

- Proteins have unique amino acid sequences, that are specified by genes.
- Each of the 20 amino acids incorporated in the proteins is encoded by one or more *specific* sequences of three nucleotides in DNA or RNA.

The Primary Structure

- Understanding of the primary structure of proteins is important because many genetic diseases result in proteins with abnormal amino acid sequences which cause loss or impairment of normal function.

The Primary Structure

- The defect can range from a single change in the amino acid sequence (as in **sickle cell anemia**) to deletion of a larger portion of the polypeptide chain (as in most cases of **Duchenne** muscular dystrophy)

The Primary Structure

- In Sickle cell anemia a **Glutamate** residue is replaced by **Valine** at position 6 of beta chain of Hemoglobin, resulting in decreased functioning and increased fragility of Hb.

The Primary Structure

- Duchenne Muscular dystrophy results from a large segment of gene deletion in X chromosome, resulting in small truncated **dystrophin** protein.

The Primary Structure

- Thus we know that if the primary structure is altered, the function of the protein may also be changed

END

The Secondary Structure

- Secondary structure, the folding of short (3- to 30-residue), contiguous segments of polypeptide into **geometrically ordered units**

The Secondary Structure

- The term secondary structure refers to any chosen segment of a polypeptide chain and describes the local spatial arrangement of its main-chain atoms, without regard to its relationship to other segments

The Secondary Structure

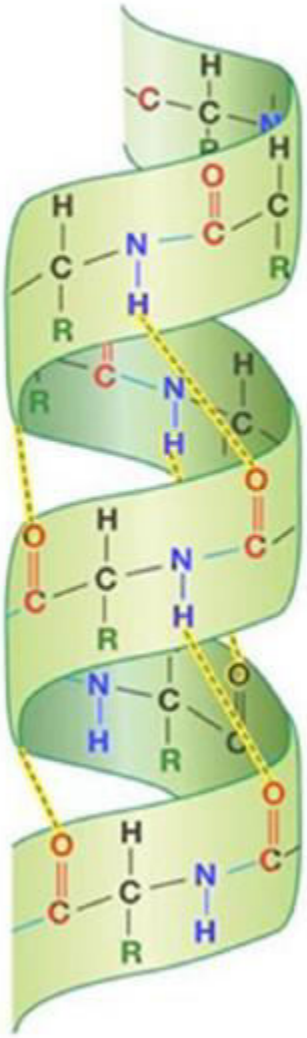
- There are a few types of secondary structure that are particularly stable and occur widely in proteins
- The most prominent are the **α helix and β conformations**

The Secondary Structure

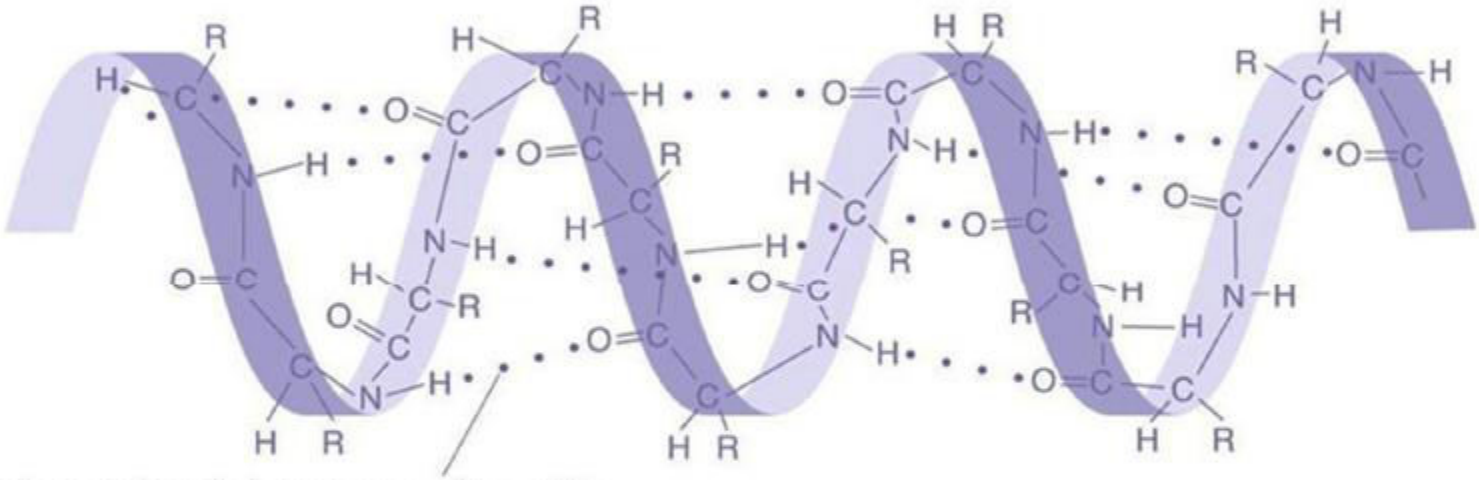
The α Helix is a common protein secondary structure

- The simplest arrangement the polypeptide chain can assume, given its rigid peptide bonds (but free rotation around its other, single bonds), is a helical structure, called the α helix

The Secondary Structure



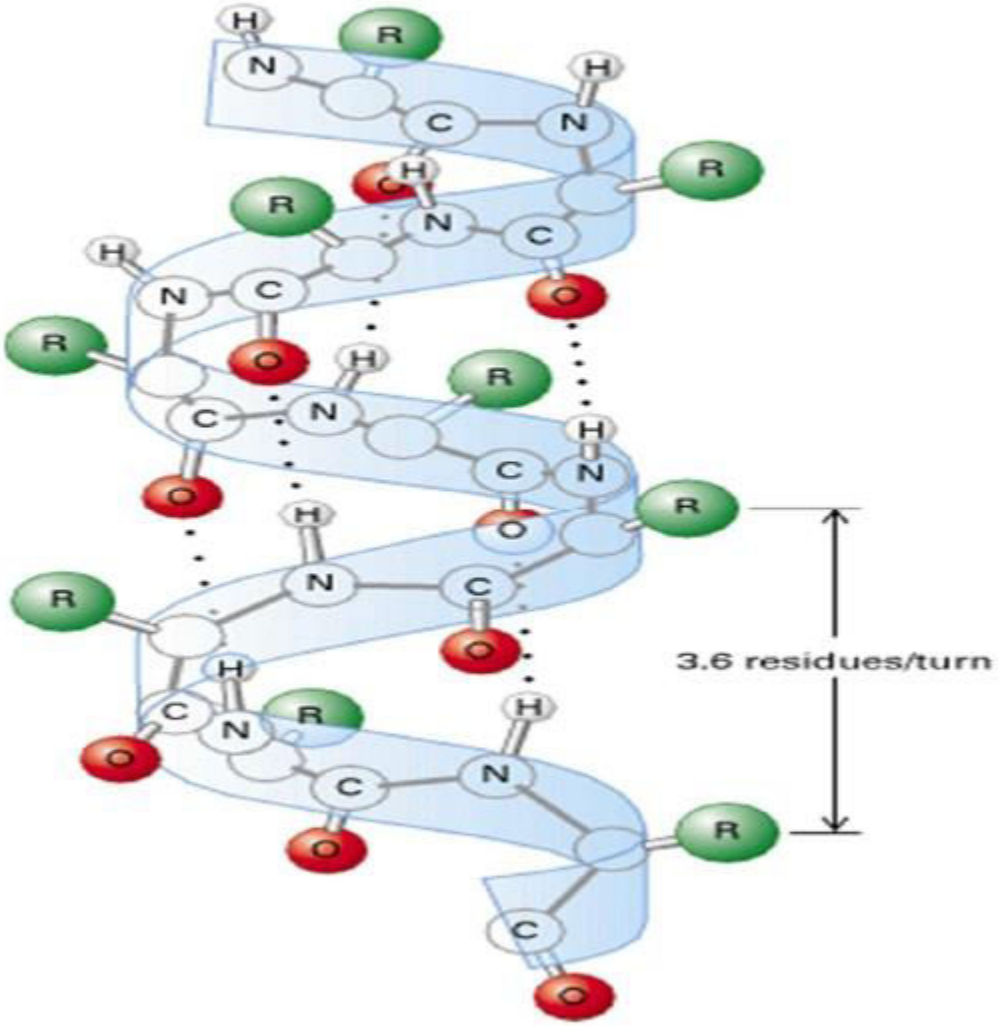
The Secondary Structure



Hydrogen bonds between amino acids at different locations in polypeptide chain

α helix

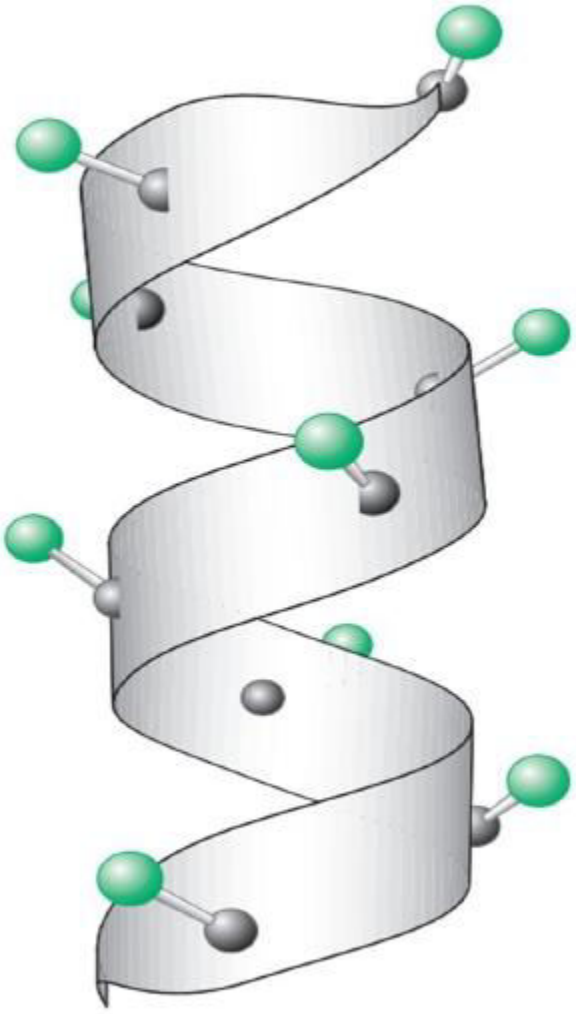
The Secondary Structure

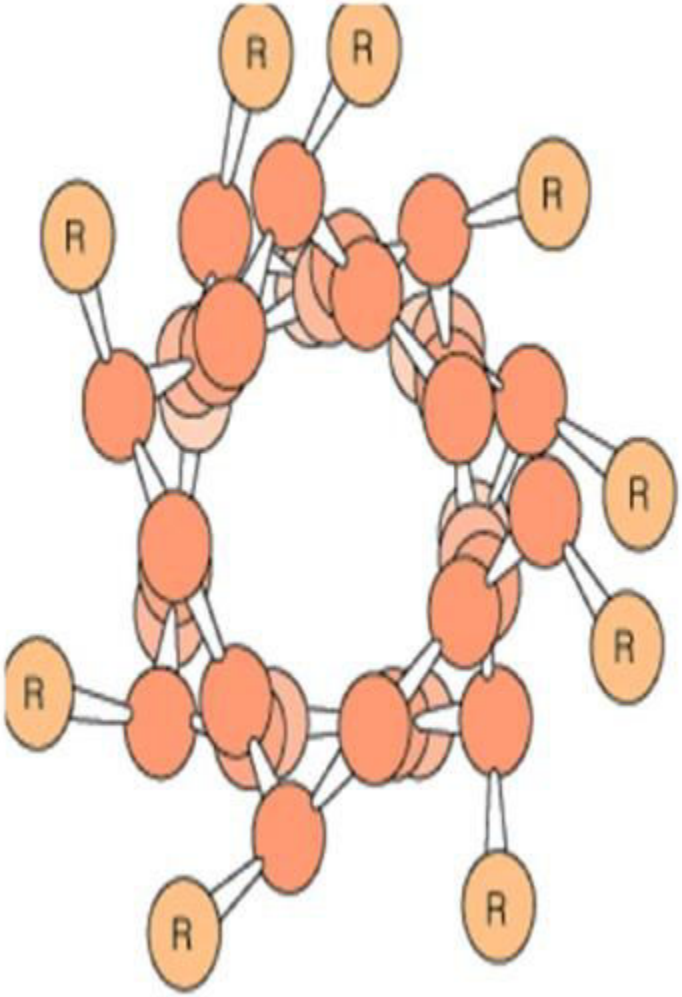


The Secondary Structure

- Within the α helix, every peptide bond (except those close to each end of the helix) participates in such hydrogen bonding

The Secondary Structure





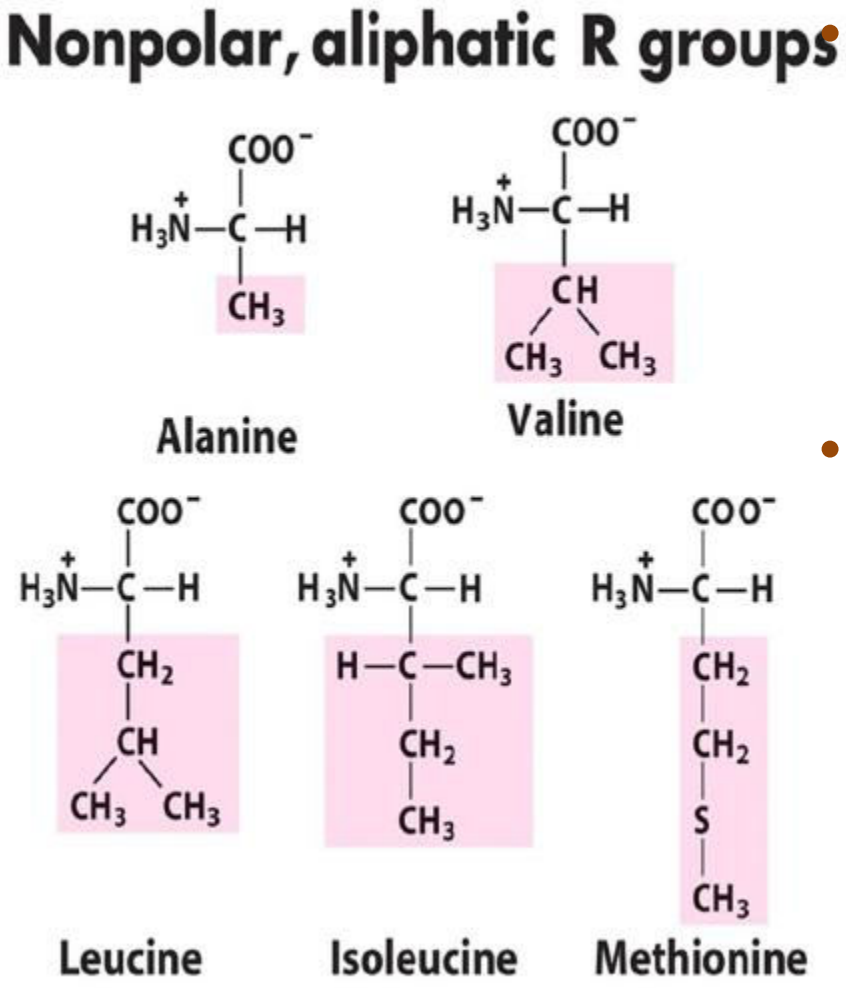
The Secondary Structure

- Amino acid residues included in alpha helix have an **intrinsic propensity** to form an alpha helix, reflecting the properties of the R group.

The Secondary Structure

- Amino acids most commonly found in alpha helices are **non-polar** with aliphatic side chains
- **Alanine** shows the greatest tendency to form α helices in most experimental model systems.

The Secondary Structure

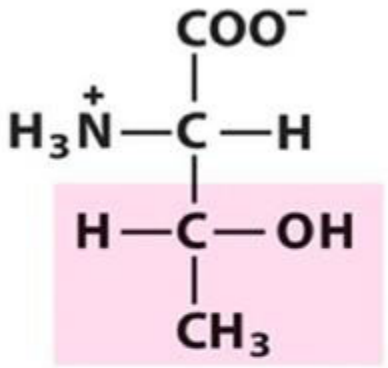


- Amino acids most commonly found in alpha helices are **non-polar** with aliphatic side chains
- **Alanine** shows the greatest tendency to form α helices in most experimental model systems.

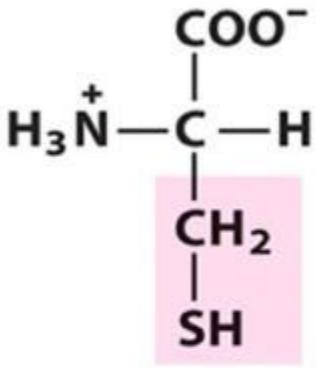
The Secondary Structure

- The bulk and shape of **Ser, Thr**, and **Cys** residues can destabilize an α helix if they are close together in the chain.

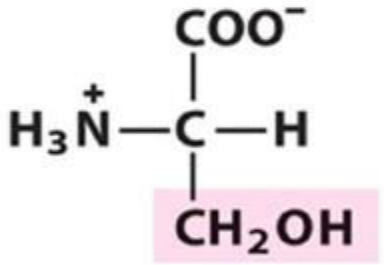
The Secondary Structure



Threonine



Cysteine



Serine

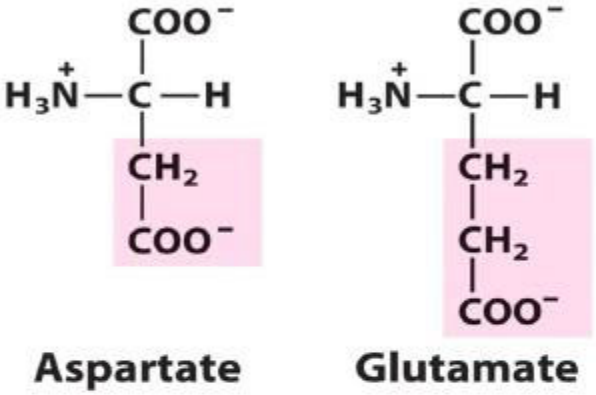
- The bulk and shape of **Ser, Thr**, and **Cys** residues can destabilize an α helix if they are close together in the chain.

The Secondary Structure

- Amino acids with charged R groups such as **Aspartate, Lysine** residues can also disrupt the helix.

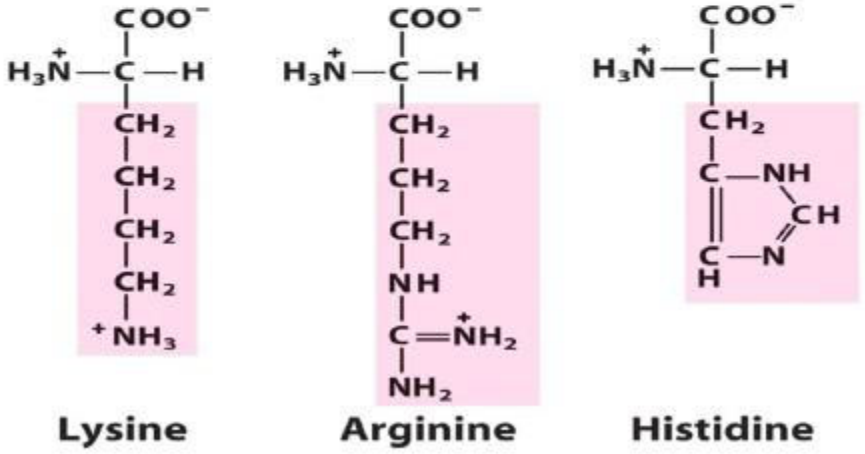
The Secondary Structure

Negatively charged R groups

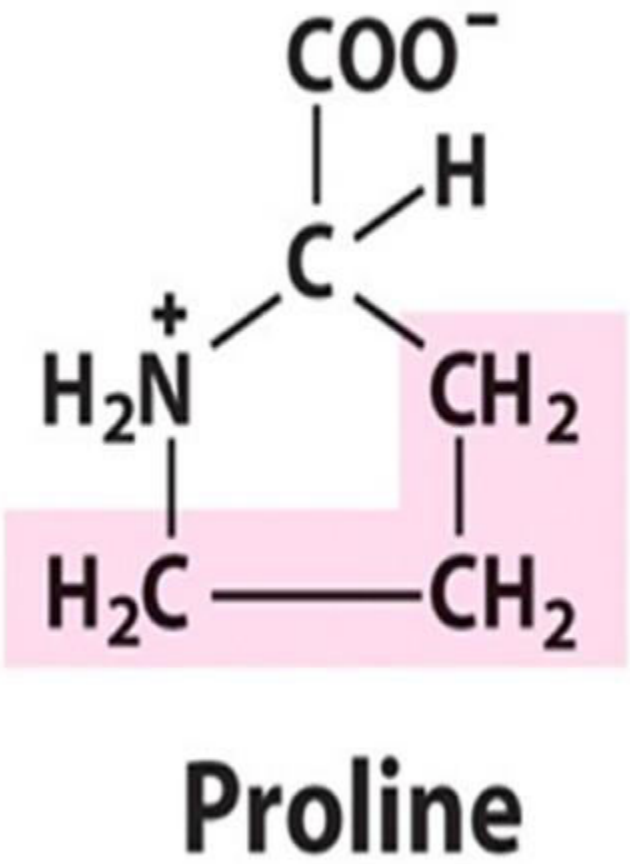


- Amino acids with charged R groups such as **Aspartate, Lysine** residues can also disrupt the helix.

Positively charged R groups



The Secondary Structure



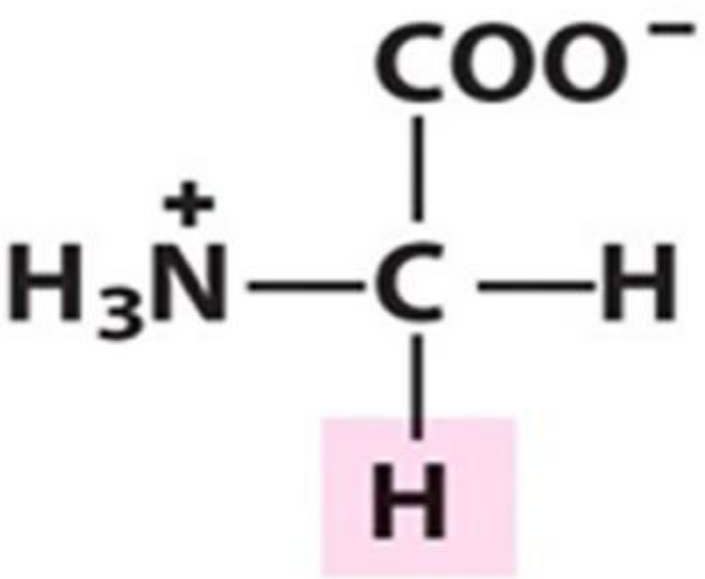
Proline

- In **proline**, the nitrogen atom is part of a rigid ring, and rotation about the N-C bond is not possible
- Thus, a proline residue introduces a destabilizing kink in an α helix

The Secondary Structure

- **Glycine** occurs infrequently in α helices for a different reason.
- It has more conformational flexibility than the other amino acid residues

The Secondary Structure



Glycine

Polymers of glycine tend to take up coiled structures quite different from an α helix

The Secondary Structure

- **A very diverse group of proteins contains α-helices**
- The alpha-helical content of proteins ranges widely, from nearly none to almost 100%

The Secondary Structure

- For example, the **keratins** are a family of closely related, fibrous proteins whose structure is nearly entirely α-helical.
- They are a major component of tissues such as hair, nails, hoofs and skin.

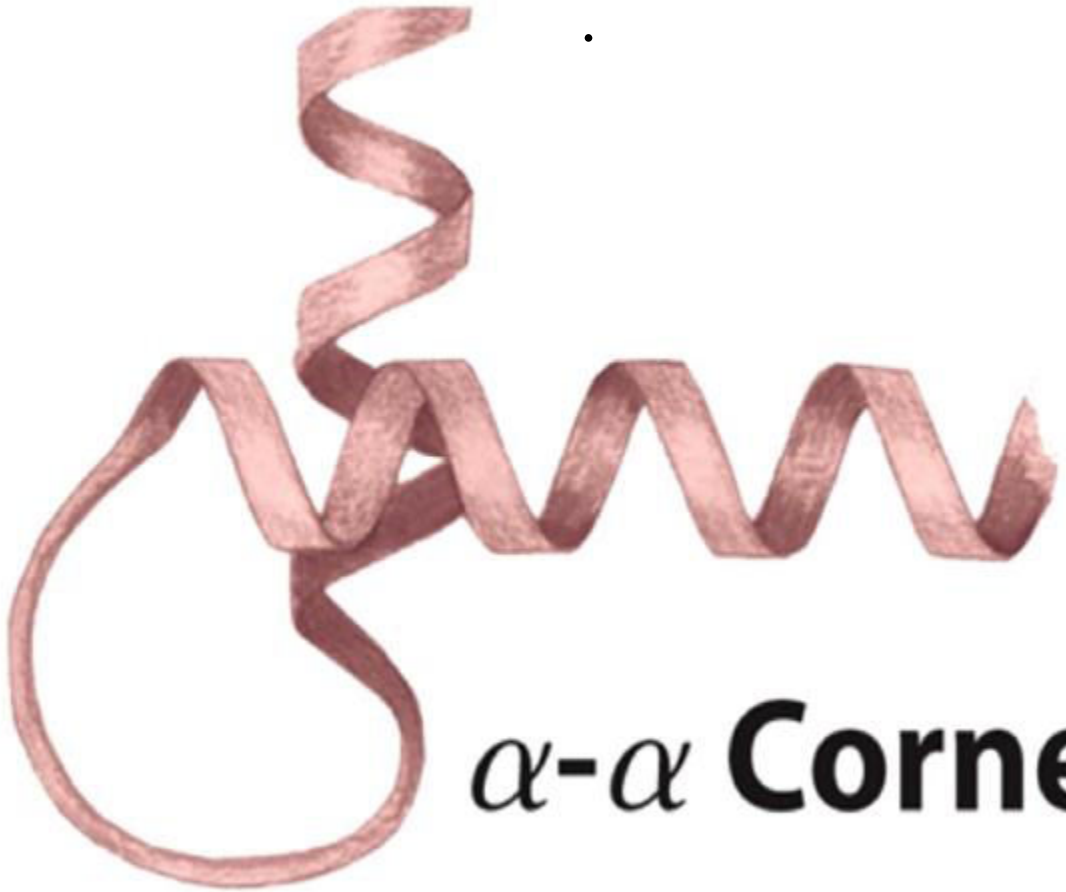
The Secondary Structure

- In contrast to keratin, myoglobin, whose structure is also highly α-helical, is a globular, flexible molecule.

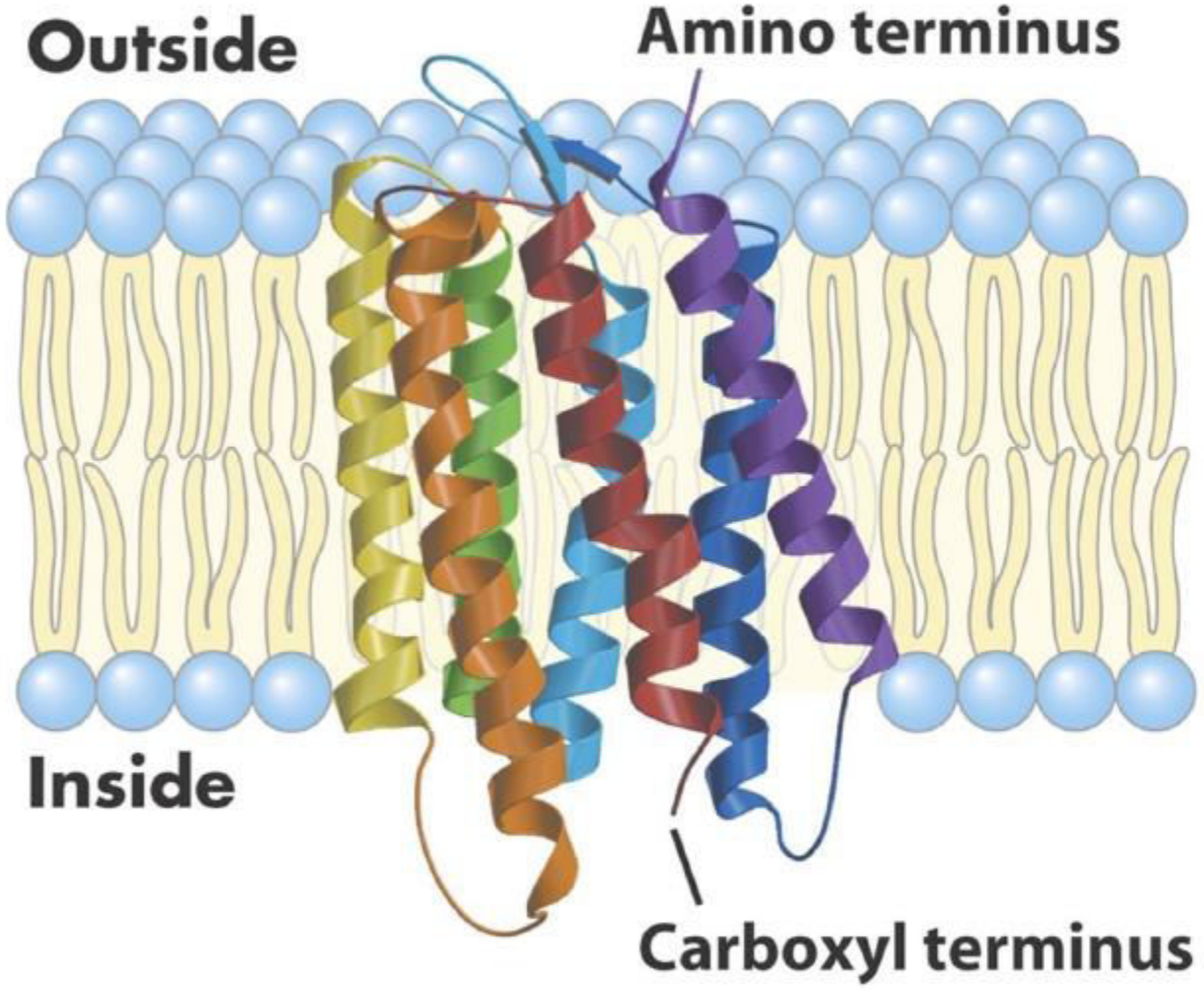
The Secondary Structure

- Nearly all trans membrane proteins contain alpha helices in their membrane spanning domains.

The Secondary Structure

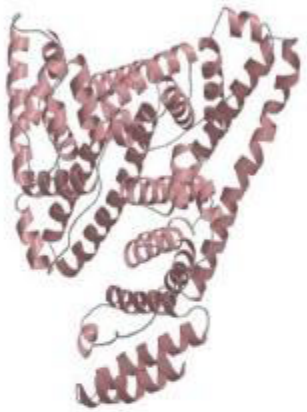


The Secondary Structure



The Secondary Structure

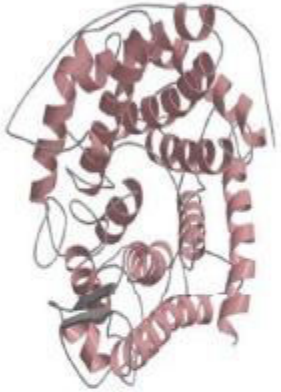
All α



1AO6
Serum albumin
Serum albumin
Serum albumin
Serum albumin
Human (*Homo sapiens*)



1BCF
Ferritin-like
Ferritin-like
Ferritin
Bacterioferritin (cytochrome b_1)
Escherichia coli



1GAI
α/α toroid
Glycosyltransferases of the superhelical fold
Glucoamylase
Glucoamylase
Aspergillus awamori, variant x100



1ENH
DNA-binding 3-helical bundle
Homeodomain-like
Homeodomain
engrailed Homeodomain
Drosophila melanogaster

The Secondary Structure

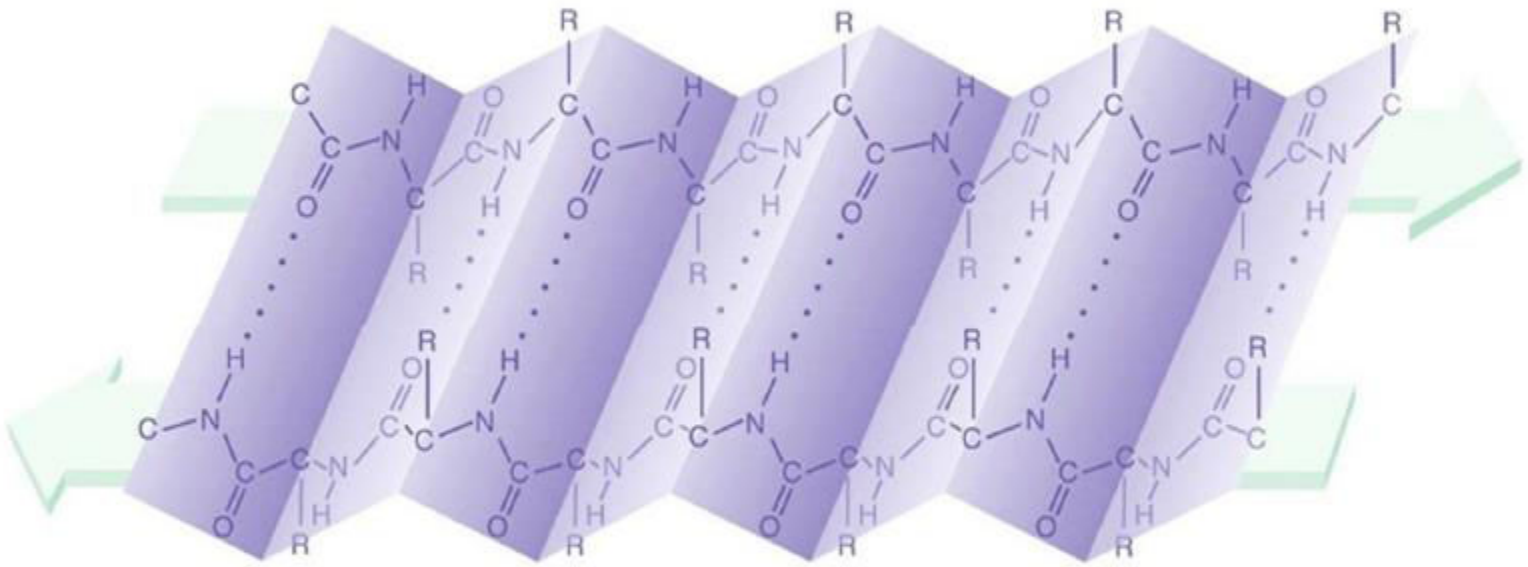
The β Conformation organizes polypeptide Chains in β sheets

β Sheets are a second type of regular secondary structure that maximizes hydrogen bonding between the peptide backbones.

The Secondary Structure

- The backbone of the polypeptide chain is extended into a zigzag rather than helical structure
- The zigzag polypeptide chains can be arranged side by side to form a structure resembling a series of pleats

The Secondary Structure



Pleated sheet

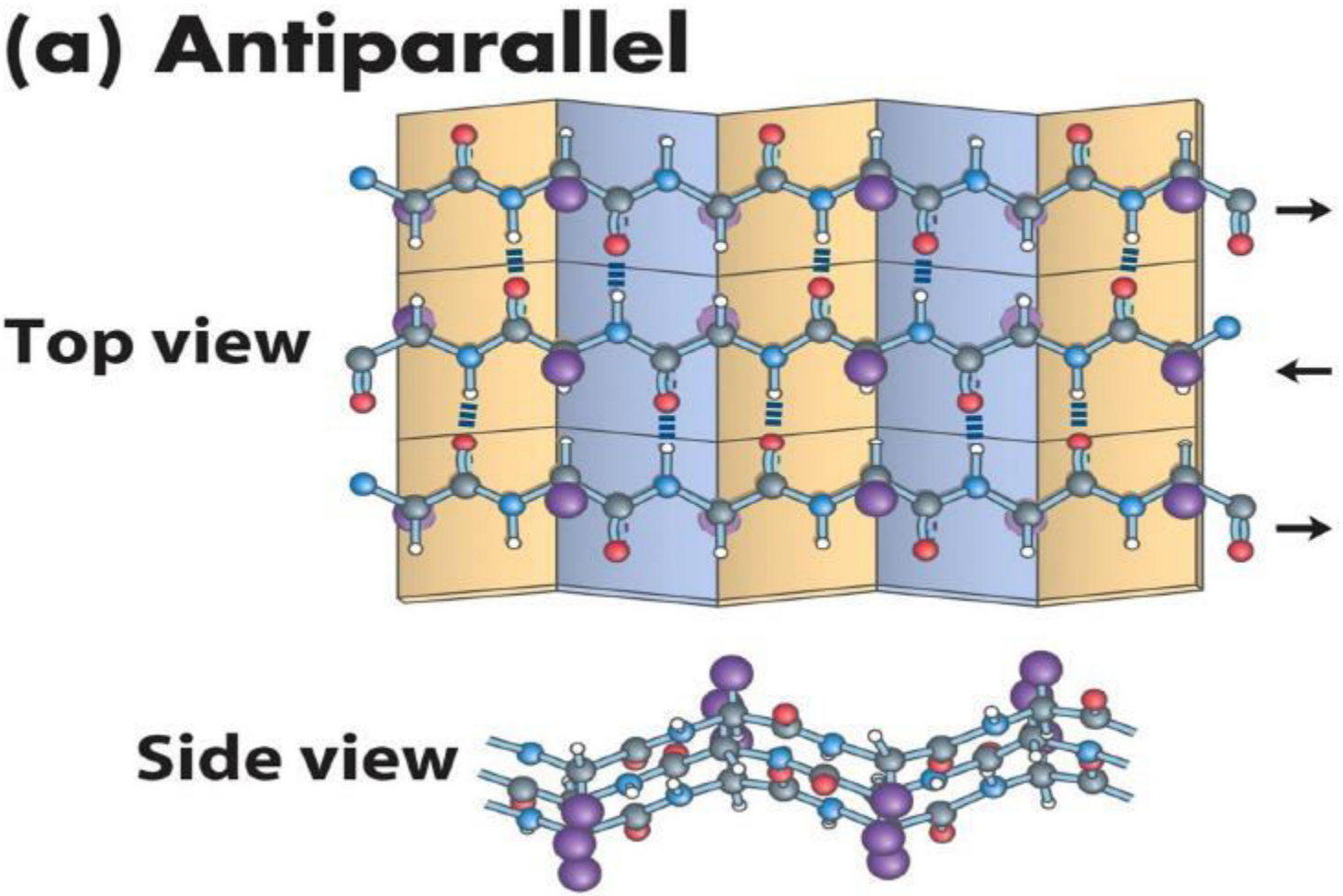
The Secondary Structure

- In this arrangement called a **β sheet**, hydrogen bonds form between adjacent segments of polypeptide chain

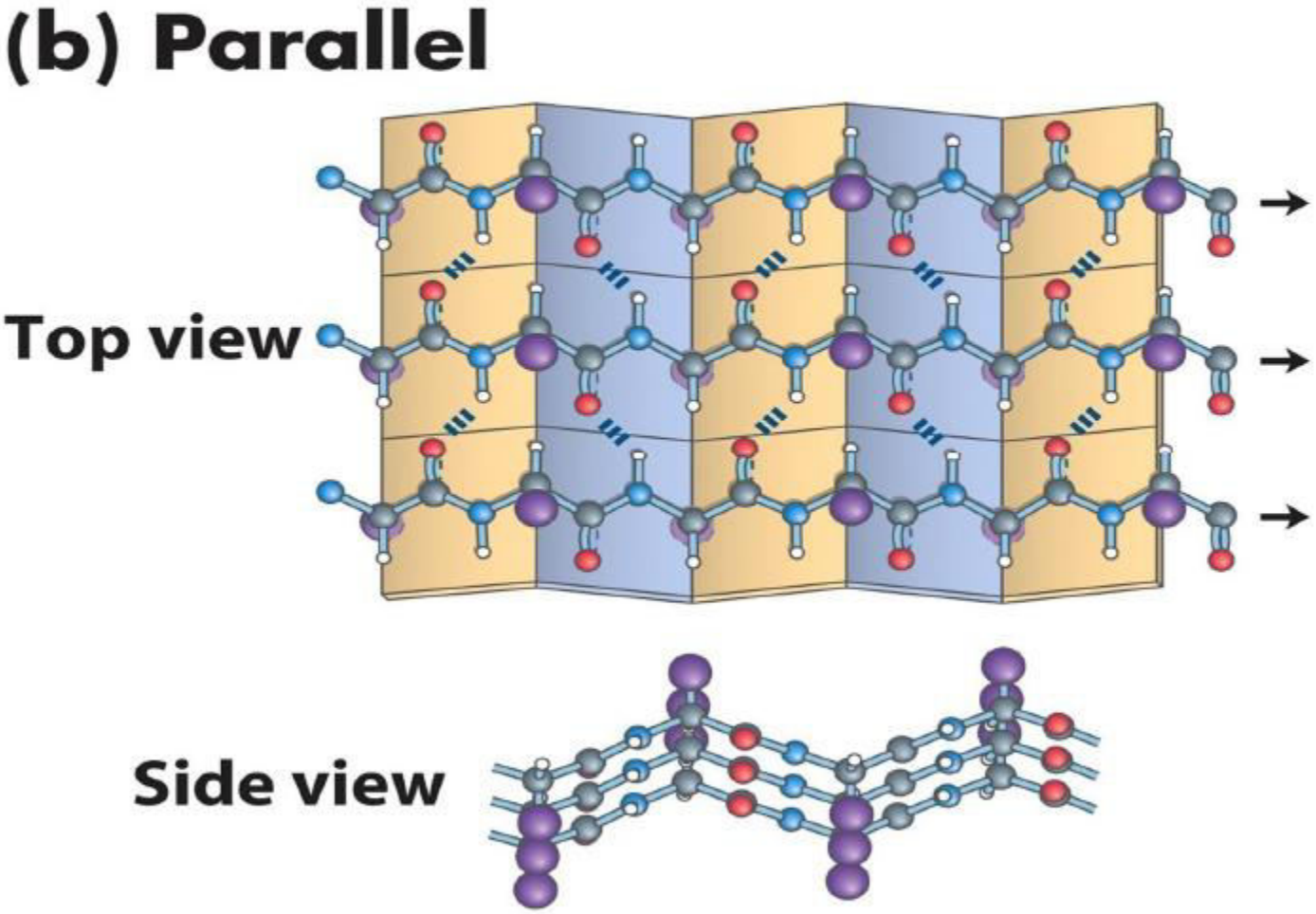
The Secondary Structure

- The -pleated sheet is described as **parallel** if the polypeptide strands run in the same direction (as defined by their amino and carboxy terminals) and **anti-parallel** if they run in opposite direction

The Secondary Structure



The Secondary Structure



The Secondary Structure

- **Antiparallel** strands are often the same polypeptide chain **folded back** on itself, with simple hairpin turns or long runs of polypeptide chain connecting the strands

The Secondary Structure

- The amino acid side chains(R groups) of each polypeptide strand alternate between extending above and below the plane of the β-sheet.
- Most sheets are not perfectly flat but tend to have a twist.

The Secondary Structure

- The individual segments that form a **β sheet** are usually nearby on the polypeptide chain, but can also be quite distant from each other in the linear sequence of the peptide and may even be in different polypeptide chains.

The Secondary Structure

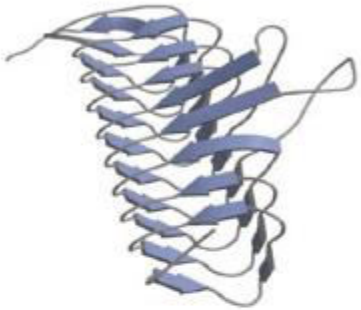
- Clusters of twisted strands of sheet form the core of many **globular proteins**.

The Secondary Structure

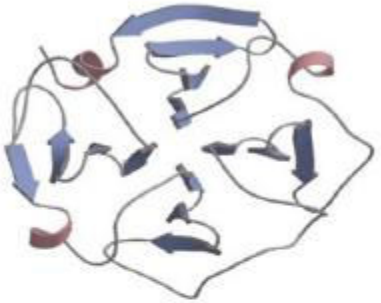
All β



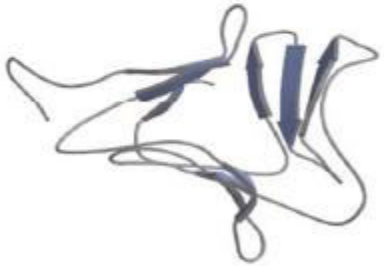
1HOE
α -Amylase inhibitor
α -Amylase inhibitor
α -Amylase inhibitor
HOE-467A
Streptomyces tendae 4158



1LXA
Single-stranded left-handed β helix
Trimeric LpxA-like enzymes
UDP *N*-acetylglucosamine acyltransferase
UDP *N*-acetylglucosamine acyltransferase
Escherichia coli



1PEX
Four-bladed β propeller
Hemopexin-like domain
Hemopexin-like domain
Collagenase-3 (MMP-13), carboxyl-terminal domain
Human (*Homo sapiens*)



1JPC
β-Prism II
α-D-Mannose-specific plant lectins
α-D-Mannose-specific plant lectins
Lectin (agglutinin)
Snowdrop (*Galanthus nivalis*)

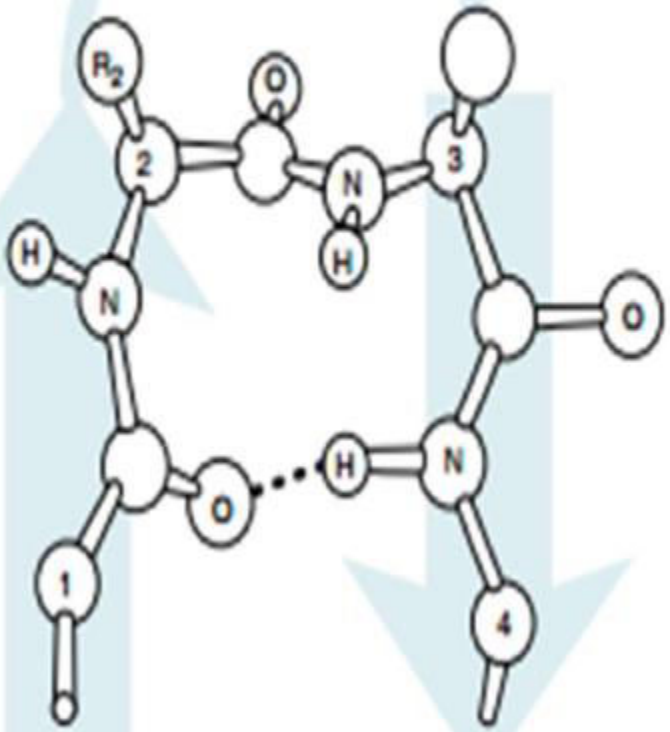


1CD8
Immunoglobulin-like β sandwich
Immunoglobulin
Antibody variable domain-like
CD8
Human (*Homo sapiens*)

The Secondary Structure

- In contrast to regular repeating element in the twist of an alpha Helix or a pleat of Beta sheet, **bends, loops,** and **turns are** non- regular secondary structures that do not have a repeating element.

The Secondary Structure

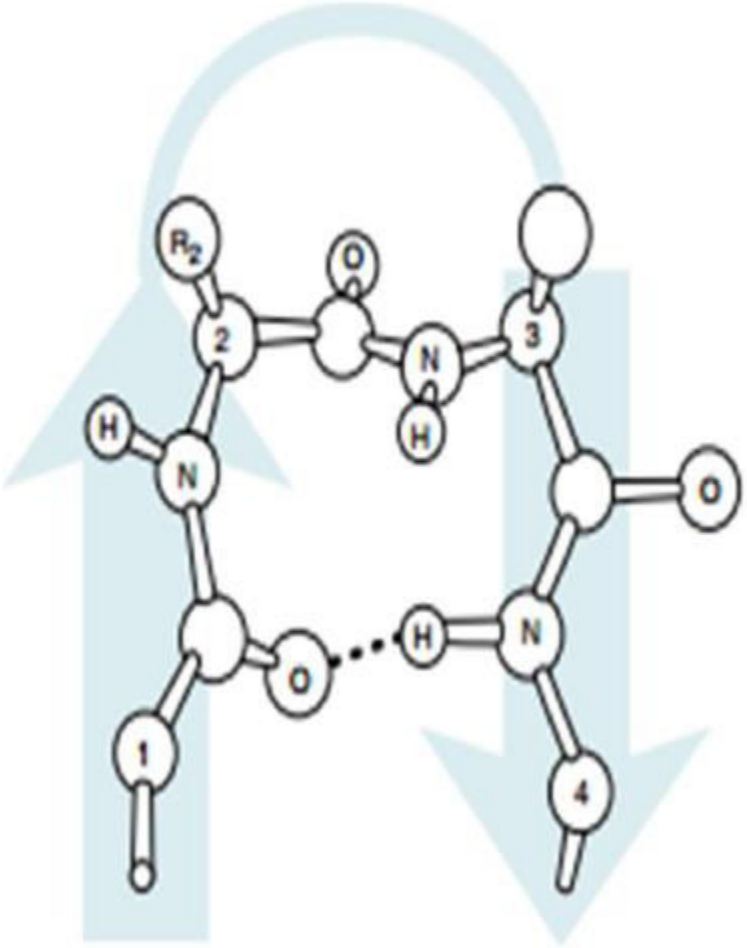


- **Beta-bends** are short regions usually involving four successive amino acid residues.
- They often connect strands of antiparallel beta-sheets.

The Secondary Structure

- β-Bends reverse the direction of a polypeptide chain, helping it form a compact, globular shape.
- The structure is a 180 degree turn involving four amino acid residues

The Secondary Structure

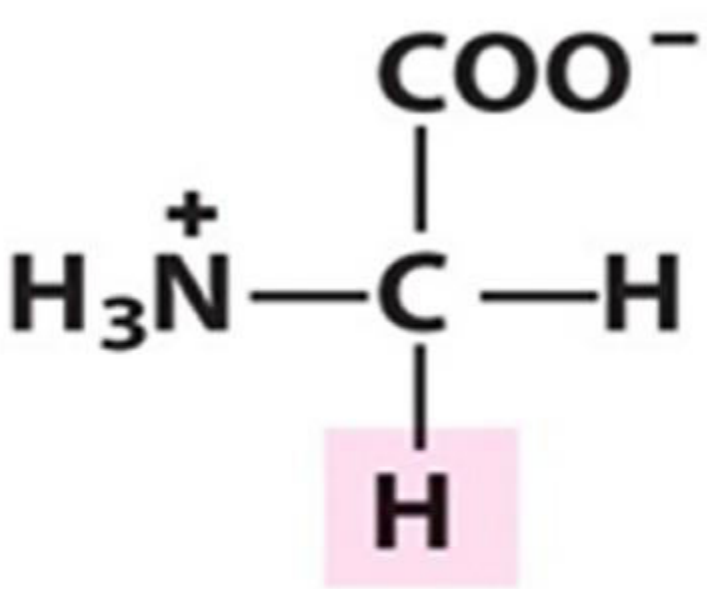


- The carbonyl oxygen of the first residue forms a hydrogen bond with the amino-group hydrogen of the fourth residue.

The Secondary Structure

- β-Bends were given this name because they often connect successive strands of antiparallel β-sheets

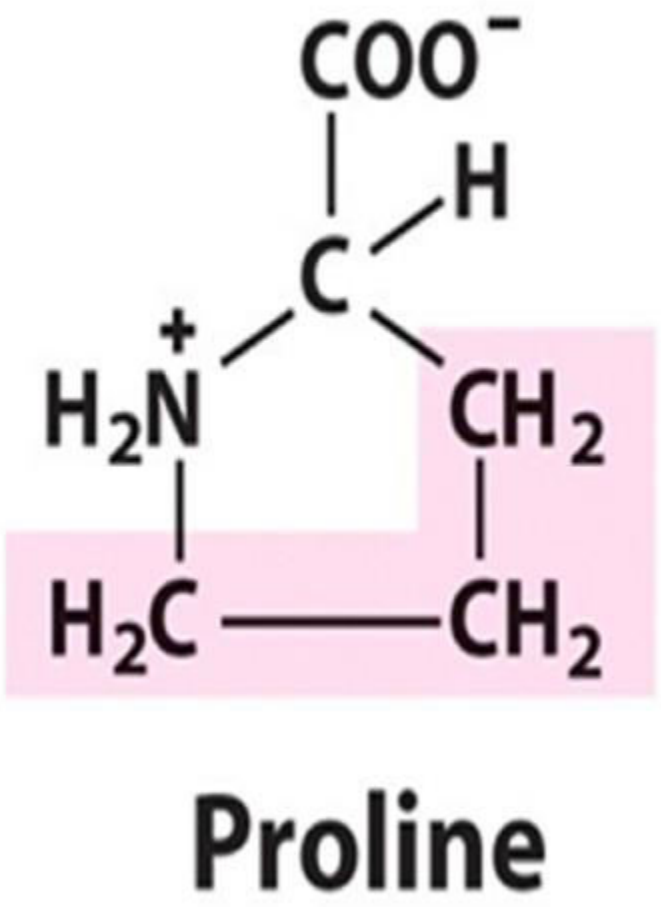
The Secondary Structure



Glycine

- Glycine and Proline residues often occur in β turns
- Glycine because it is small and flexible

The Secondary Structure



Proline

- Proline because peptide bonds involving the imino nitrogen readily assume the cis configuration, a form that is particularly amenable to a tight turn

The Secondary Structure

- A **motif** also called a **super secondary structure** is simply a recognizable folding pattern involving two or more elements of secondary structure and the connection(s) between them.

The Secondary Structure

- Structural motifs are intermediate between secondary and tertiary structures.

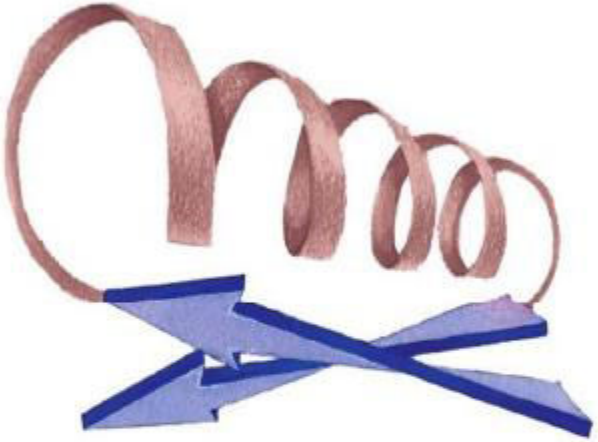
The Secondary Structure

- A motif can be very simple, such as two elements of secondary structure folded against each other, and represent only a small part of a protein
- An example is a **β-α-β loop**.

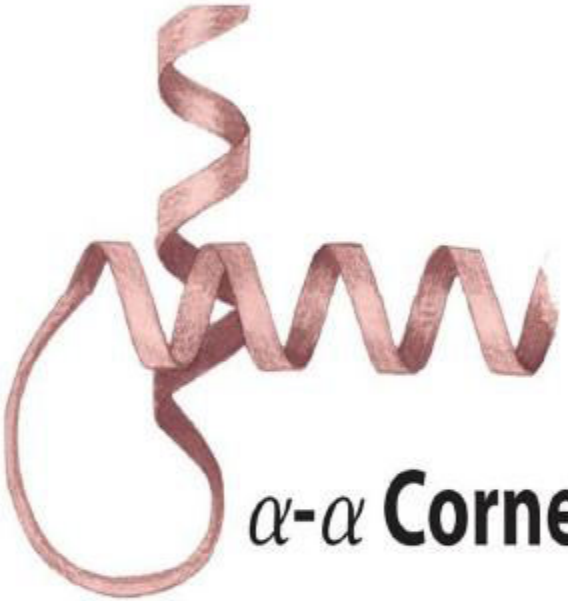
The Secondary Structure

- A motif can also be a very elaborate structure involving scores of protein segments folded together, such as the **β barrel**

The Secondary Structure



β-α-β Loop

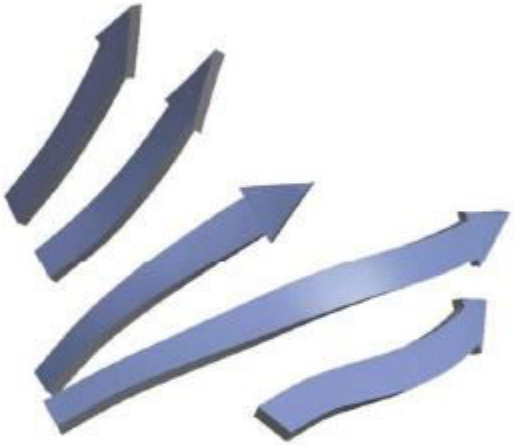


α-α Corner

The Secondary Structure



β Barrel



Twisted β sheet

The Tertiary Structure

The tertiary structure

- The tertiary structure of a protein is the folding pattern of the secondary structural elements into a three-dimensional conformation.

The Tertiary Structure

- It indicates, in three-dimensional space, how secondary structural features—helices, sheets, bends, turns, and loops—assemble to form domains and how these domains relate spatially to one another.

The Tertiary Structure

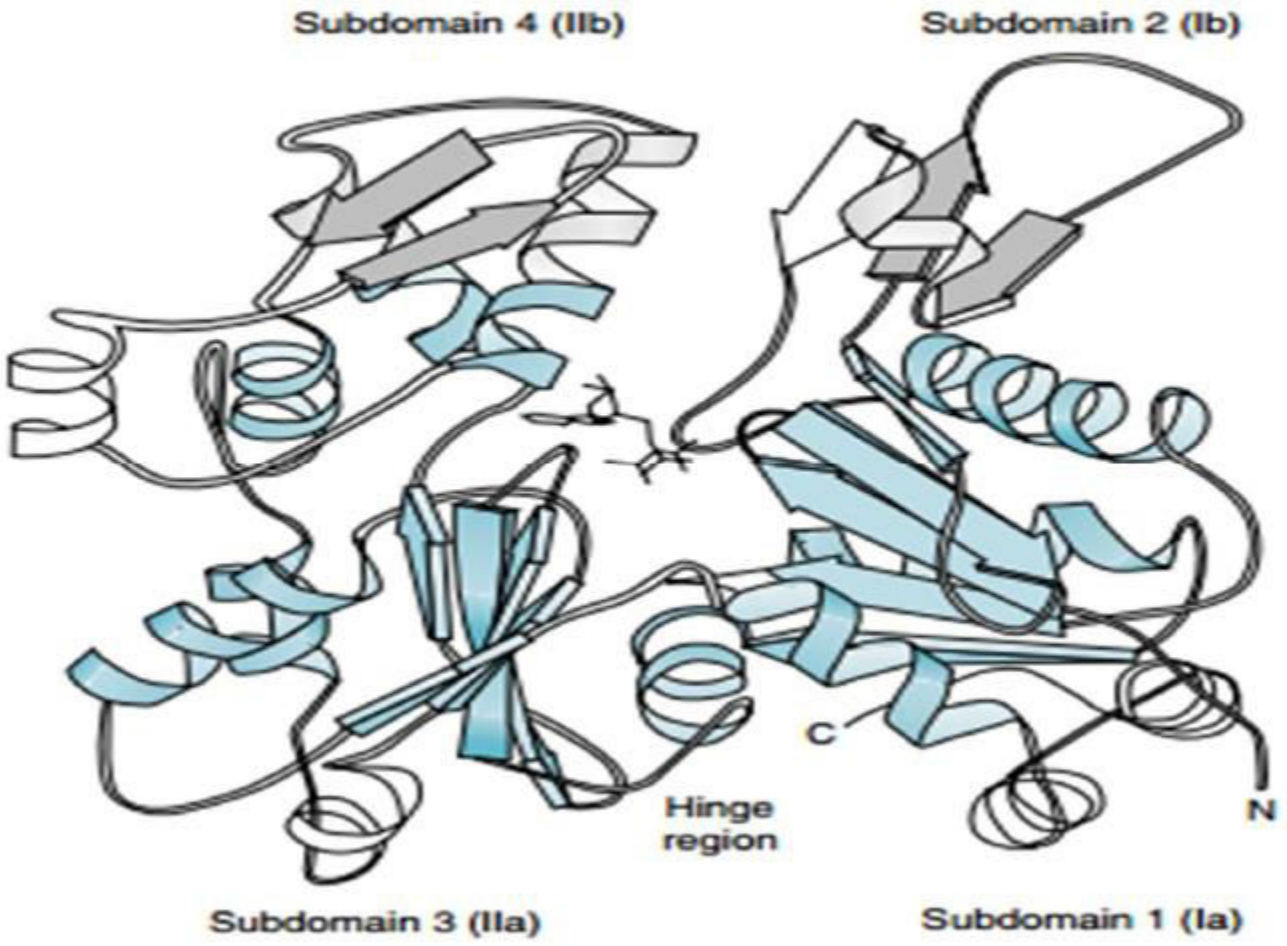
Domains in the Tertiary Structure

- The tertiary structure of large complex proteins is often described in terms of physically independent regions called structural domains.

The Tertiary Structure

- A domain is a section of protein structure sufficient to perform a particular chemical or physical task such as binding of a substrate or other ligand.

The Tertiary Structure



The Tertiary Structure

- Each domain is formed from a continuous sequence of amino acids in the polypeptide chain that are folded into a three-dimensional structure independently of the rest of the protein.

The Tertiary Structure

- Two domains are connected through a simpler structure like a loop.

END

The Tertiary Structure

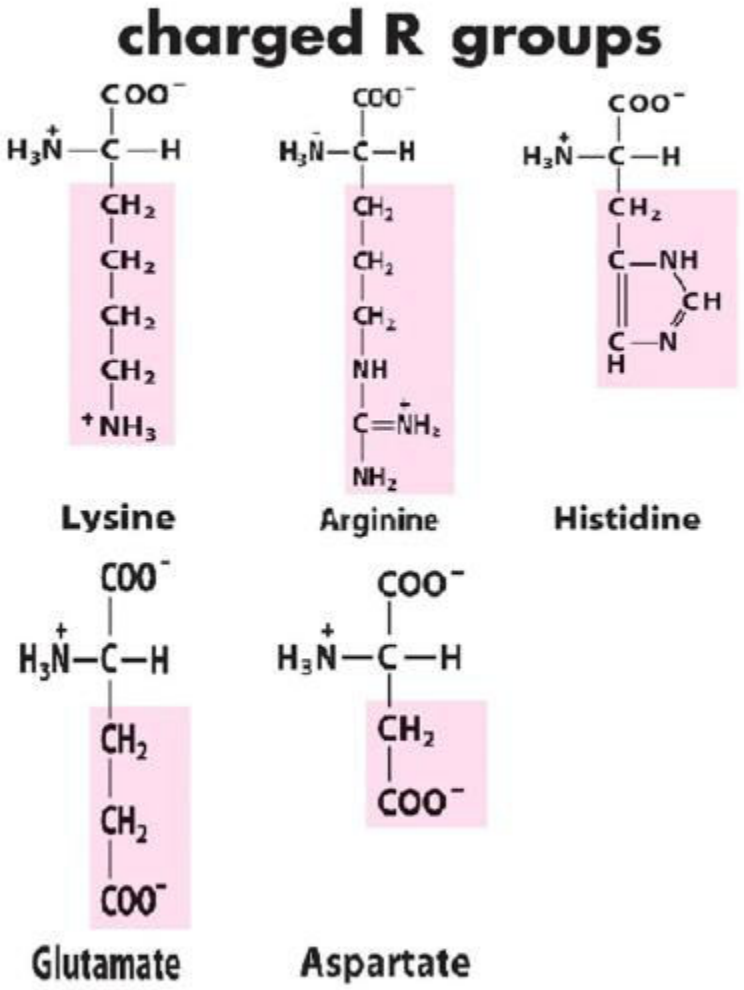
- Molecular forces that stabilise three dimensional tertiary structure include:
- Electrostatic Interactions
- Hydrophobic interactions
- Disulphide bridges
- Hydrogen bonding

The Tertiary Structure

Electrostatic Interactions

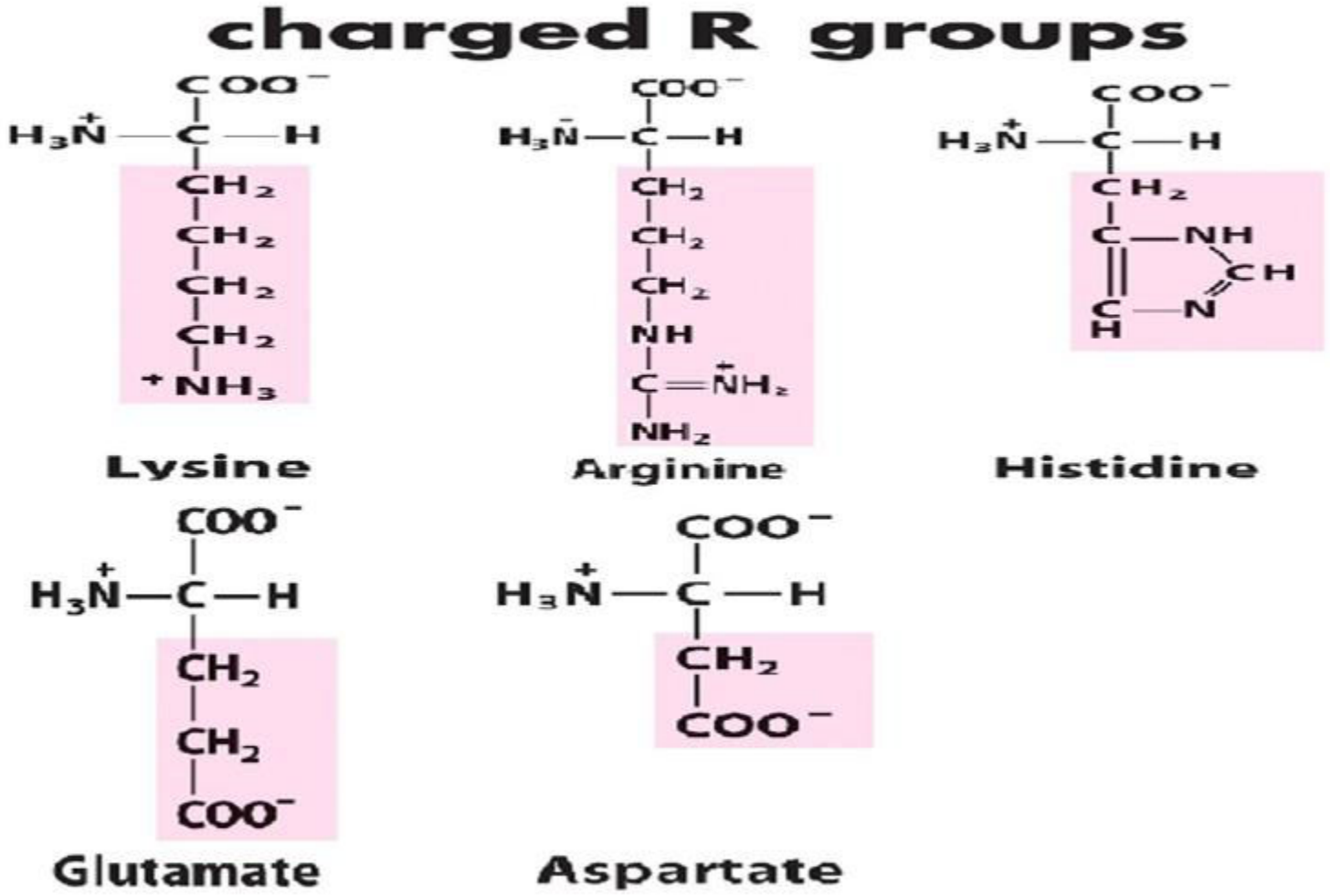
- The **positive** charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with **negatively** charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

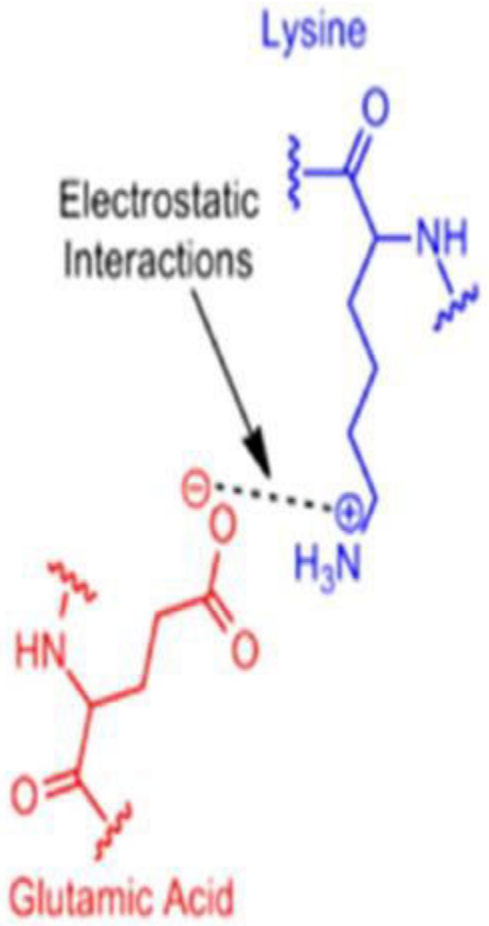


- **Electrostatic Interactions**
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The Tertiary Structure

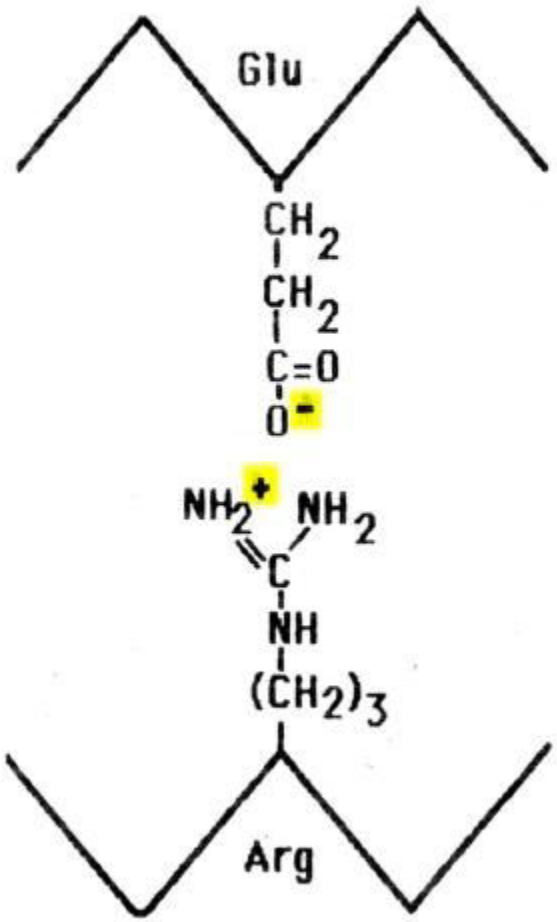


The Tertiary Structure



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The Tertiary Structure



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The Tertiary Structure

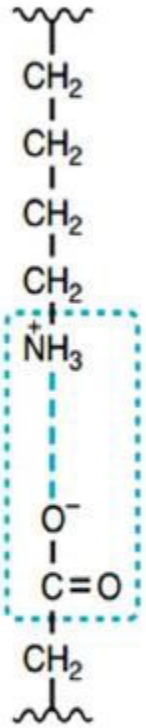


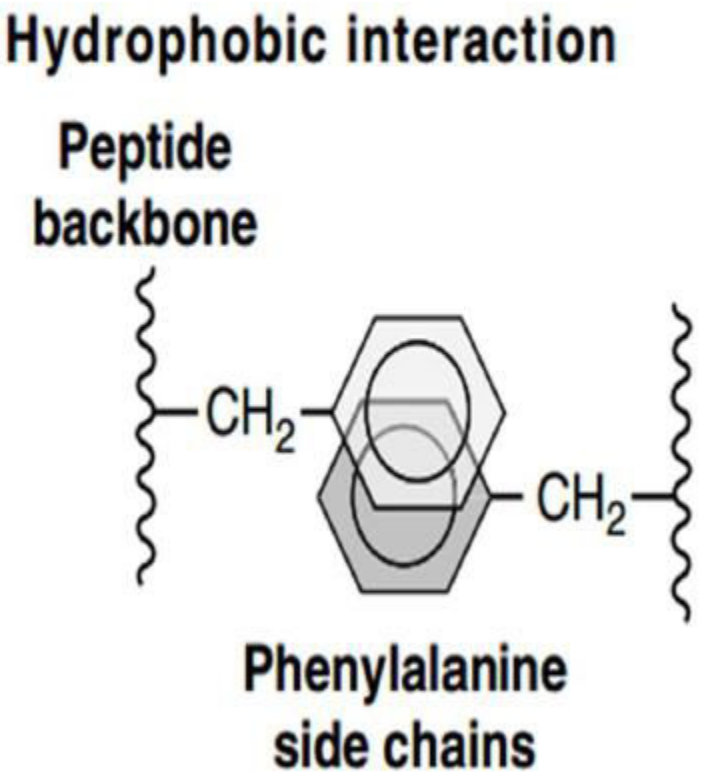
Fig. Electrostatic interaction between the positively charged side chain of lysine and the negatively charged side chain of aspartate.

- The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

- **Hydrophobic Interactions**
- Hydrophobic Interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

The Tertiary Structure



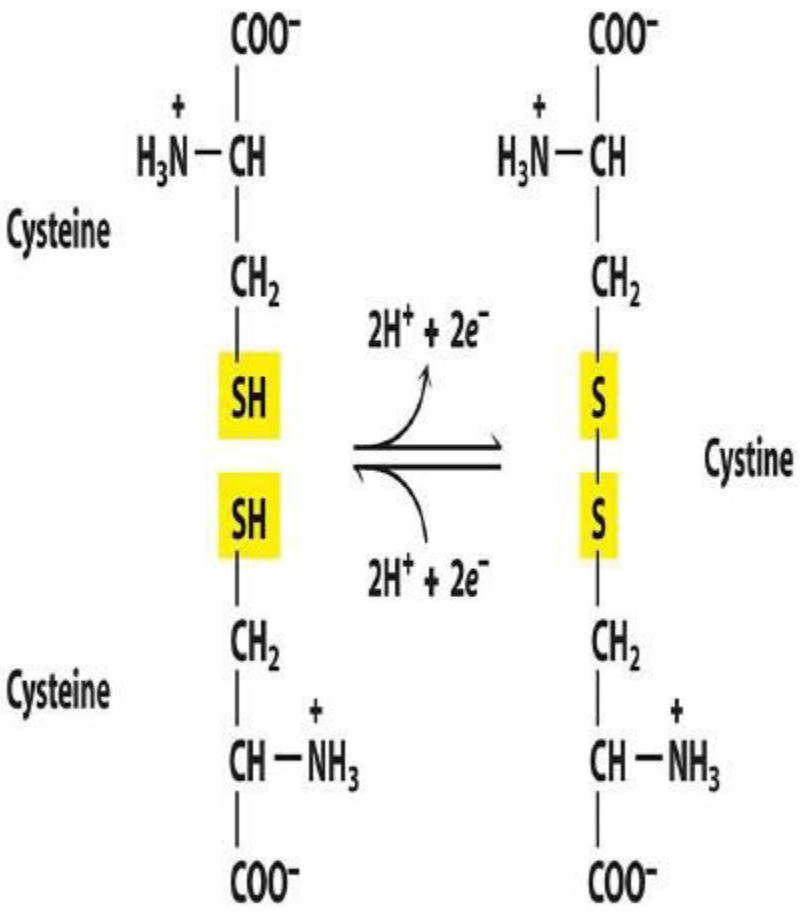
- Hydrophobic interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

The Tertiary Structure

Disulfide Bridges

- The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

The Tertiary Structure



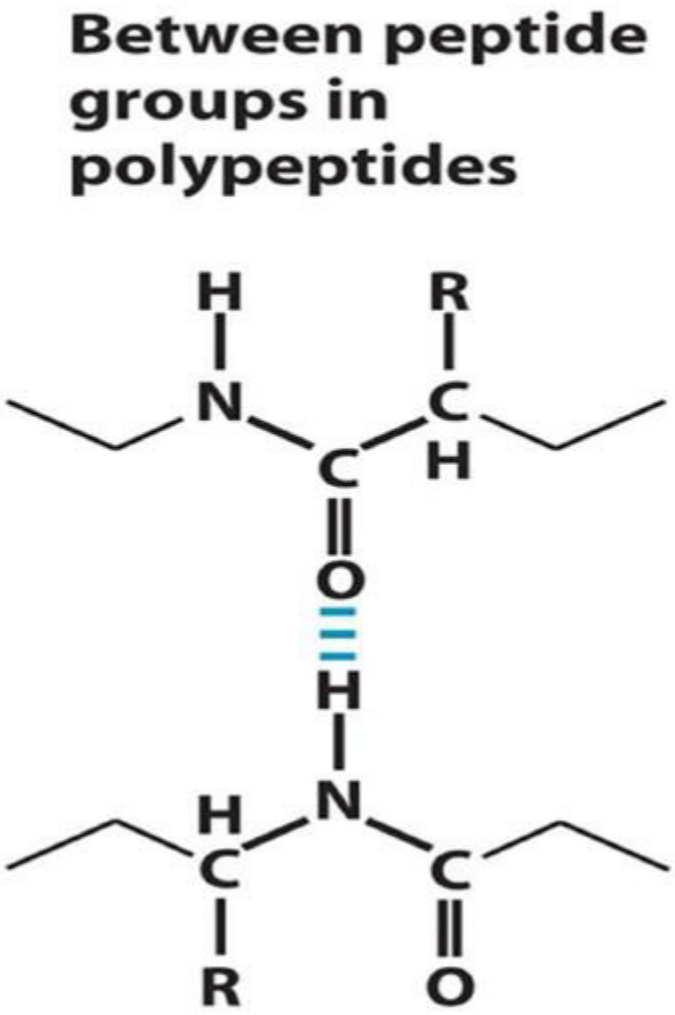
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The Tertiary Structure

Hydrogen bonds

- Hydrogen bonds in which a hydrogen atom is shared by a nitrogen (or oxygen) in the peptide backbone and an oxygen atom in an amino acid side chain.

The Tertiary Structure



- Hydrogen bonds in which a hydrogen atom is shared by a nitrogen (or oxygen) in the peptide backbone and an oxygen atom in an amino acid side chain.

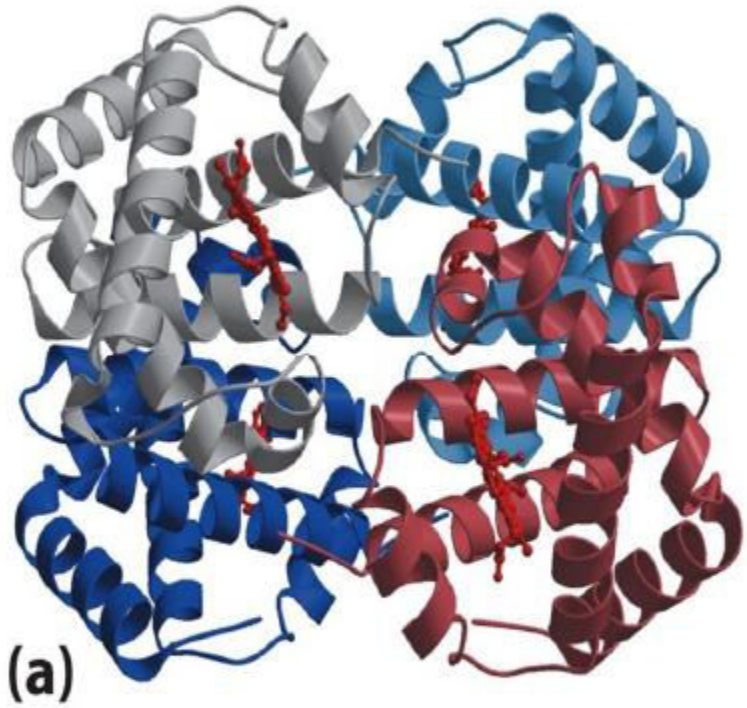
The Quaternary Structure

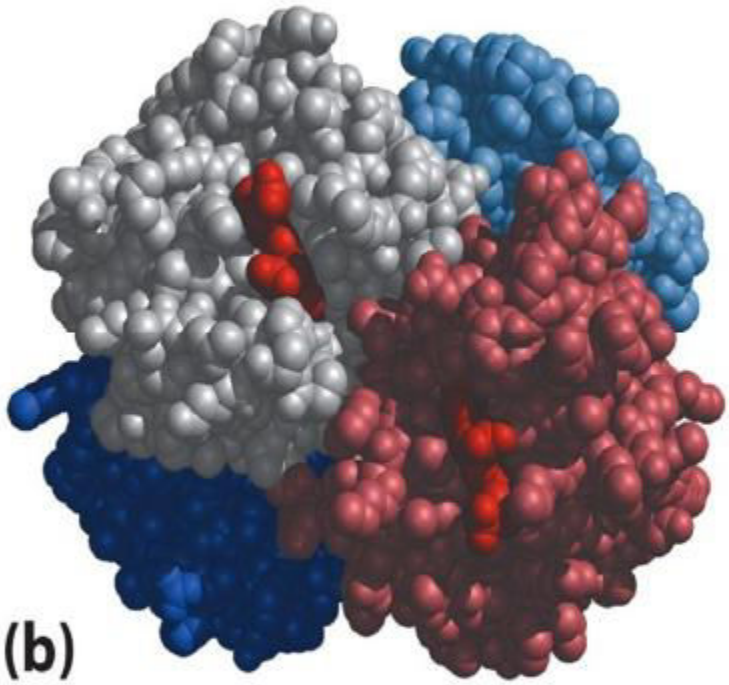
- Some proteins contain two or more separate polypeptide chains, or subunits, which may be identical or different.

The Quaternary Structure

- The arrangement of these protein subunits in three-dimensional complexes constitutes **quaternary structure.**

The Quaternary Structure





The Quaternary Structure

- Many proteins have multiple polypeptide subunits (from two to hundreds)
- The association of polypeptide chains can serve a variety of functions

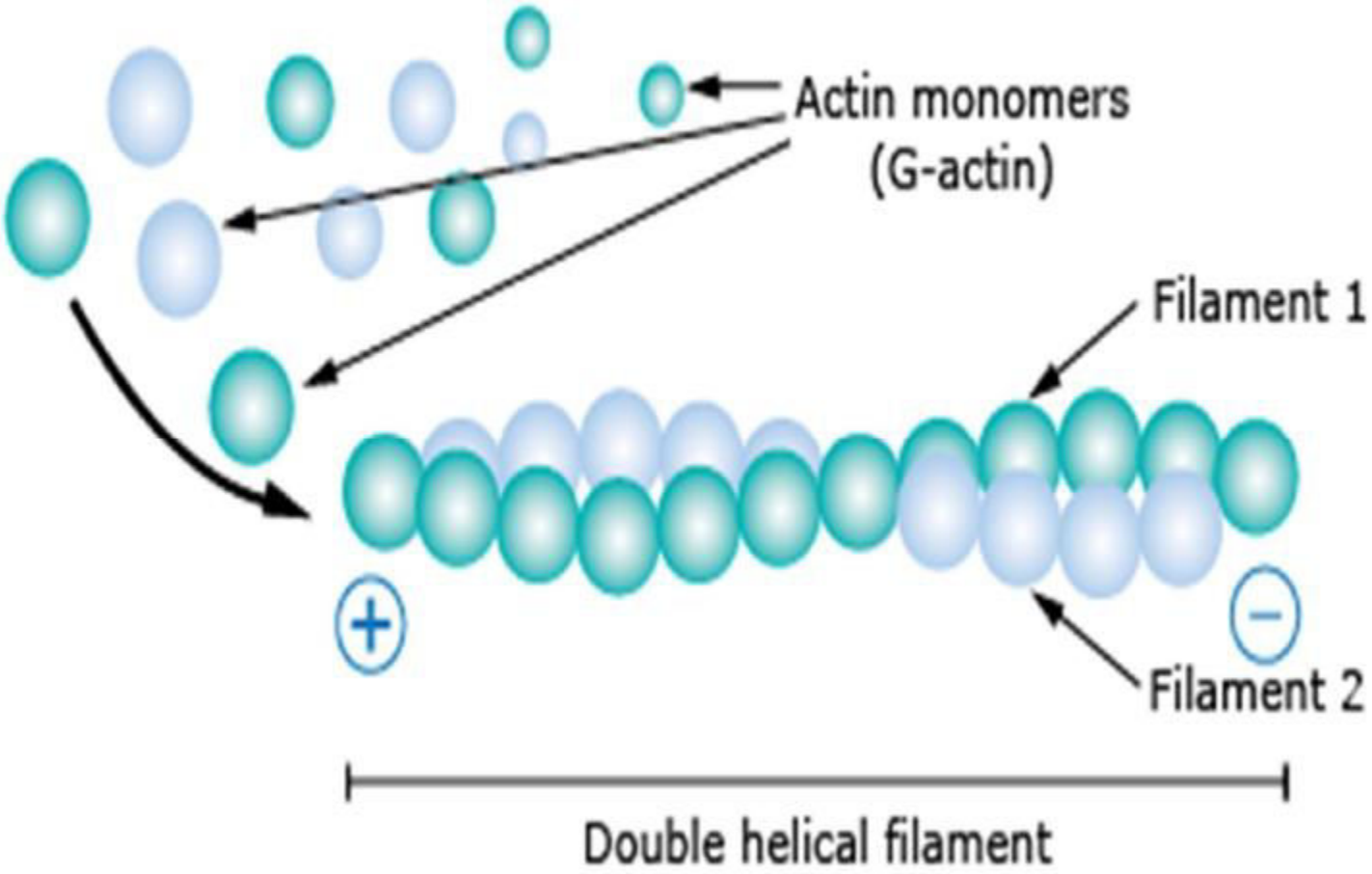
The Quaternary Structure

- A multisubunit protein is also referred to as a **multimer,** with the prefixes "homo" or "hetero" used to describe identical or different subunits, respectively.

The Quaternary Structure

- Most multimers have identical subunits
- or repeating groups of nonidentical subunits, usually in symmetric arrangements.

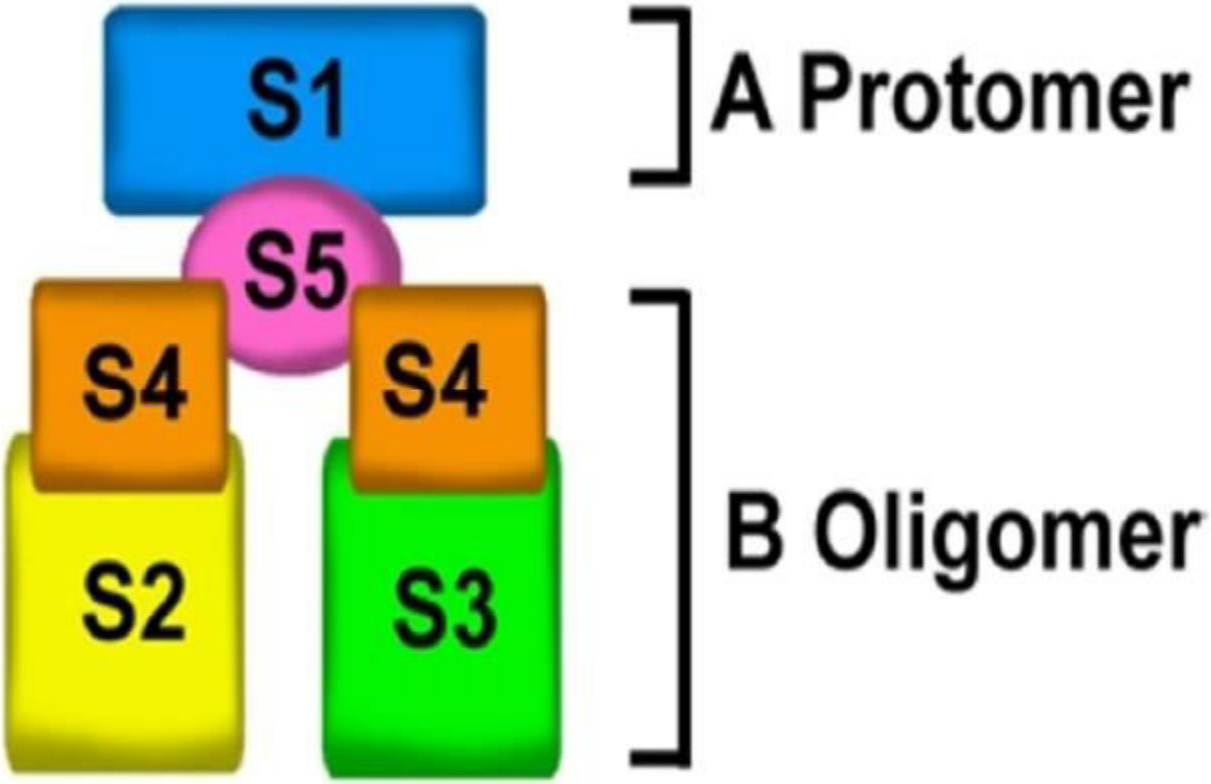
The Quaternary Structure



The Quaternary Structure

- The repeating structural unit in such a multimeric protein is **protomer**
- Assembly into a multisubunit structure increases the stability of a protein.

The Quaternary Structure



The Quaternary Structure

Perks of The Quaternary structure

- The increase in size increases the number of possible interactions between amino acid residues and therefore makes it more difficult for a protein to unfold and refold.

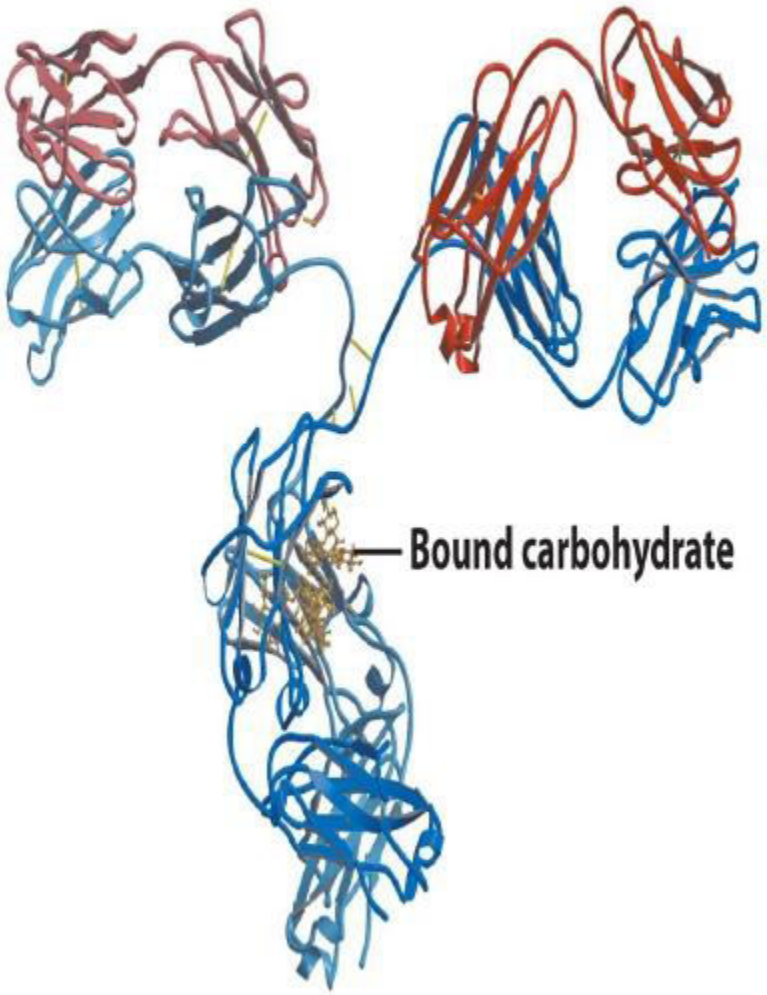
The Quaternary Structure

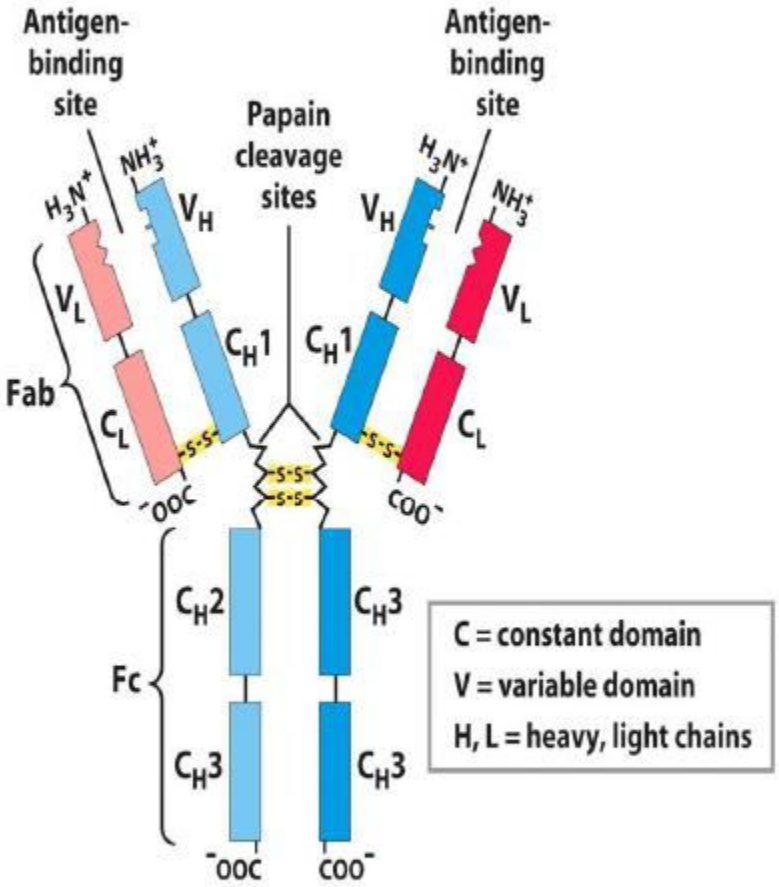
- A multisubunit structure has many advantages besides increased stability.
- It may enable the protein to exhibit cooperativity between subunits in binding ligands (illustrated later with hemoglobin)

The Quaternary Structure

- Or it may form binding sites with a high affinity for large molecules (illustrated with antigen binding to the immunoglobulin molecule IgG).

The Quaternary Structure





The Quaternary Structure

- An additional advantage of a multi-subunit structure is that the different subunits can have different activities and cooperate in a common function.

The Quaternary Structure

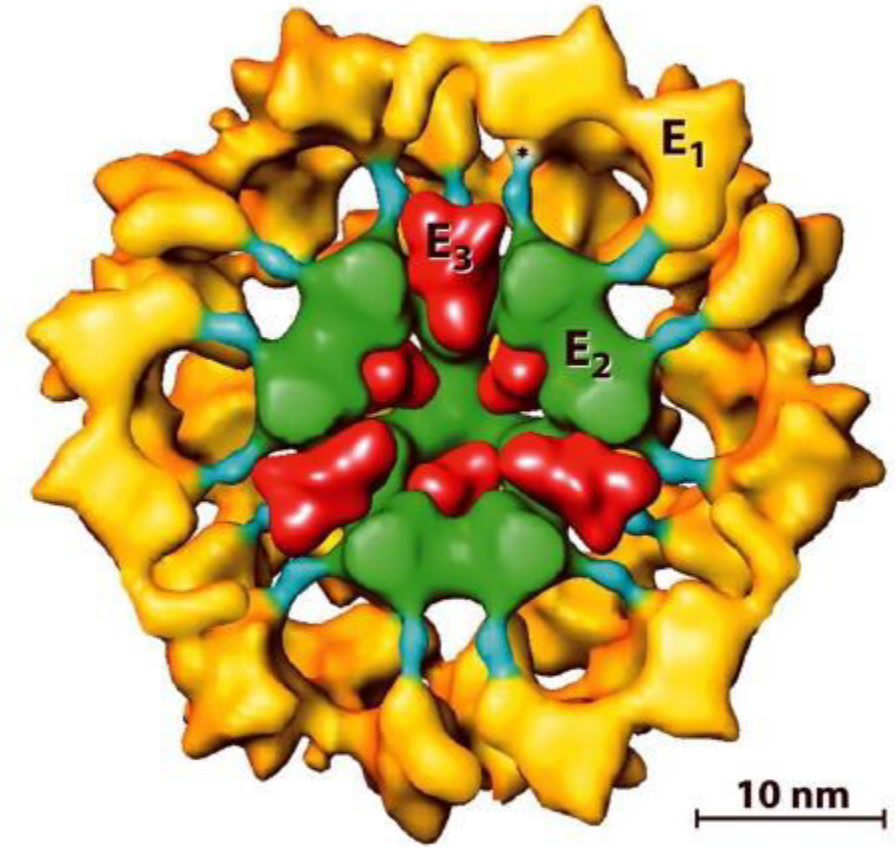
- For example enzymes that have regulatory subunits or exist as multiprotein complex, such as Pyruvate Dehydrogenase.

The Quaternary Structure

E1 , pyruvate dehydrogenase;

E2, dihydrolipoyl transacetylase; and

E3,dihydrolipoyl dehydrogenase



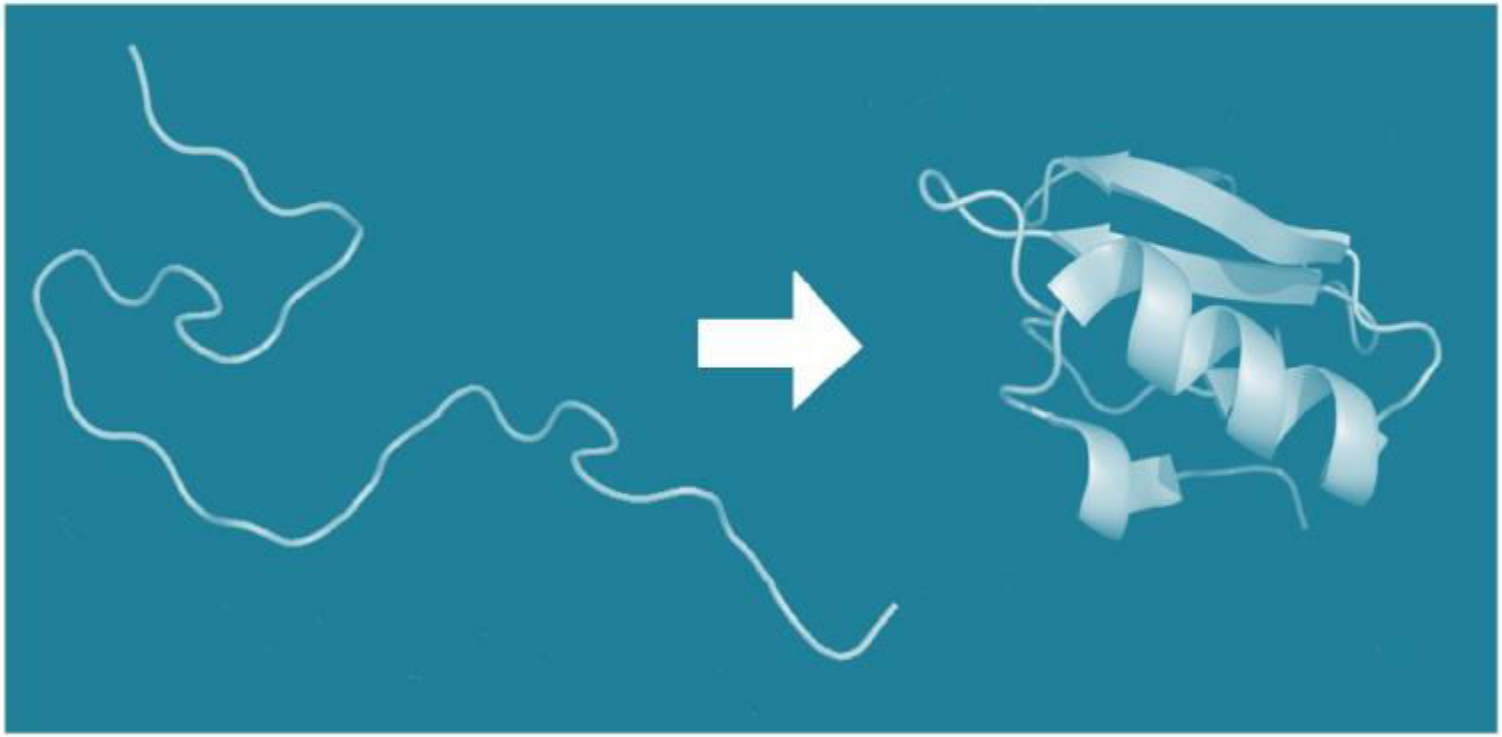
Protein Folding

Protein folding is the process by which a protein structure assumes its functional shape or conformation.

Protein Folding

It is the physical process by which a polypeptide folds into its characteristic and functional three-dimensional structure from random coil

Protein Folding



Protein Folding

- The overall three-dimensional structure of a protein must meet certain requirements to enable the protein to function in the cell or extracellular medium of the body.

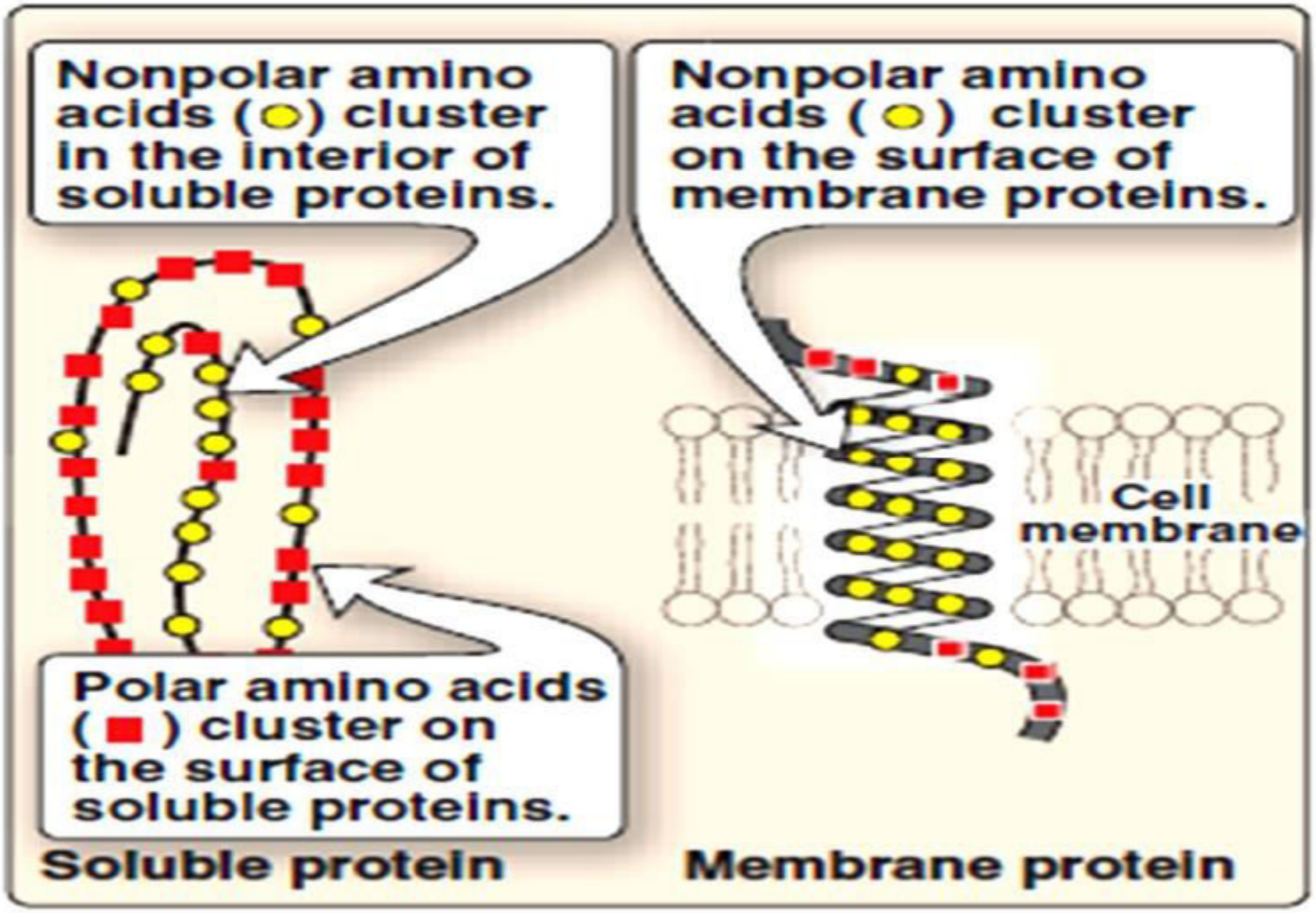
Protein Folding

- The first requirement is the creation of a binding site that is specific for just one molecule, or a group of molecules with similar structural properties.
- The specific binding sites of a protein usually define its functional role.

Protein Folding

- Moreover, The three dimensional structure must have an external surface appropriate for its environment (e.g., plasma proteins contain polar amino acids on the surface to remain soluble in an aqueous environment.)

Protein Folding



Protein Folding

- Proteins are dynamic molecules that can fold into their functionally competent conformation in milliseconds

Protein Folding

- A protein is called a native protein if its amino acid composition and molecular conformation are unchanged from that found in natural states.

Protein Folding

- Folding into the native state does not involve a haphazard search of all possible structures.
- The native conformation of a Protein is dictated thermodynamically.

Protein Folding

- The number of potential conformations of even a relatively small—15-kDa—polypeptide is unbelievably vast
- Proteins are guided through this vast labyrinth of possibilities by thermodynamics

Protein Folding

- Thermodynamic stability is not evenly distributed over the structure of a protein molecule.
- It has regions of high and low stability; variations in the stability of regions within a protein are often essential for protein to function.

END

Protein Folding

- Proteins are dynamic molecules that can fold into their functionally competent conformation in milliseconds

Protein Folding

- A protein is called a native protein if its amino acid composition and molecular conformation are unchanged from that found in natural states
- And which is operative and functional

Protein Folding

- Folding into the native state does not involve a haphazard search of all possible structures.

It not random but programmed

Protein Folding

- Protein folding in cells takes place in an orderly and guided fashion.

- Protein folding generally occurs via a stepwise process.

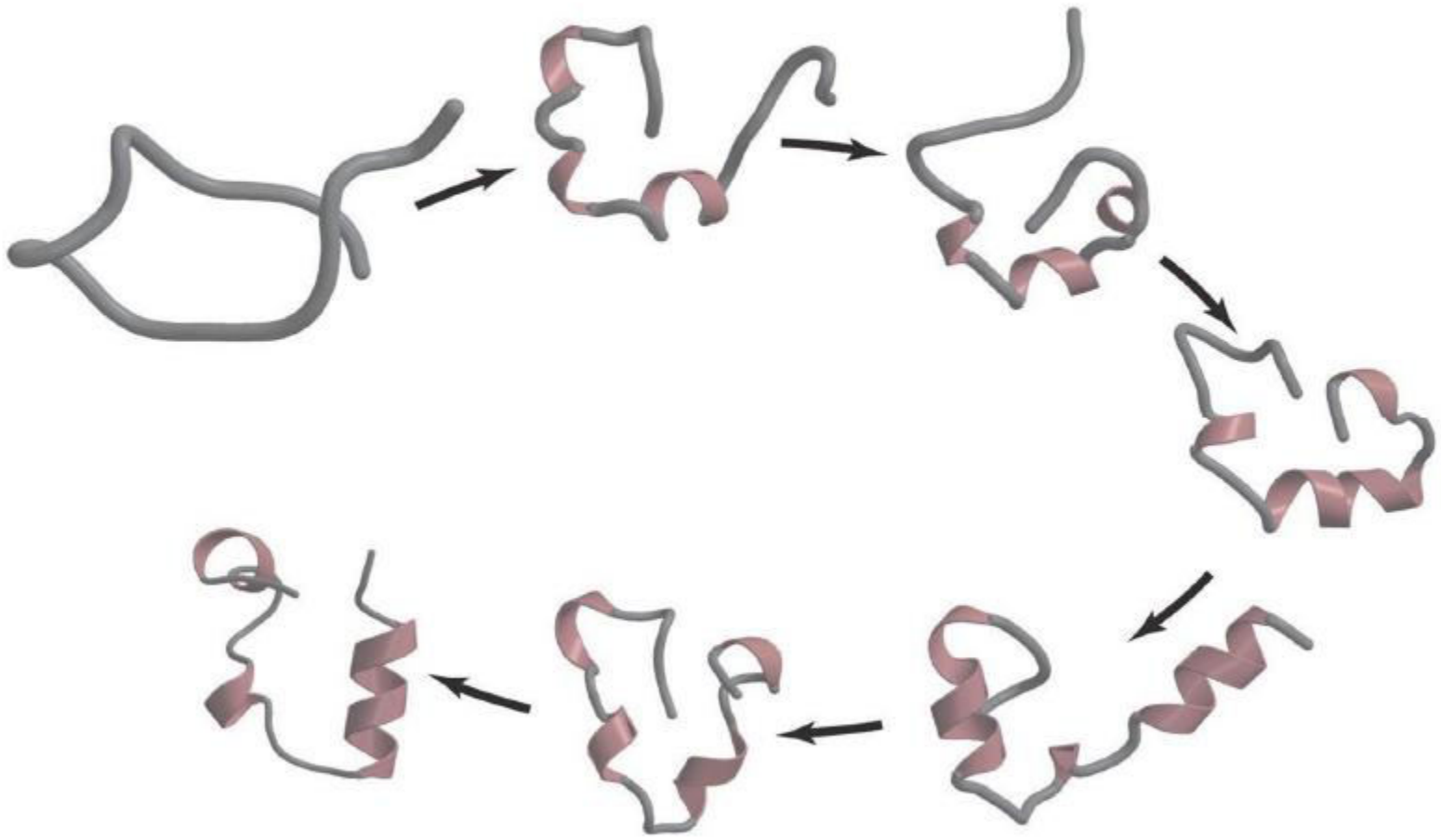
Protein Folding

- In the first stage, as the newly synthesized polypeptide emerges from the ribosome, short segments fold into secondary structural units that provide local regions of organized structure

Protein Folding

- Then, each element of secondary or super-secondary structure facilitates proper folding by directing the folding process toward the native conformation and away from unproductive alternatives.

Protein Folding



Protein Folding

- This step-wise folding of the proteins is dictated by thermodynamics resulting in a formation of native form, which is also the most energetically favoured form.

END

Protein Folding

- The native conformation of a protein is dictated thermodynamically.

Protein Folding

- The biologically relevant—or native—conformation of a protein generally is that which is most energetically favored

Protein Folding

- The three dimensional structure of a native protein in its normal physiological state is the one in which the Gibbs free energy of the whole system is lowest.

Protein Folding

- The free energy change for a process at constant pressure is:
- **$\Delta G = \Delta H - T\,\Delta S$**

- **ΔG = Free Energy**

- **ΔH = Enthalpy**

- **ΔS= Entropy**

Entropy is a measure of the randomness or disorder in a system. **Entropy of a system always tend to increase to a maximum** value.

Protein Folding

- Folding of a globular protein is a thermodynamically favored process, i.e. ΔG must be negative.
- $\mathbf{\Delta G = \Delta H - T\,\Delta S}$
- The folding process involves going from a multitude of random-coil conformations to a single folded structure.
- The folding process involves a decrease in randomness and thus a decrease in entropy $-\Delta S$ and an overall positive contribution to ΔG. This decrease in entropy is termed "conformational entropy".

Protein Folding

- Difference in energy (free energy) between folded (native) and unfolded (denatured) state is small, 5-15 kcal/mol.

Protein Folding

- It is very difficult to determine how all factors blend together to give overall $\Delta G_{folding}$
  - Use of averages contributions, but
  - Each protein is unique

Protein Folding

- Large stabilization factors,
- Large destabilization factors,
- But small difference between them

- $\Delta G = \Delta H - T\,\Delta S$

Electrostatic.Ionization/disulfide bonds, Specific hydrogen bonding, Hydrophobic effect

Protein Folding

- Small ΔG is necessary because too large a free energy change would mean a very stable protein, one that would never change.

Protein Folding

- However, structural flexibility is important to protein function, and proteins need to be degraded.

Protein Folding

- Enthalpy and entropy differences balance each other, and $\Delta \mathbf{G}$ is a small positive number.

END

Protein Folding

- The native state has a smaller Gibbs free energy than the denatured state. The stability of the protein depends on the solvent–solvent, protein–solvent, and protein–protein interactions.

Protein Folding

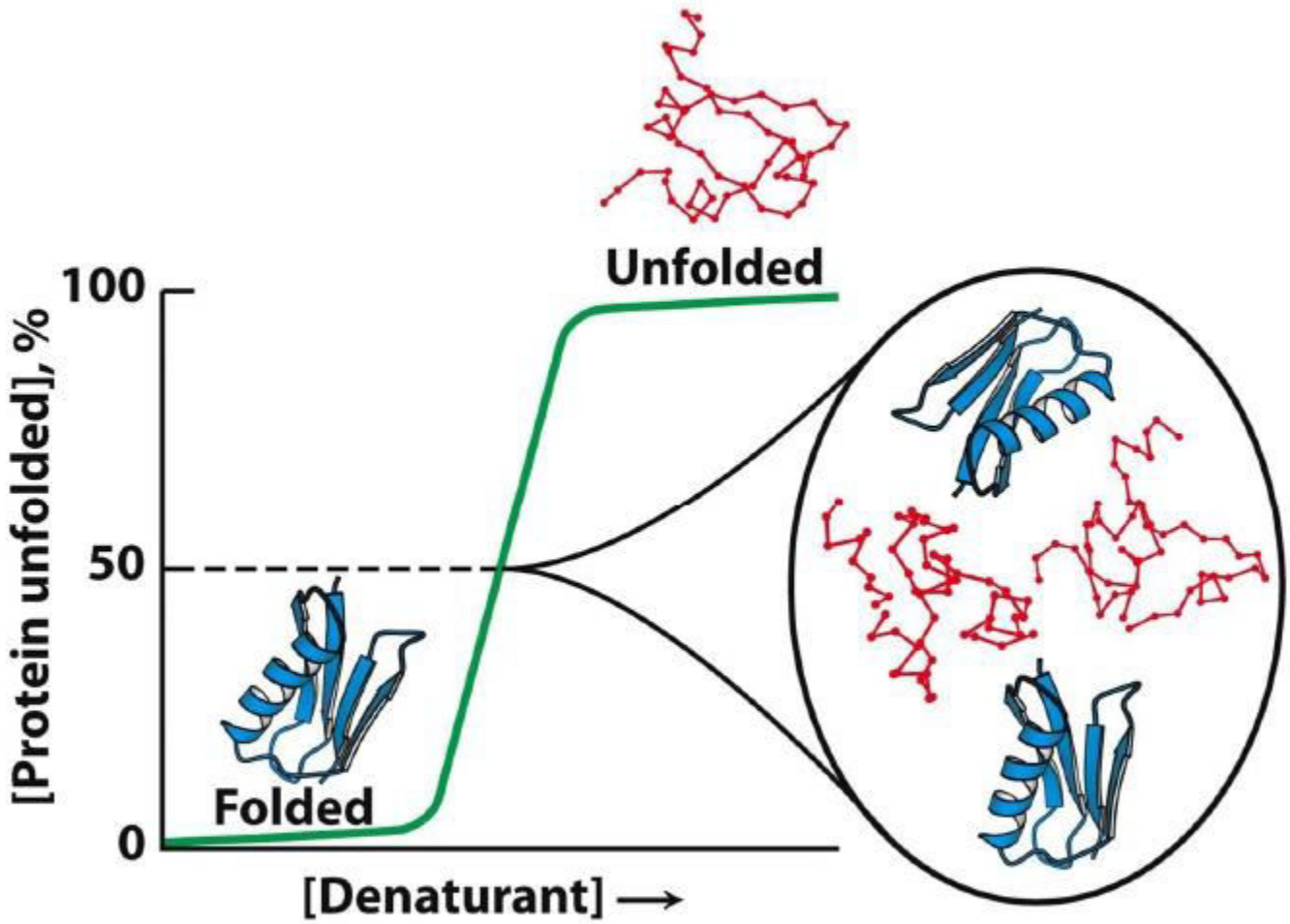


Figure 2.57
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Protein Folding

- The free energy of a protein molecule is influenced by
- (1) the hydrophobicity,
- (2) hydrogen bonds,
- (3) electrostatic interactions, and
- (4) the conformational entropy due to the restricted motion

Protein Folding

- **Entropy** is a measure of the randomness or disorder in a system.
- According to Second Law of Thermodynamics: **The entropy of a system will tend to increase to a maximum** value.

Protein Folding

The folding process involves a decrease in randomness and thus a decrease in entropy - ΔS and an overall positive contribution to ΔG. This decrease in entropy is termed "conformational entropy".

Protein Folding

- the hydrophobic effect and hydrogen bonds are the major stabilizing contributions which counteract entropy.

Protein Folding

- Starting from the unfolded protein, the polypeptide chain has to fold partially in order to bring together the residues that need to form the contacts stabilizing the native structure.

Protein Folding

- The constrained polypeptide chain has smaller entropy, which means higher Gibbs free energy.
- As native contacts form, the enthalpy term decreases, the protein is stabilized.

Protein Folding

- Thermodynamic stability is not evenly distributed over the structure of a protein molecule.
- It has regions of high and low stability; variations in the stability of regions within a protein are often essential for protein to function.

Protein Folding

- The rate-limiting step in the folding process is the formation of the transition state, i.e., the conformation that has the highest Gibbs free energy on the folding pathway

END

Protein Folding

- In the cell, not all proteins fold into their native conformation on their own.
- As the protein folds and refolds while it is searching for its native low energy state, it passes through many high-energy conformations that slow the process

Protein Folding

- These high energy conformations are called kinetic barriers.
- Cells employ auxiliary proteins to overcome these kinetic barriers.

Protein Folding

- These auxillary proteins are called chaperones.
- Chaperone use energy provided by ATP hydrolysis to assist in the folding process

Protein Folding

- **Chaperones** participate in the folding of over half of mammalian proteins.
- They prevent immature folding of the nascent polypeptide and help in the final native protein formation.

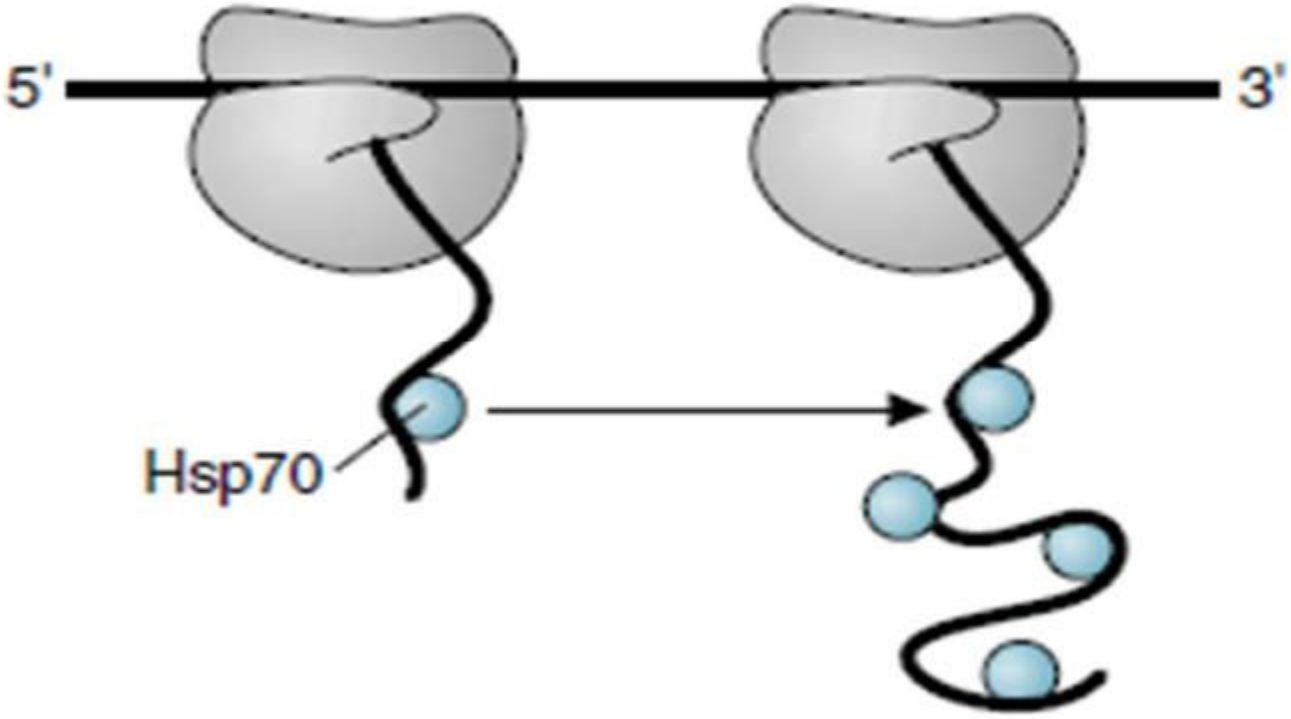
Protein Folding

- The hsp70 family of chaperones binds short sequences of hydrophobic amino acids in newly synthesized polypeptides.

Protein Folding

- Chaperones prevent aggregation, thus providing an opportunity for the formation of appropriate secondary structural elements.

Protein Folding



Protein Folding

- They also unfold proteins prior to their insertion through the membrane of mitochondria and other organelles.

END

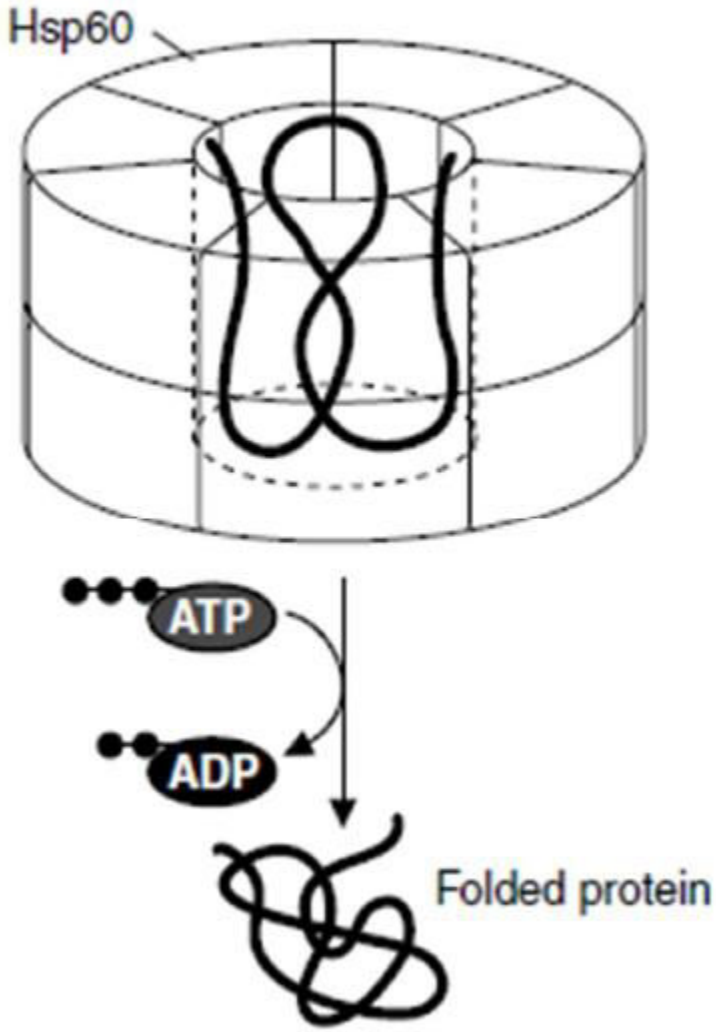
Protein Folding

- The hsp60 family of chaperones, sometimes called **chaperonins**, differ in sequence and structure from hsp70 and its homology
- Hsp60 acts later in the folding process, often together with an hsp70 chaperone

Protein Folding

- Hsp60 chaperonins form a multi-subunit barrel-shaped structure
- The unfolded protein fits into the barrel cavity that excludes water and serves as a template for the folding process.

Protein Folding



Protein Folding

- Chaperone proteins can also "rescue" unfolded proteins that have become thermodynamically trapped in a misfolded dead end by unfolding hydrophobic regions and providing a second chance to fold productively

Protein Folding

- Finally, the folding pathways of some proteins require two enzymes that catalyze isomerization reactions.

Protein Folding

- Protein disulfide isomerase (PDI) is a widely distributed enzyme that catalyzes the interchange, or shuffling, of disulflde bonds until the bonds of the native conformation are formed.

END

Protein Misfolding

- Despite the many processes that assist in protein folding, misfolding does occur.

Protein Misfolding

- Misfolding of proteins may occur spontaneously
- or caused by a mutation in a particular gene, producing an altered protein.

Protein Misfolding

- In addition some apparently normal proteins can, after abnormal proteolytic cleavage, take on a unique conformational state
- That leads to the formation of long fibrillar protein assemblies consisting of β pleated sheets

Protein Misfolding

- Many conditions, including
- Alzheimer's disease
- Type 2 diabetes,
- Huntington's disease and
- Parkinson's disease, arise from a common misfolding mechanism.

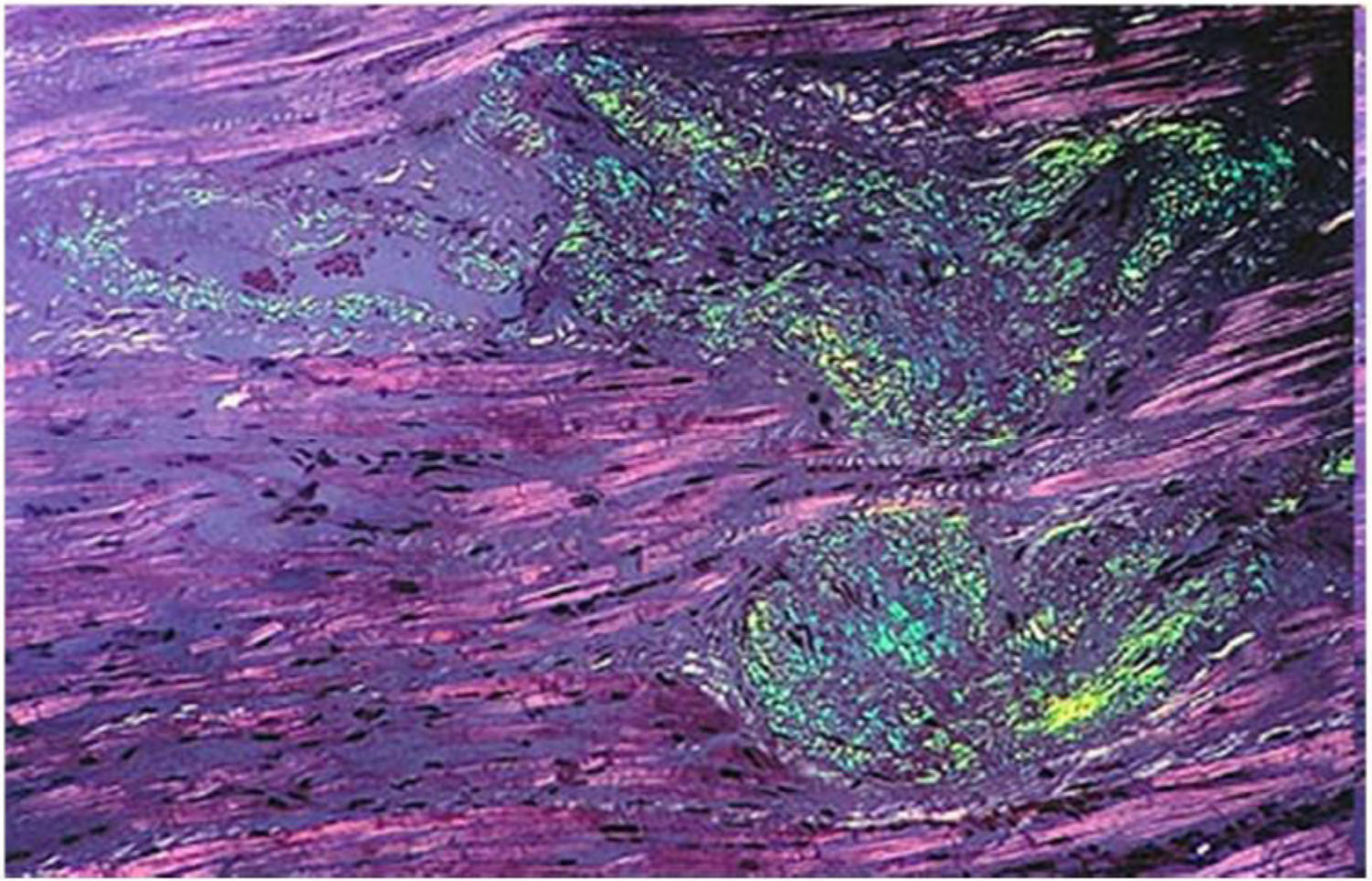
Protein Misfolding

- In most cases, a soluble protein that is normally secreted from the cell is secreted in a misfolded state
- and converted into an insoluble extracellular amyloid fiber.
- The diseases are collectively referred to as amyloidoses.

Protein Misfolding

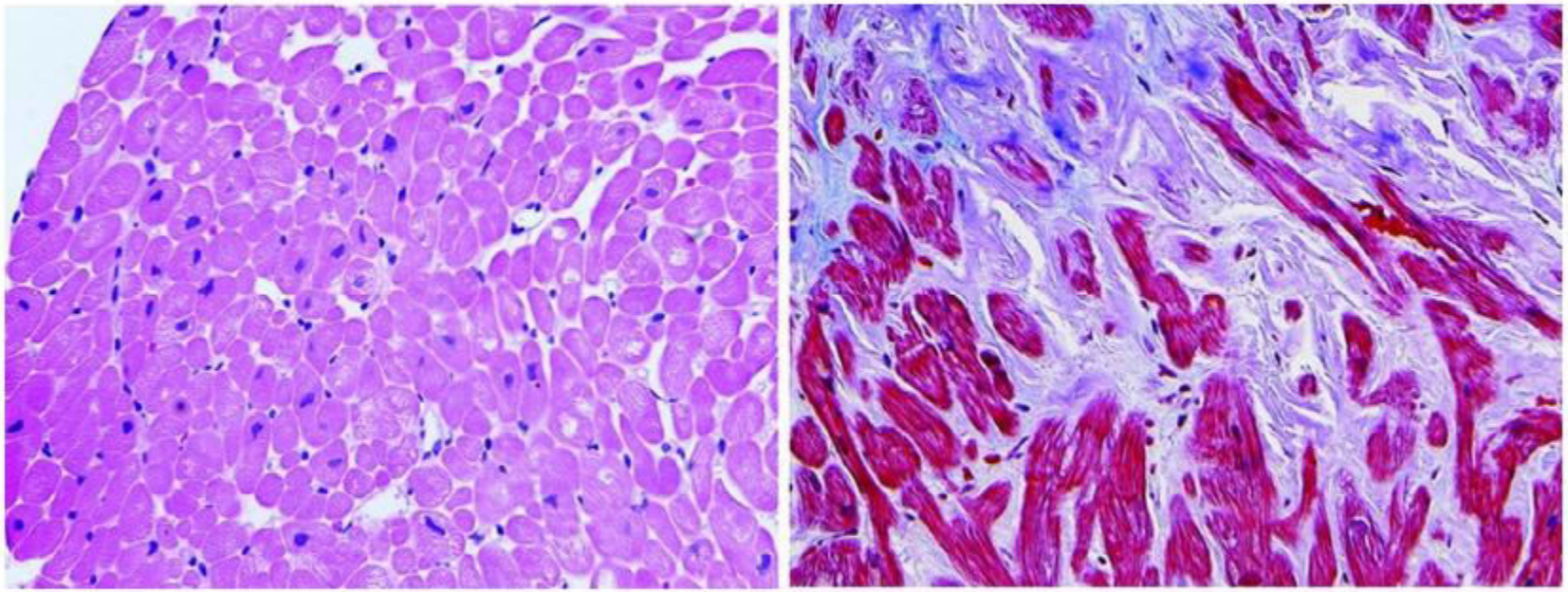
- Amyloidosis is a generic term that refers to the extracellular tissue deposition of fibrils composed of low molecular weight of a variety of proteins..

Protein Misfolding



depostion of abnormal protein in the form of amyloid in the ECM of cardiac muscle.

Protein Misfolding



Cardiac biopsies showing normal findings

extensive amyloid infiltration

Protein Misfolding

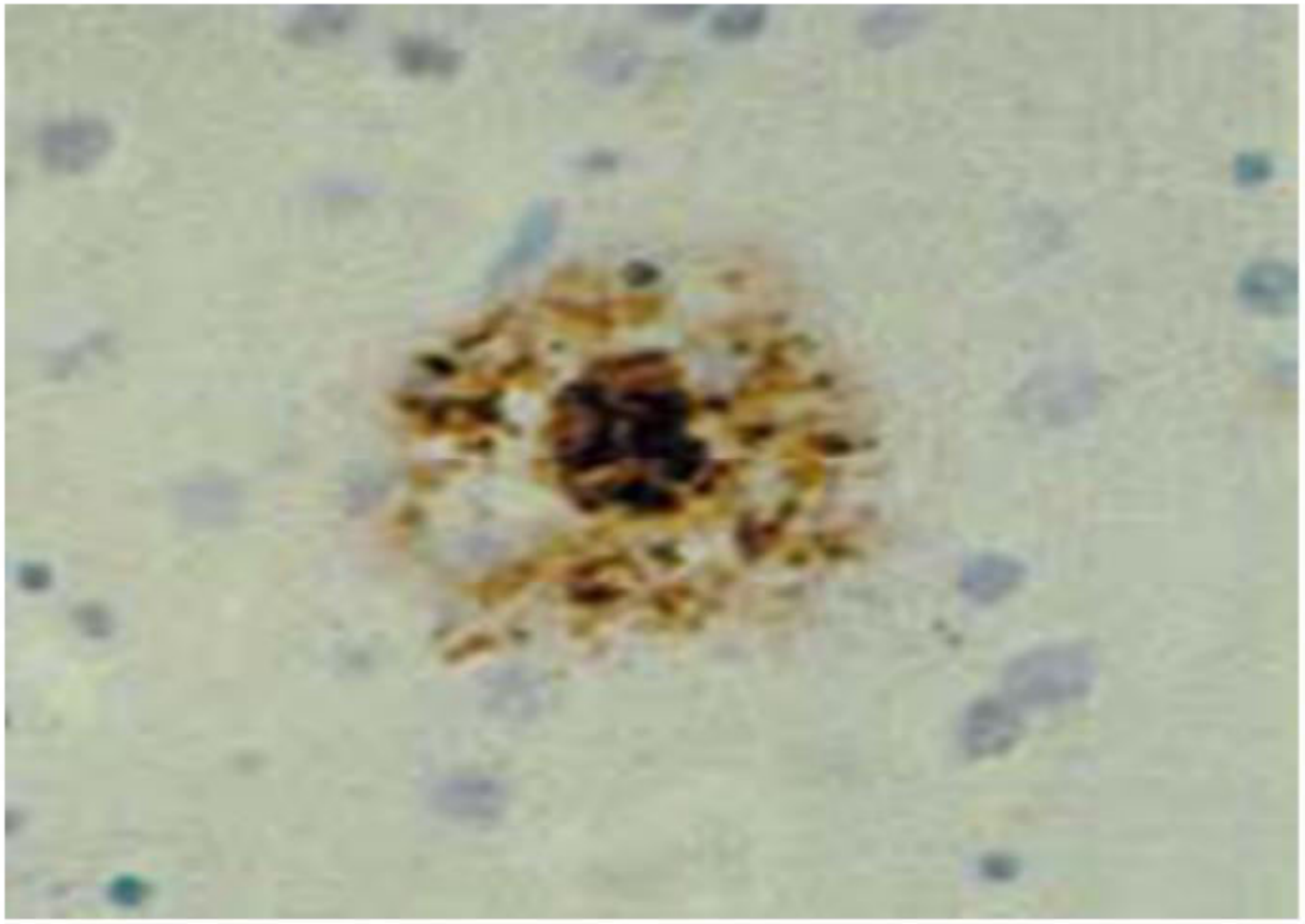
- Amyloidoses may be
  - organ specific or
  - generalized (systemic).
- Primary systemic amyloidosis is caused by deposition of fibrils consisting of misfolded immunoglobulin light chain

Protein Misfolding

- Examples of organ-specific amyloidoses include:
- **Alzheimer's disease** due to deposition of amyloid-b protein cleaved from amyloid precursor protein (APP).

Alzheimer's is a type of dementia that causes problems with memory, thinking and behavior

Protein Misfolding



Photomicrograph of amyloid plaques in a section of temporal cortex

Protein Misfolding

- Islet amyloid polypeptide (IAPP) is commonly seen in diabetes mellitus type 2 and is caused by deposition of amylin in pancreatic islets.

end

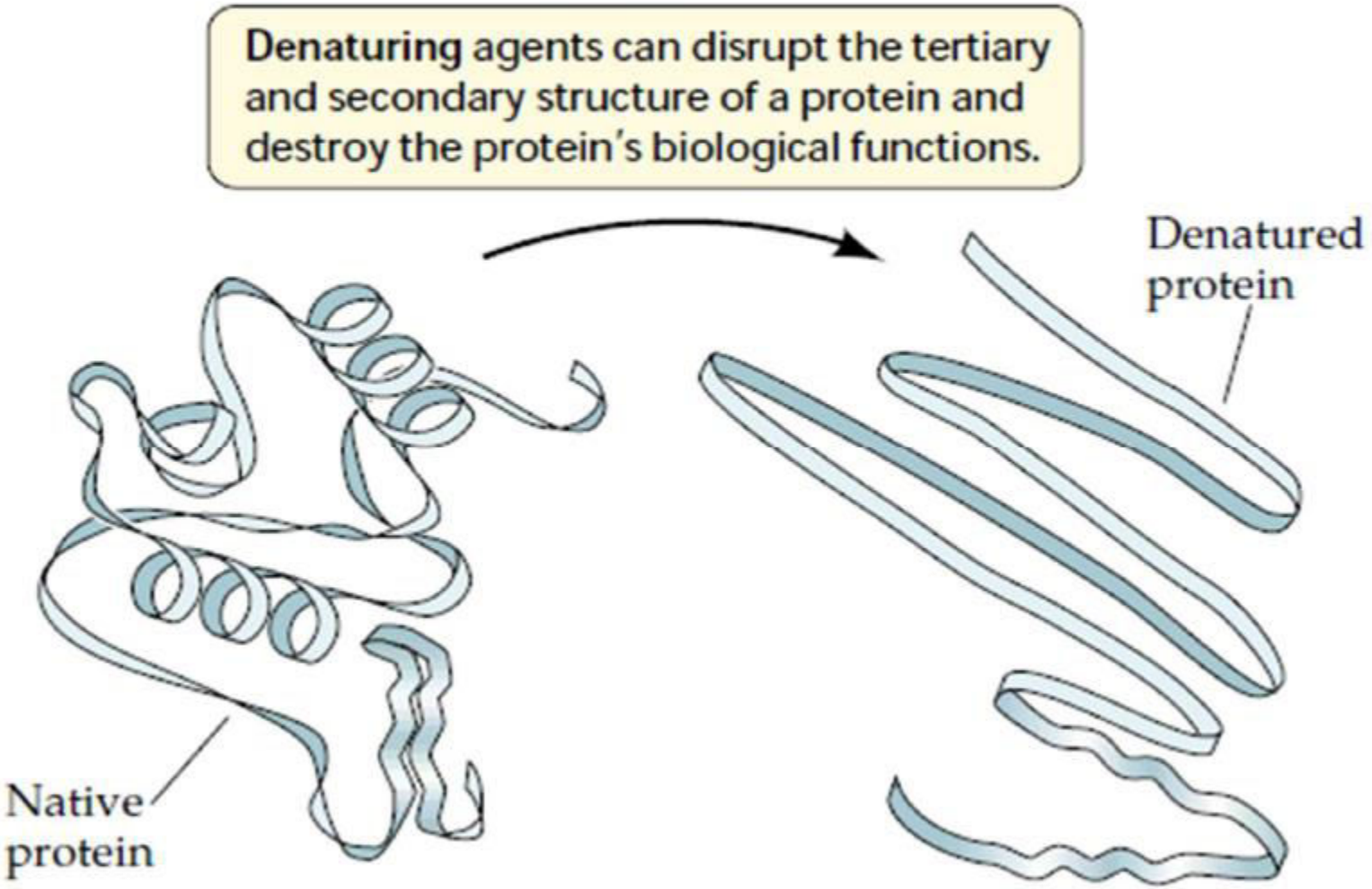
Protein Denaturation

- A loss of three-dimensional structure sufficient to cause loss of function is called denaturation

Protein Denaturation

- The denatured state does not necessarily equate with complete unfolding of the protein and randomization of conformation
- Under most conditions, denatured proteins exist in a set of partially folded state

Protein Denaturation



Protein Denaturation

- Most proteins can be denatured by heat,
- which has complex effects on the weak interactions in a protein (primarily hydrogen bonds).

Protein Denaturation

- Organic solvents, urea, and detergents act
- primarily by disrupting the hydrophobic interactions that make up the stable core of globular proteins;

Protein Denaturation

- Extremes of pH alter the net charge on the protein, causing disruption of electrostatic interactions
- And the disruption of some hydrogen bonding.

Protein Denaturation

Everyday Examples of Denaturation

- When milk curdles, the acidity increases.
- Thermal denaturation by cooking.
- Mechanical denaturation when whisking an egg.
- Perming hair breaks then reforms the disulphide bonds.

end

Protein Denaturation

- Certain proteins denatured by heat, extremes of pH, or denaturing reagents
- Regain their native structure and their biological activity if returned to conditions in which the native con-formation is stable.

Very High or Low pH. At Very High Temp.By Heavy Metal Ions.By Small Polar Molecules

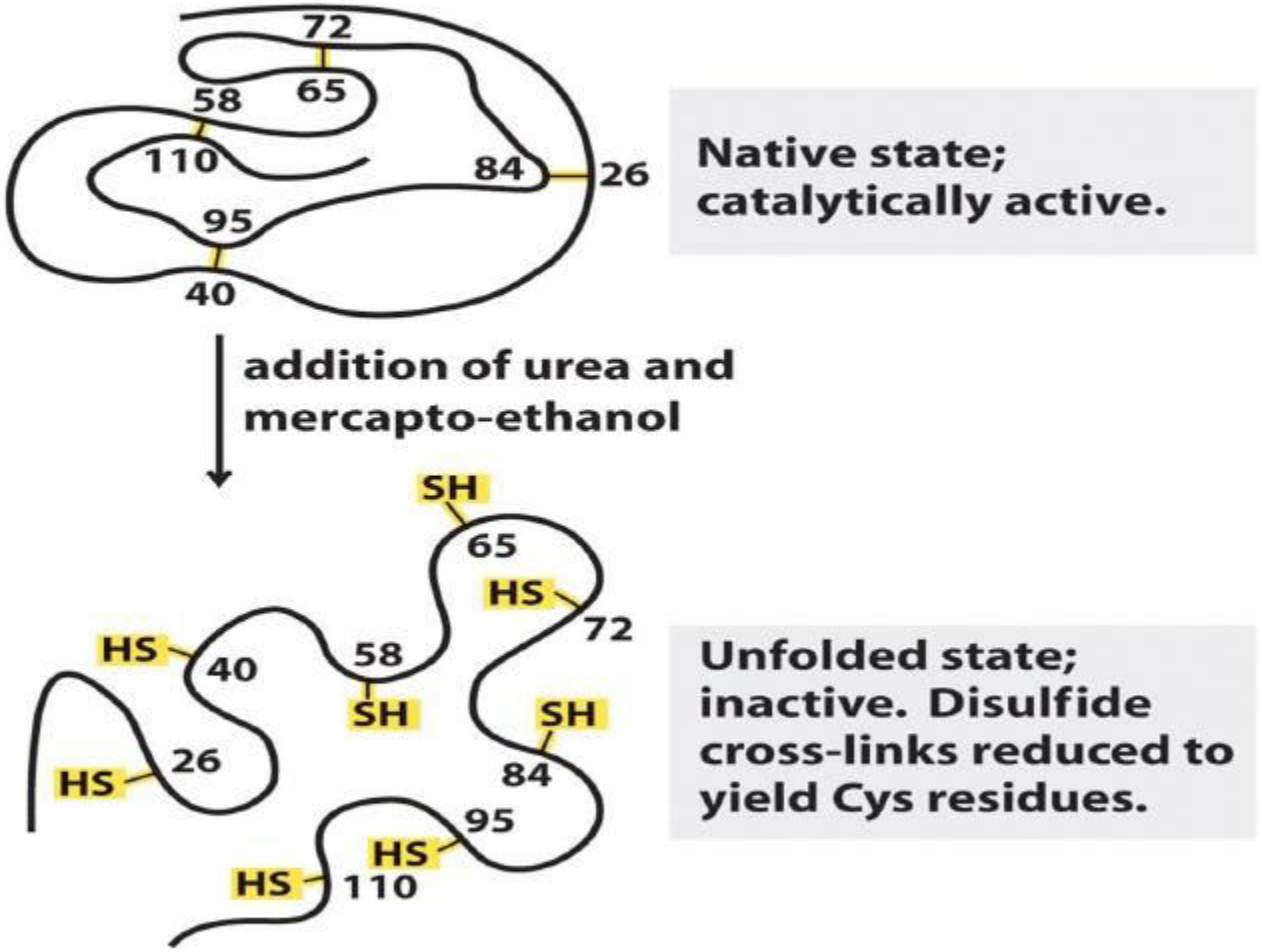
Protein Denaturation

- This process is called renaturation
- Unfolded proteins generally retain a number of contacts and regions of secondary structure that facilitate the refolding process

Protein Denaturation

- A classic example is the denaturation and renaturation of ribonuclease A
- Purified ribonuclease A denatures completely in a concentrated urea solution in the presence of a reducing agent.

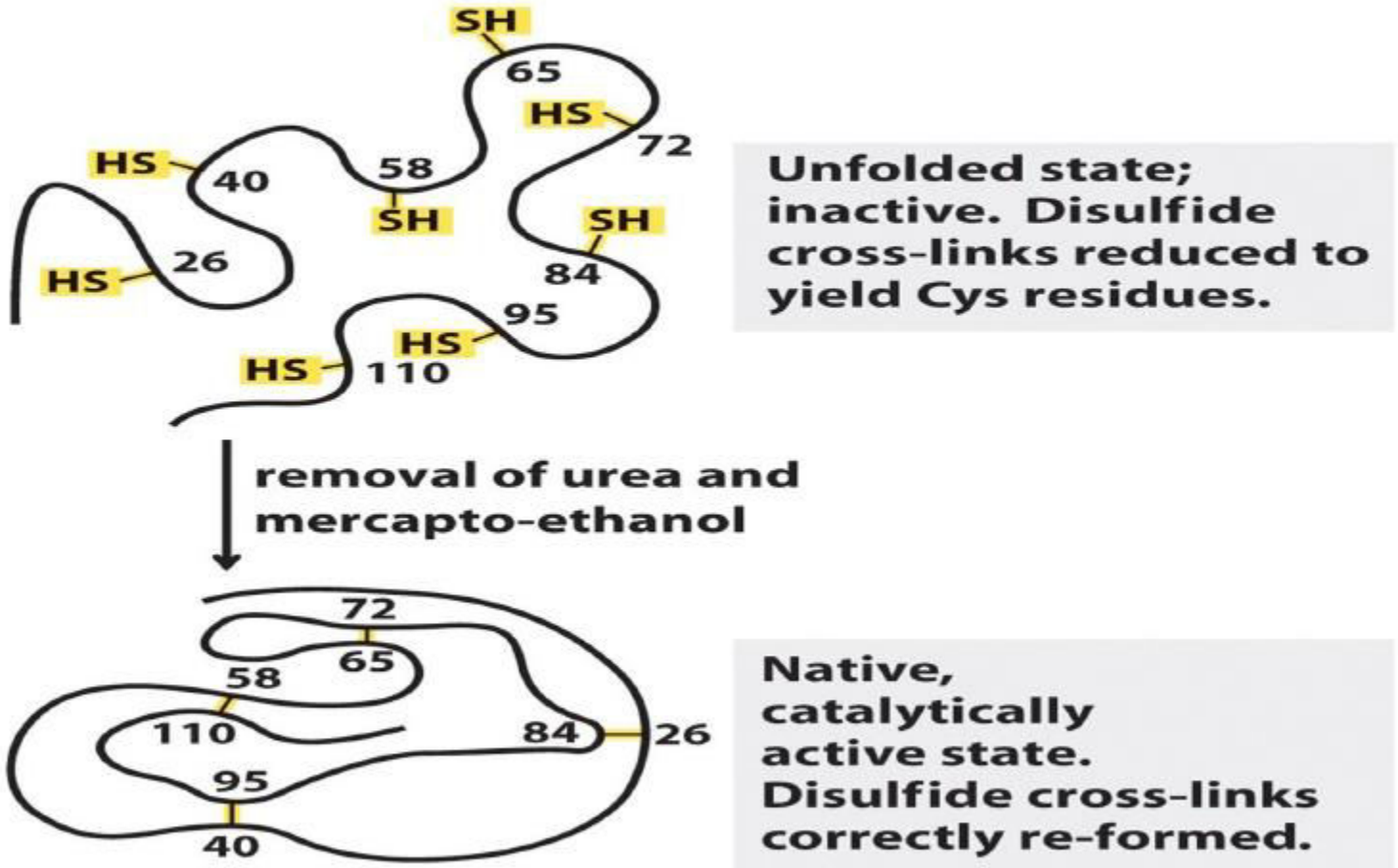
Protein Denaturation



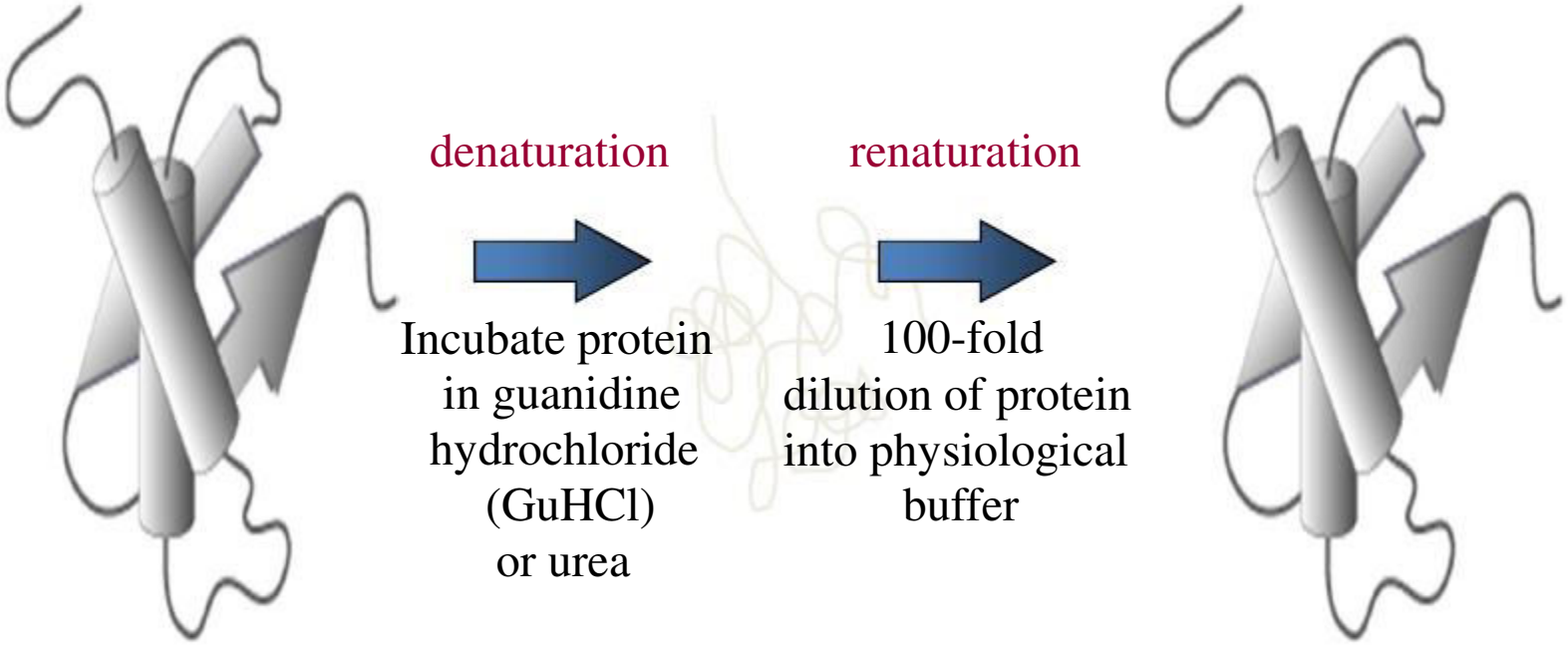
Protein Denaturation

- When the urea and the reducing agent are removed, the randomly coiled, denatured ribonuclease spontaneously refolds into its correct tertiary structure, restoring its catalytic activity

Protein Denaturation



Protein Denaturation



Protein Denaturation

- This proofs that denaturation of some proteins is reversible.
- Which tells us that primary structure of a protein determines its three-dimensional conformation.

end

Reversible binding of a protein

Reversible binding of
a protein to a ligand
Oxygen-Binding Proteins

Reversible binding of a protein

- The functions of many proteins involve the reversible binding of other molecules.
- A molecule bound reversibly by a protein is called a **ligand.**
- A ligand may be any kind of molecule, including another protein.

For example, in oxygen transport oxygen is the ligand that binds with hemoglobin, antigen

Reversible binding of a protein

- The transient nature of protein ligand interactions is critical to life, allowing an organism to respond rapidly and reversibly to changing environmental and metabolic circumstances.

ligand is able to bind to proteins by weak interactions such as ionic bonds, hydrogen

Reversible binding of a protein

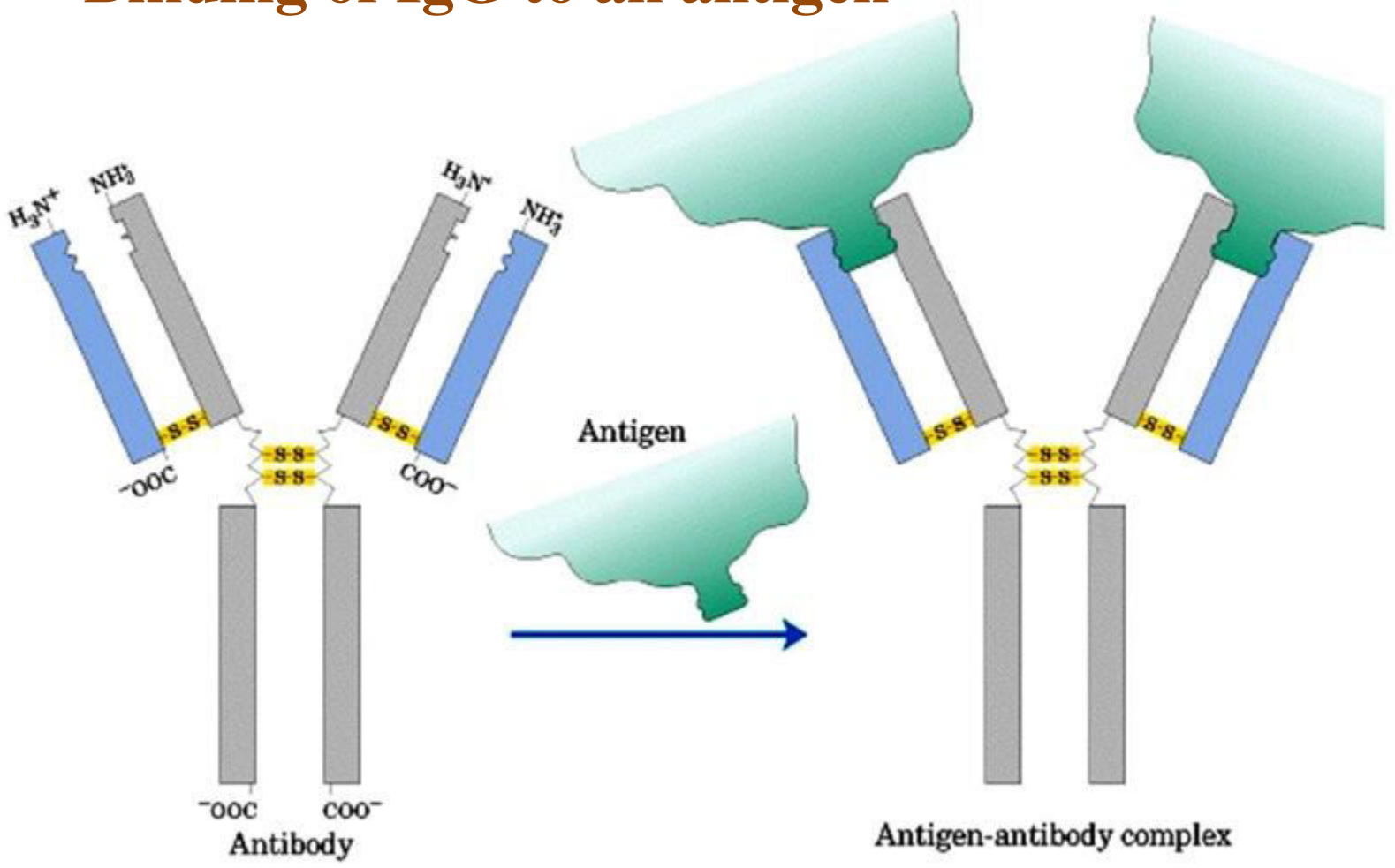
- A ligand binds at a site on the protein called **the binding site**, which is complementary to the ligand in
  - size,
  - shape,
  - charge,
  - hydrophobicity

Reversible binding of a protein

- This *complementarity* of protein ligand binding gives the interaction a great deal of specificity.

Reversible binding of a protein

Binding of IgG to an antigen



Reversible binding of a protein

- The protein can discriminate among thousands of different molecules in its environment and selectively bind only one or a few similar compounds.

Reversible binding of a protein

- A given protein may have separate binding sites for several different ligands

Reversible binding of a protein

- The binding of a protein and Iigand is often coupled to a conformational change in the protein that makes the binding site more complementary to the ligand, permitting tighter binding

Reversible binding of a protein

- The structural adaptation that occurs between protein and Iigand is called **induced fit.**

Reversible binding of a protein

Summary

1. Reversible binding of ligands is essential for the function of gloubular proteins
   - – Specificity of ligands and binding sites
   - – Ligand binding is often coupled to conformational changes, sometimes quite dramatic (Induced Fit)

Reversible binding of a protein

Summary

1. In multisubunit proteins, conformational changes in one subunit can affect the others (Cooperativity)
   - Interactions can be regulated
2. Illustrated by:
   - Hemoglobin, antibodies, and muscle contraction

end

Reversible binding of a protein

- **Oxygen-Binding Proteins**
- **Myoglobin and hemoglobin**

Reversible binding of a protein

- These two molecules illustrate almost every aspect of that most central of biochemical processes:
- The reversible binding of a ligand to a protein

Reversible binding of a protein

- Oxygen is poorly soluble in aqueous solutions and cannot be carried to tissues in sufficient quantity if it is simply dissolved in blood plasma.

Reversible binding of a protein

- Diffusion of oxygen through tissues is also ineffective over distances greater than a few millimeters
- The evolution of larger, multicellular animals depended on the evolution of proteins that could transport and store oxygen

Reversible binding of a protein

- Myoglobin is composed of a single polypeptide chain that has one O_2 binding site.
- Hemoglobin is a tetramer in which each subunit has a strong sequence homology to myoglobin and contains an O_2binding site.

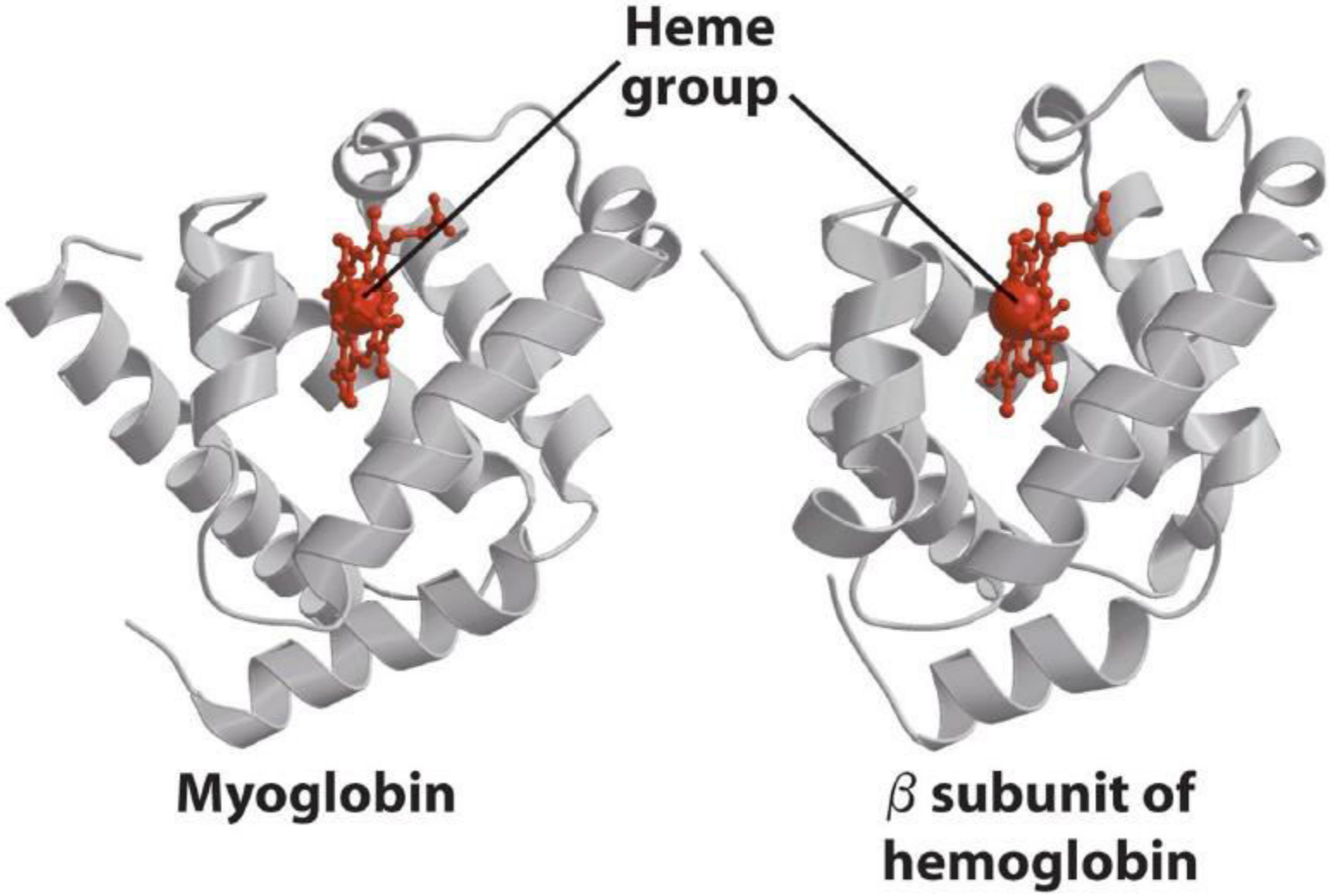
Reversible binding of a protein

- Myoglobin and hemoglobin are two oxygen-binding proteins with a very similar
- primary structure

Reversible binding of a protein

- Hemoglobin is a tetramer composed of two different types of subunits (2α and 2β polypeptide chains, referred to as two αβ protomers
- Each subunit has a strong sequence homology to myoglobin.

Reversible binding of a protein



Reversible binding of a protein

- We will compare the;
- structure and
- functional relation of each protein in the subsequent discussion to see how both these proteins are suited to perform their required function in their relative sites in the body.

end

Reversible binding of a protein

- The tertiary structure of oxygen binding globins consists of;
- Eight α-helices.
- The helices create a hydrophobic O_2 binding pocket containing;
- tightly bound heme with an iron atom **Ferrous (Fe^{2+})** in its center.

Reversible binding of a protein

- This iron bound heme is an example of prosthetic group.
- **A prosthetic group** is a compound permanently associated with a protein and contributes to the protein's function.

The prosthetic group may be organic (vitamin, sugar, or lipid) or inorganic (like metal ion),

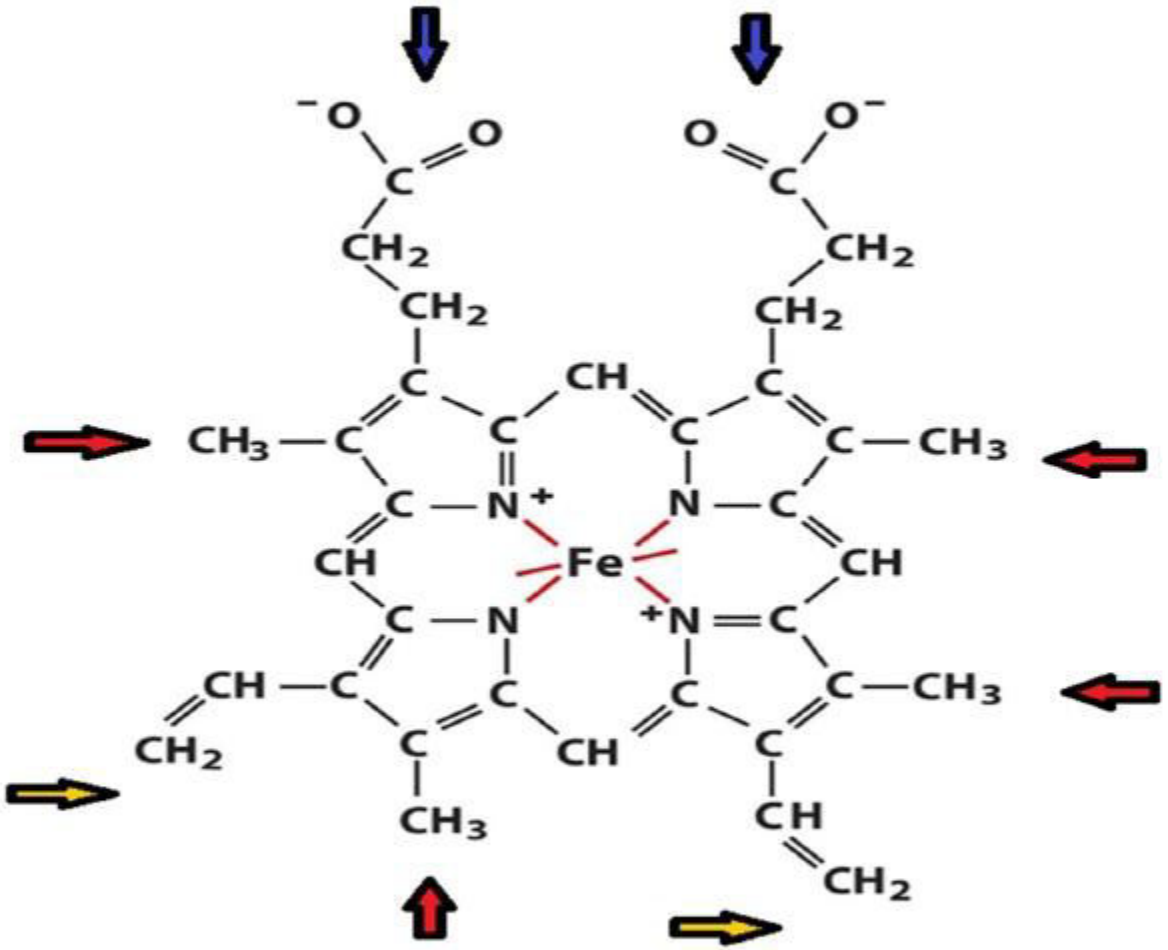
Reversible binding of a protein

- Heme consists of a complex organic ring structure, **protoporphyrin IX**, to which is bound a single iron atom in its **ferrous (Fe^{2+}) state**

Reversible binding of a protein

- The iron atom has six bonds,
- four to nitrogen atoms that are part of the flat porphyrin ring system
- and two perpendicular to the porphyrin

Reversible binding of a protein



Negatively charged propionate (blue) groups= arginine and histidine side hydrophobic methyl(red) and

Reversible binding of a protein

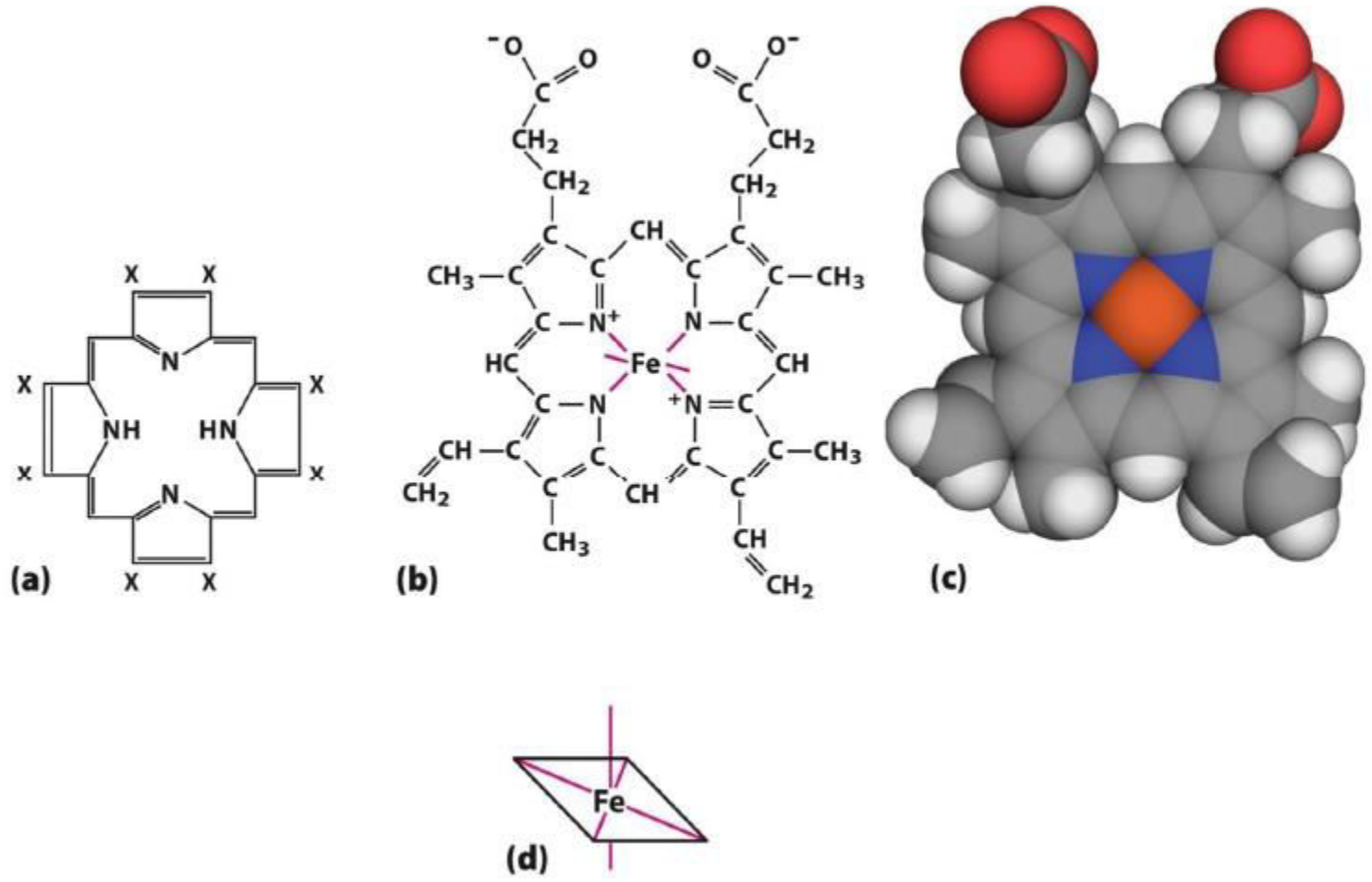


Figure 5-1
Lehninger Principles of Biochemistry, Sixth Edition
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Reversible binding of a protein

- Negatively charged propionate groups on the porphyrin ring interact with;
- Arginine and
- Histidine side chains from the hemoglobin.

Reversible binding of a protein

- The hydrophobic;
- methyl and
- vinyl groups
- interact with hydrophobic amino acid side chains from hemoglobin.

END

Reversible binding of a protein

- Iron in the *Fe^{2+} state binds oxygen reversibly*
- But in the Fe^{3+} (Ferric) state it does not bind oxygen

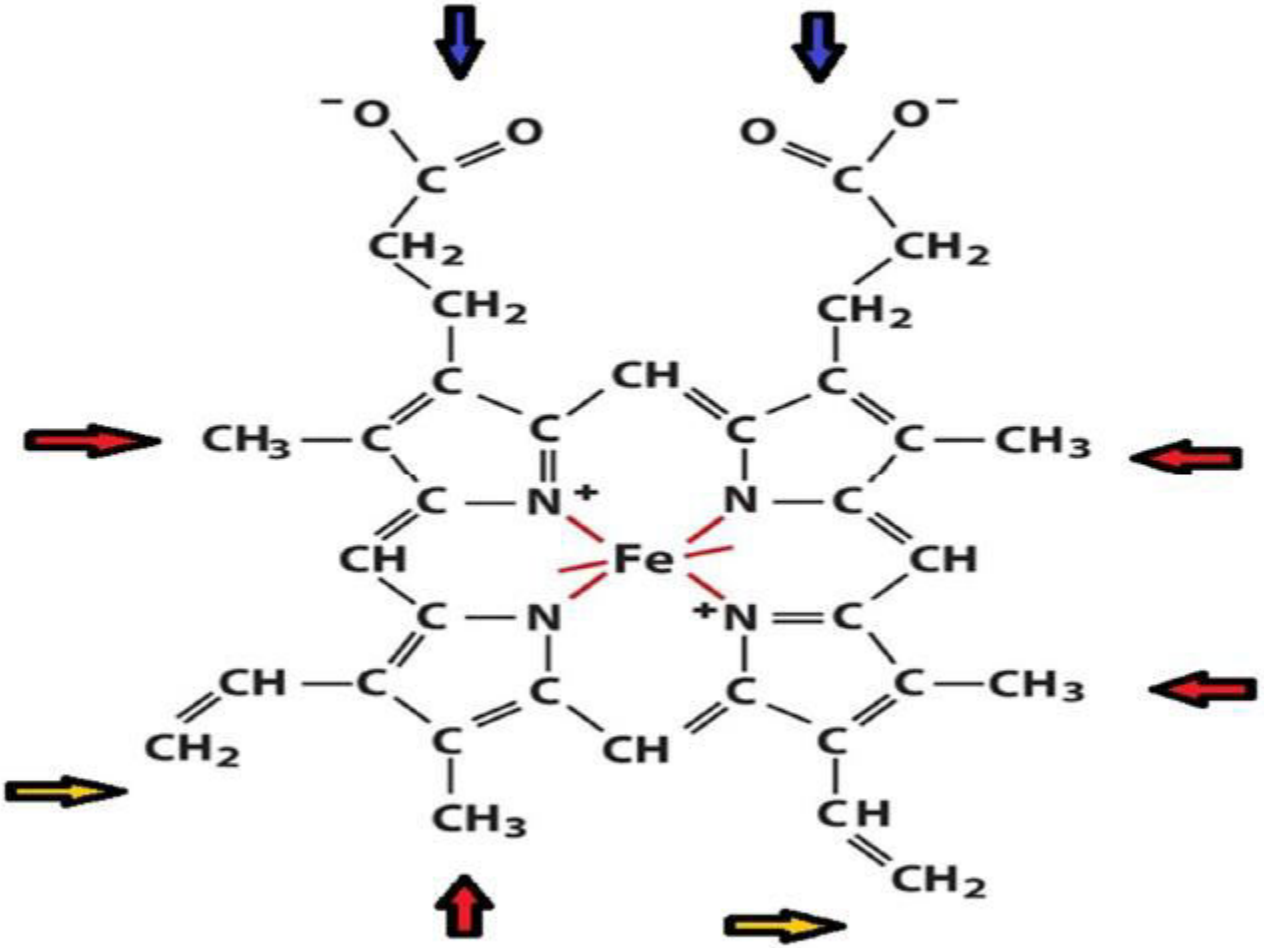
Reversible binding of a protein

- The iron atom of heme has six coordination bonds
- four in the plane of, and bonded to, the flat porphyrin ring system with Nitrogen
- Where as two perpendicular to it.

Reversible binding of a protein

- Therefore free heme molecules (heme not bound to protein) leave Fe^{2+} with two "open“ coordination bonds

Reversible binding of a protein

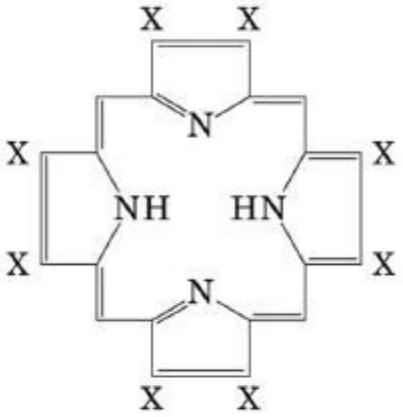


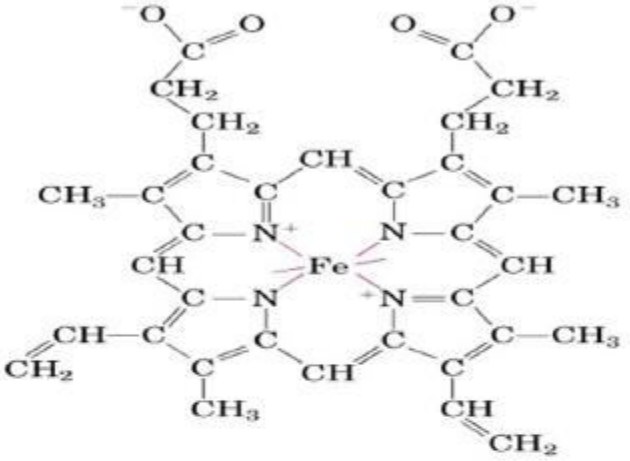
Negatively charged propionate (blue) groups= arginine and histidine side hydrophobic methyl(red) and vinyl(yellow) interact with hydrophobic amino acid side chains

Reversible binding of a protein

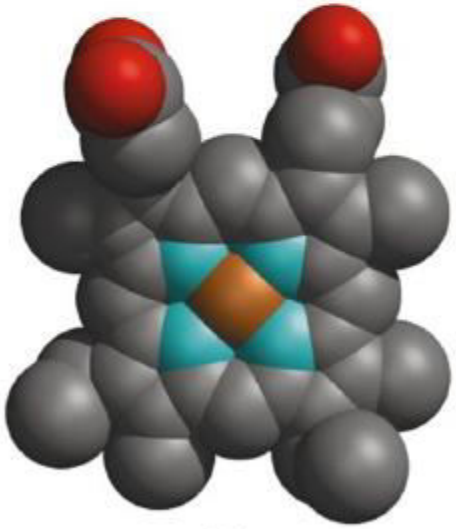
Heme:

protoporphyrin

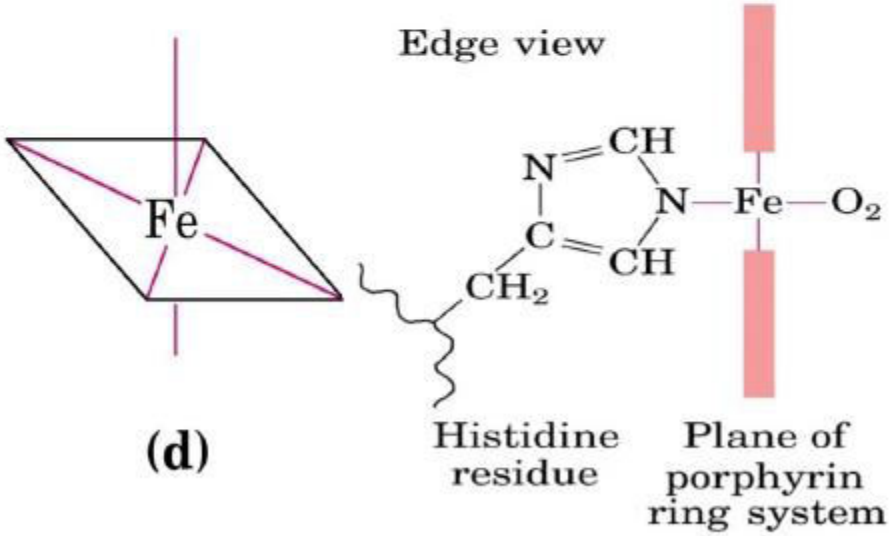




(a)

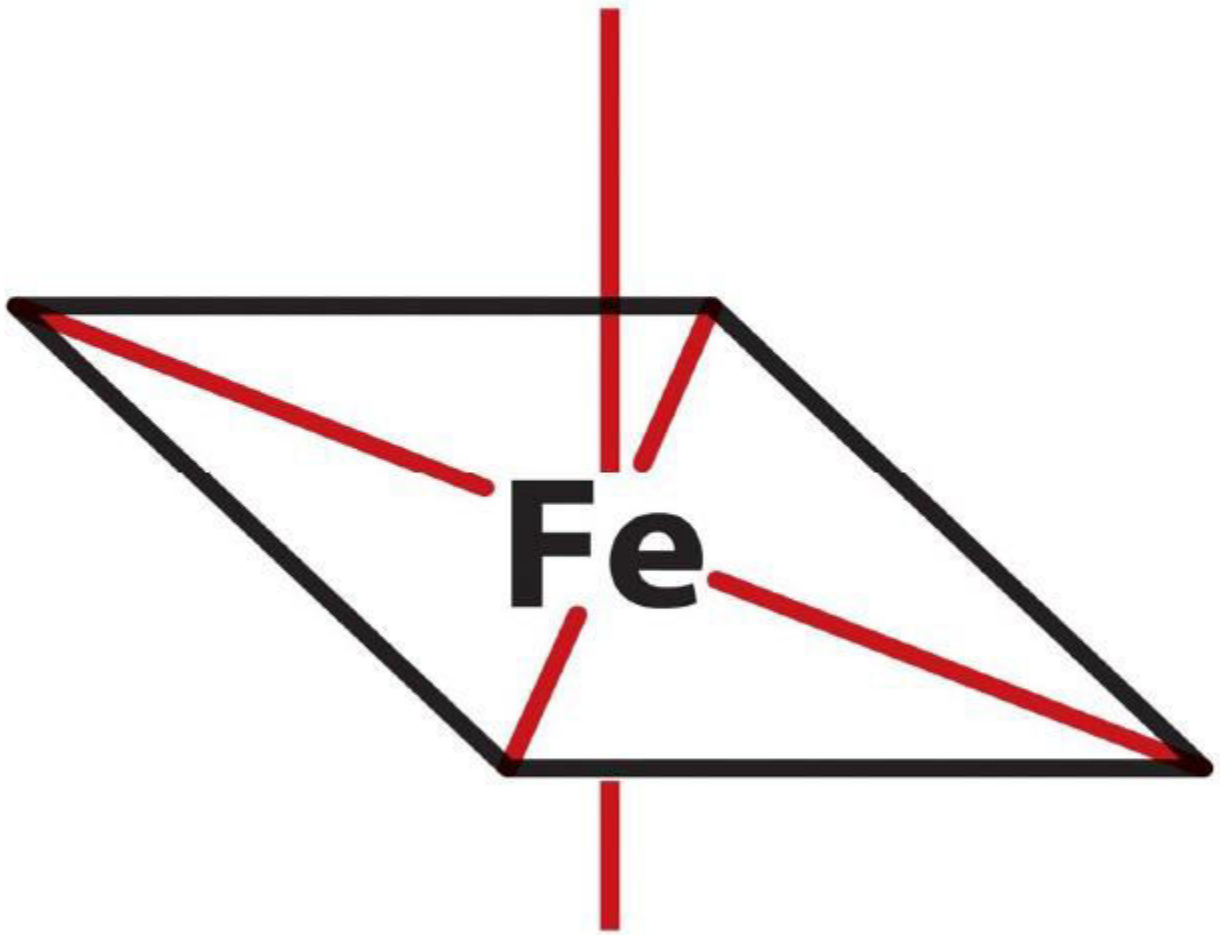


(c)



(d)

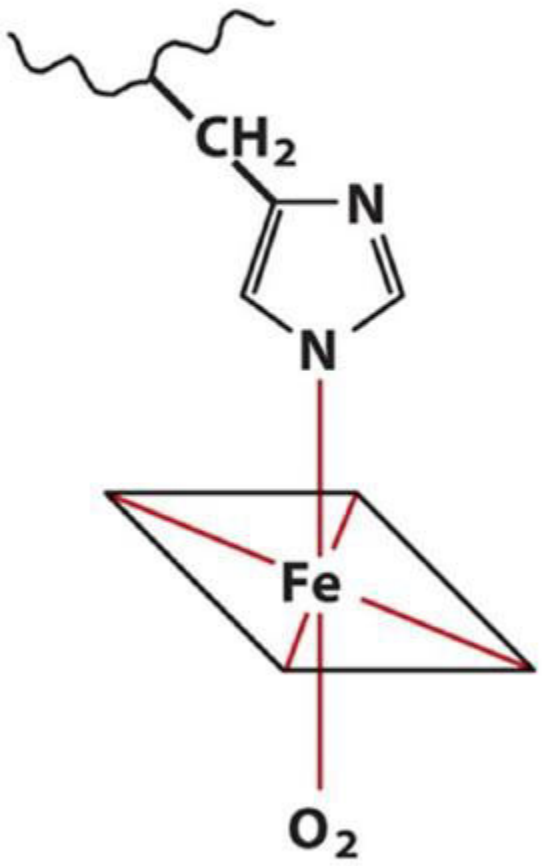
Reversible binding of a protein



Reversible binding of a protein

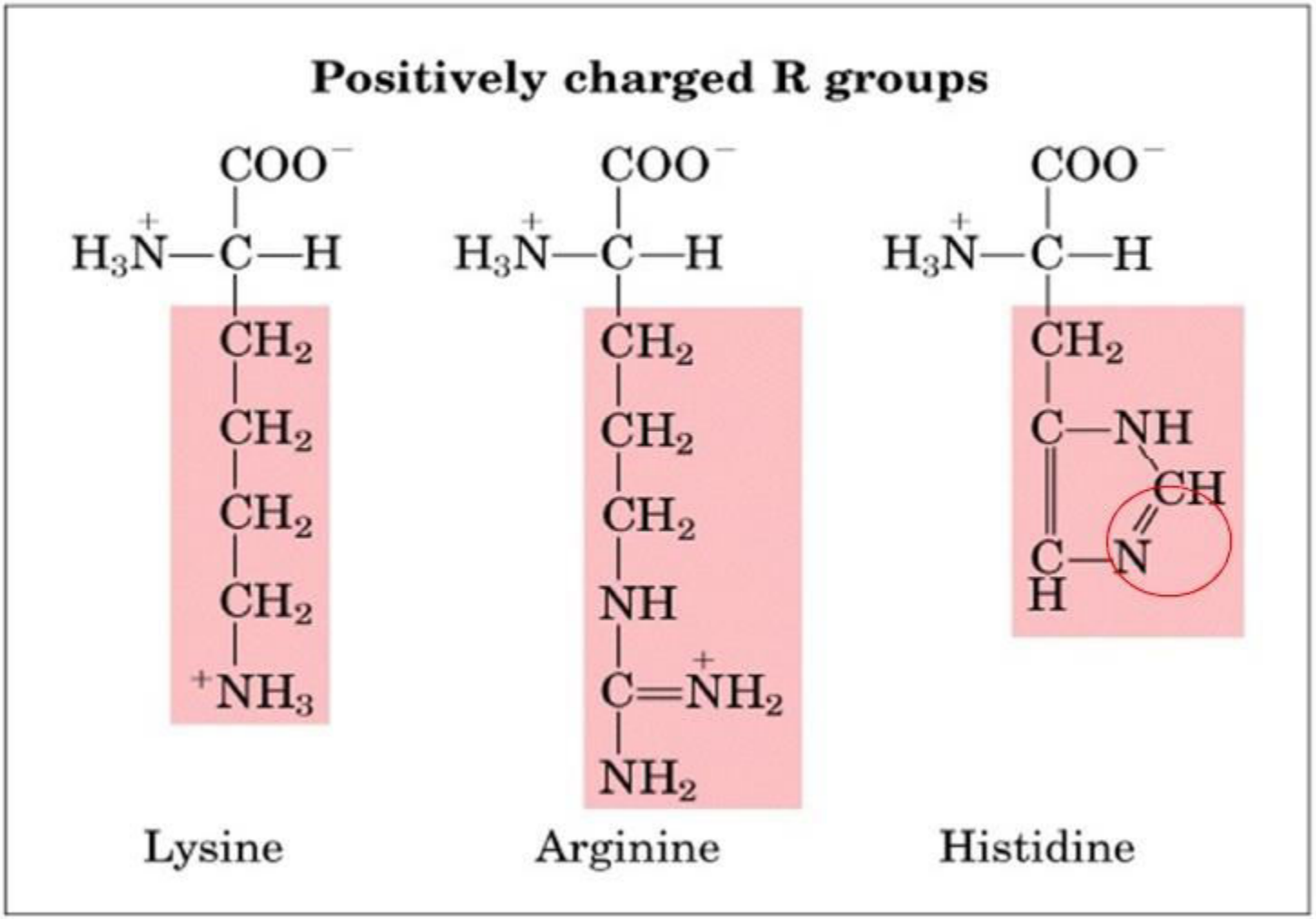
- **One** of these two coordination bonds is occupied by a side chain nitrogen of a **His residue**
- The **other** is the binding site for molecular oxygen **$(O)_2$**

Reversible binding of a protein



- **One** of these two coordination bonds is occupied by a side chain nitrogen of a **His residue**
- The **other** is the binding site for molecular oxygen **$(O)_2$**

Reversible binding of a protein



Negatively charged propionate (blue) groups= arginine and histidine side

END

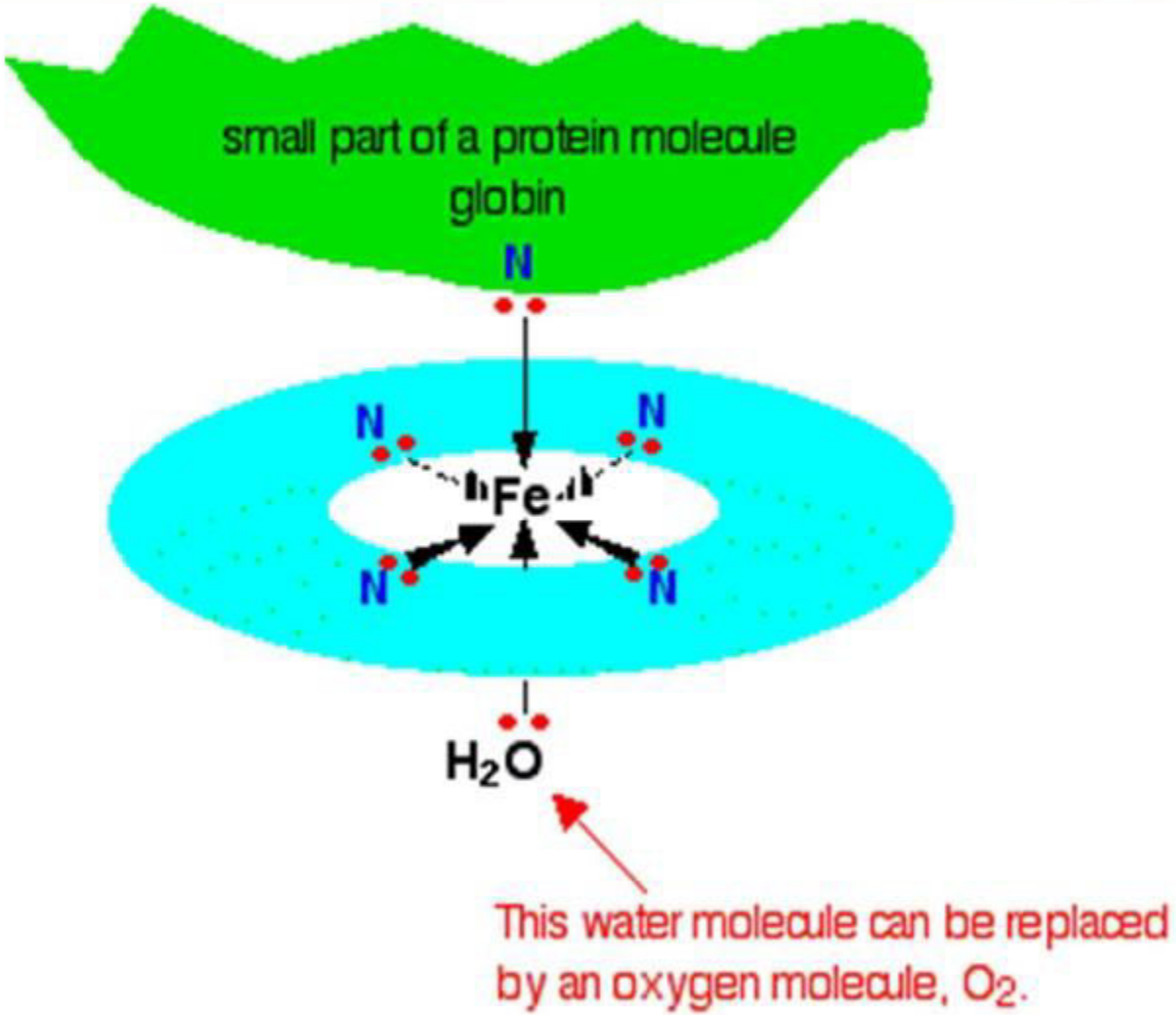
Reversible binding of a protein

- Iron in the *Fe^{2+} state binds oxygen reversibly*
- But in the Fe^{3+} (Ferric) state it does not bind oxygen

Reversible binding of a protein

- The coordinated nitrogen atoms (which have an electron-donating character) help in **preventing** the conversion of the heme iron to the ferric (Fe^{3+}) state

Reversible binding of a protein

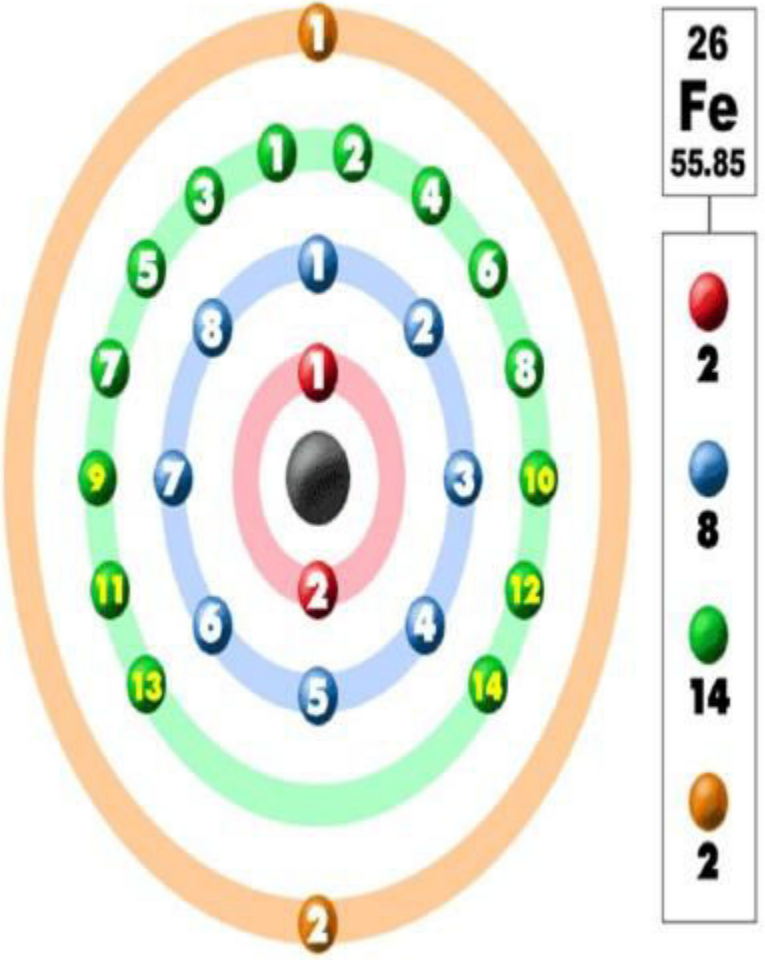


Reversible binding of a protein

- The conversion Fe^{2+} to Fe^{3+} is much more likely in *free* heme,
- Sequestering each heme deep with in the protein structure further helps to prevent this reaction.

Reversible binding of a protein

Iron is a transition metal & because its orbitals are so close energy wise they tend to give up either 2 or 3 electrons at time



Reversible binding of a protein

Problem of Octet rule

- Octet rule cannot be used for type II cations (most of the transition and inner transition elements).

Fe^{2+} Fe^{3+}

Cu^{1+} Cu^{2+}

Reversible binding of a protein

- The physiological oxidation state of an iron atom has a positive two charge (**ferrous ion**) instead of three charge (**ferric ion**)
- And it is too large to fit into the plane of protoporphyrin

Reversible binding of a protein

- However, when iron oxidized from ferrous ion (Fe^{2+}) to ferric ion (Fe^{3+}), there is loss of one extra electron, forces between protons and electrons increases so that the electron cloud will penetrate more towards to the nucleus

Reversible binding of a protein

- As a result, the ferric ion (Fe^{3+}) has a smaller size then ferrous ion (Fe^{2+}) and fits into the protoporphyrin plane

- This deep impregnation of Fe^{3+} in the protoporphyrin results in inability of Fe^{3+} to bind with oxygen.

Reversible binding of a protein

- Some small molecules, such as carbon monoxide **(CO)** and nitric oxide **(NO),** coordinate to heme iron with greater affinity than does **O_2**

Reversible binding of a protein

- When a molecule of CO is bound to heme, O_2 is excluded, that is why CO is highly toxic to aerobic organisms
- By surrounding and sequestering heme, oxygen-binding proteins regulate the access of CO to the heme iron.

END

Reversible binding of a protein

- About 78% of the amino acid residues in the **myoglobin** are found in **α helices** with bends in between.
- The helical segments are named A through H.(eight)
- The bends are designated AB, CD, EF, and so forth.

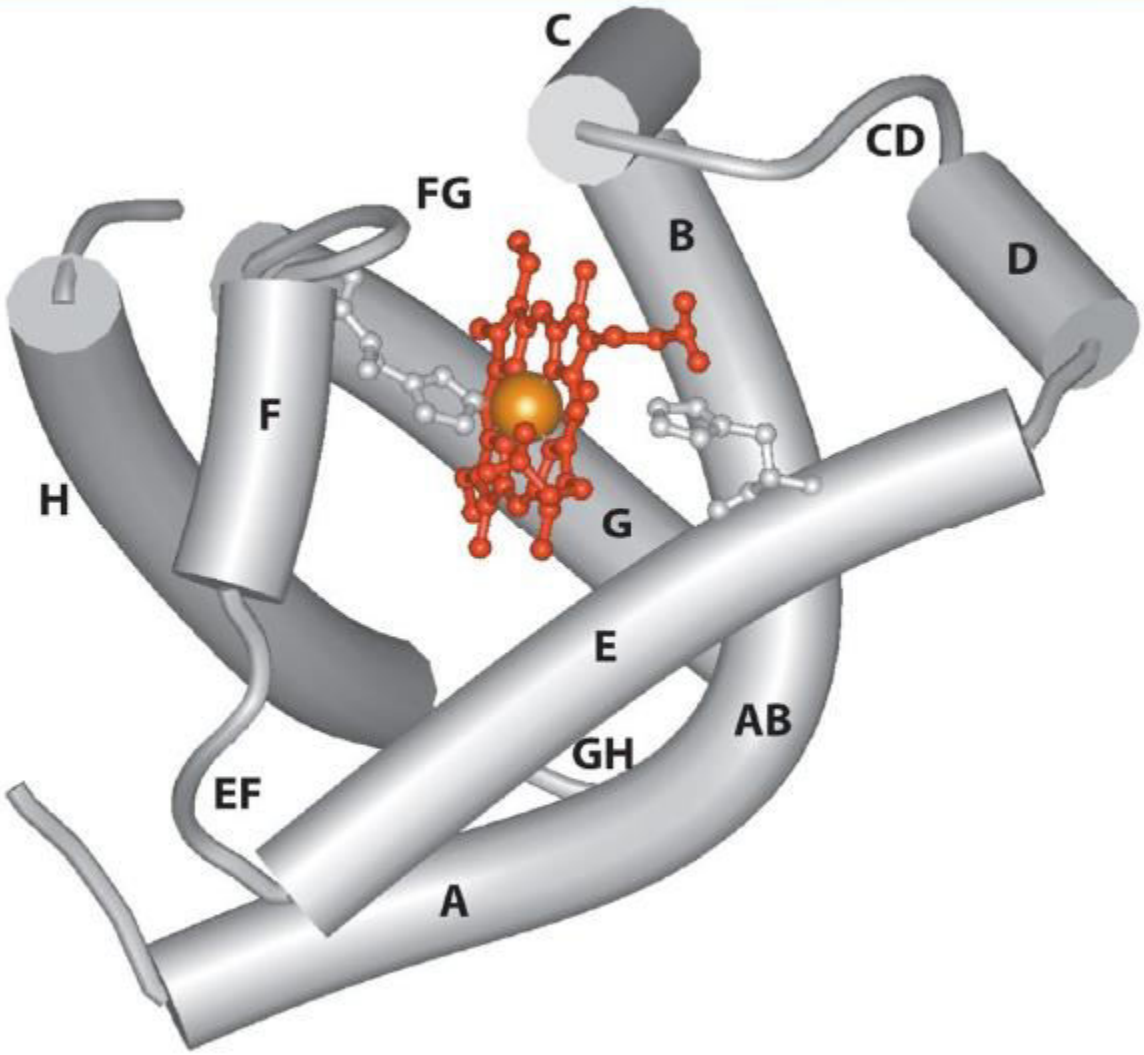
Reversible binding of a protein

- An individual amino acid residue is designated either by its position in the amino acid sequence
- or by its location in the sequence of a particular α-helical segment

Reversible binding of a protein

- For example, the His residue coordinated to the heme in myoglobin, **His93** (the 93rd residue from the amino-terminal end of the myoglobin polypeptide sequence), is also called **His F8** (the 8th residue in a helix F)

Reversible binding of a protein



Reversible binding of a protein

- Oxygen stored in red muscle myoglobin is released during O_2 deprivation (eg, severe exercise) for use in muscle mitochondria for aerobic synthesis of ATP.

Reversible binding of a protein

- The surface of myoglobin is polar, important for interacting with polar aqueous environment of cytosol.

Reversible binding of a protein

- The interior contains only nonpolar residues such as
- Leu,
- Val,
- Phe, and
- Met.
- with two notable exceptions—

Reversible binding of a protein

- The exceptions are **His E7** and **His F8**, the seventh and eighth residues in helices E and F,
- Which lie close to the heme iron, where they stabilize heme and help in O_2 binding.

END

Histidine is a generally considered to be a *polar* amino acid

Reversible binding of a protein

- The cervice (Pocket)created by nonpolar amino acids in the interior of myoglobin create a binding pocket for heme.

Leu, Val, Phe, and Met.

Reversible binding of a protein

- **The proximal histidine F8**, binds directly to the iron of heme
- Whereas **distal histidine E7**, does not directly interact with the heme, but helps stabilize the binding of oxygen to the ferrous iron.

Reversible binding of a protein

- In myoglobin and hemoglobin, heme is covalently linked with **histidine F8**(eighth residue of F helix). because of covalent bond this histidine is **closer to heme** iron and named as proximal histidine (closer histidine),

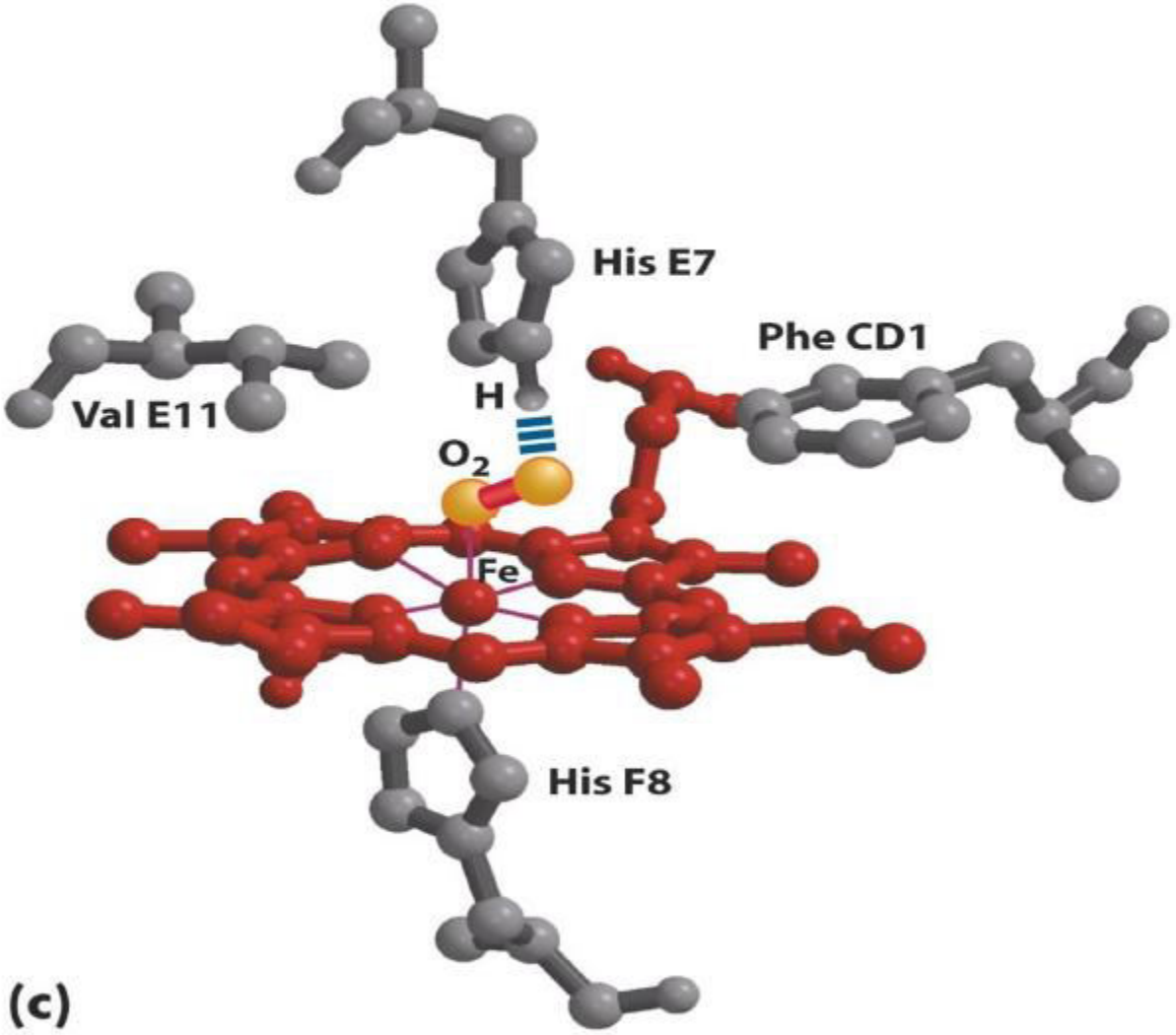
Reversible binding of a protein

- While other key histidine which responsible for stabilization of oxygen in **E7** (seventh residue of E helix) is **far from heme iron** so named as distal

Reversible binding of a protein

- The protein, or globin portion of myoglobin thus creates a special microenvironment for the heme that permits the reversible binding of one oxygen molecule (oxygenation)

Reversible binding of a protein



Reversible binding of a protein

- The physiological importance of oxygen stabilization of the distal histidine (E7) lies in the fact that it prevents the binding of CO which is a low level product of cellular metabolism.

END

Reversible binding of a protein

Structure and function of hemoglobin

- Hemoglobin is found exclusively in red blood cells, where its main function is to transport oxygen from the lungs to the capillaries of the tissues

Reversible binding of a protein

- **Hemoglobin A,** the major hemoglobin in adults, is composed of four polypeptide chains
  - **Two alpha (α) chains**
  - **Two beta (β) chains**

HbF: In infants, made up of 2 alpha chains and 2 gamma chains. **HbA2:** ($alpha_{2}delta_{2}$)

Reversible binding of a protein

- Which can be seen as a protein of two identical dimers **$(\alpha\beta)_1$ and $(\alpha\beta)_2$**
- The two polypeptide chains within each dimer are held tightly together, primarily by hydrophobic interactions

Reversible binding of a protein

- Inter-chain hydrophobic interactions form strong associations between α-subunits and β-subunits in the dimers

Reversible binding of a protein

- Ionic and hydrogen bonds also occur between the members of the dimer
- In contrast, the two dimers are able to move with respect to each other, being held together primarily by polar bonds

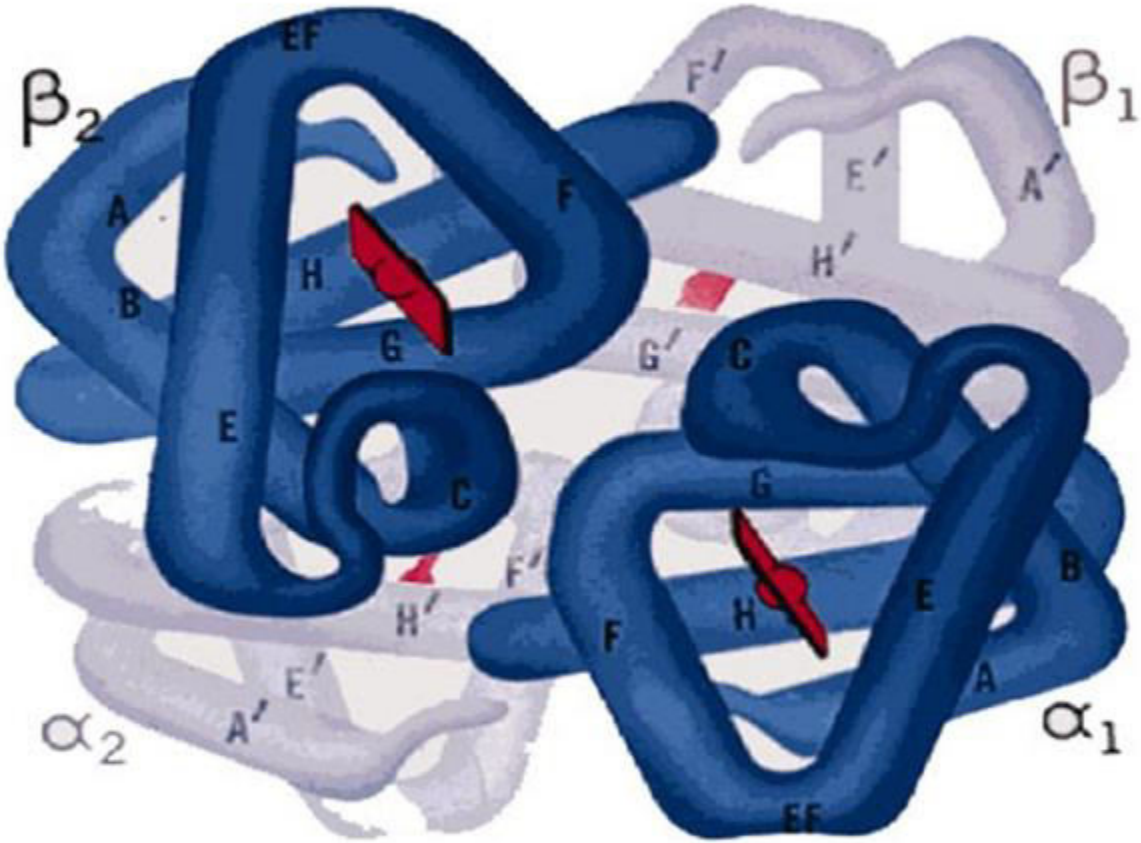
Reversible binding of a protein

- Each subunit has stretches of α-helical structure,
- And a binding pocket similar to that described for myoglobin

Reversible binding of a protein

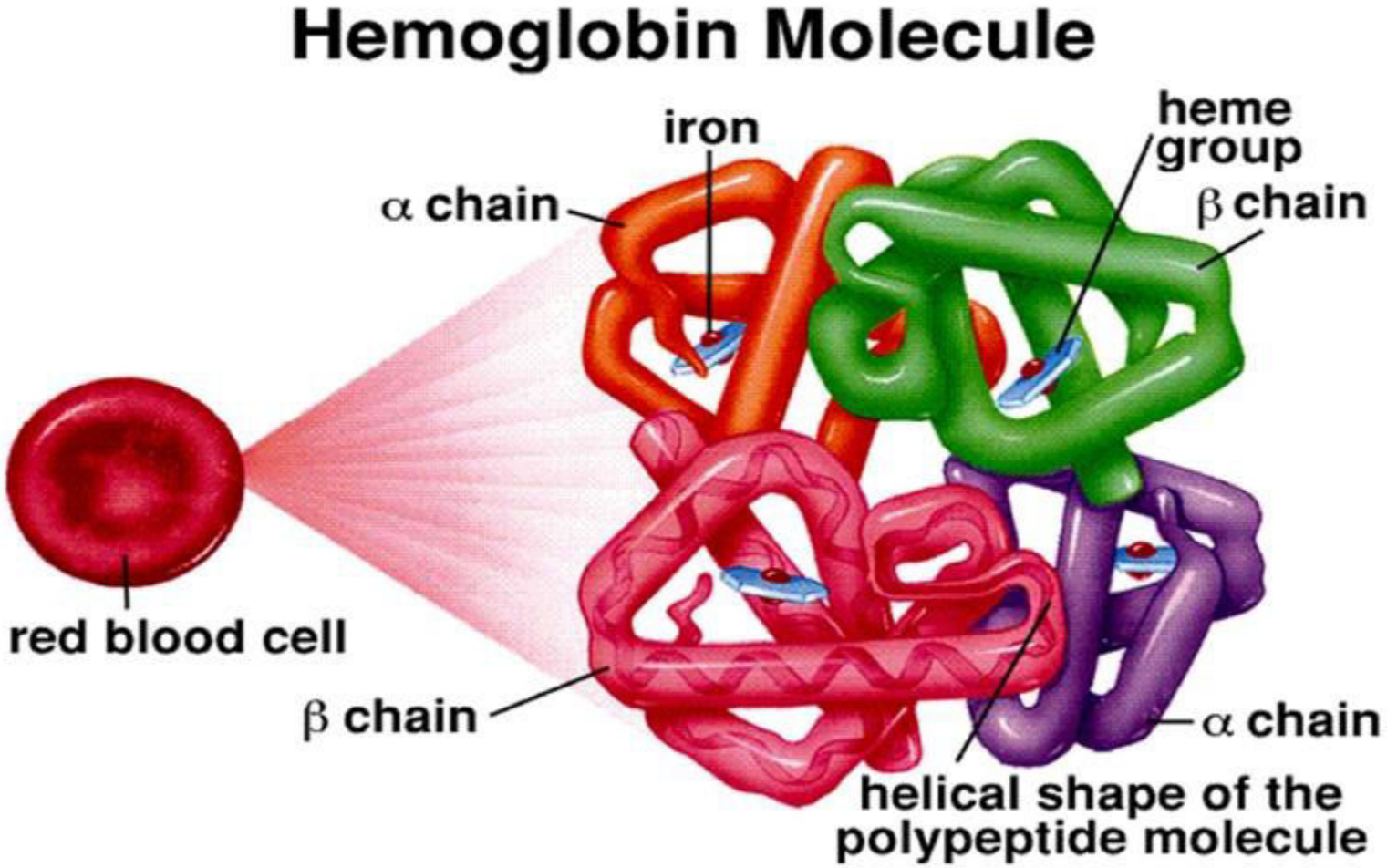
- Each subunit has stretches of α-helical structure,
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Reversible binding of a protein



Reversible binding of a protein

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Reversible binding of a protein

- However, the tetrameric hemoglobin molecule is structurally and functionally more complex than myoglobin

Reversible binding of a protein

- For example, hemoglobin can transport CO_2 from the tissues to the lungs, and carry four molecules of O_2 from the lungs to the cells of the body

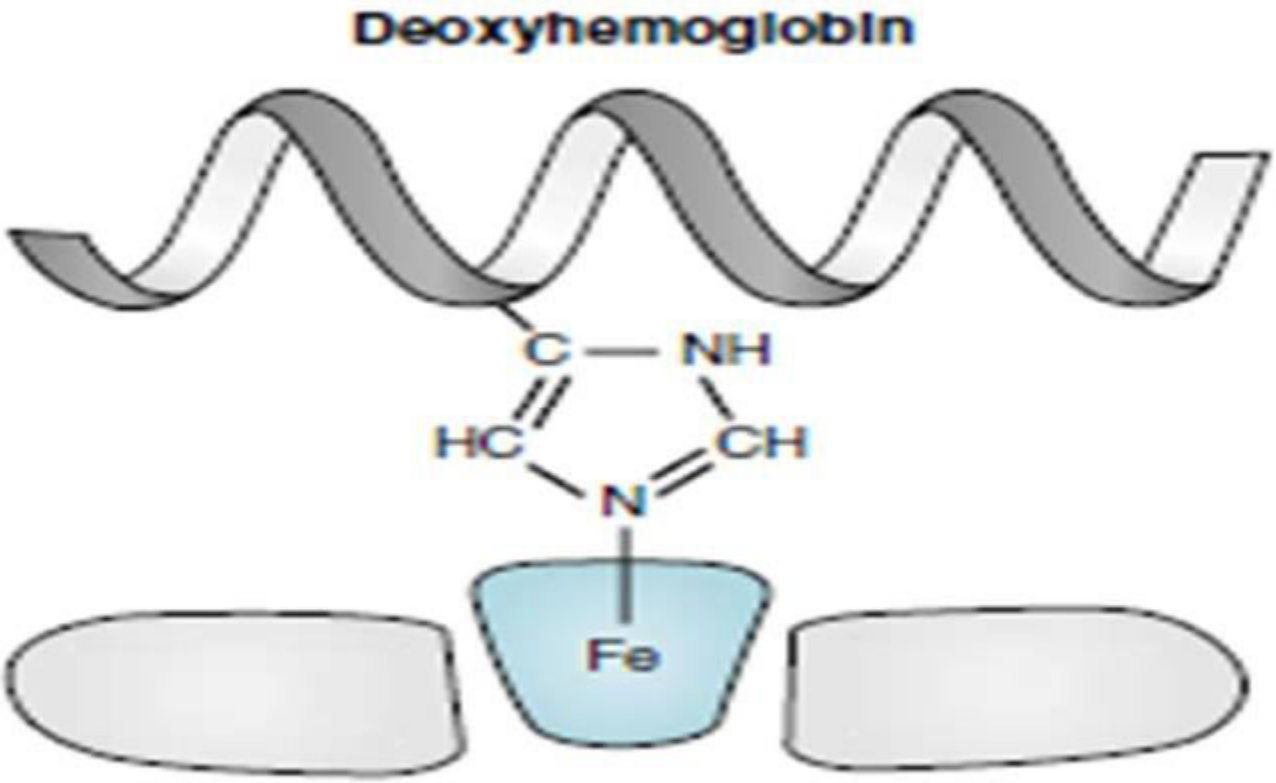
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Reversible binding of a protein

- The proximal **His (F8** histidine)of myoglobin and hemoglobin is sterically repelled by the heme porphyrin ring.
- Thus, when the His binds to the **Fe^{2+}** in the middle of the ring, it pulls the **Fe^{2+}** above the plane of the ring.

The interior contains only nonpolar residues such as Leu,Val, Phe, Met.with exceptions of His (polar)

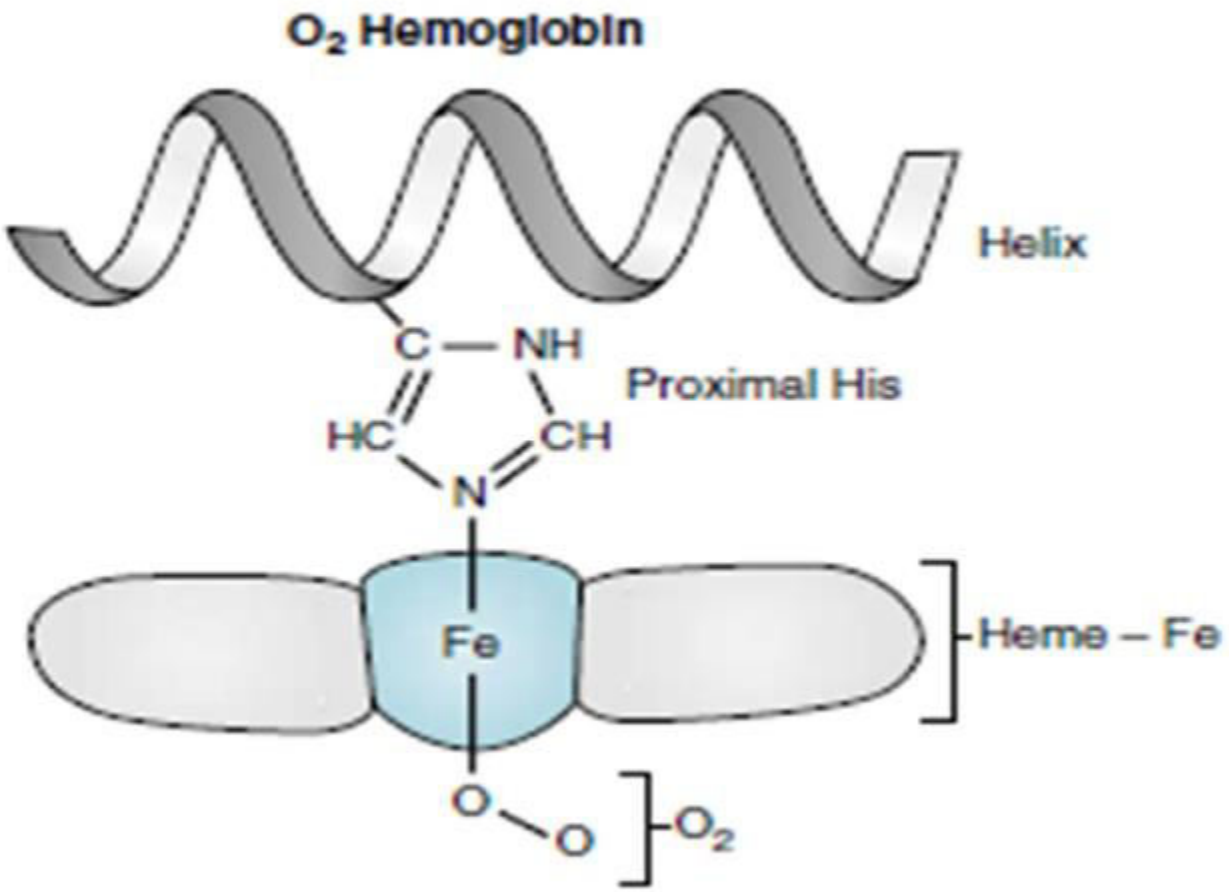
Reversible binding of a protein



Reversible binding of a protein

- When **oxygen** binds on the other side of the ring, it **pulls** the **Fe^{2+}** back into the plane of the ring.
- The **pull of O_2** binding **moves** the proximal **histidine** toward the porphyrin ring, which moves the helix containing the proximal histidine.

Reversible binding of a protein



Reversible binding of a protein

- This conformational change has no effect on the **function** of **myoglobin**.

Reversible binding of a protein

- However, in **Hemoglobin** this change is imperative for co-operative binding of oxygen to different subunits of hemoglobin.

Reversible binding of a protein

- Shifting of amino acids by the oxygenation of one of the heme groups in the protein, (due to pull of porphyrin through His), alters the structure of the interfaces between the four subunits. This results in the change of shape of the whole protein.

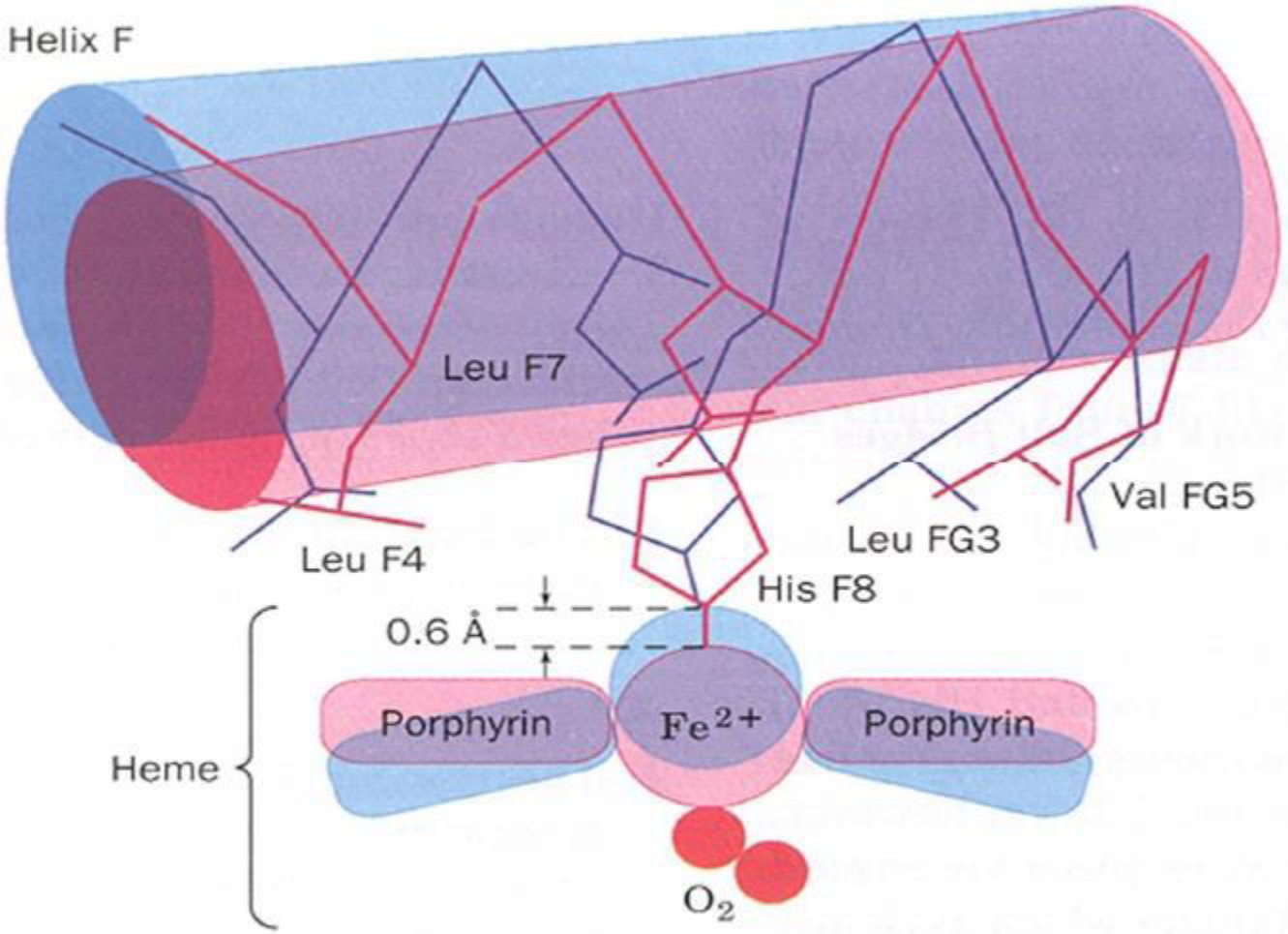
Reversible binding of a protein

- In the new shape, it is easier for the other three heme groups to become oxygenated.

Reversible binding of a protein

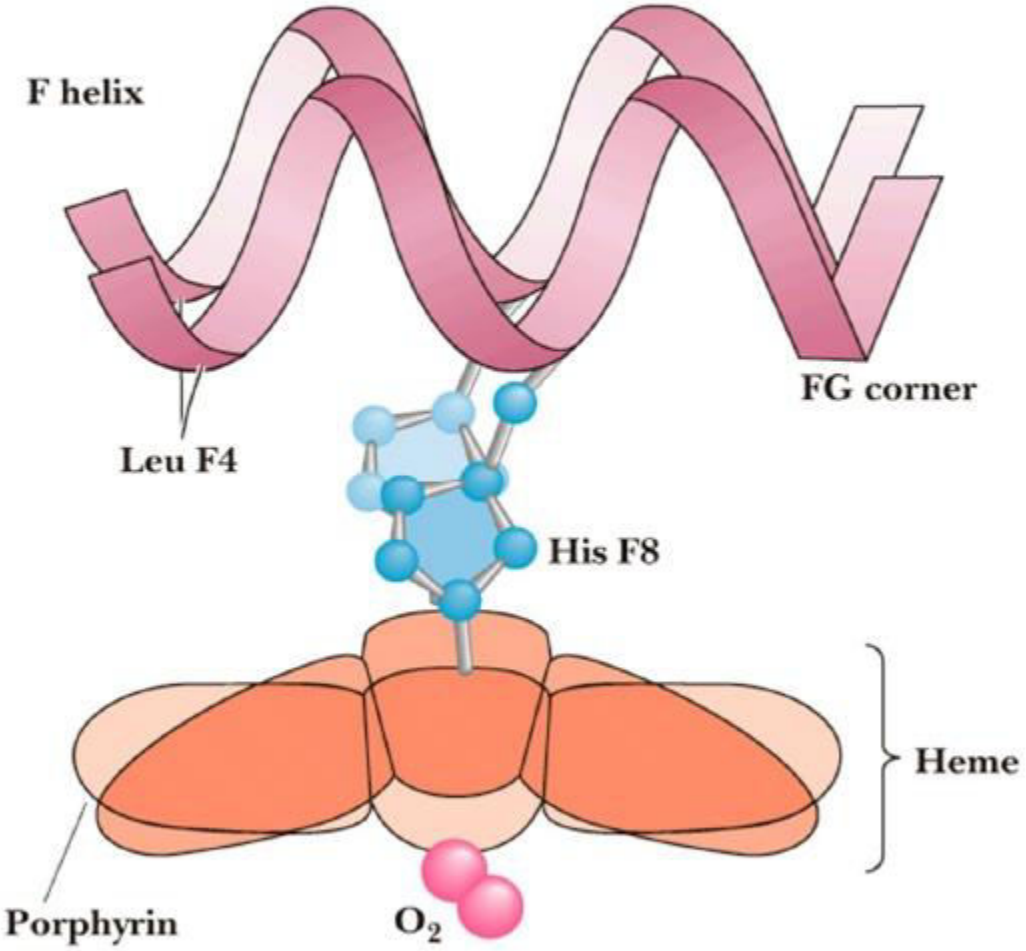
- Thus, the binding of one molecule of O_2 to hemoglobin enhances the ability of hemoglobin to bind more O_2 molecules.
- This property of hemoglobin is known as "cooperative binding."

Reversible binding of a protein



Red : Oxygenated **Blue: Deoxygenated**

Reversible binding of a protein



Reversible binding of a protein

- Cooperative binding of oxygen means that the binding of oxygen to one heme group enables an oxygen binding to the second heme group of the same hemoglobin molecule.

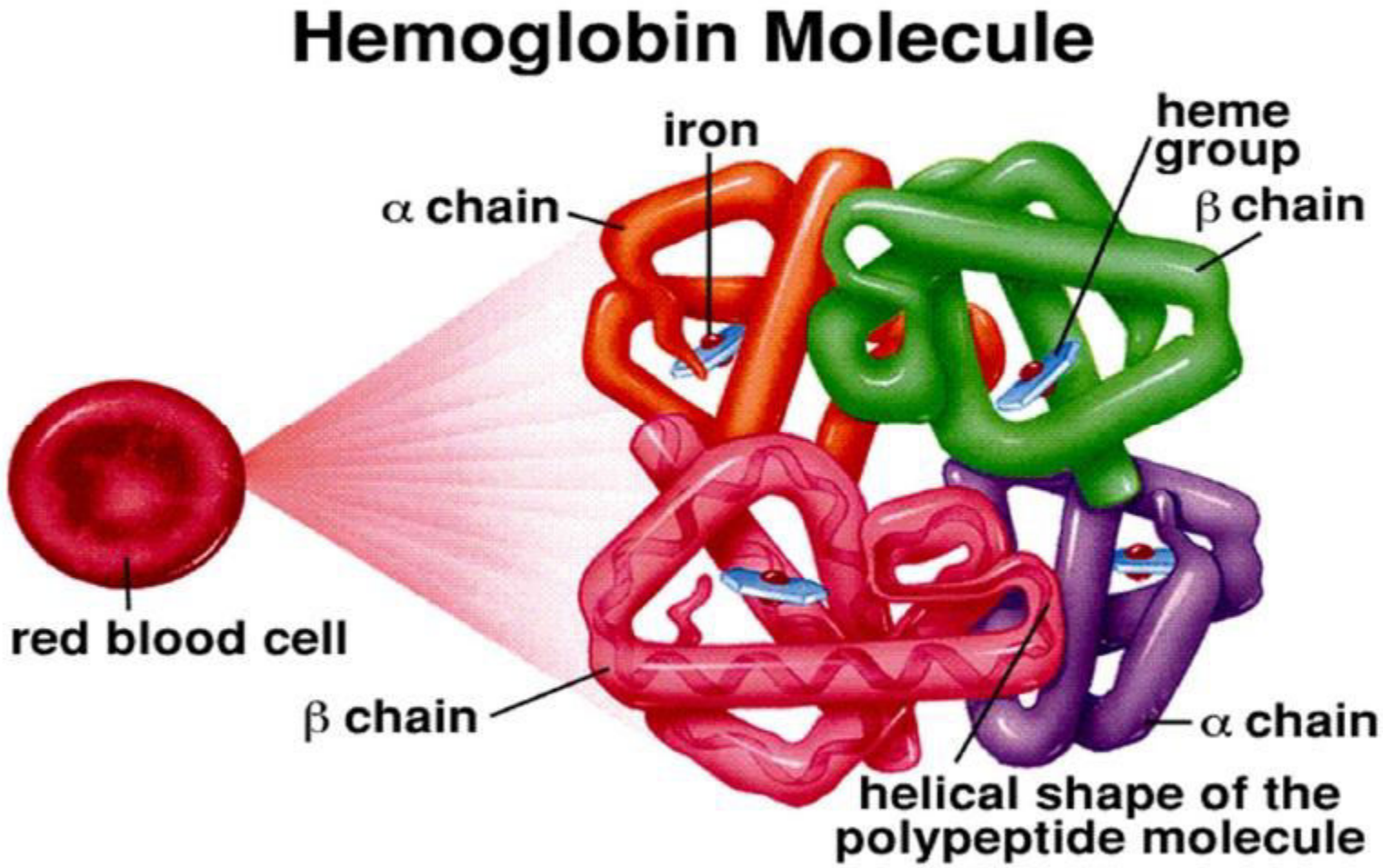
Reversible binding of a protein

- Conversely, the oxygen release from one heme group facilitates the release of oxygen from other heme groups in the same hemoglobin molecule.

END

Reversible binding of a protein

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END

Reversible binding of a protein

- In **hemoglobin**, the movement of one helix after oxygen binding leads to the movement of other helices in that subunit, including one in a corner of the subunit that is in contact with a different subunit through ionic interactions.

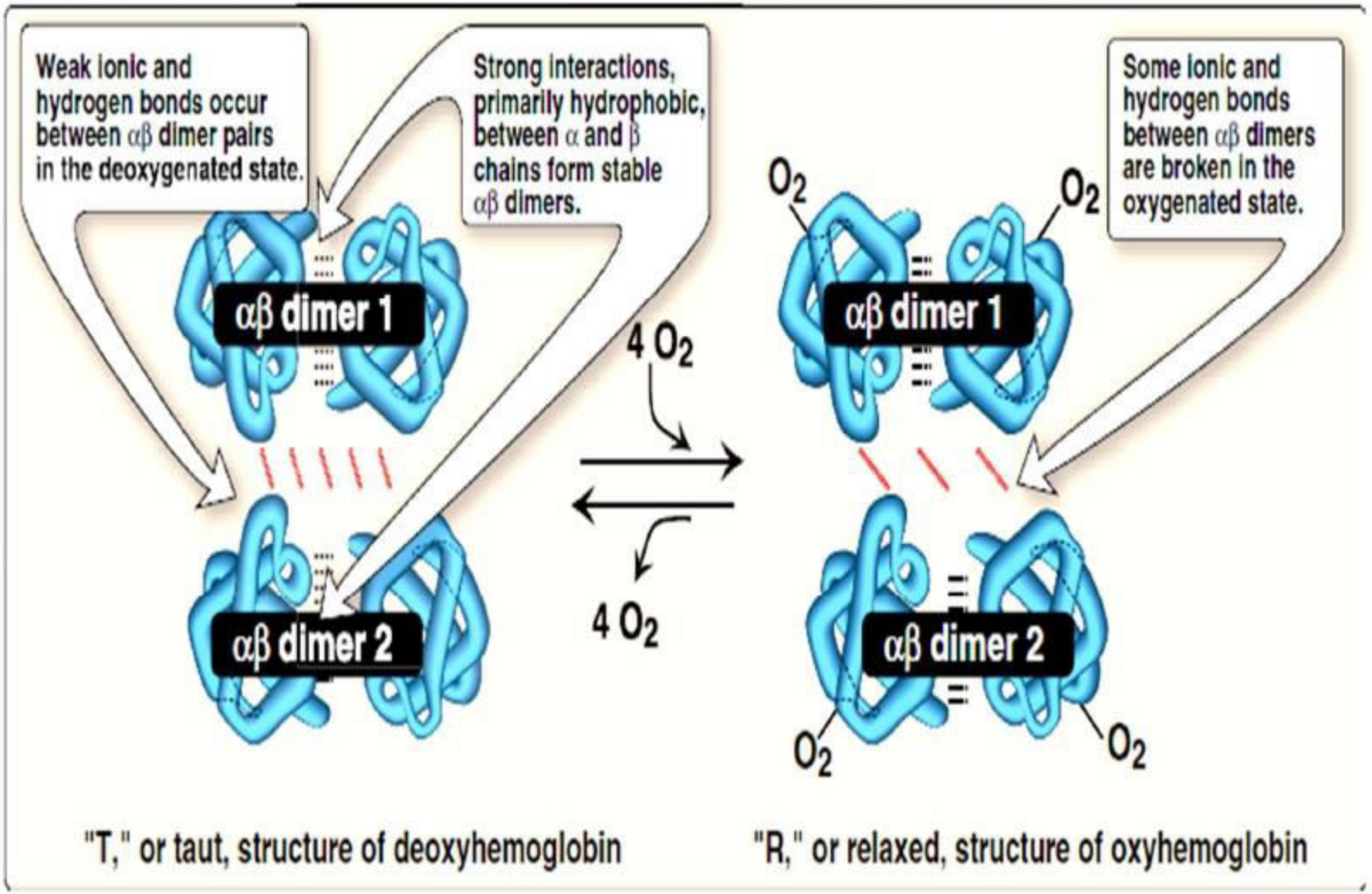
Reversible binding of a protein

- The loss of these ionic interactions, then induces **conformational** changes in all other subunits.

Reversible binding of a protein

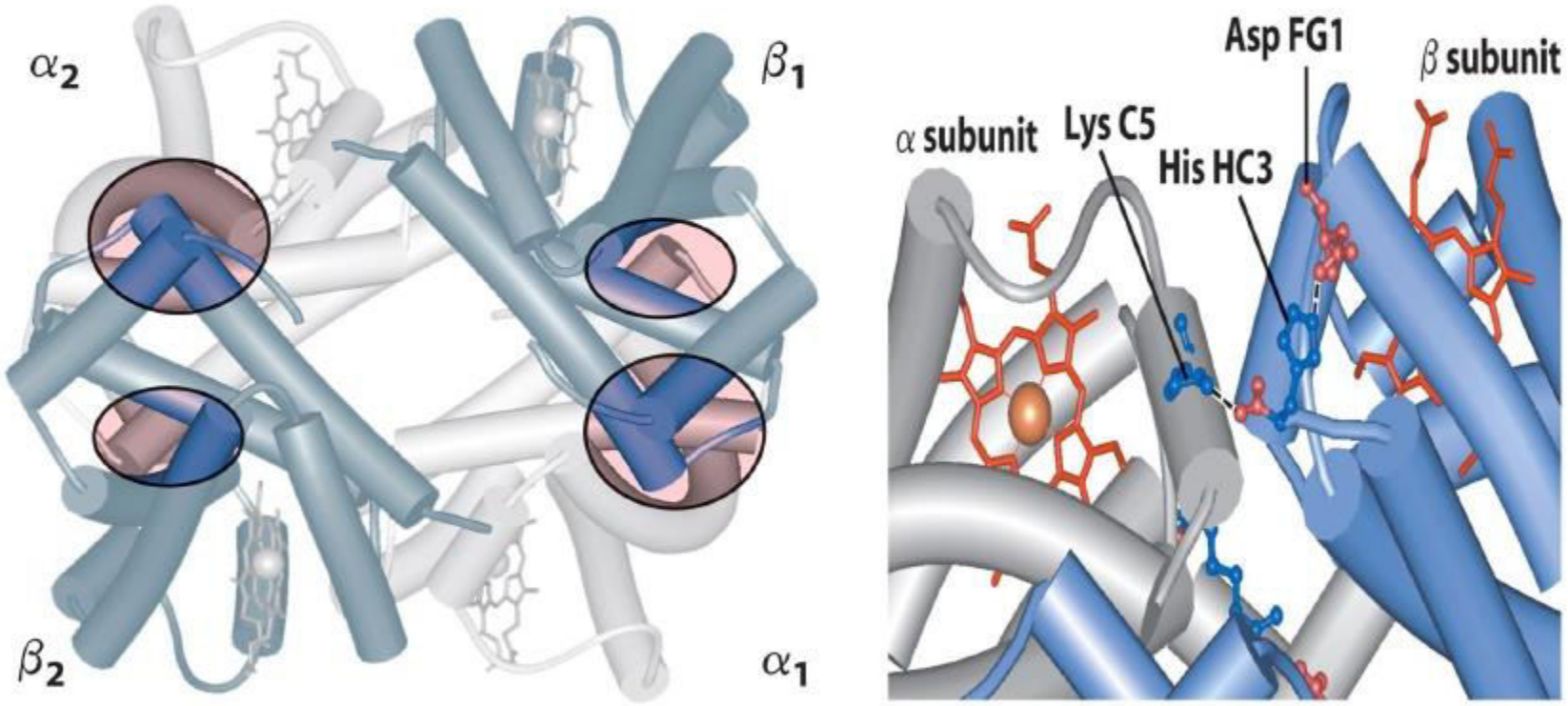
- And all four subunits may change in a concerted manner from their original conformation to a new conformation structure bringing cooperativity in oxygen binding in hemoglobin.

Reversible binding of a protein



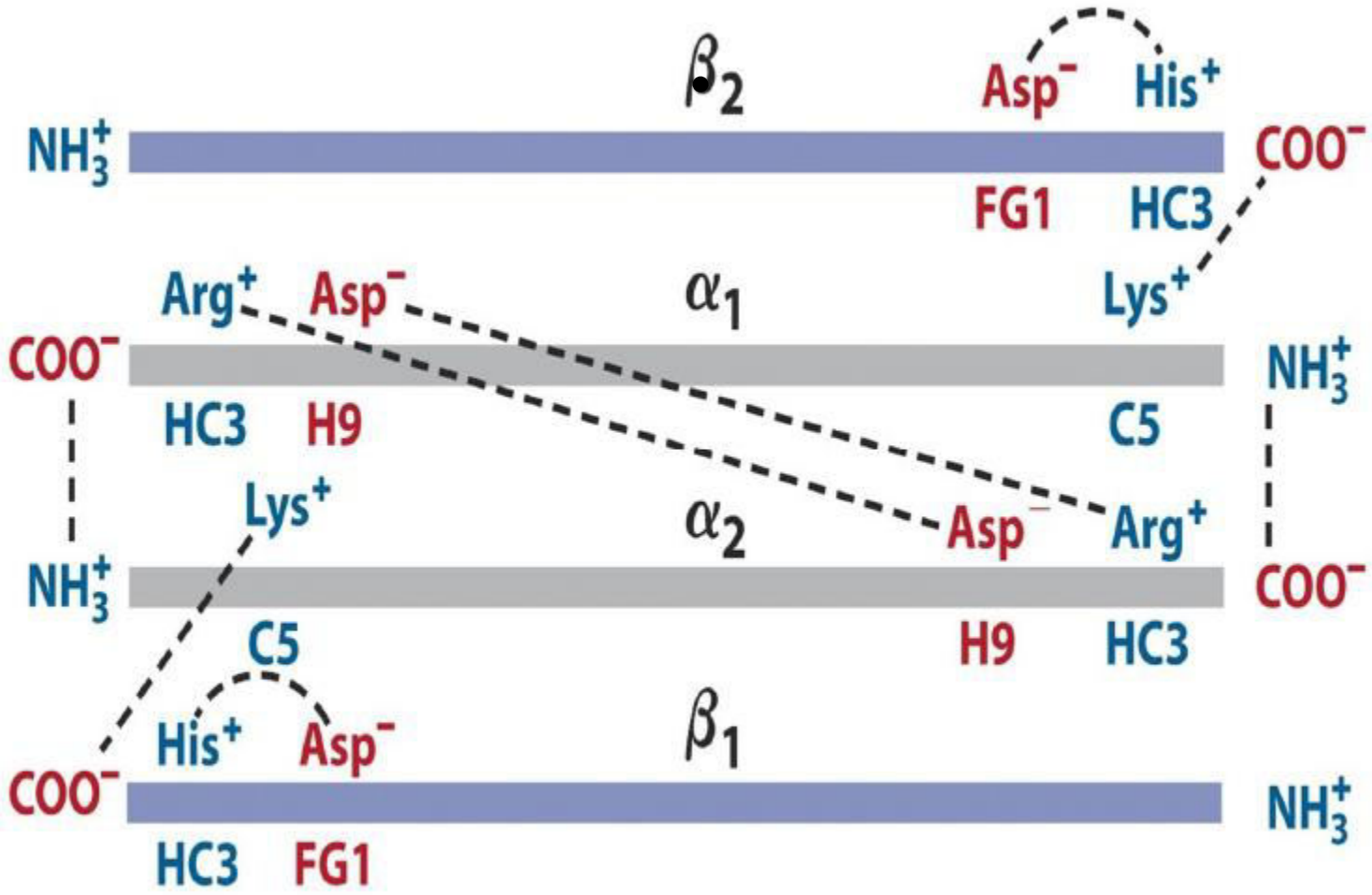
As described previously, the binding of oxygen pulls the heme, which causes conformational change in the Polypeptide change, resulting in the loss of some of the ionic interactions in the different subunits of Hb thus increasing the affinity of the Hb for the further O2 addition.

Reversible binding of a protein



Left: Hemoglobin Structure Showing Inter chain Contact Points
Right : One of the contact points magnified, showing Lys, Asp and His.

Reversible binding of a protein



Contact Points in Primary Structure:These amino acids which are far apart from each other in primary sequence become close in the final folding and give rise to characteristic interactions In oxygenated and

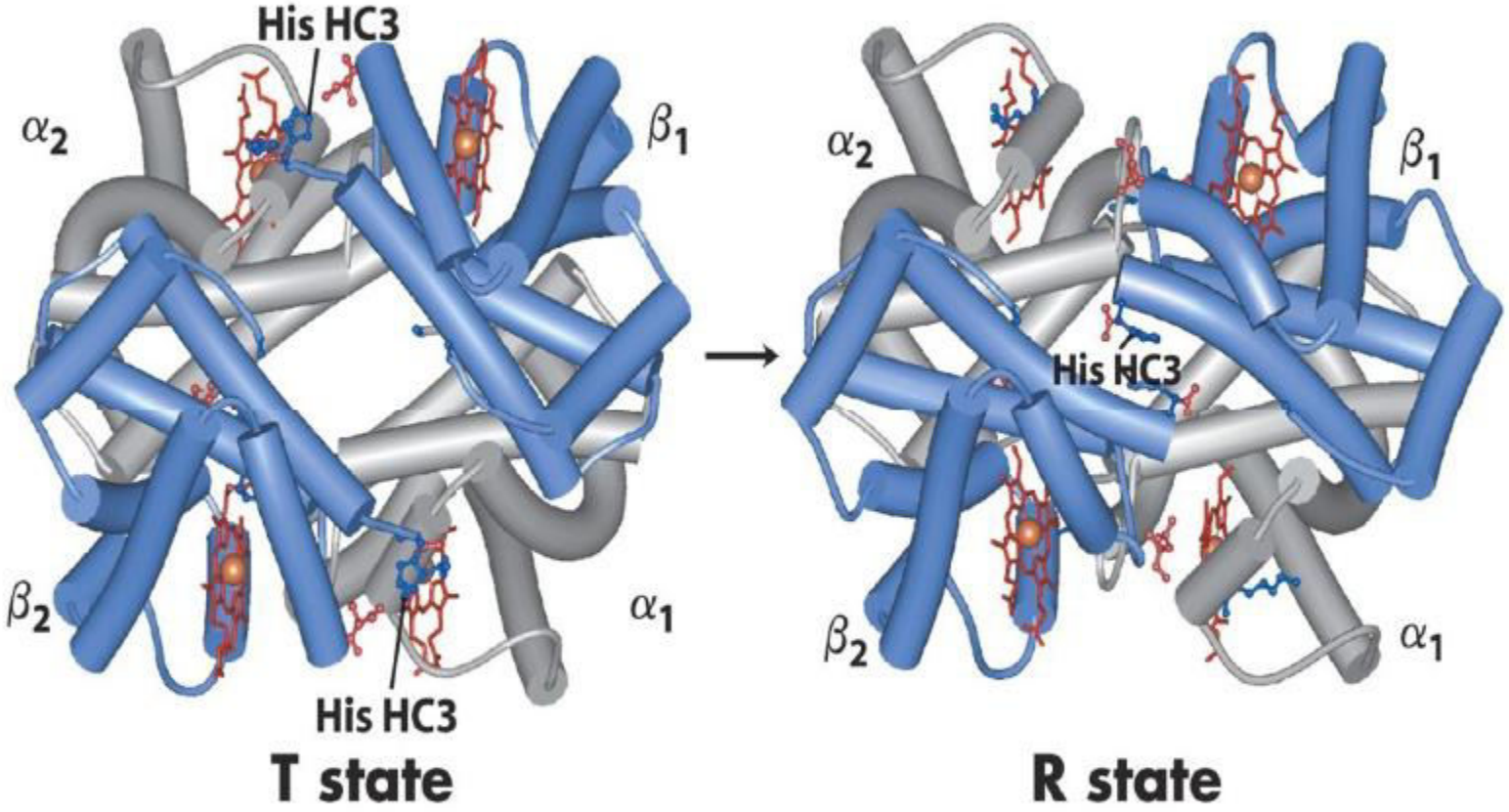
Reversible binding of a protein

- The conformational change of hemoglobin is usually described as changing from a T (tense) state with low affinity for O_2 to an **R** (relaxed) state with a high affinity for O_2.

T= Taut: low oxygen affinity;

R=Relaxed: high oxygen affinity

Reversible binding of a protein



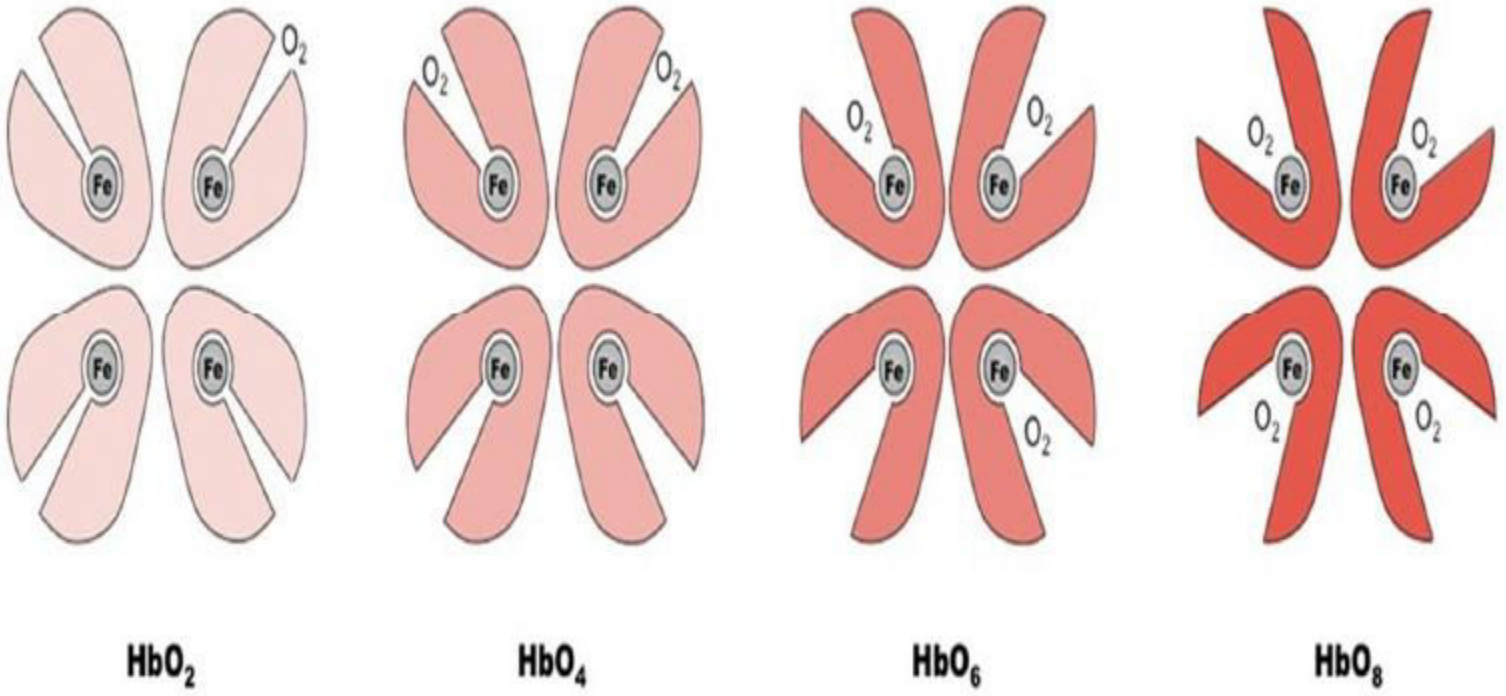
T= Taut: low oxygen affinity;

R=Relaxed: high oxygen affinity

Reversible binding of a protein

- In the T subunits, the binding sites are hindered
- And in the R state the binding sites are open.

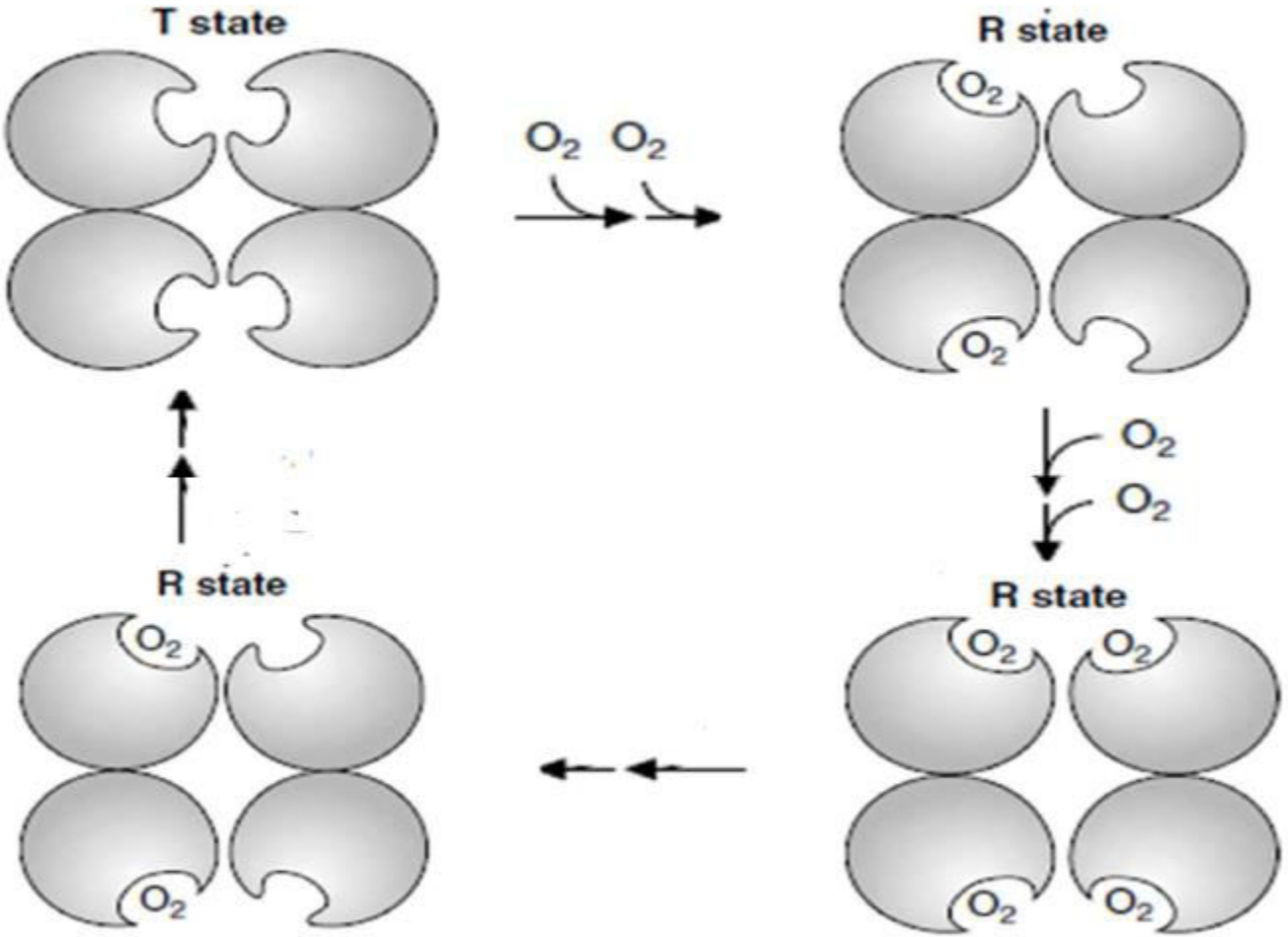
Reversible binding of a protein



Reversible binding of a protein

- Each successive addition of O_2 shifts the equilibrium further toward the R state, thus addition of oxygen in the lungs.
- Similarly, removal of one oxygen facilitates the removal of subsequent oxygen molecule in the tissues.

Reversible binding of a protein



Reversible binding of a protein

Comparison of oxygen binding to myoglobin and hemoglobin

- Myoglobin can bind only one molecule of oxygen.
- Because it contains only one heme group

Reversible binding of a protein

- In contrast, hemoglobin can bind four oxygen molecules-one at each of its four heme groups

Reversible binding of a protein

- The **degree of saturation (Y)** of these oxygen-binding sites on all myoglobin or hemoglobin molecules can vary between zero (all sites are empty) and 100 percent

Reversible binding of a protein

Oxygen dissociation curve

- A plot of **degree of saturation (Y)** measured at different partial pressures of oxygen **(pO_2)** is called *the oxygen dissociation curve*

Reversible binding of a protein

- The curves for myoglobin and hemoglobin show important differences
- This graph illustrates that myoglobin has a higher oxygen affinity than does hemoglobin

Reversible binding of a protein

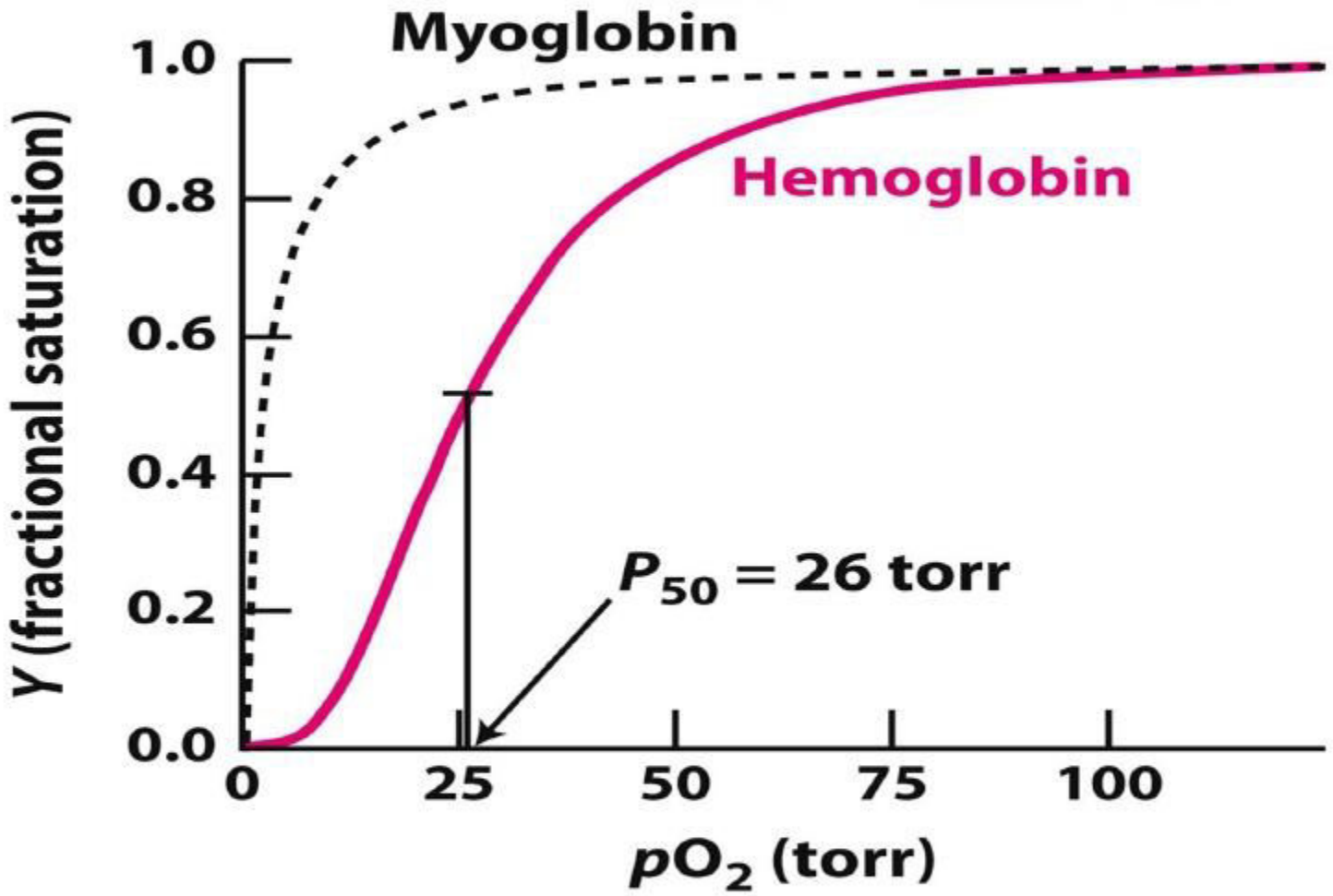


Figure 7.8
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1 torr= 1mmHg

Reversible binding of a protein

- Furthermore, the hyperbolic versus **sigmoid** curve, in myoglobin and **hemoglobin** respectively, also shows that binding of oxygen is co-operative in hemoglobin as compared to myoglobin.

Reversible binding of a protein

- The ***partial pressure of oxygen needed to achieve half-saturation of the binding sites is called (P_{50}).***
- P_{50} is approximately 1 mm Hg for myoglobin and 26.6mm Hg for hemoglobin

Reversible binding of a protein

- The partial pressure of oxygen in the blood at which the

hemoglobin is 50% saturated, is known as the **P_{50}**.

- The **P_{50}** is a conventional measure of **hemoglobin** affinity for oxygen.

Reversible binding of a protein

- Values of P_{50} are negatively correlated with substrate affinity, with lower values of P_{50} corresponding to high affinity and *vice versa*.

Reversible binding of a protein

- This means that the **higher** the oxygen **affinity** (that is the more tightly oxygen binds), the **lower** the **P_{50}**

Reversible binding of a protein

Myoglobin

- The oxygen dissociation curve for myoglobin has a **hyperbolic shape**
- This reflects the fact that myoglobin reversibly binds a single molecule of oxygen

Reversible binding of a protein

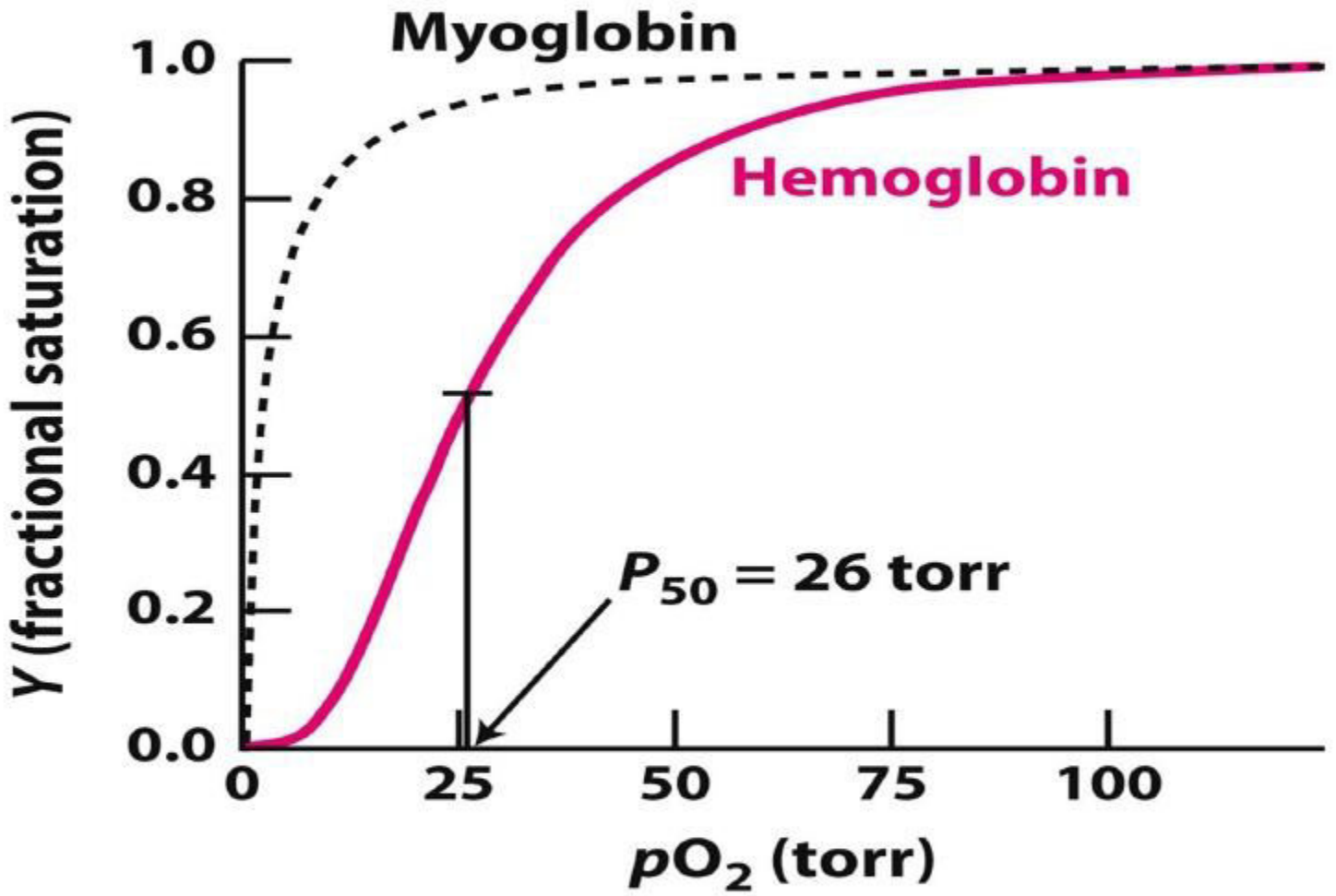


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Reversible binding of a protein

Myoglobin

- Myoglobin is designed to release and bind oxygen at very low partial pressure.

Reversible binding of a protein

- Oxygenated (MbO_2)and deoxygenated (Mb) myoglobin exist in a simple equilibrium:

$$Mb + O_2 \longleftrightarrow MbO_2$$

Reversible binding of a protein

- The equilibrium is shifted to the **right** or to the **left** as oxygen is **added** to or **removed** from the system
- Myoglobin is designed to bind oxygen released by hemoglobin at the low pO_2 found in muscle

Reversible binding of a protein

- Myoglobin, in turn, releases oxygen within the muscle cell in response to in oxygen demand during strenuous exercise.

Reversible binding of a protein

- When strenuous exercise lowers the pO_2 of muscle tissue to about **5 mm Hg**, myoglobin releases O_2 for mitochondrial synthesis of ATP, permitting continued muscular activity

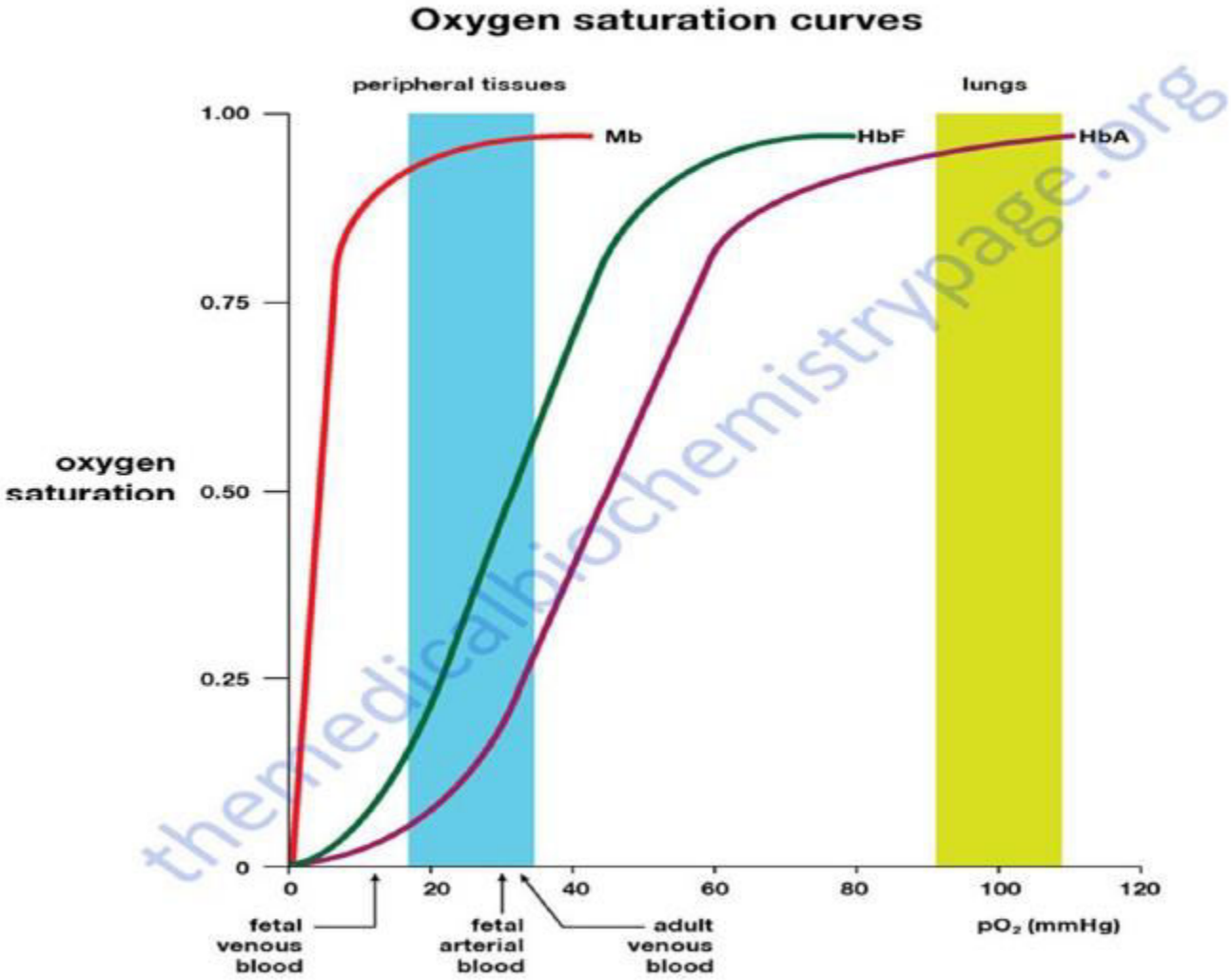
Reversible binding of a protein

- Myoglobin can load oxygen readily at the pO_2 of the lung capillary bed (100 mm Hg)

Reversible binding of a protein

- However, since myoglobin releases its bound O_2 only at the pO_2 values typically encountered in active muscle (20 mm Hg), it represents an ineffective vehicle for delivery of O_2 to tissues ($P O_2$= 40mmHg)

Reversible binding of a protein



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Reversible binding of a protein

- **Hemoglobin Dissociation Curve**

- Hemoglobin must bind oxygen efficiently in the lungs, where the pO_2 is about 100 mmHg,
- And release oxygen in the tissues, where the p O_2 is about 35 to 40 mmHg

Reversible binding of a protein

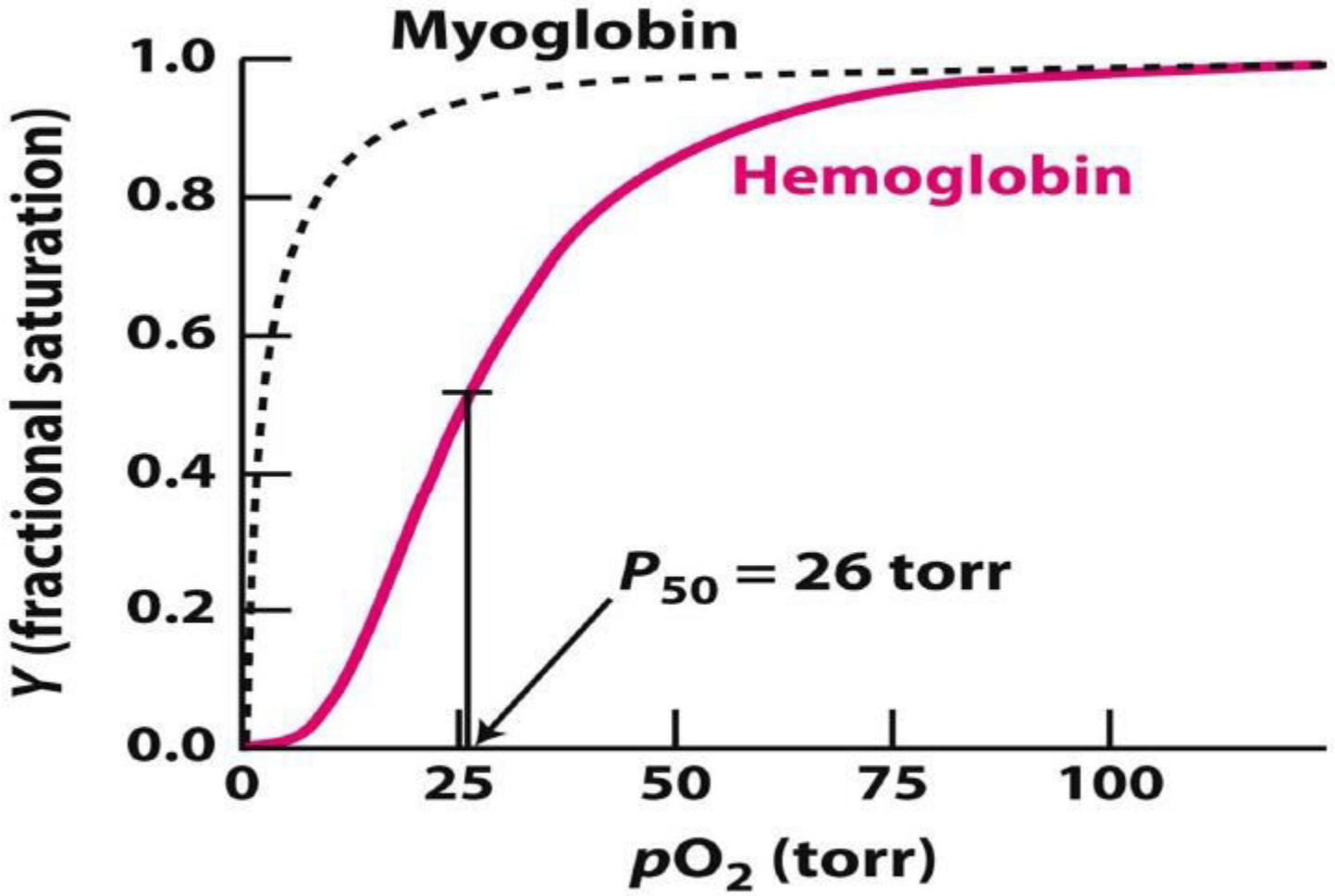
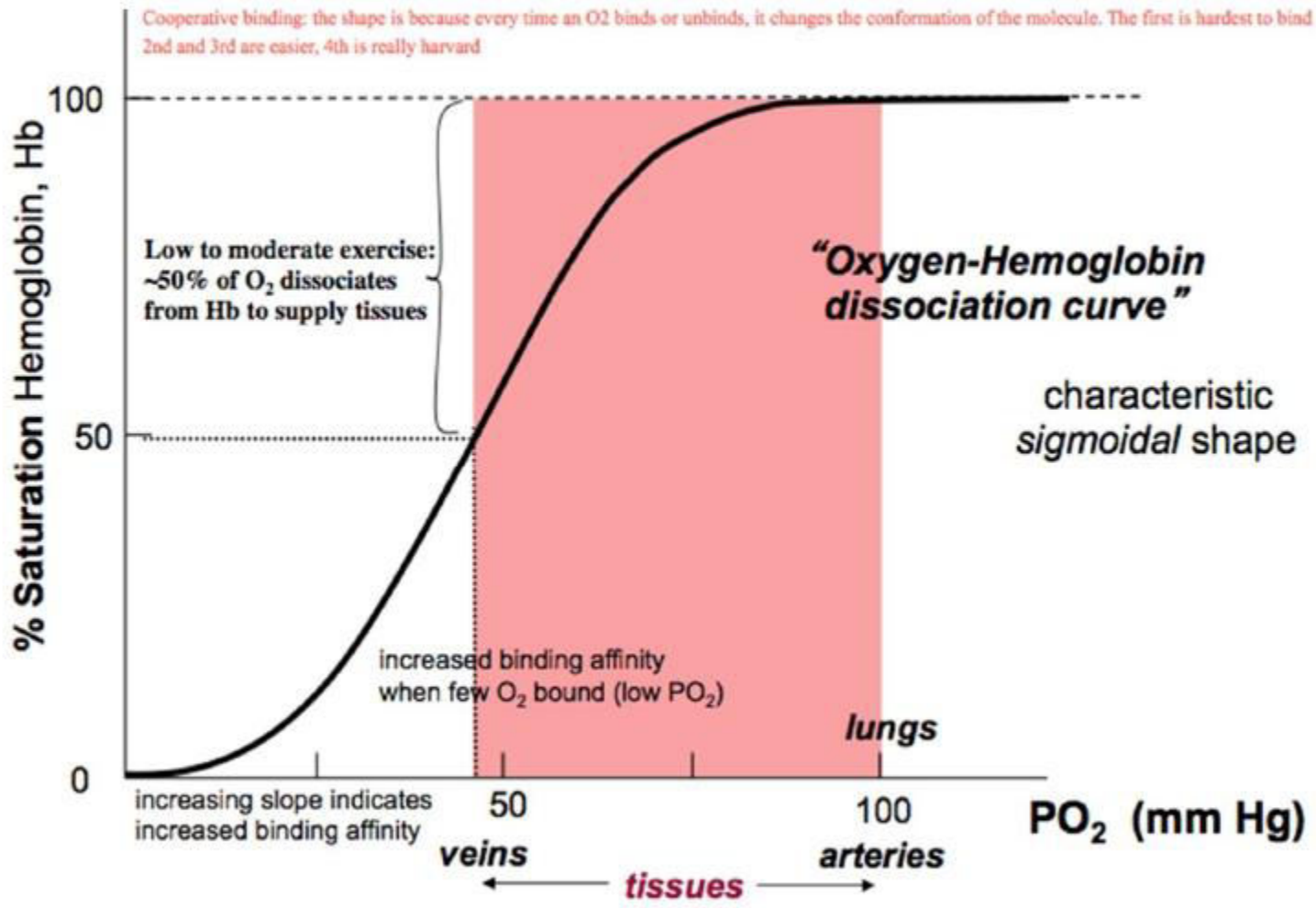


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1 torr= 1mmHg

Reversible binding of a protein



Reversible binding of a protein

- It is this **sigmoidal** loading and unloading of hemoglobin that makes it functionally useful as a on oxygen transporter in blood.

Sigmoidal or **sigmoid**, literally means S-shaped

Reversible binding of a protein

- The **sigmoidal** binding curve of Hemoglobin for oxygen is possible due to multi-subunit structure of hemoglobin. i.e
  - two **alpha (α)** and
  - two **beta (β)** **chains**

Reversible binding of a protein

- As described earlier binding of one O_2 molecule induces conformatory change in hemoglobin structure resulting in transition of hemoglobin from the low-affinity **T (taut) state** to the high-affinity **R (relaxed) state**

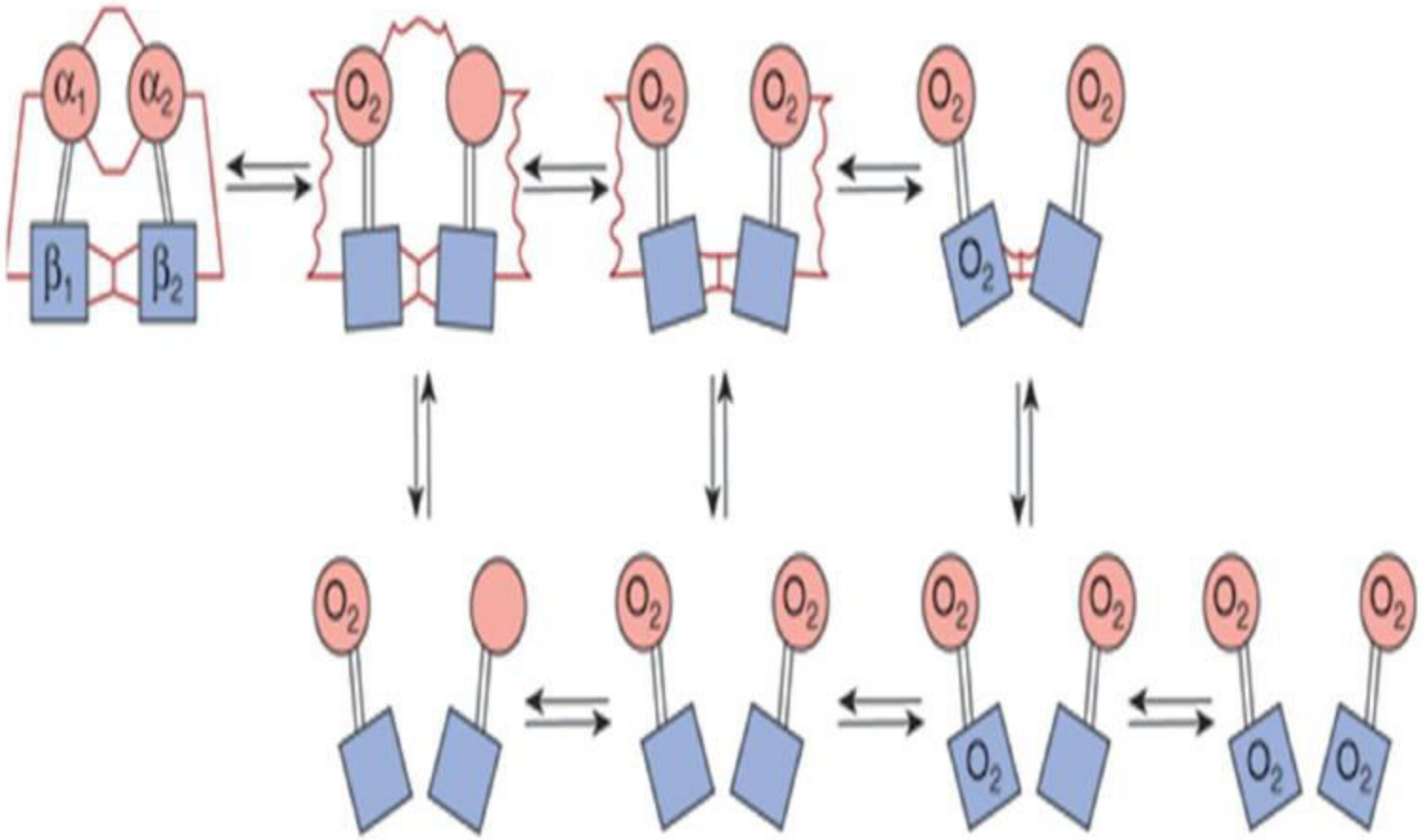
Reversible binding of a protein

- These changes significantly increase the affinity of the remaining unoxygenated hemes for O_2, as subsequent binding require the disruption of fewer ionic interactions.

Reversible binding of a protein

- The transition from T to R does not take place after a fixed number of oxygen molecules have been bound
- But becomes more probable as each successive oxygen binds

Reversible binding of a protein



Every oxygen molecule that binds to Hb increases its affinity for the binding of further oxygen molecul

Reversible binding of a protein

- This effect is referred to as **heme-heme interaction**

END

Reversible binding of a protein

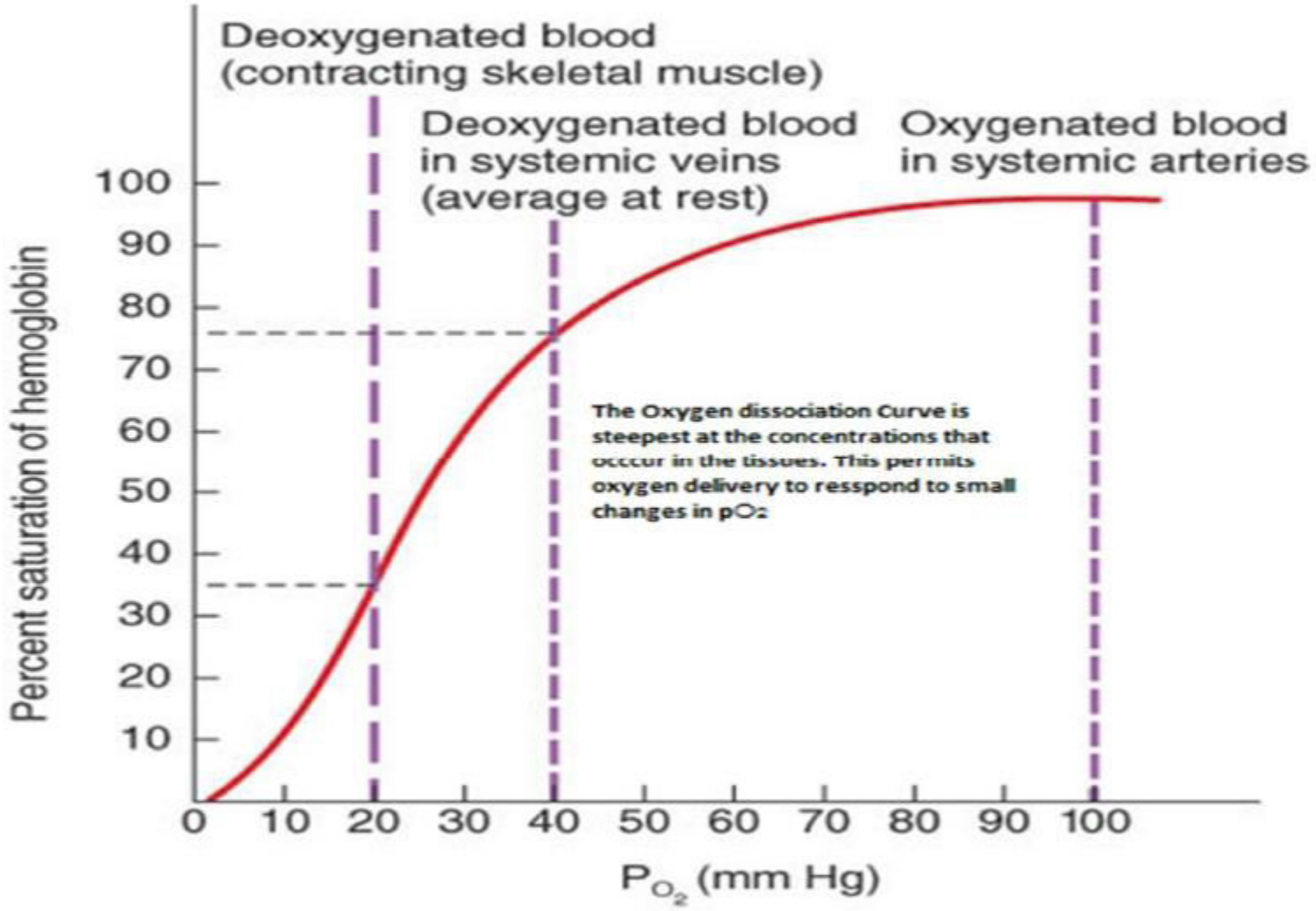
- **Hemoglobin Dissociation Curve**

- Although it is more difficult for the first oxygen molecule to bind to hemoglobin.

Reversible binding of a protein

- The subsequent binding of oxygen occurs with high affinity, as shown by the steep upward curve in the region near 20 to 30 mm Hg

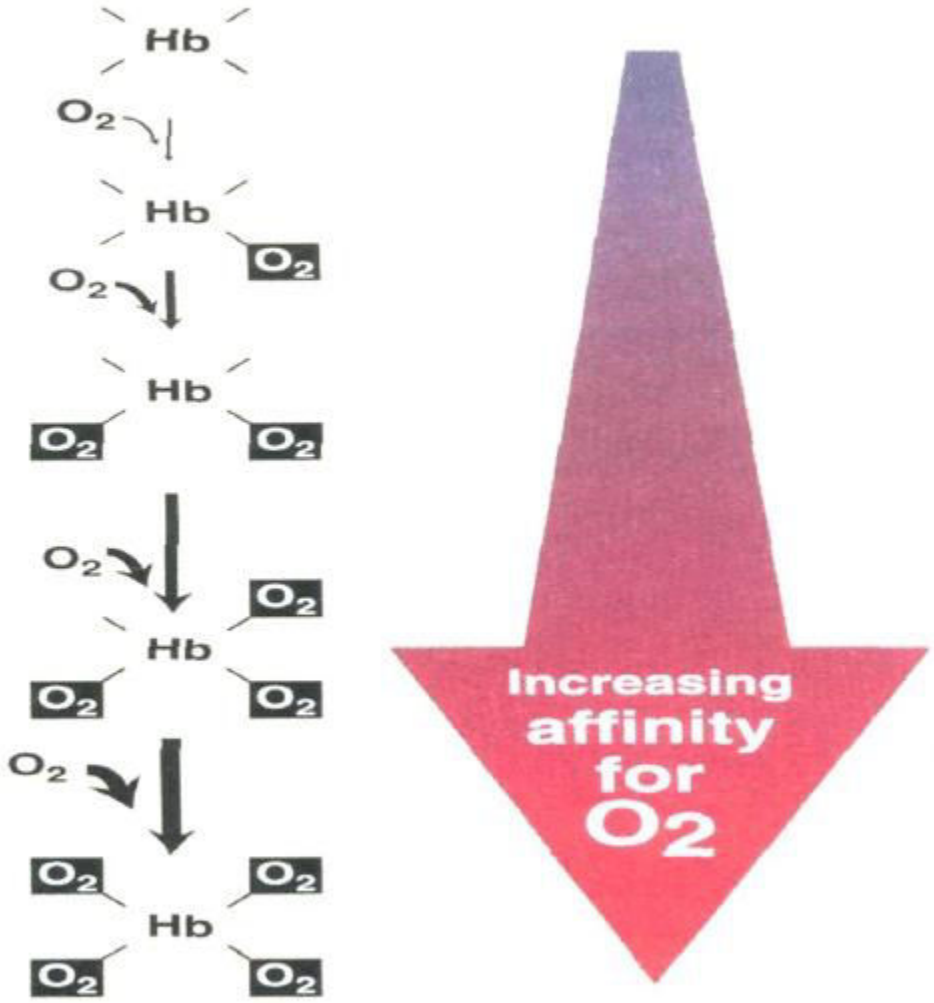
Reversible binding of a protein



Reversible binding of a protein

- Cooperative binding of oxygen by the four subunits of hemoglobin means that the binding of an oxygen molecule at one heme group increases the oxygen affinity of the remaining heme groups in the same hemoglobin molecule

Reversible binding of a protein



Reversible binding of a protein

- The net effect is that the affinity of hemoglobin for the last oxygen bound is approximately 300 times greater than its affinity for the first oxygen bound

Reversible binding of a protein

- **In Summary,**
- Hb must bind O_2 efficiently in the lungs (pO_2= 100 mm Hg), and release O_2 in the tissues (pO_2= 40 mm Hg).

Reversible binding of a protein

- Hb solves the problem by undergoing a transition from a low-affinity state (the T state) to a high-affinity state (the R state) as more O_2 molecules are bound.

Reversible binding of a protein

- Hb has a hybrid S-shaped, or sigmoid, binding curve for O_2

END

Reversible binding of a protein

Allosteric effects

- Apart from Oxygen, ability of hemoglobin to reversibly bind oxygen is affected by
  - the **pH** of the environment
  - the pCO_2 ; and
  - the availability of **2,3-bisphospho-phoglycerate**

Reversible binding of a protein

- These are collectively called **Allosteric (other site) effectors**
- because their interaction at one site on the hemoglobin molecule affects the binding of oxygen to heme groups at other locations on the molecule

In Greek, Allo means other and Steric means Site.

Reversible binding of a protein

- The binding of oxygen to myoglobin is not influenced by the allosteric effectors of hemoglobin

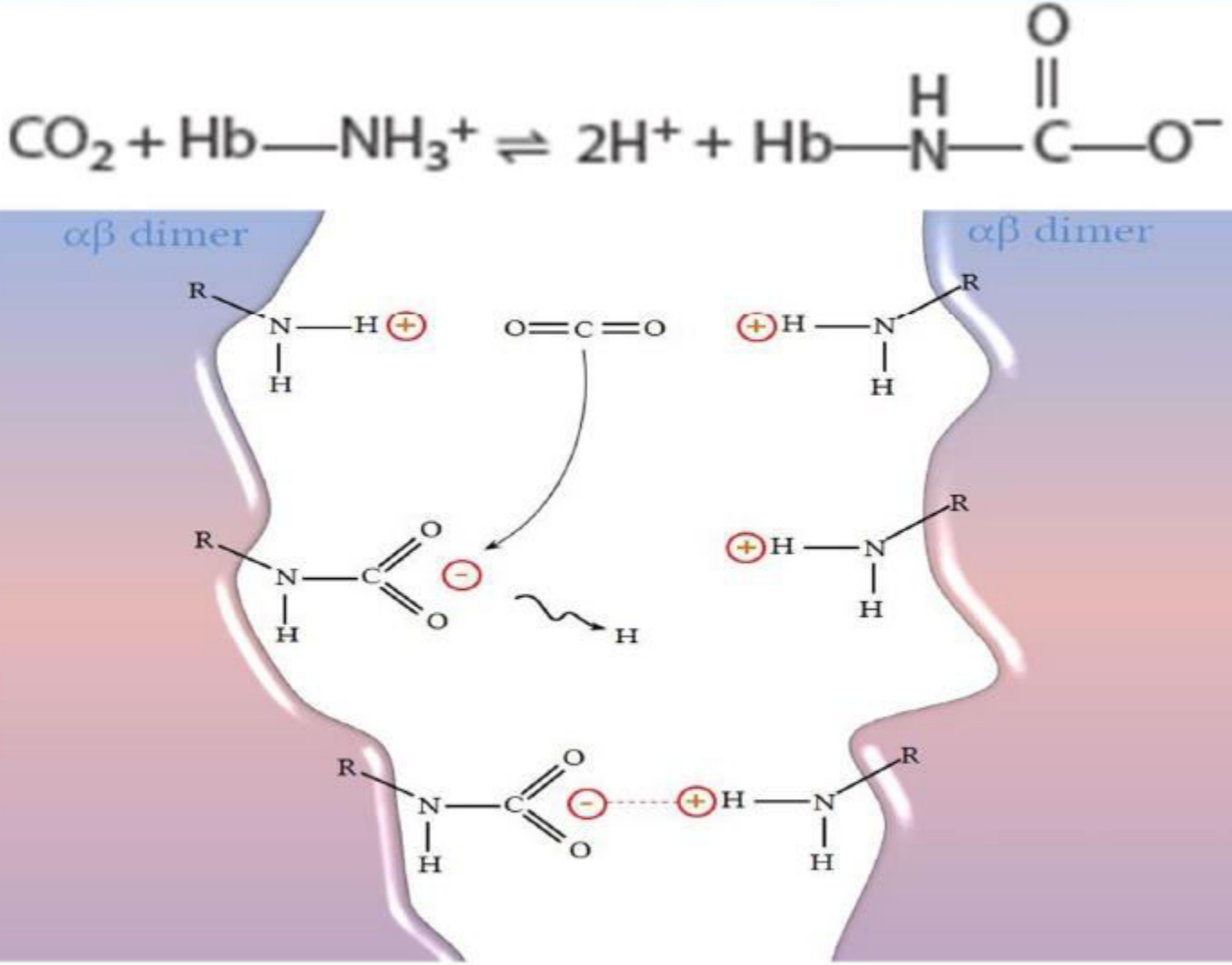
Reversible binding of a protein

- In addition to transporting O_2 from the lungs to the peripheral tissues,
- hemoglobin transports CO_2, the by product of respiration,
- And protons from peripheral tissues to the lungs

Reversible binding of a protein

- **Transport of Carbon Dioxide**
- Hemoglobin carries CO_2 as **carbamates** formed with the amino terminal nitrogens of the polypeptide chains.

Reversible binding of a protein



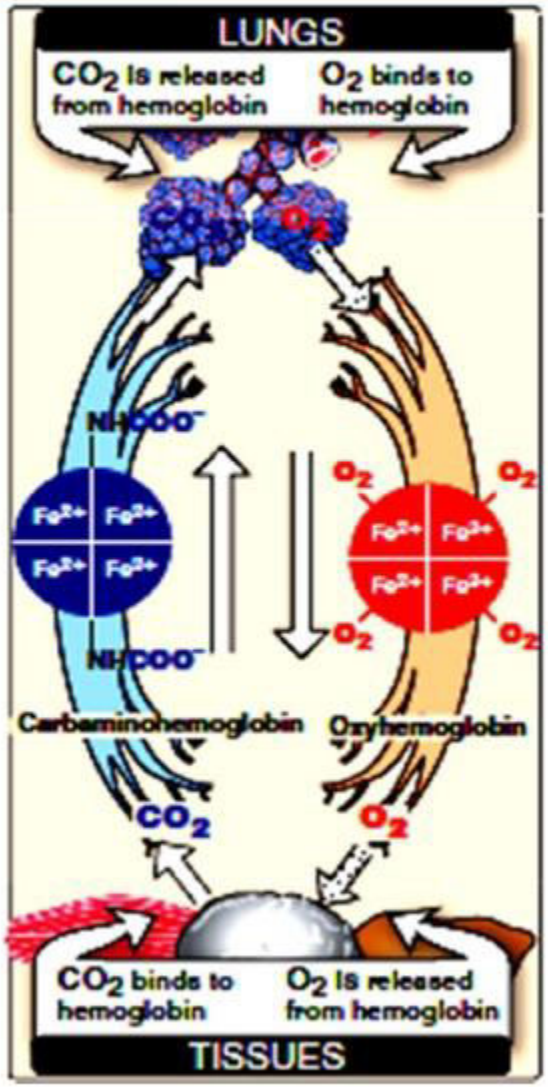
Reversible binding of a protein

- Carbamates change the charge on amino terminals from positive to negative, favoring ionic interaction formation between the α and β chains
- Hence, favoring the stabilization of taut form of deoxyhemoglobin

Reversible binding of a protein

- Hemoglobin carbamates account for about 15% of the CO_2 in venous blood

Reversible binding of a protein

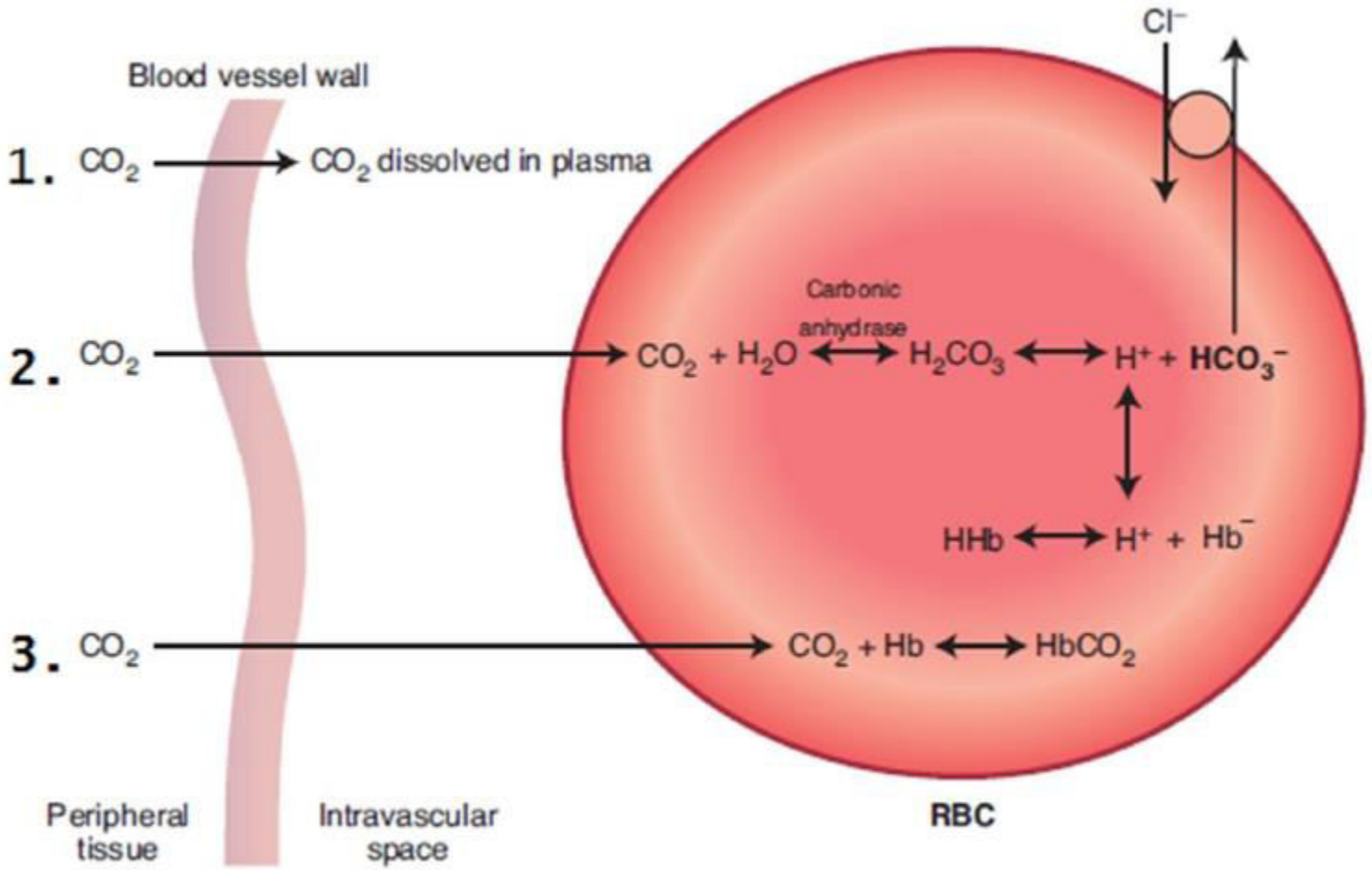


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Reversible binding of a protein

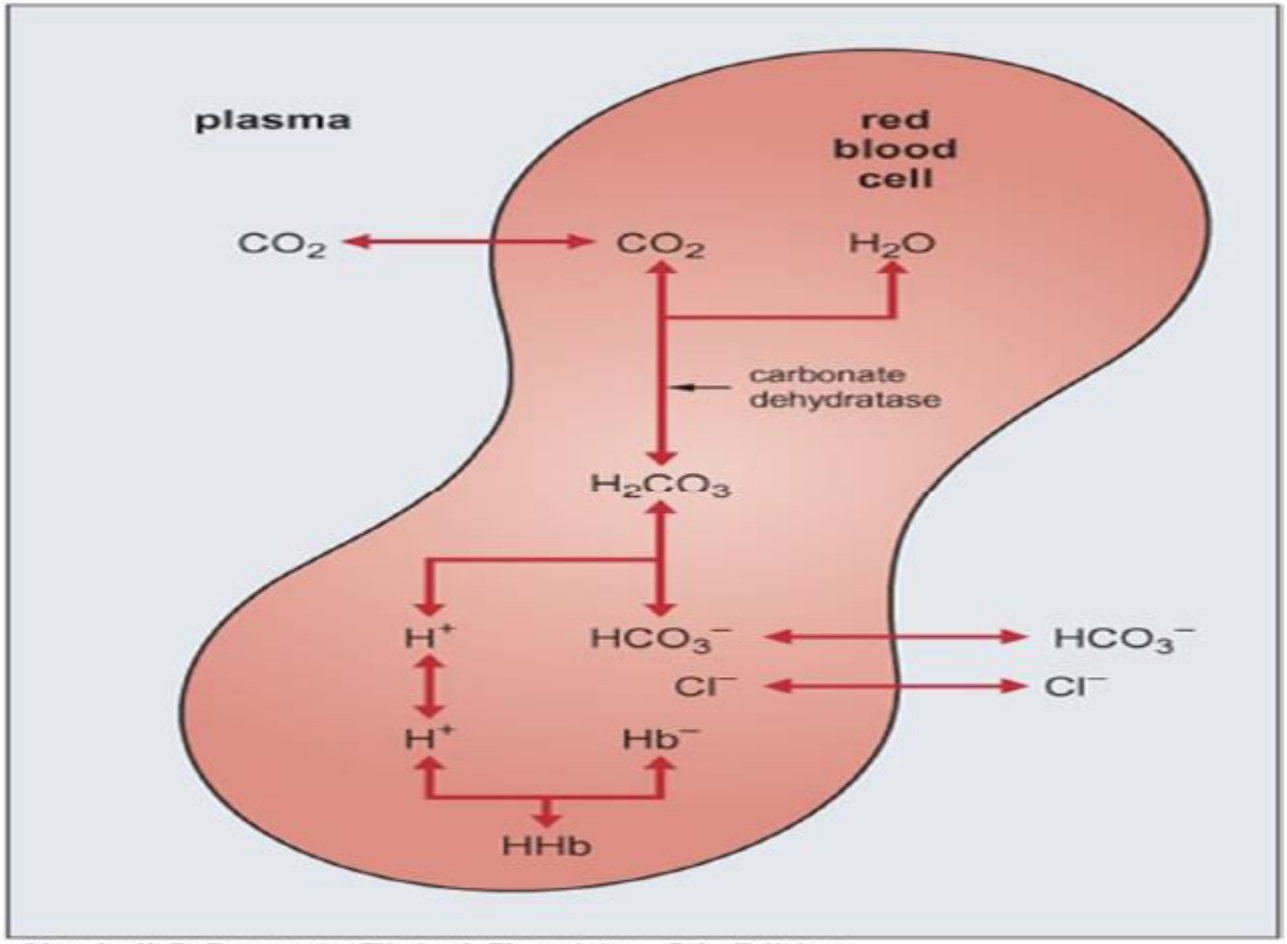
- The remainder of the CO_2 is transported as dissolved
- HCO_3^- (80 to 85%) and
- dissolved CO_2 (<5%)

Reversible binding of a protein



Modes of CO2 transport

Reversible binding of a protein



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Modes of CO2 transport

1: Dissolved CO2; **2:** Dissolved bicarbonate **3:** carbamate form

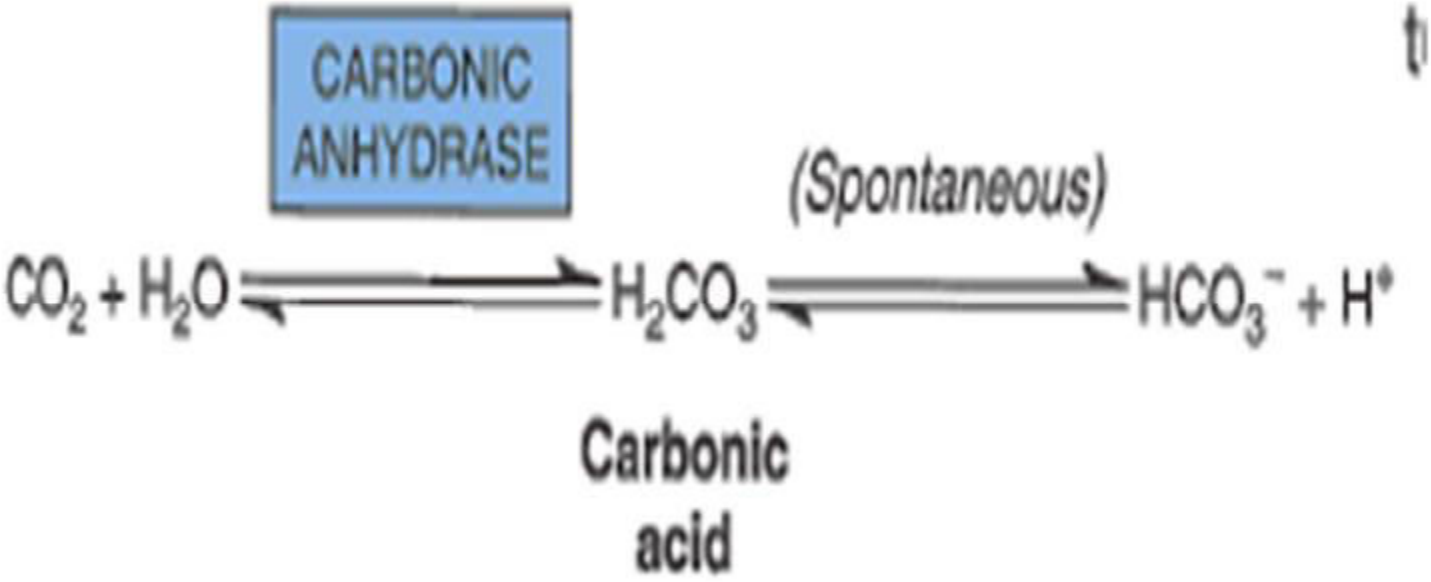
Reversible binding of a protein

- Transport of Carbon Dioxide (cont.)
- Much of the remaining CO_2 is carried as bicarbonate.
- (80 to 85%)

Reversible binding of a protein

- Bicarbonate is formed in erythrocytes by the hydration of CO_2 to carbonic acid (H_2CO_3)
- This process is catalyzed by carbonic anhydrase

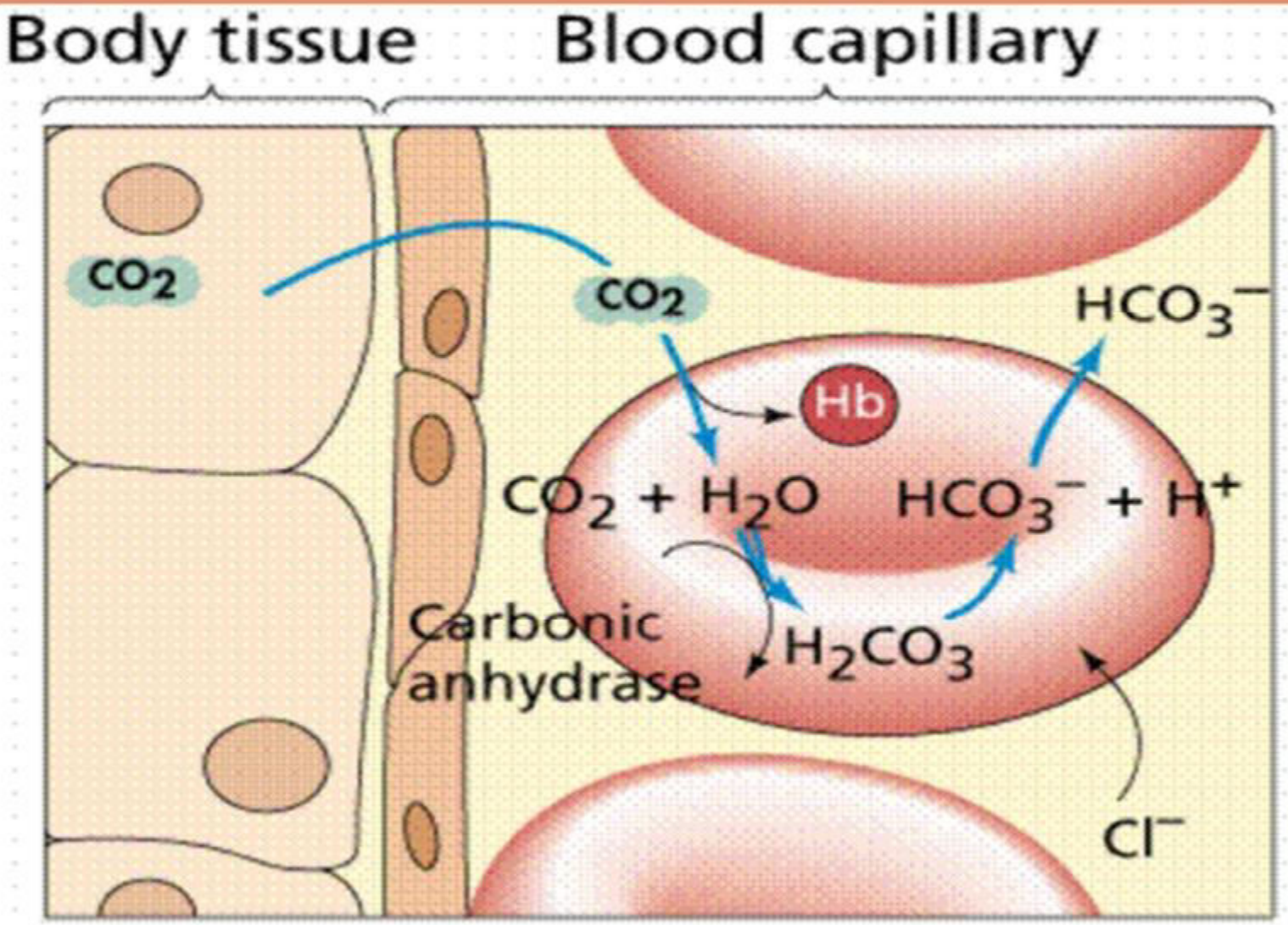
Reversible binding of a protein



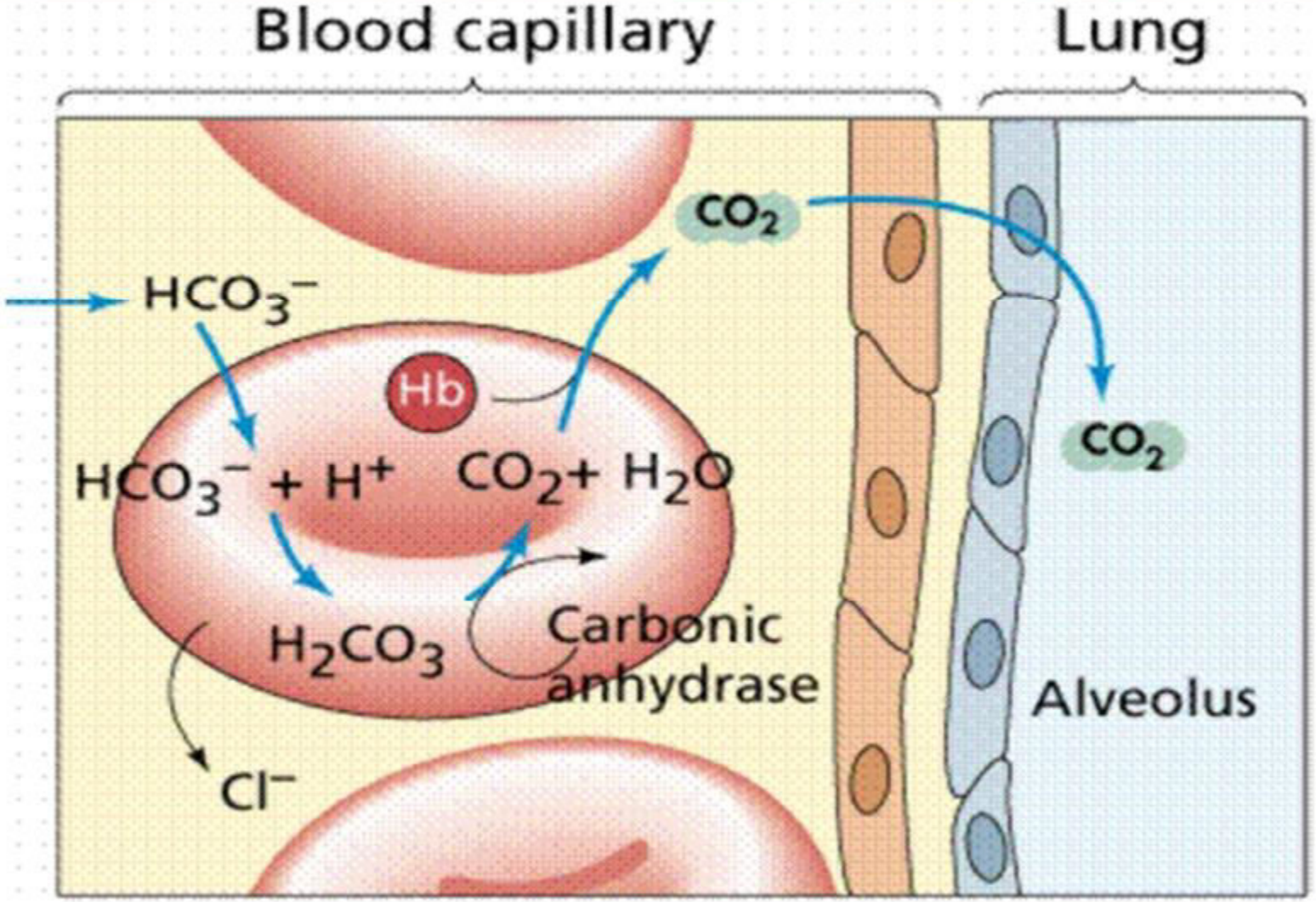
Reversible binding of a protein

- At the pH of venous blood, H_2CO_3 dissociates into;
- bicarbonate and
- a proton

Reversible binding of a protein



Reversible binding of a protein



Reversible binding of a protein

- Deoxyhemoglobin binds one proton for every two O_2 molecules released, contributing significantly to the buffering capacity of blood

Reversible binding of a protein

- Oxygen and H^+ are not bound at the same sites in hemoglobin.
- However binding of proton, helps stabilize deoxyhemoglobin in the T state.

Reversible binding of a protein

- The somewhat lower pH of peripheral tissues, aided by carbamation, stabilizes the T state and thus enhances the delivery of O_2

Reversible binding of a protein

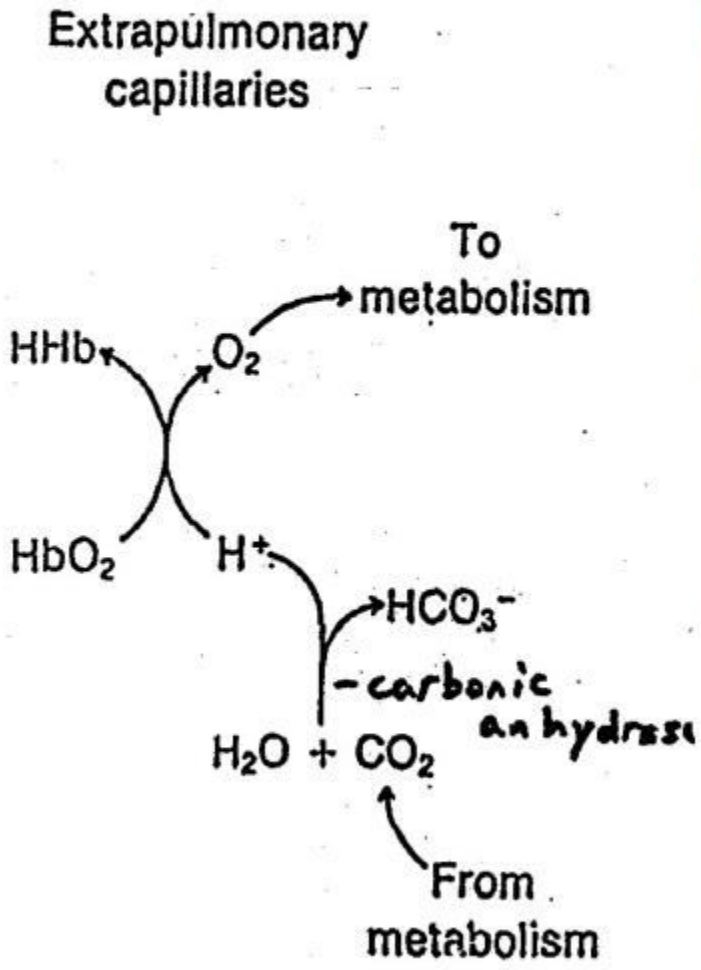
- This **reciprocal coupling** of proton and O_2 binding is termed the **Bohr effect**

END

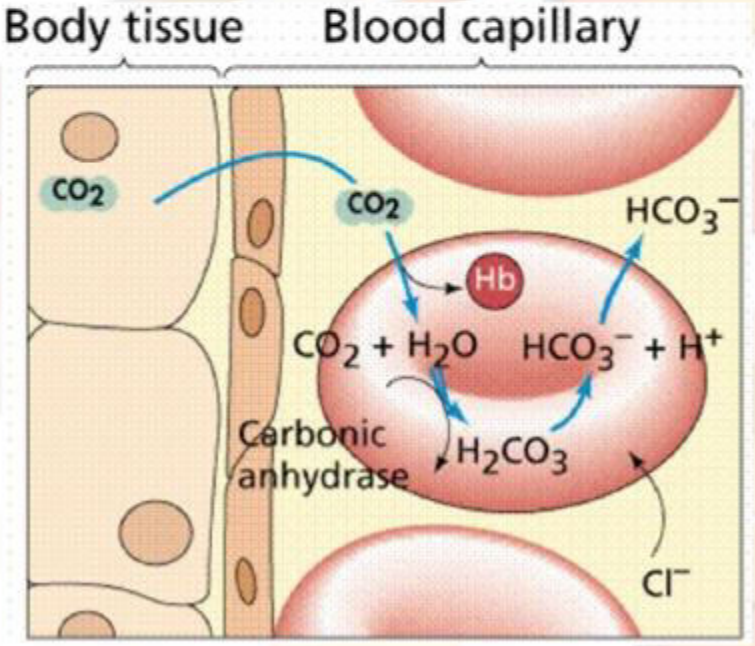
Reversible binding of a protein

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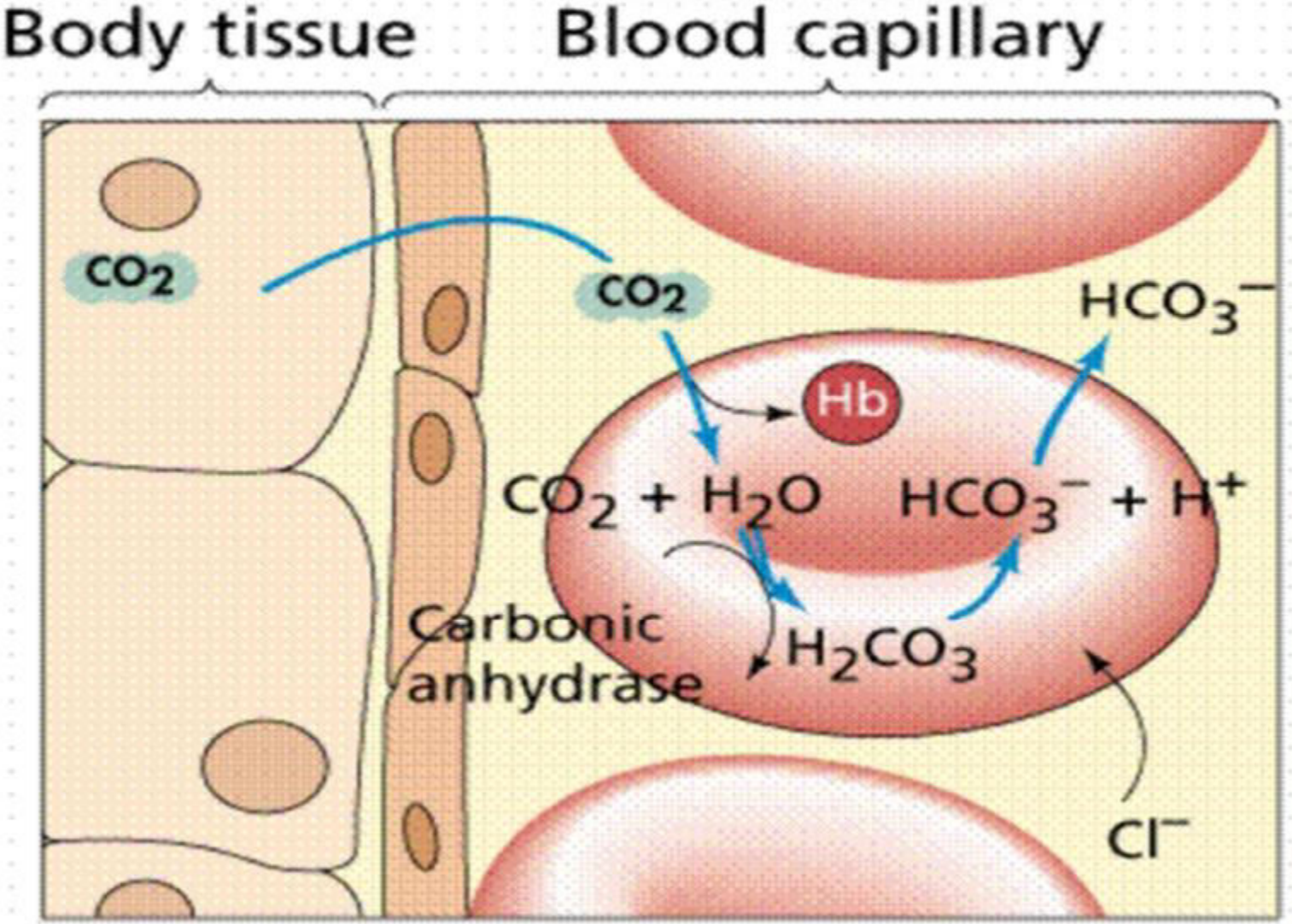
Reversible binding of a protein



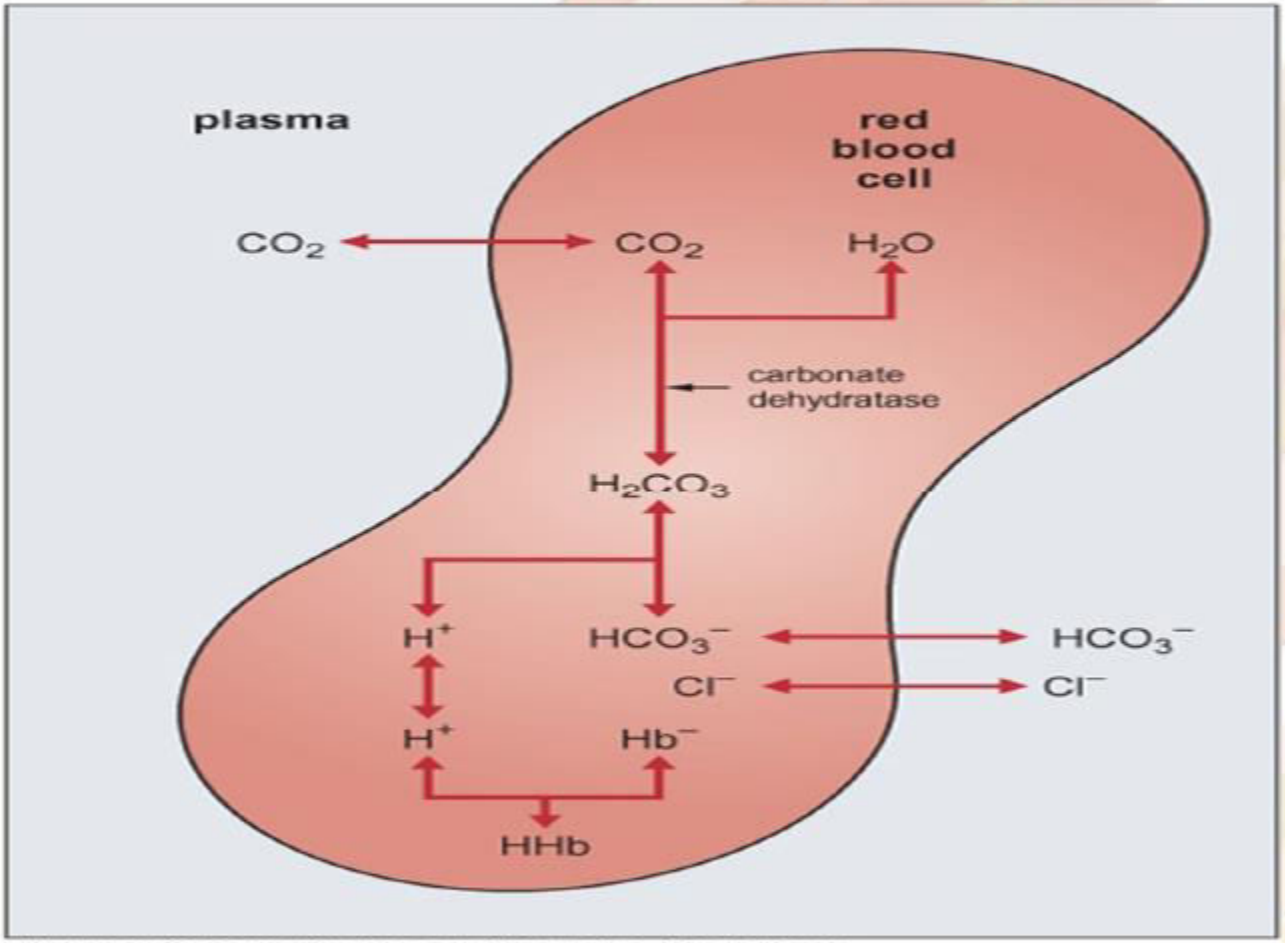
- This **reciprocal coupling** of proton and O_2 binding is termed the **Bohr effect**



Reversible binding of a protein



Reversible binding of a protein



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Chloride Shift: When bicarbonate is formed, chloride ions move into the RBCs, to maintain electroneutralityThis is referred to as **Chloride shift.**

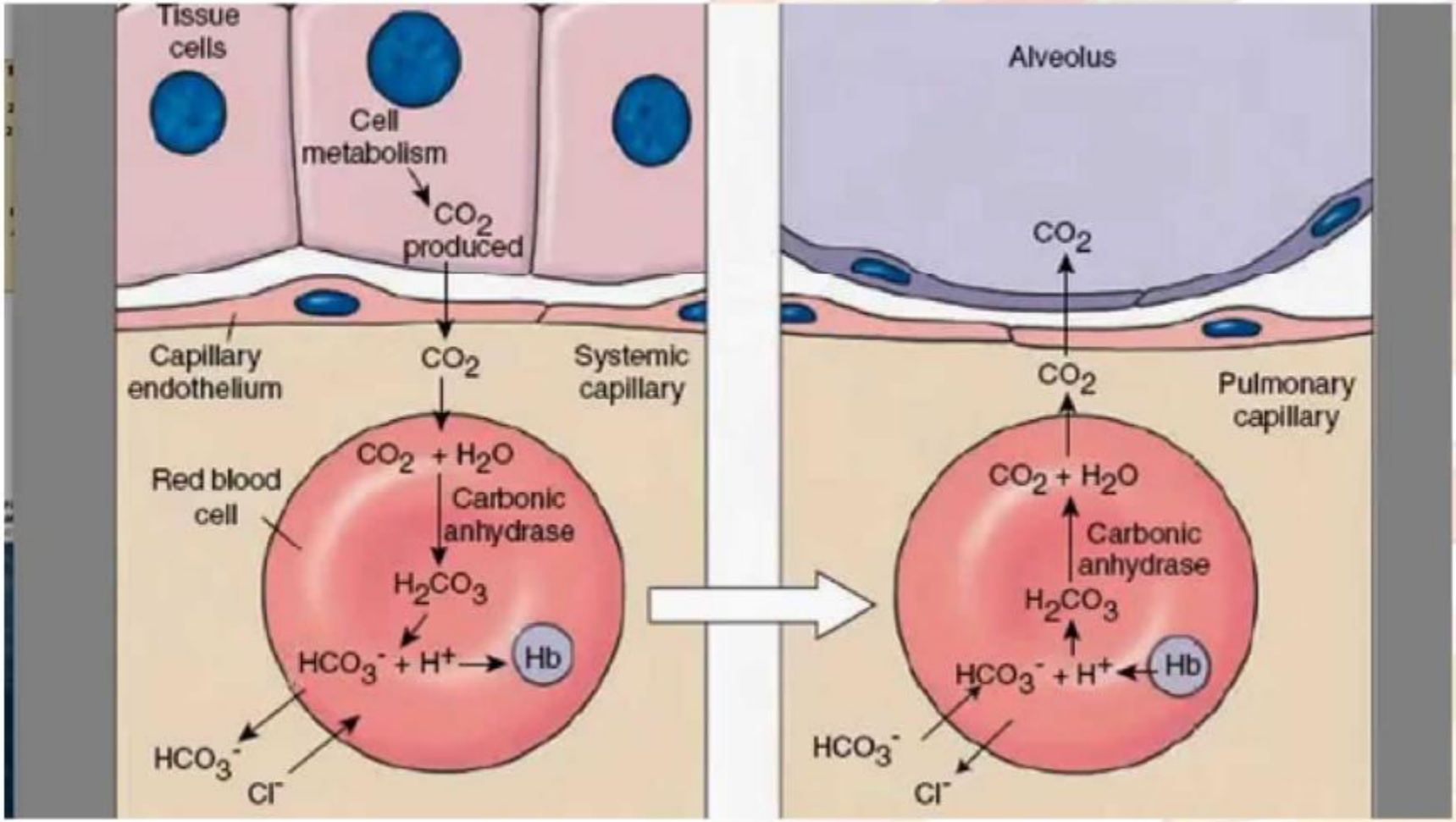
Reversible binding of a protein

- In the lungs, the process reverses
- As O_2 binds to deoxyhemoglobin, protons are released and combine with bicarbonate to form carbonic acid

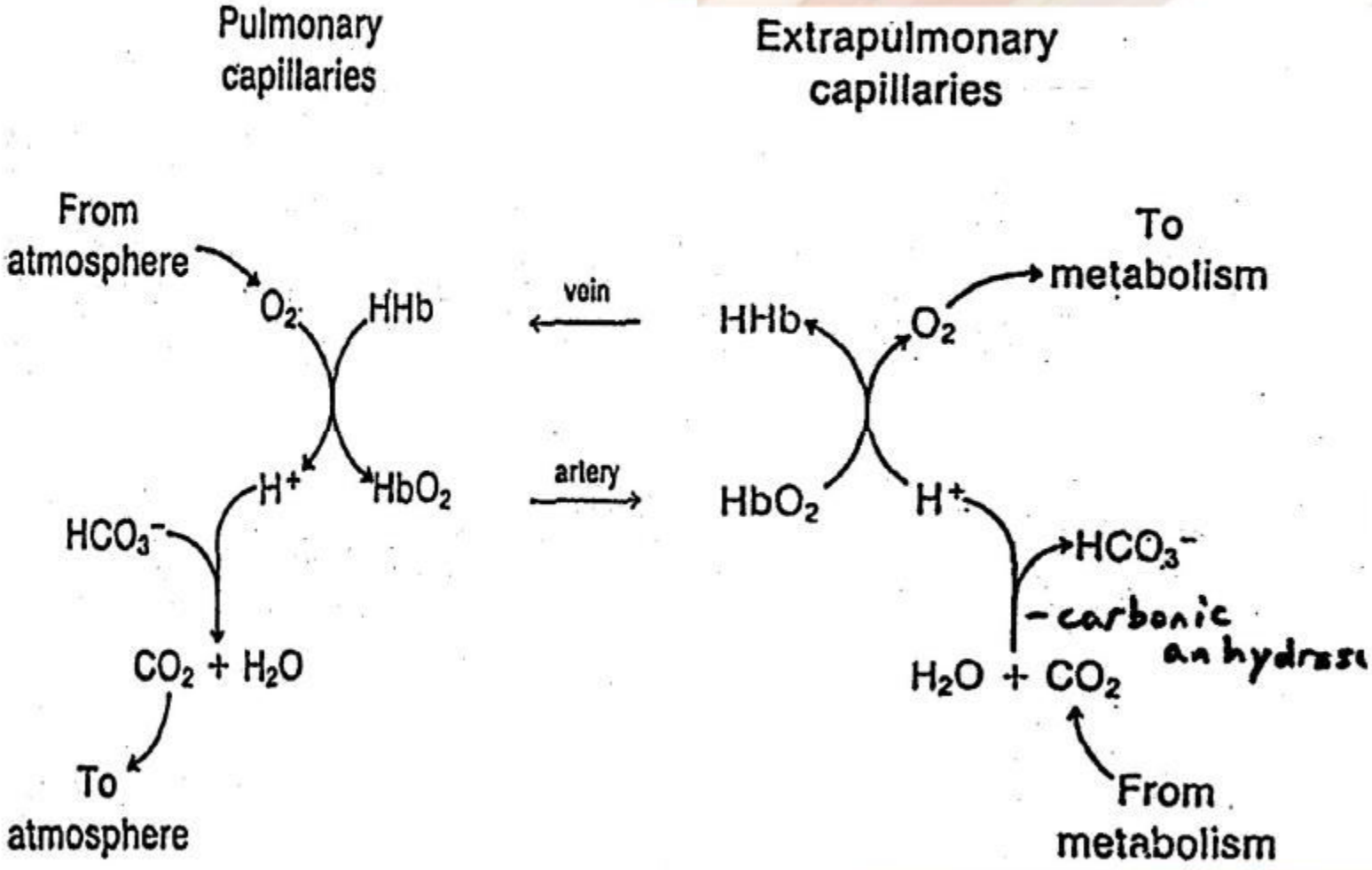
Reversible binding of a protein

- Dehydration of H_2CO_3 catalyzed by carbonic anhydrase forms CO_2, which is exhaled

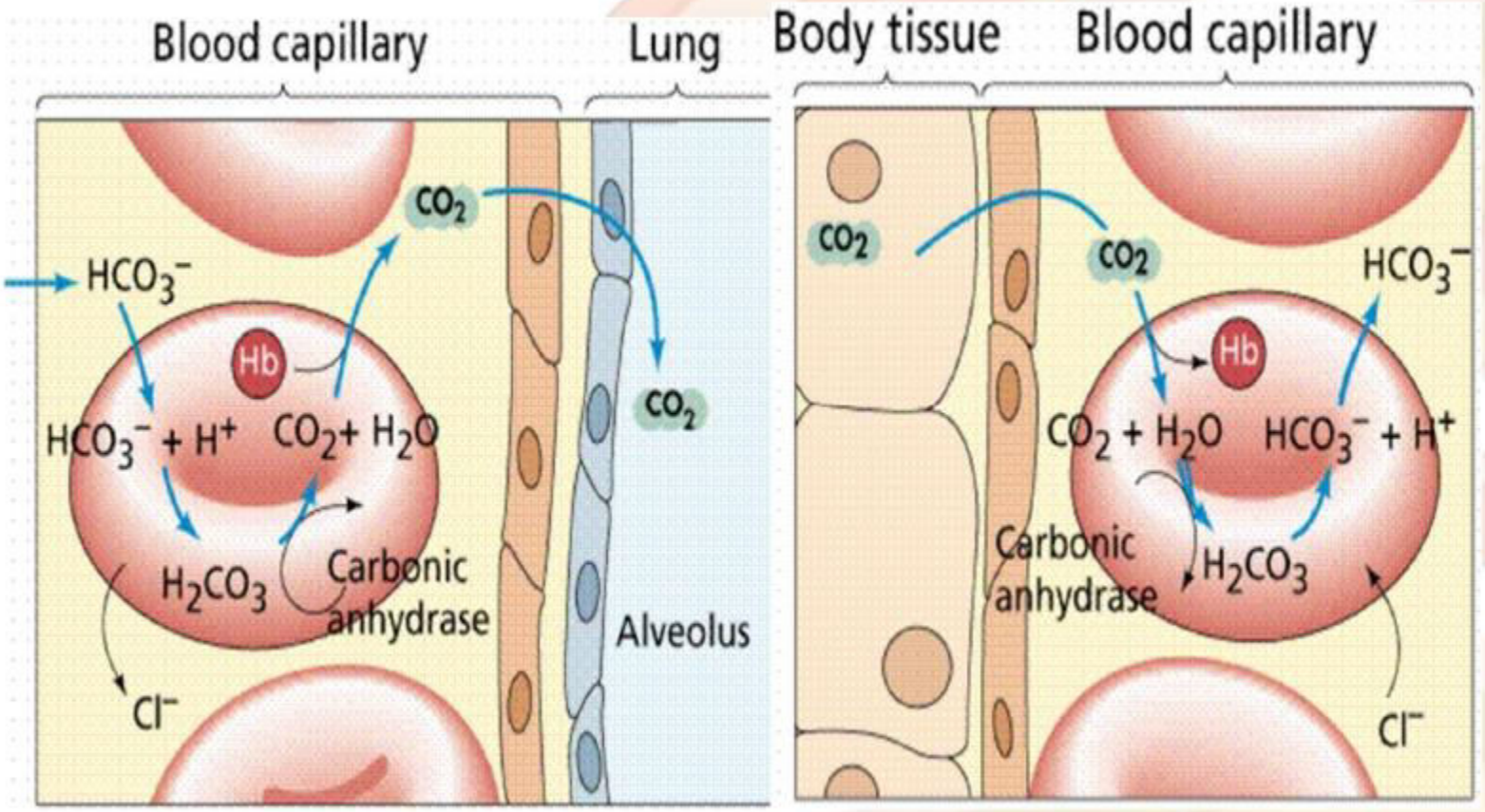
Reversible binding of a protein



Reversible binding of a protein



Reversible binding of a protein



Reversible binding of a protein

- Binding of oxygen thus drives the exhalation of CO_2

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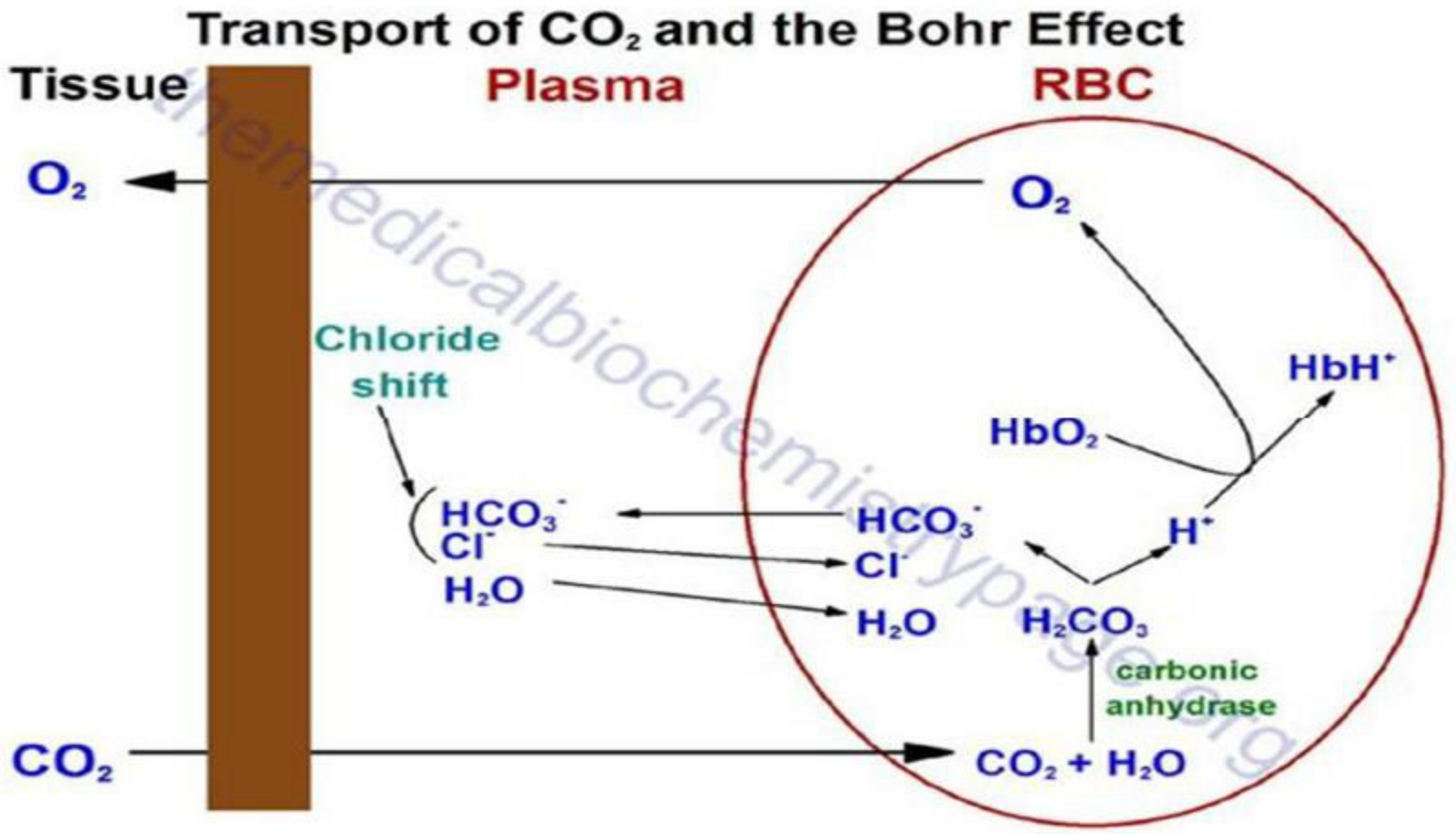
Reversible binding of a protein

- **Bohr effect**
- Hemoglobin transports about 40% of the total H^+
- And 15% to 20% of the CO_2 formed in the tissues to the lungs and the kidneys.

Reversible binding of a protein

- **Bohr effect**
- The remainder of the H^+ is absorbed by the plasma's bicarbonate buffer
- And the remainder of the CO_2 is transported as dissolved HCO_3^- and CO_2.

Reversible binding of a protein



Reversible binding of a protein

Bohr effect

- The binding of H^+ and CO_2 is inversely related to the binding of oxygen.

Reversible binding of a protein

Bohr effect

- At the relatively low pH and high CO_2 concentration of peripheral tissues, the affinity of hemoglobin for oxygen decreases as H_+ and CO_2 are bound, and O_2 is released to the tissues.

Reversible binding of a protein

Bohr effect

- Conversely, in the capillaries of the lung, as CO_2 is exhaled

Reversible binding of a protein

Bohr effect

- And the blood pH consequently rises, the affinity of hemoglobin for oxygen increases and the protein binds more O_2 for transport to the peripheral tissues.

Reversible binding of a protein

Bohr effect

- This effect of pH and CO_2 concentration on the binding and release of oxygen by hemoglobin is called the **Bohr effect**, after Christian Bohr, the Danish Physiologist who discovered it in 1904

Reversible binding of a protein

- **OxyHemoglobin Dissociation Curve**
- **Bohr effect** decreased affinity of hemoglobin for oxygen caused by an increase of carbon dioxide pH etc.

Reversible binding of a protein

- **OxyHemoglobin Dissociation Curve**
- Therefore oxyhemoglobin dissociation curve is displaced to the right because of higher partial pressure of carbon dioxide and lower pH.

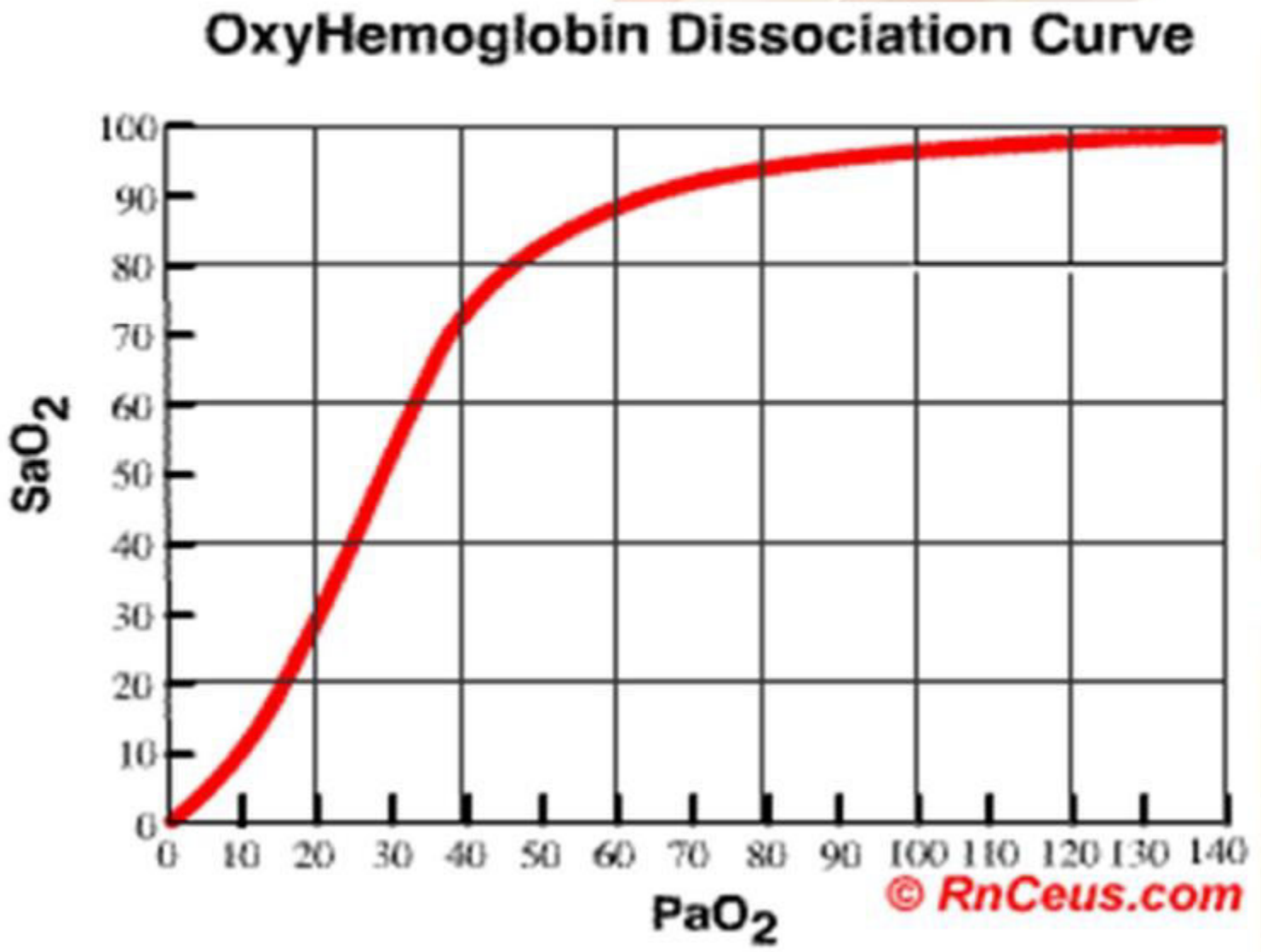
Reversible binding of a protein

- **OxyHemoglobin Dissociation Curve**
- This curve describes the relationship between available oxygen and amount of oxygen carried by hemoglobin.
- The horizontal axis is **PaO_2**, or the amount of oxygen available.

Reversible binding of a protein

- The vertical axis is **SaO_2**, or the amount of hemoglobin saturated with oxygen.

Reversible binding of a protein



Reversible binding of a protein

- Once the **PaO_2** reaches 60 mm Hg the curve is almost flat, indicating there is little change in saturation above this point.
  - So, **PaO_2** of 60 or more is usually considered adequate.

Reversible binding of a protein

- But, at less than 60 mm Hg the curve is very steep, and small changes in the **PaO_2** greatly reduce the **SaO_2**.

Reversible binding of a protein

- The term "affinity" is used to describe oxygen's attraction to hemoglobin binding sites. Affinity changes with:
  - variation in pH,
  - temperature,
  - **CO_2** and,
  - **2,3,-BPG**

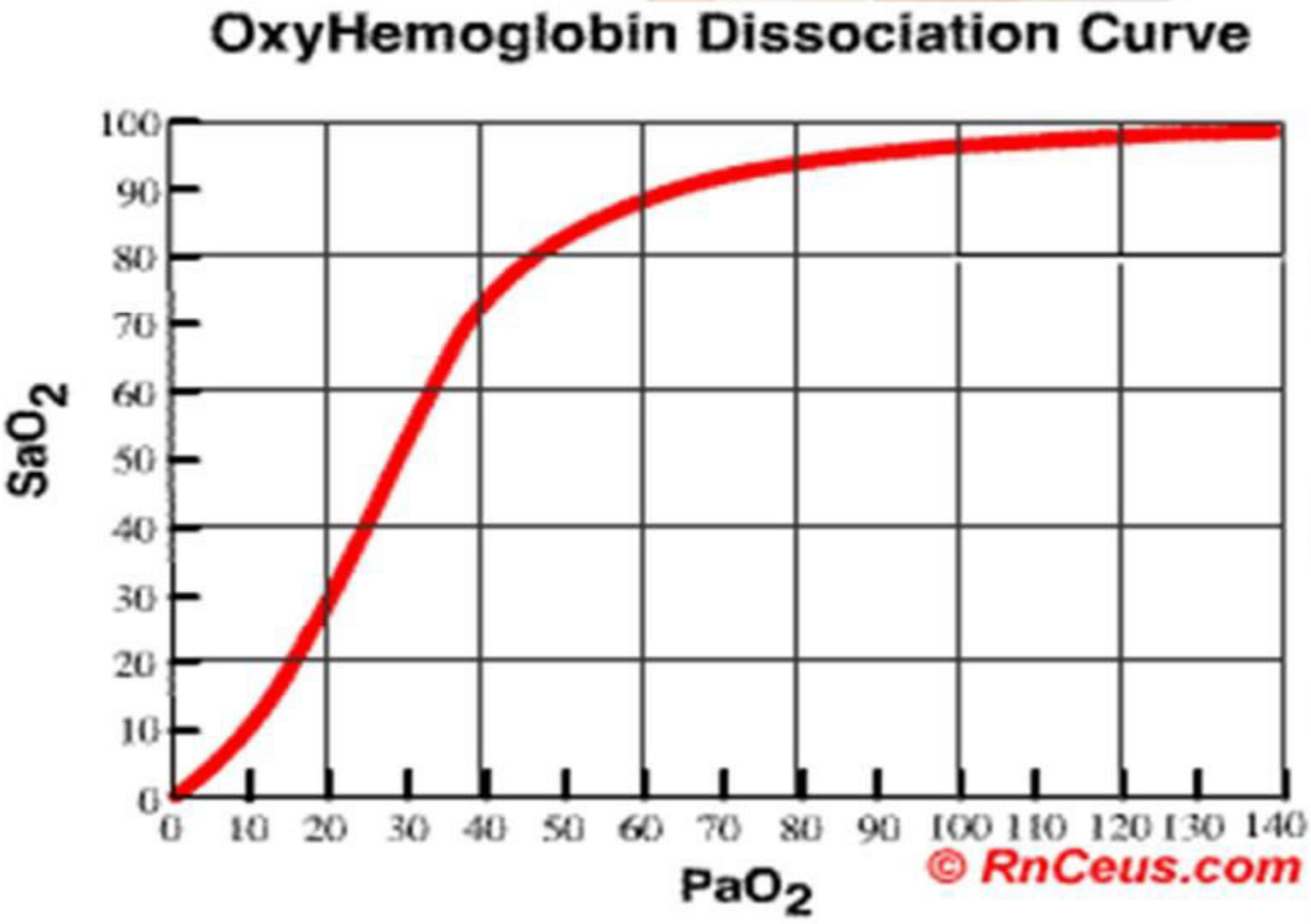
Reversible binding of a protein

- These are the metabolic by-products which competes with O_2 for binding sites.

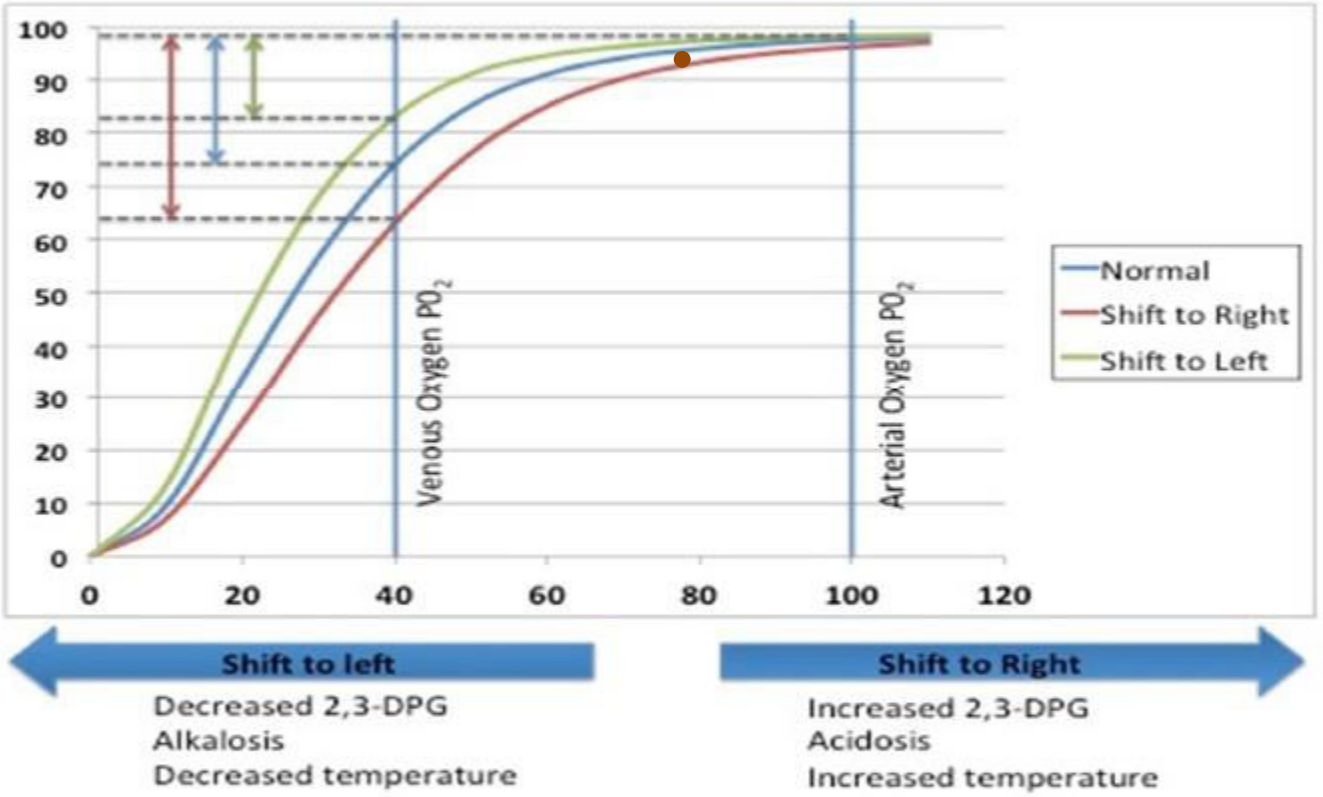
Reversible binding of a protein

- Traditionally the curve starts with:
  - pH at 7.4,
  - temperature at 37 Centigrade, and
  - **PaO_2** at 40.
- Changes from these values are called "shifts

Reversible binding of a protein



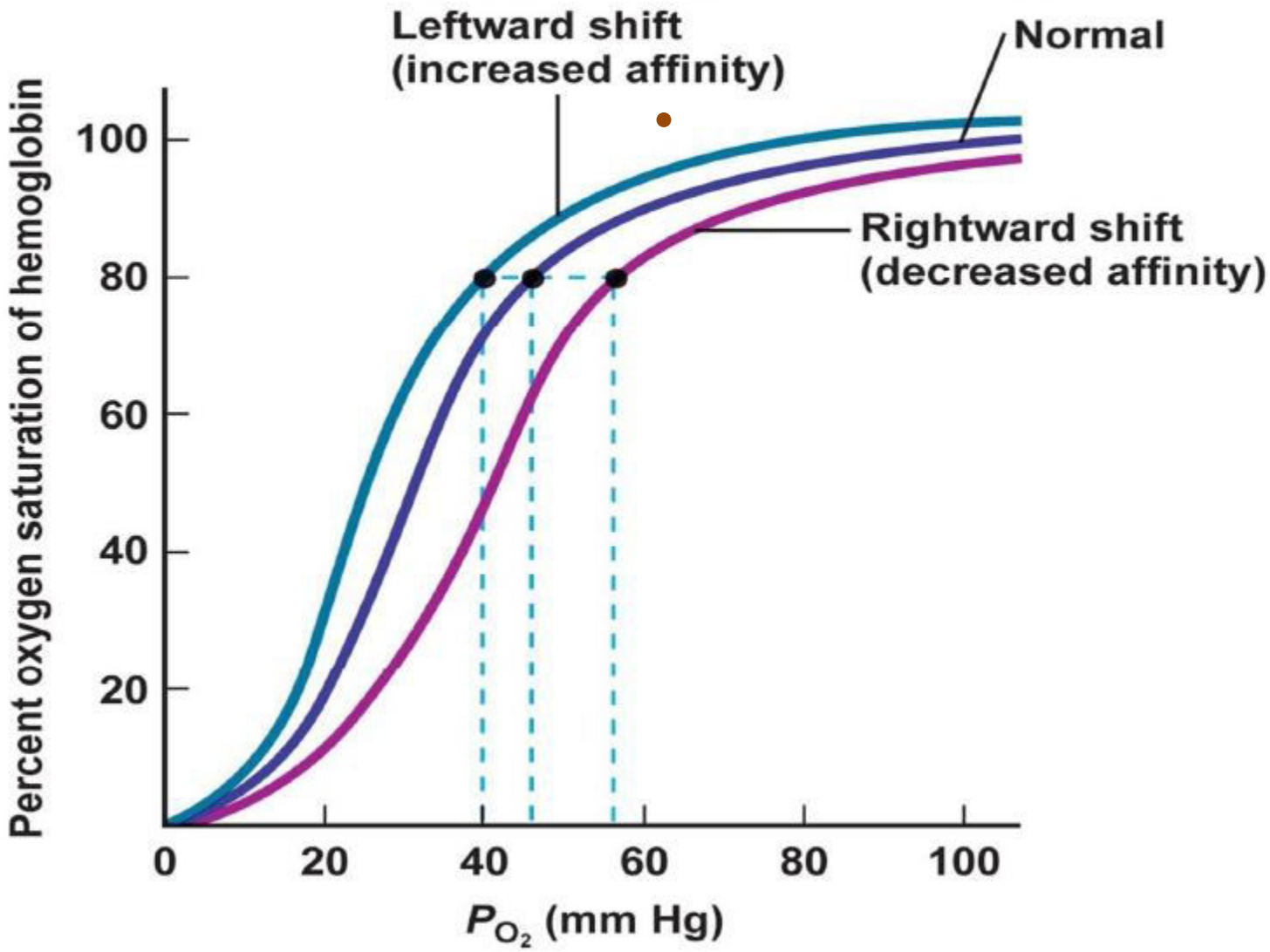
Reversible binding of a protein



A left shift will increase oxygen's affinity for

A right shift decreases oxygen's affinity for hemoglobin

Reversible binding of a protein



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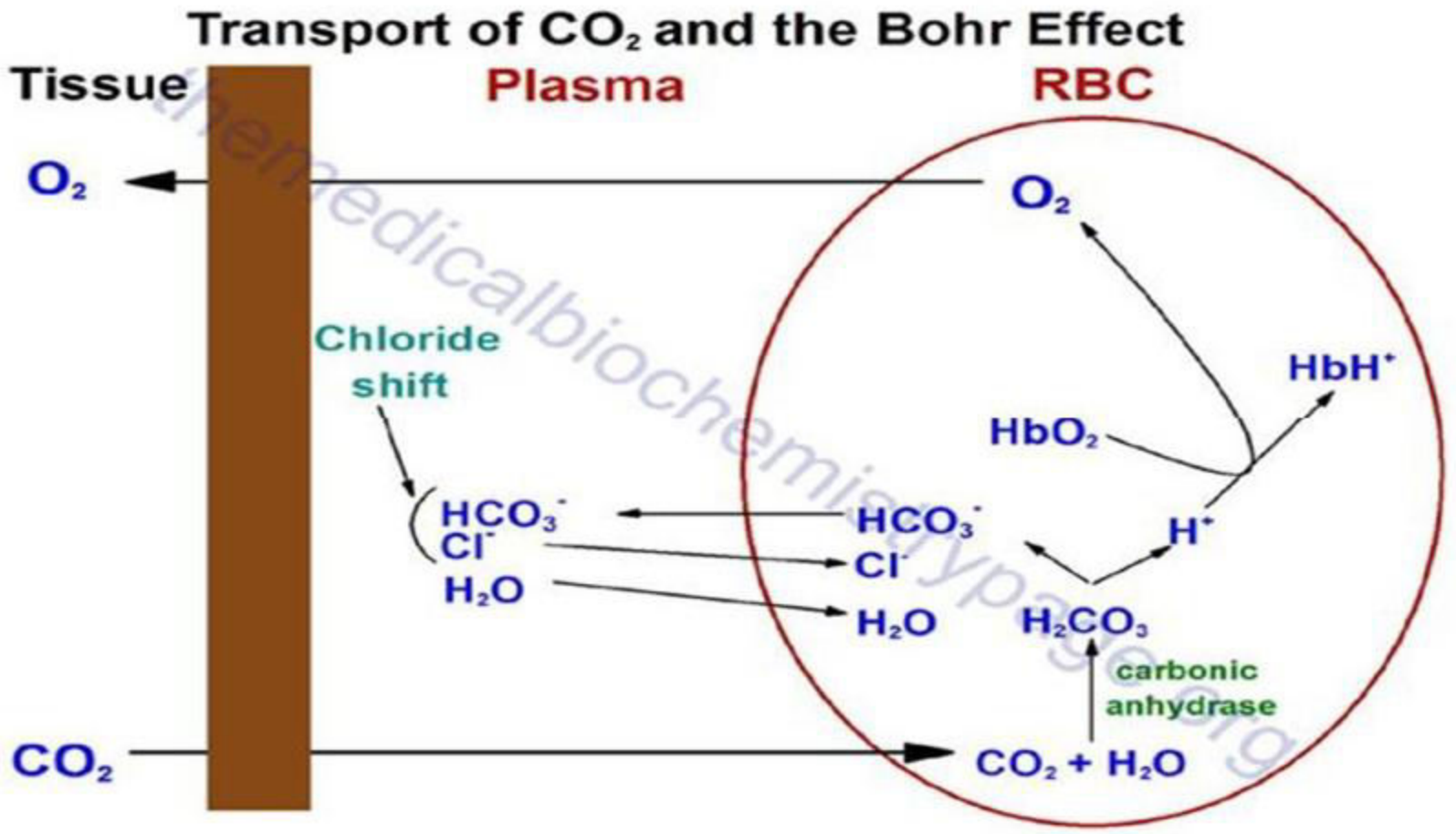
Reversible binding of a protein

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Reversible binding of a protein

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Reversible binding of a protein



Reversible binding of a protein

Bohr effect

- The binding of H^+ and CO_2 is inversely related to the binding of oxygen.

Reversible binding of a protein

Bohr effect

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Reversible binding of a protein

Bohr effect

- Conversely, in the capillaries of the lung, as CO_2 is exhaled

Reversible binding of a protein

Bohr effect

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Reversible binding of a protein

Bohr effect

- This effect of pH and CO_2 concentration on the binding and release of oxygen by hemoglobin is called the **Bohr effect**, after Christian Bohr, the Danish Physiologist who discovered it in 1904

Reversible binding of a protein

- **Proton Binding** (Cont.)
- **In the tissues**,

  as the concentration of H^+ rises, (i.e.

  decrease in pH), protonation of His HC3 occurs.

Reversible binding of a protein

- This protonation promotes **release** of oxygen by favoring a transition to the **T state**, subsequently resulting in oxygen unloading.

Reversible binding of a protein

- **In the lungs,**
- the pH of blood is higher because CO_2 is being exhaled,
- And hemoglobin is being oxygenated due to higher partial pressure of oxygen in alveoli.

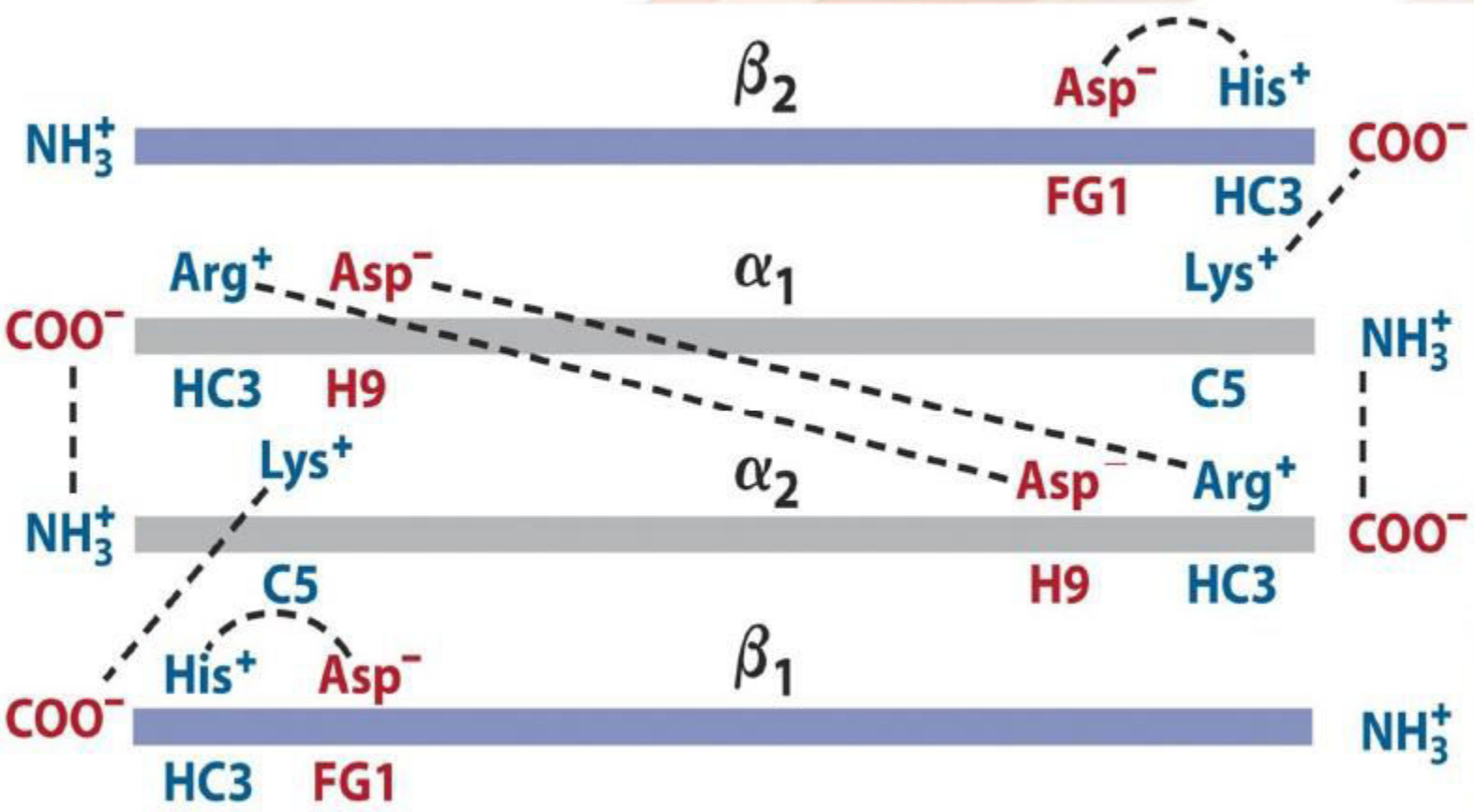
Reversible binding of a protein

- The increase in pH,
- exhalation of CO_2
- and oxygenation of Hb;
- all result in; **deprotonation** of His HC3 of Hb.

Reversible binding of a protein

- This disrupts the ionic interactions that stabilize the taut state
- And a new stabilized state of **relaxed** Hb is formed, resulting in oxygen loading.

Reversible binding of a protein



Some ion pairs that stabilize the T state of deoxyhemoglobin: Interactions between the ion pairs His HC3 and Asp FG.1 of the Beta subunit and between Lys C5 of the alpha Subunit and His HC3 of the Beta subunit are shown with dashed line.

Reversible binding of a protein

- Binding of O_2, protons and CO_2 to the Hb is a dynamic process, in which relative concentrations of each of the compounds in the different compartments of the body dictate the binding of each other.

Reversible binding of a protein

- Thus we have seen that the four polypeptide chains of hemoglobin communicate with each other about not only O_2 binding to their heme groups
- But also H^+ binding to specific amino acid residues

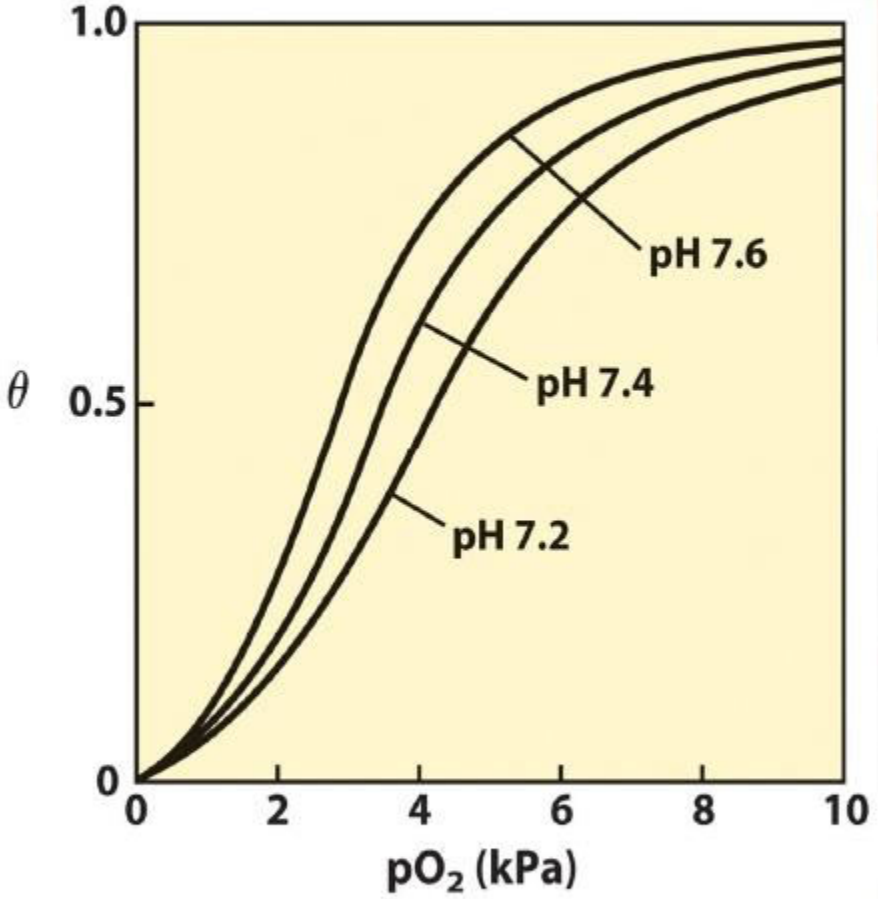
Reversible binding of a protein

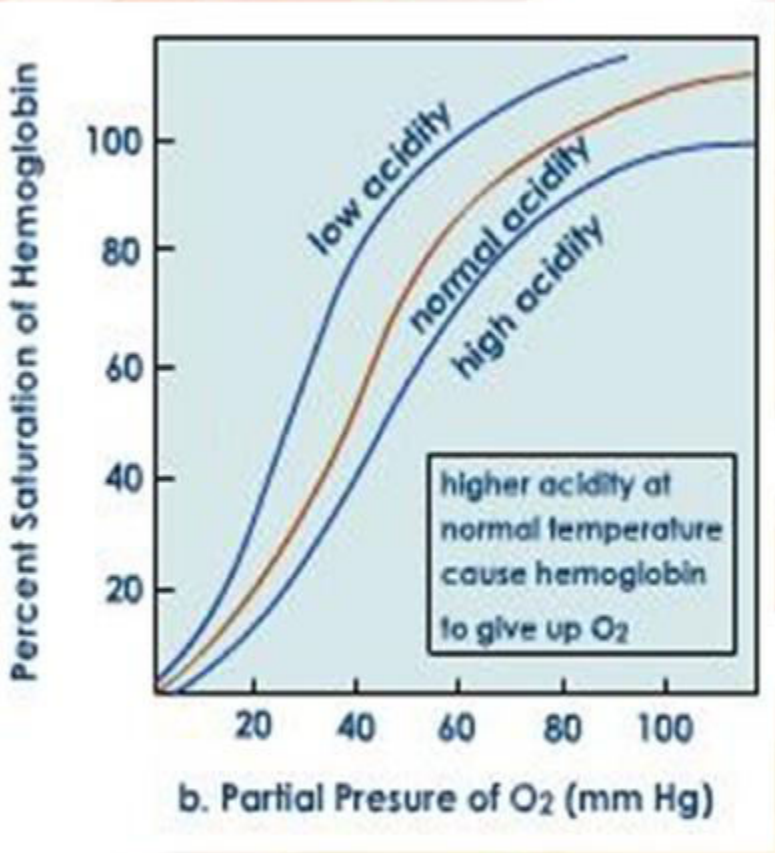
- **Proton Binding (cont.)**
- An increase in protons and\or a higher pCO_2 shifts the equilibrium to the right
- Therefore favoring deoxyhemoglobin T form.

Reversible binding of a protein

- Whereas a decrease in pCO_2 and\or a decrease in protons shifts the equilibrium to the left
- Conversely, an increase in pO_2 promotes proton release

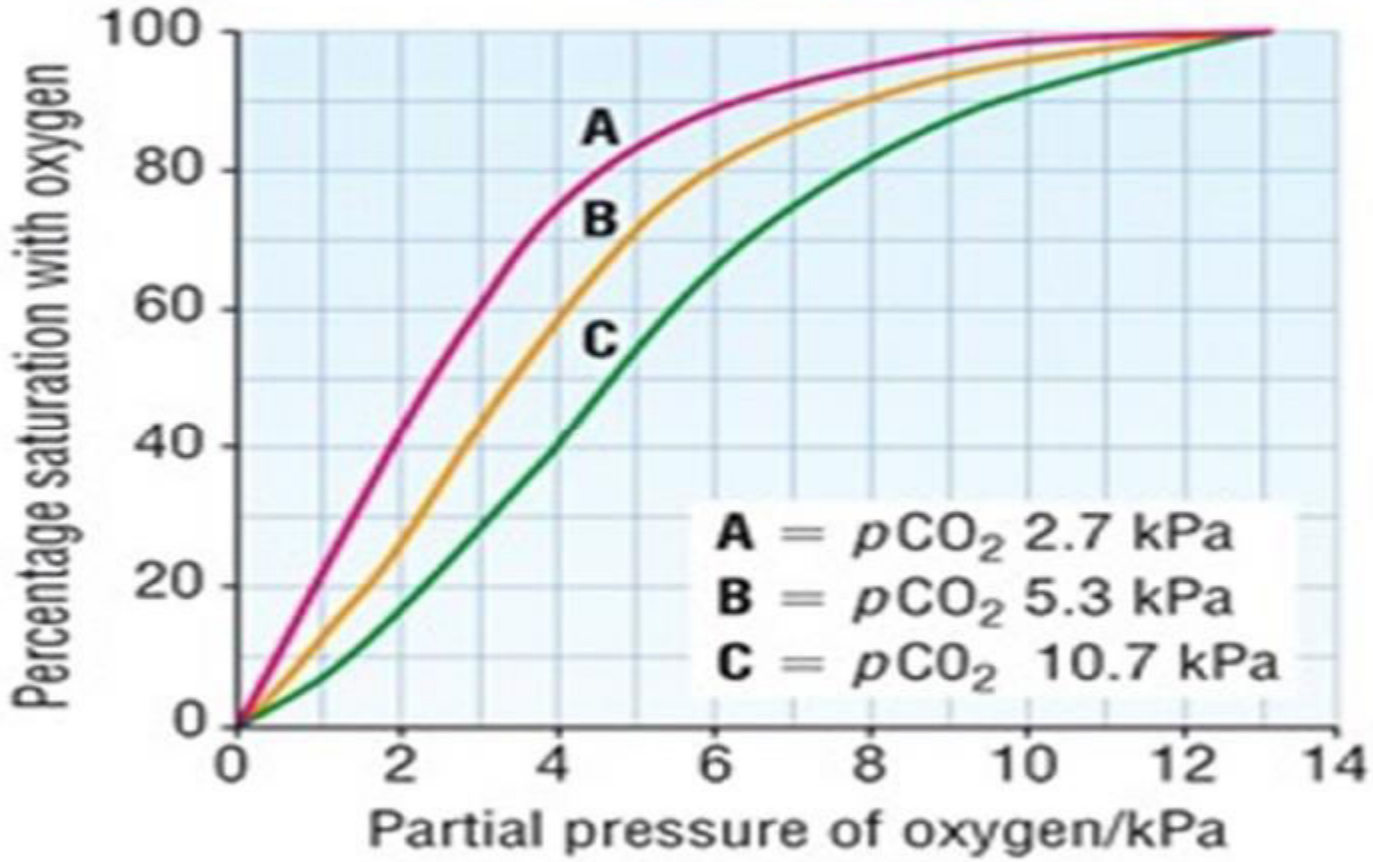
Reversible binding of a protein





Decreased pH and increased CO2 shift the curve to right, i.e. . *increased* O2 unloading :
Increased pH and decreased CO2 shift the curve to left i.e. *decreased* O2 unloading.

Reversible binding of a protein



Decreased pH and increased CO2 shift the curve to right, i.e. . *increased* O2 unloading :
Increased pH and decreased CO2 shift the curve to left i.e. *decreased* O2 unloading.

Reversible binding of a protein

- The differential pH gradient (lungs having a higher pH, tissues a lower pH) favors the unloading of oxygen in the peripheral tissues,
- And the loading of oxygen in the lung

Reversible binding of a protein

- In Summary,
- Binding of oxygen to facilitate other molecule of oxygen is called positive co-operativity.

Reversible binding of a protein

- Whereas, binding of carbon dioxide and proton to release oxygen is called negative allostery, in which;
- increased CO_2 and H_+ ion favor oxygen unloading.

Reversible binding of a protein

- Thus we see that the four polypeptide chains of hemoglobin communicate with each other about not only O_2binding to their heme groups

Reversible binding of a protein

- But also H^+ binding to specific amino acid residues and CO_2 binding to amino terminal.

Reversible binding of a protein

- It is the capacity to communicate ligand binding information from one polypeptide subunit to the others that makes the hemoglobin molecule so beautifully adapted to integrate the transport of O_2, CO_2 and H^+ by RBCs.

Reversible binding of a protein

- And there is still more to the story.
- Many more interactions of different amino acids, are being identified as new techniques are employed to study these effects, which are out of the scope of this course.

Reversible binding of a protein

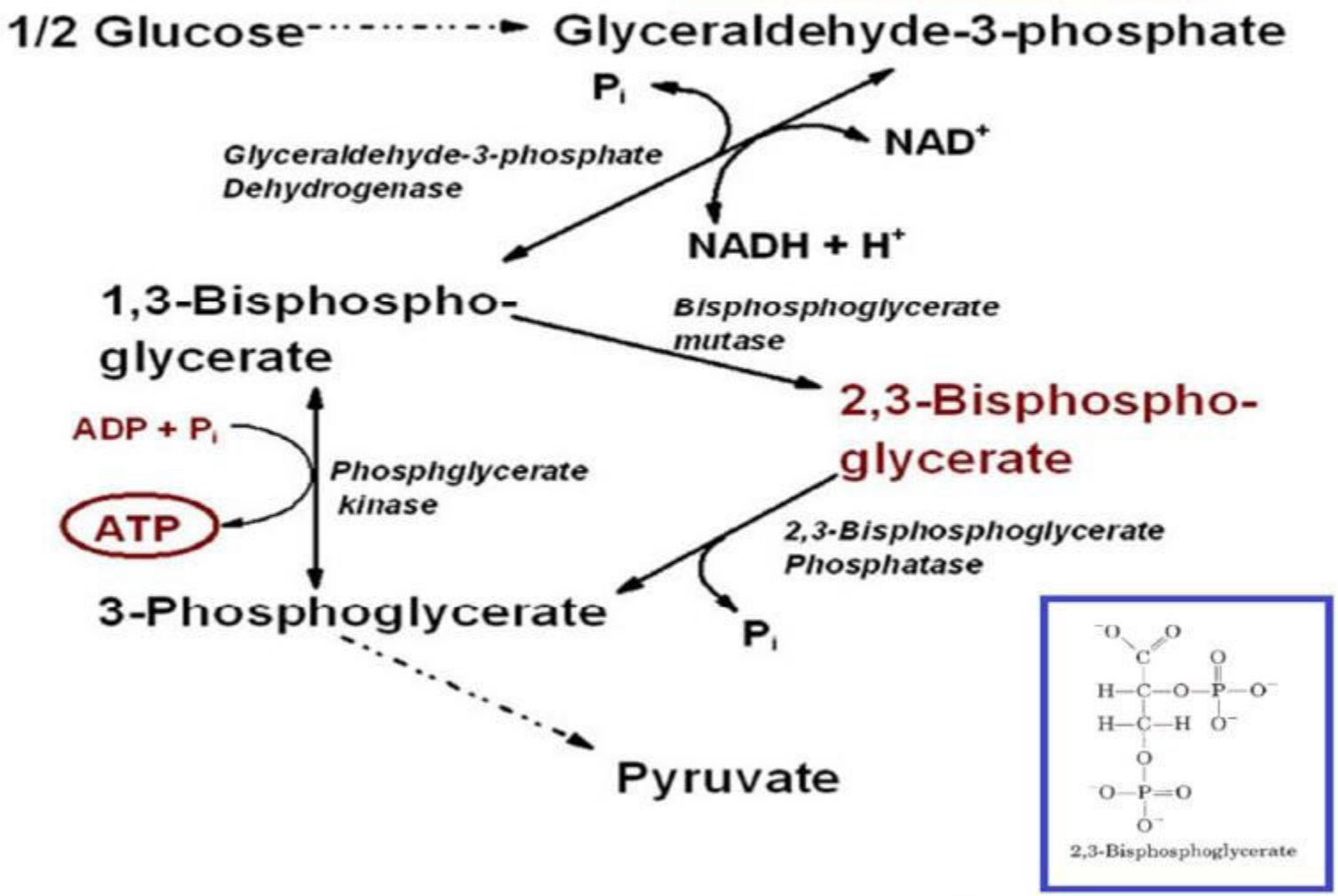
- **2,3 bisphosphoglycerate**
- The interaction of 2,3 bisphosphoglycerate (BPG) with hemoglobin further refines the function of hemoglobin
- and provides example of heterotropic allosteric modulation.

Apart from Oxygen, ability of hemoglobin to reversibly bind oxygen is affected by pH of the environment, pCO_2& the availability of 2,3-bisphospho-phoglycerate

Reversible binding of a protein

- 2, 3 BPG is an alternate product of glycolysis and its concentration increases in RBCs in states of low oxygen delivery to the tissues.
- It is a negatively charged molecule which increases the stability of taut form of hemoglobin.

Reversible binding of a protein



Reversible binding of a protein

- 2,3-Bisphosphoglycerate is known to greatly reduce the affinity of hemoglobin for oxygen.
- There is an inverse relationship between the binding of O_2 and the binding of BPG.

Reversible binding of a protein

- BPG binds at a site distant from the oxygen-binding site
- And therefore regulates the O_2-binding affinity of hemoglobin in relation to the pO_2 in the lungs.

Reversible binding of a protein

- BPG is important in the physiological adaptation to the lower pO_2 at high altitudes

Reversible binding of a protein

- At sea level, the binding of oxygen to hemoglobin is regulated such that the amount of oxygen delivered to the tissues is nearly 40% of the maximum that could be carried by the blood

Reversible binding of a protein

- At high altitude of hills and mountains, where the pO_2 is considerably lower (due to low atmospheric pressure).
- The delivery of oxygen to the tissues is now reduced.

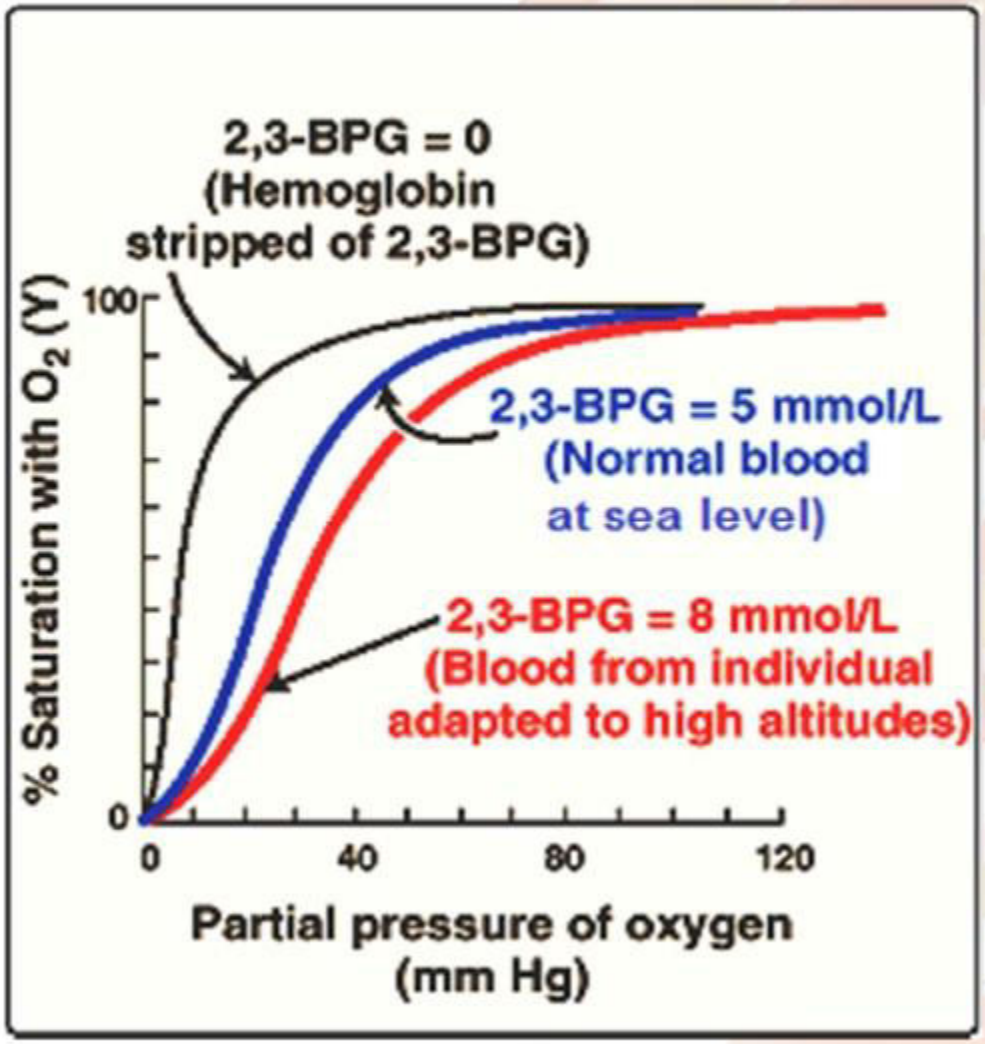
Reversible binding of a protein

- However, after just a few hours at the higher altitude, the BPG concentration in the blood begins to rise, leading to a decrease in the affinity of hemoglobin for oxygen.

Reversible binding of a protein

- This adjustment in the BPG level has only a small effect on the binding of oxygen in the lungs
- But a considerable effect on the release of oxygen in the tissues shown by increased oxygen unloading at the peripheral tissues.

Reversible binding of a protein



Increase in BPG concentration in the RBC shifts the dissociation curve to the right, i.e. increased O_2 unloading. BPG concentration also increases in those conditions in which there is decreased O_2 delivery to tissues, such as in anemia and respiratory diseases.

Reversible binding of a protein

- As a result, the delivery of oxygen to the tissues is restored to nearly 40% of the oxygen that can be transported by the blood.

Reversible binding of a protein

- **2,3 bisphosphoglycerate**
- **(Count.)**

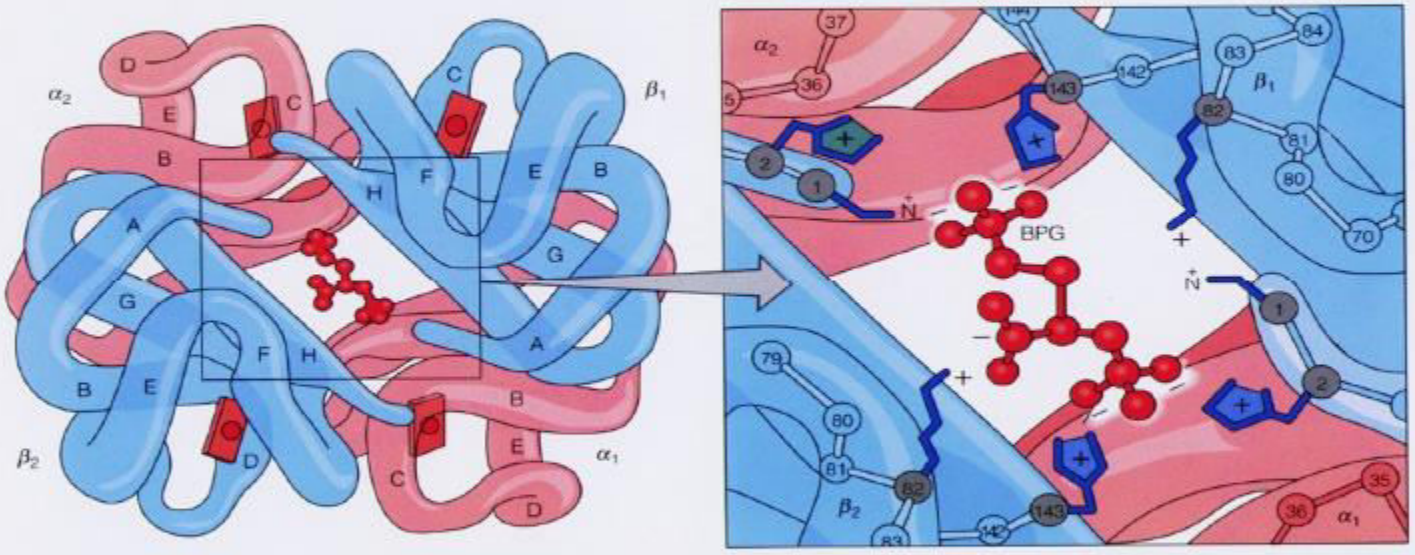
- The site of BPG binding to Hb is the cavity between the **beta** subunits.

Reversible binding of a protein

- This cavity is lined with positively charged amino acid residues that interact with the negatively charged groups of BPG.

Reversible binding of a protein

Figure 7.18 Binding of 2, 3-bisphosphoglycerate to deoxyhemoglobin



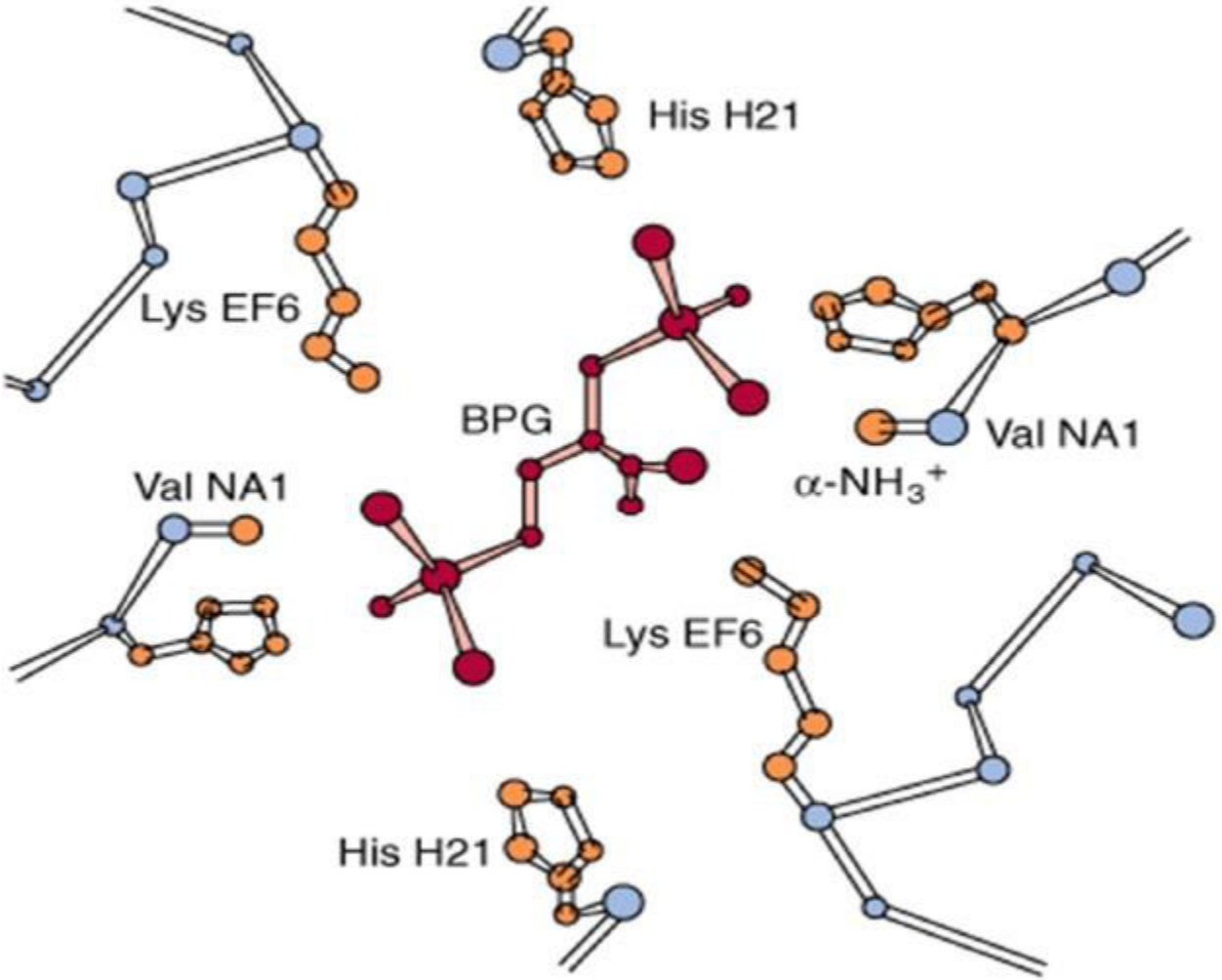
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Reversible binding of a protein

• BPG forms ionic interactions with the terminal amino groups of **both chains** via Val NA1 and with Lys EF6 and His H21.

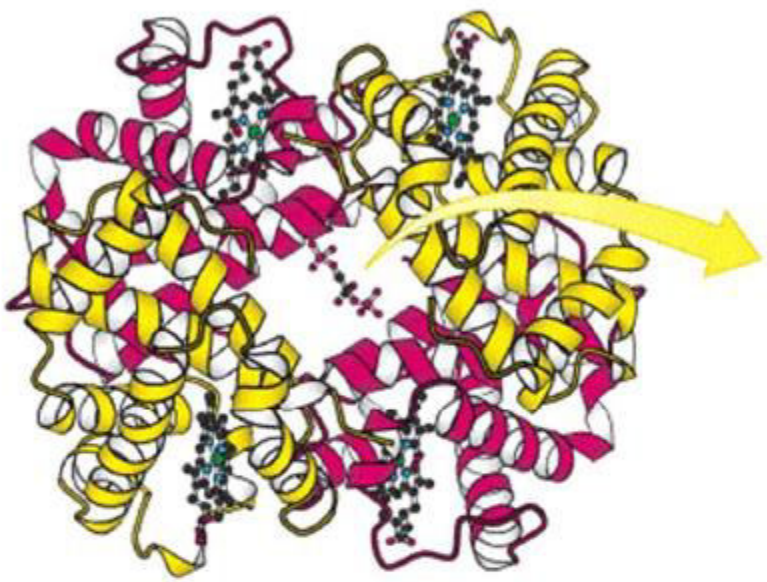
NA is a non-helical portion between the start of **amino terminal** and start of **A helix** of **beta chain.**
Val NA1 is the first amino acid in this **non-helical** portion.

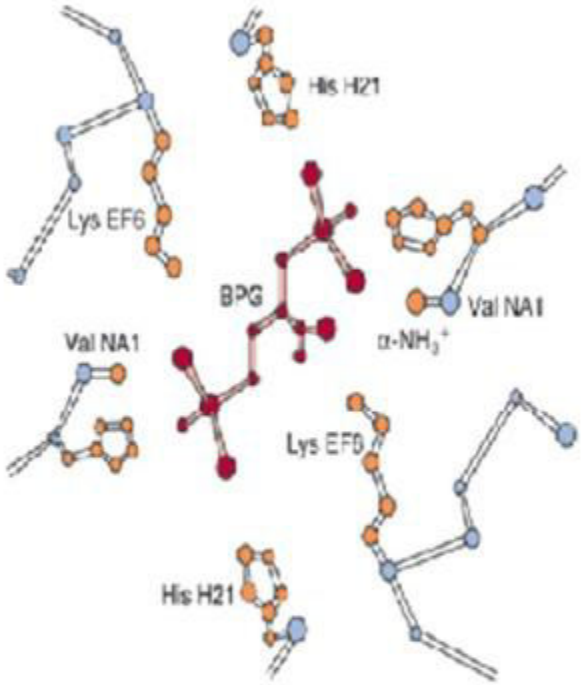
Reversible binding of a protein



NA is a non-helical portion between the start of **amino terminal** and start of **A helix** of **beta chain.**

Reversible binding of a protein





Reversible binding of a protein

- BPG therefore stabilizes deoxygenated (T state) hemoglobin by forming additional ionic interactions that must be broken prior to conversion to the R state.

Reversible binding of a protein

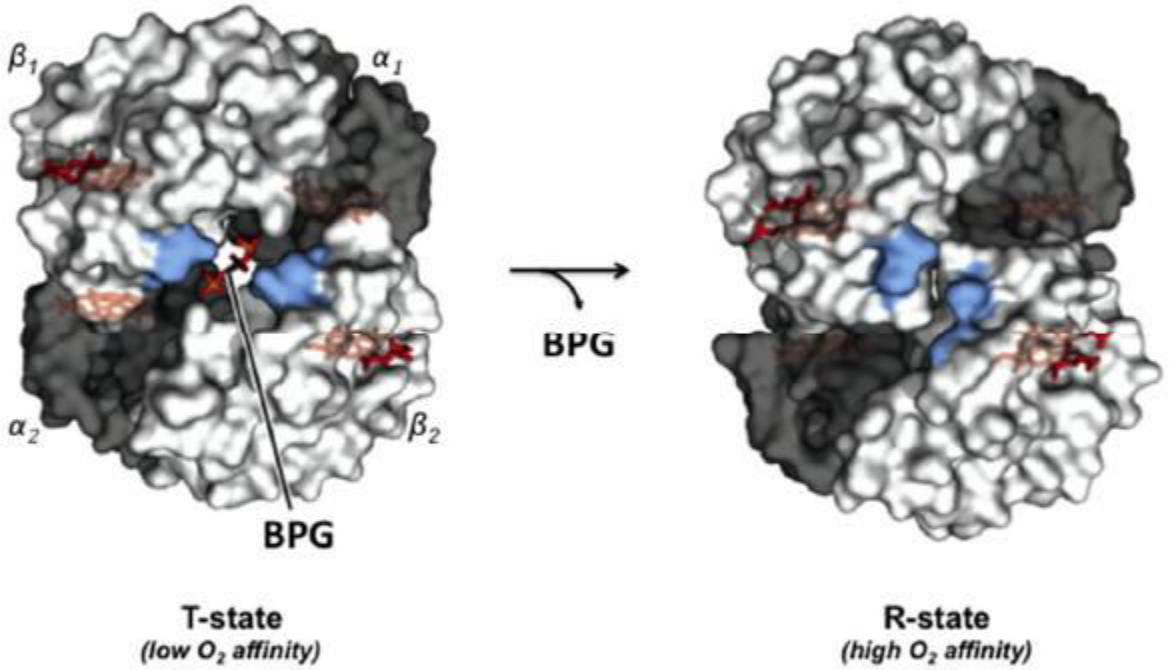
- Unlike oxygen, only one molecule of BPG is bound to each hemoglobin tetramer

- BPG lowers hemoglobin's affinity for oxygen by stabilizing the T state

Reversible binding of a protein

- The transition to the R state narrows the binding pocket.
- Therefor BPG, precluding BPG binding.

Reversible binding of a protein



Reversible binding of a protein

- Residue H21 of the γ subunit of HbF is Ser rather than His which cannot form ionic interactions with BPG
- Therefore, BPG binds more weakly to HbF than to HbA.

Reversible binding of a protein

- The lower stabilization afforded to the T state by BPG accounts for HbF having a higher affinity for O_2 than HbA

Reversible binding of a protein

Sickle Cell Anemia

- Sickle Cell Anemia is a molecular disease of hemoglobin.
- It demonstrates the importance of amino acid sequence in hierarchical structure of globular proteins, and thus their biological functions

Reversible binding of a protein

Sickle cell anemia is a genetic disorder caused by a single nucleotide alteration (a **point mutation**) in the β-globin gene.

- This results in the production of altered hemoglobin.

Reversible binding of a protein

- The point mutation in the DNA sequence result in the replacement of;
- Glutamate residue with Valine residue
- At position 6 of the β chain of hemoglobin.

Reversible binding of a protein

HBB Sequence in Normal Adult Hemoglobin (Hb A):

Nucleotide	CTG	ACT	CCT	GAG	GAG	AAG	TCT
Amino Acid	Leu	Thr	Pro	Glu	Glu	Lys	Ser
	3			6			9

HBB Sequence in Mutant Adult Hemoglobin (Hb S):

Nucleotide	CTG	ACT	CCT	GTG	GAG	AAG	TCT
Amino Acid	Leu	Thr	Pro	Val	Glu	Lys	Ser
	3			6			9

Reversible binding of a protein

- Because humans generally have two copies of β-globin gene.
- An individual may have two copies of mutant allele (i.e. **homozygous**)
- or one copy of each of normal and mutant allele

  (i.e. **heterozygous**)

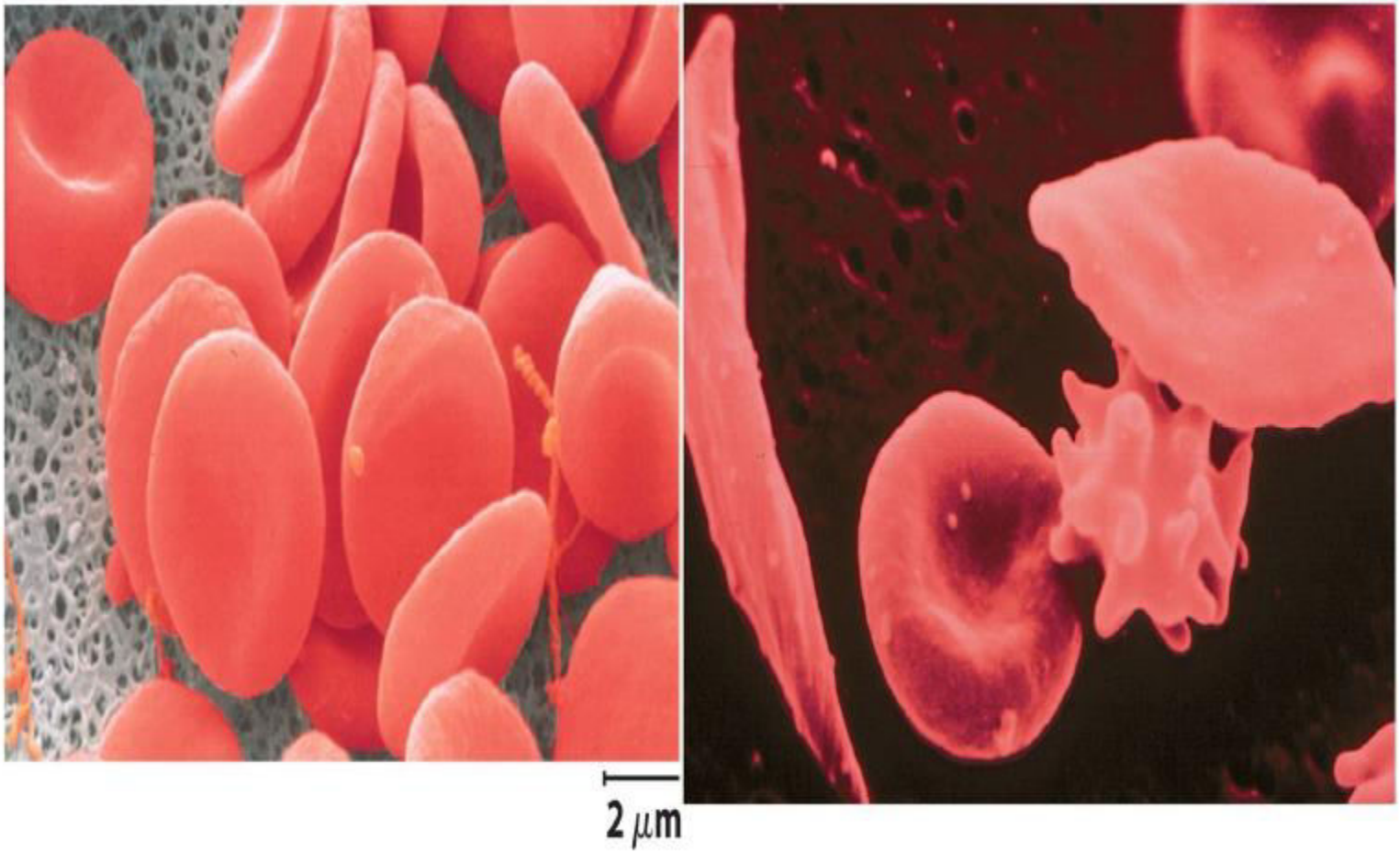
Reversible binding of a protein

- Sickle-cell anemia occurs in individuals who inherit the allele for sickle-cell hemoglobin from **both** parents.
- The resulting hemoglobin $\alpha_2\beta_{s2}$, is referred to as **HbS**

Reversible binding of a protein

- The RBCs of these individuals are fewer and also abnormal.
- In addition to an unusually large number of immature cells, the blood contains many long, thin, sickle shaped RBCs.

Reversible binding of a protein



Normal: left

sickle cell: right

Reversible binding of a protein

- When HbS is deoxygenated, it becomes insoluble and forms polymers that aggregate into tubular fibers.
- The insoluble fibers of deoxygenated hemoglobin S cause the deformed, sickle shape of the RBCS.

Reversible binding of a protein

- The proportion of sickled cells increases greatly as blood is deoxygenated.

- Normal hemoglobin (hemoglobin A) remains soluble on deoxygenation.

END

Reversible binding of a protein

Sickle Cell Anemia

(Count.)

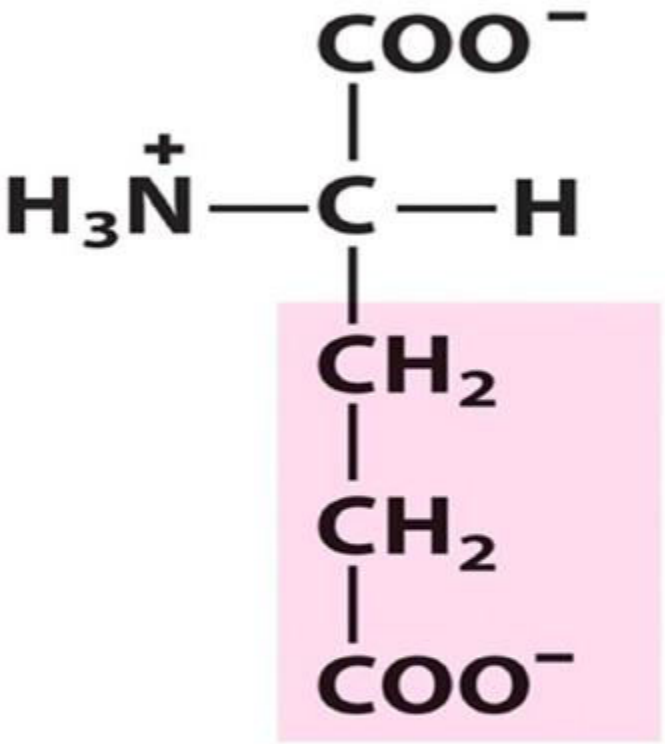
- The altered properties of hemoglobin S result from the replacement of Val instead of a Glu residue at position **6** in the two **β** chains.

Reversible binding of a protein

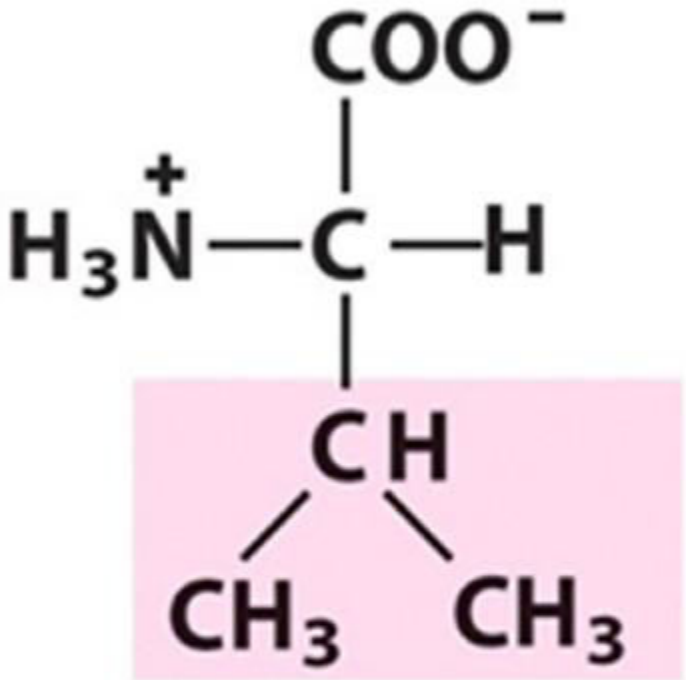
- The R group of valine is neutral, whereas glutamate has a negative charge at pH 7.4.

- Hemoglobin S therefore has two fewer negative charges than hemoglobin A.

Reversible binding of a protein



Glutamate

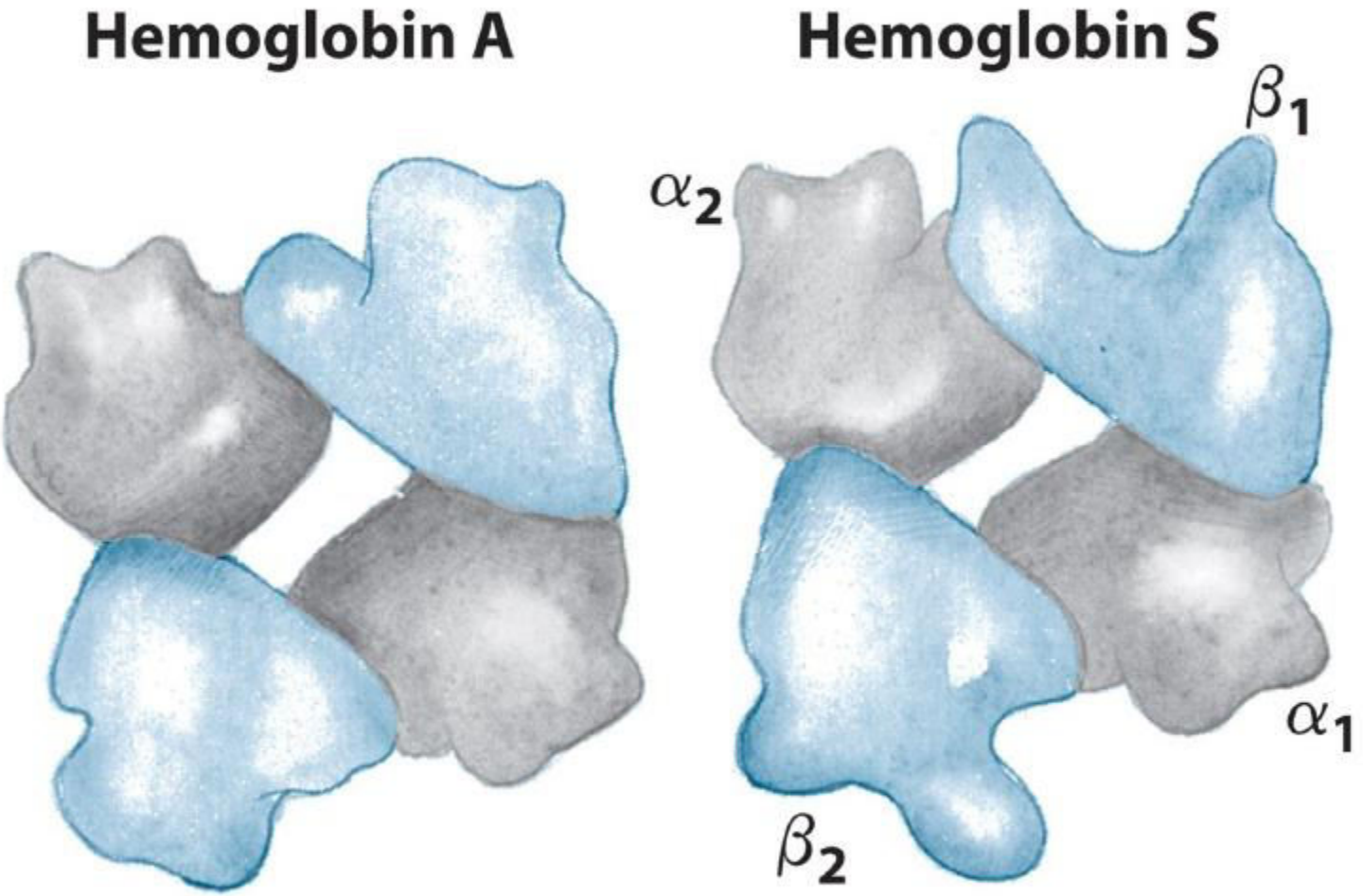


Valine

Reversible binding of a protein

- Replacement of the GIu residue by Val creates a "sticky" hydrophobic contact point at position 6 of the β chain, which is on the outer surface of the molecule

Reversible binding of a protein

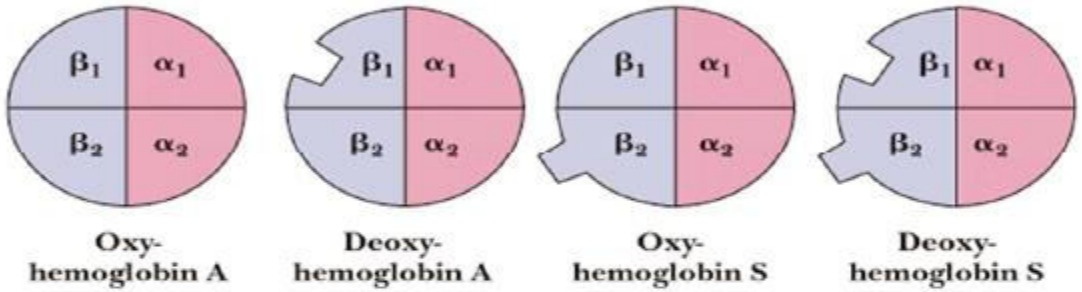


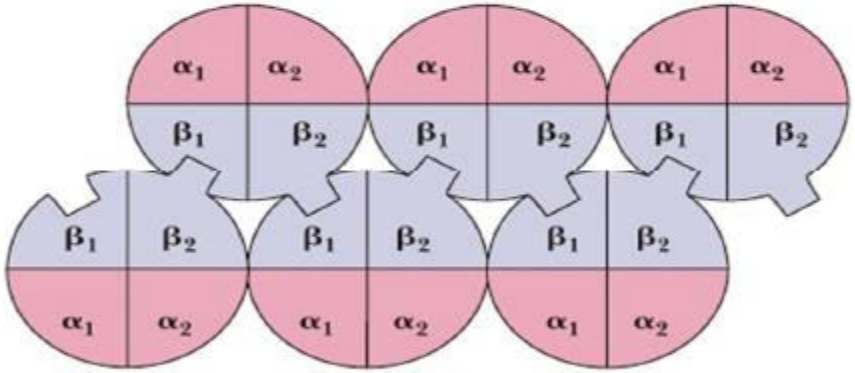
Reversible binding of a protein

- These sticky spots cause **deoxyhemoglobin** S molecules to associate abnormally with each other,
- Therefore forming the long, fibrous aggregates characteristic of this disorder.

Reversible binding of a protein

Garrett & Grisham: Biochemistry, 2/e
Figure 15.40





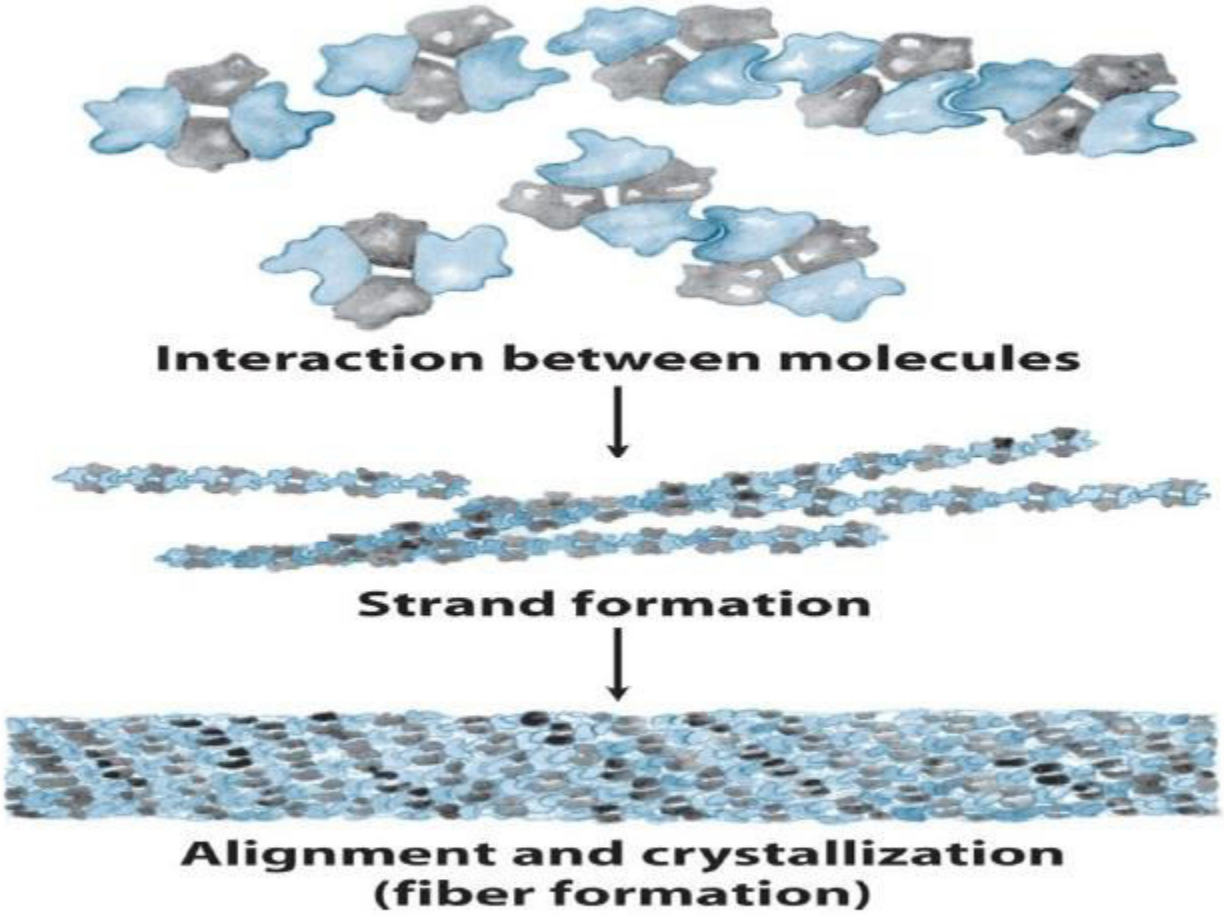
Deoxyhemoglobin S polymerizes into filaments

Saunders College Publishing

Reversible binding of a protein

- The polymerization of Hb S inside the cytosol of RBCs make RBCs sickle shaped and fragile..

Reversible binding of a protein



Reversible binding of a protein

- Normal Hb A does not have this sticky hydrophobic pocket and therefore is soluble in the cytosol.

Reversible binding of a protein

- The lifetime of an RBC homozygous for HbS is approximately 20 days, compared with 120 days for normal red blood cells.

Reversible binding of a protein

- Sickled cells are very fragile and rupture easily; this results in anemia.
- Moreover, these cells have less ability to squeeze through small blood vessels resulting in the occlusion of micro blood vessels in the body, i.e. capillaries.

Reversible binding of a protein

- Sickle-cell anemia, occurs in individuals homozygous for the HbS allele,
- Individuals who receive the HbS allele from only one parent (i.e. heterozygous) experience a milder condition called sickle-cell trait.

Reversible binding of a protein

- Only about 1% of their RBCs become sickled on deoxygenation.

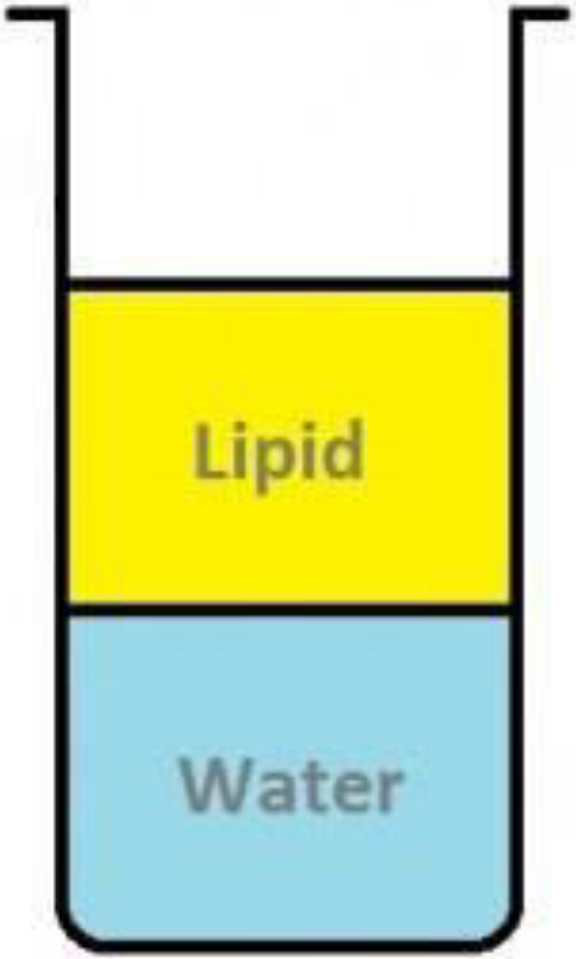
Lipid Chemistry

- The lipids are a heterogeneous group of compounds, including fats, oils, steroids, waxes,
- their derivatives
- and precursors,
- that are related more by their physical than by their chemical properties.

Lipid Chemistry

- They have the common property of being
- (1) **Insoluble in water**
- (2) **Soluble in nonpolar solvents** such as ether and chloroform.

Lipid Chemistry



Lipids are characterized by their hydrophobicity and their immiscibility in water

Lipid Chemistry

- The biological functions of the lipids are as diverse as their chemistry.

Lipid Chemistry

- Fats and oils are the principal stored form of energy in many organisms.
- Phospholipids and sterols are major structural elements of biological membranes.

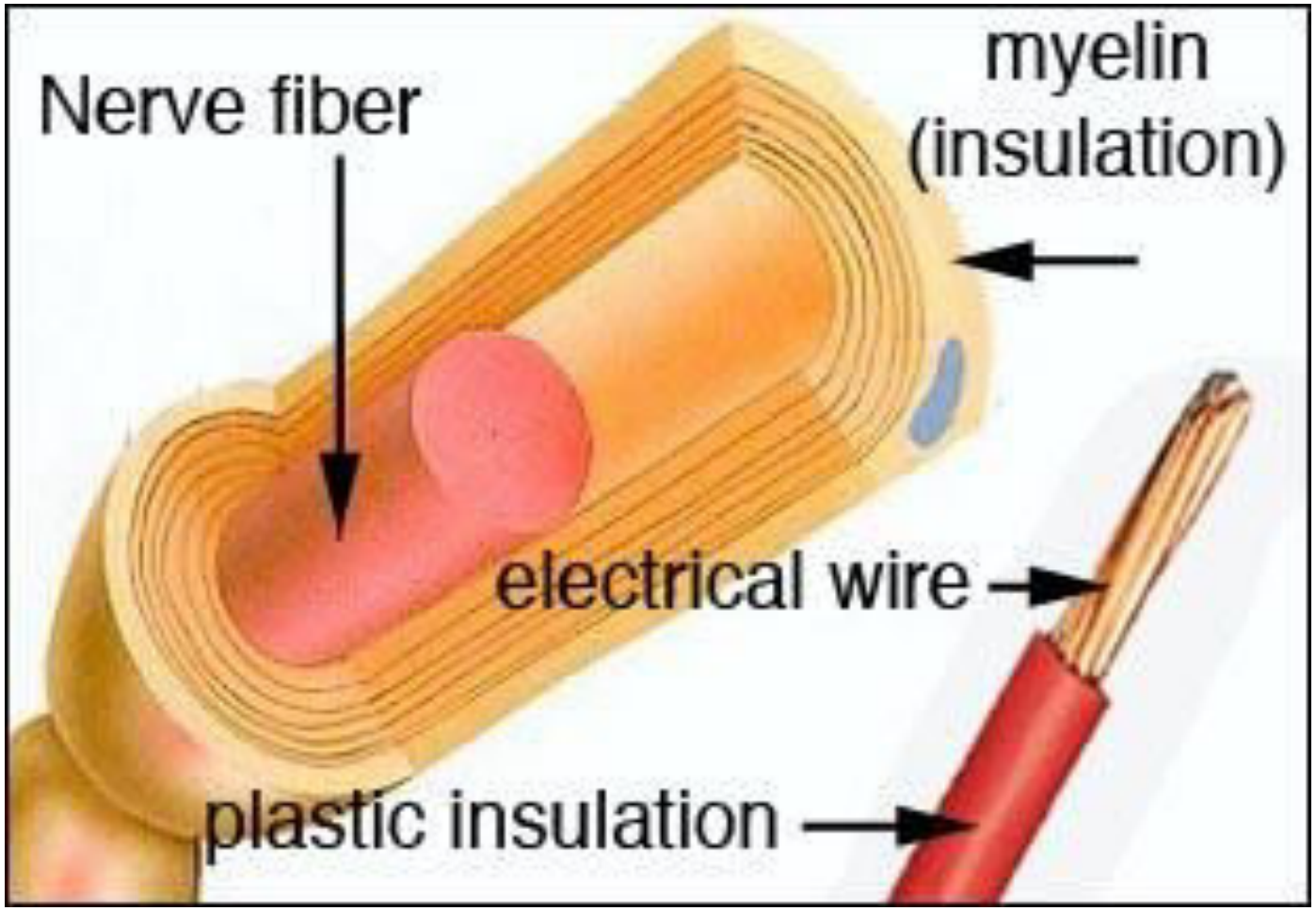
Lipid Chemistry

- Other lipids, present in relatively small quantities, play crucial roles as; enzyme cofactors, electron carriers, hydrophobic anchors for proteins, emulsifying agents in the digestive tract, hormones, and cellular messengers

Lipid Chemistry

- **Adipose tissue Fat** also serves as a thermal insulator in the subcutaneous tissues and around certain organs.
- Nonpolar lipids act as **electrical insulators**, allowing rapid propagation of depolarization waves along **myelinated nerves**

Lipid Chemistry



Myelination with FAT greatly reduce the loss of charges from nerve fibre, resulting in a High speed transmission of nerve impulse

Lipid Chemistry

- The fat-soluble vitamins serve as co enzymes for many important reactions in the body.
- Combinations of lipid and protein (lipoproteins) serve as the means of **transporting lipids** in the blood

Lipid Chemistry

- Importantly, lipids provide the hydrophobic barrier that permits partitioning the aqueous contents of cells and subcellular structures as; phospholipids and sterols are the major structural elements of biological membranes.

END

Lipid Chemistry

CLASSIFICATIONS OF LIPIDS

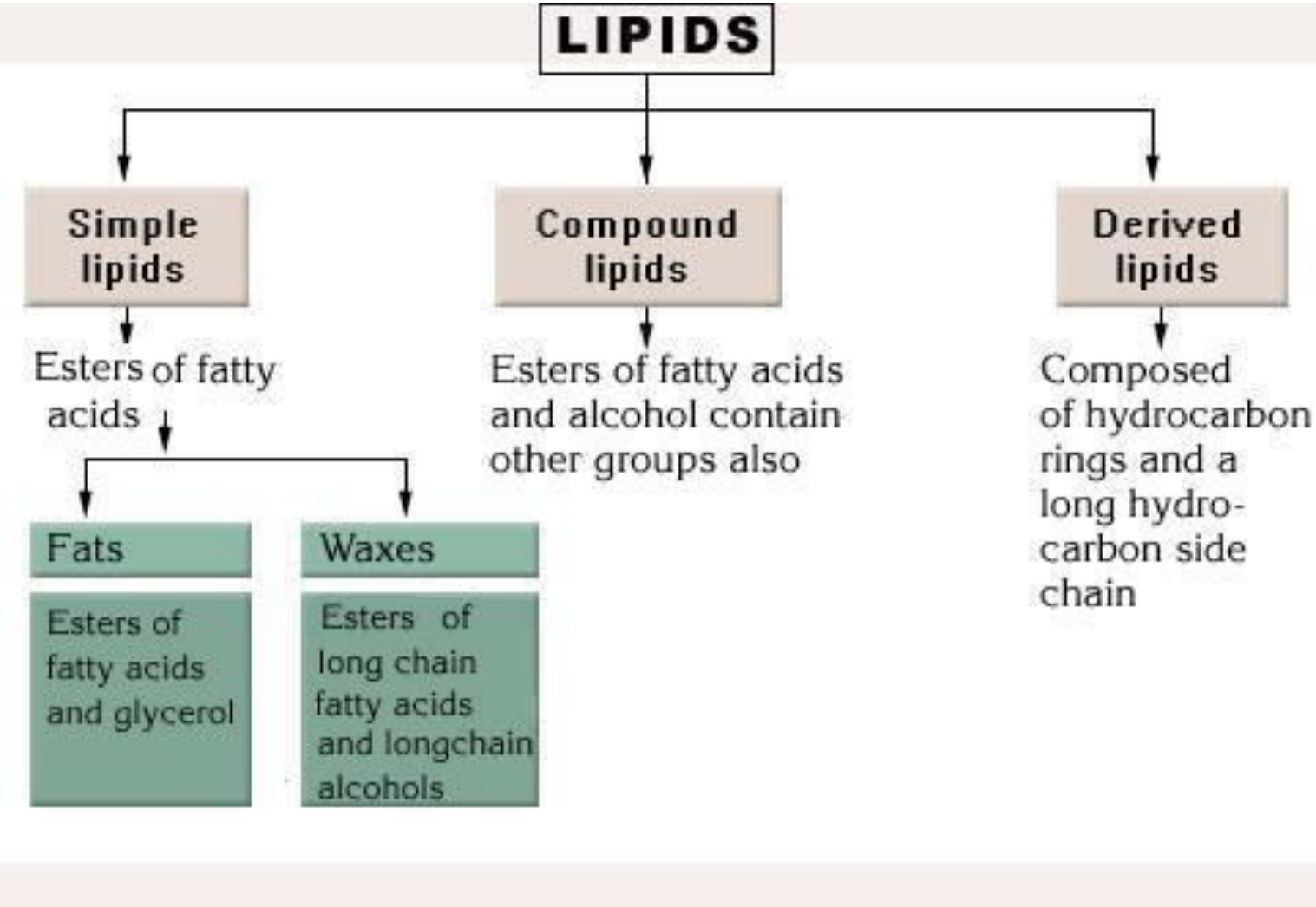
- As mentioned previously, lipids are characterized by their hydrophobicity.
- The precursors and derivatives of such compounds are included in classification of lipids.

Lipid Chemistry

- Lipids are classified in following three major groups:

1. Simple lipids
2. Complex Lipids and
3. Precursor and Derived Lipids

Lipid Chemistry



Lipid Chemistry

1. Simple lipids:

Esters of fatty acids with various alcohols

These contain:

a. Fats (and Oils) and

b. Waxes.

Lipid Chemistry

a. **Fats:** Esters of fatty acids with glycerol (**Oils** are fats in the liquid state)

b. **Waxes:** Esters of fatty acids with higher molecular weight monohydric alcohols. (having one OH group)

Lipid Chemistry

2. Complex lipids: Esters of fatty acids containing groups in addition to an alcohol

and a fatty acid.

- These include:

a) Phospholipids

b) Glycolipids

c) Other Complex lipids

Lipid Chemistry

- **a. Phospholipids**: Lipids containing, in addition to fatty acids and an alcohol, a phosphoric acid residue

Lipid Chemistry

- Phospholipids frequently have nitrogen containing bases and other substituent:

- **Glycerophospholipids** the alcohol is glycerol.

- **Sphingophospholipids** the alcohol is sphingosine.

Lipid Chemistry

b. Glycolipids (glycosphingolipids):

- Lipids containing a;
- fatty acid,
- sphingosine, and
- carbohydrate

Lipid Chemistry

C. Other complex lipids: Lipids such as

- sulfolipids
- aminolipids
- lipoproteins may also be placed in this category

Lipid Chemistry

3.Precursor and derived lipids:

- These include
  - fatty acids
  - glycerol
  - steroids
  - other alcohols
  - fatty aldehydes
  - ketone bodies
  - lipid-soluble vitamins

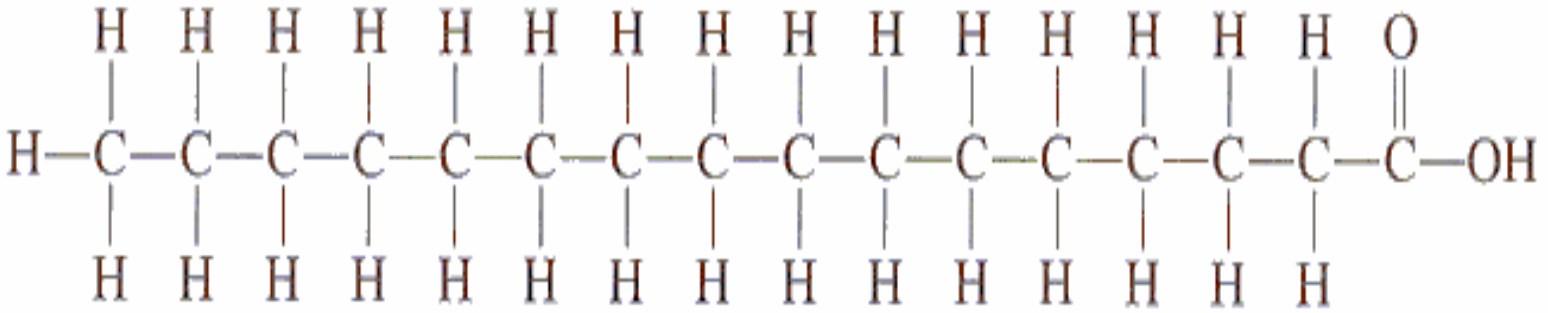
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Lipid Chemistry

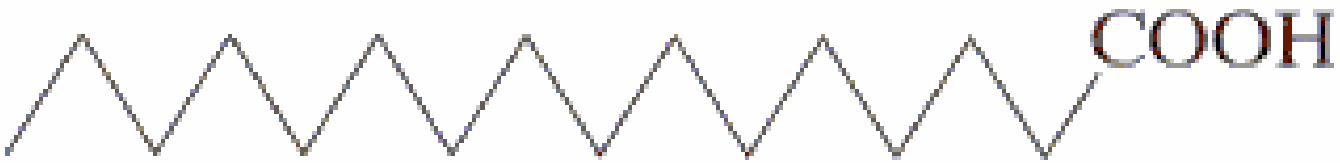
Fatty Acids

- The fats and oils used almost universally as stored forms of energy in living organisms are derivatives of fatty acids
- Fatty acids are aliphatic carboxylic acids with hydrocarbon chains.

Lipid Chemistry



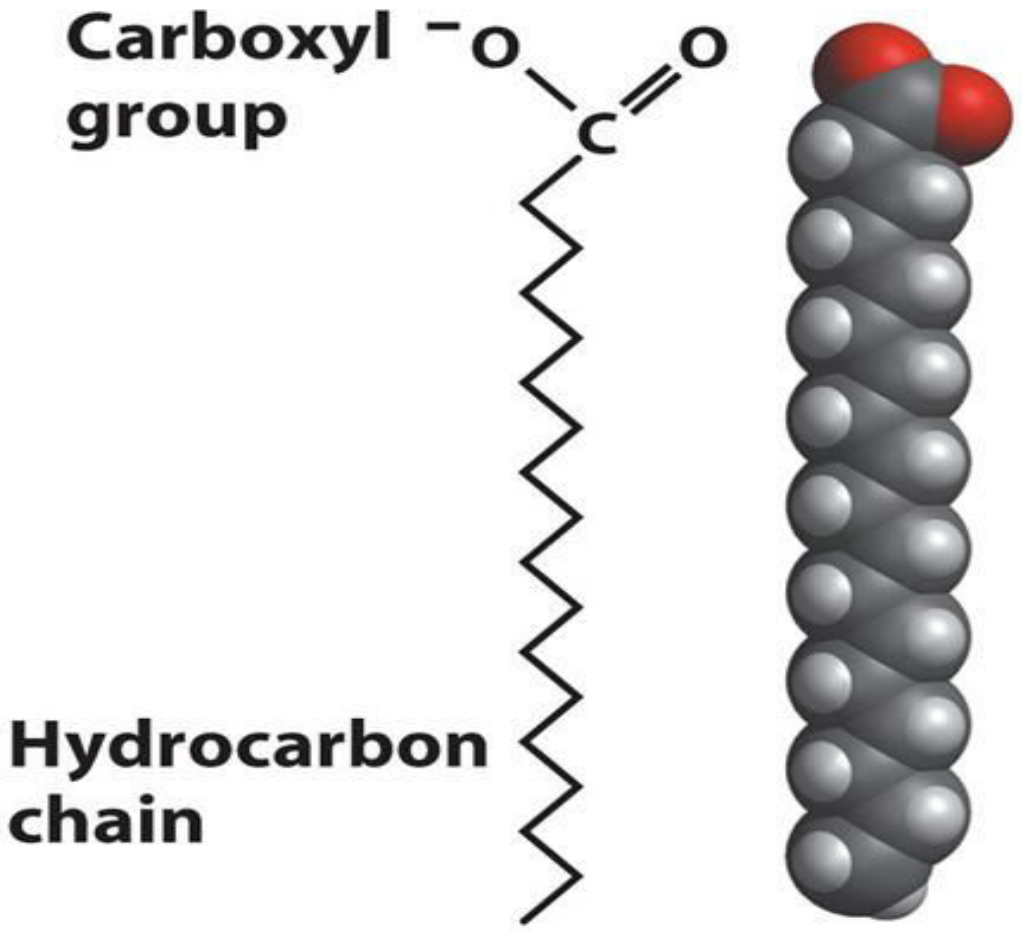
Hexadecanoic acid (palmitic acid)



Basic structure of Fatty Acid: Long chain of hydrocarbon terminating in the carboxylic group.

Hexa: **6**: Deca: **10** Hexadecanoic acid: **16** carbon Fatty acid.

Lipid Chemistry



Lipid Chemistry

- Fatty acids occur in the body mainly as esters in **natural fats and oils**,
- But are found in the unesterified form as **free fatty acids**, a transport form in the plasma.

Lipid Chemistry

- Fatty acids that occur in natural fats usually contain an even number of carbon atoms ranging from 4 to 36 carbons (C_4 to C_{36})

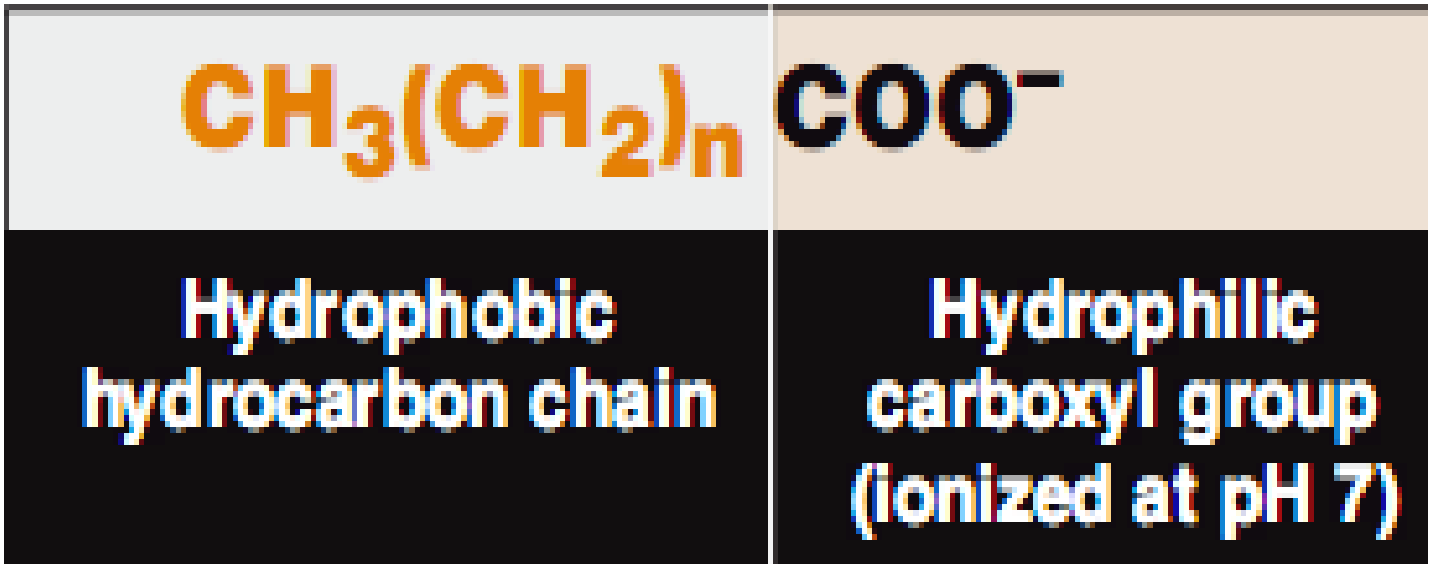
- Fatty acids are named after the corresponding hydrocarbons.

Lipid Chemistry

- A fatty acid consists of
- a hydrophobic hydrocarbon chain with
- a terminal carboxyl group that has a pKa of about 4.8

- At physiologic pH, the terminal carboxyl group (-COOH) ionizes, becoming $-COO^-$

Lipid Chemistry



Lipid Chemistry

- This anionic group has an affinity for water, giving the fatty acid its **amphipathic nature**

- (having both a hydrophilic and a hydrophobic region)

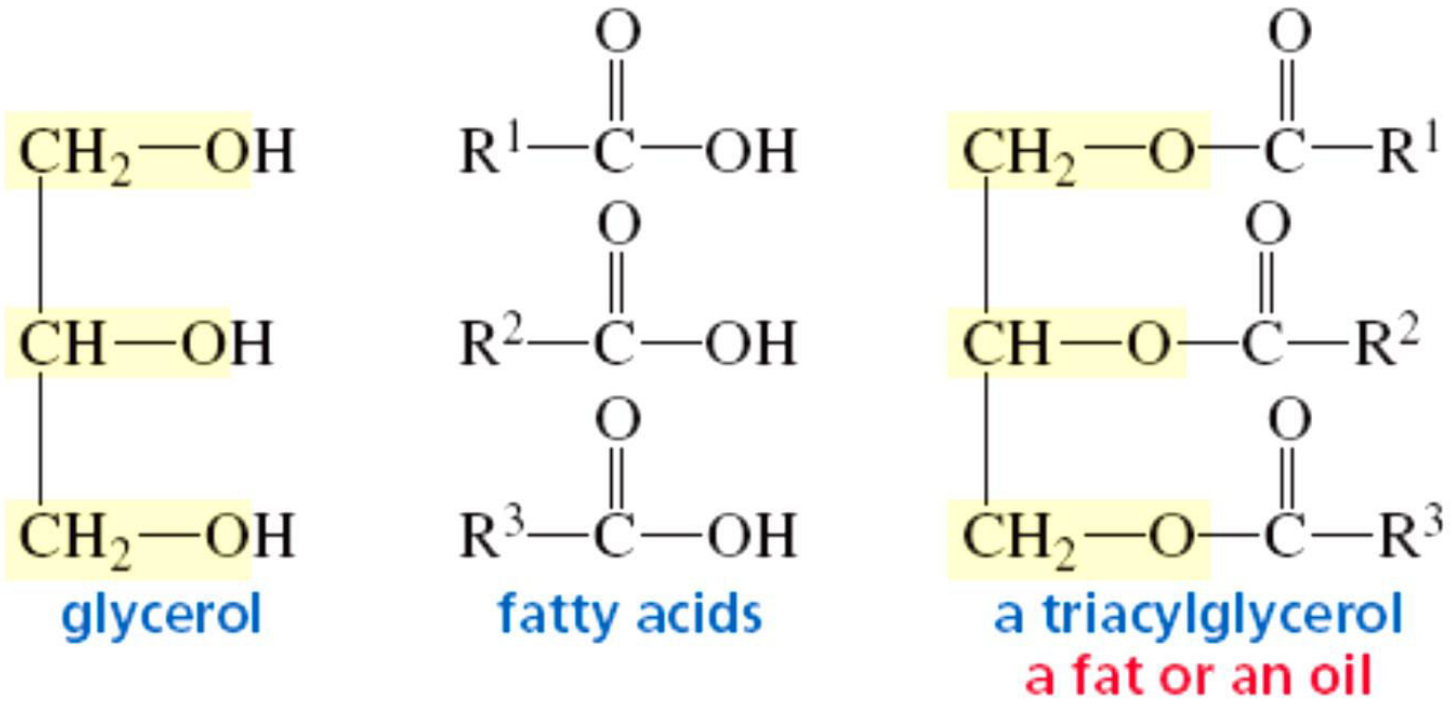
Lipid Chemistry

- However**, for long-chain fatty acids** (LCFA), the **hydrophobic portion is predominant**
- These molecules are highly water-insoluble, and must be **transported** in the circulation in association with protein.

Lipid Chemistry

- More than ninety percent of the fatty acids found in plasma are in the form of;
- fatty acid esters (primarily triacylglycerol, cholesteryl esters, and phospholipids) **contained** in circulating lipoprotein particles

Lipid Chemistry



Triacylglycerol: An example of Fatty acid containing compound, **R** denotes the hydrocarbon chain.

Lipid Chemistry

- **Unesterified** fatty acids are **transported** in the circulation in association with **albumin**.
- Low levels of free fatty acids occur in all tissues,
- But substantial amounts sometimes can be found in the plasma, particularly during fasting.

END

Lipid Chemistry

Classification Of Fatty Acids:

- Fatty Acids are classified on the basis of :
- Hydrocarbon Chain Length
- Degree of Saturation
- Dietary Requirement

Classification on the basis o dietary requirement include essential and non-essential fatty acids.

Lipid Chemistry

- Classification On The Basis Of Hydrocarbon Chain Length

These include

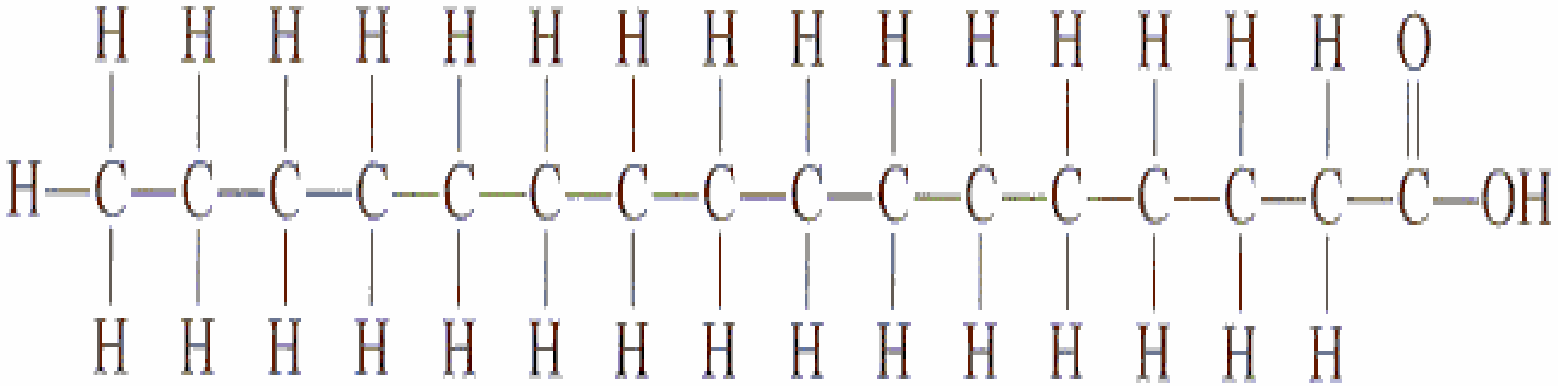
- Short Chain Fatty Acids (2-4 C)
- Medium Chain Fatty Acids (6-12 C)
- Long Chain Fatty Acids (14-18 C)
- Very Long Chain Fatty Acids (18 and more C)

Lipid Chemistry

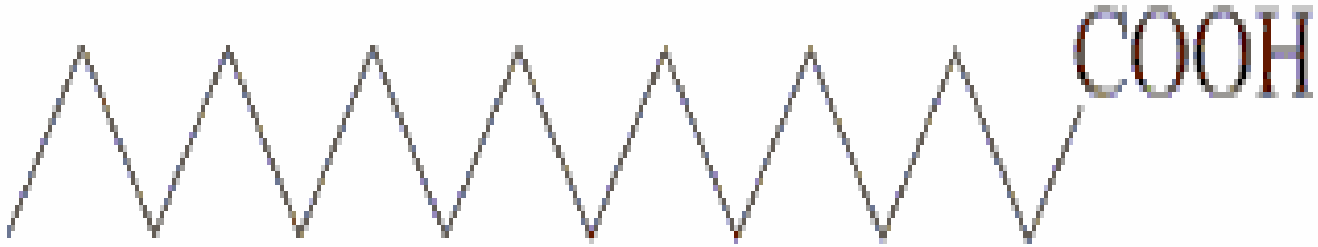
Saturation of fatty acids

- Fatty acid chains may contain
- no double bonds, that is be **saturated**, or
- contain one or more double bonds that is, be **mono- or polyunsaturated**

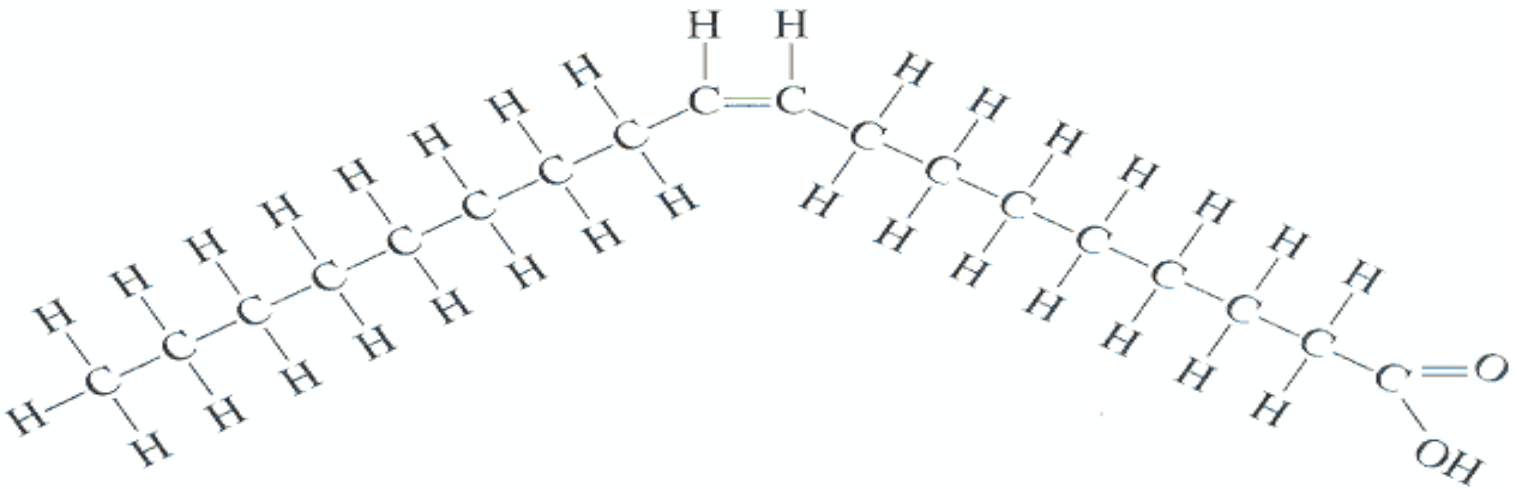
Lipid Chemistry

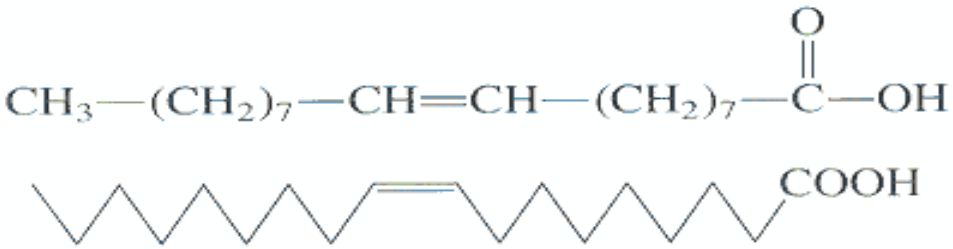


Hexadecanoic acid (palmitic acid)



Lipid Chemistry



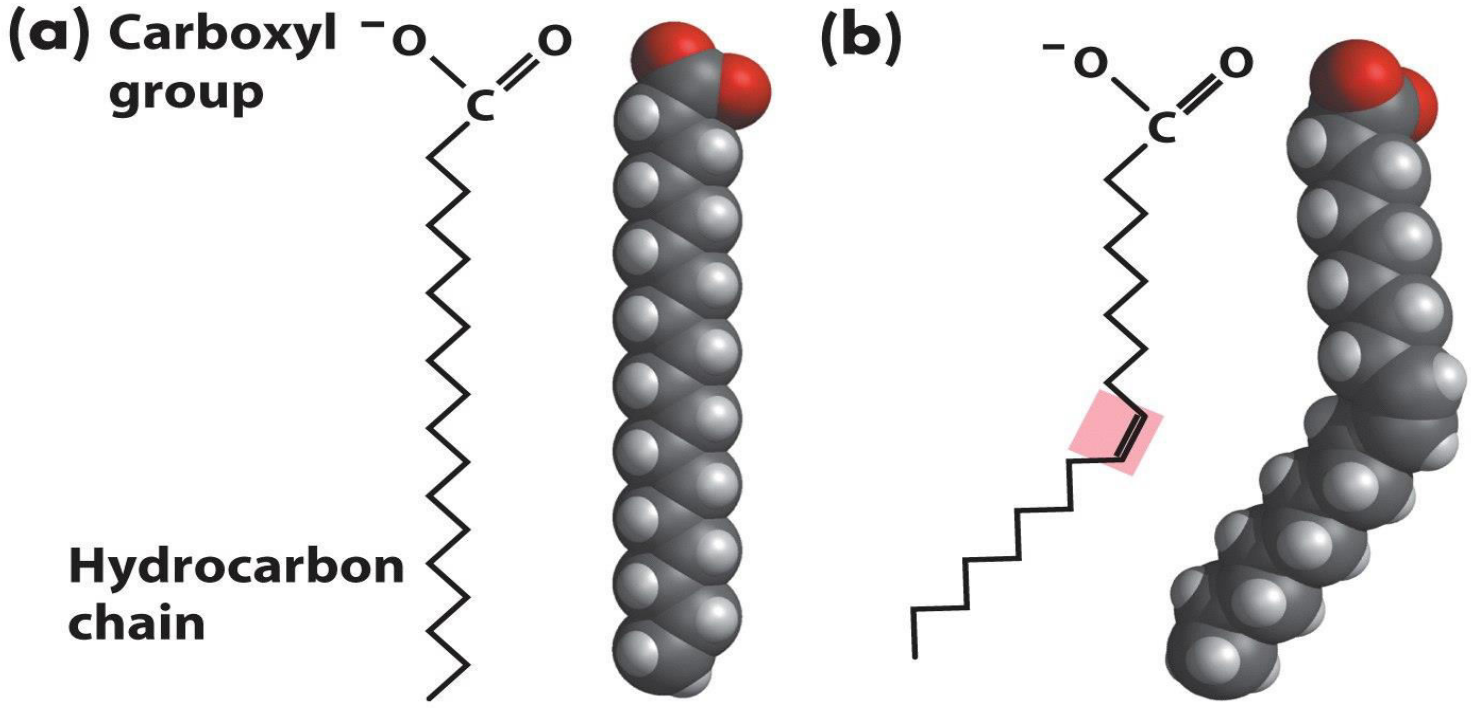


Oleic acid – 18-carbon

Lipid Chemistry

- When double bonds are present, they are **nearly always** in the **cis** rather than in the trans configuration.

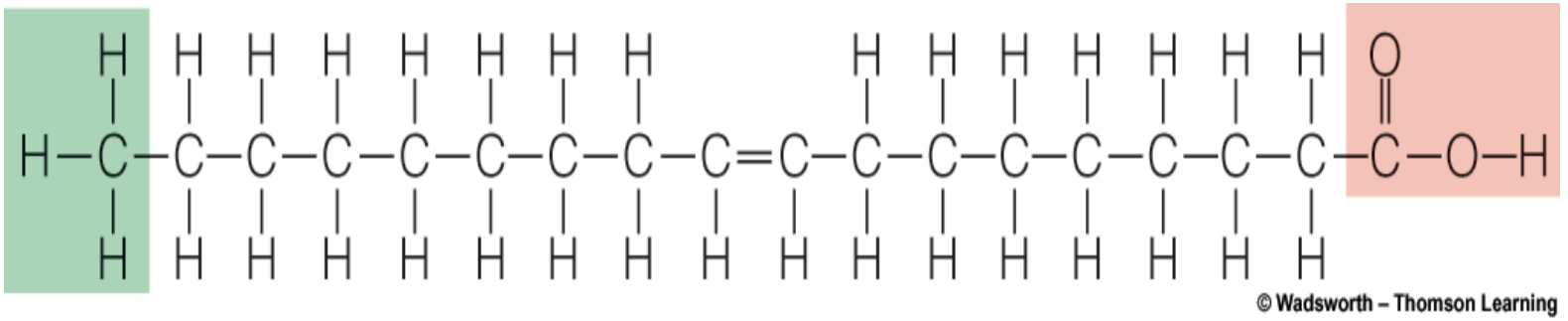
Lipid Chemistry



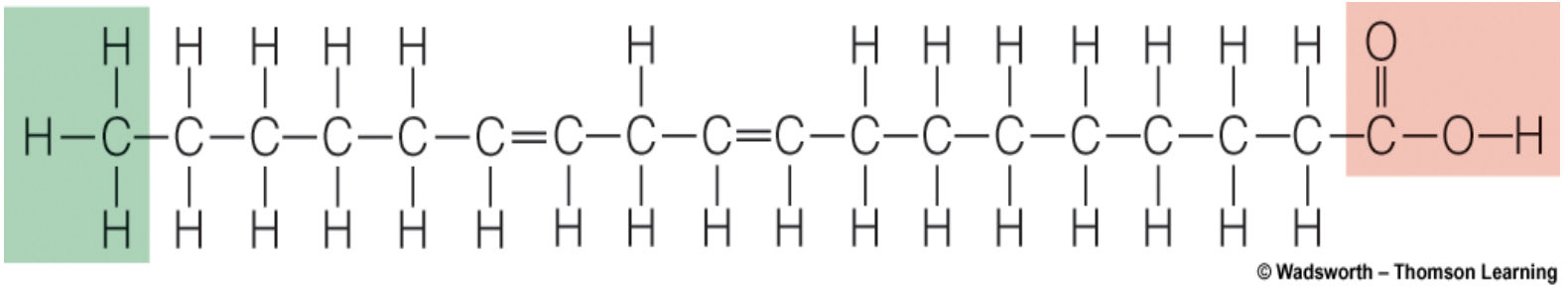
Lipid Chemistry

- If the fatty acid has two or more double bonds they are always spaced at three carbon intervals.

Lipid Chemistry



Oleic acid – 18-carbon, monounsaturated



Linoleic acid – 18-carbon, polyunsaturated

Lipid Chemistry

- Classification of Fatty Acid on Dietary Basis
- Fatty acids that are required by the human body but
- cannot be synthesized in the body, and
- therefore must be obtained from food, are called essential fatty acids.

Lipid Chemistry

- In contrast to amino acids, there are only two essential Fatty Acids:
- **Linoleic Acid**
- **Alpha Linolenic Acid**

Lipid Chemistry

- Humans lack the ability to introduce double bonds in fatty acids beyond carbons 9 and 10 as counted from the carboxylic acid side.
- **Linoleic Acid** has two double bonds at **C9 and C12**.

Lipid Chemistry

- Similarly, **α-linolenic acid** has three double bonds at **C 9, 12 and 15.**

- Therefore, both of these fatty acids can not be synthesized in the body and must be obtained in the diet.

Lipid Chemistry

- The rest of the Fatty Acids fall into the category of **non-essential fatty Acids.**

END

Lipid Chemistry

Nomenclature

- The most frequently used systematic nomenclature names the fatty acid after the hydrocarbon with the same number and arrangement of carbon atoms, with

-oic being substituted for the final **–e**

Lipid Chemistry

- Thus, saturated acids end in **–anoic.** Octane becomes Octanoic Acid.
- Unsaturated acids with double bonds end in **-enoic,** e.g. octadecenoic acid (C18) also called oleic acid.

Lipid Chemistry

- A simplified nomenclature (more aptly called as symbolic nomenclature) for unbranched fatty acids specifies
- the chain length and
- number of double bonds,
- separated by a colon.

Lipid Chemistry

- The carbon atoms are numbered, beginning with the **carboxyl carbon** as C 1
- The number **before the colon** indicates the total number of carbons in the chain,
- Those **after the colon** indicate the numbers and positions of double bonds.

Lipid Chemistry

- The 16-carbon saturated palmitic acid is abbreviated

  16:0,

- The 18-carbon oleic acid, with one double bond, is

  18:1

Lipid Chemistry

- The family of polyunsaturated fatty acids (PUFAs) with a double bond between the third and fourth carbon from the **methyl end** of the chain are of special importance in human nutrition

Lipid Chemistry

- Because the physiological role of PUFAs is related more to the position of the first double bond near the methyl end of the chain than to the carboxyl end, an alternative nomenclature is sometimes used for these fatty acids.

Lipid Chemistry

- The carbon of the terminal methyl group is called the **-ω** regardless of the chain length.

Lipid Chemistry

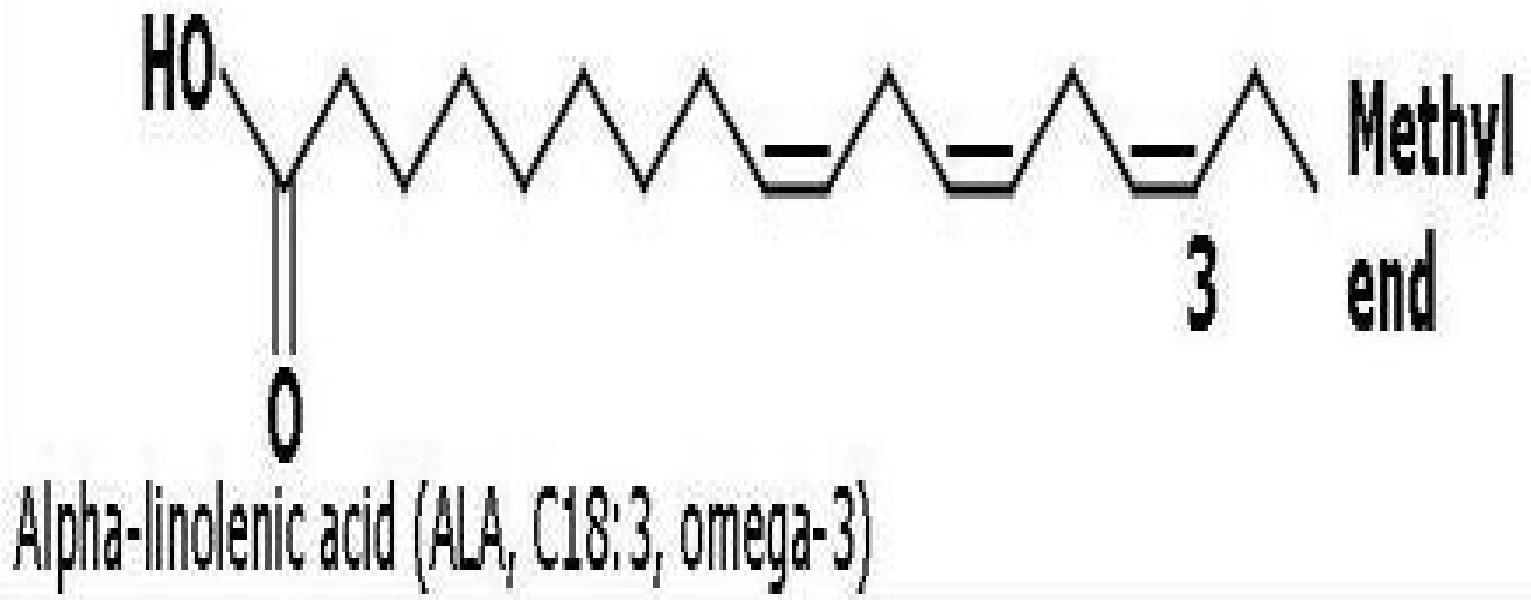
- Arachidonic acid **$20{:}4\Delta^{5,8,11,14}$** is referred to as an **ω-6 fatty acid** because the closest double bond to the ω end begins six carbons from that end.

Lipid Chemistry

- The essential fatty acid **linoliec acid**, **18:2 $\Delta^{9, 12}$** is **ω-6 fatty acid.**

- In contrast, **α-linolenic acid**, **18:3$\Delta^{9, 12, 15}$** is an **ω-3 fatty acid**

Lipid Chemistry



Alpha-linolenic acid (ALA, C18:3, omega-3)

END

Lipid Chemistry

- The carbon to which the carboxyl group is attached, carbon 2 is also called the **α-carbon**,
- carbon 3 is the **β-carbon**, and
- carbon 4 is the **γ-carbon**

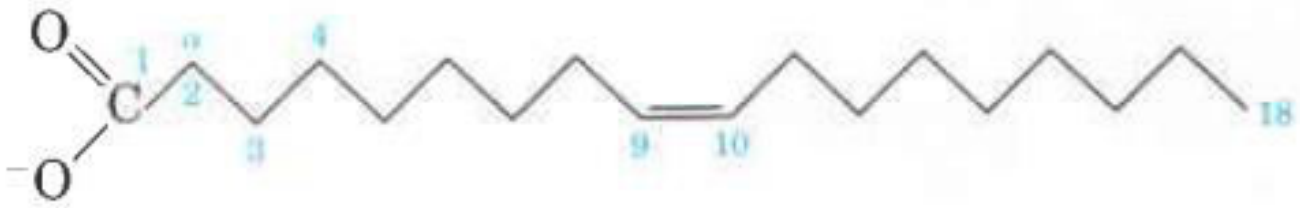
Lipid Chemistry

- The **positions of any double bonds** are specified relative to the carboxyl carbon by superscript numbers following Δ (delta)

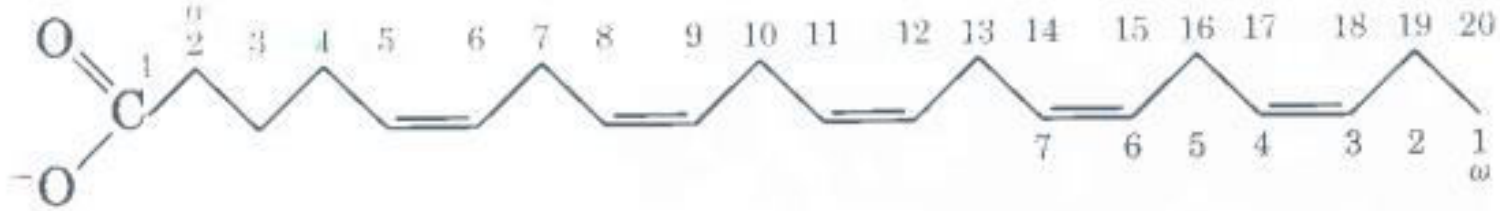
Lipid Chemistry

- A 20-carbon fatty acid with two double bonds between C-9 and C-10 (C-1 being the carboxyl carbon) and another between C-12 and C-13 is designated
- **$20{:}2\ \Delta^{9,12}$**

Lipid Chemistry



(a) 18:1(Δ^9) *cis*-9-Octadecenoic acid



(b) 20:5($\Delta^{5,8,11,14,17}$) Eicosapentaenoic acid (EPA), an omega-3 fatty acid

Lipid Chemistry

- **Physical and Chemical Properties of Fatty Acids**
- These are dictated by
- length of hydrocarbon chain and
- degree of saturation and unsaturation of Fatty Acids.

Lipid Chemistry

- **The melting points T_m** of even-numbered carbon fatty acids **increase** with increasing chain length and

- **decrease** with increasing unsaturation.

Lipid Chemistry

- A triacylglycerol containing three saturated fatty acids of 12 carbons or more is solid at body temperature, whereas

- if the fatty acid residues are 18:2, i.e. unsaturated it is liquid to below 0°C

Lipid Chemistry

Higher Tm

Lower Tm

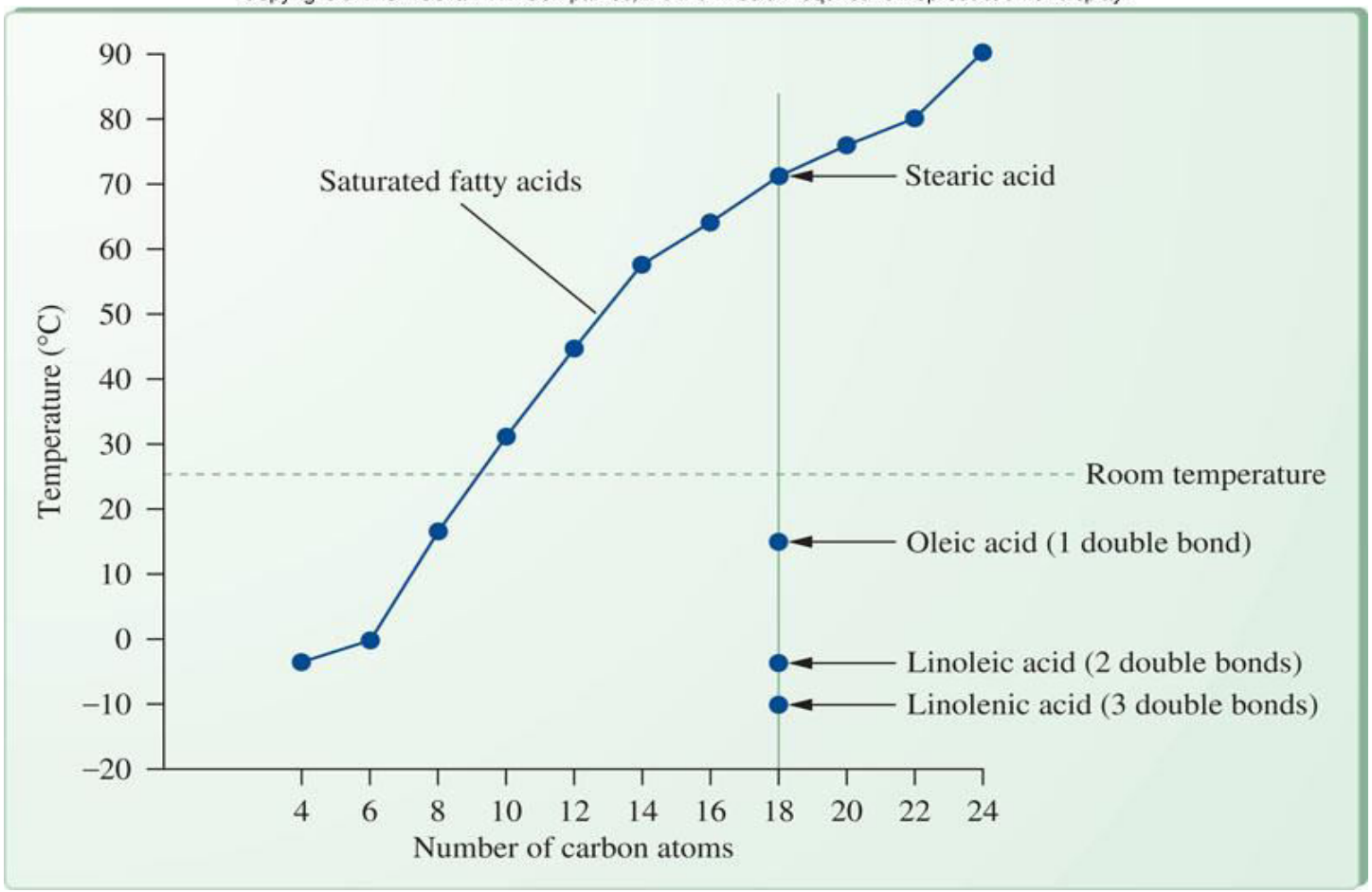
Carbon Atoms: Double Bonds	Common Name	Melting Point (°C)
Saturated Fatty Acids		
12:0	Lauric acid	44
14:0	Myristic acid	58
16:0	Palmitic acid	63
18:0	Stearic acid	70
20:0	Arachidic acid	77
Unsaturated Fatty Acids		
16:1	Palmitoleic acid	1
18:1	Oleic acid	16
18:2	Linoleic acid	-5
18:3	Linolenic acid	-11
20:4	Arachidonic acid	-49

Lipid Chemistry

- At room temperature (25^0C), the saturated fatty acids from 12:0 to 24:0 have a waxy consistency, whereas
- unsaturated fatty acids of these lengths are oily liquids.

Lipid Chemistry

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Increasing chain length increases the melting temp. Stearic acid, Oleic acid, Linoleic acid and Linolenic acid all have 18 C, but with increasing no of Double bonds, which decreases the Tm

Lipid Chemistry

- This difference in melting points is due to different degrees of packing of the fatty acid molecules due to;

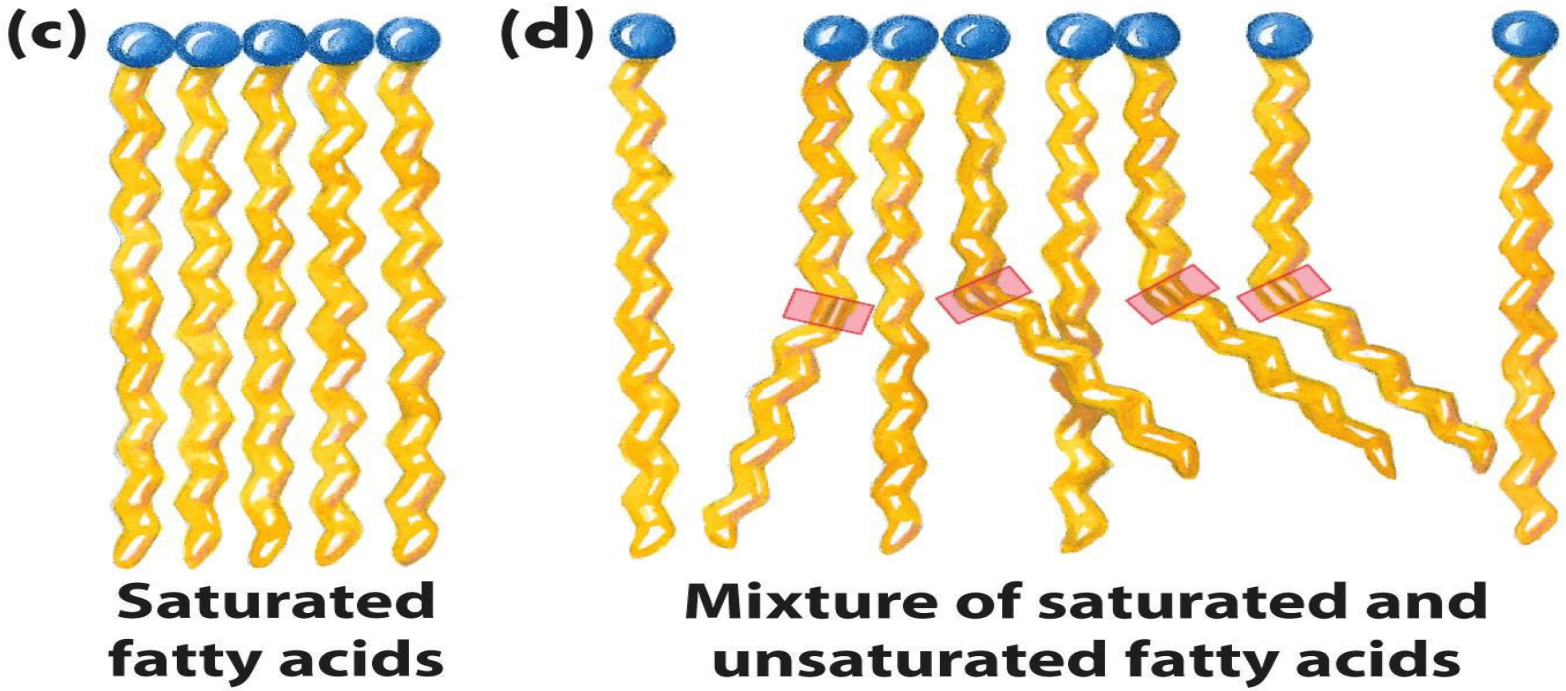
- different proportions of saturated and unsaturated Fatty Acids

Lipid Chemistry

- Saturated fatty acids are more tightly packed as compared to

- Unsaturated Fatty Acids; which have a less compact packing.

Lipid Chemistry



Lipid Chemistry

- The more compact packing;
- the more attractive molecular forces between Fatty Acid molecules and vice versa.

Lipid Chemistry

- Hence, saturated Fatty Acids have high Tm and
- unsaturated Fatty Acids have low Tm.

- Therefore saturated and Unsaturated Fatty Acids help to Shape Foods

END

Lipid Chemistry

- **Physical and Chemical Properties of Fatty Acids** (Contd.)
- **The melting points** (Contd.)
- The loose packing of unsaturated fatty acids is due to;
- cis isomerism of unsaturated fatty acids.

Lipid Chemistry

- Unsaturated fatty acids show
- Geometric isomerism, i.e.
- They are either

- **cis** or

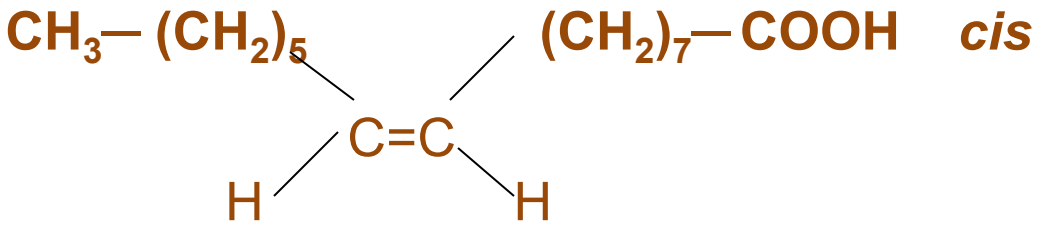
- **trans**

Lipid Chemistry

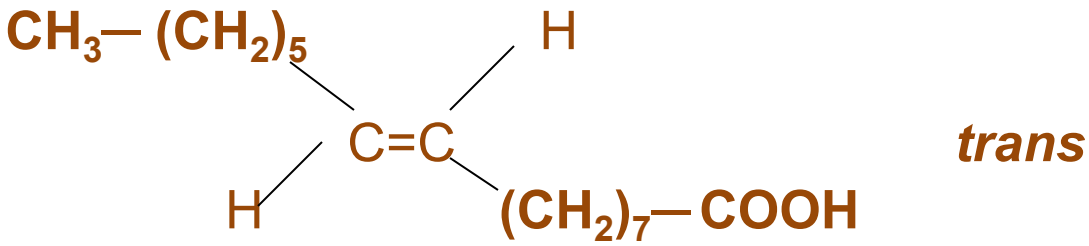
- When double bonds are present, they are nearly always in the **cis** rather than in the trans configuration.
- A **cis** configuration means that the two hydrogen atoms adjacent to the double bond stick out on the same side of the chain.

Lipid Chemistry

Unsaturated fatty acids can be cis with bulky groups on same side of C=C.

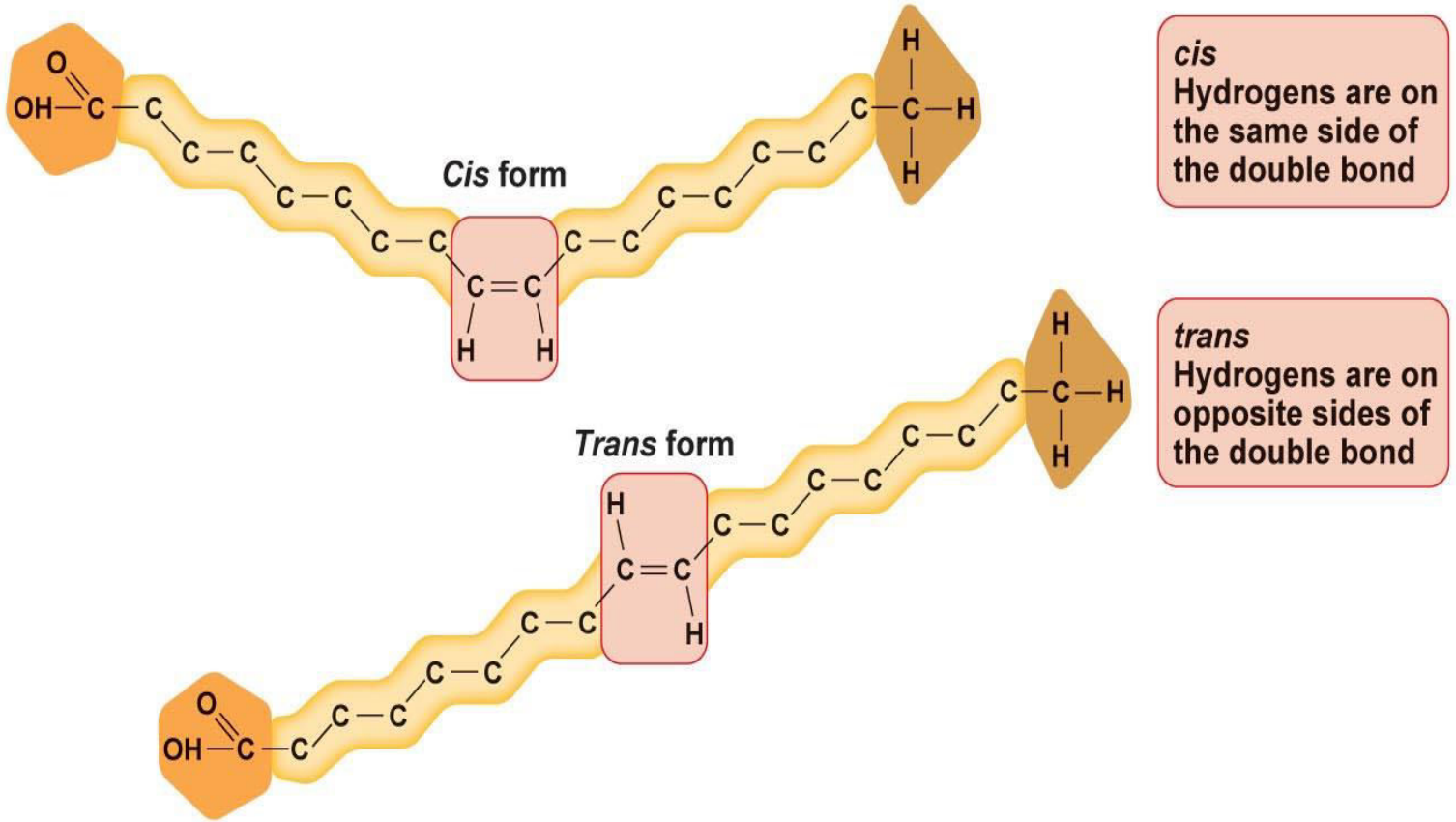


trans have bulky groups on opposite sides of C=C.



Cis configuration leads to kink, which causes loose packing, have low Tm.

Lipid Chemistry



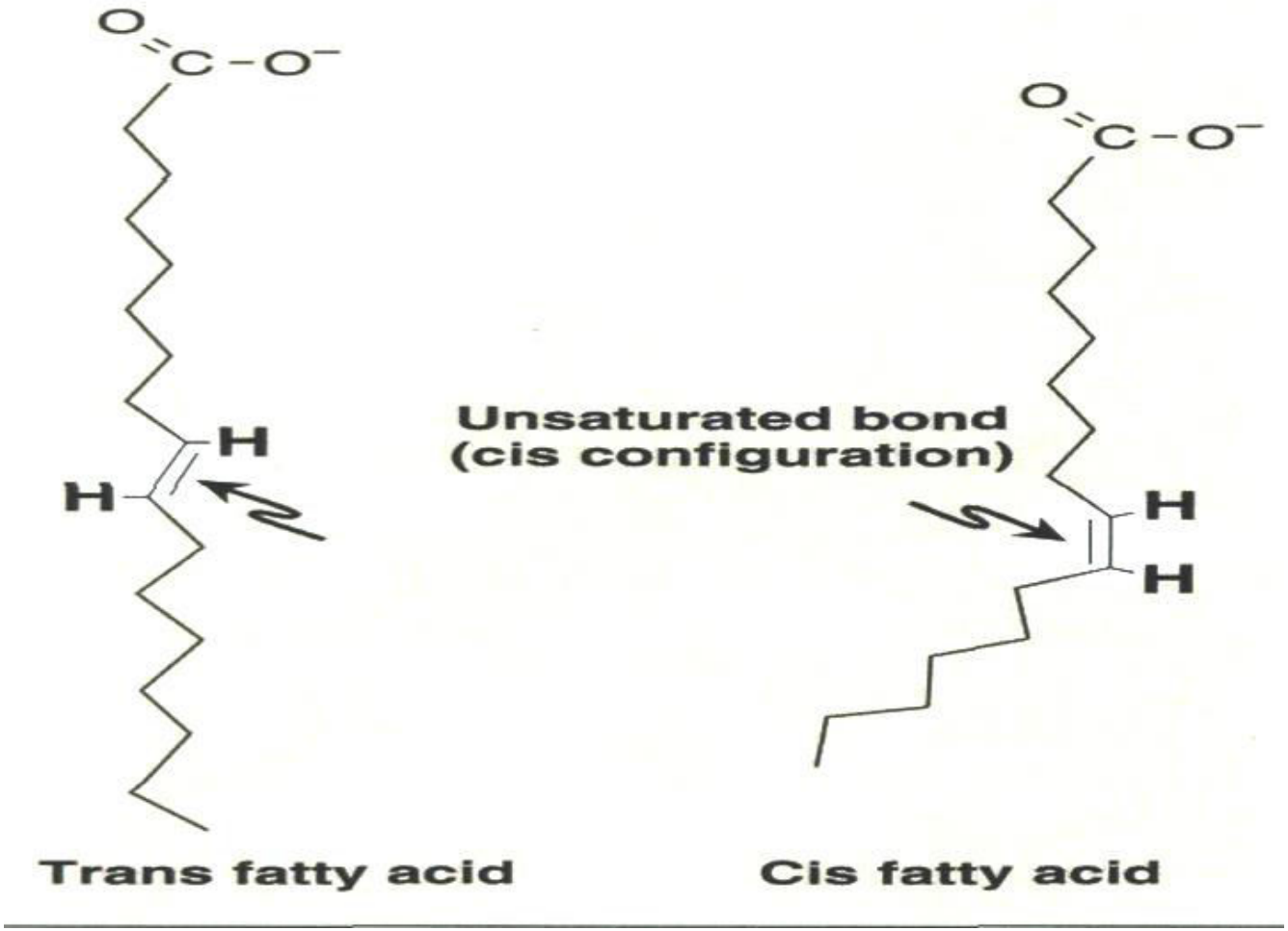
cis configuration,the molecules bent120 degrees at the double bond.Trans 110 degree

Lipid Chemistry

- In unsaturated fatty acids, a cis double bond forces a

- kink in the hydrocarbon chain

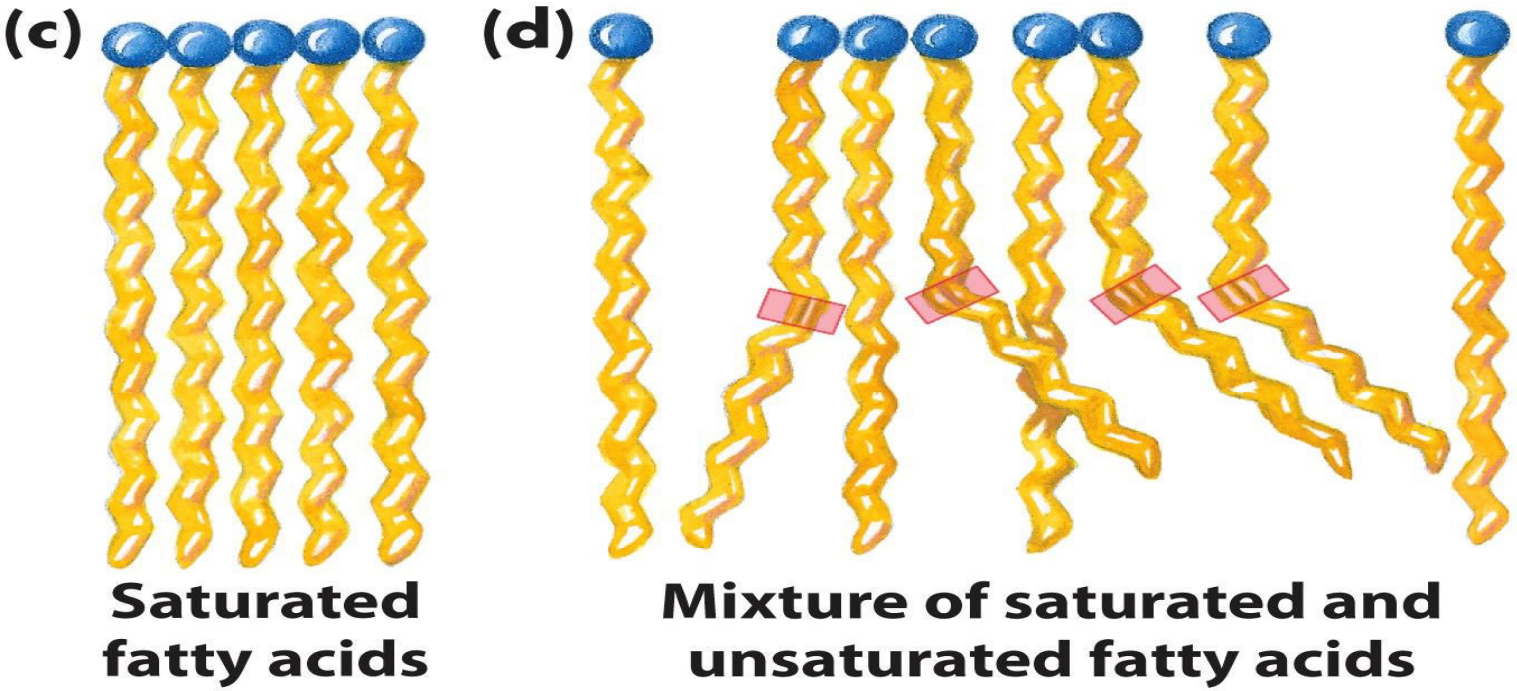
Lipid Chemistry



Lipid Chemistry

- Fatty acids with such kinks
- cannot pack together as tightly as fully saturated fatty acids,
- resulting in weak interaction with each other.

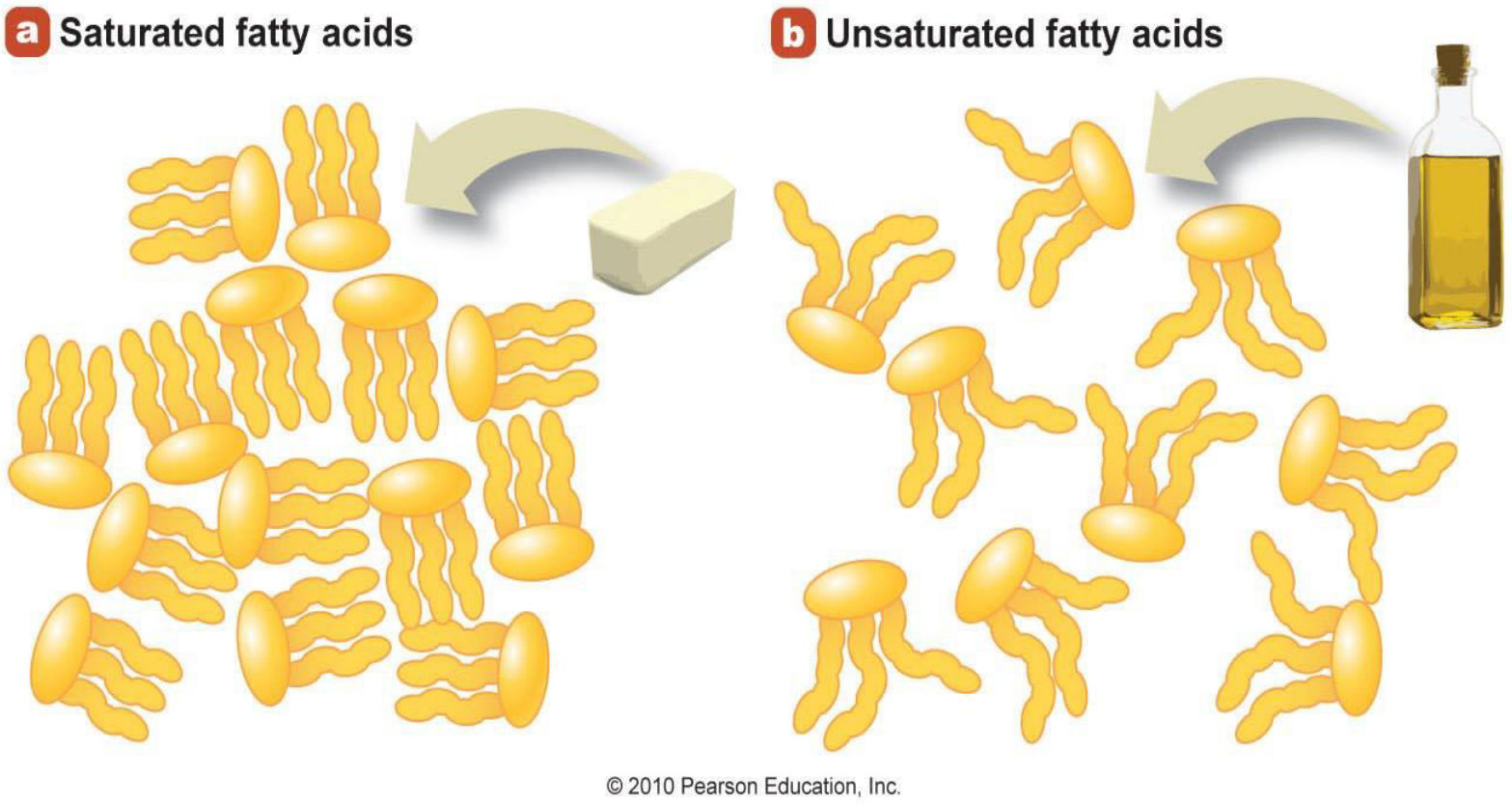
Lipid Chemistry



Lipid Chemistry

- In practice, natural acylglycerols contain a mixture of fatty acids tailored to suit their functional roles
- The membrane lipids, which must be fluid at all environmental temperatures, are more unsaturated than storage lipids.

Lipid Chemistry



Saturated; All the carbons on the fatty acid are bound to hydrogen. Solid at room tmp.
Higher melting point

Lipid Chemistry

- *Trans* fatty acids are present in certain foods, arising as a by-product of the saturation of fatty acids during hydrogenation, or "**hardening," of natural oils** in the manufacture of margarine.

Lipid Chemistry

- An additional small contribution comes from the **ingestion of ruminant fat** that contains *trans* fatty acids arising from the action of microorganisms in the rumen.

END

Lipid Chemistry

- **Physical and Chemical Properties of Fatty Acids** (Contd.)
- **Water Solubility**
- The presence of nonpolar hydrocarbon chain results in the poor water solubility of fatty acids.

Lipid Chemistry

- The larger the fatty acyl chain
- and the fewer the double bonds,
- the lower is the solubility in water.

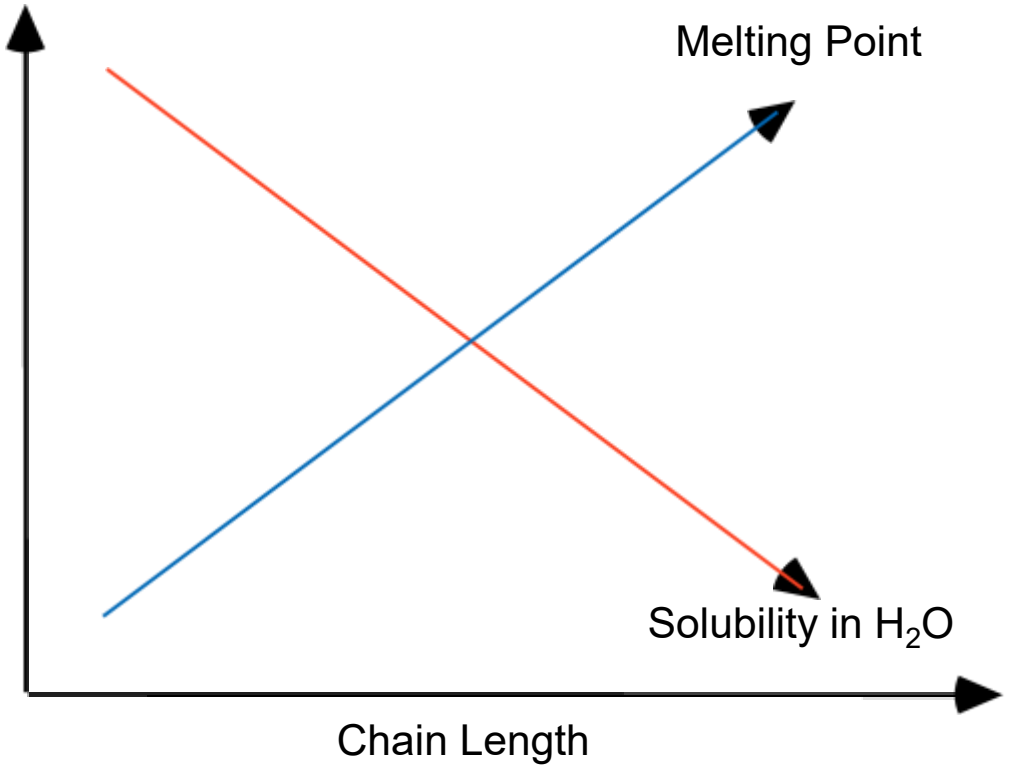
fewer the double bonds, lower the unsaturation and lower the solubility

Lipid Chemistry

- The carboxylic acid group is polar and
- ionized at neutral pH and
- accounts for the slight solubility of
- short-chain fatty acids in water

Lipid Chemistry

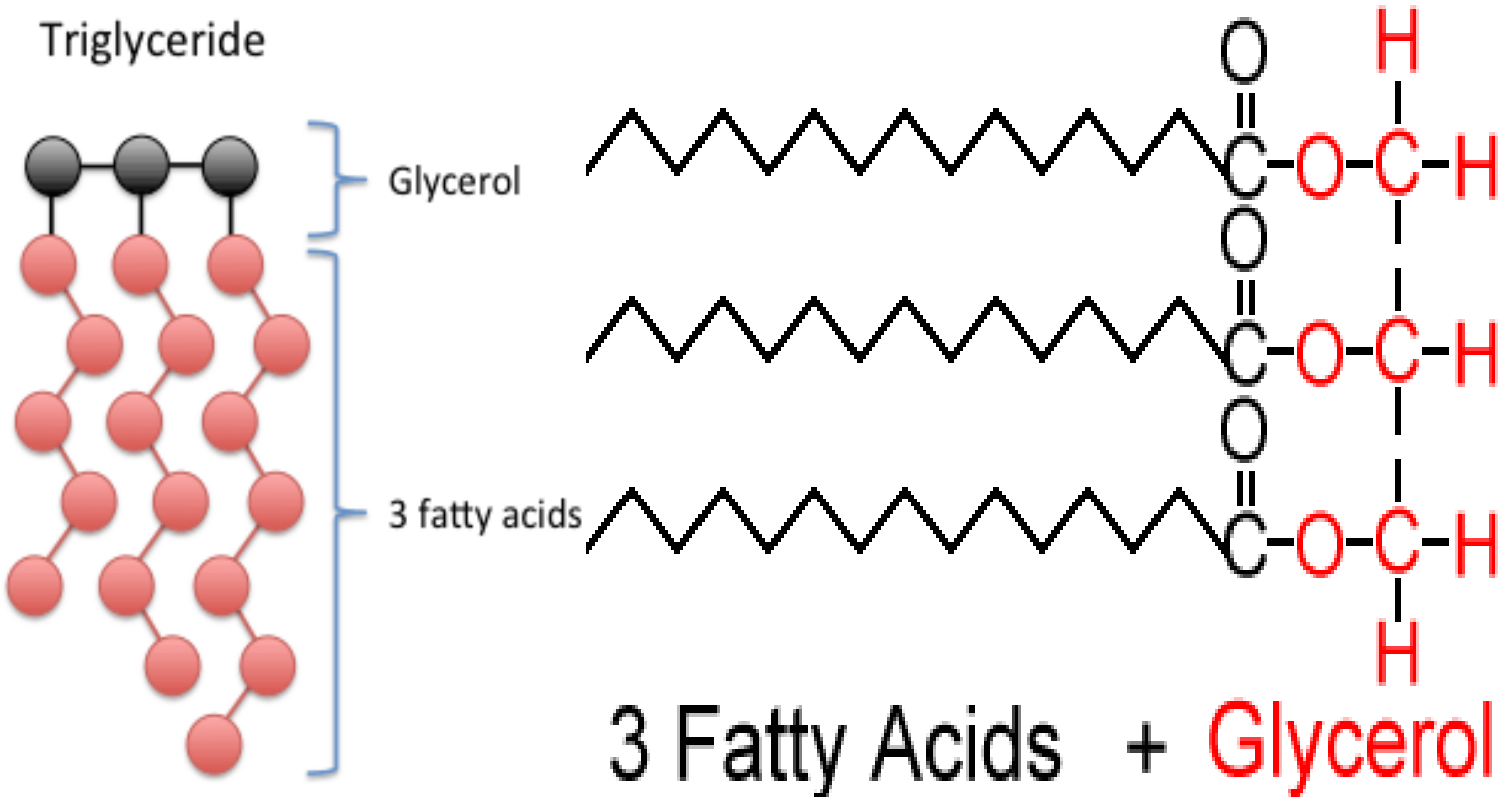
Melting Points and Solubility in Water of Fatty Acids



Lipid Chemistry

- **Formation of Esters**
- In combination with alcohol Fatty Acids form Esters.
- For example, fatty acids react with glycerol to form mono, di and tri glycerides.

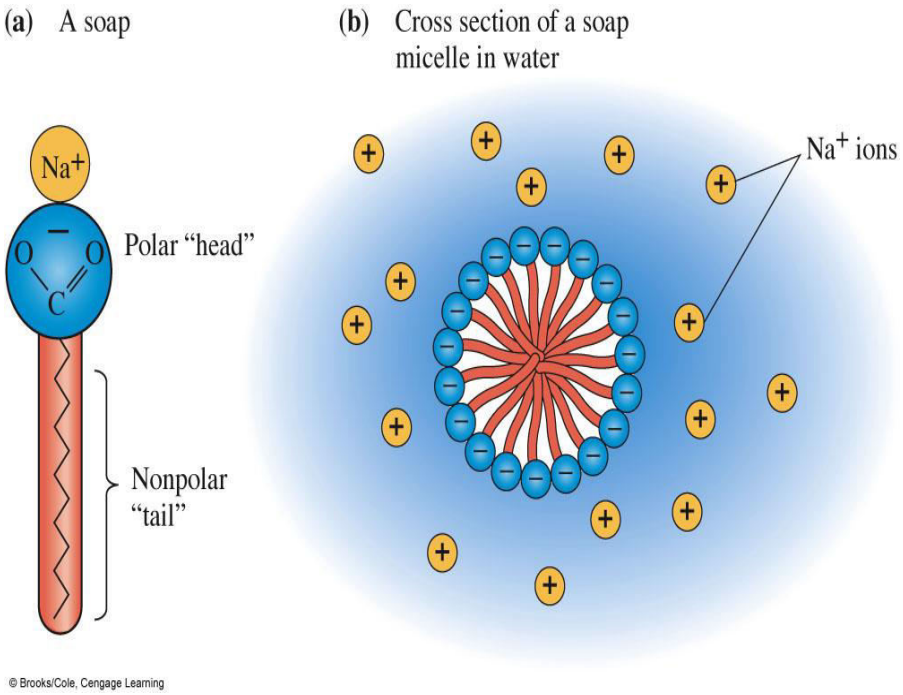
Lipid Chemistry



Lipid Chemistry

- **Physical and Chemical Properties of Fatty Acids** (Contd.)
  - **Formation of Salts**
  - Fatty Acids forms salts with alkali metals and alkaline earth metals.
  - Salts of Na, K, Ca and Mg are called soaps.

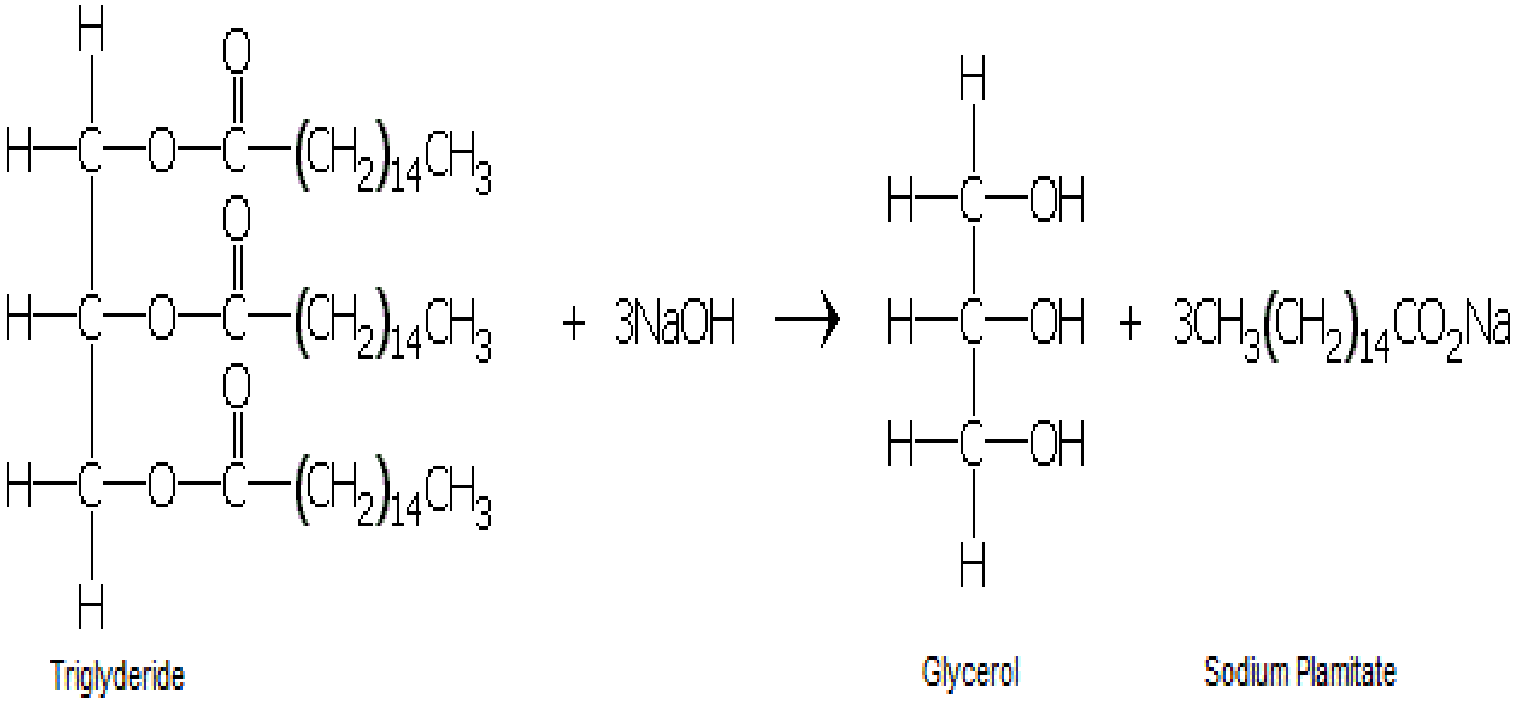
Lipid Chemistry



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- A soap micelle: nonpolar (hydrophobic) hydrocarbon chains cluster in the inside and polar (hydrophilic) carboxylate groups lie on the surface.

Lipid Chemistry



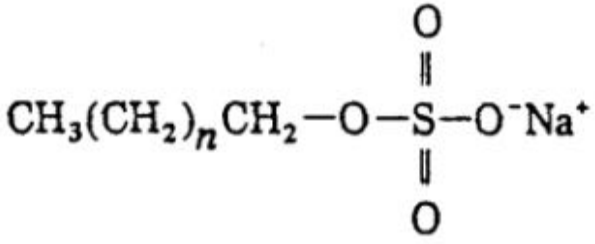
Lipid Chemistry

- The Na and K salts are water soluble,
- Whereas, Ca and Mg salts are insoluble.
- K salts are too soluble and are used in liquid soap preparations.

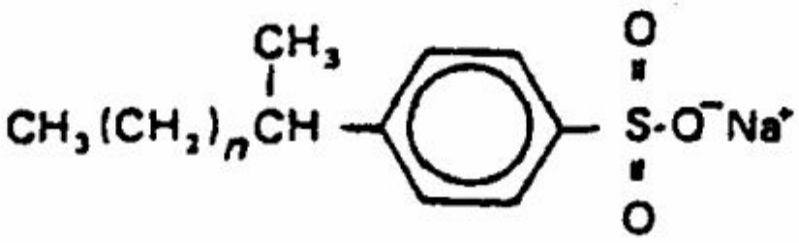
Lipid Chemistry

- **Formation of Detergents**
- Reduction of the carboxyl group of fatty acids produce alkyl alcohols
- which can be sulfated (OSO_3H) or sulfonated (SO_3H) to produce detergents.

Lipid Chemistry

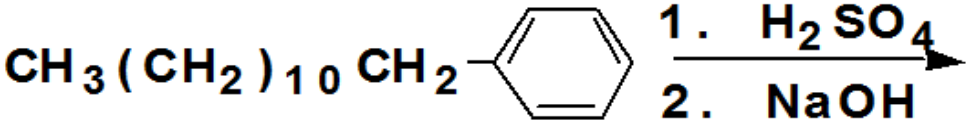


a sodium alkyl sulfate

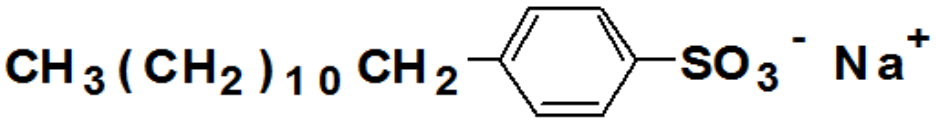


a sodium alkylbenzene sulfonate

Lipid Chemistry



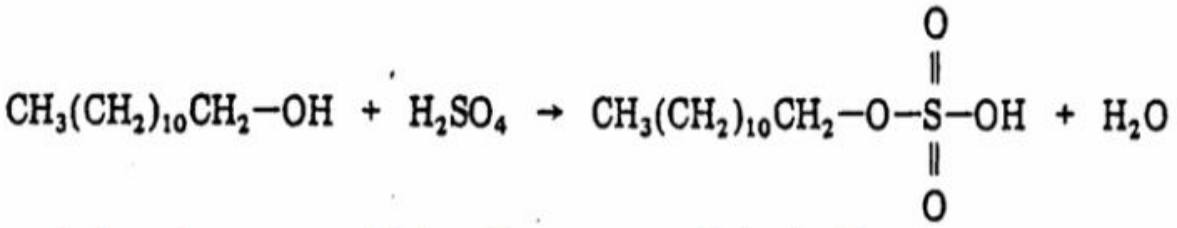
Dodecylbenzene



Sodium 4-dodecylbenzenesulfonate
(an anionic detergent)

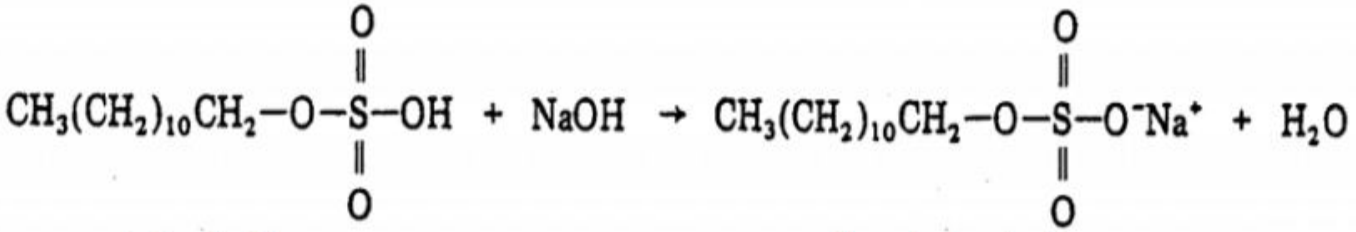
Lipid Chemistry

A synthetic detergent, a sodium alkyl sulfate called sodium dodecylsulfate, can be prepared by reacting dodecyl alcohol (dodecanol) with sulfuric acid.



dodecanol sulfuric acid dodecylsulfate

The resulting dodecylsulfate is converted to the sodium salt by a reaction with sodium hydroxide.



dodecylsulfate sodium dodecylsulfate

Lauric acid or **dodecanoic acid**, the saturated fatty acid with a 12-carbon atom chain, thus falling into themedium chain fatty acids

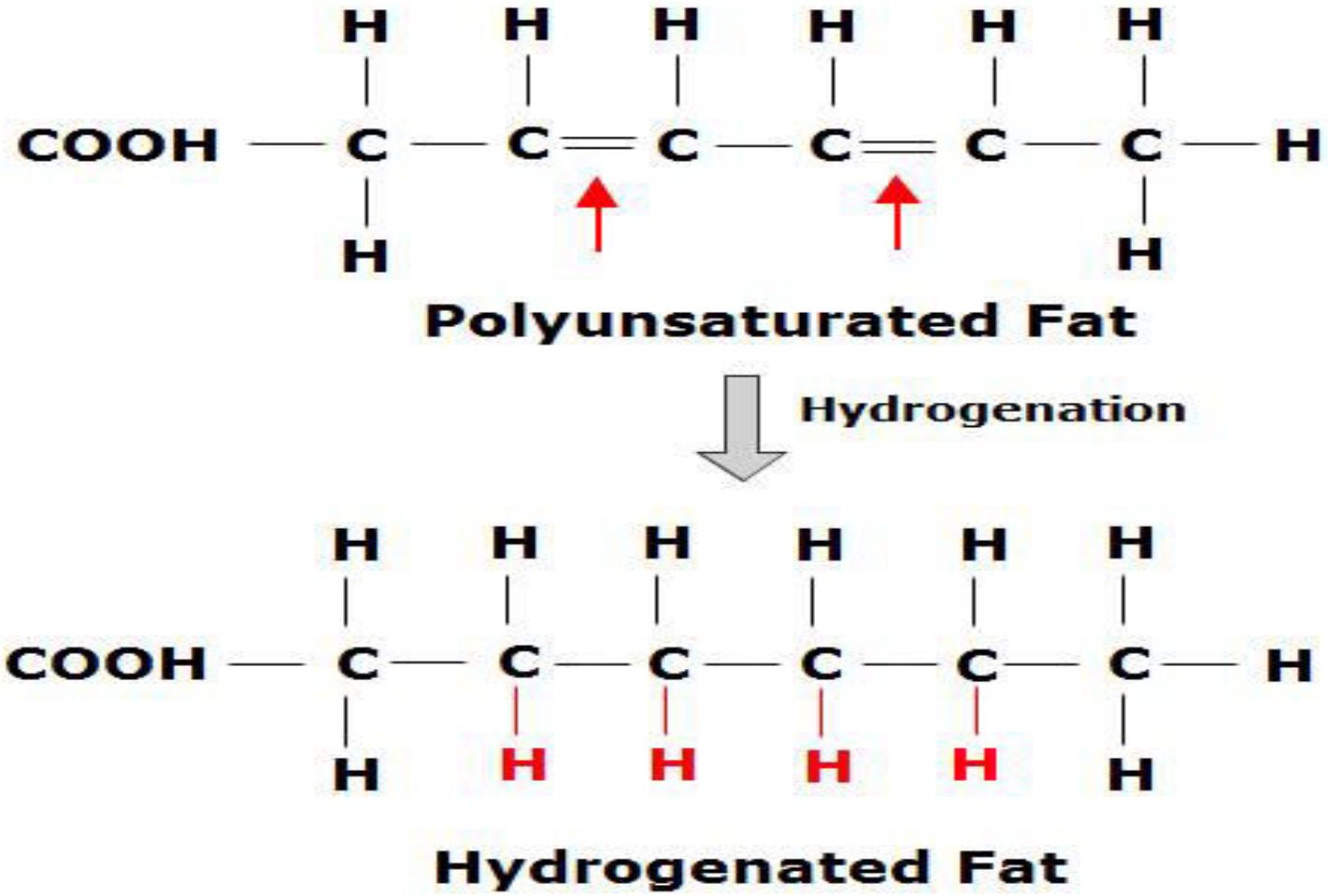
END

Lipid Chemistry

- **Special Reactions of Unsaturated Fatty Acids**
- **Hydrogenation**

- It is the addition of Hydrogen at the
- double bonds of unsaturated fatty acids.

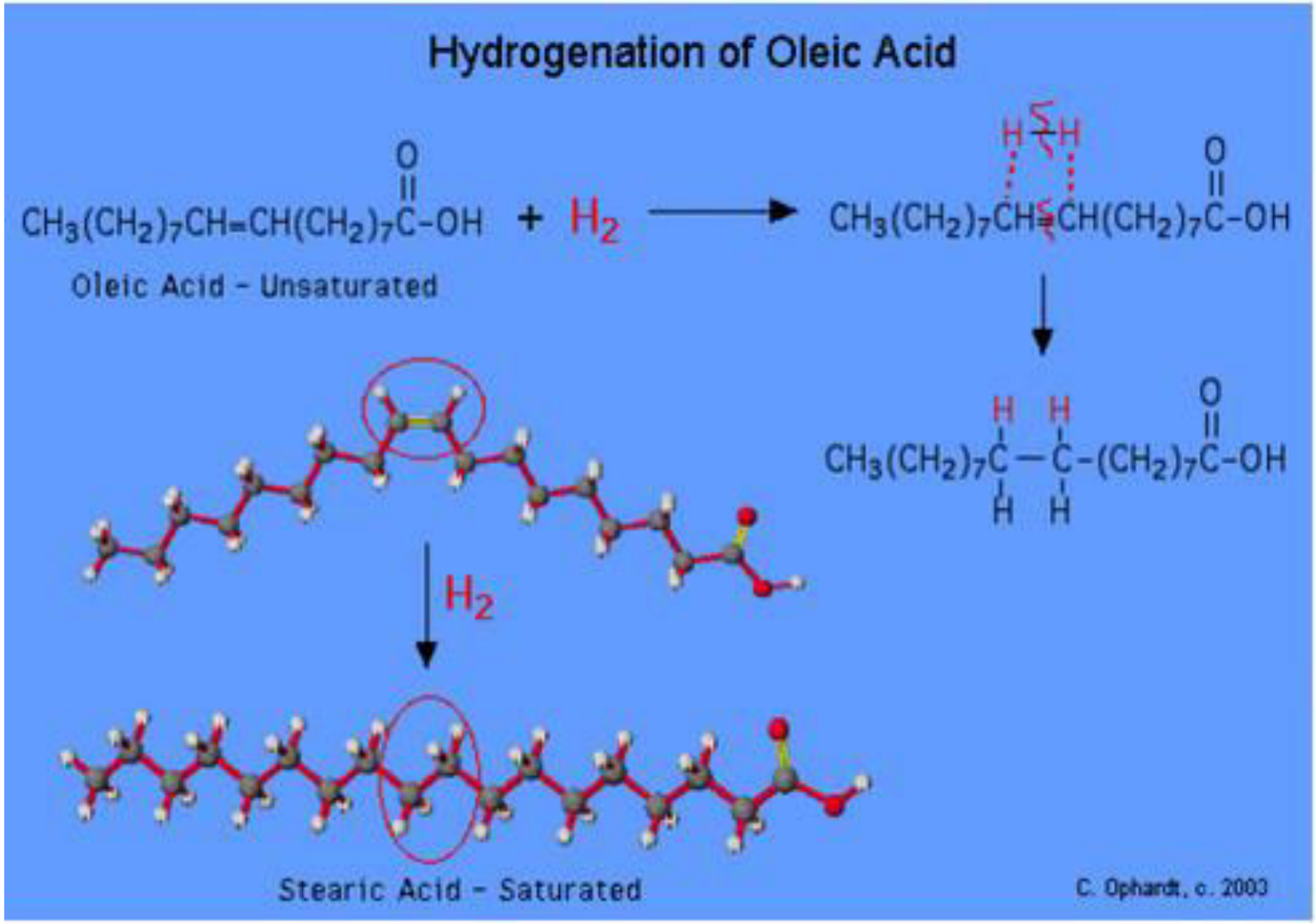
Lipid Chemistry



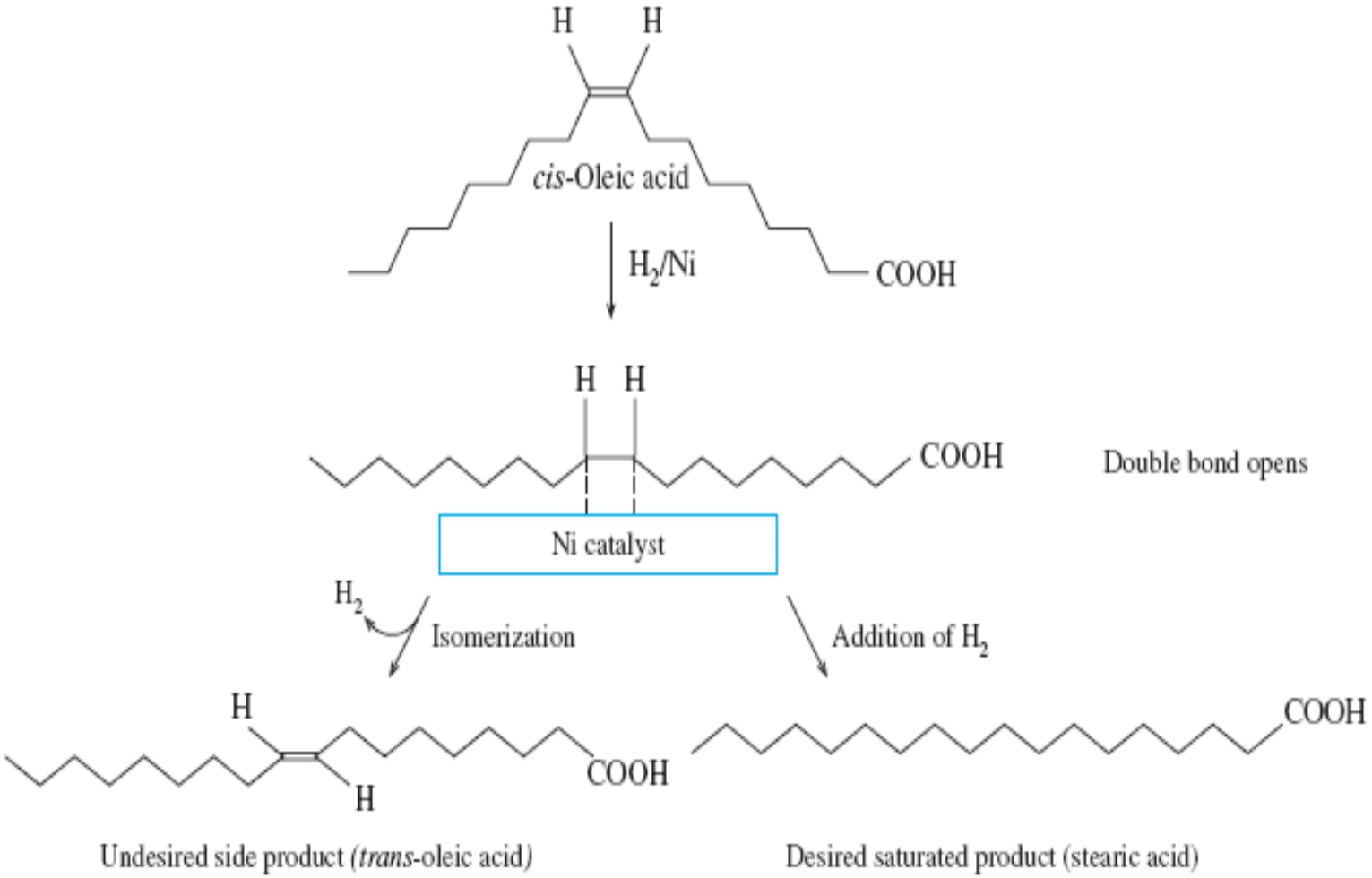
Lipid Chemistry

- Hydrogenation converts
- Unsaturated fatty acids to
- Saturated fatty acids.

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- Margarines are vegetable oils
- treated with partial hydrogenation to
- form semi- solid consistency ‘butter’

Lipid Chemistry

- Hydrogenated fats are used by many commercial food producers to
  - Provide rich texture
  - Increase shelf live
  - Increase melting point
  - Resistance to oxidation and flavour deterioration

Lipid Chemistry

- Partial hydrogenation has an undesirable, effect:
- some cis double bonds are converted
- to trans double bonds.

Lipid Chemistry

- There is a strong evidence that dietary intake of trans fatty acids
- (often referred to as "trans fats")
- leads to a higher incidence of cardio-vascular disease,
- therefore these fats should be avoided in the diet.

Lipid Chemistry

- **Halogenation**
- Similar to hydrogen,
- Halogens such as chlorine, bromine and iodine can also be added to double bonds in
- unsaturated fatty acids.

Lipid Chemistry

- Degree of halogenation is a
- good index of

- degree of unsaturation of Fatty Acids

Lipid Chemistry

- The number of grams of iodine which will be absorbed by 100 grams of a fat is termed its iodine number.

END

Lipid Chemistry

- **Polyunsaturated Fatty Acids Of Biological Importance**

- Polyunsaturated fatty acids contain more than one double bond

Lipid Chemistry

- These include:
- Linoleic acid (18: $2\Delta^{9, 12}$)
- α-Linolenic acid (18 : $3\Delta^{9, 12, 15}$)
- Arachidonic acid series (20 : $4\Delta^{5, 8, 11, 14}$)

Lipid Chemistry

- Of these
- Linoleic acid and
- α-Linolenic acid are essential fatty acids.

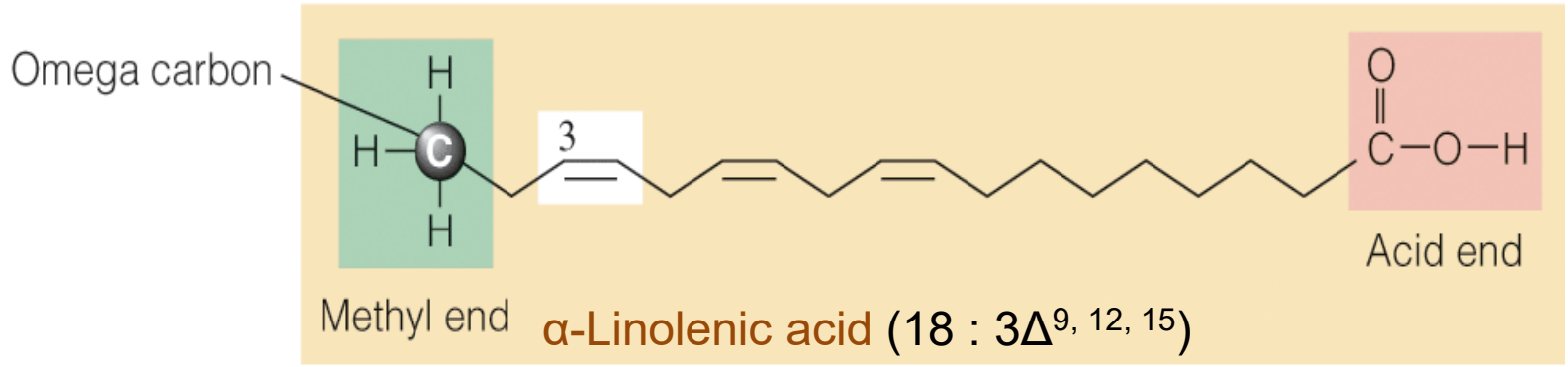
- Whereas, arachidonic acid is synthesized from both Linoleic acid and α-Linolenic acid.

Lipid Chemistry

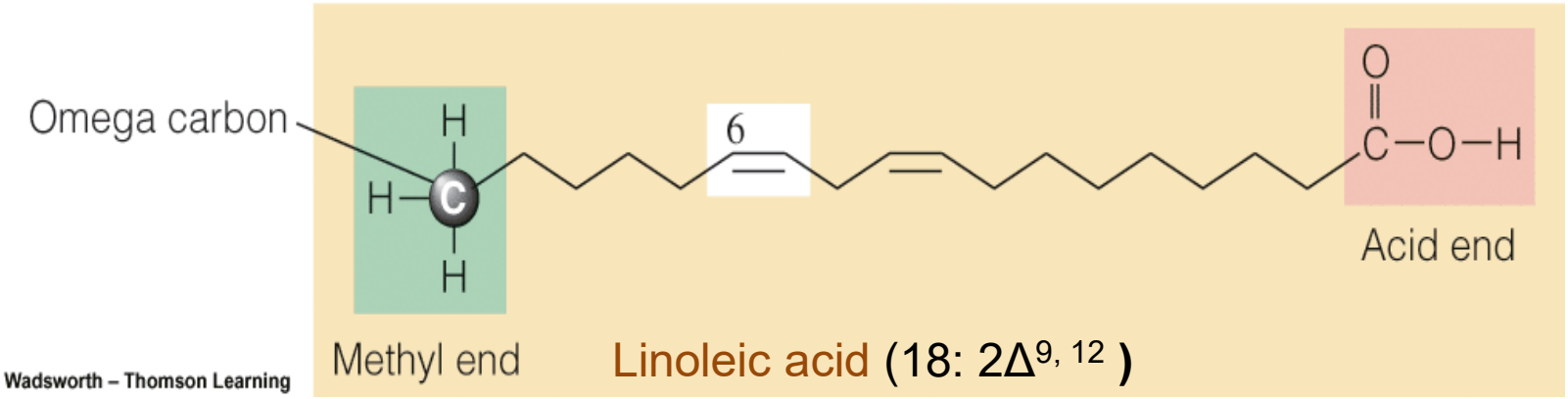
- Arachidonic acid becomes essential if linoleic acid is deficient in the diet

Lipid Chemistry

Linolenic acid, an omega-3 fatty acid

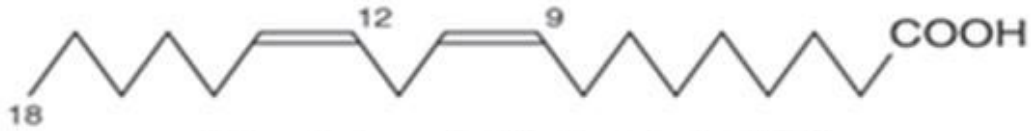


Linoleic acid, an omega-6 fatty acid

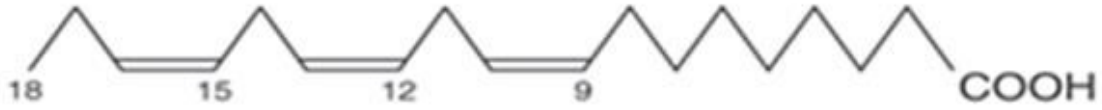


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Lipid Chemistry



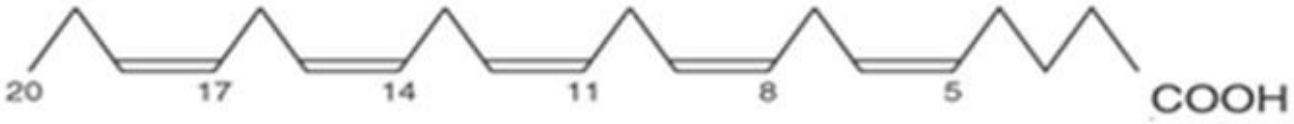
*Linoleic acid (ω6, 18:2, $\Delta^{9,12}$)



*α-Linolenic acid (ω3, 18:3, $\Delta^{9,12,15}$)



*Arachidonic acid (ω6, 20:4, $\Delta^{5,8,11,14}$)



Eicosapentaenoic acid (ω3, 20:5, $\Delta^{5,8,11,14,17}$)

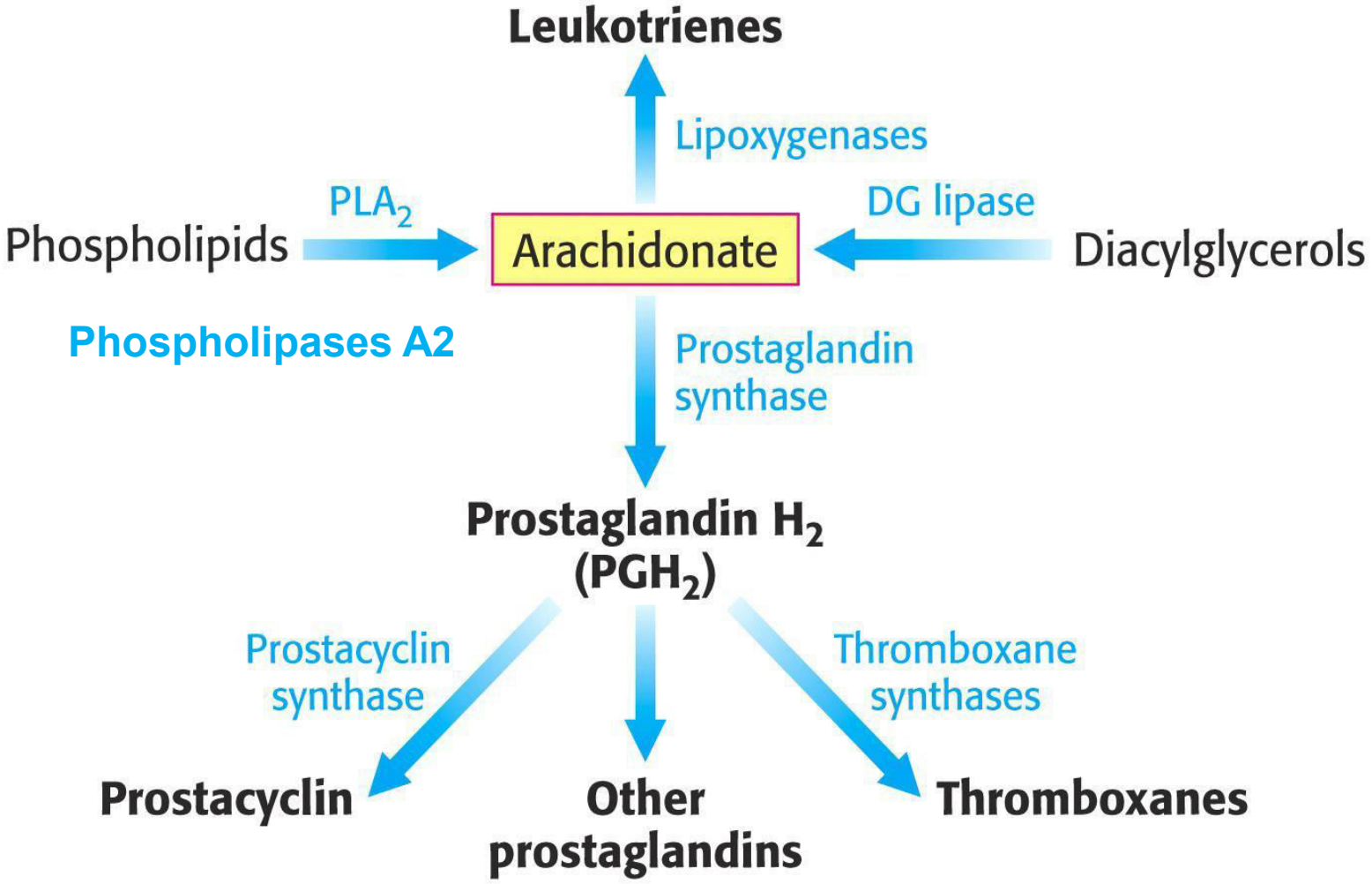
Lipid Chemistry

- Arachidonic acid is the precursor of paracrine hormones called **Eicosanoids.**
- Eicosanoids are chemical messengers which act over a short distance and include:

- Prostaglandins
- Leukotrienes and
- Thromboxanes

Chemical signals secreted by cells can act over varying distances

Lipid Chemistry

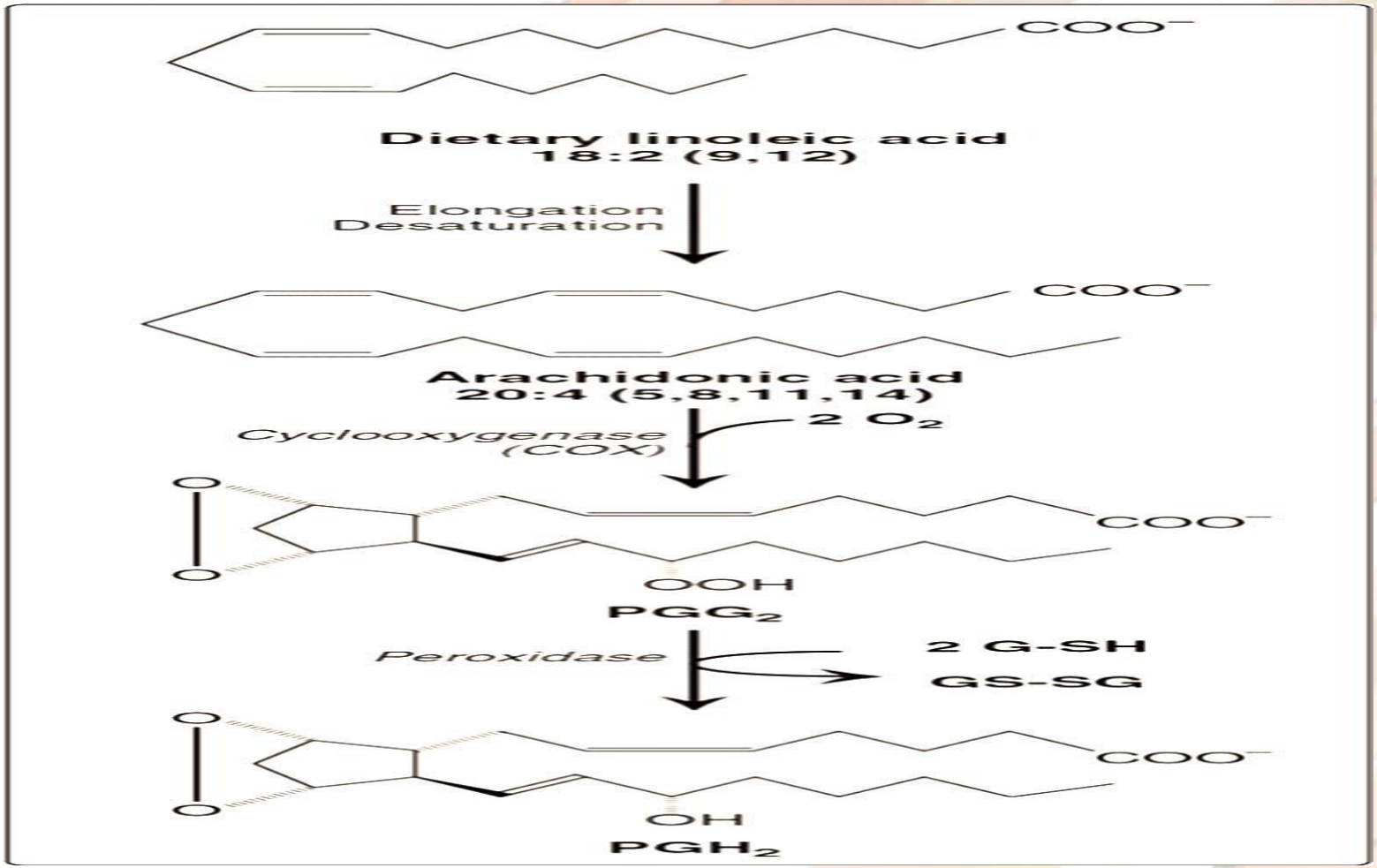


Prostaglandin synthase is more commonly called as *cyclooxgenase*

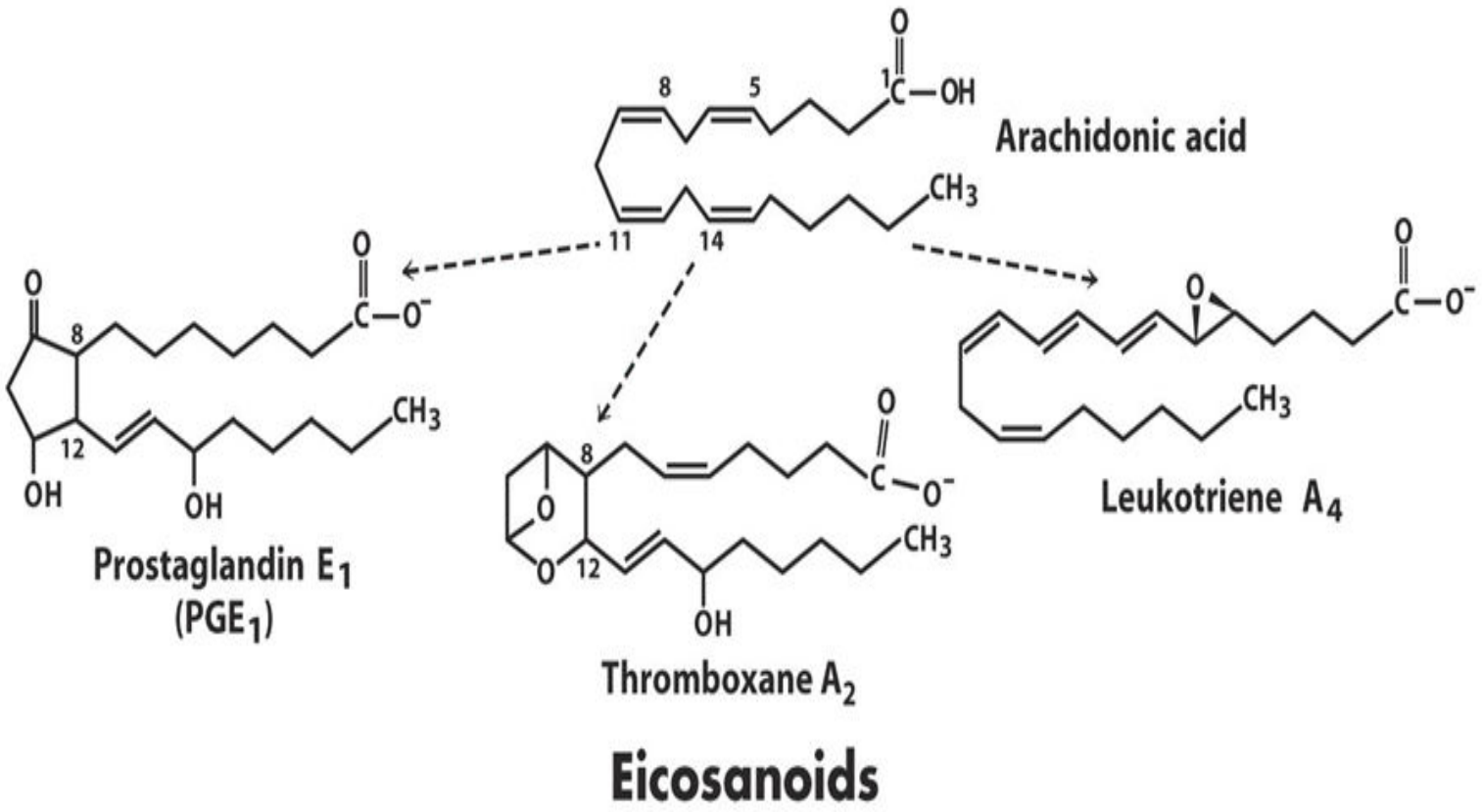
Lipid Chemistry

- These eicosanoids are semi cyclic structures.
- They exert complex control over many bodily systems; in
- inflammation
- immunity, and as
- messengers in the central nervous system.

Lipid Chemistry



Lipid Chemistry



Effort By

Amaan Khan

Midterm Syllabus Finished